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CompTIA® Network+ N10-009 Cert Guide

Anthony Sequeira, CCIE No. 15626



CompTIA® Network+ N10-009 Cert Guide

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Glossary of Key Terms

About the Author

Anthony Sequeira (CCIE No. 15626) began his IT career in 1994 with IBM in Tampa, Florida. He quickly formed his own computer consultancy, Computer Solutions, and then discovered his true passion: teaching and writing about networking technologies. Anthony lectured to massive audiences around the world while working for Mastering Computers. Anthony has never been happier in his career than he is now, as a senior technical instructor for ACI Learning. ACI is a leader in audit, cybersecurity, and IT pro training in self-paced and instructor-led formats.

Dedication

This book is dedicated to my daughter, Bella Sequeira, who inspires me to do great work every day.

Acknowledgments

I cannot thank Nancy Davis and Chris Cleveland enough for their patience as I created this latest edition of the text. Also, huge thanks to my editors, Chris Crayton and Bill McManus. Their work on this text improved it dramatically.

About the Technical Reviewer

Chris Crayton is a technical consultant, trainer, author, and industry-leading technical editor. He has worked as a computer technology and networking instructor, information security director, network administrator, network engineer, and PC specialist. Chris has authored several print and online books on PC repair, CompTIA A+, CompTIA Security+, and Microsoft Windows. He has also served as technical editor and content contributor on numerous technical titles for several of the leading publishing companies. He holds numerous industry certifications, has been recognized with many professional and teaching awards, and has served as a state-level SkillsUSA final competition judge. Chris tech edited and contributed to this book to make it better for students and those wishing to better their lives.

We Want to Hear from You!

As the reader of this book, *you* are our most important critic and commentator. We value your opinion and want to know what we're doing right, what we could do better, what areas you'd like to see us publish in, and any other words of wisdom you're willing to pass our way.

We welcome your comments. You can email or write to let us know what you did or didn't like about this book—as well as what we can do to make our books better.

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When you write, please be sure to include this book's title and author as well as your name and email address. We will carefully review your comments and share them with the author and editors who worked on the book.

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Introduction

The CompTIA Network+ certification is a popular certification for those entering the computer networking field. Although many vendor-specific networking certifications are popular in the industry, the CompTIA Network+ certification is unique in that it is vendor neutral. It does not focus its content on the techniques and technologies of any one specific network vendor. The CompTIA Network+ certification often acts as a stepping-stone to more specialized and vendor-specific certifications, such as those offered by Cisco Systems.

On the CompTIA Network+ exam, the topics are mostly generic in that they can apply to networking equipment regardless of vendor. Although the CompTIA Network+ certification is vendor neutral, network software and systems are implemented by multiple independent vendors. Therefore, several of the exercises, examples, and simulations in this book include using particular vendors' configurations and technologies, such as Microsoft Windows operating systems or Cisco Systems routers and switches. More detailed training for a specific vendor's software and hardware can be found in books and training specific to that vendor.

Who Should Read This Book?

This book was written with two audiences in mind: those who want to learn all they can about networking technology and those who want to pass the CompTIA Network+ exam. I think that both groups are going to be very impressed with the breadth of technologies this book details. Although it would be impossible to cover every topic in networking today, this book manages to cover all the massive areas that make networking an exciting field that many people want to learn.

Readers will range from people who are attempting to attain a position in the IT field to people who want to keep their skills sharp or perhaps retain their job when facing a company policy that mandates they take the new exams. This book is also for those who want to acquire additional certifications beyond the Network+ certification (for example, the Cisco Certified Network Associate [CCNA] certification and beyond). The book is designed to enable an easy transition to future certification studies.

Resources

This book comes with a wealth of digital resources to help you review, practice, and assess your knowledge. The end of each chapter contains a review section that references several of these tools, and you should be sure to use them as you complete each chapter to help reinforce what you are learning. You can use them again after

you finish the book to help review and make sure you are fully prepared for the exam.

Here's a list of resources available on the companion website:

- Interactive glossary flash card application
- Interactive exam essentials appendix
- The Pearson Test Prep (PTP) practice test app
- Video training on key exam topics
- Memory table review exercises and answer keys
- A study planner tool
- Instructions to redeem your Network+ certification exam voucher, which provides a 10% discount on the exam

To access the companion website, follow these steps:

- Step 1.** Go to <https://www.pearsonITcertification.com/register> by December 31, 2027.
- Step 2.** Either log in to your account if you have an existing account already or create a new account.
- Step 3.** Enter the ISBN of this book (**9780135367889**) and click **Submit**.
- Step 4.** Answer the challenge questions to validate your purchase.
- Step 5.** In your account page, click the **Registered Products** tab and then click the **Access Bonus Content** link.

How to Access the Pearson Test Prep (PTP) App

You have two options for installing and using the Pearson Test Prep application: a web app and a desktop app. To use the Pearson Test Prep application, start by finding the registration code that comes with the book. You can find the code in these ways:

- You can get your access code by registering the print ISBN (9780135367889) on [pearsonITcertification.com/register](https://www.pearsonITcertification.com/register). Make sure to use the print book ISBN regardless of whether you purchased an eBook or the print book. Once you register the book, your access code will be populated on your account page under the Registered Products tab. Instructions for how to redeem the code are available on the book's companion website by clicking the Access Bonus Content link.

- **Premium Edition:** If you purchase the Premium Edition eBook and Practice Test directly from the Pearson IT Certification website, the code will be populated on your account page after purchase. Just log in at pearsonITcertification.com, click Account to see details of your account, and click the digital purchases tab.

NOTE After you register your book, your code can always be found in your account under the Registered Products tab.

Once you have the access code, to find instructions about both the PTP web app and the desktop app, follow these steps:

- Step 1.** Open this book's companion website as shown earlier on the first page of the book, under the "Companion Website and Pearson Test Prep Access Code" heading.
- Step 2.** Click the **Practice Exams** button.
- Step 3.** Follow the instructions listed there for both installing the desktop app and for using the web app.

Note that if you want to use the web app only at this point, just navigate to pearsonstestprep.com, log in using the same credentials used to register your book or purchase the Premium Edition, and register this book's practice tests using the registration code you just found. The process should take only a couple of minutes.

Customizing Your Exams

Once you are in the exam settings screen, you can choose to take exams in one of three modes:

- **Study mode:** Enables you to fully customize your exams and review answers as you are taking the exam. This is typically the mode you would use first to assess your knowledge and identify information gaps.
- **Practice Exam mode:** Locks certain customization options, as it is presenting a realistic exam experience. Use this mode when you are preparing to test your exam readiness.
- **Flash Card mode:** Strips out the answers and presents you with only the question stem. This mode is great for late-stage preparation when you really want to challenge yourself to provide answers without the benefit of seeing

multiple-choice options. This mode does not provide the detailed score reports that the other two modes do, so you should not use it if you are trying to identify knowledge gaps.

In addition to these three modes, you will be able to select the source of your questions. You can choose to take exams that cover all of the chapters or you can narrow your selection to just a single chapter or the chapters that make up specific parts in the book. All chapters are selected by default. If you want to narrow your focus to individual chapters, simply deselect all the chapters and then select only those on which you wish to focus in the Objectives area.

You can also select the exam banks on which to focus. Each exam bank comes complete with a full exam of questions that cover topics in every chapter. You can have the test engine serve up exams from all test banks or just from one individual bank by selecting the desired banks in the exam bank area.

There are several other customizations you can make to your exam from the exam settings screen, such as the time of the exam, the number of questions served up, whether to randomize questions and answers, whether to show the number of correct answers for multiple-answer questions, and whether to serve up only specific types of questions. You can also create custom test banks by selecting only questions that you have marked or questions on which you have added notes.

Updating Your Exams

If you are using the online version of the Pearson Test Prep software, you should always have access to the latest version of the software as well as the exam data. If you are using the Windows desktop version, every time you launch the software while connected to the Internet, it checks if there are any updates to your exam data and automatically downloads any changes that were made since the last time you used the software.

Sometimes, due to many factors, the exam data may not fully download when you activate your exam. If you find that figures or exhibits are missing, you may need to manually update your exams. To update a particular exam you have already activated and downloaded, simply click the **Tools** tab and click the **Update Products** button. Again, this is only an issue with the desktop Windows application.

If you wish to check for updates to the Pearson Test Prep exam engine software, Windows desktop version, simply click the **Tools** tab and click the **Update Application** button. This ensures that you are running the latest version of the software engine.

Goals and Methods

The goal of this book is to assist you in learning and understanding the technologies covered in the Network+ N10-009 blueprint from CompTIA. This book also helps you prepare for the N10-009 version of the CompTIA Network+ exam.

To aid you in mastering and understanding the Network+ certification objectives, this book uses the following methods:

- **Opening topics list:** This list spells out the Network+ objectives and topics that are covered in the chapter.
- **Foundation topics:** At the heart of a chapter, the sections under “Foundation Topics” explain the topics from hands-on and theory-based standpoints. These sections include in-depth descriptions, tables, and figures that build your knowledge so that you can pass the N10-009 exam. Each chapter is broken into multiple sections.
- **Key topics:** The “Review All Key Topics” section indicates important figures, tables, and lists of information that you need to review for the exam. Key Topic icons are sprinkled throughout each chapter, and a table at the end of each chapter lists the important parts of the text called out by these icons.
- **Memory tables:** You can find memory tables and their answer key on the book’s companion website in Appendixes B and C, respectively. Use them to help memorize important information.
- **Key terms:** Key terms without definitions are listed at the end of each chapter. Write down the definition of each term and check your work against the definitions in the Glossary. On the companion website, you will find a flash card application with all the glossary terms separated by chapter, and you can use it to study key terms as well.
- **Practice exams:** As previously described, this book comes complete with several full-length practice exams available to you in the Pearson Test Prep practice test software, which you can download and install from the companion website. The Pearson Test Prep software is also available to you online, at www.PearsonTestPrep.com. Follow the directions at the beginning of the book under “Companion Website and Pearson Test Prep Access Code.” Be sure to run through the questions in exam bank 1 as you complete each chapter in study mode. When you have completed the book, take a full practice test using exam bank 2 questions in practice exam mode to test your exam readiness.

For current information about the CompTIA Network+ certification exam, visit <https://www.comptia.org/certifications/network>.

Strategies for Exam Preparation

This book comes with a study planner tool on the companion website. It is a spreadsheet that helps you keep track of the activities you need to perform in each chapter and helps you organize your exam preparation tasks. As you read the chapters in this book, jot down notes with key concepts or configurations in the study planner. Each chapter ends with a summary and series of exam preparation tasks to help you reinforce what you have learned. These tasks include review exercises such as reviewing key topics, completing memory tables, defining key terms, answering review questions, and performing exercises. Make sure you perform these tasks as you complete each chapter to improve your retention of the material and record your progress in the study planner.

The book concludes with Chapter 26, “Final Preparation,” which offers you guidance on your final exam preparation and provides you with some helpful exam advice. Make sure you read over that chapter to help assess your exam readiness and identify areas where you need to focus your review.

Download the current exam objectives by submitting a form on the following web page: <https://www.comptia.org/certifications/network>.

Use the practice exams, which are included on this book’s companion website. As you work through the practice exams, use the practice test software reporting features to note the areas where you lack confidence and then review the related concepts. After you review those areas, work through the practice exams a second time and rate your skills. Keep in mind that the more you work through the practice exams, the more familiar the questions become, and the less accurately the practice exams judge your skills.

After you work through the practice exams a second time and feel confident with your skills, schedule the real CompTIA Network+ exam (N10-009).

CompTIA Network+ Exam Topics

Table I-1 lists general exam topics (*objectives*) and specific topics under each general topic (*subobjectives*) for the CompTIA Network+ N10-009 exam. This table lists the primary chapter in which each exam topic is covered. Note that many objectives and subobjectives are interrelated and are addressed in multiple chapters in the book.

Table I-1 CompTIA Network+ Exam Topics

Chapter	N10-009 Exam Objective	N10-009 Exam Subobjective
Chapter 1: The OSI Model and Encapsulation	1.0 Networking Concepts	1.1 Explain concepts related to the Open Systems Interconnection (OSI) reference model.
Chapter 2: Networking Appliances, Applications, and Functions	1.0 Networking Concepts	1.2 Compare and contrast networking appliances, applications, and functions.
Chapter 3: Cloud Concepts	1.0 Networking Concepts	1.3 Summarize cloud concepts and connectivity options.
Chapter 4: Networking Ports, Protocols, Services, and Traffic Types	1.0 Networking Concepts	1.4 Explain common networking ports, protocols, services, and traffic types.
Chapter 5: Transmission Media and Transceivers	1.0 Networking Concepts	1.5 Compare and contrast transmission media and transceivers.
Chapter 6: Network Topologies, Architectures, and Types	1.0 Networking Concepts	1.6 Compare and contrast network topologies, architectures, and types.
Chapter 7: IPv4 Addressing	1.0 Networking Concepts	1.7 Given a scenario, use appropriate IPv4 network addressing.
Chapter 8: Evolving Use Cases	1.0 Networking Concepts	1.8 Summarize evolving use cases for modern network environments.
Chapter 9: Routing Technologies	2.0 Network Implementation	2.1 Explain characteristics of routing technologies.
Chapter 10: Ethernet Switching Technologies	2.0 Network Implementation	2.2 Given a scenario, configure switching technologies and features.
Chapter 11: Configure Wireless Devices and Technologies	2.0 Network Implementation	2.3 Given a scenario, select and configure wireless devices and technologies.
Chapter 12: Physical Installations	2.0 Network Implementation	2.4 Explain important factors of physical installations.
Chapter 13: Organizational Processes and Procedures	3.0 Network Operations	3.1 Explain the purpose of organizational processes and procedures.
Chapter 14: Network Monitoring	3.0 Network Operations	3.2 Given a scenario, use network monitoring technologies.
Chapter 15: Disaster Recovery	3.0 Network Operations	3.3 Explain disaster recovery (DR) concepts.

Chapter	N10-009 Exam Objective	N10-009 Exam Subobjective
Chapter 16: IPv4 and IPv6 Network Services	3.0 Network Operations	3.4 Given a scenario, implement IPv4 and IPv6 network services.
Chapter 17: Network Access and Management Methods	3.0 Network Operations	3.5 Compare and contrast network access and management methods.
Chapter 18: Network Security Concepts	4.0 Network Security	4.1 Explain the importance of basic network security concepts.
Chapter 19: Types of Network Attacks	4.0 Network Security	4.2 Summarize various types of attacks and their impact to the network.
Chapter 20: Network Security Features	4.0 Network Security	4.3 Given a scenario, apply network security features, defense techniques, and solutions.
Chapter 21: A Network Troubleshooting Methodology	5.0 Network Troubleshooting	5.1 Explain the troubleshooting methodology.
Chapter 22: Troubleshoot Common Cabling Problems	5.0 Network Troubleshooting	5.2 Given a scenario, troubleshoot common cabling and physical interface issues.
Chapter 23: Troubleshoot Common Issues with Network Services	5.0 Network Troubleshooting	5.3 Given a scenario, troubleshoot common issues with network services.
Chapter 24: Troubleshoot Common Performance Issues	5.0 Network Troubleshooting	5.4 Given a scenario, troubleshoot common performance issues.
Chapter 25: Network Troubleshooting Tools	5.0 Network Troubleshooting	5.5 Given a scenario, use the appropriate tool or protocol to solve networking issues.

How This Book Is Organized

Although this book could be read cover to cover, it is designed to be flexible and allow you to easily move between chapters and sections of chapters to cover just the material that you need more work with. However, if you do intend to read all the chapters, the order in the book is an excellent sequence to use:

- **Chapter 1: The OSI Model and Encapsulation**—This chapter details the OSI model and its seven layers. This chapter also presents the encapsulation and deencapsulation processes associated with this important model.
- **Chapter 2: Networking Appliances, Applications, and Functions**—This chapter presents commonly used physical and virtual appliances in use in

networks today. This chapter also covers commonly used applications and network functions.

- **Chapter 3: Cloud Concepts**—This chapter covers some of the most important cloud concepts used in modern networking. Just some of the topics covered include deployment models, service models, and cloud connectivity options.
- **Chapter 4: Networking Ports, Protocols, Services, and Traffic Types**—This chapter ensures you are familiar with some of the most popular ports, protocols, and services in use today. This chapter also covers the traffic types of unicast, multicast, anycast, and broadcast.
- **Chapter 5: Transmission Media and Transceivers**—This chapter is all about the wireless and wired connections we make in modern networks. This chapter also covers the transceivers and connector types we often see in our networks today.
- **Chapter 6: Network Topologies, Architectures, and Types**—This chapter covers common network topologies, including mesh, hybrid, hub and spoke, and more. This chapter also covers common traffic flow types found in today's networking environments.
- **Chapter 7: IPv4 Addressing**—This chapter is all about IPv4 addressing. This includes coverage of public versus private IP addressing, subnetting, and the IPv4 address classes.
- **Chapter 8: Evolving Use Cases**—This chapter is all about some of the more cutting-edge technologies appearing in our most modern of networks today. Just some of the technologies that are covered include software-defined networking, Virtual Extensible Local Area Network (VXLAN), and IPv6 addressing.
- **Chapter 9: Routing Technologies**—This chapter tackles technologies that are specific to routing in our networks. This includes such topics as static versus dynamic routing, route selection, address translation, and first hop redundancy protocols.
- **Chapter 10: Ethernet Switching Technologies**—This chapter describes technologies related to switching in today's networks. This includes such topics as VLANs, switch interface configurations, and Spanning Tree Protocol.
- **Chapter 11: Configure Wireless Devices and Technologies**—This chapter examines topics related to Wi-Fi in networks today. This includes topics such as frequency options, network types, encryption, and many more.

- **Chapter 12: Physical Installations**—This chapter examines several important factors you should consider when you are planning and operating physical installations for networking equipment. This chapter includes topics like power and environmental factors.
- **Chapter 13: Organizational Processes and Procedures**—This chapter focuses on the purpose of organizational processes and procedures. Topics like documentation, life-cycle management, change management, and configuration management are all covered.
- **Chapter 14: Network Monitoring**—This chapter presents many different options and methods when it comes to monitoring modern networks. Topics include SNMP, SIEMs, and many more.
- **Chapter 15: Disaster Recovery**—This chapter focuses on disaster recovery (DR) topics. This includes things like DR metrics, DR sites, and high-availability approaches.
- **Chapter 16: IPv4 and IPv6 Network Services**—This chapter presents many examples of popular IPv4 and IPv6 services found in networks today. Topics in this chapter include dynamic addressing options, name resolution services, and time protocols.
- **Chapter 17: Network Access and Management Methods**—This chapter presents various options available today for network access and network management. Topics include VPNs, connection methods, and specific technologies like jump boxes.
- **Chapter 18: Network Security Concepts**—This chapter is a big one. Why? It tackles the hugely important topic of basic network security concepts. Here you will learn about things like logical security, physical security, and common security terminology.
- **Chapter 19: Types of Network Attacks**—What are some of the most common network attacks in use today? This chapter addresses this question head on and covers such topics as DoS and DDoS attacks, social engineering attacks, and many, many more.
- **Chapter 20: Network Security Features**—This chapter covers network security features, defense techniques, and solutions. This includes a discussion of device hardening, network access controls, key management, security rules, and zones.
- **Chapter 21: A Network Troubleshooting Methodology**—This chapter provides guidance on a well-planned and effective troubleshooting methodology. This methodology begins with problem identification and ends with the documentation of findings.

- **Chapter 22: Troubleshoot Common Cabling Problems**—This chapter covers troubleshooting common cabling and physical interface issues. The chapter focuses on three main areas: cable issues, interface issues, and hardware issues.
- **Chapter 23: Troubleshoot Common Issues with Network Services**—This chapter focuses on troubleshooting common issues with network services. This includes the two main areas of switching- and routing-based services.
- **Chapter 24: Troubleshoot Common Performance Issues**—These days, a network that is performing poorly can be nearly as disruptive as a network that is not functioning at all. This chapter guides you through troubleshooting the most common of performance issues. Both wired and wireless networks are discussed.
- **Chapter 25: Network Troubleshooting Tools**—This chapter focuses on the use of network troubleshooting tools that can help you solve common issues. This includes the three main categories of tools: software tools, hardware tools, and basic networking device commands.

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Figures 2-2 to 2-18, 4-1 to 4-4, 5-2, 6-1 to 6-10, 7-13, 8-1, 8-3 to 8-5, 9-1 to 9-16, 10-1, 10-4 to 10-8, 10-10 to 10-17, 11-1 to 11-7, 11-11, 14-1, 14-2, 14-4 to 14-6, 15-1 to 15-4, 15-6, 16-1, 16-3, 17-1, 17-2, 17-4, 18-1, 18-2, 19-1, 24-1 courtesy of Cisco Systems, Inc.



This chapter covers the following topics related to Objective 1.1 (Explain concepts related to the Open Systems Interconnection [OSI] reference model) of the CompTIA Network+ N10-009 certification exam:

- OSI model
 - Layer 1—Physical
 - Layer 2—Data link
 - Layer 3—Network
 - Layer 4—Transport
 - Layer 5—Session
 - Layer 6—Presentation
 - Layer 7—Application

The OSI Model and Encapsulation

Way back in 1977, the International Organization for Standardization (ISO) developed a subcommittee to focus on the interoperability of multivendor communications systems. This is fancy language for getting network “stuff” to communicate with other network “stuff,” even if different companies made the network “stuff.” What sprang from this subcommittee was the *Open Systems Interconnection (OSI) reference model* (also referred to as the *OSI model* or the *OSI stack*). Thanks to this model, you can talk about any networking technology and categorize that technology as residing at one or more of the seven layers of the model.

This chapter defines those seven layers and offers examples of what you might find at each layer. It also contrasts the OSI model with another model—the TCP/IP stack, also known as the Department of Defense (DoD) model—that focuses on Internet Protocol (IP) communications.

Foundation Topics

The Purpose of Reference Models

Throughout this book, various protocols and devices that play a role in your network (and your networking career) are introduced. To better understand how a technology fits in, it helps to have a common point of reference against which various technologies from different vendors can be compared. The OSI model provides us an ideal point of reference for our learning and discovery.

Understanding the OSI model is also very useful in troubleshooting networks. In fact, Chapter 21, “A Network Troubleshooting Methodology,” provides some concrete methods for using the OSI model in conjunction with thorough network troubleshooting.

One of the most common ways of categorizing the function of a network technology is to say at what layer (or layers) of the OSI model that technology runs. Understanding how that technology performs a certain function at a certain layer of the OSI model helps you determine whether one device is going to be able to communicate with another device, which might or might not be using a similar technology, at that layer of the OSI reference model.

For example, when your end-user device connects to a web server on the Internet, your service provider assigns your device an IP address. Similarly, the web server to which you are communicating has an IP address. As described in this chapter, an IP address lives at Layer 3 (the network layer) of the OSI model. Because your device and the web server use a common protocol (that is, IP) at Layer 3, they are capable of communicating with one another.

Notice also in this example that you are interested in receiving the data from the web server, which will be web pages filled with text and graphics and maybe even videos. This is the information you are really after. You (typically) do not care about the IP addresses in use or any of the other information required by the network devices to make this transfer happen. In technical terms, you are interested in the *payload* of the packets sent from the web server. The *payload* provides a simple and generic method of describing the data itself, which is separate and distinct from any of the other information required for proper transmission.

Personally, I have been in the computer-networking industry since 1996, and I have had the OSI model explained in many classes I have attended and books I have read. From this, I have taken away a collection of metaphors to help describe the operation of the different layers of the OSI model. Some of the metaphors involve sending a letter from one location to another or placing a message in a series of

envelopes. These are often excellent metaphors for encapsulation and decapsulation (covered later in this chapter), but they do not work all that well for the OSI model in general. My favorite way to describe the OSI model is to simply think of it as being analogous to a bookcase, such as the one shown in Figure 1-1.



Figure 1-1 A Bookcase Is Analogous to the OSI Model

If you were to look at this or any other bookcase in my home office, you would see that I have organized diverse types of books on different shelves. One shelf holds my collection of technical books, another shelf holds the books I wrote for Pearson and other publishers, another shelf holds books regarding self-improvement and finance. I have grouped similar books together on each shelf, just as the OSI model groups similar protocols and functions together in a layer.

A common pitfall my readers meet when studying the OSI model is to try to neatly fit all the devices and protocols in their network into one of the OSI model's seven layers. However, not every technology fits perfectly into these layers. In fact, some networks might not have any technologies running at one or more of these layers. This reminds me of my favorite statement about the OSI model. It comes from Rich Seifert's book *The Switch Book*. In that book, Rich reminds us that the OSI model is a *reference* model, not a *reverence* model. That is, no cosmic law states that all technologies must cleanly plug into the model. So, as you discover the characteristics of

the OSI model layers throughout this chapter, remember that these layers are like shelves for organizing similar protocols and functions, not immutable laws.

The OSI Model

As previously described, the OSI model consists of seven layers:

Key Topic

- **Layer 1:** The physical layer
- **Layer 2:** The data link layer
- **Layer 3:** The network layer
- **Layer 4:** The transport layer
- **Layer 5:** The session layer
- **Layer 6:** The presentation layer
- **Layer 7:** The application layer

Graphically, we depict these layers with Layer 1 at the bottom of the stack, as shown in Figure 1-2.

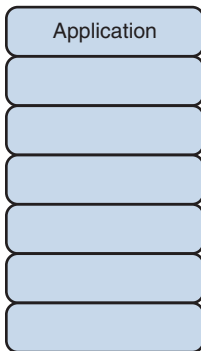


Figure 1-2 The OSI Stack

NOTE Various mnemonics are available to help memorize these layers in their proper order. A top-down (that is, starting at the top of the stack with Layer 7 and working your way down to Layer 1) memory aid is *All People Seem To Need Data Processing*. Another common technique is *Please Do Not Throw Sausage Pizza Away*, which begins at Layer 1 and works up to Layer 7.

At the physical layer, binary expressions (that is, a series of 1s and 0s) represent data. A binary expression is created using bits, where a bit is a single 1 or a single 0. At upper layers, however, bits are grouped together, into what is known as a **protocol data unit (PDU)** or a *data service unit*.

Engineers tend to use the term *packet* generically to refer to these PDUs. However, PDUs might have an added name, depending on their OSI layer. Figure 1-3 illustrates these PDU names. A common memory aid for these PDUs is *Some People Fear Birthdays*, where the *S* in *Some* reminds us of the *S* in *Segments*. The *P* in *People* reminds us of the *P* in *Packets*, and the *F* in *Fear* reflects the *F* in *Frames*. Finally, the *B* in *Birthdays* reminds us of the *B* in *Bits*.

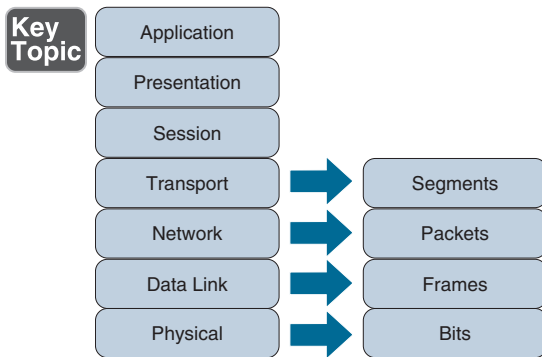


Figure 1-3 PDU Names

Layer 1: The Physical Layer

The concern of the **physical layer**, as shown in Figure 1-4, is the transmission of bits on the network along with the physical and electrical characteristics of the network.

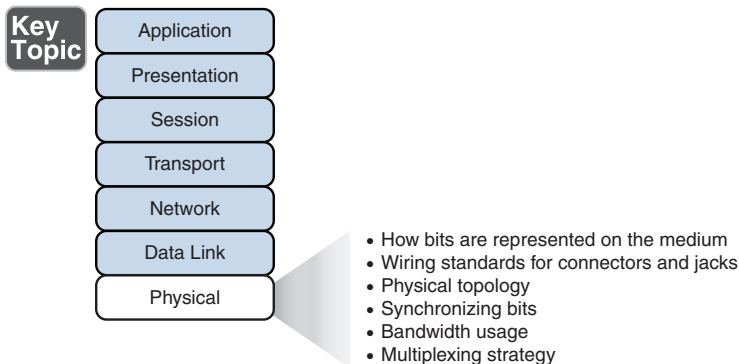


Figure 1-4 Layer 1: The Physical Layer

The physical layer defines the following:

- **How to represent bits on the medium:** Data on a computer network is represented as a binary expression. Chapter 7, “IPv4 Addressing,” discusses binary in much more detail. Electrical voltage (on copper wiring) or light (carried via fiber-optic cabling) can represent these 1s and 0s.

For example, the presence or absence of voltage on a wire portrays a binary 1 or a binary 0, respectively, as illustrated in Figure 1-5. Similarly, the presence or absence of light on a fiber-optic cable renders a 1 or 0 in binary. This type of approach is called ***current state modulation***.

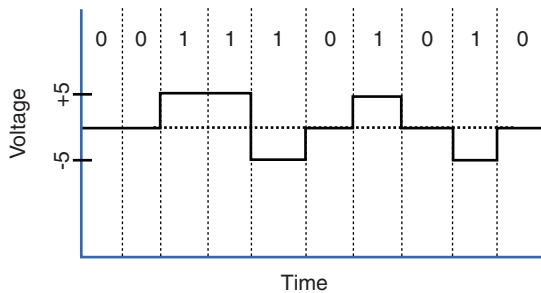


Figure 1-5 Current State Modulation

An alternative approach to portraying binary data is ***state transition modulation***, as shown in Figure 1-6, where the transition between voltages or the presence of light shows a binary value.

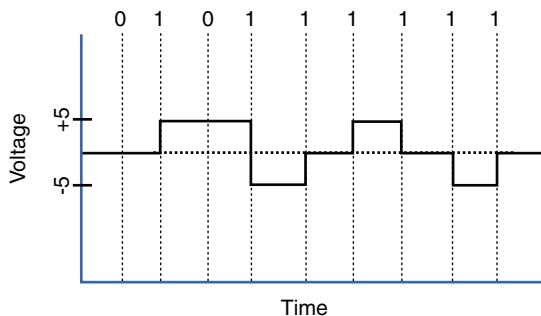


Figure 1-6 Transition Modulation

NOTE Other modulation types you might be familiar with from radio include amplitude modulation (AM) and frequency modulation (FM). AM uses a variation in a waveform’s amplitude (that is, signal strength) to portray the original signal. FM uses a variation in frequency to stand for the original signal.

- **Wiring standards for connectors and jacks:** Chapter 5, “Transmission Media and Transceivers,” describes several standards for network connectors. For example, the TIA/EIA-568-B standard describes how to wire an RJ-45 connector for use on a 100BASE-TX Ethernet network, as shown in Figure 1-7.

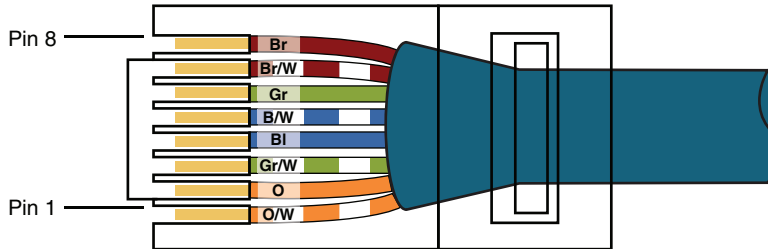


Figure 1-7 TIA/EIA-568-B Wiring Standard for an RJ-45 Connector

- **Physical topology:** Layer 1 devices view a network as a physical topology (as opposed to a logical topology). Examples of a physical topology include star and hub-and-spoke topologies, as described in Chapter 6, “Network Topologies, Architectures, and Types.”
- **Synchronizing bits:** For two networked devices to successfully communicate at the physical layer, they must agree on when one bit stops and another bit starts. Specifically, the devices need a method to synchronize the bits. Two basic approaches to bit synchronization are *asynchronous* and *synchronous* synchronization:
 - **Asynchronous:** With this approach, a sender states that it is about to start transmitting by sending a start bit to the receiver. When the receiver sees this, it starts its own internal clock to measure the next bits. After the sender transmits its data, it sends a stop bit to say that it has finished its transmission.
 - **Synchronous:** This approach synchronizes the internal clocks of the sender and the receiver to ensure that they agree on when bits begin and end. A common approach to make this synchronization happen is to use an external clock (for example, a clock provided by a service provider). The sender and receiver then reference this external clock.
- **Bandwidth usage:** The two fundamental approaches to bandwidth usage on a network are *broadband* and *baseband*:
 - **Broadband:** Broadband technologies divide the bandwidth available on a medium (for example, copper or fiber-optic cabling) into different

channels. A sender can then transmit different communication streams over the various channels. For example, consider frequency-division multiplexing (FDM) used by a cable modem. Specifically, a cable modem uses certain ranges of frequencies on the cable coming into your home from the local cable company to carry incoming data, another range of frequencies for outgoing data, and several other frequency ranges for various TV stations.

- **Baseband:** Baseband technologies use all the available frequencies on a medium to send data. Ethernet is an example of a networking technology that uses baseband.
- **Multiplexing strategy:** Multiplexing allows multiple communications sessions to share the same physical medium. Cable TV, as previously mentioned, allows you to receive multiple channels over a single physical medium (for example, a coaxial cable plugged into the back of your television). Here are some of the most common approaches to multiplexing:
 - ***Time-division multiplexing (TDM):*** TDM supports different communication sessions (for example, different telephone conversations in a telephony network) on the same physical medium by causing the sessions to take turns. For a brief period, defined as a *time slot*, data from the first session is sent, followed by data from the second session. This continues until all sessions have had a turn, and the process repeats.
 - **Statistical time-division multiplexing (StatTDM):** A downside to TDM is that each communication session receives its own time slot, even if one of the sessions does not have any data to send at the moment. To make more efficient use of available bandwidth, StatTDM dynamically assigns time slots to communications sessions on an as-needed basis.
 - **Frequency-division multiplexing (FDM):** FDM divides a medium's frequency range into channels, and different communication sessions send their data over different channels. As previously described, this approach to bandwidth usage is called *broadband*.
 - **Orthogonal frequency-division multiplexing (OFDM):** OFDM encodes digital data onto multiple carrier frequencies. OFDM is very popular today and is used in wideband digital communication. This makes OFDM useful in applications such as digital television and audio broadcasting, DSL Internet access, wireless networks, powerline networks, and 4G/5G mobile communications.

- **Orthogonal frequency-division multiple access (OFDMA):** OFDMA is a multiuser version of the popular OFDM digital modulation scheme. It divides a wireless channel into subchannels to allow simultaneous data transmission from multiple users, enhancing network efficiency and reducing latency.

Examples of devices defined by physical layer standards include legacy hubs, wireless access points, and network cabling.

NOTE Hubs are not used in modern computer networks. So why are we even bothering to mention them? Well, they really did help give rise to our modern switches. A hub interconnects PCs in a LAN; it is considered a physical layer device because it takes bits coming in on one port and retransmits those bits out all other ports. At no point does the hub learn any addressing information from the data as our modern switches do.

Layer 2: The Data Link Layer

The *data link layer* is concerned with the following:

- Packaging data into frames and transmitting those frames on the network
- Ensuring that frames do not exceed the *maximum transmission unit (MTU)* of the physical media
- Performing error detection/correction
- Uniquely finding network devices with addresses
- Handling flow control

NOTE Network interfaces use the MTU to define the largest packet size the interface will forward. For example, a 1500-byte packet could not be forwarded via a router interface with an MTU of 1470 bytes. In Chapter 22, “Troubleshoot Common Cabling Problems,” you will learn about *giants* and *runts*—packets that are too large or too small for the network on which they reside.

Data link layer processes, collectively referred to as *data link control (DLC)*, are illustrated in Figure 1-8.

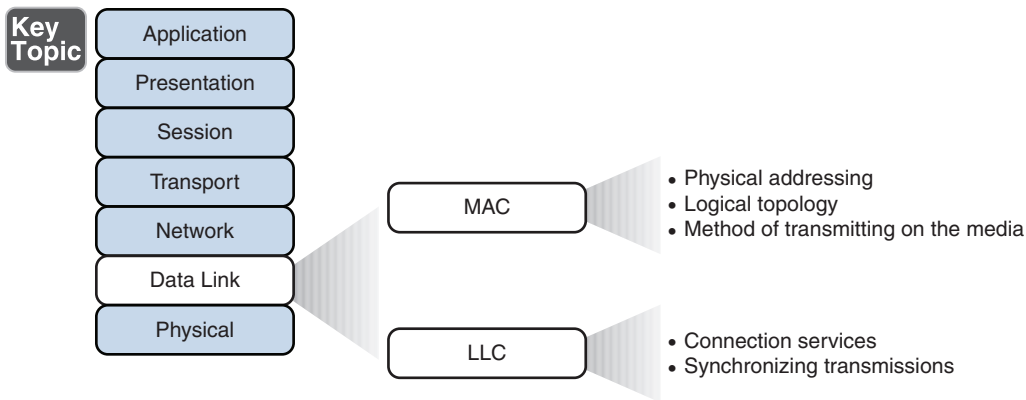


Figure 1-8 Layer 2: The Data Link Layer

In fact, the data link layer is distinct from the other layers in that it has two sublayers: MAC and LLC.

Media Access Control

Characteristics of the Media Access Control (MAC) sublayer of the data link layer include the following:

- **Physical addressing:** A common example of a Layer 2 address is a MAC address, which is a 48-bit address assigned to a device's network interface card (NIC). MAC addresses are written in hexadecimal notation (for example, 58:55:ca:eb:27:83). The first 24 bits of the 48-bit address are the *vendor code*. The IEEE Registration Authority assigns a manufacturer one or more unique vendor codes. You can use the list of vendor codes at <https://standards-oui.ieee.org/oui/oui.txt> to identify the manufacturer of a networking device, based on the first half of the device's MAC address. The last 24 bits of a MAC address are assigned by the manufacturer, and they act as a serial number for the device. No two MAC addresses in the world should have the same value.
- **Logical topology:** Layer 2 devices view a network as a logical (as opposed to physical) topology. Chapter 6 has lots more to say regarding physical and logical topologies.
- **Method of transmitting on the media:** With several devices connected to a network, there needs to be some strategy for deciding when a device sends on the media. Otherwise, multiple devices might send at the same time and interfere with one another's transmissions.

Logical Link Control

Characteristics of the Logical Link Control (LLC) sublayer of the data link layer include the following:

- **Connection services:** When a device on a network receives a message from another device on the network, that recipient device can give feedback to the sender in the form of an acknowledgment message. The two main functions provided by these acknowledgment messages are as follows:
 - **Flow control:** Limits the amount of data a sender can send at one time; this prevents the sender from overwhelming the receiver with too much information.
 - **Error control:** Allows the recipient of data to let the sender know whether the expected data frame was not received or whether it was received but is corrupted. The recipient figures out whether the data frame is corrupt by mathematically calculating a checksum of the data received. If the calculated checksum does not match the checksum received with the data frame, the recipient of the data draws the conclusion that the data frame is corrupted and can then notify the sender via an acknowledgment message.
- **Synchronizing transmissions:** Senders and receivers of data frames need to coordinate when a data frame is being transmitted and should be received. The three methods of performing this synchronization are detailed here:
 - **Isochronous:** With isochronous transmission, network devices look to a common device in the network as a clock source, which creates fixed-length time slots. Network devices can determine how much free space, if any, is available within a time slot and then insert data into an available time slot. A time slot can accommodate more than one data frame. Isochronous transmission does not need to provide clocking at the beginning of a data string (as does synchronous transmission) or for every data frame (as does asynchronous transmission). As a result, isochronous transmission uses little overhead compared to asynchronous or synchronous transmission methods.
 - **Asynchronous:** With asynchronous transmission, network devices reference their own internal clocks, and network devices do not need to synchronize their clocks. Instead, the sender places a start bit at the beginning of each data frame and a stop bit at the end of each data frame. These start and stop bits tell the receiver when to monitor the medium for the presence of bits.

An additional bit, called the *parity bit*, might also be added to the end of each byte in a frame to detect an error in the frame. For example, if even parity error detection (as opposed to odd parity error detection) is used, the parity bit (with a value of either 0 or 1) would be added to the end of a byte, causing the total number of 1s in the data frame to be an even number. If the receiver of a byte is configured for even parity error detection and receives a byte where the total number of bits (including the parity bit) is even, the receiver can conclude that the byte was not corrupted during transmission.

NOTE Using a parity bit to detect errors might not be effective if a byte has more than one error (that is, if more than one bit has been changed from its original value).

- **Synchronous:** With synchronous transmission, two network devices that want to communicate between themselves must agree on a clocking method to show the beginning and ending of data frames. One approach to providing this clocking is to use a separate communications channel over which a clock signal is sent. Another approach relies on specific bit combinations or control characters to indicate the beginning of a frame or a byte of data.

Like asynchronous transmissions, synchronous transmissions can perform error detection. However, rather than using parity bits, synchronous communication runs a mathematical algorithm on the data to create a *cyclic redundancy check (CRC)*. If the sender and the receiver calculate the same CRC value for the same chunk of data, the receiver can conclude that the data was not corrupted during transmission.

Examples of devices defined by data link layer standards include switches, bridges, and NICs.

NOTE NICs are not entirely defined at the data link layer because they are partially based on physical layer standards, such as a NIC's network connector.

Layer 3: The Network Layer

The *network layer*, as shown in Figure 1-9, is primarily concerned with forwarding data based on logical addresses.

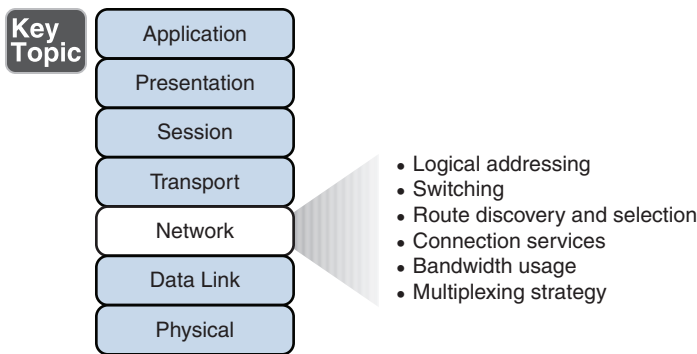


Figure 1-9 Layer 3: The Network Layer

Although many network administrators think of routing and IP addressing when they hear about the network layer, this layer is actually responsible for a variety of tasks:

- **Logical addressing:** Whereas the data link layer uses physical addresses to make forwarding decisions, the network layer uses logical addressing to make forwarding decisions. By far the most widely deployed routed protocol is Internet Protocol (IP). Chapter 7 discusses IP addressing in detail.
- **Switching:** Engineers often associate the term *switching* with Layer 2 technologies; however, the concept of switching also exists at Layer 3. Switching, at its essence, is making decisions about how data should be forwarded. At Layer 3, three common switching techniques exist:
 - **Packet switching:** With packet switching, a data stream is divided into packets. Each packet has a Layer 3 header that includes source and destination Layer 3 addresses. Another term for packet switching is *routing*, which is discussed in more detail in Chapter 9, “Routing Technologies.”
 - **Circuit switching:** Circuit switching dynamically brings up a dedicated communication link between two parties for those parties to communicate.

As a simple example of circuit switching, think of making a phone call from your home to a business. In fact, let’s go “old school” and pretend you have a traditional landline servicing your phone, the telephone company’s switching equipment interconnects your home phone with the phone system of the business you are calling. This interconnection (that is, *circuit*) exists only for the duration of the phone call.

- **Message switching:** Unlike packet switching and circuit switching technologies, message switching is usually not well suited for real-time

applications because of the delay involved. Specifically, with message switching, a data stream is divided into messages. Each message is tagged with a destination address, and the messages travel from one network device to another network device on the way to their destination. Because these devices might briefly store the messages before forwarding them, a network using message switching is sometimes called a *store-and-forward* network. Metaphorically, you could visualize message switching like routing an email message, where the email message might be briefly stored on an email server before being forwarded to the recipient.

- **Route discovery and selection:** Because Layer 3 devices make forwarding decisions based on logical network addresses, a Layer 3 device might need to know how to reach various network addresses. For example, a common Layer 3 device is a router. A router can maintain a routing table indicating how to forward a packet based on the packet's destination network address.

A router can have its routing table populated via manual configuration (that is, by entering static routes), via a dynamic routing protocol (for example, OSPF or EIGRP), or simply by being directly connected to certain networks.

NOTE Routing protocols are discussed in Chapter 9.

- **Connection services:** Just as the data link layer offers connection services for flow control and error control, connection services also exist at the network layer. Connection services at the network layer can improve the communication reliability if the data link's LLC sublayer is not performing connection services. The following functions are performed by connection services at the network layer:
 - **Flow control (also known as congestion control):** Helps prevent a sender from sending data more rapidly than the receiver is capable of receiving it.
 - **Packet reordering:** Allows packets to be placed in the proper sequence as they are sent to the receiver. This might be necessary because some networks support load balancing, where multiple links are used to send packets between two devices. Because multiple links exist, packets might arrive out of order.

Examples of devices found at the network layer include routers and multilayer switches. The most common Layer 3 protocol in use, and the protocol on which the Internet is based, is IPv4. However, IPv6 is beginning to be more common on networks today.

Layer 4: The Transport Layer

The **transport layer**, as shown in Figure 1-10, acts as a dividing line between the upper layers and lower layers of the OSI model. Specifically, messages are taken from upper layers (Layers 5–7) and are encapsulated into segments for transmission to the lower layers (Layers 1–3). Similarly, data streams coming from lower layers are decapsulated and sent to Layer 5 (the session layer), or some other upper layer, depending on the protocol.

Two common transport layer protocols are TCP and UDP:

- **Transmission Control Protocol (TCP):** TCP is a connection-oriented transport protocol. Connection-oriented transport protocols offer reliable transport, in that if a segment is dropped, the sender can detect the drop and retransmit the dropped segment. Specifically, a receiver acknowledges segments that it receives. Based on those acknowledgments, a sender can decide which segments were successfully received and which segments need to be transmitted again.
- **User Datagram Protocol (UDP):** UDP is a connectionless transport protocol. Connectionless transport protocols offer unreliable transport, in that if a segment is dropped, the sender is unaware of the drop, and no retransmission occurs.

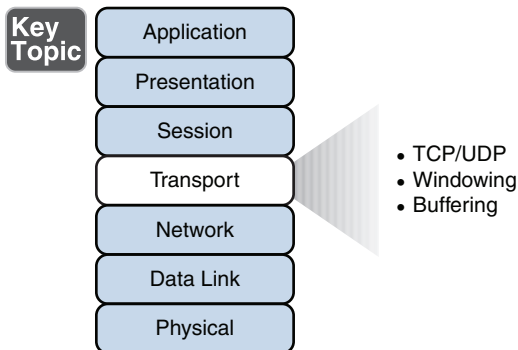


Figure 1-10 Layer 4: The Transport Layer

Just as Layer 2 and Layer 3 offer flow control services, flow control services also exist at Layer 4. Two common flow control approaches at Layer 4 are windowing and buffering:

- **Windowing:** TCP communication uses windowing, in that one or more segments are sent at one time, and a receiver can attest to the receipt of all the segments in a window with a single acknowledgment. In some cases, as

illustrated in Figure 1-11, TCP uses a sliding window, where the window size begins with one segment. If there is a successful acknowledgment of that one segment (that is, the receiver sends an acknowledgment asking for the next segment), the window size doubles to two segments. Upon successful receipt of those two segments, the next window holds four segments. This exponential increase in window size continues until the receiver does not acknowledge successful receipt of all segments within a certain amount of time—known as the *round-trip time* (RTT), which is sometimes called *real transfer time*—or until a configured maximum window size is reached.

- **Buffering:** With buffering, a device (for example, a router) uses a chunk of memory (sometimes called a *buffer* or a *queue*) to store segments if bandwidth is not available to send those segments. A queue has finite capacity, however, and can overflow (that is, drop segments) in the event of sustained network congestion.

Key Topic

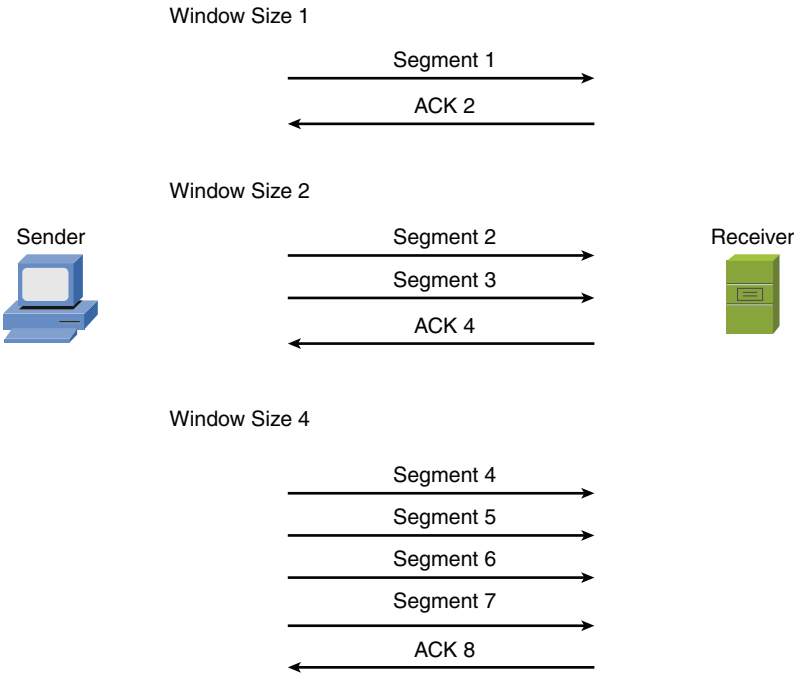


Figure 1-11 TCP Sliding Window

In addition to TCP and UDP, Internet Control Message Protocol (ICMP) is another transport layer protocol you are likely to meet. ICMP is used by utilities

such as ping and traceroute, which are discussed in Chapter 25, “Network Troubleshooting Tools.”

Layer 5: The Session Layer

The *session layer*, as shown in Figure 1-12, is responsible for setting up, maintaining, and tearing down sessions. You can think of a session as a conversation that needs to be treated separately from other sessions to avoid the intermingling of data from different conversations.

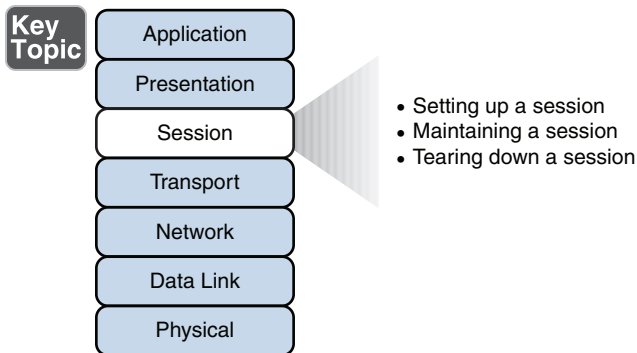


Figure 1-12 Layer 5: The Session Layer

Here is a detailed look at the functions of the session layer:

- **Setting up a session:** Examples of the procedures involved in setting up a session include the following:
 - Checking user credentials (for example, username and password)
 - Assigning numbers to a session’s communication flows to uniquely find each one
 - Negotiating services needed during the session
 - Negotiating which device begins sending data
- **Maintaining a session:** Examples of the procedures involved in supporting a session include the following:
 - Transferring data
 - Reestablishing a disconnected session
 - Acknowledging receipt of data

- **Tearing down a session:** A session can be disconnected based on agreement of the devices in the session. Alternatively, a session might be torn down because one party disconnects (either intentionally or because of an error condition). If one party disconnects, the other party can detect a loss of communication with that party and tear down its side of the session.

Session Initiation Protocol (SIP) is an example of a session layer protocol, which can help set up, support, and tear down a voice or video connection. Keep in mind, however, that not every network application neatly maps directly to all seven layers of the OSI model.

Layer 6: The Presentation Layer

The *presentation layer*, as shown in Figure 1-13, formats the data being exchanged and secures that data with encryption.

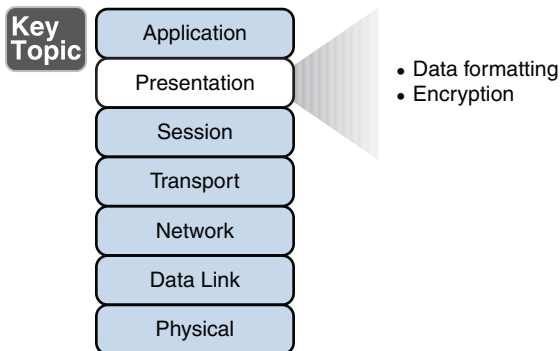


Figure 1-13 Layer 6: The Presentation Layer

The following list describes the functions involved in data formatting and encryption in more detail:

- **Data formatting:** As an example of how the presentation layer handles data formatting, consider how text is formatted. Some applications might format text using American Standard Code for Information Interchange (ASCII), while other applications might format text using Extended Binary Coded Decimal Interchange Code (EBCDIC). The presentation layer handles formatting the text (or other types of data, such as multimedia or graphics files) in a format that allows compatibility between the communicating devices.

- **Encryption:** Imagine that you are sending sensitive information over a network (for example, your credit card number or bank password). If a malicious user were to intercept your transmission, they might be able to obtain this sensitive information. To add a layer of security for such transmissions, encryption can be used to scramble (encrypt) the data in such a way that if the data were intercepted, a third party would not be able to unscramble (decrypt) it. However, the intended recipient would be able to decrypt the transmission.

Encryption is discussed in detail in Chapter 18, “Network Security Concepts.”

Layer 7: The Application Layer

The *application layer*, as shown in Figure 1-14, gives application services to a network. An important (and often-misunderstood) concept is that end-user applications (such as Microsoft Word) live at the application layer. Instead, the application layer supports services used by end-user applications. For example, email is an application layer service that does exist at the application layer, whereas Microsoft Outlook (an example of an email client) is an end-user application that does not live at the application layer. Another function of the application layer is advertising available services.

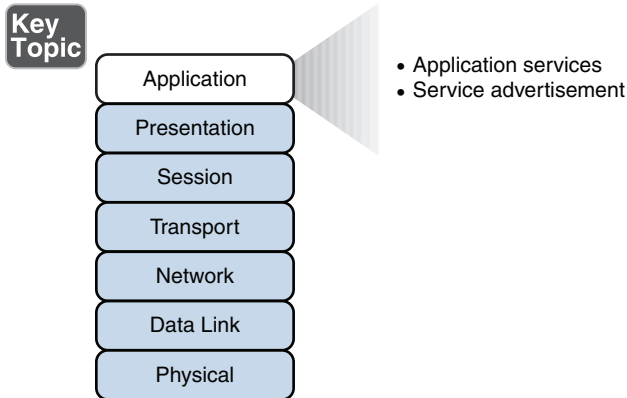


Figure 1-14 Layer 7: The Application Layer

The following list describes the functions of the application layer in more detail:

- **Application services:** Examples of the application services living at the application layer include file sharing and email.
- **Service advertisement:** Some applications' services (for example, some networked printers) periodically send out advertisements, making their availability known to other devices on the network. Other services, however,

register themselves and their services with a centralized directory (for example, Microsoft Active Directory), which can be queried by other network devices seeking such services.

Recall that even though the application layer is numbered as Layer 7, it is at the top of the OSI stack because its networking functions are closest to the end user.

The TCP/IP Stack

The ISO developed the OSI reference model to be generic, in terms of what protocols and technologies could be categorized by the model. However, most of the traffic on the Internet (and traffic on corporate networks) is based on the TCP/IP protocol suite. Therefore, a more relevant model for many network designers and administrators to reference is a model developed by the U.S. Department of Defense (DoD). This model is known as the *DoD model*, or the **TCP/IP stack**.

NOTE An older protocol known as Network Control Protocol (NCP) was similar to TCP/IP. NCP was used on ARPANET (the predecessor to the Internet), and it provided features like those offered by the TCP/IP suite of protocols on the Internet, although they were not as robust.

Layers of the TCP/IP Stack

The TCP/IP stack has only four defined layers, as opposed to the seven layers of the OSI model. Figure 1-15 contrasts these two models.

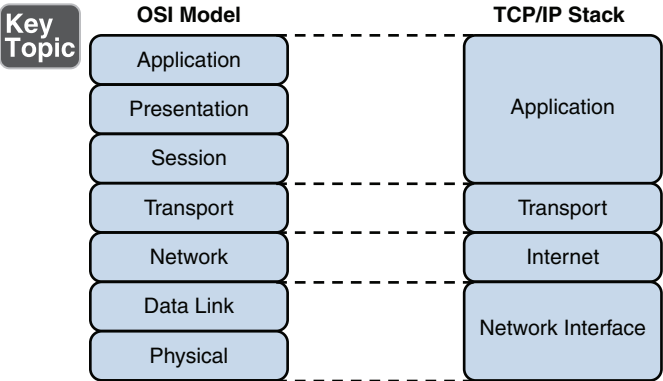


Figure 1-15 TCP/IP Stack

The TCP/IP stack is composed of the following layers:

- **Network interface:** The TCP/IP stack’s *network interface layer* encompasses the technologies offered by Layers 1 and 2 (the physical and data link layers) of the OSI model.

NOTE Some literature refers to the network interface layer as the *network access layer*.

- **Internet:** The *Internet layer* of the TCP/IP stack maps to Layer 3 (the network layer) of the OSI model. Although multiple routed protocols live at the OSI model’s network layer, the Internet layer of the TCP/IP stack focuses on IP as the protocol to be routed through a network. Figure 1-16 shows the format of an IP version 4 (IPv4) packet.

**Key
Topic**

Version	Header Length	Type of Service	Total Length	
Identification			IP Flags	Fragment Offset
TTL	Protocol		Header Checksum	
Source Address				
Destination Address				
IP Option (Variable Length)				

Figure 1-16 IPv4 Packet Format

Notice that there are fields in the IP packet header for both source and destination IP addresses. The Protocol field shows the transport layer protocol from which the packet was sent or to which the packet should be sent. Also of note is the Time-to-Live (TTL) field. The value in this field is decremented by 1 every time this packet is routed from one IP network to another (that is, when it passes through a router). If the TTL value ever reaches 0, the packet is discarded from the network. This behavior helps prevent routing loops, and we will revisit this concept in Chapter 2, “Networking Appliances, Applications,

and Functions.” As a common practice, the OSI layer numbers 1, 2, and 3 are still used when referring to physical, data link, and network layers of the TCP/IP stack, even though the TCP/IP stack does not explicitly separate the physical and data link layers.

- **Transport:** The *transport layer* of the TCP/IP stack maps to Layer 4 (the transport layer) of the OSI model. The two primary protocols found at the TCP/IP stack’s transport layer are TCP and UDP.

Figure 1-17 details the structure of a TCP segment. Notice the fields for source and destination ports. As described later in this chapter, these ports identify to which upper-layer protocol data should be forwarded or from which upper-layer protocol the data is being sent.

Key
Topic

Source Port		Destination Port	
Sequence Number			
Acknowledgment Number			
Offset	Reserved	TCP Flags	Window
Checksum		Urgent Pointer	
TCP Options Option (Optional)			

Figure 1-17 TCP Segment Format

Also notice the field for window size. The value in this field determines how many bytes a device can receive before expecting an acknowledgment. As previously described, this feature offers flow control.

The header of a TCP segment also contains sequence numbers for segments. With sequence numbering, if segments arrive out of order, the recipient can put them back in the proper order based on the sequence numbers.

The acknowledgment number in the header shows the next sequence number the receiver expects to receive. This is a way for the receiver to let the sender know that all segments up to and including that point have been received. Due

to the sequencing and acknowledgments, TCP is considered to be a *connection-oriented* transport layer protocol.

NOTE You might have noticed that both the *IP header* and the *TCP header* make use of a Flags field. Including this field is a very common technique in networking to permit the data unit to convey specific settings. For example, the IP header uses the IP Flags field (3 bits) to help manage (or prevent) fragmentation. *TCP flags* are used to indicate a particular connection state or provide additional information. They are often used for troubleshooting purposes or to control how a particular connection is handled.

Figure 1-18 presents the structure of a UDP segment. UDP is a connection-less, unreliable protocol. UDP lacks the sequence numbering, window size, and acknowledgment numbering present in the header of a TCP segment. The UDP segment's header simply contains source and destination port numbers, a UDP checksum (which is an optional field used to detect transmission errors), and the segment length (measured in bytes).

**Key
Topic**

Source Port	Destination Port
UDP Length	UDP Checksum

Figure 1-18 UDP Segment Format

Because a *UDP header* is much smaller than a TCP header, UDP is a good candidate for the transport layer protocol for applications that need to maximize bandwidth and do not require acknowledgments (for example, audio or video streams).

- **Application:** The biggest difference between the TCP/IP stack and the OSI model is at the TCP/IP stack's *application layer*. This layer addresses concepts described by Layers 5, 6, and 7 (the session, presentation, and application layers) of the OSI model.

With the reduced complexity of a four-layer model like the TCP/IP stack, network designers and administrators can more easily categorize a given networking technology into a specific layer. For example, although SIP was shown earlier as a session layer protocol within the OSI model, you would have to know more about the behavior of SIP to properly categorize it in that model. However, with the TCP/IP stack, you could quickly figure out that SIP is a higher-level protocol that gets

encapsulated inside TCP, and you could thus classify SIP in the application layer of the TCP/IP stack.

Common Application Protocols in the TCP/IP Stack

Application layer protocols in the TCP/IP stack are identifiable by unique port numbers. For example, when you enter a web address in an Internet browser, you are (by default) communicating with that remote web address using TCP port 80. Specifically, Hypertext Transfer Protocol (HTTP), which is the protocol used by web servers, uses TCP port 80. Therefore, the data you send to that remote web server has a destination port number of 80. That data is encapsulated into a TCP segment at the transport layer. That segment is further encapsulated into a packet at the Internet layer and sent out on the network using an underlying network interface layer technology such as Ethernet.

NOTE Thanks to awareness of network security today, you do not see HTTP (port 80) actually being used on the Internet very much anymore. It has been replaced with a secured version, HTTPS, which uses TCP port 443 in its operation.

Consider the example illustrated in Figure 1-19. When you send traffic to the remote website, the packet you send out to the network needs not only the destination IP address (172.16.1.2 in this example) of the web server and the destination port number for HTTP (that is, 80) but also the source IP address of your computer (10.1.1.1 in this example). Because your computer is not acting as a web server, its port is not 80. Instead, your computer selects a source port number greater than 1023. In this example, let's imagine that the client PC selects the source port 1248.

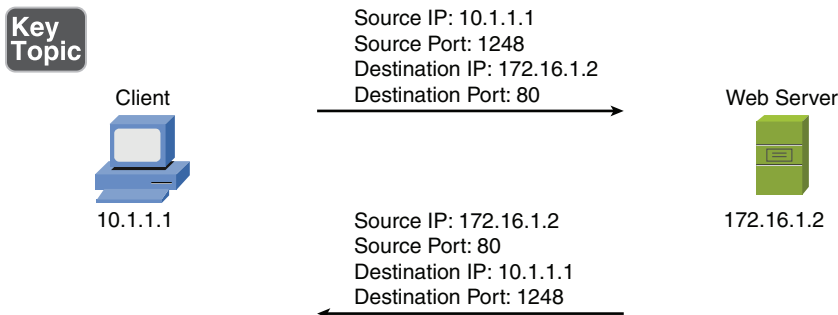


Figure 1-19 Example: Port Numbers and IP Addresses

Notice that when the web server sends content back, the IP addresses and port numbers have now switched, with the web server as the source and your PC as the

destination. With both source and destination port numbers, along with source and destination IP addresses, two-way communication becomes possible.

NOTE Ports numbered 1023 and below are called *well-known* ports, and ports numbered above 1023 are called *ephemeral* ports. The maximum value of a port is 65535. Well-known port number assignments are available at <https://www.iana.org/assignments/port-numbers>.

Real-World Case Study

Bob, a manager of the networking team at Acme, Inc., is paying extra attention to the specific words he uses as he talks to his team in preparation for the implementation of the network. When referring to transport protocols such as the connection-oriented TCP and the connectionless UDP, the word Bob uses to describe those protocol data units is *segment*.

In discussing the applications that Acme, Inc. will be using over its network, Bob notes that many of these applications will be using TCP at the transport layer. This includes HTTPS for secure web traffic and Simple Mail Transfer Protocol (SMTP) and Internet Message Access Protocol (IMAP) for email services.

The company will use the Secure Shell (SSH) protocol, which also uses TCP at the transport layer, as a secure method to remotely connect to and manage its network devices. A common connectionless UDP protocol is Domain Name System (DNS), which will be used thousands of times a day to translate a friendly name like <https://www.pearson.com> to an IP address that is reachable over the network. Another protocol based on UDP that will be used often is Dynamic Control Host Protocol (DHCP), which assigns client computers on the network an IP address that is required for sending and receiving Layer 3 packets.

For the traffic on the LAN, the Ethernet cables and electronic signals being sent as bits going over those cables represent Layer 1 from an OSI perspective. On the LAN, they will be using Ethernet technology, and as a result, the Layer 2 frames that are sent on the LAN will be encapsulated and sent as Ethernet Layer 2 frames.

For datagrams being sent across the serial WAN connections provided by the service provider, it is likely that protocols and technologies such as High-Level Data Link Control (HDLC), Layer 2 Tunneling Protocol (L2TP), Point-to-Point Protocol (PPP), Point-to-Point Tunneling Protocol (PPTP), Spanning Tree Protocol (STP), and virtual LANs (VLANs) are in operation at Layer 2 (the data link layer). On both the LAN and the WAN, at Layer 3 (the network layer), IPv4 will be used

for host addressing and defining networks. The same Layer 1, Layer 2, and Layer 3 infrastructure is also capable of transporting IPv6, if desired.

Inside the Layer 3 IP headers, each packet contains the source and destination addresses, in addition to the information to tell the receiving network device about which Layer 4 transport protocol is encapsulated or carried inside the Layer 3 packet. When a network device receives the packet and opens it up to look at the contents, this process is called *decapsulation*. As the recipient decapsulates and looks at the Layer 4 information, it identifies the application layer protocol or service being used. A segment going to a web server is likely to have a TCP destination port of 80 or 443, depending on whether encryption is being used for a secure connection. A DNS request uses a UDP destination port of 53.

Summary

Here are the main topics covered in this chapter:

- The ISO's OSI reference model consists of seven layers: physical (Layer 1), data link (Layer 2), network (Layer 3), transport (Layer 4), session (Layer 5), presentation (Layer 6), and application (Layer 7). The purpose of each layer was presented, along with examples of technologies present at the individual layers, as it pertains to networking.
- The TCP/IP stack was presented as an alternative model to the OSI reference model. The TCP/IP stack consists of four layers: network interface, Internet, transport, and application. These layers were compared with the seven layers of the OSI model.
- Data encapsulation and decapsulation within the OSI model context were covered.
- This chapter also discussed how port numbers are used to associate data at the transport layer with a proper application layer protocol.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 1-1 lists these key topics and the page number where each is found.

**Key
Topic****Table 1-1** Key Topics for Chapter 1

Key Topic Element	Description	Page Number
List	Layers of the OSI model	6
Figure 1-3	PDU Names	7
Figure 1-4	Layer 1: The Physical Layer	7
Figure 1-8	Layer 2: The Data Link Layer	12
Figure 1-9	Layer 3: The Network Layer	15
Figure 1-10	Layer 4: The Transport Layer	17
Figure 1-11	TCP Sliding Window	18
Figure 1-12	Layer 5: The Session Layer	19
Figure 1-13	Layer 6: The Presentation Layer	20
Figure 1-14	Layer 7: The Application Layer	21
Figure 1-15	TCP/IP Stack	22
Figure 1-16	IPv4 Packet Format	23
Figure 1-17	TCP Segment Format	24
Figure 1-18	UDP Segment Format	25
Figure 1-19	Example: Port Numbers and IP Addresses	26

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

application layer (OSI model), application layer (TCP/IP stack), current state modulation, cyclic redundancy check (CRC), data link layer, Internet layer, IP header, maximum transmission unit (MTU), network interface layer, network layer, Open Systems Interconnection (OSI) reference model, payload, physical layer, presentation layer, protocol data unit (PDU), session layer, state transition modulation, TCP header, TCP flags, TCP/IP stack, time-division multiplexing (TDM), Transmission Control Protocol (TCP), transport layer (OSI model), transport layer (TCP/IP stack), UDP header, User Datagram Protocol (UDP)

Additional Resources

A Review of the OSI Model for Ethical Hackers: <https://youtu.be/S2OM0IYZcR0?si=uIqE7nV4cNBEX0Wm>

The OSI Model Challenge: <https://ajsnetworking.com/osiquiz1>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. Which layer of the OSI reference model contains the MAC and LLC sublayers?
 - a. Network layer
 - b. Transport layer
 - c. Physical layer
 - d. Data link layer
2. Which approach to bandwidth usage consumes all the available frequencies on a medium to transmit data?
 - a. Broadband
 - b. Baseband
 - c. Time-division multiplexing
 - d. Simplex
3. Windowing is provided at what layer of the OSI reference model?
 - a. Data link layer
 - b. Network layer
 - c. Transport layer
 - d. Physical layer
4. IP addresses reside at which layer of the OSI reference model?
 - a. Network layer
 - b. Session layer
 - c. Data link layer
 - d. Transport layer

5. Which of the following is a connectionless transport layer protocol?
 - a. IP
 - b. TCP
 - c. UDP
 - d. SIP
6. What setting ultimately controls the size of packets that are moving through the modern network?
 - a. TTL
 - b. MTU
 - c. SSH
 - d. CSMA/CD
7. What is the range of well-known TCP and UDP ports?
 - a. Below 2048
 - b. Below 1024
 - c. 16,384–32,768
 - d. Above 8192
8. What port number is used by HTTPS?
 - a. 80
 - b. 443
 - c. 69
 - d. 23
9. What value is decremented by 1 for each router hop on the network?
 - a. Count
 - b. Type
 - c. TTL
 - d. Dead timer
10. Windowing is a technology that applies to which transport layer protocol?
 - a. UDP
 - b. FTP

- c. ICMP
 - d. TCP
- 11. What happens to data as it moves from the upper layers to the lower layers of the OSI model on a host system?
 - a. The data moves from the physical layer to the application layer.
 - b. The data is encapsulated with a header at the beginning and a trailer at the end.
 - c. The header and trailer are stripped off through decapsulation.
 - d. The data is sent in groups of segments that require two acknowledgments.
- 12. Which layer of the OSI reference model is responsible for ensuring that frames do not exceed the maximum transmission unit (MTU) of the physical media?
 - a. Network layer
 - b. Transport layer
 - c. Physical layer
 - d. Data link layer

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This chapter covers the following topics related to Objective 1.2 (Compare and contrast networking appliances, applications, and functions) of the CompTIA Network+ N10-009 certification exam:

- Physical and virtual appliances
 - Router
 - Switch
 - Firewall
 - Intrusion detection system (IDS)/intrusion prevention system (IPS)
 - Load balancer
 - Proxy
 - Network-attached storage (NAS)
 - Storage area network (SAN)
 - Wireless
 - Access point (AP)
 - Controller
- Applications
 - Content delivery network (CDN)
- Functions
 - Virtual private network (VPN)
 - Quality of service (QoS)
 - Time to live (TTL)

Networking Appliances, Applications, and Functions

Modern networks can include a daunting number of devices, and it is your job to understand the function of each device and how it works with the others. To create a network, these devices need some sort of interconnection, using one of a variety of media types. This chapter begins by delving into the many networking devices that make up a typical enterprise (or even small home office) network.

Next, this chapter explores some typical applications and functions that the network can perform. This includes a discussion of such popular features like virtual private networks (VPNs) and content delivery networks (CDNs).

Foundation Topics

Physical and Virtual Appliances

This chapter begins with a fairly detailed look at the networking devices that work together to determine the success or failure of your IT infrastructure. You might not have every one of these devices in your network, but knowledge of each helps set the stage for some of the more advanced networking technologies discussed in later chapters.

Routers

A **router** is a Layer 3 device, which means it makes forwarding decisions based on logical network address (for example, IP address) information. Although a router is considered to be a Layer 3 device it has the capability to consider high-layer traffic parameters, such as quality of service (QoS) settings, in making forwarding decisions.

Routers help us dramatically when designing networks. This is because they are so efficient at segmenting larger networks. Specifically, routers do not forward broadcast packets by default. This means that each port on a router automatically creates a separate broadcast domain.

If this were not enough, each port also creates a separate collision domain. What is a collision domain? A collision domain represents an area on a LAN on which there can be only one transmission at a time. Thanks to the elimination of legacy hubs and bridges in the modern network, our networks today feature microsegmentation. This means we eliminate the possibility of frame collisions by automatically creating tiny collision domains for each port.

Figure 2-1 shows how each port on a router is a separate collision domain and a separate broadcast domain. Notice this figure also shows two Layer 2 switches (covered next in this chapter). These devices do not automatically create broadcast domains by default, but they do help microsegment the network. Each port on the switch represents a collision domain just like a router.

Switches

There are actually many different types of **switches** found in networks today. Let's begin with the most basic type, a **Layer 2 Ethernet switch**.

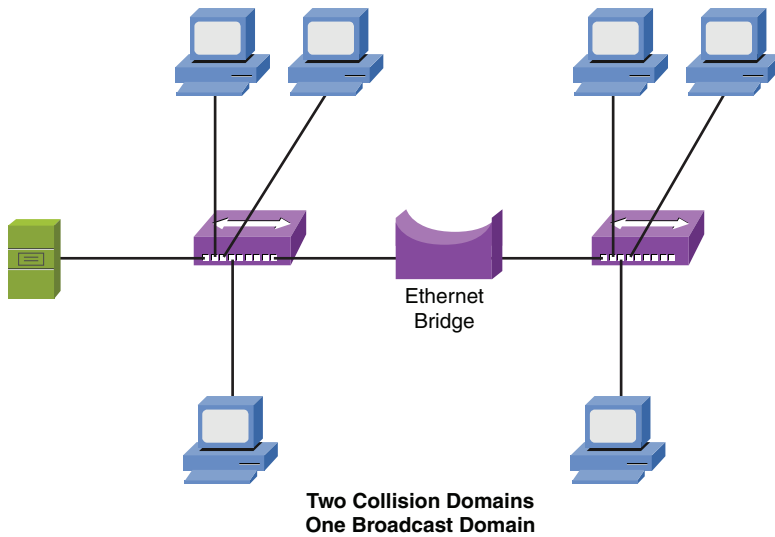


Figure 2-1 Router Broadcast and Collision Domains

NOTE Why do we call the device a Layer 2 switch? If you read Chapter 1 of this text you probably guessed it. Layer 2 switches operate at Layer 2 of the OSI model—the data link layer.

Layer 2 switches dynamically learn the MAC addresses attached to various ports by looking at the source MAC addresses on frames coming into a port. For example, if switch port Gigabit Ethernet 1/1 received a frame with source MAC address DDDD.DDDD.DDDD, the switch could conclude that MAC address DDDD.DDDD.DDDD resides off port Gigabit Ethernet 1/1. In the future, if the switch receives a frame destined for MAC address DDDD.DDDD.DDDD, the switch would only send that frame out port Gigabit Ethernet 1/1.

Initially, however, a switch is unaware of what MAC addresses reside off which ports (unless MAC addresses have been statically configured). Therefore, when a switch receives a frame destined for a MAC address not yet present in the switch's MAC address table, the switch floods that frame out all the switch ports except the port on which the frame was received. Similarly, broadcast frames (that is, frames with destination MAC address FFFF.FFFF.FFFF) are always flooded out all switch ports except the port on which the frame was received. The reason broadcast frames are always flooded is that no endpoint will have MAC address FFFF.FFFF.FFFF, which means that the FFFF.FFFF.FFFF MAC address will never be learned in a switch's MAC address table.

To illustrate how a switch's MAC address table becomes populated, consider an endpoint named PC1 that wants to form an SSH connection with a server. Also, assume that PC1 and its server both reside on the same subnet (that is, no routing is required to get traffic between PC1 and its server). Before PC1 can establish an SSH session to its server, PC1 needs to know the IP address (that is, the Layer 3 address) and the MAC address (Layer 2 address) of the server. The IP address of the server is typically known or is resolved via a Domain Name System (DNS) lookup. In this example, assume that the server's IP address is known. To properly form an SSH segment, however, PC1 needs to know the server's Layer 2 MAC address. If PC1 does not already have the server's MAC address in its Address Resolution Protocol (ARP) cache, PC1 can send an ARP request in an attempt to learn the server's MAC address, as shown in Figure 2-2.

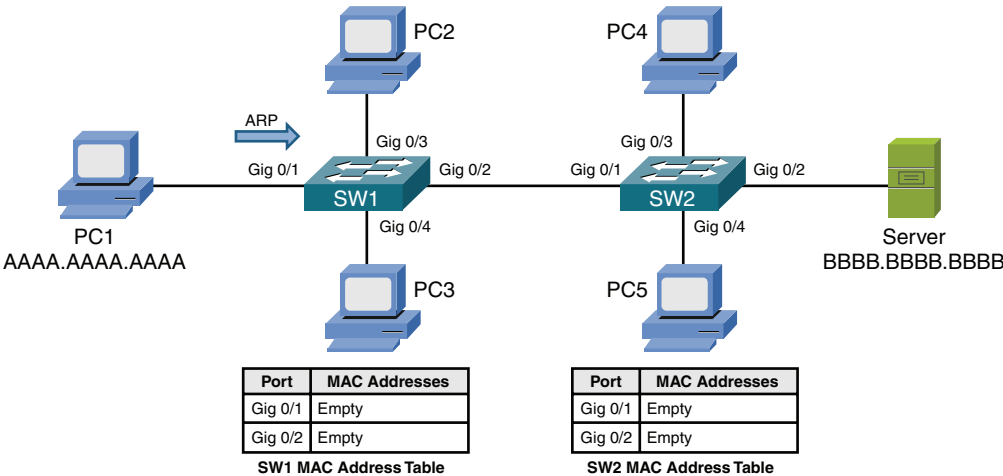


Figure 2-2 Endpoint Sending an ARP Request

When switch SW1 sees PC1's ARP request enter port Gigabit 0/1, PC1's MAC address AAAA.AAAA.AAAA is added to switch SW1's MAC address table. Also, because the ARP request is a broadcast, its destination MAC address is FFFF.FFFF.FFFF. Because the MAC address FFFF.FFFF.FFFF is not known to switch SW1's MAC address table, switch SW1 floods a copy of the incoming frame out all switch ports other than the port on which the frame was received, as shown in Figure 2-3.

When switch SW2 receives the ARP request over its Gig 0/1 trunk port, the source MAC address AAAA.AAAA.AAAA is added to switch SW2's MAC address table, as illustrated in Figure 2-4. Also, using behavior similar to that of switch SW1, switch SW2 floods the broadcast.

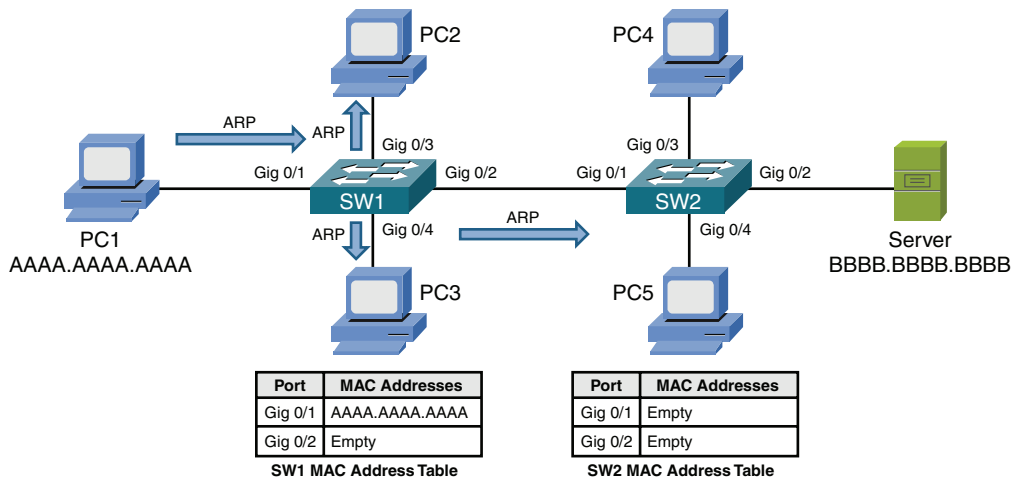


Figure 2-3 Switch SW1 Flooding the ARP Request

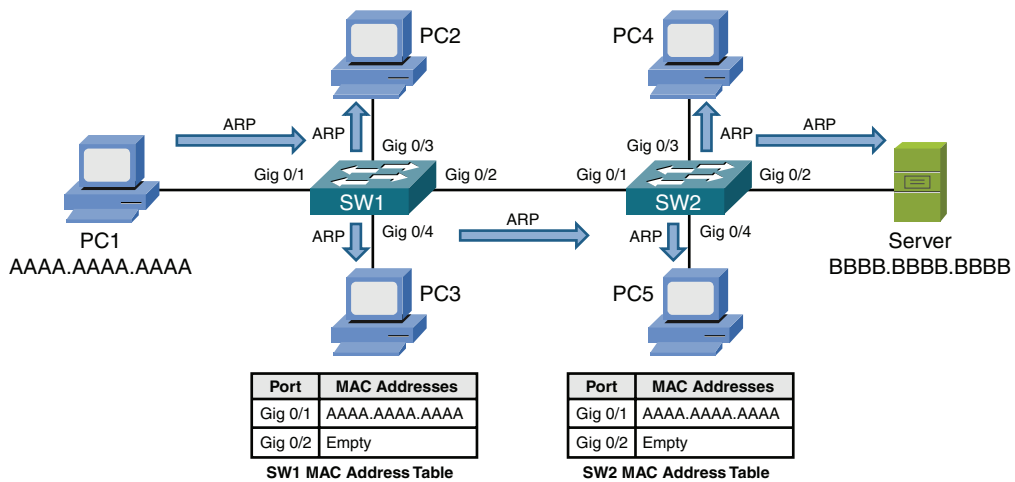


Figure 2-4 Switch SW2 Flooding the ARP Request

The server receives the ARP request and responds with an ARP reply, as shown in Figure 2-5. Unlike the ARP request, however, the ARP reply frame is not a broadcast frame. The ARP reply, in this example, has a destination MAC address AAAA.AAAA.AAAA, which makes the ARP reply a unicast frame.

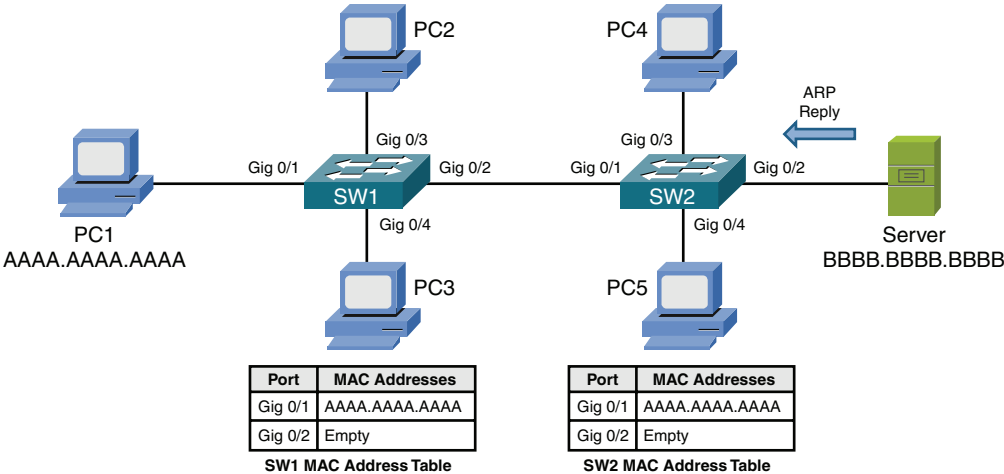


Figure 2-5 ARP Reply Sent from a Server

Upon receiving the ARP reply from the server, switch SW2 adds the server's MAC address BBBB.BBBB.BBBB to its MAC address table, as shown in Figure 2-6. Also, the ARP reply is only sent out port Gig 0/1 because switch SW1 knows that the destination MAC address AAAA.AAAA.AAAA is available off port Gig 0/1.

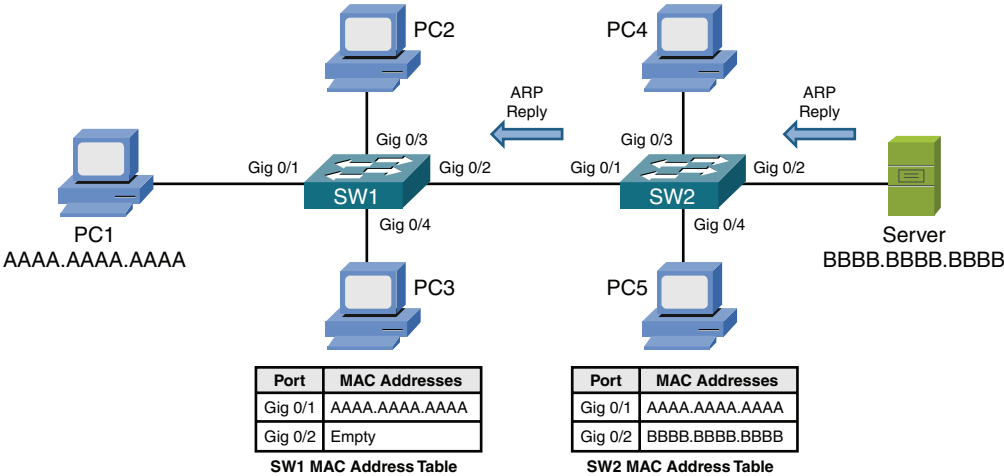


Figure 2-6 Switch SW2 Forwarding the ARP Reply

When receiving the ARP reply in its Gig 0/2 port, switch SW1 adds the server's MAC address BBBB.BBBB.BBBB to its MAC address table. Also, like switch SW2, switch SW1 now has an entry in its MAC address table for the frame's destination

MAC address AAAA.AAAA.AAAA. Therefore, switch SW1 forwards the ARP reply out port Gig 0/1 to the endpoint of PC1, as illustrated in Figure 2-7.

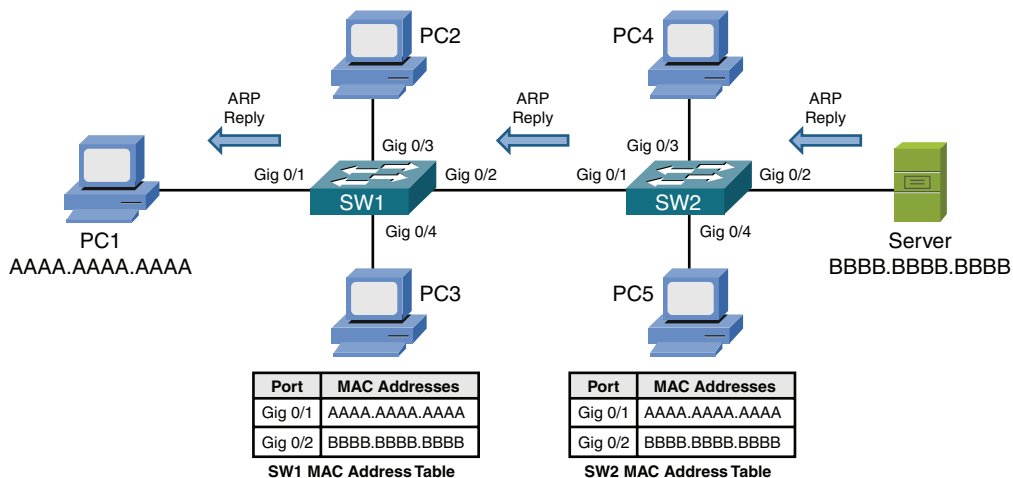


Figure 2-7 Switch SW1 Forwarding the ARP Reply

After receiving the server's ARP reply, PC1 knows the MAC address of the server. Therefore, PC1 can now properly construct an SSH segment destined for the server, as depicted in Figure 2-8.

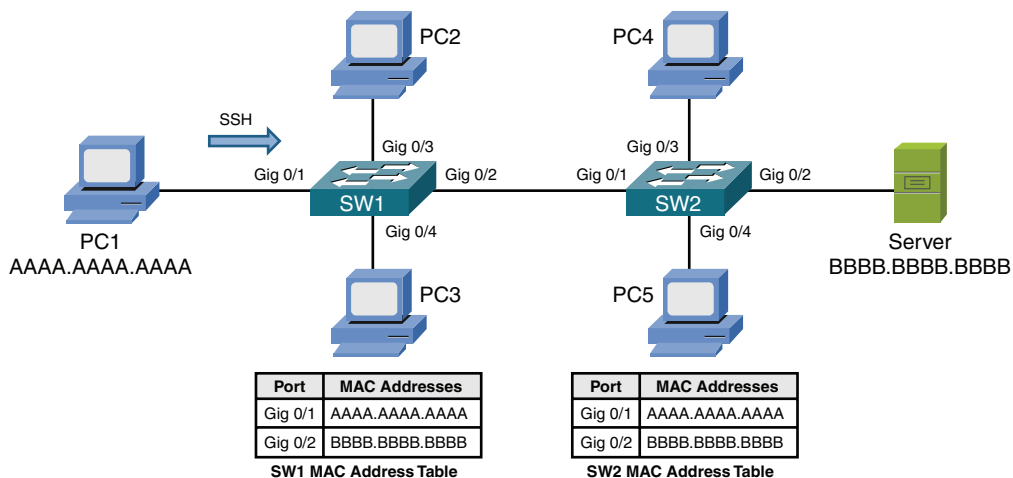


Figure 2-8 PC1 Sending an SSH Segment

Switch SW1 has the server’s MAC address BBBB.BBBB.BBBB in its MAC address table. Therefore, when switch SW1 receives the SSH segment from PC1, that segment is forwarded out of switch SW1’s Gig 0/2 port, as shown in Figure 2-9.

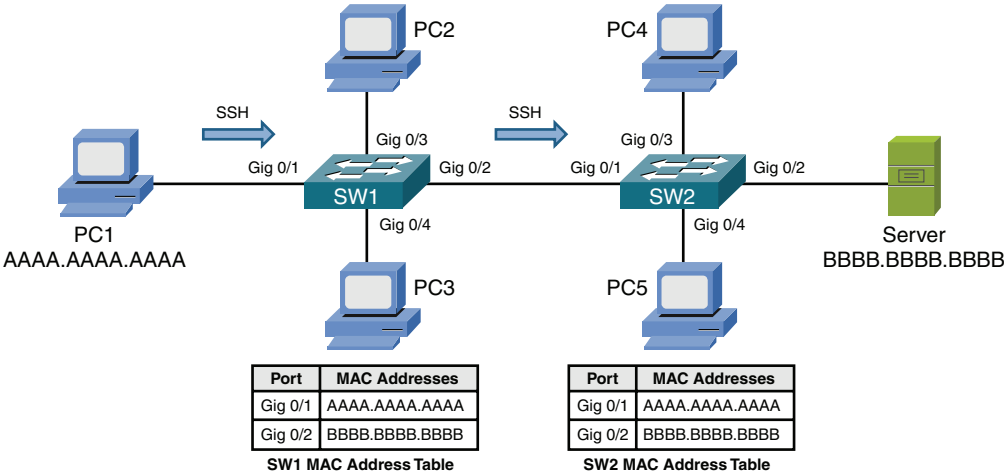


Figure 2-9 Switch SW1 Forwarding the SSH Segment

With behavior similar to that of switch SW1, switch SW2 forwards the SSH segment out of its Gig 0/2 port. This forwarding, shown in Figure 2-10, is possible because switch SW2 has an entry for the segment’s destination MAC address BBBB.BBBB.BBBB in its MAC address table.

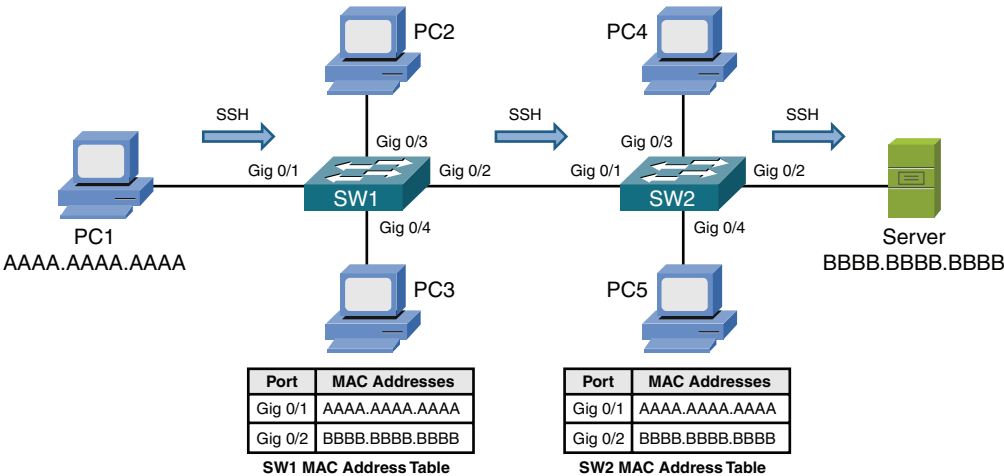


Figure 2-10 Switch SW2 Forwarding the SSH Segment

Finally, the server responds to PC1, and a bidirectional SSH session is established between PC1 and the server, as illustrated in Figure 2-11. Because PC1 learned the server's MAC address as a result of its earlier ARP request and stored that result in its local ARP cache, the transmission of subsequent Telnet segments does not require additional ARP requests. However, if unused for a period of time, entries in a PC's ARP cache can time out. Therefore, the PC would have to broadcast another ARP frame if it needed to send traffic to the same destination IP address. The sending of the additional ARP frames adds a small amount of delay when reestablishing a session with that destination IP address.

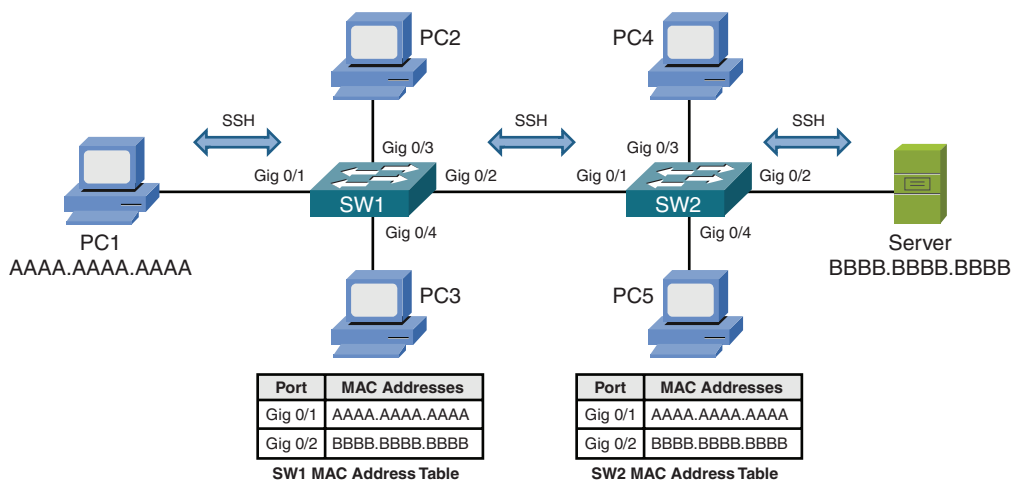


Figure 2-11 Bidirectional SSH Session Between PC1 and the Server

As shown in Figure 2-12, each port on a switch represents a separate collision domain. Also, all ports on a switch belong to the same broadcast domain, with one exception: when the ports on a switch have been divided into separate virtual LANs (VLANs). Remember that each VLAN represents a separate broadcast domain, and for traffic to travel from one VLAN to another, that traffic must be routed by a Layer 3 device.

Whereas a Layer 2 switch makes forwarding decisions based on MAC address information, a **multilayer switch** can make forwarding decisions based on upper-layer information. For example, a multilayer switch could function as a router and make forwarding decisions based on destination IP address information. Some literature refers to a multilayer switch as a **Layer 3 capable switch** because of the switch's capability to make forwarding decisions like a router. The term **multilayer switch** is more accurate, however, because many multilayer switches have policy-based routing features that allow upper-layer information (for example, application port numbers) to be used in making forwarding decisions.

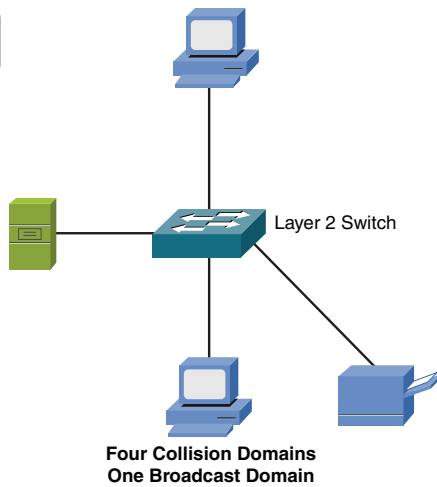
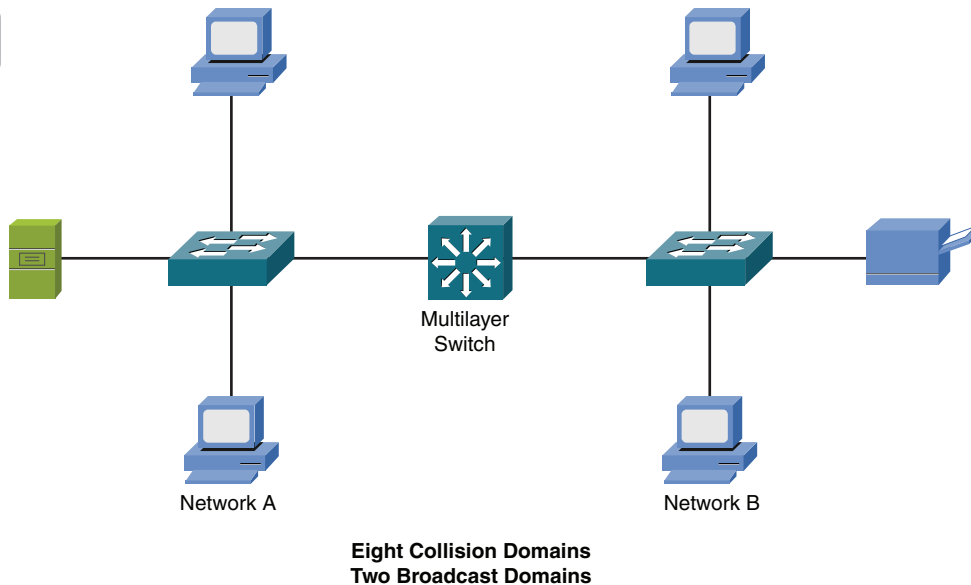
**Key
Topic****Figure 2-12** Switch Collision and Broadcast Domains

Figure 2-13 makes the point that a multilayer switch can be used to interconnect not just network segments but also entire networks. Remember that logical Layer 3 IP addresses are used to assign network devices to different logical networks. For traffic to travel between two networked devices that belong to different networks, that traffic must be *routed*. (That is, a device, such as a multilayer switch, has to make a forwarding decision based on Layer 3 information.)

**Key
Topic****Figure 2-13** Multilayer Ethernet Switch

As on a Layer 2 switch, each port on a multilayer switch represents a separate collision domain; however, a characteristic of a multilayer switch (and a router) is that it can become a boundary of a broadcast domain. Although all ports on a Layer 2 switch belong to the same broadcast domain, if configured as such, all ports on a multilayer switch can belong to different broadcast domains.

Firewalls

A network *firewall* is most often implemented as a network security appliance, but this is certainly not the only “form factor” a firewall can take on. If you are a Windows Operating System (OS) user, you have probably noticed the software firewall included by default in your OS. As depicted in Figure 2-14, a firewall appliance stands guard at the door of your network, protecting it from malicious Internet traffic.

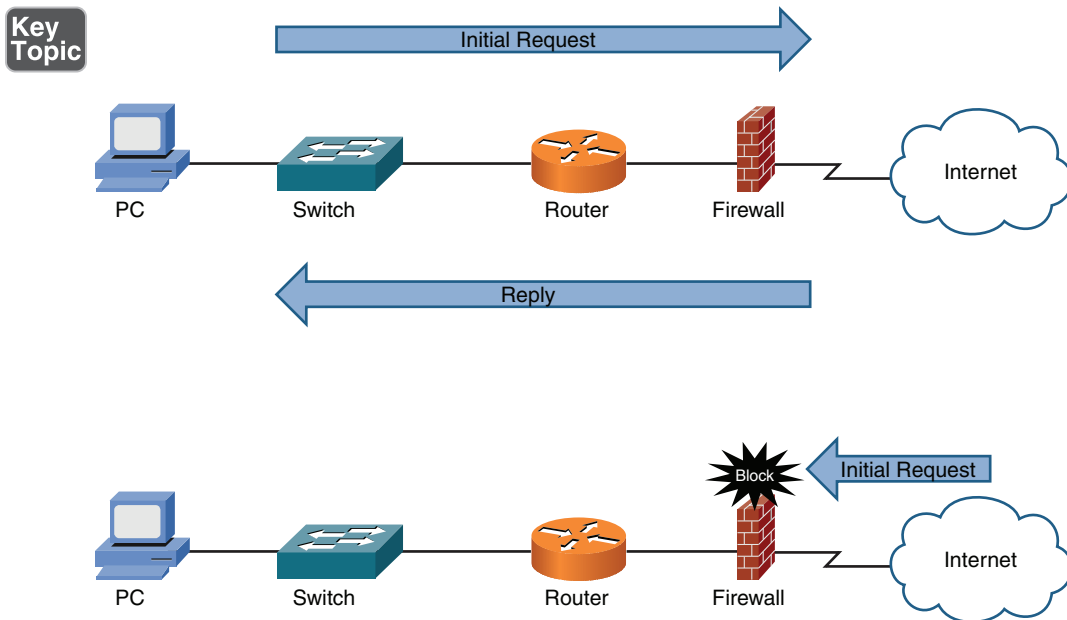


Figure 2-14 Firewall

For example, a *stateful firewall* allows traffic to originate from an inside network (that is, a trusted network) and go out to the Internet (an untrusted network). Likewise, return traffic coming back from the Internet to the inside network is allowed by the firewall. However, if traffic originates from a device on the Internet (that is, not returning traffic), the firewall blocks that traffic.

Intrusion Detection System (IDS)/Intrusion Prevention System (IPS)

When an attacker launches an attack against a network, *intrusion detection system (IDS)* and *intrusion prevention system (IPS)* technologies are often able to recognize the attack and respond appropriately. Attacks might be recognizable by comparing incoming data streams against a database of well-known attack signatures. Other mechanisms for detecting attacks include policy-based and anomaly-based approaches.

NOTE Be sure you understand the difference between an IDS and an IPS. An IDS is a software application or hardware device that monitors a network or system for malicious or non-policy-related activity and reports violations to a centralized management system. Notice its key function is *detection*. An IPS is a network device that continually scans the network, looking for inappropriate activity. It can stop potential threats. Notice its key function is *prevention*.

In addition to dedicated network-based intrusion prevention system (NIPS) sensors, IPS software can be installed on a host to provide a host-based intrusion prevention system (HIPS) or host-based intrusion detection system (HIDS) solution.

IDS Versus IPS

Both IDS and IPS devices can recognize network attacks; they differ primarily in their network placement. Specifically, whereas an IDS device receives a copy of traffic to be analyzed, an IPS device resides inline with the traffic, as illustrated in Figure 2-15.

Because the analyzed traffic does not flow through the IDS device, the IDS device is considered to be *passive*, and the IPS device is considered to be *active*. Both the IDS and the IPS devices can send alerts to, for example, a management station. Although an IDS device can also communicate with a security appliance or a router to prevent subsequent attack packets, the initially offending traffic reaches its destination. Conversely, an IPS device can drop the traffic inline, thus preventing even the first malicious packet from reaching its intended target.

The previous discussion of IDS versus IPS devices might seem to suggest that IPS devices should always be used instead of IDS devices. However, in some network environments, these two solutions complement one another. For example, an IDS device can add value to a network that already employs an IPS device by verifying that the IPS device is still operational. The IDS device might also identify suspicious traffic and send alerts about that traffic without having the IPS device drop the traffic.

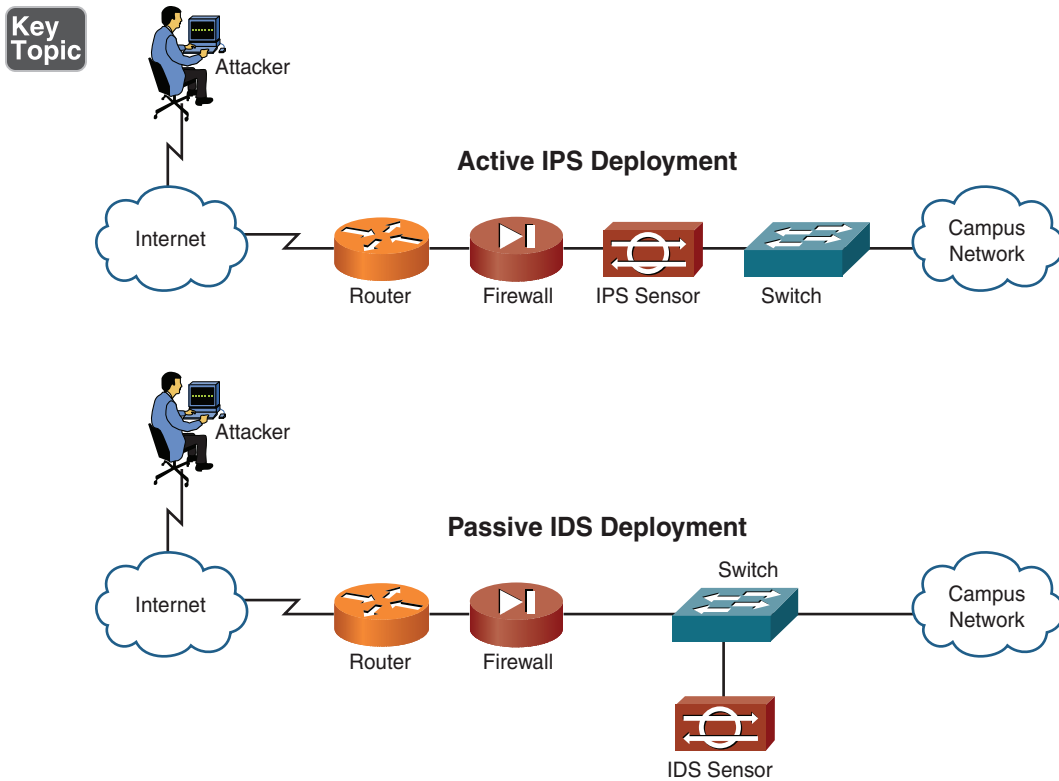


Figure 2-15 IDS and IPS Network Placement

IDS and IPS Device Categories

IDS and IPS devices can be categorized based on how they detect malicious traffic. Alternatively, IPS devices can be categorized based on whether they run on a network device or on a host.

Consider the following approaches to detecting malicious traffic:

Key Topic

- Signature-based detection
- Policy-based detection
- Anomaly-based detection

The following sections describe these methods in detail.

Signature-Based Detection

The primary method used to detect and prevent attacks using IDS or IPS technologies is signature based. A signature could be a string of bytes that, in a certain context, triggers detection. For example, attacks against a web server typically take the form of URLs. Therefore, URLs could be searched for a certain string that would identify an attack against a web server.

As another example, the IDS or IPS device could search for a pattern in the MIME header of an email message. However, because signature-based IDS/IPS is, as its name suggests, based on signatures, the administrator needs to routinely update the signature files.

Policy-Based Detection

Another approach to IDS/IPS detection is policy based. With a policy-based approach, the IDS/IPS device needs a specific declaration of the security policy. For example, you could write a network access policy that identifies which networks can communicate with other networks. The IDS/IPS device could then recognize out-of-profile traffic that does not conform to the policy and report that activity. Policy-based detection could also identify unencrypted channels and plaintext credentials and insecure protocols such as Telnet, SNMPv1, SNMPv2, HTTP, FTP, SLIP, and TFTP. Secure protocols such as SSH, SNMPv3, TLS/SSL, HTTPS, SFTP, and IPsec should be used when possible to protect the confidentiality of the data flows on the network.

Anomaly-Based Detection

A third approach to detecting or preventing malicious traffic is anomaly based. This approach is prone to false positives because a *normal* condition is difficult to measurably define. However, there are a couple of options for detecting anomalies:

- **Statistical anomaly detection:** This approach involves watching network traffic patterns over a period of time and dynamically building a baseline. Then, if traffic patterns significantly vary from the baseline, an alarm can be triggered.
- **Nonstatistical anomaly detection:** This approach allows an administrator to define what traffic patterns are supposed to look like. However, imagine that Microsoft releases a large update for its Windows 10 OS, and your company has hundreds of computers that are configured to automatically download that

service pack. If multiple employees turn on their computers at approximately the same time tomorrow morning, and multiple copies of the service pack simultaneously start to download from <https://www.microsoft.com>, the IDS/IPS device might consider that traffic pattern to be significantly outside the baseline. As a result, the nonstatistical anomaly detection approach could lead to a false positive (that is, an alarm being triggered in the absence of malicious traffic). However, an anomaly-based IPS may be able to indicate abnormal behavior, compared to the baseline of normal activity, which could assist you in discovering a new type of attack that is being used against your network. A zero-day attack is an attack that exploits a previously unknown vulnerability.

NOTE Anomaly-based detection is also known as *behavior-based detection*.

Load Balancer

A **load balancer** is a network device or software component that distributes incoming network traffic across multiple servers, systems, or resources in a network, ensuring optimal utilization of resources, high availability, and improved performance. Load balancers operate at the application layer (Layer 7) or transport layer (Layer 4) of the OSI model and use various algorithms, such as round-robin, least connections, or weighted distribution, to evenly distribute incoming requests or connections among the backend servers.

Notice how load balancers are important specialized network devices that can assist with performance and high availability of key network devices and services. For example, a load balancer can sit in front of a fleet of web servers in a data center. When a large volume of requests comes in from clients for the content of those web servers, the load balancer can ensure that the requests are intelligently distributed to the web servers. Often this intelligent distribution can be based on the resources available for each web server. More requests can be sent to those web servers that have the most available resources of the group. As you might expect, the load balancer can also check the health of the nodes periodically and intelligently prevent the forwarding of web requests to servers that are sickly.

Proxy Servers

Some clients are configured to forward their packets, which are seemingly destined for the Internet, to a **proxy server**. The proxy server receives the client's request, and on behalf of that client (that is, as that client's proxy), the proxy server sends the request out to the Internet. When a reply is received from the Internet, the proxy

server forwards the response on to the client. Figure 2-16 illustrates the operation of a proxy server.

**Key
Topic**

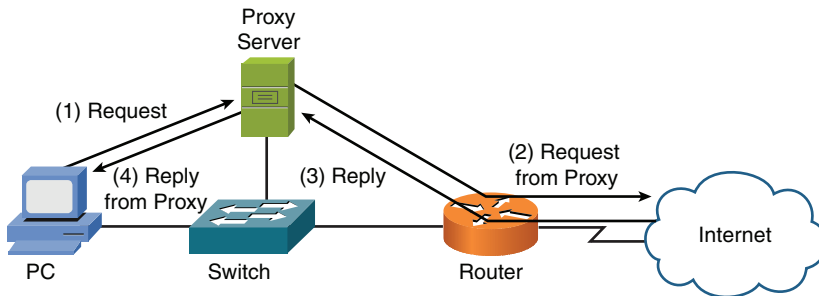


Figure 2-16 Proxy Server Operation

What possible benefit could come from such an arrangement? Security is one benefit. Specifically, because all requests going out to the Internet are sourced from the proxy server, the IP addresses of network devices inside the trusted network are hidden from the Internet.

Yet another benefit could come in the form of bandwidth savings because many proxy servers perform content caching. For example, without a proxy server, if multiple clients all visited the same website, the same graphics from the home page of the website would be downloaded multiple times (one time for each client visiting the website). However, with a proxy server performing content caching, when the first client navigates to a website on the Internet, and the Internet-based web server returns its content, the proxy server not only forwards this content to the client requesting the web page but also stores a copy of the content on its hard drive. Then, when a subsequent client points its web browser to the same website, after the proxy server determines that the page has not changed, the proxy server can locally serve up the content to the client, without having to once again consume Internet bandwidth to download all the graphic elements from the Internet-based website.

As a final example of a proxy server benefit, some proxy servers can perform content filtering to restrict clients from accessing certain URLs. For example, many companies use content filtering to prevent their employees from accessing popular social networking sites and promote productivity.

NOTE A reverse proxy receives requests on behalf of a server or servers and replies back to the clients on behalf of those servers. Reverse proxies can also be used with load-balancing and caching to better utilize a group of servers providing scalability and high availability.

Network-attached Storage (NAS)

In networks today, there is a greater need than ever for digital storage capabilities. Think about your own numerous personal files that need to be stored. It is no surprise, therefore, that many different technologies exist today to make storing files on the network simple and seamless.

One option today is to use *network-attached storage (NAS)*. A NAS device is a specialized type of storage device that is designed to provide centralized storage and file sharing capabilities to multiple users and devices within a network. Unlike traditional storage solutions that are directly attached to individual computers, NAS devices connect to a local area network or wide area network, allowing them to be accessed by any device connected to the network. NAS devices typically consist of one or more hard drives configured in a Redundant Array of Independent Disks (RAID) array for data redundancy and increased reliability.

NAS devices can range from simple single-drive enclosures suitable for home use to enterprise-grade systems capable of storing petabytes of data. They can also support various protocols and services, including file sharing protocols like Server Message Block (SMB) and Network File System (NFS), as well as backup and synchronization services. Additionally, many NAS devices offer other features such as media streaming, remote access, and even hosting applications like email servers or web servers, making them versatile solutions for both home and business environments.

Key Topic

Storage Area Networks (SANs)

As described earlier, networks will often use a NAS where disk storage can be delivered as a service over the network. As this approach evolved to include more efficiency and more features, the idea for a network dedicated to storing data was born.

A *storage area network (SAN)* is a high-speed, very specialized network that is designed to store massive amounts of data and make that data available quickly when clients and/or servers request it.

Originally, Fibre Channel was the technology that ruled the SAN space. Unlike Ethernet, Fibre Channel is designed for very low-latency, guaranteed lossless connectivity. In contrast, with UDP in an Ethernet environment, it is easy to contend with packet loss and delays. This is not the case in a SAN environment.

To make SANs even more accessible and flexible, they began to be created to support Internet Small Computer System Interface (iSCSI). With iSCSI, which is an IP-based technology for network storage, a client using the storage is referred to as an *initiator*, and the system providing the iSCSI storage is called the iSCSI *target*. Networks supporting iSCSI are often configured to support larger-than-normal frame sizes, referred to as *jumbo frames*.

NOTE Less commonly encountered these days in SANs is InfiniBand (IB), a communication technology that permits high-speed, low-latency communications between supercomputers.

Network convergence is often the goal today. For example, voice and video and data networks are successfully combined today. These very different types of communications can all be sent over the same network. Converging different forms of traffic on a network extended to SAN and LAN traffic is thanks to a remarkable invention: Fibre Channel over Ethernet (FCoE). Just like it sounds, this technology allows Ethernet to encapsulate Fibre Channel frames and carry them over a high-speed Ethernet infrastructure. Modifications were made to the Ethernet standards for FCoE to provide the level of service required in the SAN for Fibre Channel.

NOTE As you might guess, Ethernet needs to be truly high speed in order to have a chance at providing FCoE services. In the Cisco implementation, 10-Gbps Ethernet is required for FCoE! This might sound crazy and too difficult to accommodate, but keep in mind that FCoE is mostly used with data centers, where 10-Gbps and even 40-Gbps Ethernet links have become common.

Access Points (APs)

If you examine the sample wireless network topology shown in Figure 2-17, you will notice that wireless clients gain access to a wired network by communicating via radio waves with a wireless *access point (AP)*. The AP can then be hardwired to a LAN.

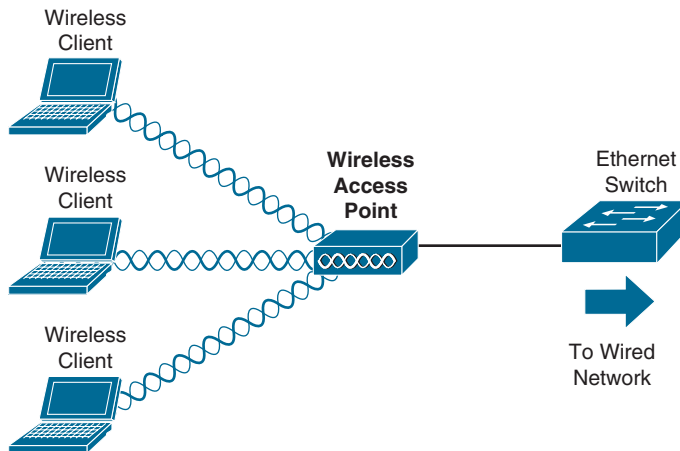


Figure 2-17 A Wireless Access Point in a Network

Wireless LANs include multiple standards that support various transmission speeds and security features. However, you need to understand, at this point, that all wireless devices connecting to the same AP are considered to be on the same *shared network segment*, which means that only one device can send data to and receive data from an AP at any one time.

Controllers

Access points tend to come in two major types: autonomous and lightweight. Lightweight access points do not have the control plane intelligence built in to perform their functions for the network. These devices are controlled by controllers, or more formally known as *wireless LAN controllers (WLCs)*. WLCs are specialized network devices that permit the central control and management of large numbers of lightweight access points. WLCs simplify the administration of your access points, and they can also assist you dramatically in the monitoring and ongoing maintenance of the wireless infrastructure.

Networking Device Summary

Table 2-1 provides a summary of the networking devices discussed in this chapter.

Key Topic

Table 2-1 Networking Device Summary

Device	Description	Key Points
Router	Connects networks	A router uses the software-configured network address to make forwarding decisions.
Switch	Connects devices on a twisted-pair network	A switch forwards data to its destination by using the MAC address embedded in each packet. It forwards data only to nodes that need to receive it.
Multilayer switch	Functions as a switch or router	Operates on Layers 2 and 3 of the OSI model as a switch and can perform router functionality.
Firewall	Provides controlled data access between networks	Firewalls can be hardware or software based. They are an essential part of a network's security strategy.
IDS/IPS	Detects and prevents intrusions	Monitors the network and attempts to detect/prevent intrusion attempts.
Load balancer	Distributes network load	Load balancing increases redundancy and performance by distributing the load to multiple servers.

Device	Description	Key Points
Proxy	Sends resource requests on behalf of network clients	Can add to the security and bandwidth efficiency of the network.
Network-attached storage (NAS)	Centralized storage device that connects to a network	Provides scalability and flexibility for the large amounts of storage today.
Storage area network (SAN)	A high-speed network that connects storage devices to servers	Allows multiple servers to easily and efficiently access large amounts of storage.
Access point (AP)	Enables wireless devices to connect to wireless and wired networks	Should provide the latest in security standards to help reduce security breaches.
Controller	Used with branch/remote office deployments for wireless authentication	When an AP boots, it authenticates with a controller before it can start working as an AP.

Applications and Functions

Now that we have examined just some of the physical and virtual appliances we might encounter in our home and business networks, let's discuss just a few of the many applications and functions you might encounter in modern networks.

Content Delivery Network (CDN)

A *content delivery network (CDN)* is a sophisticated infrastructure comprised of multiple servers distributed across various geographic locations, strategically positioned to deliver web content and digital media to users with optimal speed and reliability. For example, when a user requests access to content hosted on a web server, the CDN automatically identifies the user's location and routes the request to the nearest server within its network. By serving content from servers near the user, CDNs can significantly reduce latency and minimize the time it takes for content to reach the end-user device, enhancing the overall user experience.

CDNs can also help mitigate the risk of server overload and improve scalability by distributing the load of content delivery across multiple servers. This distributed architecture enables CDNs to handle high volumes of traffic efficiently, particularly during peak usage periods or sudden spikes in demand. CDNs can also offer advanced caching mechanisms that store copies of frequently accessed content on edge servers, further optimizing performance and reducing the strain on the servers of origin.

Virtual Private Network (VPN)

Companies with locations spread across multiple sites often require secure communications between those sites. One option is to purchase multiple WAN connections to interconnect those sites. Sometimes, however, a more cost-effective option is to create secure connections through an untrusted network, such as the Internet. Such a secure tunnel is called a *virtual private network (VPN)*.

VPNs provide a layer of privacy and security by encrypting data transmitted between the user's device and the VPN server, protecting it from potential eavesdropping or interception by third parties. This encryption ensures that even if data is intercepted, it remains unreadable and secure. Today, VPNs are commonly used by businesses to provide remote workers with secure access to the organization's internal network resources, such as files, applications, and databases, while ensuring data confidentiality and integrity. You will learn a lot more about VPNs in Chapter 17, "Network Access and Management Methods."

Quality of Service (QoS)

While the main concern of your network is to ensure that data packets reach their rightful destinations, it is the job of *quality of service (QoS)* to ensure that packets do not suffer from long delays (latency) or, worse, dropped packets.

QoS is actually a suite of technologies that allows you to strategically optimize network performance for select traffic types. For example, in today's converged networks (that is, networks simultaneously transporting voice, video, and data), some applications (for example, voice) might be more intolerant of delay (or *latency*) than other applications; for example, an FTP file transfer is less latency sensitive than a VoIP call. Fortunately, through the use of QoS technologies, you can identify which traffic types need to be sent first, how much bandwidth to allocate to various traffic types, which traffic types should be dropped first in the event of congestion, and how to make the most efficient use of the relatively limited bandwidth of an IP WAN.

NOTE Do not get confused by the many uses we have for the word *converged* in networking. It all depends on the context. For example, when speaking about the network in general and what data it can carry, a converged network is one that includes multiple forms of traffic—for example, VoIP and data traffic. When we are speaking of a single routing protocol, converged means the device has learned of all the updates that have been in the routing protocol's information.

A lack of bandwidth is the overshadowing issue for most network quality problems. Specifically, when there is a lack of bandwidth, packets might suffer from one or more of the symptoms listed in Table 2-2.

Key
Topic

Table 2-2 Three Categories of Quality Issues

Issue	Description
Delay	Delay is the time required for a packet to travel from source to destination. You might have witnessed delay on the evening news when the news anchor is talking via satellite to a foreign news correspondent. Because of the satellite delay, the conversation begins to feel unnatural.
Jitter	Jitter is the uneven arrival of packets. For example, imagine a VoIP conversation where packet 1 arrives at a destination router. Then, 20 ms later, packet 2 arrives. After another 70 ms, packet 3 arrives, and then packet 4 arrives 20 ms behind packet 3. This variation in arrival times (that is, <i>variable delay</i>) is not due to dropped packets, but the jitter might be interpreted by the listener as dropped packets.
Drops	Packet drops occur when a link is congested and a router's interface queue overflows. Some types of traffic, such as UDP traffic carrying voice packets, are not retransmitted if packets are dropped.

Fortunately, QoS features available on many routers and switches can recognize important traffic and then treat that traffic in a special way. For example, you might want to allocate 128 Kbps of bandwidth for your VoIP traffic and give that traffic priority treatment.

Consider water flowing through a series of pipes with varying diameters. The water's flow rate through those pipes is limited to the water's flow rate through the pipe with the smallest diameter. Similarly, as a packet travels from source to destination, its effective bandwidth is the bandwidth of the slowest link along that path. For example, in Figure 2-18, notice that the slowest link speed is 256 Kbps. This weakest link becomes the effective bandwidth between client and server.

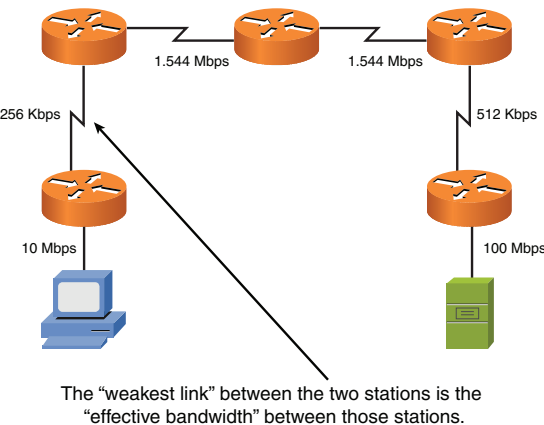


Figure 2-18 Effective Bandwidth of 256 Kbps

Because the primary challenge of QoS is a lack of bandwidth, the logical question is, “How do we increase available bandwidth?” A knee-jerk response to that question is often “Add more bandwidth.” However, more bandwidth often comes at a relatively high cost.

Think of your network as a highway system in a large city. During rush hour, the lanes of the highway are congested; during other periods of the day, the lanes might be underutilized. Instead of just building more lanes to accommodate peak traffic rates, the highway engineers might add a carpool lane to give higher priority to cars with two or more occupants. Similarly, you can use QoS features to give your mission-critical applications higher-priority treatment in times of network congestion.



Time to Live (TTL)

As you will learn more about in Chapter 9, “Routing Technologies,” a key function in the modern network is to provide a *time to live (TTL)* for communication packets. The TTL function plays a crucial role in preventing packets from circulating endlessly and potentially congesting the network. TTL is a field in the header of IP packets that indicates the maximum number of hops, or network devices, that a packet can pass through before being discarded. As a packet travels through a network, each router it encounters decrements the TTL value by one. If the TTL reaches zero before the packet reaches its destination, the router discards the packet and sends an Internet Control Message Protocol (ICMP) message back to the sender, notifying them of the packet’s expiration. This is just one of the many functions our networks simply could not live without.

Real-World Case Study

Acme, Inc. is growing dramatically and has decided to build a new headquarters from scratch. This has presented Acme with the opportunity to implement a modernized infrastructure with some of the latest physical and virtual appliances.

Acme, Inc. plans to deploy high-performance routers and switches throughout the facility to facilitate fast and reliable data transmission. These devices form the backbone of Acme’s network, ensuring seamless connectivity between different departments and locations.

Security of the HQ network has become a growing concern. As such, next-generation firewalls (NGFWs) will be deployed to protect Acme’s network from cyber threats such as malware, ransomware, and unauthorized access. Advanced intrusion detection and prevention systems will also be implemented to monitor network traffic in real time and mitigate potential security breaches.

Inside the new HQ building, a few of the users will have corporate-issued mobile devices designed to help facilitate work tasks. To accommodate network access for these mobile users, wireless APs, which are physically connected through cabling to the switches on each floor, will be used. To consolidate hardware, multilayer switches will be used to provide not only Layer 2 forwarding of frames based on MAC addresses but also Layer 3 forwarding of packets based on IP addresses (routing).

Acme, Inc. also plans to support a dedicated storage area network to provide centralized storage for Acme's critical data and applications. This should allow for efficient data management, backup, and recovery while ensuring high availability and scalability.

Finally, a robust VPN will also be installed in the new headquarters facility to allow users who are connected to the Internet from their home or other locations to build a secure VPN remote access connection to the corporate headquarters. Instead of buying a dedicated VPN device, Acme, Inc. is going to use a firewall that has this VPN capability integrated as part of its services.

Summary

Here are the main topics covered in this chapter:

- This chapter contrasted the roles of various networking device infrastructure components, including switches, routers, and wireless LAN controllers.
- This chapter provided examples of specialized network devices and explained how they can enhance a network. These devices included firewalls, proxy servers, and IPS and IDS devices.
- This chapter presented some of the popular applications and functions found in modern networks today, such as content delivery networks (CDNs), virtual private networks (VPNs), and quality of service (QoS).

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 2-3 lists these key topics and the page number where each is found.

**Table 2-3** Key Topics for Chapter 2

Key Topic Element	Description	Page Number
Figure 2-1	Router Broadcast and Collision Domains	37
Figure 2-12	Switch Collision and Broadcast Domains	44
Figure 2-13	Multilayer Ethernet Switch	44
Figure 2-14	Firewall	45
Figure 2-15	IDS/IPS Network Placement	47
List	Approaches to detecting malicious traffic	47
Figure 2-16	Proxy Server Operation	50
Section	Storage Area Networks (SANs)	51
Table 2-1	Networking Device Summary	53
Table 2-2	Three Categories of Quality Issues	56
Section	Time to Live (TTL)	57

Complete Tables and Lists from Memory

Print a copy of Appendix B, “Memory Tables,” or at least the section for this chapter, and complete as many of the tables as possible from memory. Appendix C, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

access point (AP), content delivery network (CDN), firewall, intrusion detection system (IDS), intrusion prevention system (IPS), Layer 2 Ethernet switch, Layer 3 capable switch, load balancer, multilayer switch, network-attached storage (NAS), proxy server, quality of service (QoS), router, storage area network (SAN), switch, time to live (TTL), virtual private network (VPN), wireless LAN controller (WLC)

Additional Resources

What Is a Switch in Networking: <https://youtu.be/NDVImep5UBI?si=KfjYEgAwZuh1023g>

Hub, Switch, & Router Explained: https://youtu.be/1z0ULvg_pW8?si=7PUV83vMuy8AWqsr

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. A router operates at what layer of the OSI model?
 - a. Layer 1
 - b. Layer 2
 - c. Layer 3
 - d. Layer 4
2. Which network infrastructure device primarily makes forwarding decisions based on Layer 2 MAC addresses?
 - a. Router
 - b. Switch
 - c. Hub
 - d. Multilayer switch
3. A router primarily makes its forwarding decisions based on what address?
 - a. Destination MAC address
 - b. Source IP address
 - c. Source MAC address
 - d. Destination IP address
4. What does a switch do by default when it receives a frame that possesses an unknown destination MAC address?
 - a. The switch floods the frame out all ports, except the port on which it was received.
 - b. The switch drops the frame.
 - c. The switch buffers the frame.
 - d. The switch returns the frame to the device that originated it.
5. In a router that has 12 ports, how many broadcast domains does the router have?
 - a. None
 - b. 1
 - c. 2
 - d. 12

6. In a switch that has 12 ports, how many collision domains does the switch have?
 - a. None
 - b. 1
 - c. 2
 - d. 12
7. What device is used to connect wireless clients to the wireless and wired networks within an organization?
 - a. Firewall
 - b. Controller
 - c. Load balancer
 - d. Access point
8. What networking device often uses signatures to help protect a network from known malicious attacks?
 - a. Wireless LAN controller
 - b. Switch
 - c. Proxy server
 - d. IPS
9. What device can act as both a content cache and a URL filter and is typically configured in the client browser of an end-user system?
 - a. Wireless LAN controller
 - b. Access point
 - c. Proxy server
 - d. IPS
10. You are using a network firewall in your enterprise infrastructure. This firewall requires you to define which connections are permitted outbound. Once you do so, the appropriate and expected return data flows are dynamically allowed through the firewall. What firewall characteristic does this scenario describe?
 - a. Stateless
 - b. Stateful
 - c. Deterministic
 - d. Zone-based

11. Which of the following is a network device that continually scans the network looking for attack activity and can actually stop potential threats?
 - a. IDS
 - b. IPS
 - c. Load balancer
 - d. FCoE

12. Which of the following is a specialized type of storage device that is designed to provide centralized storage and file sharing capabilities to multiple users and devices within a network?
 - a. NAS
 - b. Access point
 - c. Controller
 - d. WLC

13. Which of the following network functions enables administrators to better predict bandwidth use, monitor that usage, and control the usage to ensure that bandwidth is available to the applications that need it?
 - a. Firewall
 - b. QoS
 - c. CDN
 - d. TTL

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This chapter covers the following topics related to Objective 1.3 (Summarize cloud concepts and connectivity options) of the CompTIA Network+ N10-009 certification exam:

- Network functions virtualization (NFV)
- Virtual private cloud (VPC)
- Network security groups
- Network security lists
- Cloud gateways
 - Internet gateway
 - Network address translation (NAT) gateway
- Cloud connectivity options
 - VPN
 - Direct Connect
- Deployment models
 - Public
 - Private
 - Hybrid
- Service models
 - Software as a service (SaaS)
 - Infrastructure as a service (IaaS)
 - Platform as a service (PaaS)
- Scalability
- Elasticity
- Multitenancy

Cloud Concepts

It is not just a fad. It is not just marketing hype. Cloud computing is the future, and it is exploding in usage and usefulness. In this chapter, you will learn about some of the most foundational and critical topics in the vast field of cloud computing today.

Foundation Topics

Network Functions Virtualization (NFV)

One of the technologies that has helped cloud computing become a very popular reality is virtualization. While virtualization of servers was the initial push, these days we are seeing virtualization move into all aspects of Information Technology (IT). *Network functions virtualization (NFV)* is a great example of this. NFV abstracts and virtualizes important network functions that would traditionally be implemented by physical network appliances. Examples of network functions that can be virtualized with NFV are numerous and include

- Routing
- Switching
- Firewalling
- Load balancing
- Content delivery networking (CDN)
- Network monitoring

Thanks to NFV, the complex functionalities of the preceding network functions are decoupled from proprietary hardware and implemented as software applications. While this fact alone presents many advantages, IT administrators love the fact that they can run these applications on standard commercial off-the-shelf (COTS) servers. This decoupling of the network functions from specific hardware enables network operators to dynamically deploy, manage, and orchestrate network services with greater agility and efficiency, while also facilitating the optimization of resource utilization and operational costs. Specifically, just some of the advantages of NFV include

- **Doing more with less:** NFV typically permits dramatic increases in workload capacity with less power consumption, a smaller data center footprint, and reduced cooling requirements.
- **Reduced vendor lock-in:** Because NFV fosters the use of COTS servers, organizations can avoid being locked into proprietary hardware solutions from a single vendor.
- **Increased flexibility:** NFV tends to increase the agility of network operations because NFV facilitates quick changes to the network infrastructure, especially when compared to the traditional, nonvirtualized environments.

Cloud Networking Components

If an organization is interested in building IT solutions in the cloud, it is going to need key networking components and is also going to want to easily secure these resources. In this section, let's examine how an organization can accomplish both of these requirements with the cloud services of a leader in cloud technology—Amazon Web Services (AWS).

NOTE While this section focuses on AWS-specific examples, it is important to realize that other major cloud providers offer similar services. The specific names just vary.

Virtual Private Cloud (VPC)

The AWS *virtual private cloud (VPC)* lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including the selection of your IP address range, creation of subnets, and configuration of route tables and network gateways. You can use both IPv4 and IPv6 in VPCs for secure and easy access to resources and applications. Figure 3-1 shows elements inside a VPC.

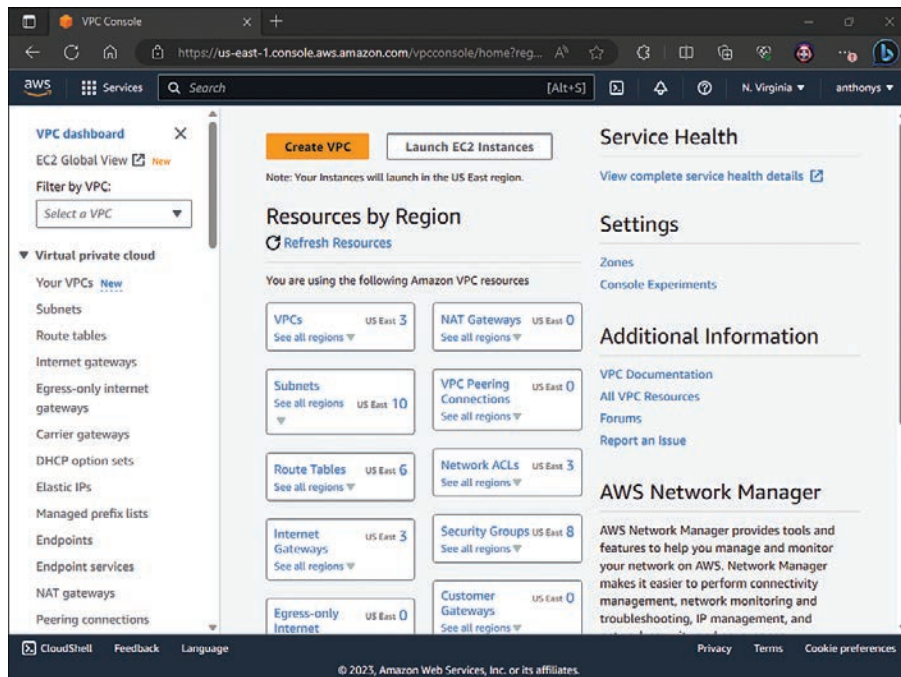


Figure 3-1 Some Components of an AWS VPC

NOTE AWS is not the only public cloud vendor to offer virtual private cloud functionality. Other major public cloud providers like Google Cloud Platform (GCP) and Microsoft Azure allow VPC functionality. Be aware that in Microsoft Azure, the VPC-like service is named Virtual Networks.

Network Security Groups

Because the security of resources in the cloud is a prime concern for both cloud customers and cloud providers, it is no big surprise that AWS provides built-in firewalls for use with its virtual compute resources. These *network security groups* help you easily control the accessibility of your virtual machines and their resources. Figure 3-2 shows an example of a network security group (named sg-f76c8d80) assigned to a virtual network interface in AWS.

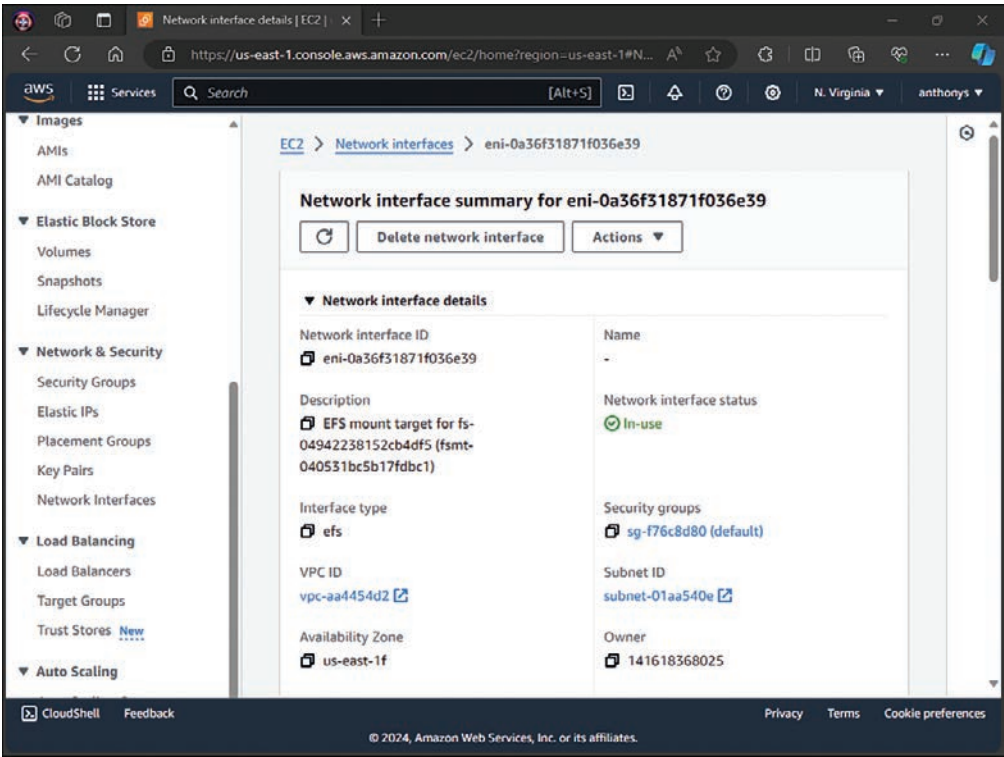


Figure 3-2 A Security Group Assigned to a vNIC

NOTE AWS refers to network security groups as simply security groups and often abbreviates them as SGs.

Consider the case of a web tier available in the AWS VPC. The AWS customer can configure the security group for this tier to permit HTTP and HTTPS traffic from customers using the web tier, and at the same time they can permit their team of support engineers to access the web tier using SSH and RDP. All other protocol attempts at accessing the web tier are denied by the security group.

Note that AWS applies a security group by attaching the security group to the virtual network interface (vNIC) of a virtual machine. This allows the security group to easily control (firewall) the traffic entering and exiting the virtual machine instance.

Network Security Lists

To further enhance cloud security, AWS also implements a *network security list* functionality (like other cloud vendors). AWS specifically terms its solution *network access control lists (NACLs)*. NACLs allow cloud customers to control access to their VPC subnets. NACLs are stateless constructs, which means customers must configure inbound and outbound rules, as there is no automatic recognition of state with traffic flows, and there are no automated access entries.

Students are often puzzled about why these “firewalls” exist when there are already security groups inside AWS. Remember that security groups are attached to virtual network interfaces of virtual machine instances and can control traffic to and from the virtual machines. NACLs are attached to subnets in a VPC to control traffic into and out of the entire subnet. Having multiple levels of built-in firewalls within the AWS network infrastructure permits fine-grained control of security.

Figure 3-3 shows an example of a NACL inside AWS.

Cloud Gateways

A *cloud gateway* is network infrastructure component or service that provides secure and efficient connectivity between on-premises networks or devices and cloud-based resources, applications, or services. Cloud gateways act as intermediaries or bridges, facilitating communication, data exchange, and access control between the local network and cloud environments, such as public cloud platforms, software as a service (SaaS) applications, or hybrid cloud deployments.

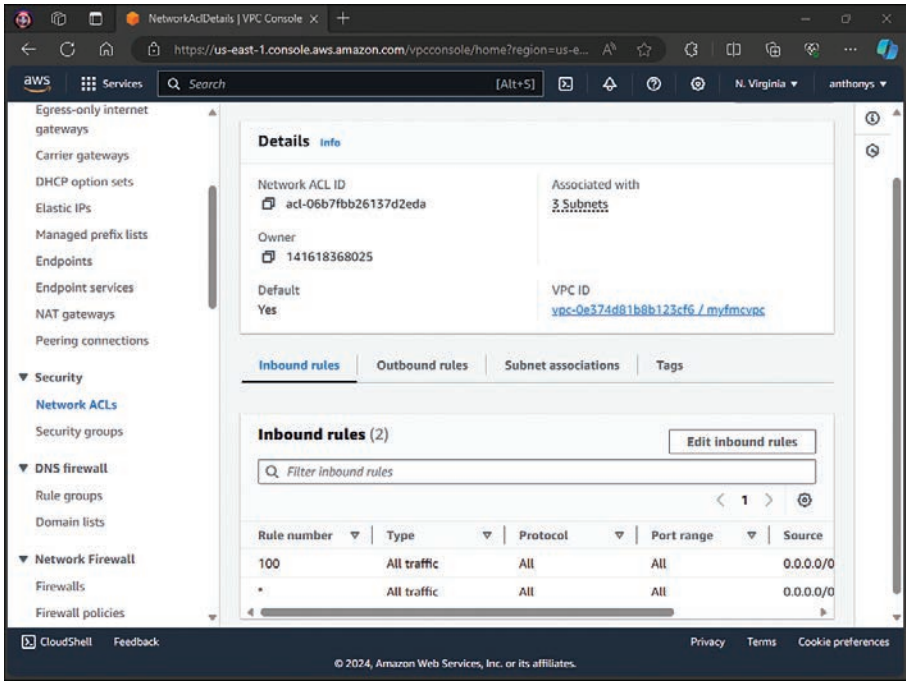


Figure 3-3 A NACL in AWS

There are several different types of specific cloud gateways commonly seen in cloud environments. These include the following:

- Internet gateway:** An Internet gateway serves as a pivotal entry and exit point for network traffic between the cloud environment and the broader Internet. As a virtualized component within the AWS infrastructure, the internet gateway seamlessly connects resources hosted in virtual private clouds to external networks. This connectivity enables bidirectional communication while ensuring security and scalability. By facilitating the flow of data between AWS services and the Internet, the Internet gateway allows organizations to build, deploy, and manage cloud-based applications with flexibility and efficiency, while also providing robust access control mechanisms and integration options for establishing secure connections with on-premises infrastructure.
- Network address translation (NAT) gateway:** While an Internet gateway in AWS seamlessly facilitates Internet connectivity, it does not provide fine-grained control over the network address translations required to translate private-use-only IP addresses with publicly routable IP addresses. Should a customer want to control these NAT translations, AWS (and its competitors) offers a NAT gateway. Acting as a managed service, the NAT gateway

translates outgoing traffic from private IP addresses of instances within the VPC into public IP addresses, allowing them to communicate with external networks while concealing their internal addresses.

Deployment Models

How do we implement (or deploy) cloud services most often today? In this section, we examine several different deployment models for cloud. Virtualized services and solutions are often offered by service providers as *cloud computing*. A company purchasing cloud computing services has several options to choose from:

Key Topic

- **Private cloud:** With this model, the cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers, which might be business units. It might be owned, managed, and operated by the organization, a third party, or some combination of both, and it might exist on or off premises. The key is that this cloud is not for use by multiple organizations.
- **Public cloud:** With this model, the cloud infrastructure is provisioned for open use by the general public. It might be owned, managed, and operated by a business, an academic institution, a government organization, or some combination of the three. It exists on the premises of the cloud provider, and these premises are typically located all over the globe to facilitate reduced latency from any location. This is the model presented by AWS and its main competitors, Microsoft Azure and Google Cloud Platform.
- **Hybrid cloud:** With this model, the cloud infrastructure is a composition of two or more distinct cloud infrastructures (private or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability. This is a widespread deployment model today. For example, an organization might build a private cloud for operational financial transactions while relying on a public cloud model for encrypted archiving of legacy transactions that must be maintained for compliance reasons.

NOTE While the main cloud deployment models are private, public, and hybrid, there is also the **community cloud** model. Community cloud services are used by individuals, companies, or entities with similar interests. A community cloud is typically a partnership between many different cloud vendors or organizations. An excellent example of a community cloud is cloud.gov in the United States. This community cloud provides a wide variety of services to employees of the U.S. government. These services are offered in the community cloud by several different public cloud vendors.

When an organization requires scalability, wants reduced costs, lacks in-house administrative personnel trained in cloud technologies, and has a complex network, a public cloud is the best deployment model to use. The advantages of using a public cloud include the following:

- Lower infrastructure, maintenance, and administrative costs
- Greater hardware efficiency
- Reduced implementation times
- Availability of short-term usage

When an organization needs to maintain strict control of business-critical data or is in a highly regulated businesses, such as the financial industry, a private cloud is typically the best choice.

A hybrid cloud environment is the best choice when an organization offers services that need to be configured for diverse vertical markets or has varying needs. Hybrid clouds are also ideal for organizations migrating to public clouds because they can ensure interoperability and efficiencies across the different environments.

Service Models

Cloud services are typically offered using one of several “as a service” models, which are usually offered on a pay-as-you-go basis. This is similar to the model used with utilities in the home (for example, electricity, Internet).

The following are a few of the service models that are available as part of cloud computing:

Key Topic

- **Infrastructure as a service (IaaS):** With IaaS, the company rents virtualized servers (which are hosted by a service provider) and runs specific applications on those servers. IaaS is made simple thanks to public cloud providers like AWS, Microsoft Azure, and Google Cloud Platform.
- **Software as a service (SaaS):** With SaaS, the details of the servers are hidden from the customer, and the customer’s experience is similar to the experience of using a web-based application. Examples of SaaS in use today include Gmail, Microsoft 365, and Google Docs.
- **Platform as a service (PaaS):** PaaS provides a development platform for companies that are developing applications and want to focus on creating the software without having to worry about the servers and infrastructure that are being used for that development. AWS Elastic Beanstalk and Google App Engine are examples of public cloud PaaS offerings.

NOTE Countless “as a service” designations have been established. For example, *desktop as a service (DaaS)* refers to cloud-hosted OS desktops made available to cloud clients. The term *everything as a service (XaaS)* is often used to describe the vast array of cloud services available now that the use of cloud approaches has exploded in popularity.

Key Cloud Concepts

We could easily fill a book this size with details of cloud technology, but for the CompTIA Network+ exam, you need to know the following key aspects:

Key Topic

- Cloud connectivity options
- Multitenancy
- Elasticity
- Scalability

Cloud Connectivity Options

As more and more technology has appeared in the cloud (both public and private), secured connectivity has become crucial. Secure protocols such as HTTPS, TLS, and SSH are necessary when accessing most cloud resources. Fortunately, massive cloud providers such as AWS make it simple to securely connect using a wide variety of methods, including hardware *virtual private network (VPN)* appliances located at your corporate or home office. Although your applications and servers might be sharing physical equipment with other customers of AWS, great pains are taken to ensure that this multitenancy (described in the next section) does not come at the cost of security.

AWS and other large public cloud providers go a step further, allowing customers to purchase AWS *Direct Connect* circuits. These circuits are private, dedicated leased lines directly into the AWS infrastructure. When you use AWS Direct Connect, your traffic does not commingle with that of other AWS customers or other Internet users at all. Of course, these types of connections are more expensive than other options.

NOTE The first thought most network engineers have when thinking about a service like AWS Direct Connect is the added security. Another huge benefit, however, is the predictable, low-latency performance such a private connection can offer.

Multitenancy

If you ask engineers why they are hesitant to move to the public cloud, they are likely to say that they are worried about security. When they react this way, it is almost always because they are thinking about *multitenancy*. This is a fancy term for the fact that the physical servers in the public cloud infrastructure are hosting workloads for many different customers. Your virtual machines (VMs) might be located on the same physical server as the VMs of your biggest competitor. Multitenancy requires you to have great faith that the cloud provider has done excellent setup of security work to ensure that the VMs of one customer are completely hidden from other customers.

NOTE When you have the money to spend and need the certainty, you can purchase a fully dedicated hosting solution. In such a situation, the cloud provider leases to you your own dedicated hardware in its cloud—and this hardware is not shared with other customers.

Key Topic

Elasticity

Elasticity is another extremely compelling reason to move to the cloud. Elasticity means that a cloud solution can scale (up or down or in or out) as needed based on demand. Scaling up or down means adding or subtracting resources to provide larger or smaller systems. Scaling in or out refers to allowing systems to clone themselves or terminate themselves, as needed.

NOTE The National Institute of Standards and Technology (NIST) includes rapid elasticity as one of the five key characteristics that define what *cloud* truly means. The other four characteristics are on-demand self-service, broad network access, resource pooling, and measured service. If you are interested in reading this important document for yourself, check out NIST SP 800-145.

Elasticity is truly incredible. When your cloud offers you rapid elasticity (and it should!), your network infrastructure and services offered can grow and shrink along with demand. This is incredibly exciting for many reasons, but perhaps the two biggest reasons are the fact that you do not have to worry (as much) about your infrastructure failing under a large load, and you do not have to worry about over-provisioned resources when demand is low.

Scalability

It should be no surprise that if your cloud solution can grow and shrink as needed (elasticity), it has no problem demonstrating great scalability. *Scalability* refers to a solution's ability to grow with need or demand. Cloud technology today can do even better than this: It can grow or shrink as needed.

Real-World Case Study

Acme, Inc. is considering a move to the cloud with some key services it is having problems scaling from its on-prem, local data center. Specifically, Acme, Inc. is considering a public cloud implementation for the streaming of new training videos globally. Acme, Inc. needs these movies to be delivered in a format that is based on the bandwidth of the access device.

To facilitate this new public cloud initiative, IT personnel from Acme, Inc. have created a virtual private cloud in AWS and constructed the necessary infrastructure to support the deployment and conversion of video content globally. Acme is thrilled to be taking advantage of a hybrid cloud deployment model. It is utilizing its on-premises resources right alongside its cloud resources to facilitate a much-improved global delivery network.

Acme, Inc. is also investigating the changes in the security posture of the organization with this move. Specifically, Acme is exploring its security responsibilities in this new solution compared to the security responsibilities of the cloud provider. Acme personnel are carefully constructing security groups and network access control lists to accommodate careful, secured access of cloud resources.

Summary

Here are the main topics covered in this chapter:

- This chapter covered network functions virtualization and described this emerging trend in detail.
- This chapter described some of the key cloud networking components found today, including virtual private clouds, network security groups, network security lists, and cloud gateways.
- This chapter covered the deployment models often seen with cloud in networking today. These include private, public, hybrid, and community clouds.
- This chapter also covered the service models seen today with cloud, including SaaS, IaaS, and PaaS.
- Finally, this chapter covered key cloud concepts you should be aware of, including connectivity options, multitenancy, elasticity, and scalability.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 3-1 lists these key topics and the page number where each is found.

Key
Topic

Table 3-1 Key Topics for Chapter 3

Key Topic Element	Description	Page Number
List	Deployment models	71
List	“As a service” models	72
List	Key cloud concepts	73
Section	Elasticity	74

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

cloud gateway, community cloud, desktop as a service (DaaS), Direct Connect, elasticity, hybrid cloud, infrastructure as a service (IaaS), Internet gateway, multitenancy, network address translation (NAT) gateway, network functions virtualization (NFV), network security group, network security list, platform as a service (PaaS), private cloud, public cloud, scalability, software as a service (SaaS), virtual private cloud (VPC), virtual private network (VPN)

Additional Resources

Google Cloud Platform: Deploying Cloud Marketplace Solutions to Google Kubernetes Engine (GKE): <https://www.ajsnetworking.com/gcp-compute/>

Cloud Versus Outsourcing: <https://www.ajsnetworking.com/comptia-cloud-essentials-cloud-vs-outsourcing/>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. What cloud component is assigned to a virtual machine and acts as a firewall, controlling traffic into and out of the VM?
 - a. Virtual private cloud
 - b. Network security list
 - c. Network security group
 - d. Internet gateway
2. What type of cloud deployment model features many customers consuming resources from a single provider?
 - a. Public
 - b. Private
 - c. Community
 - d. Hybrid
3. What type of “as a service” model features a powerful cloud environment targeted at developers who need to test and deploy updates to their applications?
 - a. IaaS
 - b. SaaS
 - c. PaaS
 - d. DaaS
4. Which cloud characteristic refers to the ability to dynamically scale resources as needed during times of great demand as well as in times of low demand?
 - a. Scalability
 - b. Elasticity
 - c. On-demand
 - d. Centralized
5. Which of the following are popular options when it comes to ensuring security of cloud-based data in transit? (Choose two.)
 - a. Public Internet
 - b. VPN

- c. Direct private connection
 - d. RDP
- 6. Which term refers to your company's data being stored on shared physical servers in the public cloud infrastructure?
 - a. Hybrid
 - b. Multitenancy
 - c. Elasticity
 - d. Orchestration
- 7. Which of the following enables you to provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network?
 - a. VPC
 - b. NFV
 - c. Internet gateway
 - d. NAT gateway
- 8. Which of the following are the best examples of SaaS?
 - a. AWS, Microsoft Azure, and Google Cloud Platform
 - b. AWS Elastic Beanstalk, Google App Engine, and IBM Cloud
 - c. AWS S3 and Azure Virtual Networks
 - d. Gmail and Microsoft 365

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This chapter covers the following topics related to Objective 1.4 (Explain common networking ports, protocols, services, and traffic types) of the CompTIA Network+ N10-009 certification exam:

- File Transfer Protocol (FTP): Ports 20/21
- Secure File Transfer Protocol (SFTP): Port 22
- Secure Shell (SSH): Port 22
- Telnet: Port 23
- Simple Mail Transfer Protocol (SMTP): Port 25
- Domain Name System (DNS): Port 53
- Dynamic Host Configuration Protocol (DHCP): Ports 67/68
- Trivial File Transfer Protocol (TFTP): Port 69
- Hypertext Transfer Protocol (HTTP): Port 80
- Network Time Protocol (NTP): Port 123
- Simple Network Management Protocol (SNMP): Ports 161/162
- Lightweight Directory Access Protocol (LDAP): Port 389
- Hypertext Transfer Protocol Secure (HTTPS): Port 443
- Server Message Block (SMB): Port 445
- Syslog: Port 514
- Simple Mail Transfer Protocol Secure (SMTPS): Port 587
- Lightweight Directory Access Protocol over SSL (LDAPS): Port 636
- Structured Query Language (SQL) Server: Port 1433
- Remote Desktop Protocol (RDP): Port 3389
- Session Initiation Protocol (SIP): Ports 5060/5061
- Internet Protocol (IP) types
 - Internet Control Message Protocol (ICMP)
 - Transmission Control Protocol (TCP)
 - User Datagram Protocol (UDP)
 - Generic Routing Encapsulation (GRE)
 - Internet Protocol Security (IPsec)
 - Authentication Header (AH)
 - Encapsulating Security Payload (ESP)
 - Internet Key Exchange (IKE)
- Traffic types
 - Unicast
 - Multicast
 - Anycast
 - Broadcast

Networking Ports, Protocols, Services, and Traffic Types

This chapter might initially look like a reference chapter, and it can certainly serve that function, but this chapter is much more than that. This chapter also helps prepare you for more in-depth explanations of many of these protocols in later chapters of this book. Additionally, this chapter teaches you about the various IP protocol types in use today and the traffic types that help make our networks so incredibly useful.

Foundation Topics

Ports and Protocols

I know it might be very intimidating to look at this section of the chapter and realize just how many different protocols (and ports) are needed in a typical network today. Resist the urge to panic! Through study and practice, you will master these different protocols and the ports they use.

As you will quickly learn in this chapter, there are several common networking ports that are used frequently. Ports 0 through 1023 are defined as well-known ports. Registered ports are from 1024 to 49151. The remainder of the ports from 49152 to 65535 are used dynamically by applications.

FTP (File Transfer Protocol)

File Transfer Protocol (FTP) can be used to transfer files with a remote host (which typically requires authentication of user credentials). It uses TCP ports 20 and 21 in its operation. Unfortunately, FTP users authenticate using cleartext mechanisms. It is one of many protocols that lacks security (and is becoming rarely used as a result).

SFTP

Secure File Transfer Protocol (SFTP) provides FTP file transfer service over an SSH connection. Just like SSH, this secure protocol uses TCP port 22 in its operation.

SSH

Secure Shell (SSH) is used to securely connect to a remote host (typically via a terminal emulator). SSH has become the new de facto standard method of remote access for devices that use a command-line interface. SSH uses TCP port 22 in its operation.

Telnet

Telnet is used to connect to a remote host (typically via a terminal emulator). This remote access protocol should be used only in a lab or practice environment because it does not provide security. The alternative to this protocol that you should use in production is SSH. Telnet uses TCP port 23 in its operation.

SMTP

Simple Mail Transfer Protocol (SMTP) is used for sending email throughout a network. This protocol uses TCP port 25 in its operation.

DNS (Domain Name System)

Domain Name System (DNS) is used to translate domain names, such as www.pearsonitcertification.com, into IP addresses, such as 165.193.123.44. DNS uses a hierarchical namespace that enables the database of hostname-to-IP address mappings to be distributed across multiple servers. DNS uses both TCP and UDP in its operation. The port used in both cases is port 53.

DHCP (Dynamic Host Configuration Protocol)

You can use *Dynamic Host Configuration Protocol (DHCP)* to dynamically assign IP address information (for example, IP address, subnet mask, DNS server IP address, and default gateway IP address) to a network device. DHCP uses UDP ports 67 and 68 in its operation.

TFTP

Trivial File Transfer Protocol (TFTP) transfers files with a remote host (and does not require authentication of user credentials). TFTP uses UDP port 69 in its operation.

HTTP

Hypertext Transfer Protocol (HTTP) retrieves content from a web server. This protocol operates using TCP port 80. Many web server administrators liked to move the protocol to port 8080 in an attempt to circumvent the massive number of attacks against the default port 80. HTTP lacks security, and as a result, today, it is hard to find websites that are running HTTP. HTTPS is now the de facto standard for web servers.

NTP

Network devices use *Network Time Protocol (NTP)* to synchronize their clocks with a time server (that is, an NTP server). This protocol relies on UDP port 123 in its operation.

SNMP

Simple Network Management Protocol (SNMP) is used to monitor and manage network devices. SNMP uses UDP ports 161 and 162 in its operation. You will learn many details about SNMP in Chapter 14, “Network Monitoring.”

LDAP

Lightweight Directory Access Protocol (LDAP) provides directory services (for example, a user directory that includes username, password, email, and phone number information) to network clients. LDAP is used to access and query compliant directory services systems, such as Microsoft Active Directory. TCP port 389 is used in its operation.

HTTPS

Hypertext Transfer Protocol Secure (HTTPS) is used to securely retrieve content from a web server. This secure version of HTTP uses TCP port 443 in its operation. HTTPS was originally made possible by the *Secure Sockets Layer (SSL)* protocol. *Transport Layer Security (TLS)* is the latest version of this technology. Because TLS is closely related to SSL, we often simply say that HTTPS is made possible by SSL/TLS.

SMB

Server Message Block (SMB) is used to share files, printers, and other network resources. This protocol was originally used by Microsoft clients only. Since then, UNIX and Linux clients can install software to also use SMB for file sharing. Older versions of SMB used both UDP and TCP port 3020. Today, newer versions of SMB (after Windows 2000) use port 445 on top of a TCP stack in order to communicate over the Internet.

Syslog

Syslog is the standard used by network devices (and other computer systems) to report and send log messages on status information and health events to network administrators. In fact, to centrally store these messages, they are often sent to a Syslog server. This data is often called *machine data*. Storing and analyzing this information, which may include driver failures, device conflicts, read/write errors, timeouts, and bad block errors, can be very important. Syslog uses UDP port 514 in its operation.

SMTPS

Simple Mail Transfer Protocol Secure (SMTPS) uses TCP port 587 in its operation and is the secure version of SMTP. SMTPS uses Transport Layer Security to provide authentication of the communication partners along with data integrity and confidentiality by wrapping SMTP data in TLS. This is like how HTTPS wraps HTTP data inside TLS.

LDAPS

Lightweight Directory Access Protocol over SSL (LDAPS) is the secure version of LDAP. It operates over TCP port 636.

Structured Query Language (SQL) Server

SQL (Structured Query Language) is a standardized programming language used to manage and manipulate relational databases by performing tasks such as querying, updating, and managing data. You can use this SQL language via a specialized server to run powerful queries against data that is stored in databases. A *Structured Query Language (SQL) Server* can use TCP and UDP in its operation. Port 1433 was reserved for this Microsoft invention.

RDP

Remote Desktop Protocol (RDP) is a Microsoft protocol that allows a user to view and control the desktop of a remote computer. RDP is a presentation layer protocol that supports a Windows-based Remote Desktop Connection (RDC) between an RDP client (formerly known as Windows Terminal Client) and a server. This remote access technology uses both TCP and UDP port 3389 in its operation.

SIP

Session Initiation Protocol (SIP) is used to create and end sessions for one or more media connections, including Voice over IP (VoIP) calls. SIP can use both TCP and UDP ports 5060 and 5061 in its operation.

Protocol/Port Summary

Table 4-1 provides a summary of protocols and their assigned ports

**Key
Topic****Table 4-1** Port Assignments for Commonly Used Protocols

Protocol	Description	Port Assignment
DHCP (Dynamic Host Configuration Protocol)	Dynamically assigns IP address information (for example, IP address, subnet mask, DNS server IP address, and default gateway IP address) to a network device	UDP 67, 68
DNS (Domain Name System)	Resolves domain names to corresponding IP addresses	TCP/UDP 53
FTP (File Transfer Protocol)	An insecure protocol used to transfer files with a remote host (typically requires authentication of user credentials)	TCP 20, 21
HTTP (Hypertext Transfer Protocol)	An insecure protocol used to retrieve content from a web server	TCP 80
HTTPS (Hypertext Transfer Protocol Secure)	Used to securely retrieve content from a web server	TCP 443
LDAP (Lightweight Directory Access Protocol)	Provides directory services (for example, a user directory that includes username, password, email, and phone number information) to network clients	TCP 389
LDAPS (Lightweight Directory Access Protocol over SSL)	The secure version of LDAP	TCP 636
NTP (Network Time Protocol)	Used by a network device to synchronize its clock with a time server (NTP server)	UDP 123
RDP (Remote Desktop Protocol)	A Microsoft protocol that allows a user to view and control the desktop of a remote computer	TCP/UDP 3389
SFTP (Secure File Transfer Protocol)	Provides FTP file transfer service over an SSH connection	TCP 22
SIP (Session Initiation Protocol)	Used to create and end sessions for one or more media connections, including Voice over IP (VoIP) calls	TCP/UDP 5060, 5061
SMB (Server Message Block)	Used to share files, printers, and other network resources	TCP 445
SMTP (Simple Mail Transfer Protocol)	Used for sending email throughout the network	TCP 25
SMTPS (Simple Mail Transfer Protocol Secure)	Allows for secure communication between email clients and mail servers, ensuring that email messages are encrypted during transmission	TCP 587

Protocol	Description	Port Assignment
SNMP (Simple Network Management Protocol)	Used to monitor and manage network devices; SNMPv3 is the latest and most secure version of SNMP and allows for the use of strong cryptographic algorithms, such as HMAC-SHA, HMAC-MD5, and AES, to ensure data integrity and confidentiality	UDP 161, 162
SQL (Structured Query Language) Server	A server that specializes in SQL and is used to run powerful queries against data that is stored in databases	TCP/UDP 1433
SSH (Secure Shell)	Used to securely connect to a remote host (typically via a terminal emulator)	TCP 22
Syslog	The standard used by network devices (and other computer systems) to report on status information and events	UDP 514
Telnet	An insecure protocol used to connect to a remote host (typically via a terminal emulator)	TCP 23
TFTP (Trivial File Transfer Protocol)	Used to transfer files with a remote host (does not require authentication of user credentials)	UDP 69

Internet Protocol (IP) Types

As you have probably realized, there are many, many protocols that are considered part of the TCP/IP protocol suite. This section provides an overview of several of the IP protocol types used within TCP/IP. Many of these protocols are discussed in detail in other chapters.

Remember, **Internet Protocol (IP)** is a network layer protocol, documented in RFC 791, that offers a connectionless internetwork service. IP provides features for addressing, packet fragmentation and reassembly, type-of-service specification, and security.

Internet Control Message Protocol (ICMP)

In addition to TCP and UDP, **Internet Control Message Protocol (ICMP)** is another transport layer protocol you are likely to meet. ICMP is a network layer Internet protocol documented in RFC 792 that reports errors and provides other information relevant to IP packet processing. It is used by utilities such as ping and traceroute.

Transmission Control Protocol (TCP)

The transport layer of the OSI model offers an important protocol for reliable transport—*Transmission Control Protocol (TCP)*. What does *reliable transport* mean, in networking terms? It means that if a segment is dropped, the sender can detect that drop and retransmit the dropped segment. Specifically, a receiver acknowledges segments that it receives. Based on those acknowledgments, a sender can decide which segments were successfully received and which segments need to be transmitted again.

User Datagram Protocol (UDP)

User Datagram Protocol (UDP) can be considered an opposite approach to TCP. Unlike TCP, UDP offers unreliable transport. This means that if a segment is dropped, the sender is unaware of the drop, and no retransmission occurs. Note that UDP is another option at the transport layer of the OSI model. This works as an ideal alternative to TCP.

NOTE The reliability of TCP versus the unreliability of UDP is often referred to as *connectionless* versus *connection-oriented* communications. You might use the analogy of a phone call versus a postcard. With a phone call (which represents TCP), as you send messages, you ensure that the connection remains strong and that communications are clear. For example, you might ask “can you hear me now?” as you move to a different part of the house for a better cell connection. In fact, you begin the connection with a “hello” and end the connection with a “goodbye.” UDP operates more like a postcard: There is no connection maintained at all. You drop the postcard in the mail and hope that it reaches its destination.

Generic Routing Encapsulation (GRE)

Generic Routing Encapsulation (GRE), as its name implies, is a multipurpose, simple tunneling protocol that you might call on in a wide variety of circumstances. For example, let’s say you have traffic that you want to protect over an IPsec tunnel, but this traffic is multicast traffic. IPsec does not support securing this type of traffic. That is where GRE can come into play. The multicast can be encapsulated inside GRE, and then this GRE traffic can be protected by IPsec. This is just one example of many where GRE can come in handy.

Internet Protocol Security (IPsec)

Internet Protocol Security (IPsec) is a complex suite of protocols that is used to create secured connections between network systems. In fact, IPsec is quickly becoming the de facto standard for VPN connections. IPsec features the use of *Authentication Header (AH)* and *Encapsulating Security Payload (ESP)*. As you would guess, AH handles authentication functions, while ESP takes care of encryption. IPsec also takes advantage of the *Internet Key Exchange (IKE)* protocol. IKE negotiates cryptographic keys and security parameters to facilitate the secure IPsec data transmissions over the Internet.

Internet Protocol (IP) Types Summary

Table 4-2 provides a summary of the TCP/IP suite protocols described so far in this chapter.



Table 4-2 Internet Protocol (IP) Types Summary

Protocol	Complete Protocol Name	Description
IPsec	Internet Protocol Security	A complex suite of protocols that is used to create secured connections between network systems.
GRE	Generic Routing Encapsulation	A tunneling protocol used to encapsulate a wide variety of network layer protocols inside virtual point-to-point links or point-to-multipoint links over an IP network.
IP	Internet Protocol	A network layer protocol, documented in RFC 791, that offers a connectionless internetwork service. IP provides features for addressing, packet fragmentation and reassembly, type-of-service specification, and security.
TCP	Transmission Control Protocol	A connection-oriented protocol that offers flow control, sequencing, and retransmission of dropped packets.
UDP	User Datagram Protocol	A connectionless alternative to TCP used for applications that do not require the functions offered by TCP.
ICMP	Internet Control Message Protocol	Sends error messages and operational information indicating issues with packet delivery in IP networks.

Protocol	Complete Protocol Name	Description
ESP	Encapsulating Security Payload	Provides for encryption in the IPsec suite of protocols.
AH	Authentication Header	Provides for authentication in the IPsec suite of protocols.
IKE	Internet Key Exchange	Negotiates cryptographic keys and security parameters for IPsec.

Traffic Types

In our IP-based networks of today, traffic flows in different manners as dictated by the various traffic types we create. In this section, we examine these different traffic types and how data moves throughout the network with each.

Unicast

Most network traffic is *unicast* in nature, meaning that traffic travels from a single source device to a single destination device. Figure 4-1 illustrates an example of a unicast transmission.

Key
Topic

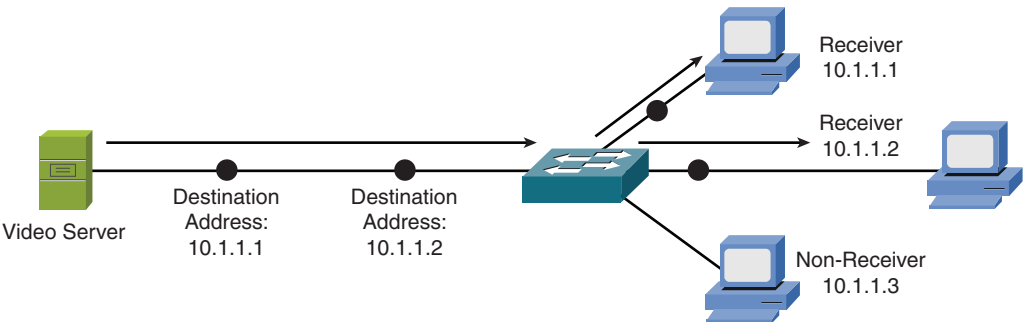


Figure 4-1 Sample Unicast Transmission

Notice the Layer 3 IP addresses shown in Figure 4-1. We delve deep into these addresses in Chapter 7, “IPv4 Addressing,” but for now it is worth noting that these are unicast IP addresses to facilitate this most common form of IP traffic.

It is at this point that you might ask, “Why do we even need other forms of traffic in the IP network?” This is a great question. The answer lies with how inefficient unicast traffic would be when we have something (let’s say a large video file) that we want to send to all devices on the network. Creating a separate copy of the video for every single recipient and then sending the data to every single network node would

be horribly inefficient. This is a big part of the reason why we have other network traffic types that we see in this section of the chapter.

Broadcast

Broadcast traffic travels from a single source to all destinations on a network (that is, a *broadcast domain*) as shown in Figure 4-2. Notice the use of a very special IP address to facilitate the broadcast traffic type—it is 255.255.255.255.

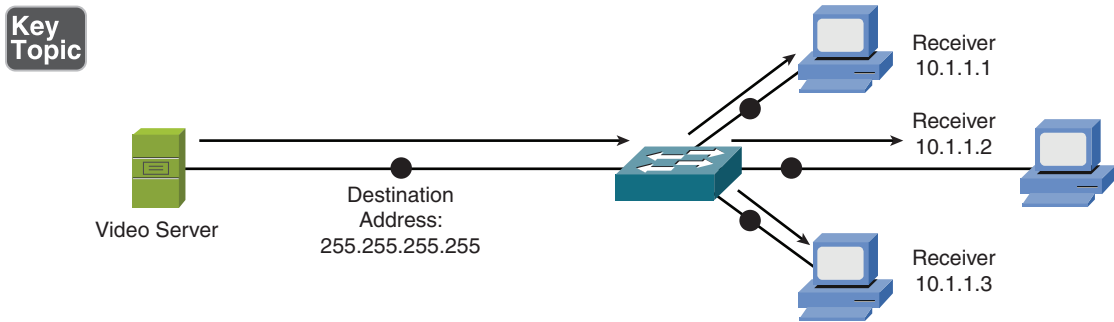


Figure 4-2 Sample Broadcast Transmission

It might seem as though the broadcast address 255.255.255.255 would reach all hosts on all interconnected networks. However, 255.255.255.255 targets all devices on a single network—specifically, the network local to the device sending a packet destined for 255.255.255.255. Another type of broadcast address is a *directed broadcast address*, which targets all devices in a remote network. For example, the address 172.16.255.255 /16 is a directed broadcast address targeting all devices in the 172.16.0.0 /16 network.

Multicast

Multicast technology offers an efficient mechanism for a single host to send traffic to multiple specific destinations. For example, say that a network has 100 users, and 20 of those users want to receive a video stream from a video server. With a unicast solution, the video server would have to send 20 individual streams, 1 for each recipient. As described earlier in this section, such a solution could consume a significant amount of network bandwidth and put a heavy processor burden on the video server.

With a broadcast solution, the video server would only have to send the video stream once; however, it would be received by every device on the local subnet, even devices not wanting to receive the video stream. Even though a lot of the devices do

not want to receive the video stream, they still must pause what they are doing and take time to check each of these unwanted packets.

As shown in Figure 4-3, multicast offers a compromise, allowing the video server to send the video stream only once and sending the video stream only to devices on the network that want to receive the stream. Multicast is possible thanks to the use of what is called a Class D IP address. A Class D address, such as 239.1.2.3, represents the address of a *multicast group*. The video server could, in this example, send a single copy of each video stream packet destined for 239.1.2.3. Devices wanting to receive the video stream could join the multicast group. Based on the device request, switches and routers in the topology could then dynamically determine out of which ports the video stream should be forwarded.

Key Topic

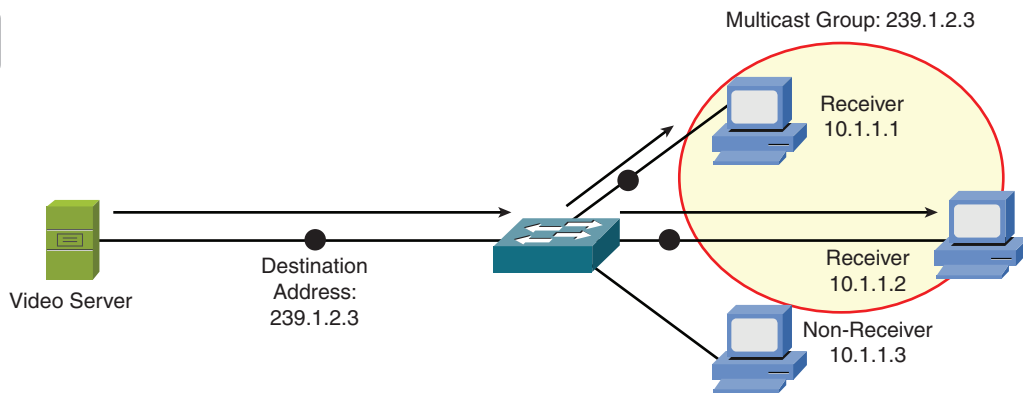


Figure 4-3 Sample Multicast Transmission

Are there other classes of IP addresses? There sure are, and you will learn all about them in Chapter 7.

Anycast

With *anycast*, a single IP address is assigned to multiple devices, as illustrated in Figure 4-4. It is a one-to-nearest (from the perspective of a router's routing table) communication flow.

Notice the IP addresses shown in Figure 4-4, which might look quite odd to you. These are IPv6 addresses. As you will learn in Chapter 8, “Evolving Use Cases,” IPv6 is the next-generation version of IP and it is becoming more and more common in our networks today. While anycast traffic is possible in IPv4, it is going to be much more commonly used in the new IPv6 protocol suite.

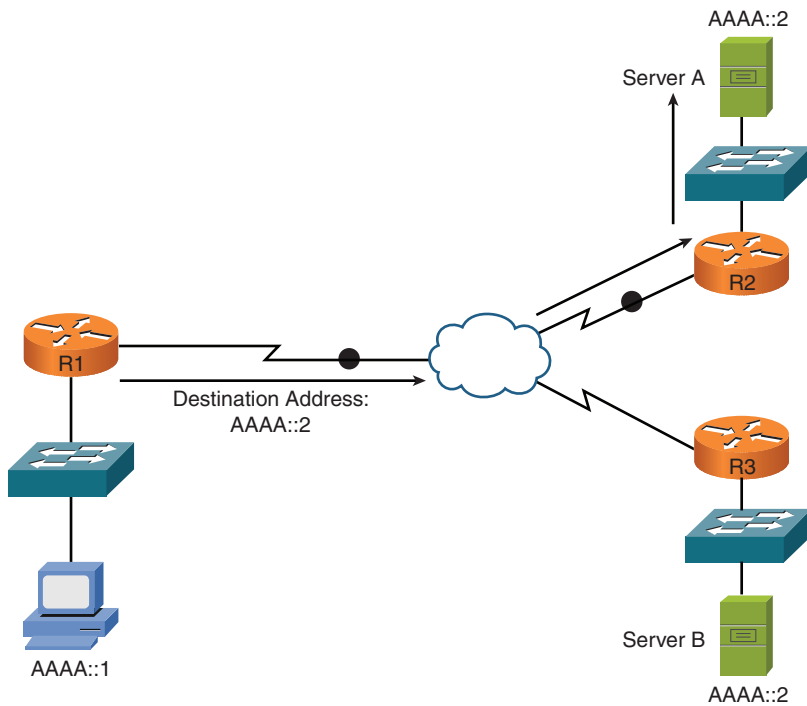


Figure 4-4 An IPv6 Anycast Example

In Figure 4-4, a client with IPv6 address AAAA::1 wants to send traffic to destination IPv6 address AAAA::2. Notice that two servers (Server A and Server B) have the IPv6 address AAAA::2. In the figure, the traffic destined for AAAA::2 is sent to Server A, via router R2, because the network on which Server A resides appears to be closer than the network on which Server B resides, from the perspective of router R1's IPv6 routing table.

Remember that you need to distinguish among these four network traffic types:

- With unicast, traffic is one-to-one communication between a single sender and a single receiver, and this is the most common type of traffic in IP networks. It is used for point-to-point communication between devices, such as client-server interactions and most Internet browsing.
- Multicast is a mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group.

- Anycast traffic is one-to-nearest communication from a sender to the nearest of multiple receivers. Multiple devices share the same anycast IP address, but packets are routed to the nearest (or best) destination based on routing metrics such as shortest path or lowest latency.
- A broadcast is at the opposite end of the spectrum from a unicast. Broadcast traffic is one-to-all communication from a single sender to all devices within a specific network segment. Each packet in a broadcast transmission is addressed to a special broadcast IP address (e.g., 255.255.255.255 in IPv4) or a Layer 2 broadcast address (e.g., MAC address FF:FF:FF:FF:FF:FF).

NOTE IPv6 was engineered to provide many advantages over the IPv4 protocol. One of the massive advantages is that broadcast frames and packets from IPv4 do not exist in an IPv6-only network. IPv6 uses only unicasts, multicasts, and anycasts, as described in this section. With IPv6, if you want to send a frame or packet to all nodes in the local network, you use a special all-nodes IPv6 multicast address. This completely eliminates the need for the broadcast traffic type in IPv6-only networks.

Traffic Types Summary

Table 4-3 provides a summary of the traffic types described in this chapter.



Table 4-3 Traffic Types Summary

Traffic Type	Description
Unicast	Traffic is one-to-one communication between a single sender and a single receiver, and this is the most common type of traffic in IP networks. It is used for point-to-point communication between devices, such as client-server interactions and most Internet browsing.
Multicast	A mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group.
Anycast	Anycast traffic is one-to-nearest communication from a sender to the nearest of multiple receivers. Multiple devices share the same anycast IP address, but packets are routed to the nearest (or best) destination based on routing metrics such as shortest path or lowest latency.
Broadcast	A broadcast is at the opposite end of the spectrum from a unicast. Broadcast traffic is one-to-all communication from a single sender to all devices within a specific network segment.

Summary

Here are the main topics covered in this chapter:

- The focus of this chapter was to ensure you are well versed in the most common protocols we work with in the modern network. Notice that these protocols are often recognized by their well-known port numbers. When you see 443, you should immediately think about HTTPS traffic.
- This chapter also covered the most important IP protocol types in use today, including TCP, UDP, ICMP, GRE, and IPsec.
- Finally, this chapter covered the traffic types found in computer networks today, including unicast, multicast, anycast, and broadcast.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 4-4 lists these key topics and the page number where each is found.



Table 4-4 Key Topics for Chapter 4

Key Topic Element	Description	Page Number
Table 4-1	Port Assignments for Commonly Used Protocols	86
Table 4-2	Internet Protocol (IP) Types Summary	89
Figure 4-1	Sample Unicast Transmission	90
Figure 4-2	Sample Broadcast Transmission	91
Figure 4-3	Sample Multicast Transmission	92
Figure 4-4	An IPv6 Anycast Example	93
Table 4-3	Traffic Types Summary	94

Complete Tables and Lists from Memory

Print a copy of Appendix B, “Memory Tables,” or at least the section for this chapter and complete as many of the tables as possible from memory. Appendix C, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

anycast, Authentication Header (AH), broadcast, connection-oriented, connectionless, Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP), Encapsulating Security Payload (ESP), File Transfer Protocol (FTP), Generic Routing Encapsulation (GRE), Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), Internet Control Message Protocol (ICMP), Internet Key Exchange (IKE), Internet Protocol (IP), Internet Protocol Security (IPsec), Lightweight Directory Access Protocol (LDAP), Lightweight Directory Access Protocol over SSL (LDAPS), multicast, Network Time Protocol (NTP), Remote Desktop Protocol (RDP), Secure File Transfer Protocol (SFTP), Secure Shell (SSH), Secure Sockets Layer (SSL), Server Message Block (SMB), Session Initiation Protocol (SIP), Simple Mail Transfer Protocol (SMTP), Simple Mail Transfer Protocol Secure (SMTPS), Simple Network Management Protocol (SNMP), Structured Query Language (SQL) Server, Syslog, Telnet, Transmission Control Protocol (TCP), Transport Layer Security (TLS), Trivial File Transfer Protocol (TFTP), unicast, User Datagram Protocol (UDP)

Additional Resources

IP, ICMP, UDP, and TCP: <https://www.ajsnetworking.com/udp-and-tcp/>

Ports and Protocols Quiz for Network+: https://www.ajsnetworking.com/ports_protocols_net_plus

Review Questions

The answers to these review questions are in Appendix A, “Answers to Review Questions.”

1. What protocol is considered the de facto standard when it comes to secure access to remote systems for management purposes?
 - a. Telnet
 - b. SSH
 - c. IPsec
 - d. IMAP
2. You are interested in dynamically assigning the IP address information in your IPv4-based network infrastructure. What protocol can you use to accomplish this?
 - a. DNS
 - b. TFTP
 - c. FTP
 - d. DHCP
3. What global hierarchical system is used to resolve domain names to IP addresses?
 - a. TFTP
 - b. DHCP
 - c. NTP
 - d. DNS
4. What port and protocol are used by HTTPS? (Choose two.)
 - a. TCP
 - b. UDP
 - c. 443
 - d. 123
 - e. 8080
 - f. 80

5. What are the port and protocol used by Syslog? (Choose two.)
 - a. TCP
 - b. UDP
 - c. 148
 - d. 514
 - e. 240
6. What IP protocol do ping and traceroute use in their operation?
 - a. IPsec
 - b. DNS
 - c. ICMP
 - d. DHCP
7. What port is used for SSH?
 - a. 80
 - b. 443
 - c. 22
 - d. 23
8. What protocol negotiates cryptographic keys and security parameters for IPsec?
 - a. AH
 - b. ESP
 - c. IKE
 - d. GRE
9. What type of traffic uses a destination Layer 3 address of 255.255.255.255 and is no longer used in IPv6 environments?
 - a. Unicast
 - b. Multicast
 - c. Broadcast
 - d. Anycast

10. Which of the following is a Microsoft protocol that allows a user to view and control the desktop of a remote computer?
 - a. Syslog
 - b. SNMP
 - c. NTP
 - d. RDP

11. Which is a mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group?
 - a. Unicast
 - b. Multicast
 - c. Anycast
 - d. Broadcast



This chapter covers the following topics related to Objective 1.5 (Compare and contrast transmission media and transceivers) of the CompTIA Network+ N10-009 certification exam:

- Wireless
 - 802.11 standards
 - Cellular
 - Satellite
- Wired
 - 802.3 standards
 - Single-mode vs. multimode fiber
 - Direct attach copper (DAC) cable
 - Twinaxial cable
 - Coaxial cable
 - Cable speeds
 - Plenum vs. non-plenum cable
- Transceivers
 - Protocol
 - Ethernet
 - Fibre Channel (FC)
 - Form factors
 - Small form-factor pluggable (SFP)
 - Quad small form-factor pluggable (QSFP)
- Connector types
 - Subscriber connector (SC)
 - Local connector (LC)
 - Straight tip (ST)
 - Multi-fiber push on (MPO)
 - Registered jack (RJ) 11
 - RJ45
 - F-type
 - Bayonet Neill–Concelman (BNC)

Transmission Media and Transceivers

Many modern networks have a daunting number of devices, and it is your job to understand the function of each device and how it works with the others. To create a network, these devices need some sort of interconnection. An interconnection uses one of a variety of media types. This chapter dives into the world of physical media, both media you can see (wired) and media that you cannot see (wireless). You will learn about classic media technologies that set the stage for the modern, high-speed media used in networks today. You will also learn about the connectors used for this media.

Foundation Topics

Wireless

Later in this chapter, you will learn about a variety of wireless standards, which are all variants of the IEEE **802.11 standards**. As you contrast one standard with another, a characteristic to watch out for is the frequencies at which these standards operate. Although there are some country-specific variations, certain frequency ranges (or *frequency bands*) have been reserved internationally for industrial, scientific, and medical purposes. These frequency bands are called the *ISM bands*, where ISM derives from *i*ndustrial, *s*cientific, and *m*edical.

Two of these bands are commonly used for WLANs. Specifically, WLANs use the range of frequencies in the 2.4GHz–2.5GHz range (commonly referred to as the *2.4GHz band*) or in the 5.725GHz–5.875GHz range (commonly referred to as the *5GHz band*). In fact, some WLANs support a mixed environment, where 2.4GHz devices run alongside 5GHz devices.

NOTE The latest wireless standard, 802.11ax (Wi-Fi 6), takes Wi-Fi operation to the 6GHz frequency. Wi-Fi 6 uses up to 14 additional 80MHz channels or 7 additional super-wide 160MHz channels in the 6GHz band. This new technology is ideal for applications such as high-definition video streaming and virtual reality.

Within each band are specific frequencies (or *channels*) at which wireless devices operate. To avoid interference, nearby wireless APs should use frequencies that do not overlap with one another. You can use wireless survey tools to analyze what is currently in use so you can set up a new wireless system that does not compete for the frequencies that are already in use. Those same tools can also help you identify *wireless channel* utilization in existing and new wireless networks. Regarding channel selection, merely selecting different channels is not sufficient because transmissions on one channel spill over into nearby channels. Site survey tools can collect data to show the relative strength of signals in the areas being serviced by the APs. This output can be color-coded and overlaid on top of the floor plan and is often referred to as a *heat map* of the wireless signals. Heat maps are further covered in Chapter 13, “Organizational Processes and Procedures,” as part of the critical documentation that an organization should maintain regarding its networks.

Consider, for example, the 2.4GHz band. Here, channel frequencies are separated by 5MHz (with the exception of channel 14, which has 12MHz of separation from channel 13). However, a single channel’s transmission can spread over a frequency range of 22MHz. As a result, channels must have five channels of separation

($5 \times 5\text{MHz} = 25\text{MHz}$, which is greater than 22MHz). You can see from Figure 5-1 that, in the United States, you could select the non-overlapping channels 1, 6, and 11.

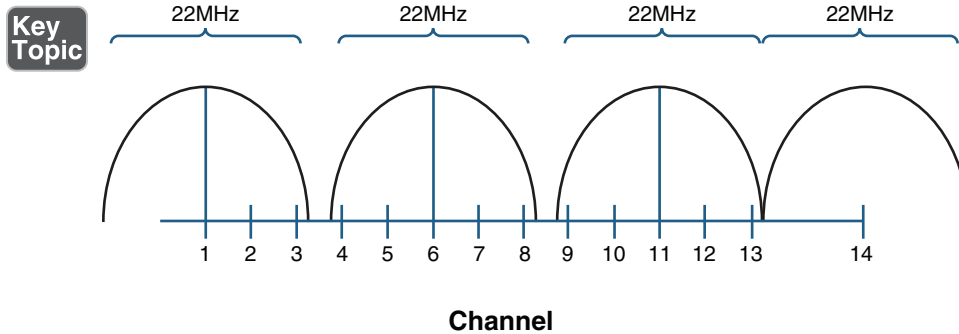


Figure 5-1 Non-overlapping Channels in the 2.4GHz Band

NOTE Remember that non-overlapping channels are a set of wireless frequency channels that do not interfere with each other. For example, in the 2.4GHz band, channels 1, 6, and 11 are non-overlapping channels for Wi-Fi networks. Even though some countries use channel 14 as a non-overlapping channel, it is not supported in the United States.

As a reference, Table 5-1 shows the specific frequencies for each of the channels in the 2.4GHz band.

Table 5-1 Channel Frequencies in the 2.4GHz Band

Channel	Frequency (GHz)	Recommended as a Non-overlapping Channel
1	2.412	Yes
2	2.417	No
3	2.422	No
4	2.427	No
5	2.432	No
6	2.437	Yes
7	2.442	No
8	2.447	No
9	2.452	No
10	2.457	No

Channel	Frequency (GHz)	Recommended as a Non-overlapping Channel
11	2.462	Yes
12	2.467	No
13	2.472	No
14	2.484	Yes (not supported in the United States)

The 5GHz band has more channels than the 2.4GHz band. Table 5-2 lists the recommended non-overlapping channels for the 5GHz band in the United States. Note that additional channels are supported in some countries.

Table 5-2 Non-overlapping Channels in the 5GHz Band Recommended for Use in the United States

Channel	Frequency (GHz)
36	5.180
40	5.200
44	5.220
48	5.240
52	5.260*
56	5.280*
60	5.300*
64	5.320*
100	5.500**
104	5.520**
108	5.540**
112	5.560**
116	5.580**
136	5.680**
140	5.700**
149	5.745
153	5.765
157	5.785
161	5.805
165	5.825

*Must support dynamic frequency selection to prevent interference with RADAR.

**Must be professionally installed.

Transmission Methods

In the previous section, you learned about the frequencies used for various wireless channels. However, you need to be aware that each of those frequencies is considered to be the *center frequency* of a channel. In actual operation, a channel uses more than one frequency, which is a transmission method called *spread spectrum*. These frequencies are, however, very close to one another, which results in a *narrowband transmission*.

Here are the three variations of spread-spectrum technology to be aware of for your study of wireless:

Key Topic

- **Direct-sequence spread spectrum (DSSS):** DSSS modulates data over an entire range of frequencies by using a series of symbols called *chips*. A chip is shorter in duration than a bit, meaning that chips are transmitted at a higher rate than is the actual data. These chips encode not only the data to be transmitted but also what appears to be random data. Although both parties involved in a DSSS communication know which chips represent actual data and which chips do not, if a third party intercepted a DSSS transmission, it would be difficult for that party to eavesdrop on the data because they would not easily know which chips represented valid bits. DSSS is more subject to environmental factors than FHSS and OFDM because of its use of an entire frequency spectrum.
- **Frequency-hopping spread spectrum (FHSS):** FHSS allows the participants in a communication to hop between predetermined frequencies. Security is enhanced because the participants can predict the next frequency to be used but a third party cannot easily predict the next frequency. FHSS can also provision extra bandwidth by simultaneously using more than one frequency.
- **Orthogonal frequency-division multiplexing (OFDM):** Whereas DSSS uses a high modulation rate for the symbols it sends, OFDM uses a relatively slow modulation rate for symbols. This slower modulation rate, combined with the simultaneous transmission of data over 52 data streams, helps OFDM support high data rates while resisting interference between the various data streams.

NOTE Wi-Fi 6 features the latest enhancements in technology and uses the latest spread-spectrum technology, called *orthogonal frequency-division multiple access (OFDMA)*. OFDMA provides sophisticated scheduling techniques to provide better performance in highly congested Wi-Fi environments.

WLAN Standards (802.11)

Most modern WLAN standards are variations of the original IEEE 802.11 standard, which was developed in 1997. This original standard supported a DSSS implementation and an FHSS implementation, both of which operated in the 2.4GHz band. However, with supported speeds of 1Mbps or 2Mbps, the original 802.11 standard lacks sufficient bandwidth to meet the needs of today's WLANs. The most popular variants of the 802.11 standard in use today are 802.11n, 802.11ac, and 802.11ax, as described in detail in the following sections.

802.11a

The *802.11a* WLAN standard, which was ratified in 1999, supported speeds as high as 54Mbps. Other supported data rates (which could be used if conditions were not suitable for the 54Mbps rate) included 6Mbps, 9Mbps, 12Mbps, 18Mbps, 24Mbps, 36Mbps, and 48Mbps. The 802.11a standard used the 5GHz band and used the OFDM transmission method. Interestingly, 802.11a never gained widespread adoption because it was not backward compatible with 802.11b, whereas 802.11g was backward compatible. However, it is worth noting that 802.11a was a possible alternative to 802.11b/g, as the 2.4GHz band was often far more crowded than the 5GHz band.

802.11b

The *802.11b* WLAN standard, which was ratified in 1999, supported speeds as high as 11Mbps. However, 5.5Mbps was another supported data rate. The 802.11b standard used the 2.4GHz band and used the DSSS transmission method.

802.11g

The *802.11g* WLAN standard, which was ratified in 2003, supported speeds as high as 54Mbps. Like 802.11a, 802.11g also supported data rates of 6Mbps, 9Mbps, 12Mbps, 18Mbps, 24Mbps, 36Mbps, and 48Mbps. However, like 802.11b, 802.11g operated in the 2.4GHz band, which allowed it to offer backward compatibility to 802.11b devices. 802.11g used either the OFDM or the DSSS transmission method.

802.11n (Wi-Fi 4)

The *802.11n* WLAN standard, which was ratified in 2009, supports a wide variety of speeds, depending on its implementation. Although the speed of an 802.11n network could exceed 300Mbps (through the use of channel bonding, which is discussed shortly), many 802.11n devices on the market have speed ratings in the

130Mbps–150Mbps range. An 802.11n WLAN can operate in the 2.4GHz band, the 5GHz band, or both simultaneously. 802.11n uses the OFDM transmission method.

One way 802.11n achieves superior throughput compared to earlier Wi-Fi standards is through the use of a technology called *multiple input, multiple output (MIMO)*.

MIMO uses multiple antennas for transmission and reception. These antennas do not interfere with one another, thanks to MIMO's use of *spatial multiplexing*, which encodes data based on the antenna from which the data will be transmitted. Both reliability and throughput are increased by MIMO's simultaneous use of multiple antennas. Chapter 11, “Configure Wireless Devices and Technologies,” will have lots more to say on the important subject of wireless antennas.

Yet another technology implemented by 802.11n is *channel bonding*. With channel bonding, two *wireless bands* are logically bonded together, forming a band with twice the bandwidth of an individual band. Some literature refers to channel bonding as *40MHz mode*, which is the bonding of two adjacent 20MHz bands into a 40MHz band.

The 802.11n High Throughput (HT) standard defines modes for ensuring that older a/b/g devices and newer 802.11n devices avoid collisions with each other.

NOTE The IEEE finally accepted the fact that using the standard identification codes for describing wireless to nontechnical end users makes no sense, and it is now using simpler names for recent standards. For example, 802.11n is called Wi-Fi 4, 802.11ac is Wi-Fi 5, and 802.11ax is Wi-Fi 6.

802.11ac (Wi-Fi 5)

802.11ac is a 5GHz standard that uses more simultaneous streams than 802.11n and features *multi-user MIMO (MU-MIMO)*. MU-MIMO is a set of advanced MIMO technologies included with IEEE 802.11ac and 802.11ax that dramatically enhances wireless throughput. MU-MIMO is an enhancement over the original MIMO technology. It allows antennas to be spread over a multitude of independent access points. Thanks to 802.11ac technology, a single 80MHz stream supports 433Mbps.

802.11ax (Wi-Fi 6)

802.11ax (Wi-Fi 6) is up to 30% faster than Wi-Fi 5 (802.11ac), but even more exciting are its lower latency, more simultaneously deliverable data, and improved power efficiency. Wi-Fi 6 is the first iteration of 802.11 to include OFDMA (which is an improvement on OFDM). OFDMA can transmit data to multiple devices at the same time. It does so by splitting traffic into smaller packets to eliminate

queueing. Also, Wi-Fi 6 adds MU-MIMO capabilities to upstream connections. The net effect of this is to allow more devices on one network at the same time.

NOTE What about Wi-Fi 7? Wi-Fi 7 is indeed being developed as I write this text. It is based on IEEE 802.11be technology and enhances Wi-Fi performance in the 2.4GHz, 5GHz, and 6GHz bands, bringing cutting-edge capabilities to enable innovations that require high throughput, lower latency, and greater reliability across home, enterprise, and industrial environments.

802.11 Standards Summary

Table 5-3 provides a reference to help you contrast the characteristics of the 802.11 standards.



Table 5-3 Characteristics of 802.11 Standards

Standard	Band	Maximum Bandwidth	Transmission Method	Maximum Range
802.11	2.4GHz	1Mbps or 2Mbps	DSSS or FHSS	20 m indoors/100 m outdoors
802.11a	5GHz	54Mbps	OFDM	35 m indoors/120 m outdoors
802.11b	2.4GHz	11Mbps	DSSS	32 m indoors/140 m outdoors
802.11g	2.4GHz	54Mbps	OFDM or DSSS	32 m indoors/140 m outdoors
802.11n	2.4GHz or 5GHz (or both)	> 300Mbps (with channel bonding)	OFDM	70 m indoors/250 m outdoors
802.11ac	5GHz	> 3Gbps (with MU-MIMO and several antennas)	OFDM	70 m indoors/250 m outdoors
802.11ax	2.4GHz, 5GHz, 6GHz	9.6Gbps	OFDMA	70 m indoors/250 m outdoors

Cellular

Others forms of wireless technologies are in a group termed *cellular* because the technology is used on cell phones (among other mobile devices). Some cellular phone technologies, such as *Long-Term Evolution (LTE)*, which supports a 100Mbps

data rate for mobile devices and a 1Gbps data rate for stationary devices, can be used to connect a mobile device such as a smartphone to the Internet.

Other technologies for cellular phones include the older 2G (Edge), which offers slow data rates. 2G was improved upon with 3G, in addition to the newer 4G, 5G, LTE, and *Evolved High-Speed Packet Access (HSPA+)*.

Tethering allows a smartphone's data connection to be used by another device, such as a laptop. Also, mobile hotspots are growing in popularity because these devices connect to a cell phone company's data network and make that data network available to nearby devices (typically, a maximum of five devices) via wireless networking technologies. For example, multiple passengers in a car can share a mobile hotspot and have Internet connectivity from their laptops or tablets while riding down the road.

Code-division multiple access (CDMA) and *Global System for Mobile Communications (GSM)* are the two major radio systems used in cell phones. GSM uses time-division multiple access (TDMA) in its operation.

Satellite

Many rural locations lack the option of connecting to an IP WAN or to the Internet via physical media (for example, a DSL modem or a broadband cable modem connection). For such locations, a *satellite* WAN connection, as illustrated in Figure 5-2, might be an option.

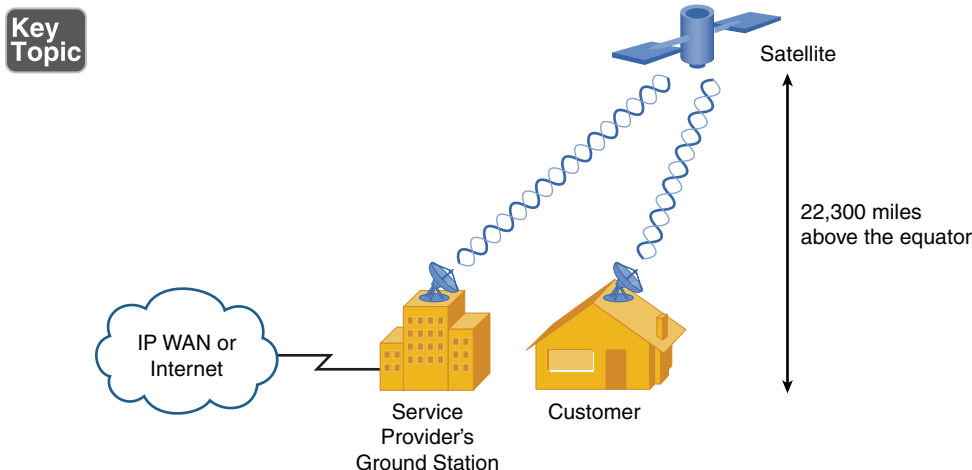


Figure 5-2 Satellite WAN Sample Topology

Most satellites used for WAN connectivity are in orbit above the earth's equator, about 22,300 miles high. Therefore, a customer in North America, for example, with a clear view of the southern sky would be able to install a satellite dish and establish a line-of-sight communication path with the orbiting satellite. The satellite would then relay transmissions back and forth between the customer's site and the service provider's ground station. The ground station could then provide connectivity, via physical media, to an IP WAN or to the Internet.

Two significant design considerations need to be taken into account:

- **Delay:** Radio waves travel at the speed of light, which is 186,000 miles per second, or 3×10^8 meters per second. This speed is specifically the speed of light (and radio waves) in a vacuum; technically, the speed of light (and radio waves) is a bit slower when traveling through air than when traveling through a vacuum, but for purposes of the following calculations, the speed of light in a vacuum is used. Although these are fast speeds, consider the distance between a customer and the satellite. If a customer were located 2000 miles north of the equator, the approximate distance between the customer site and the satellite could be calculated using the Pythagorean theorem: $d^2 = 2000^2 + 22,300^2$. Solving the equation for d , which is the distance between the customer and the satellite, yields a result of approximately 22,390 miles.

A transmission from a customer to a destination on the Internet (or an IP WAN) would have to travel from the customer to the satellite, from the satellite to the ground station, and then out to the Internet (or IP WAN). The propagation delay alone introduced by bouncing a signal off the satellite is approximately 241 ms—that is, $(22,390 \times 2) / 186,000 = .241$ seconds = 241 ms). In addition, there are other delay components, such as processing delay (by the satellite and other networking devices), making the one-way delay greater than one-fourth of a second and, therefore, the round-trip delay greater than one-half of a second. Such delays are not conducive to latency-sensitive applications such as Voice over IP (VoIP).

- **Sensitivity to weather conditions:** Because communication between a customer's satellite dish and an orbiting satellite must travel through the earth's atmosphere, weather conditions can impede communications. For example, if a thunderstorm is near the customer location, that customer might temporarily lose connectivity with their satellite.

Based on these design considerations, even though satellite WAN technology offers tremendous flexibility in terms of geographic location, more terrestrial-based solutions are preferred.

NOTE Remember, wireless technologies receive more coverage in Chapter 11.

Copper and Fiber Media and Connectors

A network is an interconnection of devices. Those interconnections occur over some type of media. The media might be physical, such as a copper or fiber-optic cable. Alternatively, the media might be the air, through which radio waves propagate (as is the case with wireless networking technologies).

This section examines copper and fiber physical media types and the connectors they commonly use.

Coaxial Cable

Coaxial cable (referred to as *coax*) consists of two conductors. As illustrated in Figure 5-3, one of the conductors is an inner insulated conductor. This inner conductor is surrounded by another conductor that is sometimes made of a metallic foil or woven wire.



Figure 5-3 Coaxial Cable

Because the inner conductor is shielded by the metallic outer conductor, coaxial cable is resistant to electromagnetic interference (EMI). For example, EMI occurs when an external signal is received on a wire and might result in a corrupted data transmission. As another example, EMI occurs when a wire acts as an antenna and radiates electromagnetic waves, which might interfere with data transmission on another cable. Coaxial cables have an associated characteristic impedance that needs to be balanced with the device (or terminator) with which the cable connects.

NOTE The term *electromagnetic interference (EMI)* is sometimes used interchangeably with the term *radio frequency interference (RFI)*.

There are three common types of coaxial cables:

- **RG-59:** Typically used for short-distance applications, such as carrying composite video between two nearby devices. This cable type has loss characteristics such that it is not right for long-distance applications. RG-59 cable has a characteristic impedance of 75 ohms.

- **RG-6:** Used by local cable companies to connect individual homes to the cable company's distribution network. Like RG-59 cable, RG-6 cable has a characteristic impedance of 75 ohms.
- **RG-58:** Has loss characteristics and distance limitations like those of RG-59. However, the characteristic impedance of RG-58 is 50 ohms, and this type of coax was popular with early 10BASE2 Ethernet networks.

Although RG-58 coaxial cable was commonplace in early computer networks (that is, 10BASE2 networks), coaxial cable's role in modern computer networks is as the media used by cable modems. Cable modems are commonly installed in residences to provide high-speed Internet access over the same connection used to receive multiple television stations.

NOTE Far less popular is *twinaxial cable*, commonly called *twinax*. This is very similar to coaxial cable, but it uses two inner conductors instead of one.

Common connectors used on coaxial cables include the following:

**Key
Topic**

- **BNC:** A *Bayonet Neill-Concelman (BNC)* connector (*British Naval Connector* in some literature) can be used for a variety of applications, including as a connector in a 10BASE2 Ethernet network. A BNC coupler could be used to connect two coaxial cables together back-to-back.
- **F-connector:** An F-connector is often used for cable TV (including cable modem) connections. Notice that some, including CompTIA, refer to it as *F-type connector*. F-type connectors are used with coaxial cable, most commonly to connect cable modems and TVs. F-type connectors are screw-type connectors.

Figure 5-4 shows what both of these connectors look like.

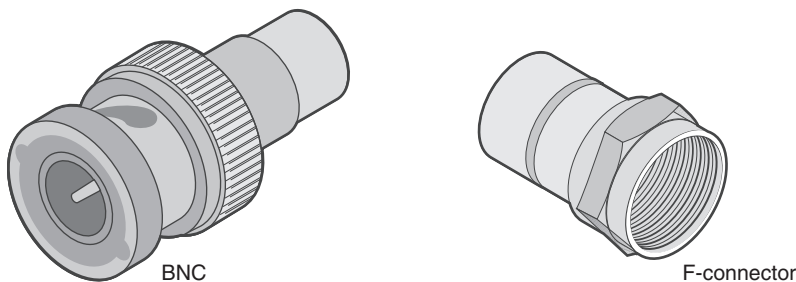


Figure 5-4 Coaxial Cable Connectors

One type of twinax copper cable you might encounter in the data center is a ***direct attach copper (DAC) cable***. This cable is used to connect ports of data center network equipment over short distances (think 1 to 2 meters). These cables feature pluggable connectors on each end. As you might expect, the type of connector varies for the equipment used. DAC cables typically consist of copper conductors enclosed in a protective jacket and are terminated with standard connectors, such as small form-factor pluggable (SFP) or quad small form-factor pluggable (QSFP) connectors, discussed later in this chapter.

Twisted-Pair Cable

Today's most popular LAN media type is ***twisted-pair*** cable, in which individually insulated copper strands are intertwined. Two categories of twisted-pair cable are shielded twisted pair (STP) and unshielded twisted pair (UTP). A UTP coupler could be used to connect two UTP cables, back-to-back. Also, for adherence to fire codes, you might need to select plenum-rated cable versus nonplenum cable as discussed later in this chapter.

To define industry-standard pinouts and color coding for twisted-pair cabling, the TIA/EIA-568 standard was developed. The first iteration of the TIA/EIA-568 standard, which was released in 1991, has come to be known as the *TIA/EIA-568A* standard.

NOTE The TIA/EIA acronym comes from Telecommunications Industry Association/Electronic Industries Alliance. The TIA is an organization that develops standards for telecommunications technologies. The EIA is now defunct.

In 2001, an updated standard was released, which became known as *TIA/EIA-568B*. Interestingly, the pinout of these two standards is the same. However, the color coding of the wiring is different. 568B is the more commonly used standard in the United States.

Shielded Twisted Pair

If wires in a cable are not twisted or shielded, the cable can act as an antenna, which might receive or transmit EMI. To help prevent this type of behavior, the wires (which are individually insulated) can be twisted together in pairs.

If the distance between the twists is less than a quarter of the wavelength of an electromagnetic waveform, the twisted pair of wires will not radiate that wavelength or receive EMI from that wavelength (in theory, if the wires were perfect conductors). However, as frequencies increase, wavelengths decrease.

One option for supporting higher frequencies is to surround a twisted pair in a metallic shielding, similar to the outer conductor in a coaxial cable. This type of cable is referred to as a *shielded twisted-pair (STP) cable*.

Figure 5-5 shows an example of STP cable. The outer conductors shield the copper strands from EMI; however, the drawback of STP is that the addition of the metallic shielding adds to the expense of the cable.

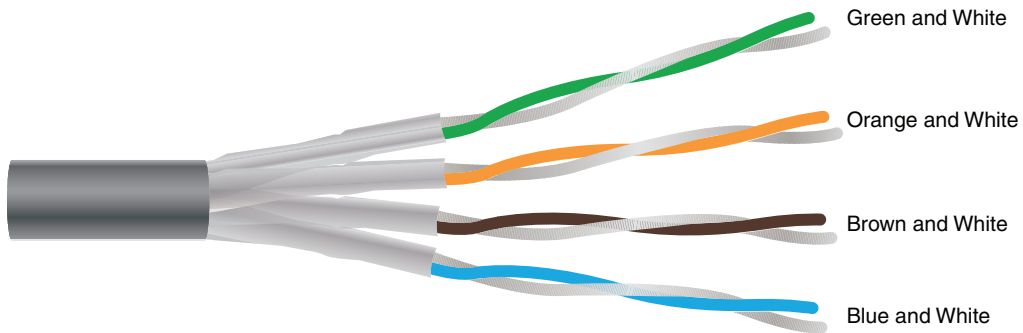


Figure 5-5 Shielded Twisted Pair

Unshielded Twisted Pair

Another way to block EMI from the copper strands making up a twisted-pair cable is to twist the strands more tightly (that is, more twists per centimeter). With the strands wrapped tightly around each other, the wires insulate each other from EMI.

Figure 5-6 illustrates an example of UTP cable. Because UTP is less expensive than STP, it has grown in popularity since the mid-1990s to become the media of choice for most LANs.

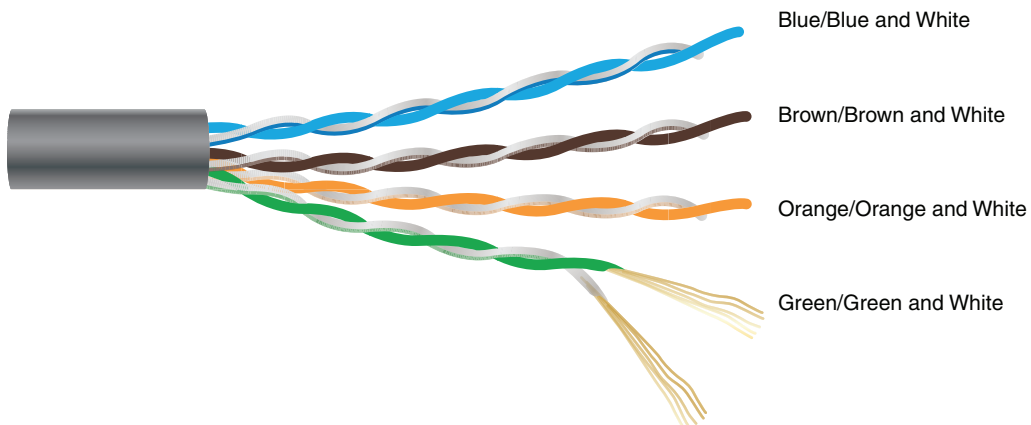


Figure 5-6 Unshielded Twisted Pair

UTP cable types vary in their data-carrying capacity, more commonly known as *cable speeds*. Common categories of UTP cabling include the following:

**Key
Topic**

- **Category 5:** Category 5 (Cat 5) cable is commonly used in *Ethernet* 100BASE-TX networks, which carry data at a rate of 100Mbps. However, Cat 5 cable can carry Asynchronous Transfer Mode (ATM) traffic at a rate of 155Mbps. ATM traffic is a data transmission method that uses fixed-size cells to carry diverse types of network traffic, including voice, video, and data, over a single network infrastructure. Most Cat 5 cables consist of four pairs of 24-gauge wires. Each pair is twisted, with a different number of twists per meter. However, on average, one pair of wires has a twist every 5 cm.
- **Category 5e:** Category 5e (Cat 5e) cable is an updated version of Cat 5 that is commonly used for 1000BASE-T networks, which carry data at a rate of 1Gbps. Cat 5e cable offers reduced crosstalk compared to Cat 5 cable. Crosstalk is electronic interference caused when two wires are too close to each other, and the adjacent cable creates interference.
- **Category 6:** Like Cat 5e cable, Category 6 (Cat 6) cable is commonly used for 1000BASE-T Ethernet networks. Some Cat 6 cable is made of thicker conductors (for example, 22-gauge or 23-gauge wire), although some Cat 6 cable is made from the same 24-gauge wire used by Cat 5 and Cat 5e. Cat 6 cable has thicker insulation and offers reduced crosstalk compared with Cat 5e.
- **Category 6a:** Category 6a (Cat 6a), or augmented Cat 6, supports twice as many frequencies as Cat 6 and can be used for 10GBASE-T networks, which can transmit data at a rate of 10 billion bits per second (10Gbps).
- **Category 7:** Cat 7 is not an IEEE standard, and it is not very popular as a result. This very strict specification supports 10Gbps over 100 m using copper media.
- **Category 8:** Cat 8 is capable of 40Gbps speeds. As you might guess, however, this speed comes at a cost. Cat 8 supports distances of only 30 to 36 m, depending on the patch cables used. These short distances and very high speeds are ideal for connections in a data center between high-speed multi-layer switches.

Although other wiring categories exist, the ones presented in this list are the categories most commonly seen in modern networks.

Most UTP cabling used in today's networks is considered to be *straight-through*, meaning that the RJ45 jacks at each end of a cable have matching pinouts. For example, pin 1 in an RJ45 jack at one end of a cable uses the same copper conductor as pin 1 in the RJ45 jack at the other end of a cable.

NOTE RJ45 connectors are covered in detail later in this chapter.

However, some network devices cannot be interconnected with a straight-through cable. For example, consider two PCs interconnected with a straight-through cable. Because the network interface cards (NICs) in these PCs use the same pair of wires for transmission and reception, when one PC sends data to the other PC, the receiving PC would receive the data on its transmission wires rather than its reception wires. For such a scenario, you can use a crossover cable, which swaps the transmit and receive wire pairs between the two ends of a cable.

NOTE A crossover cable for Ethernet devices is different from a crossover cable used for a digital T1 circuit. Specifically, an Ethernet crossover cable has a pin mapping of 1 [rfa] 3, 2 [rfa] 6, 3 [rfa] 1, and 6 [rfa] 2, whereas a T1 crossover cable has a pin mapping of 1 [rfa] 4, 2 [rfa] 5, 4 [rfa] 1, and 5 [rfa] 2. Another type of cable is the rollover cable, which is used to connect to a console port to manage a device such as a router or switch. The pin mapping for a rollover cable is 1 [bda] 8, 2 [bda] 7, 3 [bda] 6, 4 [bda] 5. The end of the cable looks like an RJ45 eight-pin connector.

NOTE A traditional port found in a PC's NIC is called a *media-dependent interface* (MDI). If a straight-through cable connects a PC's MDI port to an Ethernet switch port, the Ethernet switch port needs to swap the transmit pair of wires (that is, the wires connected to pins 1 and 2) with the receive pair of wires (that is, the wires connected to pins 3 and 6).

Therefore, a traditional port found on an Ethernet switch is called a *media-dependent interface crossover* (MDIX), and it reverses the transmit and receive pairs. However, if you want to interconnect two switches, where both switch ports used for the interconnection are MDIX ports, the cable needs to be a crossover cable.

Fortunately, most modern Ethernet switches have ports that can automatically detect whether they need to act as MDI ports or MDIX ports and make the appropriate adjustments. This eliminates the necessity of using straight-through cables for some Ethernet switch connections and crossover cables for other connections. With this *Auto-MDIX* feature, you can use either straight-through cables or crossover cables.

Twisted-Pair Cable Connectors

Common connectors used on twisted-pair cables are as follows:

- **RJ45:** A type 45 registered jack (RJ45) is an eight-pin connector found in most Ethernet networks. However, most Ethernet implementations only use four of the eight pins.



- **RJ11:** A type 11 registered jack (RJ11) has the capacity to be a six-pin connector. However, most RJ11 connectors have only two or four conductors. An RJ11 connector is found in most home telephone networks. However, most home phones use only two of the six pins.
- **DB-9 (RS-232):** A nine-pin D-subminiature (DB-9) connector is an older connector used for low-speed asynchronous serial communications, such as a PC to a serial printer, a PC to a console port of a router or switch, or a PC to an external modem. Do not confuse the DB-9 with a DB-25. The DB-25 connector was also used for the serial or parallel ports of early personal computers.

Figure 5-7 shows what these connectors look like.

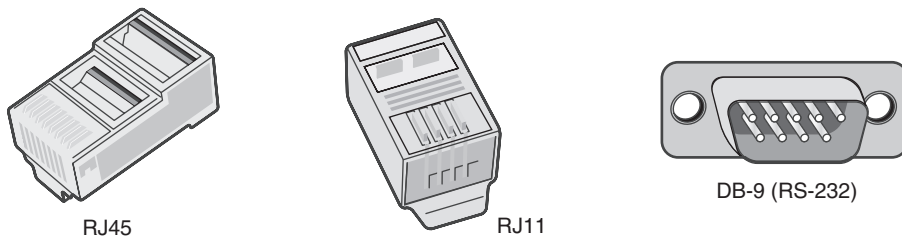


Figure 5-7 Twisted-Pair Cable Connectors

Plenum Versus Non-plenum Cable

If a twisted-pair cable is to be installed under raised flooring or in an open-air return, fire codes must be considered. For example, imagine that there was a fire in a building. If the outer insulation of a twisted-pair cable caught on fire or started to melt, it could release toxic fumes. If those toxic fumes were released in a location such as an open-air return, those fumes could be spread throughout a building, posing a huge health risk.

To mitigate the concern of pumping poisonous gas throughout a building's HVAC system, **plenum cable** can be used. The outer insulator of a plenum twisted-pair cable is not only fire retardant; in addition, some plenum cabling uses a fluorinated ethylene polymer (FEP) or a low-smoke polyvinyl chloride (PVC) to minimize dangerous fumes.

NOTE Check with your local fire codes before installing network cabling.

Fiber-Optic Cable

An alternative to copper cabling is fiber-optic cabling, which sends light (instead of electricity) through an optical fiber (typically made of glass). Using light instead of electricity makes fiber optics immune to EMI. Also, depending on the Layer 1

(physical layer) technology being used, fiber-optic cables typically have greater range (that is, a greater maximum distance between networked devices) and greater data-carrying capacity.

Lasers are often used to inject light pulses into a fiber-optic cable. However, lower-cost light-emitting diodes (LEDs) are also available. Fiber-optic cables are generally classified according to their diameter and fall into one of two categories: *multimode fiber (MMF)* and *single-mode fiber (SMF)*.

The wavelengths of light also vary between MMF and SMF cables. Usually, wavelengths of light in an MMF cable are in the range 850–1300 nm, where nm stands for nanometers. (A nanometer is one-billionth of a meter.) Conversely, the wavelengths of light in an SMF cable are usually in the range 1310–1550 nm. A fiber coupler could be used to connect two fiber cables, back-to-back.

Multimode Fiber

When a light source, such as a laser, sends light pulses into a fiber-optic cable, what keeps the light from simply passing through the glass and being dispersed into the surrounding air? The trick is that fiber-optic cables use two different types of glass. There is an inner strand of glass (that is, a *core*) surrounded by an outer *cladding* of glass, similar to the construction of the previously mentioned coaxial cable.

The light injected by a laser (or LED) enters the core, and the light is prevented from leaving that inner strand and going into the outer cladding of glass. Specifically, the indexes of refraction of these two different types of glass are so different that if the light attempts to leave the inner strand, it hits the outer cladding and bends back on itself.

To better understand this concept, consider a straw in a glass of water, as shown in Figure 5-8. Because air and water have different indexes of refraction (that is, light travels at slightly different speeds in air and water), the light that bounces off the straw and travels to our eyes is bent by the water's index of refraction. When a fiber-optic cable is manufactured, dopants are injected into the two types of glasses, making up the core and cladding to give them significantly different indexes of refraction, thus causing any light attempting to escape to be bent back into the core.

The path that light takes through a fiber-optic cable is called a *mode of propagation*. The diameter of the core in a multimode fiber is large enough to permit light to enter the core at different angles, as depicted in Figure 5-9. If light enters at a steep angle, it bounces back and forth much more frequently on its way to the far end of the cable than does light that enters the cable perpendicularly. If pulses of light representing different bits travel down the cable using different modes of propagation, it is possible that the bits (that is, the pulses of light representing the bits) will arrive

out of order at the far end (where the pulses of light, or absence of light, are interpreted as binary data by photoelectronic sensors).



Figure 5-8 Example: Refractive Index

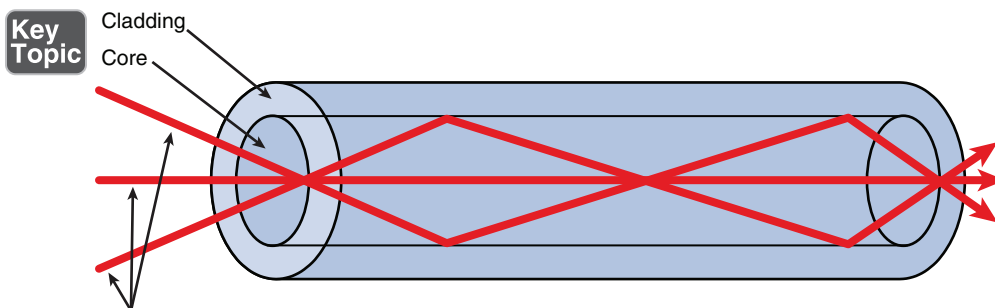


Figure 5-9 Light Propagation in Multimode Fiber

For example, say that the pulse of light representing the first bit intersects the core at a steep angle and bounces back and forth many times on its way to the far end of the cable, while the light pulse representing the second bit intersects the core perpendicularly and does not bounce back and forth very much. With all of its bouncing, the first bit has to travel further than the second bit, and so the bits might arrive out of order. This condition is known as *multimode delay distortion*. To mitigate multimode delay distortion, multimode fiber (MMF) is typically limited to shorter distances than SMF.

Single-Mode Fiber

Single-mode fiber (SMF) eliminates the issue of multimode delay distortion by having a core with a diameter so small that it permits only one mode (that is, one path) of propagation, as shown in Figure 5-10. With the issue of multimode delay distortion mitigated, SMF typically can be run for longer distances than MMF.

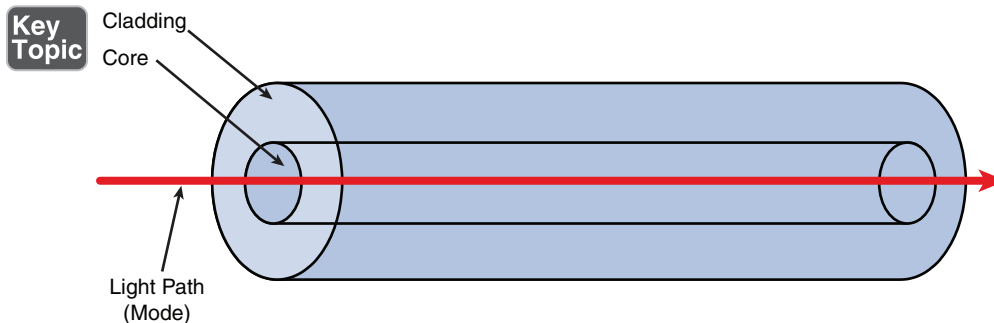


Figure 5-10 Light Propagation in Single-Mode Fiber

A potential downside to SMF, however, is cost. Because SMF has to be manufactured to very exacting tolerances, you usually pay more for a given length of single-mode fiber-optic cabling. However, for some implementations where greater distances are required, the cost is an acceptable trade-off for reaching greater distances.

Fiber-Optic Cable Connectors

Some common connectors used on fiber-optic cables are as follows:

- **ST:** A *straight tip (ST) connector* is sometimes referred to as a *bayonet connector*, because of the long tip extending from the connector. ST connectors are most commonly used with MMF. You connect an ST connector to a terminating device by pushing the connector into the terminating equipment and then twisting the connector housing to lock it in place.

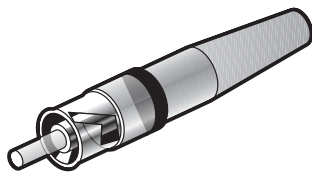
- **SC:** Different literature defines an SC connector as *subscriber connector (SC)*, *standard connector*, or *square connector*. You connect an SC connector by pushing it into the terminating device; you can remove it by pulling the connector from the terminating device. This connector type has slight variants within the industry, with the major types being APC, UPC, and MTRJ. Always consult with the vendor or an IT staff member regarding the exact requirements.
- **LC:** You connect a *Lucent connector*, *little connector*, or *local connector (LC)* to a terminating device by pushing the connector into the terminating device. You can remove it by pressing the tab on the connector and pulling it out of the terminating device.
- **MTRJ:** The most unique characteristic of a *media termination recommended jack (MTRJ)* or mechanical transfer (MT) *registered jack (RJ)* connector is that two fiber strands (a transmit strand and a receive strand) are included in a single connector. You connect an MTRJ connector by pushing it into the terminating device; you can remove it by pulling the connector from the terminating device.

Figure 5-11 shows what these connectors look like.

NOTE Yet another connector type you might encounter in a data center is the *multi-fiber push on (MPO) connector* type. It is a high-density fiber-optic connector sometimes used for its ability to streamline cabling and support high-bandwidth applications. MPO connectors can house multiple fibers—typically 12, 24, or more—in a single interface, significantly increasing the amount of data transmitted through a single connection. This design reduces physical space requirements and simplifies network scalability and maintenance.

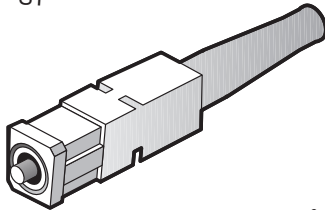
Fiber Connector Polishing Styles

Fiber-optic cables have different types of mechanical connections. The type of connection impacts the quality of the fiber-optic transmission. Listed from basic to better, the options include physical contact (PC), ultra physical contact (UPC), and angled physical contact (APC), which refer to the polishing styles of fiber-optic connectors. The different polish of the fiber-optic connectors results in different performance of the connector. The less back reflection, the better the transmission. The PC back reflection is -40 dB, the UPC back reflection is around -55 dB, and the APC back reflection is about -70 dB.

**Key
Topic**

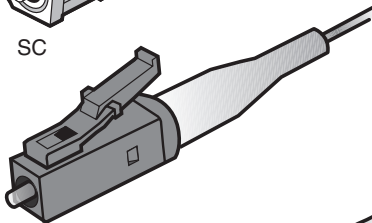
ST

The ST connector uses a half-twist bayonet type of lock.



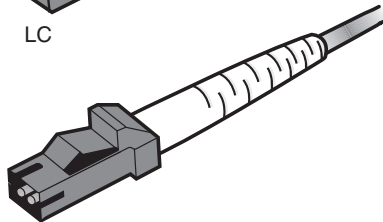
SC

The SC uses a push-pull connector similar to common audio and video plugs and sockets.



LC

LC connectors have a flange on top, similar to an RJ45 connector, that aids secure connection.



MTRJ

MTRJ is a popular connector for two fibers in a very small form factor.

Figure 5-11 Common Fiber-Optic Connectors

Ethernet and Fiber Standards (802.3)

It is time to tackle some of the most important **802.3 standards** that define our Ethernet networks of the past and those that we rely on today. A popular implementation of Ethernet, in the early days, was called 10BASE5. The 10 in 10BASE5 referred to network throughput, specifically 10Mbps (that is, 10 million [mega] bits per second). The BASE in 10BASE5 referred to baseband, as opposed to broadband. Finally, the 5 in 10BASE5 indicated the distance limitation of 500 m. The cable used in 10BASE5 networks, illustrated in Figure 5-12, was a larger diameter than most types of media. In fact, this network type became known as thicknet.

Another early Ethernet implementation was 10BASE2. From the previous analysis of 10BASE5, you might conclude that 10BASE2 was a 10Mbps baseband technology with a distance limitation of 200 meters. That is almost correct. However, 10BASE2's actual distance limitation was 185 m. The cabling used in 10BASE2 networks was significantly

thinner and therefore less expensive than 10BASE5 cabling. As a result, 10BASE2 cabling, illustrated in Figure 5-13, was known as thinnet or cheapernet.

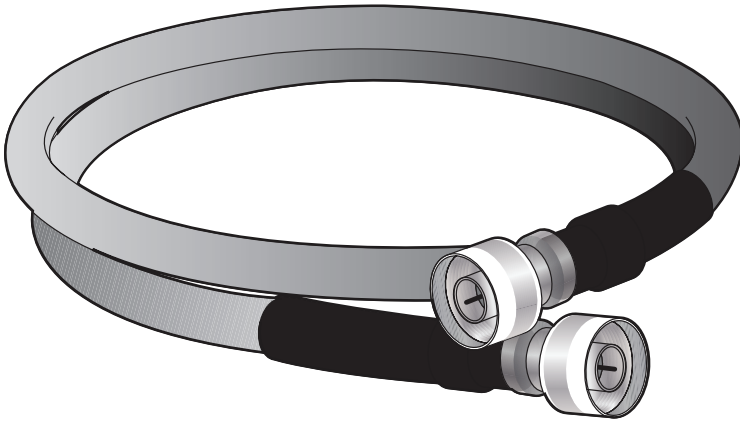


Figure 5-12 10BASE5 Cable

10BASE5 and 10BASE2 networks are rarely, if ever, seen today. The cabling used by these legacy technologies quickly faded in popularity with the advent of UTP cabling. The 10Mbps version of Ethernet that relied on UTP cabling, an example of which is provided in Figure 5-14, is known as 10BASE-T. Notice that the “T” in 10BASE-T refers to twisted-pair cabling.



Figure 5-13 Coaxial Cable Used for 10BASE2

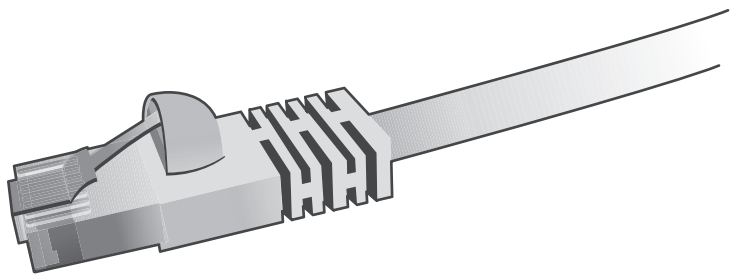


Figure 5-14 UTP Cable Used for 10BASE-T

Distance and Speed Limitations

To understand the bandwidth available on networks, you need to understand a few terms. You should already know that a *bit* refers to one of two values. These values are represented using binary math, which uses only the numbers 0 and 1. On a cable such as twisted-pair cable, a bit could be represented by the absence or presence of voltage. Fiber-optic cables, however, might represent a bit with the absence or presence of light.

The bandwidth of a network is measured in terms of how many bits the network can transmit during a 1-second period of time. For example, if a network has the capacity to send 10,000,000 (that is, 10 million) bits in a 1-second period of time, the bandwidth capacity is said to be 10 megabits (that is, millions of bits) per second (or *Mbps*). Table 5-4 defines common bandwidths supported on distinct types of Ethernet networks.

**Key
Topic**

Table 5-4 Ethernet Bandwidth Capacities

Ethernet Type	Bandwidth Capacity
Standard Ethernet	10Mbps: 10 million bits per second (that is, 10 megabits per second)
Fast Ethernet	100Mbps: 100 million bits per second (that is, 100 megabits per second)
Gigabit Ethernet	1Gbps: 1 billion bits per second (that is, 1 gigabit per second)
10-Gigabit Ethernet	10Gbps: 10 billion bits per second (that is, 10 gigabits per second)
40-Gigabit Ethernet	40Gbps: 40 billion bits per second (that is, 40 gigabits per second)
100-Gigabit Ethernet	100Gbps: 100 billion bits per second (that is, 100 gigabits per second)

The type of cabling used in an Ethernet network influences the bandwidth capacity and the distance limitation of the network. For example, fiber-optic cabling often has a higher bandwidth capacity and can be run longer distances than twisted-pair cabling.

As mentioned earlier in this chapter, because of the issue of multimode delay distortion, SMF usually has a longer distance limitation than MMF.

Although not comprehensive, Table 5-5 lists a number of Ethernet standards, along with the media type, bandwidth capacity, and distance limitation for each.

**Key
Topic**

Table 5-5 Types of Ethernet

Ethernet Standard	Media Type	Bandwidth Capacity	Distance Limitation
10BASE5	Coax (thicknet)	10Mbps	500 m
10BASE2	Coax (thinnet)	10Mbps	185 m
10BASE-T	Cat 3 (or higher) UTP	10Mbps	100 m
100BASE-TX	Cat 5 (or higher) UTP	100Mbps	100 m
100BASE-FX	MMF	100Mbps	2 km
100BASE-SX	MMF	100Mbps	300 m
1000BASE-T	Cat 5e (or higher) UTP	1Gbps	100 m
1000BASE-TX	Cat 6 (or higher) UTP	1Gbps	100 m
1000BASE-SX	MMF	1Gbps	550 m
1000BASE-LX	SMF	1Gbps	5 km
1000BASE-LH	SMF	1Gbps	10 km
1000BASE-ZX	SMF	1Gbps	70 km
10GBASE-SR	MMF	10Gbps	26–400 m
10GBASE-LR	SMF	10Gbps	10–25 km
10GBASE-ER	SMF	10Gbps	40 km
10GBASE-SW	MMF	10Gbps	300 m
10GBASE-LW	SMF	10Gbps	10 km
10GBASE-EW	SMF	10Gbps	40 km
10GBASE-T	Cat 6a (or higher)	10Gbps	100 m
40GBASE-T	Cat 8	40Gbps	30 m

Ethernet Standard	Media Type	Bandwidth Capacity	Distance Limitation
100GBASE-SR10	MMF	100Gbps	125 m
100GBASE-LR4	SMF	100Gbps	10 km
100GBASE-ER4	SMF	100Gbps	40 km

NOTE Two often-confused terms are 100BASE-T and 100BASE-TX. 100BASE-T is not a specific standard. Rather, 100BASE-T is a category of standards and includes 100BASE-T2 (which uses two pairs of wires in a Cat 3 cable), 100BASE-T4 (which uses four pairs of wires in a Cat 3 cable), and 100BASE-TX. 100BASE-T2 and 100BASE-T4 were early implementations of 100Mbps Ethernet and are no longer used. Therefore, you can generally use the terms 100BASE-T and 100BASE-TX interchangeably.

Similarly, the term 1000BASE-X is not a specific standard. Rather, 1000BASE-X refers to all Ethernet technologies that transmit data at a rate of 1Gbps over fiber-optic cabling. Additional and creative ways of using Ethernet technology include IEEE 1901-2013, which could be used for Ethernet over HDMI cables and Ethernet over existing power lines to avoid having to run a separate cabling just for networking.

Transceivers

We like to think of a *transceiver* as a connector for our cable to our networking device. But where does this fancy term come from? The term *transceiver* means that the connector combines both a transmitter and a receiver in the single, efficient unit.

When you want to uplink one Ethernet switch to another, you might need different connectors (for example, for MMF, SMF, or UTP) for different installations. Fortunately, some Ethernet switches have one or more empty slots in which you can insert a gigabit interface converter (GBIC). GBICs are interfaces that have a bandwidth capacity of 1Gbps and are available with MMF, SMF, and UTP connectors. This allows you to have flexibility in the uplink technology you use in an Ethernet switch.

A smaller variant of a regular GBIC is the *small form-factor pluggable (SFP)*, which is sometimes called a *mini-GBIC*. And to show the variety of transceivers you might encounter today, even this SFP has many variations, including the following:

- Enhanced form-factor pluggable (SFP+)
- *Quad small form-factor pluggable (QSFP)*
- Enhanced quad small form-factor pluggable (QSFP+)

NOTE Ethernet networks are not the only networks that rely on transceivers. We also use them in *Fibre Channel (FC)* networks. You might recall from Chapter 2, “Networking Appliances, Applications, and Functions,” that FC is often found in storage area networks (SANs). These SANs dedicate themselves to the fast storage and fast retrieval of the data companies rely on.

Multiplexing in Fiber-Optic Networks

Remember, as mentioned in Chapter 1, “The OSI Model and Encapsulation,” that multiplexing allows multiple communications sessions to share the same physical medium. At this point, you need to be familiar with three different approaches common with fiber networking:

- **Dense wavelength-division multiplexing (DWDM):** DWDM uses as many as 32 light wavelengths on a single fiber, where each wavelength can support as many as 160 simultaneous transmissions using more than eight active wavelengths per fiber.
- **Coarse wavelength-division multiplexing (CWDM):** CWDM uses fewer than eight active wavelengths per fiber.
- **Bidirectional wavelength-division multiplexing (WDM):** This approach multiplexes a number of optical carrier signals onto a single optical fiber by using different wavelengths. Using this technique enables bidirectional communications over one strand of fiber and increases the overall capacity.

Media Converters

Due to the wide variety of copper and fiber cabling used by different network devices, you might need one or more *media converters*. Examples of media converters include the following:

- MMF to Ethernet
- SMF to Ethernet
- Fiber to coaxial
- SMF to MMF

Real-World Case Study

The network engineers at Acme, Inc. are analyzing their physical media implementations and the performance they are achieving in their Ethernet-based LAN. The decision has been made to retain portions of the Cat 5e-based media, which is capable of 1Gbps speeds. Other areas of the network, particularly those that must deal with aggregate bandwidth demands, are being upgraded to Cat 8. Remember, Category 8 is capable of 40Gbps speeds. Cat 8 supports distances of only 30 to 36 m, depending on the patch cables used.

For Acme, these short distances and very high speeds are ideal for connections between high-speed multilayer switches that make up a good part of the distribution and core layers of the LAN.

The engineers are also examining the wireless technologies in use, focusing on evaluating the efficiency of the current Wi-Fi standards, identifying potential interference sources, and considering the implementation of advanced protocols such as Wi-Fi 6 to enhance network performance, coverage, and security in the environment.

Summary

Here are the main topics covered in this chapter:

- This chapter began by exploring key concepts about wireless networking.
- There are many different options today when it comes to copper and fiber media and connectors. This chapter explored transmission media, connector types, and transceivers in great detail.
- This chapter also covered the different options for multiplexing in fiber-optic environments.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 5-6 lists these key topics and the page number where each is found.

**Key
Topic****Table 5-6** Key Topics for Chapter 5

Key Topic Element	Description	Page Number
Figure 5-1	Non-overlapping Channels in the 2.4GHz Band	103
List	Variations of spread-spectrum technology	105
Table 5-3	Characteristics of 802.11 Standards	108
Figure 5-2	Satellite WAN Sample Topology	109
List	Common connectors used with coaxial cables	112
List	Common categories of UTP cabling	115
List	Common connectors used with twisted-pair cables	116
Figure 5-9	Light Propagation in Multimode Fiber	119
Figure 5-10	Light Propagation in Single-Mode Fiber	120
Figure 5-11	Common Fiber-Optic Connectors	122
Table 5-4	Ethernet Bandwidth Capacities	124
Table 5-5	Types of Ethernet	125

Complete Tables and Lists from Memory

Print a copy of Appendix B, “Memory Tables,” or at least the section for this chapter and complete as many of the tables as possible from memory. Appendix C, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

802.3 standards, 802.11 standards, Bayonet Neill-Concelman (BNC), cable speeds, cellular, coaxial cable, direct attach copper (DAC) cable, Ethernet, F-type connector, Fibre Channel (FC), local connector (LC), media converter, multi-fiber push on (MPO) connector, multimode fiber (MMF), plenum cable, quad small form-factor pluggable (QSFP), registered jack (RJ), RJ11, RJ45, satellite, single-mode fiber (SMF), small form-factor pluggable (SFP), straight tip (ST) connector, subscriber connector (SC), transceiver, twinaxial cable, twisted pair

Additional Resources

Wi-Fi Standards (and how they work): <https://youtu.be/q64AZjPfa0Y>

Fundamentals of Fiber Optic Cabling: <https://youtu.be/-VYhfR8Fv2I>

What is 5G? <https://www.cisco.com/c/en/us/solutions/what-is-5g.html>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. Which of the following categories of UTP cabling are commonly used for 1000BASE-T networks? (Choose two.)
 - a. Cat 5
 - b. Cat 5e
 - c. Cat 6
 - d. Cat 6f
2. Which type of cable might be required for installation in a drop ceiling that is used as an open-air return duct?
 - a. Riser
 - b. Plenum
 - c. Multimode
 - d. Twinaxial
3. Which of the following is the eight-pin connector found in most Ethernet networks?
 - a. RJ11
 - b. RJ45
 - c. DB-9
 - d. ST
4. What are the two major categories of fiber-optic media? (Choose two.)
 - a. Straight-through
 - b. Single-mode
 - c. Unshielded
 - d. Multimode

5. What is the speed of Fast Ethernet?
 - a. 1Mbps
 - b. 100Mbps
 - c. 1Gbps
 - d. 10Gbps
6. In what networking use case do we often find Fibre Channel (FC)?
 - a. A wide area network
 - b. A local area network
 - c. Satellite topologies
 - d. A storage area network
7. What Ethernet UTP cable was the first standard cable to carry data at speeds of 1Gbps?
 - a. Cat 4
 - b. Cat 5
 - c. Cat 5e
 - d. Cat 6
8. What 802.11 standard is also known as Wi-Fi 6?
 - a. 802.11g
 - b. 802.11n
 - c. 802.11ax
 - d. 802.11ac
9. Which standards were developed to define industry-standard pinouts and color coding for twisted-pair cabling? (Choose two.)
 - a. CWDM
 - b. TIA/EIA-568A
 - c. DWDM
 - d. TIA/EIA-568B

10. What fiber type would you use if the requirements to be met included an SMF media type, 1Gbps bandwidth capacity, and a distance limitation of 5 km?
 - a. 1000BASE-LX
 - b. 1000BASE-SX
 - c. 1000BASE-TX
 - d. Thicknet
11. In the 2.4GHz frequency band, which are considered non-overlapping channels?
 - a. 36, 40, 44
 - b. 48, 52
 - c. 56, 60
 - d. 1, 6, 11
12. Which of the following is a set of advanced technologies included with IEEE 802.11ac and 802.11ax that dramatically enhances wireless throughput?
 - a. MIMO
 - b. MU-MIMO
 - c. CDMA
 - d. GSM
13. Which type of fiber-optic cable is used for longer distances?
 - a. SMF
 - b. MMF
 - c. MPO
 - d. ST
14. Which category of Ethernet cable is capable of 40Gbps speeds and ideal for connections in a data center between high-speed multilayer switches?
 - a. Cat 5e
 - b. Cat 6a
 - c. Cat 7
 - d. Cat 8

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This chapter covers the following topics related to Objective 1.6 (Compare and contrast network topologies, architectures, and types) of the CompTIA Network+ N10-009 certification exam:

- Mesh
- Hybrid
- Star/hub and spoke
- Spine and leaf
- Point to point
- Three-tier hierarchical model
 - Core
 - Distribution
 - Access
- Collapsed core
- Traffic flows
 - North-south
 - East-west

Network Topologies, Architectures, and Types

What comes to mind when you think of a computer network? Is it the Internet? Is it email? Is it the wireless connection that lets you print to your printer from your laptop? Is it the smart thermostat and lights in your home?

Whatever your current perception of a computer network, this chapter just might help you expand your thought process in this regard. Be aware that although you think of computer networks as interconnecting computers, today's computer networks interconnect a variety of devices in addition to just computers. Examples include game consoles, video-surveillance devices, IP-based telephones, tablets, and smartphones. Therefore, throughout this book, think of the term *computer network* as being synonymous with the more generic term *network*, because these terms are used interchangeably.

The goal of this chapter is to acquaint you with the purpose of a network and help you categorize a given network based on criteria such as geography, topology, and the location of the network's resources.

Foundation Topics

Defining a Network

The movie *Field of Dreams* featured the statement “If you build it, he will come.” This statement most certainly applies to the evolution of network-based services in modern-day networks. Computer networks are no longer relegated to allowing a group of computers to access a common set of files stored on a computer chosen as a *file server*. Instead, with the building of high-speed, highly redundant networks, network architects are seeing the wisdom of placing a variety of traffic types on a single network. Examples include voice and video, in addition to data. As you will learn in this chapter, the Internet of Things (IoT) means that just about everything wants to join your network, from the lights in your home to many of your household appliances.

One could argue that a network is the sum of its parts. So, as you begin your study of networking, you should start to gain a basic understanding of fundamental networking components, including such entities as the client, server, hub, switch, and router, as well as the media used to interconnect these devices.

The Purpose of Networks

The basic purpose of a network is to make connections. These connections might be between a PC and a printer or between a laptop and the Internet, as just a couple of examples. However, the true value of a network comes from the traffic flowing over those connections. Consider a sampling of applications that can travel over a network's connections:

- File sharing between two computers
- Video chatting between computers located in different parts of the world
- Surfing the Web (for example, to use social media sites, watch streaming video, listen to an Internet radio station, or do research for a school term paper)
- Instant messaging (IM) between computers with IM software installed
- Email
- Voice over IP (VoIP), to replace traditional telephony systems

A term given to a network transporting multiple types of traffic (for example, voice, video, and data) is a *converged network*. A converged network might offer significant cost savings to organizations that previously supported separate network infrastructures for voice, data, and video traffic. This convergence also potentially reduces staffing costs because only a single network needs to be supported, rather than separate networks for separate traffic types.

Network Types and Characteristics

As you might be sensing at this point, not all networks look the same. They vary in many ways. One criterion by which networks are classified is how geographically dispersed the network's components are. For example, a network might interconnect devices within an office, or a network might interconnect a database at a corporate headquarters location with a remote sales office on the opposite side of the globe.

Based on the geographic dispersion of network components, you can classify networks into various categories, including the following:

- Local area network (LAN)
- Wide area network (WAN)
- Wireless local area network (WLAN)
- Storage area network (SAN)
- Campus area network (CAN)
- Metropolitan area network (MAN)
- Personal area network (PAN)

LAN

A LAN interconnects network components within a local area (for example, within a building). Examples of common LAN technologies you are likely to meet include Ethernet (that is, IEEE 802.3) and wireless networks (that is, IEEE 802.11). Figure 6-1 illustrates an example of a LAN.

NOTE IEEE stands for *Institute of Electrical and Electronics Engineers*, which is an internationally recognized standards body.

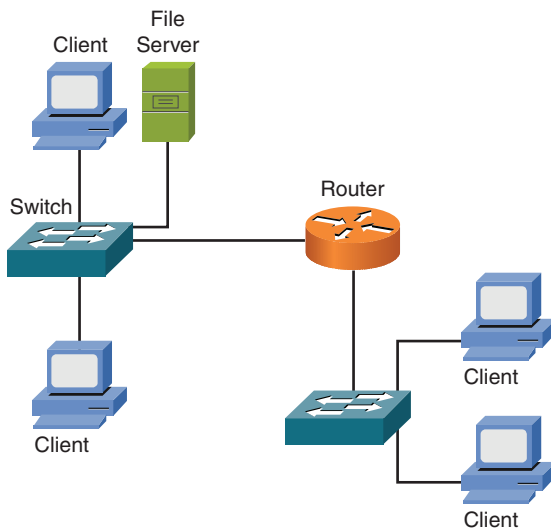


Figure 6-1 Sample LAN Topology

WAN

A WAN interconnects network components that are geographically separated. For example, a corporate headquarters might have multiple WAN connections to remote office sites. Multiprotocol Label Switching (MPLS) and Asynchronous Transfer Mode (ATM) are examples of WAN technologies. Figure 6-2 depicts a simple WAN topology, which interconnects two geographically distant locations.

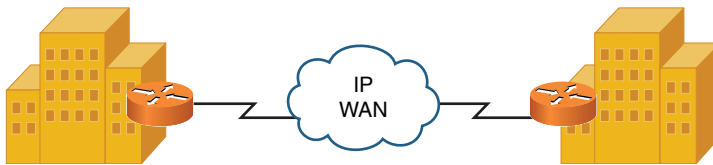


Figure 6-2 Sample WAN Topology

WLAN

A local area network made up of wireless networking devices is a wireless local area network (WLAN).

SAN

You can construct a high-speed, highly reliable network for the express purpose of transmitting stored data. This network is called a storage area network (SAN).

Other Categories of Networks

Although LAN and WAN are the most common terms used to categorize computer networks based on geography, other categories include campus area network (CAN), metropolitan area network (MAN), and personal area network (PAN).

CAN

The first time I discovered a CAN-type topology was at a major university. The university covered several square miles and had several dozen buildings. Many of these buildings were running individual LANs, and these building-centric LANs were interconnected. The interconnection of these LANs created another network type: a campus area network (CAN). Besides being common on university campuses, CANs are often used in industrial parks and business parks.

MAN

More widespread than a CAN and less widespread than a WAN, a metropolitan area network (MAN) is a network that spans a defined geographic location, such as a city or suburb. It interconnects locations scattered throughout a metropolitan area. Imagine, for example, that a business in Chicago has a location near O'Hare Airport, another location near the Navy Pier, and another location in the Willis Tower (previously known as the Sears Tower). If a service provider could interconnect those locations using a high-speed network, such as a 10Gbps (that is, 10 billion bits per second) network, the interconnection of those locations would form a MAN. One example of a MAN technology is Metro Ethernet, which features much higher speeds than the traditional WAN technologies that were used in the past to connect such locations.

PAN

A personal area network (PAN) is a network whose scale is even smaller than a LAN. For example, a connection between a PC and a digital camera via a universal serial bus (USB) cable could be considered a PAN. Another example is a PC connected to an external hard drive via a USB 3.0 or Thunderbolt connection. A PAN, however, is not necessarily a wired connection. A Bluetooth connection between your cell phone and your car's audio system is considered a wireless PAN (WPAN). The main distinction of a PAN is that its range is typically limited to just a few meters.

Networks Defined Based on Resource Location

Another way to categorize networks is based on where the network resources reside. For example, a *client/server network* is a collection of PCs all sharing files stored on a centralized server. However, if those PCs had their operating system (for example, Microsoft Windows 11 or macOS) configured for file sharing, they could share files from one another's hard drives. This is referred to as a *peer-to-peer network* because the peers (the PCs in this example) make resources available to other peers. The following sections describe client/server and peer-to-peer networks in more detail.

Client/Server Networks

Figure 6-3 illustrates an example of a *client/server network*, where a dedicated file server gives shared access to files, and a networked printer is available as a resource to the network's clients. Client/server networks are commonly used by businesses. Because resources are found on one or more servers, administration is simpler than administration of network resources on multiple peer devices.

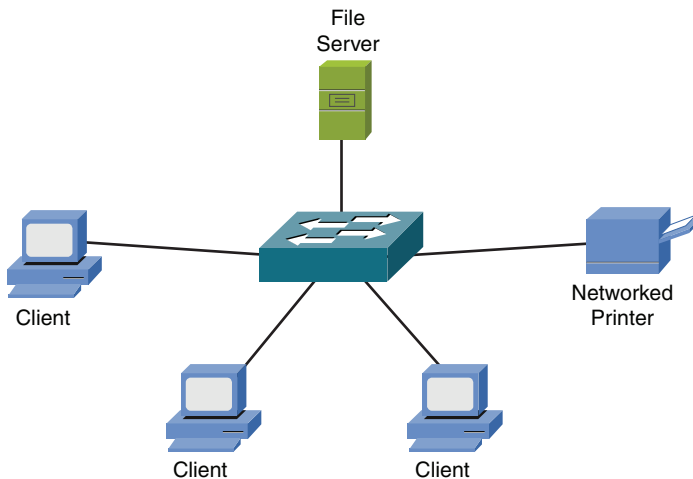


Figure 6-3 Client/Server Network Example

The performance of a client/server network can be better than that of a peer-to-peer network because resources can be located on dedicated servers rather than on a PC running a variety of end-user applications. You can simplify backups because fewer locations must be backed up. However, client/server networks come with the extra expense of dedicated server resources. Table 6-1 lists the benefits and drawbacks of client/server networks.

Table 6-1 Characteristics, Benefits, and Drawbacks of Client/Server Networks

Characteristics	Benefits	Drawbacks
Client devices (for example, PCs) share a common set of resources (for example, file or print resources) located on one or more dedicated servers.	Client/server networks can easily scale, which might require the purchase of additional client licenses.	Because multiple clients might rely on a single server for their resources, the single server can become a single point of failure in the network.
Resource sharing is made possible via dedicated server hardware and network operating systems.	Administration is simplified because parameters such as file-sharing permissions and other security settings can be administered on a server as opposed to on multiple clients.	Client/server networks can cost more than peer-to-peer networks. For example, client/server networks might require the purchase of dedicated server hardware and a network OS with an appropriate number of licenses.

NOTE A server in a client/server network could be a computer running a network operating system (NOS) such as Linux Server or one of the Microsoft Windows Server operating systems. Alternatively, a server might be a host making its file system available to remote clients via the Network File System (NFS) service, which was originally developed by Sun Microsystems.

NOTE A variant of the traditional server in a client/server network, where the server provides shared file access, is network-attached storage (NAS). A NAS device is a mass storage device that attaches directly to a network. It typically has an array of disks providing network storage capacity to the users on the network. It is a specialized file-level computer storage device connected to a network. Rather than running an advanced NOS, a NAS device usually makes files available to network clients via a service such as NFS. If the NAS sounds familiar, it is because you learned about it back in Chapter 2, “Networking Appliances, Applications, and Functions.”

Peer-to-Peer Networks

Peer-to-peer networks allow interconnected devices (for example, PCs) to share their resources with one another. Those resources could be, for example, files or printers. As an example of a peer-to-peer network, consider Figure 6-4, where each of the peers can share files on its own hard drive, and one of the peers has a directly attached printer that can be shared with the other peers in the network.

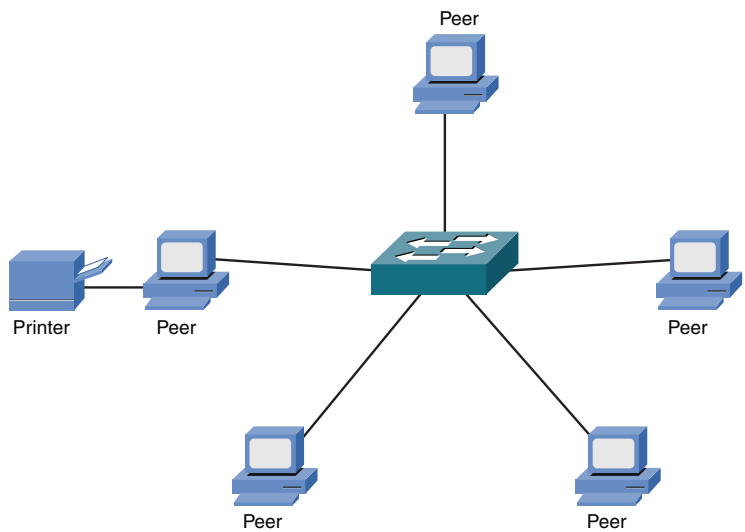


Figure 6-4 Peer-to-Peer Network Example

Peer-to-peer networks tend to be used in smaller businesses and in homes. The popularity of peer-to-peer networks is fueled in part by client operating systems that support file and print sharing. Scalability for peer-to-peer networks is a concern, however. Specifically, as the number of devices (that is, peers) increases, the administration burden increases. For example, a network administrator might have to manage file permissions on multiple devices, as opposed to on a single server. Table 6-2 lists the characteristics, benefits, and drawbacks of peer-to-peer networks.

Table 6-2 Characteristics, Benefits, and Drawbacks of a Peer-to-Peer Network

Characteristics	Benefits	Drawbacks
Client devices (for example, PCs) share their resources (for example, file and printer resources) with other client devices.	Peer-to-peer networks can be installed easily because resource sharing is made possible by the clients' operating systems, and knowledge of advanced networking operating systems is not required.	Scalability is limited because of the increased administration burden of managing multiple clients.
Resource sharing is made available through the clients' operating systems.	Peer-to-peer networks usually cost less than client/server networks because there is no requirement for dedicated server resources or advanced NOS software.	Performance might not be as strong as in a client/server network because the devices providing network resources might be performing other tasks not related to resource sharing (for example, word processing).

NOTE Some networks have characteristics of both peer-to-peer and client/server networks. For example, all PCs in a company might point to a centralized server for accessing a shared database in a client/server topology. However, these PCs might simultaneously share files and printers with one another in a peer-to-peer topology.

Cloud Networking

While we covered cloud networking in Chapter 3, “Cloud Concepts,” it is worth a reminder here in this chapter. Remember that it is quite common for businesses today to be engaged in cloud computing for some or all of their networking needs. Virtual private clouds (VPCs) are logical topologies made available to these businesses in the public cloud. The IT staff can use these VPCs to host their own cloud-based networking solutions.

Networks Defined by Topology

In addition to classifying networks based on the geographic placement of their components, another approach to classifying a network is to use the network’s topology. Looks can be deceiving, however. You need to be able to distinguish between a physical topology and a logical topology.

Physical Versus Logical Topology

Even if a network appears to be a star topology (that is, where the network components all connect to a centralized device, such as a switch), the traffic might be flowing in a circular pattern through all the network components attached to the centralized device. The actual traffic flow determines the *logical topology*, whereas the way components are physically interconnected determines the *physical topology*.

For example, consider Figure 6-5, which shows a collection of computers connected to a legacy Token Ring media access unit (MAU). From a quick inspection of Figure 6-5, you can conclude that the devices are physically connected in a star topology, where the connected devices radiate out from a centralized aggregation point (the MAU in this example).

Now contrast the physical topology in Figure 6-5 with the logical topology illustrated in Figure 6-6. Although you can see that the computers physically connect to a centralized MAU, when you examine the flow of traffic through (or in this case, around) the network, you see that the traffic flow actually loops around and around the network. The traffic flow dictates how to classify a network’s logical topology. In this instance, the logical topology is a *ring topology* because the traffic circulates around the network as if circulating around a ring.

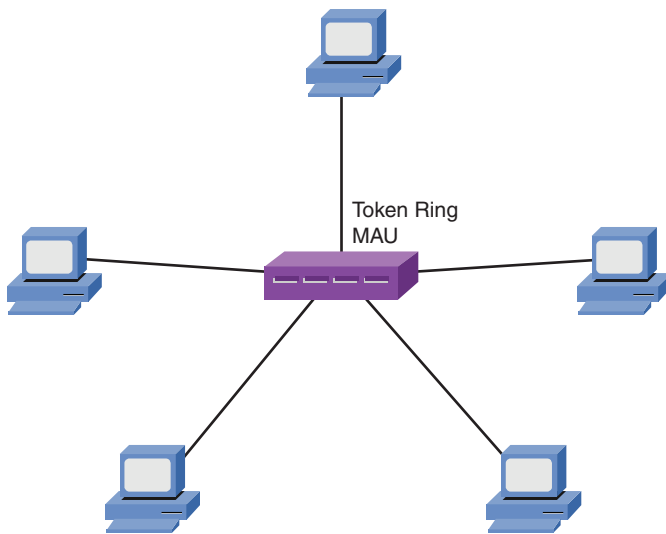


Figure 6-5 Physical Star Topology

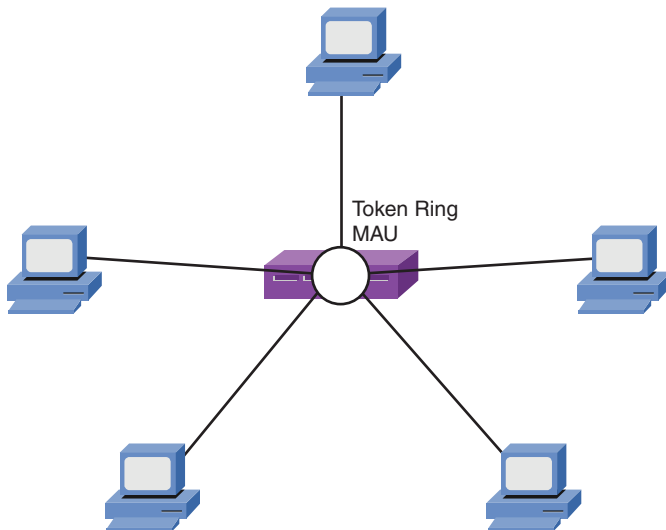


Figure 6-6 Logical Ring Topology

NOTE Before you run out and try to purchase a Token Ring network for your LAN, keep in mind that you'll only see this technology in networking museums now!

Point-to-Point Topology

Let's begin with the simplest of all network topologies, it's the point-to-point network. A *point-to-point network topology* involves a direct connection between two network nodes, facilitating exclusive communication between them. This simple and straightforward setup is commonly used for dedicated communications where security and speed are paramount, such as in leased lines or private network connections. In this topology, the communication link is dedicated to the two connected devices, minimizing latency and maximizing data transfer speeds.

Point-to-point connections are ideal for scenarios requiring reliable, high-speed, and secure data transmission, such as in data center interconnects or between network devices that handle sensitive information. However, they can be less scalable and more expensive compared to other topologies, as each new connection requires additional hardware and infrastructure.

Star Topology

Figure 6-7 shows a sample *star topology* with a hub at the center of the topology and a collection of clients individually connected to the hub. Notice that a star topology has a central point from which all attached devices radiate. In LANs in the early 1990s, that centralized device was typically a hub. Modern networks, however, usually have a switch located at the center of the star.

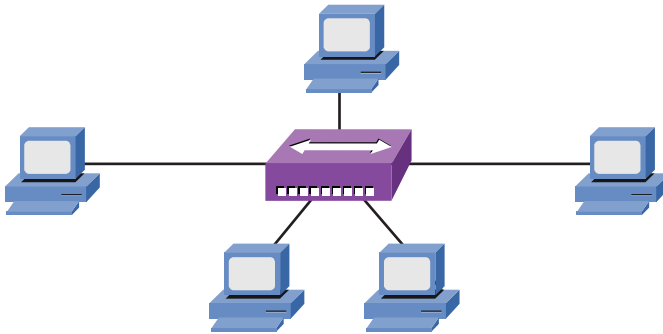


Figure 6-7 Star Topology

The star topology is the most popular physical LAN topology in use today, with an Ethernet switch at the center of the star and unshielded twisted-pair (UTP) cable used to connect from the switch ports to clients.

Table 6-3 identifies some of the primary characteristics, benefits, and drawbacks of a star topology.

Key
Topic

Table 6-3 Characteristics, Benefits, and Drawbacks of a Star Topology

Characteristics	Benefits	Drawbacks
Devices have independent connections to a central device (for example, a hub or a switch).	A cable break impacts only the device connected via the broken cable and not the entire topology.	More cable is required for a star topology because each device requires its own cable to connect back to the central device.
Star topologies are commonly used with Ethernet technologies (described in Chapter 5, “Transmission Media and Transceivers”).	Troubleshooting is relatively simple because a central device in the star topology acts as the aggregation point for all the connected devices.	Installation can take longer for a star topology because more cable runs must be installed.

Hub-and-Spoke Topology

When interconnecting multiple sites (for example, multiple corporate locations) via WAN links, a *hub-and-spoke topology* may be used, with a WAN link from each remote site (that is, a *spoke site*) to the main site (that is, the *hub site*). This approach, an example of which is shown in Figure 6-8, is similar to the star topology used in LANs.

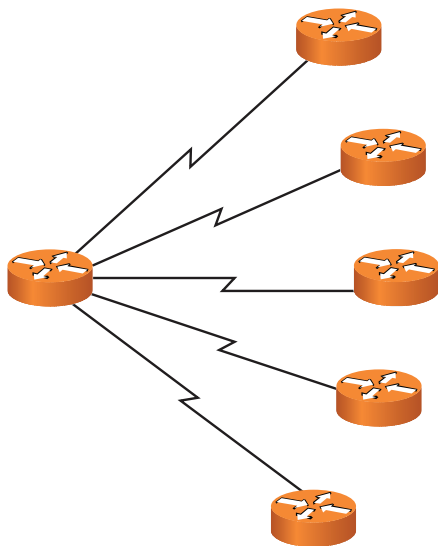


Figure 6-8 Hub-and-Spoke Topology

With WAN links, a service provider is paid a recurring fee for each link. Therefore, a hub-and-spoke topology helps minimize WAN expenses by not directly connecting any two spoke locations. If two spoke locations need to communicate with each

other, their communication is sent via the hub location. Table 6-4 describes the characteristics, benefits, and drawbacks of a hub-and-spoke WAN topology.

Key Topic

Table 6-4 Characteristics, Benefits, and Drawbacks of a Hub-and-Spoke WAN Topology

Characteristics	Benefits	Drawbacks
Each remote site (that is, a spoke) connects to a main site (that is, the hub) via a WAN link.	Costs are reduced (as compared to a full-mesh or partial-mesh topology) because a minimal number of links is used.	Suboptimal routes must be used between remote sites because all intersite communication must travel via the main site.
Communication between two remote sites travels through the hub site.	Adding one or more additional sites is easy (compared to a full-mesh or partial-mesh topology) because only one link needs to be added per site.	Because all remote sites converge on the main site, this hub site is potentially a single point of failure.
—	—	Because each remote site is reachable by only a single WAN link, the hub-and-spoke topology lacks redundancy.

Full-Mesh Topology

Whereas a hub-and-spoke WAN topology lacks redundancy and suffers from suboptimal routes, a *full-mesh topology* (sometimes just termed *mesh*), as shown in Figure 6-9, directly connects every site to every other site.

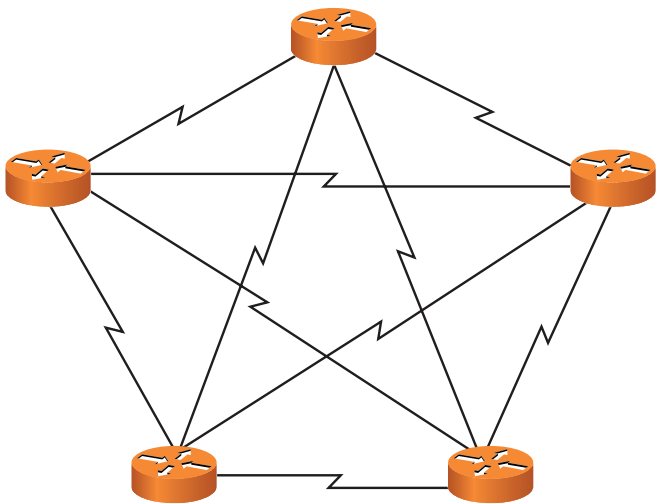


Figure 6-9 Full-Mesh Topology

Because each site connects directly to every other site, an optimal path can be selected, as opposed to relaying traffic via another site. Also, a full-mesh topology is highly fault tolerant. By inspecting Figure 6-9, you can see that multiple links in the topology could be lost, and every site might still be able to connect to every other site. Table 6-5 summarizes the characteristics, benefits, and drawbacks of a full-mesh topology.

Key
Topic

Table 6-5 Characteristics, Benefits, and Drawbacks of a Full-Mesh WAN Topology

Characteristics	Benefits	Drawbacks
Every site has a direct WAN connection to every other site.	An optimal route exists between any two sites.	A full-mesh network can be difficult and expensive to scale because the addition of one new site requires a new WAN link between the new site and every other existing site.
The number of required WAN connections can be calculated with the formula $w = n \times (n - 1) / 2$, where w = the number of WAN links and n = the number of sites. For example, a network with 10 sites would require 45 WAN connections to form a fully meshed network: $45 = 10 \times (10 - 1) / 2$.	A full-mesh network is fault tolerant because one or more links can be lost, and reachability between all sites might still be maintained.	—
—	Troubleshooting a full-mesh network is relatively easy because each link is independent of the other links.	—

Partial-Mesh Topology

A partial-mesh WAN topology, as depicted in Figure 6-10, is a *hybrid* of the previously described hub-and-spoke topology and full-mesh topology. Specifically, a *partial-mesh topology* can be designed to offer an optimal route between selected sites while avoiding the expense of interconnecting every site to every other site.

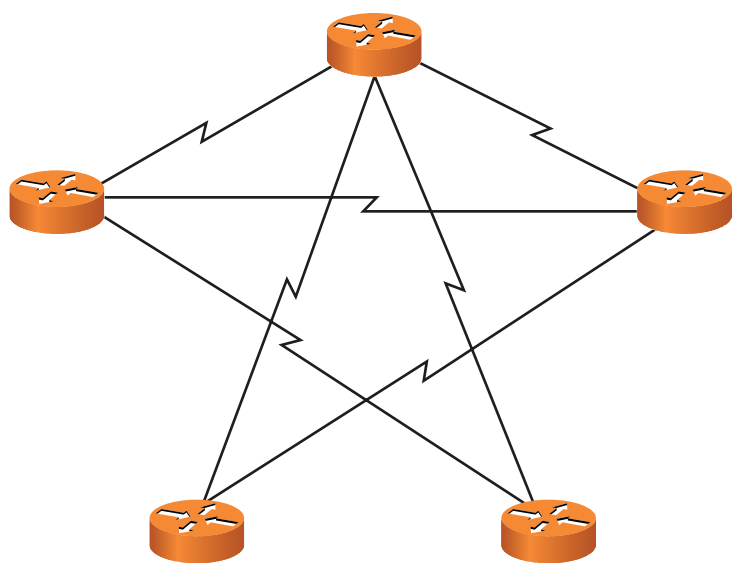


Figure 6-10 Partial-Mesh Topology

When designing a partial-mesh topology, a network designer must consider network traffic patterns and strategically add links interconnecting sites that have higher volumes of traffic between themselves. Table 6-6 highlights the characteristics, benefits, and drawbacks of a partial-mesh topology.

**Key
Topic**

Table 6-6 Characteristics, Benefits, and Drawbacks of a Partial-Mesh Topology

Characteristics	Benefits	Drawbacks
Selected sites (that is, sites with frequent intersite communication) are interconnected via direct links, whereas sites that have less-frequent communication can communicate via another site.	A partial-mesh topology provides optimal routes between selected sites with higher intersite traffic volumes while avoiding the expense of interconnecting every site to every other site.	A partial-mesh topology is less fault tolerant than a full-mesh topology.
A partial-mesh topology uses fewer links than a full-mesh topology and more links than a hub-and-spoke topology for interconnecting the same number of sites.	A partial-mesh topology is more redundant than a hub-and-spoke topology.	A partial-mesh topology is more expensive than a hub-and-spoke topology.

NOTE There are plenty of network topologies today that use a variety of different design approaches. This often results in several approaches being used in one solution. We often term this a *hybrid* approach.

**Key
Topic**

The Three-Tier Hierarchical Model

For a very long time now in networking, we have had the concept of building a network in tiers. Doing so allows you to separate the logical functions of the network and target the appropriate hardware and software at the correct areas. For example, you do not want to consider a switch that is designed to permit users to access the network in the same way that you would consider a switch for moving traffic through the core of the network. We begin this section by examining one of the most classic (and still used) models for network architecture: the classic three-tier design.

Perhaps the very first thing you need to understand about the ***three-tier hierarchical model*** (shown in Figure 6-11) is that in addition to being called the classic or classical model, it is also referred to as the hierarchical network model. Whatever you call it, notice that it is built with three layers: the ***access layer*** (which is the layer closest to the end users), the distribution layer, and the ***core layer***.

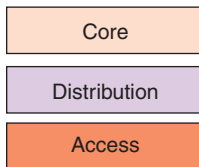


Figure 6-11 The “Classic” Three-Tier Network Architecture

Different names are often used for these layers. For example, the core layer might be called the backbone layer, and the distribution layer is often referred to as the aggregation layer. Finally, the access layer might be called the workstation or edge layer. You can clearly see what is happening here: Engineers often use names that say more about what is happening at the various layers of the model.

The Access/Edge Layer

The access/edge layer ensures that authorized users can access the network with ease and in a high-bandwidth fashion. In addition to providing a guide for what equipment is appropriate at the access layer, the classic three-tier model gives you a sense of what technologies are needed at certain layers.

Devices commonly found at the access layer include the following:

- End-user devices such as laptops, desktops, tablets, and smartphones
- Relatively inexpensive Layer 2 (data link layer) switches; notice that vendors will often refer to these devices as access layer switches
- Wireless access points (APs)

Common technologies found at the access layer include the following:

- Layer 2 switching
- Spanning Tree Protocol (STP)
- Power over Ethernet (PoE)
- Voice virtual local area network (VLAN) technologies
- Quality of service (QoS) functions
- Port security
- VLAN access control lists (VACLs)

The Distribution/Aggregation Layer

A consistent trend in networking decade after decade has been the fact that more and more of the data that users need to access is located remotely. More and more traffic is being sent outside the local network infrastructure. As cloud technology continues to take hold, remote data access will become even more common.

The *distribution/aggregation layer* is critical in the design of a network. This layer is responsible for connecting the access layer and the many devices inside it to the rest of the world and the valuable data that must be accessed.

At this layer, you typically find the following network devices:

- High-speed routers
- High-speed multilayer switches
- Firewalls
- Intrusion prevention devices
- Proxy servers

NOTE Network engineers often try to remove overhead from the access and core layers and add it to the distribution layer.

The distribution layer is famous for its use of the following technologies:

- Security
- Policy
- Routing
- Load balancing
- Redundancy
- Summarization

The Core Layer

You’ve no doubt heard that three things are important in real estate: location, location, and location. Well, when it comes to the core layer of the classic design, three things are important as well: speed, speed, and speed. The core layer is responsible for moving massive amounts of data in an enterprise network—often between a data center (or even data centers) and the main employee locations.

Even in today’s remote work environment, there are still core network devices that are crucial to a properly operating IT organization. The core layer is home to very high-speed routers and multilayer switches (or even high-speed Layer 2 switches). (Remember, it is all about speed.)

The following types of technologies exist at the core layer:

- Redundancy
- Bandwidth aggregation
- Traffic policing and/or shaping

NOTE Many organizations are just too small to really need a three-tier model. Such organizations can often successfully implement a classic two-tier model, which is often referred to as a *collapsed core design*. In this approach, the core and distribution layers are merged together for the sake of simplicity (as well as cost savings).

Spine and Leaf

The classic hierarchical model for building an enterprise network is excellent for building an enterprise network. But what about for different kinds of network locations in an organization, such as a data center? Certainly, different models might be more appropriate in a data center, where traffic is different and traffic patterns are not always as expected. Different models are used for different network locations. One that has become quite famous is called the *spine and leaf* topology or network model.

One of the reasons the spine and leaf topology has gained popularity in the data center is that vendors such as Cisco Systems rely on it for new technologies they are rolling out. Cisco created the Application Centric Infrastructure (ACI) for Software Defined Networking (SDN) in the data center. As its name directly communicates, ACI takes a very application-centric (software as a service [SaaS]) approach to the control and management of network design, bringing automation and orchestration to the SaaS engine of the private data center.

NOTE Chapter 8, “Evolving Use Cases,” covers software-defined networking in detail for you.

Figure 6-12 shows a simple spine and leaf network design.

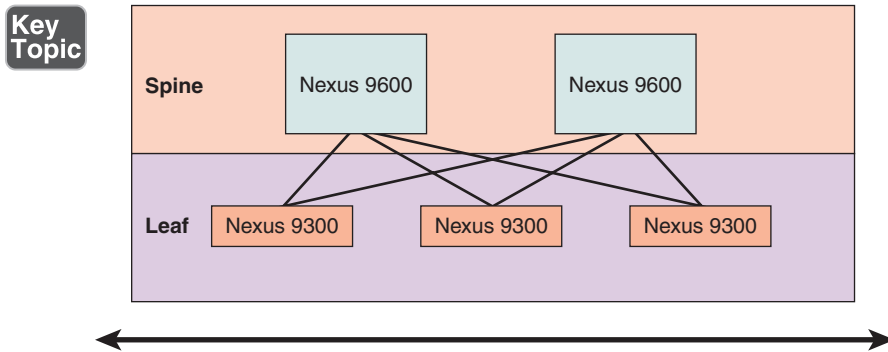


Figure 6-12 The Cisco ACI Spine and Leaf Design

Notice that this model consists of just two layers: the spine and the leaf layers.

NOTE Some engineers call the spine layer the core layer or the *backbone* layer.

One thing that can really surprise long-time Cisco fans is just how inflexible Cisco is when it comes to the spine and leaf topology that powers the Cisco ACI solution. Cisco makes it very clear exactly which devices must be used at each layer of the model. Figure 6-12 shows Cisco Nexus 9600s at the spine layer and Cisco Nexus 9300s at the leaf layer. Cisco will continue to produce different classes of devices that will fit in these layers, but you can expect that the ACI solution will always require particular Cisco devices.

There is also no flexibility in how you connect the spine and leaf topology. Each leaf node *must* connect to each spine node—and that is it. The spine nodes do not directly connect to each other, and the leaf nodes do not directly connect to each other. Rather, each leaf must connect to each spine. This inflexibility brings several benefits:

- There is no doubt about how you should connect the equipment. It is actually refreshing to be told exactly how the Gigabit Ethernet segments must be constructed, leaving no room for doubt or experimentation.
- There is completely predictable data pathing in the system. VMs (or containers) are hosted on systems that connect to leaf nodes. When a VM that connects to the leaf layer needs to communicate with another VM on the same leaf device, it does not need to transit to the spine layer; when a VM on one leaf needs to communicate to a VM on another leaf device, it is *always* one hop away; there is exactly one spine device that must be transited.
- The latency is predictable, due in large part to the predictability of the data pathing.
- Monitoring and expanding the topology is simple. If you need to connect more hosts (hardware) to connect more containers and VMs, you can just add a leaf device; if you need more overall bandwidth and processing power for the solution, you can just add a new spine.

NOTE The Cisco ACI approach is actually a variation of an earlier technological approach called top-of-rack switching, or end-of-rack switching. In this design, multiple physical components act as one network device. For example, with the end-of-rack switching design, there are physical switches in each data center rack for connectivity, but the control layer (or plane) intelligence and the “real” switching engine are located at the end of a row of racks in the data center. This larger physical device is controlling the behavior of the distributed switches in the racks. They function logically as one big device. Specifically, the smaller switches in the racks act like line cards inserted in the chassis of a bigger and more powerful switch. ACI functions similarly, with leaf nodes appearing to the solution as part of the spine devices.

Traffic Flows

When working with the various network architectures that are possible (such as three-tier versus spine and leaf), engineers will often study how the majority of traffic flows in the model. This helps them ensure they have provisioned adequate bandwidth and other resources. You should be familiar with two overall patterns:

- ***North-south traffic flows:*** In the classic three-tier model, most traffic flows are north to south (and also south to north). This describes traffic moving from the core to the access layer and vice versa.
- ***East-west traffic flows:*** In the newer spine and leaf type architecture, most traffic flows tend to be east to west (and vice versa). In this architecture (along with other data center type designs), virtual machines and/or containers communicate with each other over east-west flows.

Real-World Case Study

The headquarters for Acme, Inc. is located on a single floor of a downtown building. Acme also has two branch offices, Branch1 and Branch2, that are in remote locations. The company wants to do file sharing, instant messaging, email, and voice on its own private networks when possible. It also wants connectivity to the Internet.

At the headquarters location, Acme sets up a LAN with UTP (Cat 6a) cabling, with the clients and servers connected to a central switch. This forms a physical star topology. For connectivity between HQ and its two branch offices, the company uses a service provider (SP) for WAN connectivity.

The SP provides logical, point-to-point connections between the headquarters office and both of the branch locations. Physically, the path between the headquarters and each branch office is going through several routers in the SP's network. For the time being, Branch1 and Branch2 do not have direct connectivity to each other, so branch-to-branch traffic must pass through the headquarters site (hub and spoke).

Next year, as more funds are available, the company can add WAN connectivity directly between Branch1 and Branch2. This will change the WAN topology from hub and spoke to full mesh.

Summary

Here are the main topics covered in this chapter:

- One way to classify networks is by their geographic dispersion. Examples include LAN, WAN, CAN, MAN, and PAN. This chapter covered these major network classifications in use today.

- Another approach to classifying networks is based on a network’s topology. Examples of network types, based on topology, include star, hybrid (partial mesh), full mesh, and hub and spoke. This chapter also covered these topologies.
- This chapter contrasted client/server and peer-to-peer networks.
- This chapter covered the three-tier hierarchical model and the collapsed core design.
- This chapter also covered the spine and leaf network topology and discussed north to south versus east to west traffic flows.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 6-7 lists these key topics and the page number where each is found.



Table 6-7 Key Topics for Chapter 6

Key Topic Element	Description	Page Number
Table 6-3	Characteristics, Benefits, and Drawbacks of a Star Topology	146
Table 6-4	Characteristics, Benefits, and Drawbacks of a Hub-and-Spoke WAN Topology	147
Table 6-5	Characteristics, Benefits, and Drawbacks of a Full-Mesh WAN Topology	148
Table 6-6	Characteristics, Benefits, and Drawbacks of a Partial-Mesh Topology	149
Section	The Three-Tier Hierarchical Model	150
Figure 6-12	The Cisco ACI Spine and Leaf Design	153

Complete Tables and Lists from Memory

Print a copy of Appendix C, “Memory Tables,” or at least the section for this chapter and complete as many of the tables as possible from memory. Appendix D, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

access layer, collapsed core design, core layer, distribution/aggregation layer, east–west traffic flows, full-mesh topology, hub-and-spoke topology, hybrid, north–south traffic flows, partial-mesh topology, point-to-point network topology, spine and leaf, star topology, three-tier hierarchical model

Additional Resources

Wide Area Network (WAN) Topologies: <https://www.youtube.com/watch?v=9WkZT0YMZ70>

Leaf Spine DC Architectures: <https://www.youtube.com/watch?v=WzwpKVx9Tb0>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. What type of network is most likely to incorporate IEEE 802.3 technologies?
 - a. PAN
 - b. SAN
 - c. WLAN
 - d. LAN
2. A physical topology is dictated by the way in which components are physically interconnected. What type of topology is dictated by the way in which traffic actually flows?
 - a. Overlay
 - b. Underlay
 - c. Logical
 - d. Access

3. Which layer of the three-tier hierarchical model is most concerned with speed?
 - a. Access
 - b. Distribution
 - c. Policy
 - d. Core
4. How is connectivity for the spine and leaf topology implemented?
 - a. Each spine device connects to every other device.
 - b. Each leaf device connects to each spine device.
 - c. Each spine device connects to every other spine device.
 - d. Each leaf device must connect to each other leaf device.
5. A company has various locations in a city interconnected using Metro Ethernet connections. This is an example of what type of network?
 - a. WAN
 - b. CAN
 - c. PAN
 - d. MAN
6. A network formed by interconnecting a PC to a digital camera via a USB cable is considered what type of network?
 - a. WAN
 - b. CAN
 - c. PAN
 - d. MAN
7. Which of the following physical LAN topologies requires the most cabling?
 - a. Point-to-point
 - b. Ring
 - c. Star
 - d. WLAN

8. Which of the following topologies offers the highest level of redundancy?
 - a. Full mesh
 - b. Hub and spoke
 - c. Bus
 - d. Partial mesh
9. How many WAN links are required to create a full mesh of connections between five remote sites?
 - a. 5
 - b. 10
 - c. 15
 - d. 20
10. Which of the following are advantages of a hub-and-spoke WAN topology as compared to a full-mesh WAN topology? (Choose two.)
 - a. Lower cost
 - b. Optimal routes
 - c. More scalable
 - d. More redundancy
11. Which type of network is based on network clients sharing resources with one another?
 - a. Client/server
 - b. Client/peer
 - c. Peer-to-peer
 - d. Peer-to-server
12. Which of the following is an advantage of a peer-to-peer network as compared with a client/server network?
 - a. More scalable
 - b. Less expensive
 - c. Better performance
 - d. Simplified administration

13. What network type would help facilitate communications when large video or audio files need to be housed and transferred through the network?
 - a. WLAN
 - b. CAN
 - c. PAN
 - d. SAN

14. Which of the following traffic flows occurs primarily between servers or services deployed within the same data center or cloud environment?
 - a. East–west
 - b. North–south
 - c. Point to point
 - d. NAS

15. What are the three layers in the three-tier hierarchical model? (Choose three.)
 - a. Intermediate
 - b. Core
 - c. Distribution
 - d. Access

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This chapter covers the following topics related to Objective 1.7 (Given a scenario, use appropriate IPv4 network addressing) of the CompTIA Network+ N10-009 certification exam:

- Public vs. private
 - Automatic Private IP Addressing (APIPA)
 - RFC1918
 - Loopback/localhost
- Subnetting
 - Variable Length Subnet Mask (VLSM)
 - Classless Inter-domain Routing (CIDR)
- IPv4 address classes
 - Class A
 - Class B
 - Class C
 - Class D
 - Class E

IPv4 Addressing

When two devices on a network want to communicate, they need logical addresses (that is, Layer 3 addresses, as described in Chapter 1, “The OSI Model and Encapsulation”). Most modern networks use Internet Protocol (IP) addressing. Therefore, the focus of this chapter is IP.

While there are two versions of IP in use today, IP version 4 (IPv4) and IP version 6 (IPv6), this chapter focuses exclusively on IPv4. Even with IPv6 making strides of adoption throughout the Internet, IPv4 will be with us for a long, long time, especially when we consider how well the two versions integrate with each other.

This chapter begins with coverage of how IP addresses are represented in binary notation. You will learn about the structure of an IPv4 address and learn to distinguish between different categories of IPv4 addresses.

Next, this chapter details various options for assigning IP addresses to end stations. As you will see, one of the benefits of IP addressing is that you have flexibility in how you can subdivide a network address into multiple subnets. This discussion of subnetting is a bit mathematical, and multiple practice exercises are provided to help solidify these concepts in your mind.

Foundation Topics

Binary Numbering

Chapter 1 describes how a network transmits data as a series of binary 1s and 0s. Similarly, IP addresses are represented as series of binary digits (that is, *bits*). An IPv4 address consists of 32 bits.

NOTE An IPv6 address has a whopping 128 bits. As you will learn in Chapter 8, “Evolving Use Cases,” this four-times expansion of the bits used for addressing creates an incredible number of available addresses for future use.

Later in this chapter, you will need to be able to convert between the decimal representation of a number and that number’s binary equivalent. This skill is needed for things such as subnet mask calculations. This section describes this mathematical procedure and provides you with practice exercises.

Principles of Binary Numbering

You are accustomed to using base 10 numbering on a day-to-day basis. In a base 10 numbering system, you have 10 digits, in the range 0 through 9, at your disposal. Binary numbering, however, uses a base 2 numbering system, where there are only two digits: 0 and 1.

Because computer systems divide 32-bit IPv4 addresses into four 8-bit octets each, this discussion focuses on converting between 8-bit binary numbers and decimal numbers. To convert a binary number to decimal, you can create a table like Table 7-1.

Key
Topic

Table 7-1 Binary Conversion Table

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

Note the structure of this table. There are eight columns, representing the 8 bits in an octet. The column headings are the powers of 2, from 0 to 7, beginning in the rightmost column. Specifically, 2 raised to the power of 0 (2^0) is 1. (In fact, any number raised to the power of 0 is 1.) If you raise 2 to the first power (2^1), that equals 2, and 2 raised to the second power (that is, 2^2 , or 2×2) is 4. This continues through 2 raised to the power of 7 (that is, 2^7 , or $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$), which equals 128. You can use this table for converting binary numbers to decimal and decimal

numbers to binary. The skill of binary-to-decimal and decimal-to-binary conversion is critical for working with subnet masks, as discussed later in this chapter.

**Key
Topic****Converting a Binary Number to a Decimal Number**

To convert a binary number to a decimal number, you populate the previously described binary table with the given binary digits. Then you add up the column heading values for the columns that contain a binary 1.

For example, consider Table 7-2. Only the 128, 16, 4, and 2 columns contain a 1, and all the other columns contain a 0. If you add all the column headings containing a 1 in their column (that is, $128 + 16 + 4 + 2$), you get the result 150. Therefore, you can conclude that the binary number 10010110 equates to the decimal value 150.

Table 7-2 Binary Conversion Example 1

128	64	32	16	8	4	2	1
1	0	0	1	0	1	1	0

**Key
Topic****Converting a Decimal Number to a Binary Number**

To convert numbers from decimal to binary, starting with the leftmost column, ask the question, “Is this number equal to or greater than the column heading?” If the answer to that question is no, place a 0 in that column and move to the next column. If the answer is yes, place a 1 in that column and subtract the value of the column heading from the number you are converting. When you then move to the next column (to your right), again ask yourself, “Is this number (which is the result of your earlier subtraction) equal to or greater than the column heading?” This process continues (to the right) for all the remaining column headings.

For example, say that you want to convert the number 167 to binary. The following steps walk you through the process:

Step 1. Ask the question, “Is 167 equal to or greater than 128?” Because the answer is yes, you place a 1 in the 128 column, as shown in Table 7-3, and subtract 128 from 167, which yields the result 39.

Table 7-3 Binary Conversion Example 2: Step 1

128	64	32	16	8	4	2	1
1							

Step 2. Now that you are done with the 128 column, move (to the right) to the 64 column. Ask the question, “Is 39 equal to or greater than 64?” Because

the answer is no, you place a 0 in the 64 column, as shown in Table 7-4, and continue to the next column (the 32 column).

Table 7-4 Binary Conversion Example 2: Step 2

128	64	32	16	8	4	2	1
1	0						

Step 3. Under the 32 column, ask the question, “Is 39 equal to or greater than 32?” Because the answer is yes, you place a 1 in the 32 column, as shown in Table 7-5, and subtract 32 from 39, which yields the result 7.

Table 7-5 Binary Conversion Example 2: Step 3

128	64	32	16	8	4	2	1
1	0	1					

Step 4. Now you are under the 16 column and ask, “Is 7 equal to or greater than 16?” Because the answer is no, you place a 0 in the 16 column, as shown in Table 7-6, and move to the 8 column.

Table 7-6 Binary Conversion Example 2: Step 4

128	64	32	16	8	4	2	1
1	0	1	0				

Step 5. As with the 16 column, the number 7 is not equal to or greater than 8. So, you place a 0 in the 8 column, as shown in Table 7-7.

Table 7-7 Binary Conversion Example 2: Step 5

128	64	32	16	8	4	2	1
1	0	1	0	0			

Step 6. Because 7 is greater than or equal to 4, you place a 1 in the 4 column, as shown in Table 7-8, and subtract 4 from 7, yielding 3 as the result.

Table 7-8 Binary Conversion Example 2: Step 6

128	64	32	16	8	4	2	1
1	0	1	0	0	1		

Step 7. Now under the 2 column, you ask the question, “Is 3 greater than or equal to 2?” Because the answer is yes, you place a 1 in the 2 column, as shown in Table 7-9, and subtract 2 from 3, yielding 1 as the result.

Table 7-9 Binary Conversion Example 2: Step 7

128	64	32	16	8	4	2	1
1	0	1	0	0	1	1	

Step 8. Finally, in the rightmost column (that is, the 1 column), you ask whether the number 1 is greater than or equal to 1. Because it is, you place a 1 in the 1 column, as shown in Table 7-10.

Table 7-10 Binary Conversion Example 2: Step 8

128	64	32	16	8	4	2	1
1	0	1	0	0	1	1	1

You can now conclude that the decimal number 167 equates to the binary value 10100111. In fact, you can check your work by adding up the values for the column headings that contain a 1 in their column. In this example, the 128, 32, 4, 2, and 1 columns contain a 1. If you add these values, the result is 167 (that is, $128 + 32 + 4 + 2 + 1 = 167$).

Binary Numbering Practice

Because binary number conversion is a skill developed through practice, you will now be challenged with a few conversion exercises. The first two exercises ask you to convert a binary number to a decimal number, and the last two exercises ask you to convert a decimal number to a binary number.

Binary Conversion Exercise 1

Using Table 7-11 as a reference, convert the binary number 01101011 to a decimal number.

Table 7-11 Binary Conversion Exercise 1: Base Table

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

Write your answer here: _____

Binary Conversion Exercise 1: Solution

Given the binary number 01101011 and filling in a binary conversion table, as shown in Table 7-12, you find that the 64, 32, 8, 2, and 1 columns contain a 1. Each of the other columns contains a 0. By adding up the column headings for the columns that contain a 1 (that is, $64 + 32 + 8 + 2 + 1$), you get the decimal value 107.

Table 7-12 Binary Conversion Exercise 1: Solution Table

128	64	32	16	8	4	2	1
0	1	1	0	1	0	1	1

Binary Conversion Exercise 2

Using Table 7-13 as a reference, convert the binary number 10010100 to a decimal number.

Table 7-13 Binary Conversion Exercise 2: Base Table

128	64	32	16	8	4	2	1

Write your answer here: _____

Binary Conversion Exercise 2: Solution

Given the binary number 10010100 and filling in a binary conversion table, as shown in Table 7-14, you find that the 128, 16, and 4 columns contain a 1. Each of the other columns contains a 0. By adding up the column headings for the columns that contain a 1 (that is, $128 + 16 + 4$), you get the decimal value 148.

Table 7-14 Binary Conversion Exercise 2: Solution Table

128	64	32	16	8	4	2	1
1	0	0	1	0	1	0	0

Binary Conversion Exercise 3

Using Table 7-15 as a reference, convert the decimal number 49 to a binary number.

Table 7-15 Binary Conversion Exercise 3: Base Table

128	64	32	16	8	4	2	1

Write your answer here: _____

Binary Conversion Exercise 3: Solution

You can begin your conversion of the decimal number 49 to a binary number by asking the following questions and performing the following calculations:

1. Is 49 greater than or equal to 128? No. Put a 0 in the 128 column.
2. Is 49 greater than or equal to 64? No. Put a 0 in the 64 column.
3. Is 49 greater than or equal to 32? Yes. Put a 1 in the 32 column and subtract 32 from 49. $49 - 32 = 17$.
4. Is 17 greater than or equal to 16? Yes. Put a 1 in the 16 column and subtract 16 from 17. $17 - 16 = 1$.
5. Is 1 greater than or equal to 8? No. Put a 0 in the 8 column.
6. Is 1 greater than or equal to 4? No. Put a 0 in the 4 column.
7. Is 1 greater than or equal to 2? No. Put a 0 in the 2 column.
8. Is 1 greater than or equal to 1? Yes. Put a 1 in the 1 column.

Combining these 8 binary digits, you form the binary number 00110001, as shown in Table 7-16. Verify your work by adding the values of the column headings whose columns contain a 1. In this case, columns 32, 16, and 1 each contain a 1. By adding these values (that is, $32 + 16 + 1$), you get the value 49.

Table 7-16 Binary Conversion Exercise 3: Solution Table

128	64	32	16	8	4	2	1
0	0	1	1	0	0	0	1

Binary Conversion Exercise 4

Using Table 7-17 as a reference, convert the decimal number 236 to a binary number.

Table 7-17 Binary Conversion Exercise 4: Base Table

128	64	32	16	8	4	2	1

Write your answer here: _____

Binary Conversion Exercise 4: Solution

You can begin your conversion of the decimal number 236 to a binary number by asking the following questions and performing the following calculations:

- 1. Is 236 greater than or equal to 128? Yes. Put a 1 in the 128 column and subtract 128 from 236. $236 - 128 = 108$.
- 2. Is 108 greater than or equal to 64? Yes. Put a 1 in the 64 column and subtract 64 from 108. $108 - 64 = 44$.
- 3. Is 44 greater than or equal to 32? Yes. Put a 1 in the 32 column and subtract 32 from 44. $44 - 32 = 12$.
- 4. Is 12 greater than or equal to 16? No. Put a 0 in the 16 column.
- 5. Is 12 greater than or equal to 8? Yes. Put a 1 in the 8 column and subtract 8 from 12. $12 - 8 = 4$.
- 6. Is 4 greater than or equal to 4? Yes. Put a 1 in the 4 column and subtract 4 from 4. $4 - 4 = 0$.
- 7. Is 0 greater than or equal to 2? No. Put a 0 in the 2 column.
- 8. Is 0 greater than or equal to 1? No. Put a 0 in the 1 column.

By combining these 8 binary digits, you form the binary number 11101100, as shown in Table 7-18. You can verify your work by adding the values of the column headings whose columns contain a 1. In this case, columns 128, 64, 32, 8, and 4 each contain a 1. By adding these values (that is, $128 + 64 + 32 + 8 + 4$), you get the value 236.

Table 7-18 Binary Conversion Exercise 4: Solution Table

128	64	32	16	8	4	2	1
1	1	1	0	1	1	0	0

IPv4 Addressing

Although IPv6 is increasingly being adopted in corporate networks, IPv4 is by far the most popular Layer 3 addressing scheme in today’s networks. For brevity in this section, the term *IPv4 address* is used interchangeably with the more generic term *IP address*.

Devices on an IPv4 network use unique IP addresses to communicate with one another. Metaphorically, you can relate this to sending a letter through the postal service. You place a destination address on an envelope containing the letter, and in the upper-left corner of the envelope, you place your return address. Similarly, when

an IPv4 network device sends data on a network, it places both a destination IP address and a source IP address in the packet's IPv4 header.

IPv4 Address Structure

An IPv4 address is a 32-bit address. However, rather than write out each individual bit value, you write the address in dotted-decimal notation. Consider the IP address 10.1.2.3. Notice that this IP address is divided into four separate numbers, separated by periods. Each number represents one-fourth of the IP address. Specifically, each number represents an 8-bit portion of the 32 bits in the address. Because each of these four divisions of an IP address represents 8 bits, these divisions are called *octets*. For example, Figure 7-1 shows the binary representation of the 10.1.2.3 IP address. In Figure 7-1, notice that the 8 leftmost bits of 00001010 equate to the decimal value 10. (The calculation for this is described in the previous section.) Similarly, 00000001 in binary equates to 1 in decimal, and 00000010 in binary equals 2 in decimal. Finally, 00000011 yields the decimal value 3.

Dotted-Decimal Notation	10	1	2	3
Binary Bits	00001010	00000001	00000010	00000011
	Octet 1	Octet 2	Octet 3	Octet 4

Figure 7-1 Binary Representation of a Dotted-Decimal IP Address

Interestingly, an IP address is composed of two types of addresses: a network address and a host address. Specifically, a group of contiguous left-justified bits represent the network address, and the remaining bits (that is, a group of contiguous right-justified bits) represent the address of a host on a network. The IP address component that determines which bits refer to the network and which bits refer to the host is called the *subnet mask*. You can think of the subnet mask as a dividing line separating an IP address's 32 bits into a group of network bits (on the left) and a group of host bits (on the right).

A subnet mask typically consists of a series of contiguous 1s followed by a set of contiguous 0s. In total, a subnet mask contains 32 bits, which correspond to the 32 bits found in an IPv4 address. The 1s in a subnet mask correspond to network bits in an IPv4 address, and 0s in a subnet mask correspond to host bits in an IPv4 address.

For example, consider Figure 7-2. The 8 leftmost bits of the subnet mask are 1s, and the remaining 24 bits are 0s. As a result, the 8 leftmost bits of the IP address represent the network address, and the remaining 24 bits represent the host address.

Key Topic

Dotted-Decimal Notation	10	1	2	3
IP Address (in Binary)	00001010	00000001	00000010	00000011
Subnet Mask	11111111	00000000	00000000	00000000

Network Bits

Host Bits

Figure 7-2 Dividing an IP Address into a Network Portion and a Host Portion

When you write a network address, all host bits are set to 0s. Once again, consider the example shown in Figure 7-2. The subnet mask in this example is an *8-bit subnet mask*, meaning that the 8 leftmost bits in the subnet mask are 1s. If the remaining bits are set to 0, as shown in Figure 7-3, the network address is 10.0.0.0.

Network Address (in Dotted Decimal)	10	0	0	0
Network Address (in Binary)	00001010	00000000	00000000	00000000
Subnet Mask	11111111	00000000	00000000	00000000

Network Bits

Host Bits

Figure 7-3 Network Address Calculation

When writing a network address, or an IP address for that matter, you need to provide more detail than just a dotted-decimal representation of an IP address’s 32 bits. For example, just being told that a device has IP address 10.1.2.3 does not tell you the network on which the IP address resides. To know the network address, you need to know the subnet mask, which could be written in dotted-decimal notation or in prefix notation (also known as *slash notation*). In the example with the IP address 10.1.2.3 and an 8-bit subnet mask, the IP address could be written as 10.1.2.3 255.0.0.0 or 10.1.2.3 /8. Similarly, the network address could be written as 10.0.0.0 255.0.0.0 or 10.0.0.0 /8.

Classes of Addresses

Although for an IP address (or a network address) you need subnet mask information to determine which bits represent the network portion of the address, there are default subnet masks with which you should be familiar. The default subnet mask for a given IP address is solely determined by the value in the IP address's first octet. Table 7-19 shows the default subnet masks for various ranges of IP addresses.



Table 7-19 IP Address Classes

Address Class	Value in First Octet	Classful Mask (Dotted Decimal)	Classful Mask (Prefix Notation)
Class A	1–126	255.0.0.0	/8
Class B	128–191	255.255.0.0	/16
Class C	192–223	255.255.255.0	/24
Class D	224–239	—	—
Class E	240–255	—	—

These IP address ranges, which you should memorize, are referred to as the different *IPv4 address classes*. **Class A**, **Class B**, and **Class C** addresses are assigned to network devices. **Class D** addresses are used as destination IP addresses (that is, not assigned to devices sourcing traffic) for multicast networks, and **Class E** addresses are reserved for experimental use. The default subnet masks associated with address Classes A, B, and C are called *classful masks*.

For example, consider the IP address 172.16.40.56. If you are told that this IP address uses its classful mask, you should know that it has subnet mask 255.255.0.0, which is the classful mask for a Class B IP address. You should know that 172.16.40.56 is a Class B IP address, based on the value of the first octet (172), which falls in the Class B range 128–191.

NOTE You might have noticed that in the ranges of values in the first octet, the number 127 seems to have been skipped. The reason is that 127 is used as a *loopback* IP address, which is a locally significant IP address representing the device itself. For example, if you are working on a network device and want to verify that the device has a TCP/IP stack loaded, you can try to ping IP address 127.0.0.1. If you receive ping responses, you can conclude that the device is running a TCP/IP stack. To make this testing even more convenient, most computer operating systems automatically resolve the name *localhost* to 127.0.0.1.

The nonprofit corporation Internet Corporation for Assigned Names and Numbers (ICANN) globally manages publicly routable IP addresses. ICANN does not directly assign a block of IP addresses to your Internet service provider (ISP) but rather assigns a block of IP addresses to a regional Internet registry. One example of a regional Internet registry is the American Registry for Internet Numbers (ARIN), which acts as an Internet registry for North America.

The Internet Assigned Numbers Authority (IANA) is yet another entity responsible for IP address assignment. ICANN operates IANA and is responsible for IP address assignment outside North America.

NOTE Some literature references the *Internet Network Information Center (InterNIC)*, which was the predecessor to ICANN and existed until September 18, 1998.

When an organization is assigned one or more publicly routable IP addresses by its service provider, that organization often needs more IP addresses to accommodate all of its devices. One solution is to use *private IP addressing* within an organization, in combination with network address translation (NAT).

Specific Class A, B, and C networks have been designated for private use. Although these networks are routable (with the exception of the 169.254.0.0–169.254.255.255 address range) within the organization, ISPs do not route these private networks over the public Internet. These private IP address ranges are often referred to as *RFC1918* addresses. This is due to the Request for Comments document that helped to make them a reality in networking today. Table 7-20 shows the IP networks reserved for internal use.

Key
Topic

Table 7-20 Private IP Networks

Address Class	Address Range	Default Subnet Mask
Class A	10.0.0.0–10.255.255.255	255.0.0.0
Class B	172.16.0.0–172.31.255.255	255.255.0.0
Class B	169.254.0.0–169.254.255.255	255.255.0.0
Class C	192.168.0.0–192.168.255.255	255.255.255.0

NOTE The 169.254.0.0–169.254.255.255 address range is not routable. Addresses in this range are only usable on their local subnet and are dynamically assigned to network hosts using Automatic Private IP Addressing (APIPA), which is discussed later in this chapter.

NAT, which is available on routers, allows private IP addresses used within an organization to be translated into a pool of one or more publicly routable IP addresses.

Assigning IPv4 Addresses

At this point in the discussion, you should understand that networked devices need IP addresses. However, beyond just an IP address, what extra IP address–related information needs to be provided to a device, and how does an IP address get assigned to a device?

This section begins by discussing various parameters that might be assigned to a network device, followed by discussions covering various approaches to assigning IP addresses to devices.

IP Addressing Components

As discussed in the previous section, an IP address has two portions: a network portion and a host portion. A subnet mask is required to delineate between these two portions.

In addition, if traffic is destined for a different subnet than the subnet on which the traffic originates, a default gateway needs to be defined. A default gateway routes traffic from the sender's subnet toward the destination subnet. Chapter 9, “Routing Technologies,” covers the concept of routing.

Another consideration is that end users typically do not type in the IP address of the destination device with which they want to connect (for example, a web server on the Internet). Instead, end users typically type in fully qualified domain names (FQDNs), such as `www.ajsnetworking.com`. When connecting to devices on the public Internet, a Domain Name System (DNS) server translates an FQDN into the corresponding IP address.

For a very long time, in a company's internal network (that is, an intranet), a Microsoft Windows Internet Name Service (WINS) was used to convert the names of network devices into their corresponding IP addresses. For example, say that you attempted to navigate to the shared folder `\\server1\hrdocs`. A WINS server could be used to resolve the network device name `server1` to a corresponding IP address. The path `\\server1\hrdocs` is in *universal naming convention (UNC)* form, where you are specifying a network device name (in this case, `server1`) and a resource available on that device (in this case, `hrdocs`). Companies today use DNS even for internal network name resolution.

To summarize, network devices (for example, an end-user PC) can benefit from a variety of IP address parameters, such as the following:

- IP address
- Subnet mask
- Default gateway
- Server address

Static Configuration

A simple way of configuring a PC with, for example, IP address parameters is to statically configure that information. For example, on a PC running Microsoft Windows as the operating system, you can navigate to the Control Panel, as shown in Figure 7-4, and click **Network and Internet**.

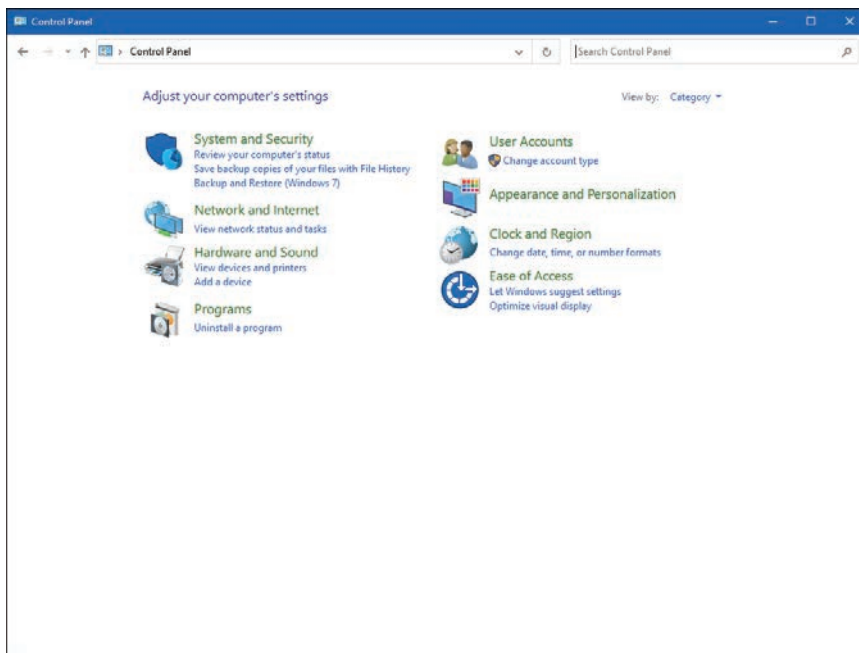


Figure 7-4 Windows Control Panel

In the Network and Internet control panel, click **Network and Sharing Center**, as shown in Figure 7-5.

You can then click the **Change adapter settings** link, as shown in Figure 7-6.

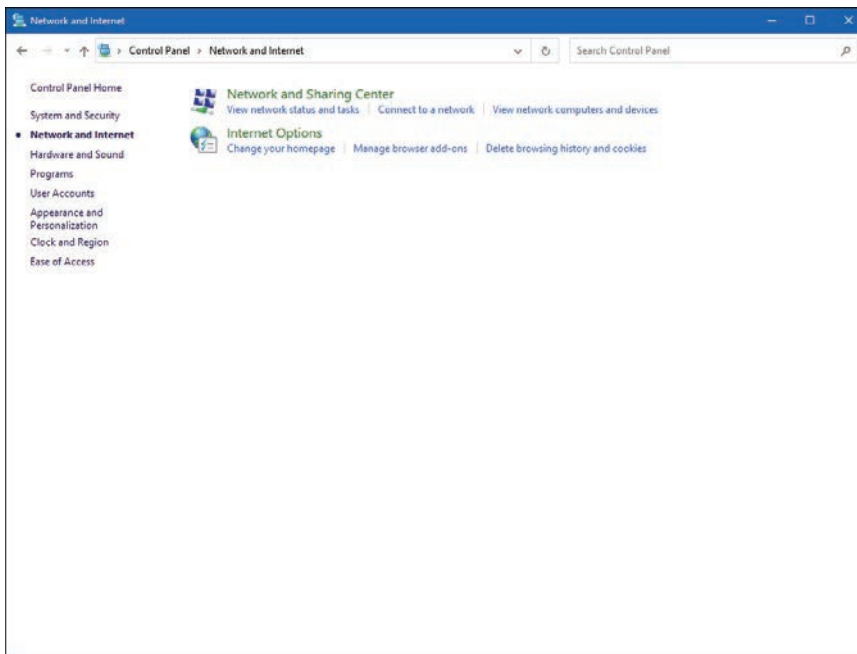


Figure 7-5 Network and Internet Control Panel

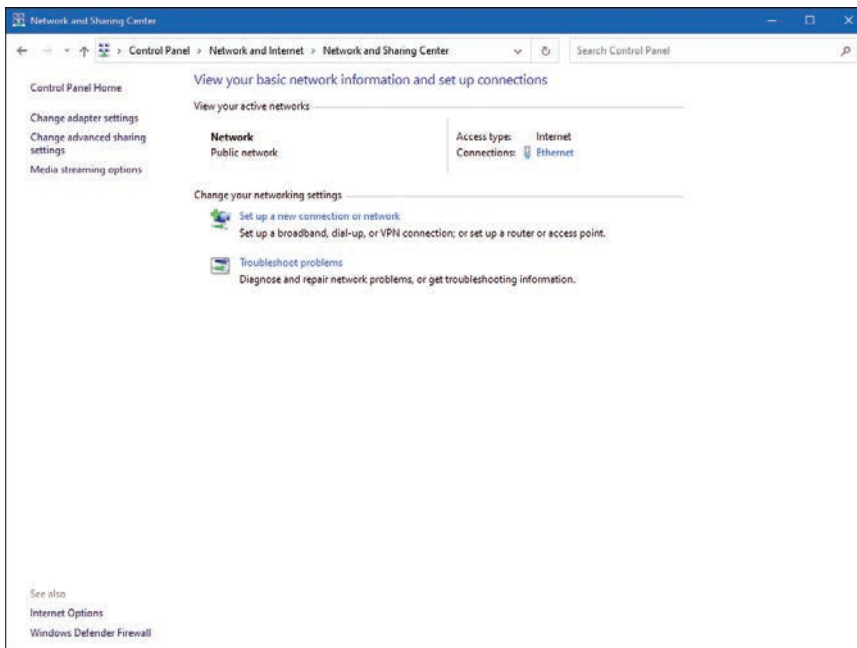


Figure 7-6 Network and Sharing Center

In the Network Connections window, double-click the network adapter whose settings you want to change, as shown in Figure 7-7.

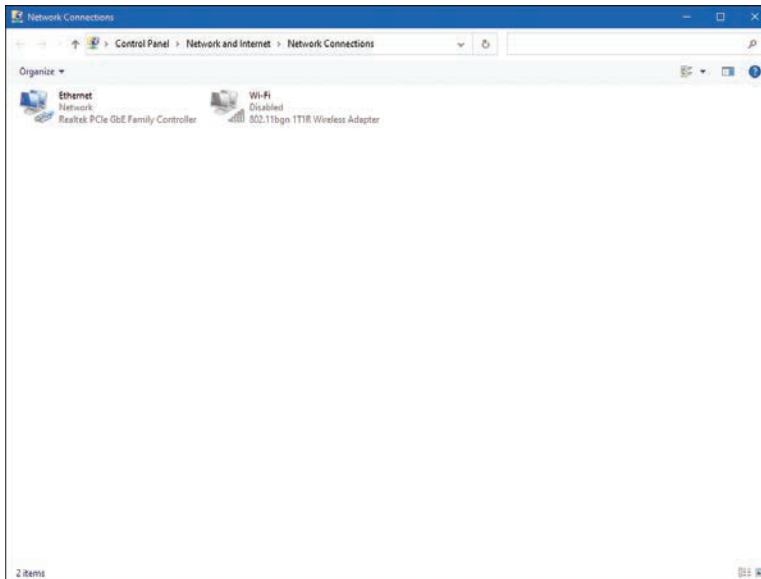


Figure 7-7 Network Connections Window

You are then taken to the Local Area Connection Status window, as shown in Figure 7-8, where you can click the **Properties** button.

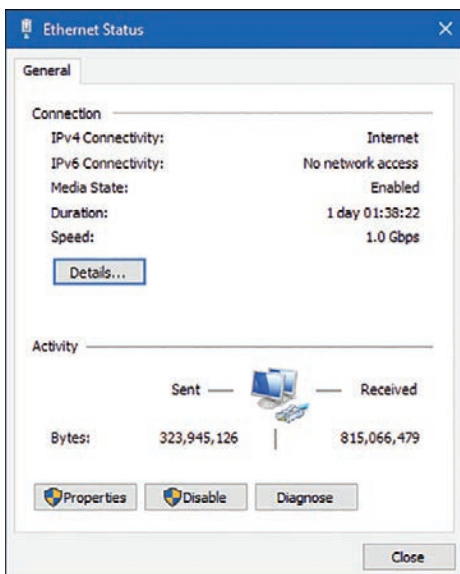


Figure 7-8 Local Area Connection Status Window

As shown in Figure 7-9, you can highlight **Internet Protocol Version 4 (TCP/IPv4)** and click the **Properties** button.

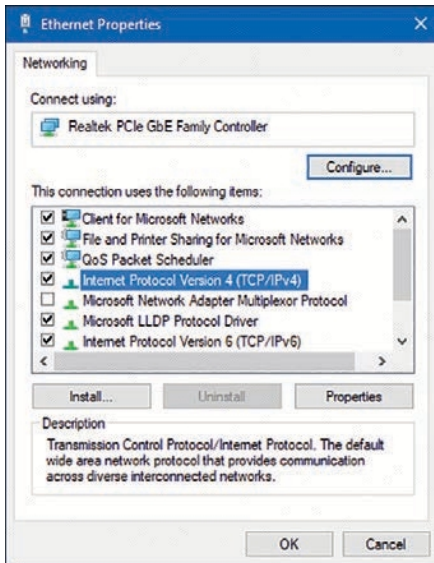


Figure 7-9 Local Area Connection Properties

You can enter an IP address, a subnet mask, a default gateway, and DNS server information into the Internet Protocol Version 4 (TCP/IPv4) Properties window, shown in Figure 7-10. Although DNS server information can be entered in this window, more advanced DNS options are available by clicking the **Advanced** button.

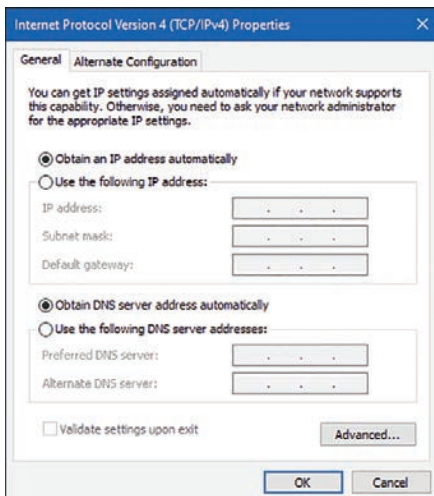


Figure 7-10 Internet Protocol Version 4 (TCP/IPv4) Properties

By clicking the **DNS** tab in the Advanced TCP/IP Settings window, as shown in Figure 7-11, you can add, remove, or reorder DNS servers, and you can adjust various other DNS parameters. Recall that a DNS server converts an FQDN to an IP address.

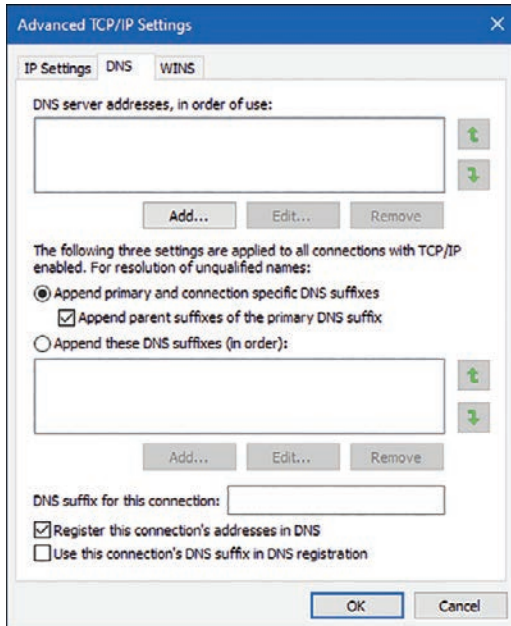


Figure 7-11 Advanced TCP/IP Settings: DNS Tab

Dynamic Configuration

Statically assigning IP address information to individual networked devices can be time-consuming, error-prone, and lacking in scalability. Instead of using static IP address assignments, many corporate networks dynamically assign IP address parameters to their devices. An early choice for performing this automatic assignment of IP addresses was the *Bootstrap Protocol (BOOTP)*. Currently, however, the most popular approach for dynamic IP address assignment is *Dynamic Host Configuration Protocol (DHCP)*. We cover DHCP in detail for you in Chapter 16, “IPv4 and IPv6 Network Services.”

Automatic Private IP Addressing

If a networked device does not have a statically configured IP address and is unable to contact a DHCP server, it still might be able to communicate on an IP network

thanks to *Automatic Private IP Addressing (APIPA)*. The APIPA feature allows a networked device to self-assign an IP address from the 169.254.0.0/16 network. Note that this address is usable only on the device's local subnet. (The IP address is not routable.)

As shown in Figure 7-12, Microsoft Windows 11 defaults to APIPA if a client is configured to automatically obtain IP address information and that client fails to obtain IP address information from a DHCP server.

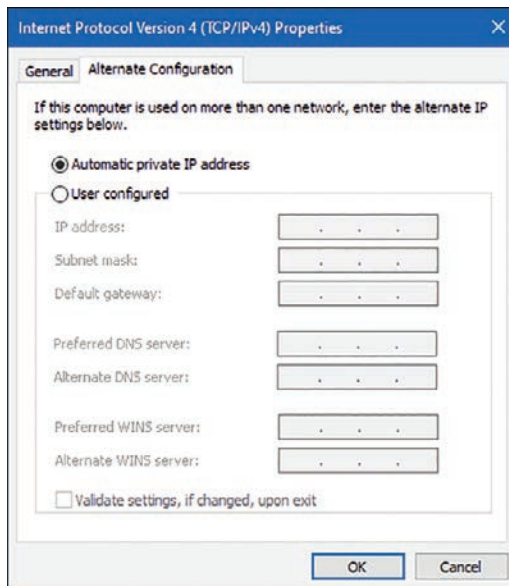


Figure 7-12 APIPA Configuration Enabled by Default

Subnetting

Earlier in this chapter, you were introduced to the purpose of a subnet mask and the default subnet masks for the various IP address classes. Default subnet masks (that is, classful subnet masks) are not always the most efficient choice. Fortunately, you can add additional network bits to a subnet mask (thereby extending the subnet mask) to create subnets within a classful network. We refer to this process as *subnetting* and this section explains why you might want to perform this process and describes how you mathematically perform subnet calculations.

Purpose of Subnetting

Consider the number of assignable IP addresses in the various classes of IP addresses shown in Table 7-21. Recall that the host bits of an IP address cannot be all 0s (which represents the network address) or all 1s (which represents the directed broadcast address). Therefore, the number of assignable IP addresses in a subnet can be determined by the following formula:

Number of assignable IP addresses in a subnet = $2^b - 2$

where *b* is the number of host bits in a subnet mask.

Key
Topic

Table 7-21 Assignable IP Addresses

Address Class	Assignable IP Addresses
Class A	16,777,214 ($2^{24} - 2$)
Class B	65,534 ($2^{16} - 2$)
Class C	254 ($2^8 - 2$)

Suppose that you decide to use a private Class B IP address (for example, 172.16.0.0/16) for your internal IP addressing. For performance reasons, you would not want to support as many as 65,534 hosts in a single broadcast domain. Therefore, a best practice with such a network address is to subnet the network (thereby extending the number of network bits in the network’s subnet mask) into additional subnetworks. In fact, you could subnet your major network address space and then further subnet one of your unused subnet addresses! This practice, known as *Variable-Length Subnet Masking (VLSM)*, enables you to design the network the best way possible in terms of the number of IP addresses required in different areas. Of course, this network also uses a variety of subnet masks to accomplish this task.

Subnet Mask Notation

As previously mentioned, the number of bits in a subnet mask can be represented in dotted-decimal notation (for example, 255.255.255.0) or in prefix notation (for example, /24). As a reference, Table 7-22 shows valid subnet masks in dotted-decimal notation and the corresponding prefix notation.

Key
Topic

Table 7-22 Dotted-Decimal and Prefix-Notation Representations for IPv4 Subnets

Dotted-Decimal Notation	Prefix Notation
255.0.0.0	/8 (classful subnet mask for Class A networks)
255.128.0.0	/9

Dotted-Decimal Notation	Prefix Notation
255.192.0.0	/10
255.224.0.0	/11
255.240.0.0	/12
255.248.0.0	/13
255.252.0.0	/14
255.254.0.0	/15
255.255.0.0	/16 (classful subnet mask for Class B networks)
255.255.128.0	/17
255.255.192.0	/18
255.255.224.0	/19
255.255.240.0	/20
255.255.248.0	/21
255.255.252.0	/22
255.255.254.0	/23
255.255.255.0	/24 (classful subnet mask for Class C networks)
255.255.255.128	/25
255.255.255.192	/26
255.255.255.224	/27
255.255.255.240	/28
255.255.255.248	/29
255.255.255.252	/30

Recall that any octet with a value of 255 contains eight 1s. Also, you should memorize valid octet values for an octet and the corresponding number of 1s (that is, continuous, left-justified 1s) in that octet, as shown in Table 7-23. Based on this information, you should be able to see the dotted-decimal notation of a subnet mask and quickly determine the corresponding prefix notation.

**Key
Topic**
Table 7-23 Subnet Octet Values

Subnet Octet Value	Number of Contiguous Left-Justified 1s
0	0
128	1
192	2

Subnet Octet Value	Number of Contiguous Left-Justified 1s
224	3
240	4
248	5
252	6
254	7
255	8

For example, consider the subnet mask 255.255.192.0. Because each of the first two octets has a value of 255, you know that you have sixteen 1s from the first two octets. You then recall that a value of 192 in the third octet requires two 1s from that octet. By adding the sixteen 1s from the first two octets to the two 1s from the third octet, you can determine that the subnet mask 255.255.192.0 has the corresponding prefix notation /18.

To help you develop the skill of making these calculations quickly, work through the following two exercises.

Subnet Notation: Practice Exercise 1

Given the subnet mask 255.255.255.248, what is the corresponding prefix notation?

Subnet Notation: Practice Exercise 1 Solution

Given the subnet mask 255.255.255.248, you should recognize that the first three octets, each containing a value of 255, represent twenty-four 1s. To those twenty-four 1s, you add five additional 1s, based on your memorization of how many contiguous, left-justified 1s in an octet are required to produce various octet values. The sum of 24 bits (from the first three octets) and the 5 bits (from the fourth octet) gives you a total of 29 bits. Therefore, you can conclude that a subnet mask with dotted-decimal notation 255.255.255.248 has equivalent prefix notation /29.

Subnet Notation: Practice Exercise 2

Given the subnet mask /17, what is the corresponding dotted-decimal notation?

Subnet Notation: Practice Exercise 2 Solution

You know that each octet contains 8 bits. So, given the subnet mask /17, you can count by 8s to determine that there are eight 1s in the first octet, eight 1s in the second octet, and one 1 in the third octet. You already knew that an octet containing all 1s has the decimal value 255. From that knowledge, you conclude that each of the first two octets has the value 255. Also, based on your memorization of Table 7-23, you know that one 1 (that is, a left-justified 1) in an octet has the decimal equivalent value 128. Therefore, you can conclude that a subnet mask with prefix notation /17 can be represented in dotted-decimal notation as 255.255.128.0.

Extending a Classful Mask

The way to take a classful network (that is, a network using a classful subnet mask) and divide that network into multiple subnets is by adding 1s to the network's classful subnet mask. However, the class of the IP address does not change, regardless of the new subnet mask. For example, if you took the 172.16.0.0/16 network and subnetted it into multiple networks using a 24-bit subnet mask (172.16.0.0/24, 172.16.1.0/24, 172.16.2.0/24, ...), those networks would still be Class B networks.

Specifically, the class of a network is entirely determined by the value of the first octet. The class of a network has nothing to do with the number of bits in a subnet, which makes this an often-misunderstood concept. For example, the network 10.2.3.0/24 has the subnet mask of a Class C network (that is, a 24-bit subnet mask). However, the 10.2.3.0/24 network is a Class A network because the value of the first octet is 10. It is simply a Class A network that happens to have a 24-bit subnet mask.

Borrowed Bits

When you add bits to a classful mask, the bits you add are referred to as *borrowed bits*. The number of borrowed bits you use determines how many subnets are created and the number of usable hosts per subnet.

Calculating the Number of Created Subnets

To determine the number of subnets created when adding bits to a classful mask, you can use the following formula:

Number of created subnets = 2^s

where s is the number of borrowed bits.



For example, let's say you subnetted the 192.168.1.0 network with a 28-bit subnet mask, and you want to determine how many subnets were created. First, you determine how many borrowed bits you have. Recall that the number of borrowed bits is the number of bits in a subnet mask beyond the classful mask. In this case, because the first octet in the network address has the value 192, you can conclude that this is a Class C network. Also recall that a Class C network has 24 bits in its classful (that is, its default) subnet mask. Because you now have a 28-bit subnet mask, the number of borrowed bits can be calculated as follows:

Key Topic

Number of borrowed bits = Bits in custom subnet mask – Bits in classful subnet mask

$$\text{Number of borrowed bits} = 28 - 24 = 4$$

Now that you know you have 4 borrowed bits, you can raise 2 to the power of 4 (2^4 , or $2 \times 2 \times 2 \times 2$), which equals 16. From this calculation, you conclude that subnetting 192.168.1.0/24 with a 28-bit subnet mask yields 16 subnets.

Calculating the Number of Available Hosts

Earlier in this section, you saw the formula for calculating the number of available (that is, assignable) host IP addresses, based on the number of host bits in a subnet mask. Here again is the formula:

Key Topic

$$\text{Number of assignable IP addresses in a subnet} = 2^b - 2$$

where b is the number of host bits in the subnet mask.

Using the previous example, let's say you want to determine the number of available host IP addresses in one of the 192.168.1.0/28 subnets. First, you need to determine the number of host bits in the subnet mask. Because you know that an IPv4 address consists of 32 bits, you can subtract the number of bits in the subnet mask (28, in this example) from 32 to determine the number of host bits:

$$\text{Number of host bits} = 32 - \text{Number of bits in subnet mask}$$

$$\text{Number of host bits} = 32 - 28 = 4$$

Now that you know the number of host bits, you can apply it to the previously presented formula:

Key Topic

$$\text{Number of assignable IP addresses in a subnet} = 2^b - 2$$

where b is the number of host bits in the subnet mask.

$$\text{Number of assignable IP addresses in a subnet} = 2^4 - 2 = 14$$

From this calculation, you can conclude that each of the 192.168.1.0/28 subnets has 14 usable IP addresses.

To reinforce your skill with these calculations, you are now challenged with a few practice exercises.

Basic Subnetting Practice: Exercise 1

Using a separate sheet of paper, solve the following scenario:

Your company has been assigned the 172.20.0.0/16 network for use at one of its sites. You need to use a subnet mask that will accommodate 47 subnets while simultaneously accommodating the maximum number of hosts per subnet. What subnet mask will you use?

Basic Subnetting Practice: Exercise 1 Solution

To determine how many borrowed bits are required to accommodate 47 subnets, you can write out a table that lists the powers of 2, as shown in Table 7-24. In fact, you might want to sketch out a similar table on the dry-erase card (or electronic notepad) you are given when you take the CompTIA Network+ exam.

Key Topic

Table 7-24 Number of Subnets Created by a Specified Number of Borrowed Bits

Borrowed Bits	Number of Subnets Created (2^s , Where s Is the Number of Borrowed Bits)
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024
11	2048
12	4096

In this example, where you want to support 47 subnets, 5 borrowed bits are not enough, and 6 borrowed bits are more than enough. Because 5 borrowed bits are not enough, you round up and use 6 borrowed bits.

The first octet in the network address 172.20.0.0 has the value 172, which means you are dealing with a Class B address. Because a Class B address has 16 bits in its classful mask, you can add the 6 borrowed bits to the 16-bit classful mask, which results in a 22-bit subnet mask.

One might argue that although a 22-bit subnet mask would accommodate 47 subnets, so would a 23-bit subnet mask or a 24-bit subnet mask. Although that is true, recall that the scenario said you should have the maximum number of hosts per subnet. This suggests that you should not use more borrowed bits than necessary. Therefore, you can conclude that to meet the scenario’s requirements, you should use a subnet mask of /22, which could also be written as 255.255.252.0.

Basic Subnetting Practice: Exercise 2

Using a separate sheet of paper, solve the following scenario:

Your company has been assigned the 172.20.0.0/16 network for use at one of its sites. You need to calculate a subnet mask that will accommodate 100 hosts per subnet while maximizing the number of available subnets. What subnet mask will you use?

Basic Subnetting Practice: Exercise 2 Solution

To determine how many host bits are required to accommodate 100 hosts, you can write out a table that shows the number of hosts supported by a specific number of hosts bits, as shown in Table 7-25. As with the previous example, you might want to sketch out a similar table on the dry-erase card (or electronic notepad) you are given when taking the Network+ exam.



Table 7-25 Number of Supported Hosts, Given a Specified Number of Host Bits

Host Bits	Number of Supported Hosts (2 ^h – 2, Where <i>h</i> Is the Number of Host Bits)
2	2
3	6
4	14
5	30
6	62

Host Bits	Number of Supported Hosts ($2^h - 2$, Where h Is the Number of Host Bits)
7	126
8	254
9	510
10	1022
11	2046
12	4094

In this example, where you want to support 100 hosts, 6 host bits are not enough, and 7 host bits are more than enough. Because 6 host bits are not enough, you round up and use 7 host bits.

Because an IPv4 address has 32 bits and you need 7 host bits, you can calculate the number of subnet bits by subtracting the 7 host bits from 32 (that is, the total number of bits in an IPv4 address). This results in a 25-bit subnet mask (that is, 32 total bits – 7 host bits = 25 subnet mask bits). Therefore, you can conclude that to meet the scenario's requirements, you should use a subnet mask of /25, which could also be written as 255.255.255.128.

Calculating New IP Address Ranges

Now that you can calculate the number of subnets created based on a given number of borrowed bits, the next logical step is to calculate the IP address ranges making up those subnets. For example, if you subnetted 172.25.0.0/16 with a 24-bit subnet mask, the resulting subnets would be as follows:

172.25.0.0/24

172.25.1.0/24

172.25.2.0/24

...

172.25.255.0/24

Let's consider how such a calculation is performed. Notice in the previous example that you count by 1 in the third octet to calculate the new networks. To decide in what octet you start counting and by what increment you count, a new term needs to be defined. The *interesting octet* is the octet that contains the last 1 in the subnet mask.

In this example, the subnet mask is a 24-bit subnet mask, which has the dotted-decimal equivalent 255.255.255.0 and the binary equivalent 11111111.11111111.11111111.00000000. From any of these subnet mask representations, you can determine that the third octet is the octet to contain the last 1 in the subnet mask. Therefore, you will be changing the value of the third octet to calculate the new networks.

Now that you know the third octet is the interesting octet, you need to know by what increment you will be counting in that octet. This increment is known as the *block size*, and you can calculate it by subtracting the subnet mask value in the interesting octet from 256. In this example, the subnet mask has the value 255 in the interesting octet (that is, the third octet). If you subtract 255 from 256, you get the result 1 (that is, $256 - 255 = 1$). The first subnet is the original network address, with all of the borrowed bits set to 0. After this first subnet, you start counting by the block size (1 in this example) in the interesting octet to calculate the remainder of the subnets.

The process just described for calculating subnets can be summarized as follows:

Key Topic

- Step 1.** Determine the interesting octet by determining the last octet in the subnet mask to contain a 1.
- Step 2.** Determine the block size by subtracting the decimal value in the subnet's interesting octet from 256.
- Step 3.** Determine the first subnet by setting all the borrowed bits (which are bits in the subnet mask beyond the bits in the classful subnet mask) to 0.
- Step 4.** Determine additional subnets by taking the first subnet and counting by the block size increment in the interesting octet.

To reinforce this procedure, consider another example. Say that a 27-bit subnet mask is applied to the network address 192.168.10.0/24. To calculate the created subnets, you can perform the following steps:

- Step 1.** The subnet mask /27 (in binary) is 11111111.11111111.11111111.11100000. The interesting octet is the fourth octet because the fourth octet contains the last 1 in the subnet mask.
- Step 2.** The decimal value of the fourth octet in the subnet mask is 224 (11100000 in decimal). Therefore, the block size is 32 ($256 - 224 = 32$).
- Step 3.** The first subnet is 192.168.10.0/27—the value of the original 192.168.10.0 network with the borrowed bits (the first 3 bits in the fourth octet) set to 0.

Step 4. Counting by 32 (the block size) in the interesting octet (the fourth octet) allows you to calculate the remaining subnets:

192.168.10.0
 192.168.10.32
 192.168.10.64
 192.168.10.96
 192.168.10.128
 192.168.10.160
 192.168.10.192
 192.168.10.224

Now that you know the subnets created from a classful network given a subnet mask, the next logical step is to determine the usable addresses within those subnets. Recall that you cannot assign an IP address to a device if all the host bits in the IP address are set to 0 because an IP address with all host bits set to 0 is the address of the subnet itself. Similarly, you cannot assign an IP address to a device if all the host bits in the IP address are set to 1 because an IP address with all host bits set to 1 is the directed broadcast address of a subnet.

By excluding the network and directed broadcast addresses from the 192.168.10.0/27 subnets (as previously calculated), you can determine the usable addresses shown in Table 7-26.



Table 7-26 Usable IP Address Ranges for the 192.168.10.0/27 Subnets

Subnet Address	Directed Broadcast Address	Usable IP Addresses
192.168.10.0	192.168.10.31	192.168.10.1–192.168.10.30
192.168.10.32	192.168.10.63	192.168.10.33–192.168.10.62
192.168.10.64	192.168.10.95	192.168.10.65–192.168.10.94
192.168.10.96	192.168.10.127	192.168.10.97–192.168.10.126
192.168.10.128	192.168.10.159	192.168.10.129–192.168.10.158
192.168.10.160	192.168.10.191	192.168.10.161–192.168.10.190
192.168.10.192	192.168.10.223	192.168.10.193–192.168.10.222
192.168.10.224	192.168.10.255	192.168.10.225–192.168.10.254

To help develop your subnet-calculation skills, you are now challenged with a few practice subnetting exercises.

Advanced Subnetting Practice: Exercise 1

Using a separate sheet of paper, solve the following scenario:

Based on your network design requirements, you determine that you should use a 26-bit subnet mask applied to your 192.168.0.0/24 network. You now need to calculate each of the created subnets. In addition, you want to know the broadcast address and the range of usable addresses for each of the created subnets.

Advanced Subnetting Practice: Exercise 1 Solution

As described earlier, you can go through the following four-step process to determine the subnet address:

- Step 1.** The subnet mask /26 (in binary) is 11111111.11111111.11111111.11000000. The interesting octet is the fourth octet because the fourth octet contains the last 1 in the subnet mask.
- Step 2.** The decimal value of the fourth octet in the subnet mask is 192 (11000000 in decimal). Therefore, the block size is 64 ($256 - 192 = 64$).
- Step 3.** The first subnet is 192.168.0.0/26—the value of the original 192.168.0.0 network with the borrowed bits (the first 2 bits in the last octet) set to 0.
- Step 4.** Counting by 64 (the block size) in the interesting octet (the fourth octet) allows you to calculate the remaining subnets, resulting in the following subnets:

192.168.0.0

192.168.0.64

192.168.0.128

192.168.0.192

The directed broadcast addresses for each of these preceding subnets can be calculated by adding 63 (that is, one less than the block size) to the interesting octet for each subnet address. Excluding the subnet addresses and directed broadcast addresses, you can calculate a range of usable addresses, the results of which are shown in Table 7-27.

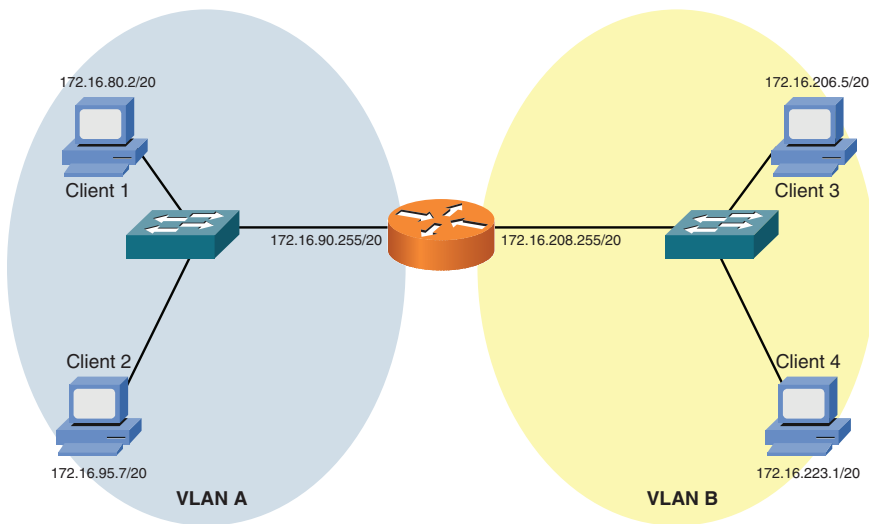
Table 7-27 Usable IP Address Ranges for the 192.168.0.0/26 Subnets

Subnet Address	Directed Broadcast Address	Usable IP Addresses
192.168.0.0	192.168.0.63	192.168.0.1–192.168.0.62
192.168.0.64	192.168.0.127	192.168.0.65–192.168.0.126
192.168.0.128	192.168.0.191	192.168.0.129–192.168.0.190
192.168.0.192	192.168.0.255	192.168.0.193–192.168.0.254

Advanced Subnetting Practice: Exercise 2

Using a separate sheet of paper, solve the following scenario:

In the network shown in Figure 7-13, the 172.16.0.0/16 network is subnetted using a 20-bit subnet mask. Notice that two VLANs (two subnets) are configured; however, one of the client PCs is assigned an IP address that is not in that PC's VLAN. Which client PC is assigned an incorrect IP address?

**Figure 7-13** Topology for Advanced Subnetting Practice: Exercise 2

Advanced Subnetting Practice: Exercise 2 Solution

To determine which client PC is assigned an IP address outside its local VLAN, you need to determine the subnets created by the 20-bit subnet mask applied to the 172.16.0.0/16 network:

- Step 1.** The interesting octet for a 20-bit subnet mask is the third octet because the third octet is the last octet to contain a 1 in the 20-bit subnet mask

(11111111.11111111.11110000.00000000, which could also be written as 255.255.240.0).

- Step 2.** The decimal value of the third octet in the subnet mask is 240. Therefore, the block size is 16 ($256 - 240 = 16$).
- Step 3.** The first 172.16.0.0/20 subnet is 172.16.0.0 (172.16.0.0/20 with the 4 borrowed bits in the third octet set to 0).
- Step 4.** Beginning with the first subnet, 172.16.0.0/20, and counting by the block size 16 in the interesting octet yields the following subnets:

172.16.0.0/20
172.16.16.0/20
172.16.32.0/20
172.16.48.0/20
172.16.64.0/20
172.16.80.0/20
172.16.96.0/20
172.16.112.0/20
172.16.128.0/20
172.16.144.0/20
172.16.160.0/20
172.16.176.0/20
172.16.192.0/20
172.16.208.0/20
172.16.224.0/20
172.16.240.0/20

Based on the IP addresses of the router interfaces, you can figure out the subnets for VLAN A and VLAN B. Specifically, the router interface in VLAN A has the IP address 172.16.90.255/20. Based on the previous listing of subnets, you can determine that this interface resides in the 172.16.80.0/20 network, whose range of usable addresses is 172.16.80.1–172.16.95.254. Then you can examine the IP addresses of

Client 1 and Client 2 to determine whether their IP addresses reside in that range of usable addresses.

Similarly, for VLAN B, the router's interface has an IP address of 172.16.208.255/20. Based on the previous subnet listing, you notice that this interface has an IP address that is part of the 172.16.208.0/20 subnet. As you did for VLAN A, you can check the IP address of Client 3 and Client 4 to decide whether their IP addresses live in VLAN B's range of usable IP addresses (that is, 172.16.208.1–172.16.223.254).

Table 7-28 shows these comparisons.

Table 7-28 IP Address Comparison for Advanced Subnetting Practice: Exercise 2

Client	VLAN	Range of Usable Addresses	Client IP Address	Is Client in Range of Usable Addresses?
Client 1	A	172.16.80.1–172.16.95.254	172.16.80.2	Yes
Client 2	A	172.16.80.1–172.16.95.254	172.16.95.7	Yes
Client 3	B	172.16.208.1–172.16.223.254	172.16.206.5	No
Client 4	B	172.16.208.1–172.16.223.254	172.16.223.1	Yes

The comparison in Table 7-28 reveals that Client 3 (with IP address 172.16.206.5) does not have an IP address in VLAN B's subnet (with the usable address range 172.16.208.1–172.16.223.254).

Additional Practice

If you want to continue practicing these concepts, make up your own subnet mask and apply it to a classful network of your choosing. Then you can calculate the created subnets, the directed broadcast IP address for each subnet, and the range of usable IP addresses for each subnet.

To check your work, you can use a subnet calculator. An example of such a calculator is the free Advanced Subnet Calculator, available for download from <https://www.solarwinds.com/free-tools/advanced-subnet-calculator>, as shown in Figure 7-14.

NOTE As you read through different networking literature, you might come across other approaches to performing subnetting. Various shortcuts exist (including the one presented in this chapter), and some approaches involve much more binary math. The purpose of this section is not to provide an exhaustive treatment of all available subnetting methods but to show a quick and easy approach to performing subnet calculations in the real world and for the Network+ certification exam.

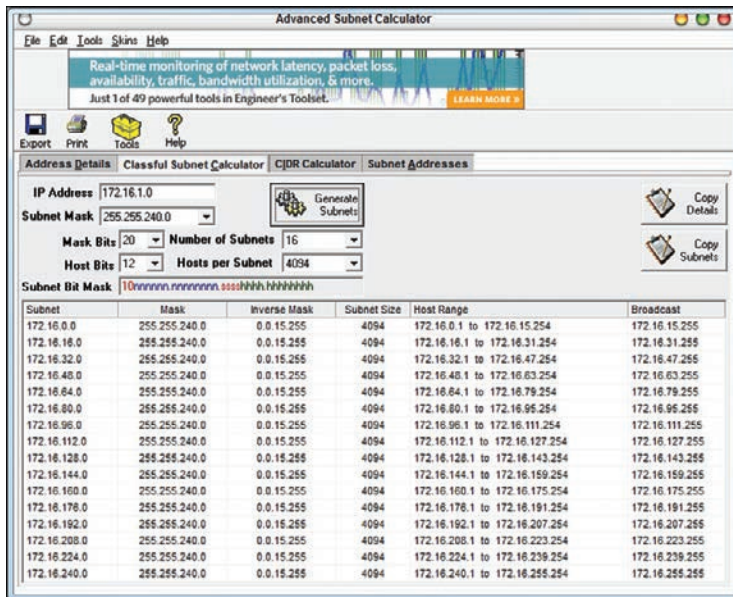


Figure 7-14 Free Advanced Subnet Calculator

Classless Inter-domain Routing

Whereas subnetting is the process of extending a classful subnet mask (that is, adding 1s to a classful mask), **classless inter-domain routing (CIDR)** does just the opposite. Specifically, CIDR shortens a classful subnet mask by removing 1s from the classful mask. As a result, CIDR allows contiguous classful networks to be aggregated. This process is sometimes called *route aggregation*.

A typical use of CIDR is by a service provider summarizing multiple Class C networks that are assigned to the provider's various customers. For example, imagine that a service provider is responsible for advertising the following Class C networks:

192.168.32.0/24

192.168.33.0/24

192.168.34.0/24

192.168.35.0/24

The service provider could advertise all four networks with the single route advertisement 192.168.32.0/22. To calculate this advertisement, convert the values in the third octet (that is, the octet where the values start to differ) to binary, as shown in

Figure 7-15. Then determine how many bits the networks have in common. The number of common bits then becomes the number of bits in the CIDR mask.

Key Topic

Network Address	1 st Octet	2 nd Octet	3 rd Octet	4 th Octet
192.168.32.0	11000000	10101000	001000 00	00000000
192.168.33.0	11000000	10101000	001000 01	00000000
192.168.34.0	11000000	10101000	001000 10	00000000
192.168.35.0	11000000	10101000	001000 11	00000000

All Networks Have 22 Bits in Common

Figure 7-15 CIDR Calculation Example

Because all four of the network addresses have the first 22 bits in common, and because setting the remaining bits to 0 (11000000.10101000.00100000.00000000) creates the network address 192.168.32.0, these networks can be summarized as 192.168.32.0/22.

Real-World Case Study

Acme, Inc. has decided to use private IP addresses for its internal LAN and for the WAN. The company will use the private block 10.0.0.0/8 and create enough subnets to cover the number of virtual local area networks (VLANs) it will be using at the headquarters site and at each of the remote offices. The association between the Layer 2 VLANs and the Layer 3 IP subnets will be one-to-one, with each VLAN having its own associated subnet.

The company will have nine VLANs and will use a couple subnets for the WAN connections. For the VLANs, the company plans to use the network mask /12, which will offer enough subnets to meet its needs based on the starting mask /8 for the Class A private address 10.0.0.0/8.

For the WAN connectivity Acme is purchasing from a service provider for connectivity between the remote branch offices and the headquarters site, the company will use /30 masks, which will allow for two hosts on each of the WAN connections. This is enough for each device at the end of the point-to-point WAN connections.

To connect its LANs to the Internet, Acme plans to use NAT, which is going to be performed by its service provider so that traffic going to the Internet will appear to be coming from a globally routable IP address and not from a private address. (You'll learn more about NAT in Chapter 9.)

Summary

Here are the main topics covered in this chapter:

- A binary math tutorial was provided in this chapter to give you a basic understanding of why binary math is necessary for working with subnet masks.
- The characteristics of IPv4 were presented in this chapter, including IPv4's address format.
- This chapter examined various approaches for assigning IP address information to network devices, including static assignment, dynamic assignment, and APIPA.
- Finally, this chapter provided multiple examples and practice exercises for various subnet calculations.

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 7-29 lists these key topics and the page number where each is found.



Table 7-29 Key Topics for Chapter 7

Key Topic Element	Description	Page Number
Table 7-1	Binary Conversion Table	164
Section	Converting a Binary Number to a Decimal Number	165
Section	Converting a Decimal Number to a Binary Number	165
Figure 7-2	Dividing an IP Address into a Network Portion and a Host Portion	172
Table 7-19	IP Address Classes	173
Table 7-20	Private IP Networks	174
Table 7-21	Assignable IP Addresses	182
Table 7-22	Dotted-Decimal and Prefix-Notation Representations for IPv4 Subnets	182
Table 7-23	Subnet Octet Values	183
Formula	Number of created subnets	185
Formula	Number of borrowed bits	186
Formula	Number of host bits	186
Formula	Number of assignable IP addresses in a subnet	186

Key Topic Element	Description	Page Number
Table 7-24	Number of Subnets Created by a Specified Number of Borrowed Bits	187
Table 7-25	Number of Supported Hosts, Given a Specified Number of Host Bits	188
Step list	Steps for calculating subnets	190
Table 7-26	Usable IP Address Ranges for the 192.168.10.0/27 Subnets	191
Figure 7-15	CIDR Calculation Example	197

Complete Tables and Lists from Memory

Print a copy of Appendix C, “Memory Tables,” or at least the section for this chapter and complete as many of the tables as possible from memory. Appendix D, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

Automatic Private IP Addressing (APIPA), Class A, Class B, Class C, Class D, Class E, classless inter-domain routing (CIDR), IPv4 address classes, localhost, loopback, private IP addressing, RFC1918, subnetting, Variable-Length Subnet Masking (VLSM)

Additional Resources

Subnetting—Hosts per Subnet: <https://www.ajsnetworking.com/subnetting>

Subnetting—What Mask to Use: <https://www.ajsnetworking.com/subnetting2>

Subnetting—“I Feel the Need, the Need for Speed”: <https://www.ajsnetworking.com/subnetting3>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

- What is the binary representation of the decimal number 117?
 - 10110101
 - 01110101

- c. 10110110
 - d. 01101001
- 2. What is the decimal equivalent of the binary number 10110100?
 - a. 114
 - b. 190
 - c. 172
 - d. 180
- 3. What is the class of IP address 10.1.2.3/24?
 - a. Class A
 - b. Class B
 - c. Class C
 - d. Class D
- 4. What is the classful subnet mask for a Class B network?
 - a. 255.255.0.0
 - b. 255.255.255.255
 - c. 255.255.255.0
 - d. 255.0.0.0
- 5. Which of the following is a dynamic approach to assigning routable IP addresses to networked devices?
 - a. CIDR
 - b. APIPA
 - c. RFC1918
 - d. DHCP
- 6. How many assignable IP addresses exist in the 172.16.1.10/27 network?
 - a. 30
 - b. 32
 - c. 14
 - d. 64

7. What is the prefix notation for the subnet mask 255.255.255.240?
 - a. /20
 - b. /24
 - c. /28
 - d. /29
8. Your company has been assigned the 192.168.30.0/24 network for use at one of its sites. You need to use a subnet mask that will accommodate seven subnets while simultaneously accommodating the maximum number of hosts per subnet. What subnet mask should you use?
 - a. /24
 - b. /26
 - c. /27
 - d. /28
9. A client with IP address 172.16.18.5/18 belongs to what network?
 - a. 172.16.0.0/18
 - b. 172.16.64.0/18
 - c. 172.16.96.0/18
 - d. 172.16.128.0/18
10. What effect does classless inter-domain routing have on the subnet mask in IP addressing?
 - a. The mask is not changed
 - b. The mask is lengthened
 - c. The mask is shortened
 - d. The mask is eliminated
11. Which of the following is not a private address range as defined in RFC1918?
 - a. 127.x.x.x
 - b. 10.x.x.x
 - c. 172.16.x.x to 172.31.x.x
 - d. 192.168.x.x



This chapter covers the following topics related to Objective 1.8 (Summarize evolving use cases for modern network environments) of the CompTIA Network+ N10-009 certification exam:

- Software-defined network (SDN) and software-defined wide area network (SD-WAN)
 - Application aware
 - Zero-touch provisioning
 - Transport agnostic
 - Central policy management
- Virtual Extensible Local Area Network (VXLAN)
 - Data center interconnect (DCI)
 - Layer 2 encapsulation
- Zero trust architecture (ZTA)
 - Policy-based authentication
 - Authorization
 - Least privilege access
- Secure Access Secure Edge (SASE)/Security Service Edge (SSE)
- Infrastructure as code (IaC)
 - Automation
 - Playbooks/templates/reusable tasks
 - Configuration drift/compliance
 - Upgrades
 - Dynamic inventories
- Source control
 - Version control
 - Central repository
 - Conflict identification
 - Branching
- IPv6 addressing
 - Mitigating address exhaustion
 - Compatibility requirements
 - Tunneling
 - Dual stack
 - NAT64

Evolving Use Cases

Are you excited to learn about some of the newer technologies featured in this version of the CompTIA Network+ exam? As the title of this chapter suggests, here we are going to focus on some of the latest evolving technologies taking the networking world by storm (and no, not a broadcast storm).

This chapter begins with a look at software-defined networking (SDN) and a very specific implementation called the software-defined wide area network (SD-WAN). As you will learn, these technologies make it much simpler to operate modern networks with all their sophisticated capabilities and features.

Next, this chapter explores the latest evolution in virtual local area networks. It is called virtual extensible local area network, or VXLAN. As you will learn, VXLAN is a network virtualization technology that encapsulates Ethernet frames in UDP packets to create a scalable Layer 2 overlay network across Layer 3 infrastructures. It enables the extension of VLANs beyond traditional boundaries, supporting large-scale cloud and data center environments by allowing for more flexible and dynamic network segmentation.

As one would guess, another important area of evolving technologies for networking is in the space of security. In this part of the chapter, we examine some of the latest advancements in network security. These include zero trust architecture (ZTA), Secure Access Secure Edge (SASE), and Security Service Edge (SSE).

Next, this chapter describes many aspects and benefits of infrastructure as code (IaC). Infrastructure as code is a method of managing and provisioning computing infrastructure through machine-readable configuration files, enabling automation and consistency across environments. Treating infrastructure configurations as code allows for version control, collaborative development, and efficient scaling of IT resources.

Although IPv4 is the most widely deployed Layer 3 addressing scheme in today's networks, its scalability limitations are causing available IPv4 addresses to quickly become depleted. Fortunately, a newer version of IP, IPv6, is scalable beyond anything you will need in your lifetime. This chapter concludes by introducing you to the fundamental characteristics of IPv6 addressing.

Foundation Topics

SDN and SD-WAN

Never before in my decades of studying and teaching computer networking have I seen more fear from students regarding the elimination of their jobs due to automation and cutting-edge technologies. In this author's opinion, artificial intelligence (AI) and computers are not going to be eliminating the need for you (a human) in the network any time soon. While *software-defined networking (SDN)* allows you to add more and more automation and orchestration to a network, there will still be a need for you and your skills.

Key Topic

Software-Defined Networking (SDN)

Software-defined networking, which has been around for a very long time, is making a huge resurgence and being implemented in many parts of large and small networks today. For example, consider your wireless LAN. Perhaps you are using lightweight access points and wireless LAN controllers (WLCs). If so, you are seeing a very strict separation of the data, management, and control planes. The WLC is the primary control plane intelligence of the solution. (The specific SDN planes of operation are covered in more detail later in this chapter.)

SDN is changing the landscape of traditional networks. A well-implemented software-defined network allows the administrator to implement features, functions, and configurations without the need to do command-line configuration on the individual network devices. The front end that the administrator interfaces with can alert the administrator to what the network is currently doing, and then, through that same graphical user interface, the administrator can indicate what he or she wants done; behind the scenes, the software-defined network implements the detailed configurations across multiple network devices.

A key component in most software-defined networking solutions is an SDN controller. This appliance-based device is responsible for distributing control plane instructions to network devices downstream for configuration and management.

While many different approaches can be taken to SDN, almost everyone agrees that the best strategy is to separate the network into different discrete planes or layers of operation:

- **Application plane:** This is where all the technology that involves the applications resides. Today, it is not uncommon for an application to be powered by tiny microservices running as containers in a heavily virtualized cloud

environment. But of course, there are plenty of other options for powering this layer. Many of them can even be much more traditional.

- **Control plane:** Although this layer of operation is often described as the “brains” of the operation, you are still the true brains of the operation. In fact, you are likely to use a “single pane of glass” solution that provides the correct application programming interface (API) calls to the controller. The controller turns these API commands into calls to the network devices in order to monitor or configure them properly. The API calls from you to the controller are referred to as *northbound* operations, and the commands from the controller to the network devices are referred to as *southbound operations*. The controller is always considered to be in the middle. Examples of control layer functions include routing and switching intelligence, and common control layer protocols include Open Shortest Path First (OSPF), Border Gateway Protocol (BGP), and Rapid Spanning Tree Protocol (RSTP).
- **Data plane:** The data plane (sometimes called the infrastructure plane) contains the hardware and software that power the enterprise. In it, you often find legacy and dated technologies. This infrastructure is now being controlled in a new and exciting way.
- **Management plane:** It is necessary to perform a lot of routine maintenance in a network, and the management plane is for these “boring” tasks. The management plane allows administrators to see their devices and traffic flows and react as needed to manage data plane behavior. This can be done automatically through configuration apps that can, for example, add more bandwidth if it looks as if edge components are getting congested. Note that the management plane manages and monitors processes across all layers of the network stack.

All the layers of operation are critically important, and each plays an important role. The layers of operation work seamlessly together as one to get the various jobs done. Figure 8-1 shows these commonly defined planes of operations with software-defined networking.

Software-Defined Wide Area Network (SD-WAN)

For many years, new technologies and improvements have been made in local area networks (LANs). Sadly, there were not many innovations in a very important part of the network—the wide area network (WAN). Thanks to software-defined networking improvements, we now have a celebrated and popular new improvement called the *software-defined wide area network (SD-WAN)*.

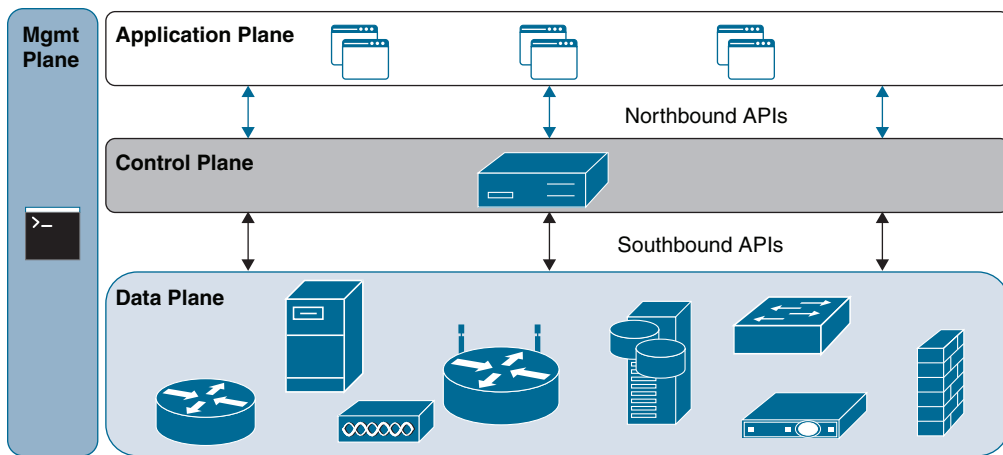


Figure 8-1 Software-Defined Networking

The SD-WAN is a transformative approach to managing and optimizing wide area networks. Unlike traditional WANs, which rely on proprietary hardware and inflexible connectivity options, SD-WAN utilizes software-defined networking principles to create a more adaptable and efficient network infrastructure. SD-WAN abstracts the network layer from the hardware, enabling centralized control and dynamic management of network traffic across multiple connection types, such as broadband, Multiprotocol Label Switching (MPLS), Long-Term Evolution (LTE), and more. This abstraction enhances performance, reduces costs, and improves overall agility, making it particularly valuable for enterprises with distributed branch locations.

One of the key features of SD-WAN is its ***application awareness***. This capability allows the network to identify and prioritize traffic based on the application, ensuring that critical applications, such as video conferencing and VoIP, receive the necessary bandwidth and low latency for optimal performance. Application awareness in SD-WAN is achieved through deep packet inspection and real-time analytics, which categorize and manage traffic flows according to predefined policies. This feature not only improves the quality of experience for end users but also enhances overall network efficiency by intelligently routing traffic based on application requirements and current network conditions.

Zero-touch provisioning (ZTP) is another significant feature of SD-WAN, simplifying the deployment and management of network devices. With ZTP, network administrators can configure and deploy new branch devices without manual intervention. This process typically involves shipping a preconfigured device to a location, where it automatically connects to the SD-WAN controller, downloads its configuration, and becomes operational with minimal human involvement. ZTP

significantly reduces deployment time and operational costs, enabling rapid scaling of the network to meet the needs of growing businesses and facilitating easier maintenance and updates.

SD-WAN is designed to be *transport agnostic*, meaning it can leverage any available connectivity option, such as broadband, MPLS, LTE, or even satellite links. This flexibility allows organizations to choose the most cost-effective and efficient connectivity for each location, without being tied to a specific provider or technology. Transport agnosticism enhances the resilience and redundancy of the network, as SD-WAN can dynamically route traffic across multiple links to maintain performance and availability, even in the event of a link failure or degradation.

Central policy management is a cornerstone of SD-WAN architecture, providing a unified platform for defining and enforcing network policies across all connected devices and locations. Through a centralized management console, administrators can easily set rules for traffic prioritization, security, and compliance, ensuring consistent policy application throughout the network. This centralized approach simplifies network management, improves security by standardizing configurations, and enables quick adjustments to network policies in response to changing business needs or threats. Central policy management also allows for real-time monitoring and analytics, providing valuable insights into network performance and usage.

Key Topic

Virtual Extensible Local Area Network (VXLAN)

Virtual Extensible Local Area Network (VXLAN) is a network virtualization technology designed to address the limitations of traditional VLANs in large-scale data center environments and the cloud. It operates by encapsulating Layer 2 Ethernet frames within Layer 3 UDP packets, enabling the extension of Layer 2 networks over a Layer 3 infrastructure. This encapsulation allows for the creation of large-scale, logical Layer 2 networks across geographically dispersed data centers, which facilitates the movement of virtual machines (VMs) and workloads without reconfiguring the underlying physical network.

At its core, VXLAN provides a way to overcome the scalability limitations of traditional VLANs, which are restricted to a maximum of 4096 segments due to the 12-bit VLAN ID field. By using a 24-bit segment identifier known as a VXLAN Network Identifier (VNI), VXLAN can support up to 16 million distinct segments. This significant increase in segmentation capacity is crucial for modern cloud environments and large enterprise data centers, where thousands of tenants and millions of isolated networks might coexist.

The key mechanism that makes VXLAN powerful is its ability to perform *Layer 2 encapsulation*. In VXLAN, a Layer 2 Ethernet frame from a VM or host is encapsulated into a Layer 3 UDP packet. This packet includes an outer IP header, which

can route across a Layer 3 network, and an outer UDP header, which facilitates the tunneling mechanism. The encapsulated packet is then transmitted over the existing Layer 3 infrastructure. This process allows for Layer 2 segments to be extended across different Layer 3 networks, creating a seamless and scalable virtual network that behaves as if all connected hosts are on the same local network. Figure 8-2 shows the Layer 2 encapsulation used with VXLAN technology.

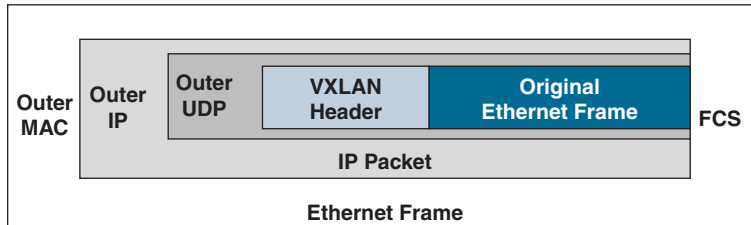


Figure 8-2 VXLAN Encapsulation

One of the primary applications of VXLAN is in *data center interconnect (DCI)*. DCI involves connecting multiple data centers to provide a unified infrastructure, allowing for efficient resource sharing, workload mobility, and disaster recovery. VXLAN is particularly suited for DCI because it enables the extension of Layer 2 networks over Layer 3 distances, thus facilitating the seamless migration of VMs and applications between data centers. This capability is crucial for businesses that need to maintain high availability and disaster resilience by distributing workloads across multiple locations.

VXLAN also integrates well with modern network management and automation tools, supporting dynamic and programmable networking. The VXLAN gateways or Virtual Tunnel Endpoints (VTEPs) play a critical role in encapsulating and decapsulating traffic and can be implemented in both hardware (switches) and software (hypervisors). This flexibility makes VXLAN an essential component in the architecture of software-defined networks (SDNs) and network functions virtualization (NFV), where it provides the necessary overlay networks that decouple virtual network management from physical network hardware.



Zero Trust Architecture (ZTA)

Zero trust architecture (ZTA) is a security model centered on the principle that no entity, whether inside or outside the network, should be trusted by default. Instead, verification is required from everyone trying to access resources within the network, ensuring robust and granular security. Unlike traditional security models that rely on a trusted internal network and a less trusted external network, ZTA treats all network traffic as untrusted, continuously validating users and devices before granting

access to sensitive data and systems. This model significantly mitigates the risk of cyber threats by ensuring that access is granted only to those who genuinely need it and are properly authenticated.

In zero trust architecture, *policy-based authentication* is a crucial component. This approach ensures that all access requests are dynamically authenticated using pre-defined security policies that consider a variety of factors. These factors include the identity of the user, the device being used, the location of the access request, and the nature of the resource being accessed. Each access attempt is subjected to rigorous authentication checks, which may include multifactor authentication (MFA) and contextual data analysis. For instance, a user attempting to access a corporate resource from an unfamiliar location or device might be required to provide additional verification to ensure they are who they claim to be. By implementing policy-based authentication, ZTA enhances security by dynamically adjusting access requirements based on the context and potential risks associated with each request.

Once authentication is successfully achieved, ZTA moves to policy-based *authorization*, which governs what authenticated users are permitted to do within the network. Authorization policies are designed to be granular and specific, ensuring that users have access only to the resources necessary for their roles and tasks. These policies are enforced in real time, continually reassessing user permissions based on their current context and behavior. For example, if a user's behavior deviates from their usual patterns, such as accessing sensitive data they don't typically handle, the system may prompt for additional verification or deny access altogether. This dynamic and context-aware approach to authorization helps prevent unauthorized access and reduces the risk of data breaches by ensuring that permissions are strictly aligned with business needs and security requirements.

Central to the ZTA model is the concept of *least privilege access*. This principle dictates that users and devices should be granted the minimum level of access required to perform their functions and no more. By limiting access rights, ZTA minimizes the potential damage that could be caused by compromised credentials or malicious actors. Implementing least privilege access involves meticulously defining user roles, responsibilities, and the associated access permissions. For instance, a financial analyst may need access to financial records but not to customer personal information, while an IT administrator might need access to system logs but not to employee payroll data. Regular reviews and adjustments of access levels are also essential to accommodate changes in roles and responsibilities, ensuring that access permissions remain tightly controlled and aligned with the principle of least privilege access.

SASE and SSE

Secure Access Secure Edge (SASE) is a transformative architectural framework designed to meet the demands of modern networking and security. It is a cloud-native service model that converges wide area networking and network security services like Secure Web Gateway (SWG), Cloud Access Security Broker (CASB), Zero Trust Network Access (ZTNA), and Firewall as a Service (FWaaS) into a single cloud-based service. This convergence allows organizations to securely connect users, devices, and applications over a global network. The SASE framework was first conceptualized by Gartner in 2019 as a response to the evolving IT landscape, where traditional network and security models were becoming increasingly insufficient for the dynamic, distributed, and cloud-centric environments.

At its core, SASE provides secure and optimized access to applications and resources regardless of the user's location. This is crucial in the current era where remote work and cloud adoption have surged, making traditional perimeter-based security models obsolete. SASE combines networking and security functionalities in a unified platform, delivered as a service from the cloud. This integration simplifies the complexity of managing multiple standalone solutions, reduces costs, and provides consistent security policies across all edges of the network, including data centers, branches, mobile users, and Internet of Things (IoT) devices.

A key aspect of SASE is its emphasis on zero trust security principles. Unlike traditional network security models that focus on defending a defined perimeter, zero trust assumes that threats can originate from both outside and inside the network. SASE implements zero trust by verifying the identity and integrity of users and devices before granting access to applications and data. This ensures that only authenticated and authorized entities can access sensitive resources, mitigating risks associated with internal and external threats. Furthermore, SASE continuously monitors and enforces security policies based on user behavior, device status, and network context to dynamically adapt to changing threat landscapes.

SASE also addresses the need for optimized network performance by integrating SD-WAN capabilities. By leveraging the global presence of SASE providers, organizations can benefit from reduced latency, improved application performance, and enhanced user experience, regardless of the user's geographical location.

SASE also supports a holistic approach to data protection and compliance. By consolidating security functions into a single framework, SASE provides comprehensive visibility and control over data flows across the network. This enables organizations to enforce data loss prevention (DLP) policies, detect and respond to threats in real time, and ensure compliance with regulatory requirements. The centralized management of security policies also simplifies the auditing process and facilitates the rapid implementation of policy changes to adapt to evolving compliance demands.

Security Service Edge (SSE) is a cloud-native cybersecurity framework that provides a comprehensive suite of security services to protect data, applications, and users in a distributed, cloud-centric environment. Introduced by Gartner as a distinct subset of the broader Secure Access Service Edge (SASE) model, SSE focuses specifically on delivering security services without encompassing the networking components such as SD-WAN. It is designed to address the challenges of modern IT architectures, where traditional perimeter-based security is inadequate for safeguarding against sophisticated cyber threats targeting a dispersed workforce and cloud-hosted resources.

SSE is particularly relevant in the context of today's hybrid and remote work environments, where employees access corporate resources from various locations and devices. Traditional security solutions that rely on a fixed perimeter are insufficient in such scenarios, as they cannot effectively protect against threats targeting remote users or cloud-hosted data. SSE addresses this challenge by extending security controls to the edge, ensuring that all users, regardless of their location, are subject to the same rigorous security policies. This approach not only improves security but also simplifies the management of security infrastructure by consolidating it into a single cloud-based service.

Another significant aspect of SSE is its focus on data protection and regulatory compliance. With the increasing volume of sensitive data being stored and processed in the cloud, organizations face greater risks of data breaches and regulatory fines. SSE helps mitigate these risks by providing advanced data protection capabilities such as encryption, data loss prevention, and threat detection. These features ensure that sensitive data is safeguarded against unauthorized access and exfiltration, and that organizations can maintain compliance with data privacy regulations like GDPR, CCPA, and HIPAA.

Infrastructure as Code (IaC)

One of the most exciting developments in technology today is ***infrastructure as code (IaC)***. When your infrastructure (often in the cloud) is all virtualized, it can be easily created (and destroyed) as well as maintained by using scripts (code). This makes it possible for you to “spin up” test environments or pilot tests with ease. Think about how much easier it is to create a duplicate site for high availability (HA) needs when using IaaS and IaC than when using physical devices.

The large public cloud providers make it simple for you to implement IaC. They provide tools (such as CloudFormation from AWS) that permit you to easily generate the code required to script the creation of useful (and even complex) infrastructures. Thanks to this capability, you can easily automate—and even orchestrate—common networking tasks that used to take weeks or months to carry out.

For example, say that you need to spin up 50 servers for a test project. Thanks to IaC, you can now do this with a few clicks of the mouse instead of using a massive (and often) expensive deployment of physical servers.

There is a difference between automation and orchestration:

- **Automation** refers to the automated completion of a task or tasks.
- **Orchestration** refers to the scheduling and monitoring of many different automations. It is, basically, automating the automation.

NOTE IaC is also known as *programmable infrastructure* to indicate that the infrastructure configuration can be incorporated into application code. IaC enables DevOps teams to test applications in production-like environments from the beginning of the development cycle.

Key components of automation in IaC include playbooks, templates, and reusable tasks, which facilitate the creation, maintenance, and scaling of infrastructure in a consistent manner. Additionally, automation helps address challenges such as configuration drift, compliance, and upgrades, and supports dynamic inventories for flexible resource management. Here are more details on the key components and advantages of automation with IaC:

**Key
Topic**

- **Playbooks:** Playbooks are a fundamental tool in IaC automation, particularly in tools like Ansible. They provide a structured way to define a series of tasks that automate the provisioning, configuration, and management of infrastructure. Playbooks are written in YAML and describe the desired state of the infrastructure in a declarative manner. This allows for complex workflows to be automated, such as deploying applications, configuring servers, and managing network devices.

NOTE Ansible is a software tool that enables infrastructure as code. It is open source and includes modules for software provisioning, configuration management, and application deployment functionality. YAML is a human-readable data serialization language. It is commonly used for configuration files and in applications where data is being stored or transmitted.

- **Templates:** Templates in IaC are used to define infrastructure resources in a reusable and consistent manner. Tools like Terraform and AWS CloudFormation utilize templates to describe cloud resources and their relationships. These templates can include variables, allowing for parameterization and flexibility in resource configurations. For example, a template might define a

virtual machine with specific attributes like instance type, security groups, and attached storage. By using templates, organizations can ensure that infrastructure components are created with a consistent configuration across different environments, reducing the risk of configuration errors and making it easier to replicate and scale infrastructure.

- **Reusable tasks:** Reusable tasks are a key aspect of IaC automation that promote efficiency and maintainability. In tools like Ansible, reusable tasks can be defined in roles, which are collections of tasks, variables, and templates organized in a structured format. Roles can be shared across multiple playbooks and projects, allowing for the reuse of common configurations and deployment steps. For example, a role might encapsulate the tasks required to set up a web server, including installing packages, configuring services, and managing firewall rules.
- **Configuration drift:** Configuration drift occurs when the actual state of the infrastructure deviates from the desired state defined in the IaC. This can happen due to manual changes, system updates, or environmental factors. Automation in IaC helps mitigate configuration drift by regularly applying the desired state to the infrastructure. Tools like Terraform and Ansible can perform periodic checks and reapply configurations to ensure consistency. This not only helps maintain the reliability and predictability of the infrastructure but also reduces the time and effort required to troubleshoot and resolve issues caused by drift. Automation ensures that the infrastructure remains aligned with the defined state, minimizing the risks associated with unintended changes.
- **Compliance:** Compliance with industry standards and regulatory requirements is a critical aspect of infrastructure management. Automation in IaC enables organizations to enforce compliance by embedding policies and controls directly into the infrastructure code. For example, security configurations, access controls, and data protection measures can be defined in the IaC templates and playbooks. Automated tools can continuously monitor the infrastructure for compliance with these policies, generating reports and alerts when deviations occur.
- **Upgrades:** Upgrading infrastructure components, such as software versions, operating systems, and hardware configurations, can be a complex and error-prone process. IaC automation simplifies upgrades by allowing organizations to define the desired state of the infrastructure, including the required versions and configurations. Upgrades can be tested in a staging environment using the same IaC definitions before being applied to production, reducing the risk of disruptions.

- **Dynamic inventories:** Dynamic inventories are a feature of IaC automation that allows the infrastructure to be dynamically discovered and managed based on current configurations and states. This is particularly useful in cloud environments, where resources can be created and terminated frequently. Tools like Ansible support dynamic inventories, which can query cloud providers or other data sources to generate an up-to-date list of resources for configuration management tasks.

Because the C in IaC stands for code, it is of no surprise that IaC systems tend to take advantage of *source control* systems. These systems tend to feature the following:

- **Version control:** Version control is at the heart of source control systems and is vital for managing IaC. It enables teams to track changes to infrastructure code over time, maintaining a history of modifications, additions, and deletions. Each change is recorded with a unique identifier, often called a *commit*, along with metadata such as the author, timestamp, and a message describing the change.
- **Central repository:** A central repository in a source control system acts as the single source of truth for all infrastructure code. This repository stores the master copies of the code and provides a central location where all team members can access, contribute to, and collaborate on the infrastructure codebase.
- **Conflict identification:** Conflict identification is an essential feature of source control systems, especially in collaborative IaC environments, where multiple team members may work on the same code simultaneously. Conflicts occur when changes made by different users overlap or are incompatible with each other.
- **Branching:** Branching is a powerful feature of source control systems that allows teams to create isolated copies of the codebase for different purposes. This is particularly useful in IaC environments for managing multiple streams of development and experimentation without affecting the main codebase.

IP Version 6

With the global proliferation of IP-based networks, available IPv4 addresses are rapidly becoming exhausted. Fortunately, IPv6 provides enough IP addresses for many generations to come. This section introduces *IPv6 addressing* with a deep dive into IPv6's address structure and a discussion of some of its unique characteristics.

Need for IPv6

With the worldwide depletion of IP version 4 (IPv4) addresses, many organizations have migrated, are in the process of migrating, or are considering migrating their IPv4 addresses to IPv6 addresses. IPv6 dramatically increases the number of available IP addresses. In fact, IPv6 offers approximately 5×10^{28} IP addresses for each person on the planet.

Beyond the increased address space, IPv6 offers many other features:

- Simplified header:
 - The IPv4 header uses 12 fields.
 - The IPv6 header uses 5 fields.
- No broadcasts
- No fragmentation (performs MTU discovery for each session)
- Can coexist with IPv4 during a transition:
 - **Dual stack** (running IPv4 and IPv6 simultaneously on a network interface or device)
 - IPv6 over IPv4 (tunneling IPv6 over an IPv4 tunnel)

Even if you are designing a network based on IPv4 addressing, it is a good practice to consider how readily an IPv6 addressing scheme could be overlaid on that network at some point in the future. Using Teredo **tunneling**, an IPv6 host could provide IPv6 connectivity even when the host is directly connected to an IPv4-only network. Miredo is a client that can be used to implement the Teredo protocol and is included in many versions of Linux. IPv6/IPv4 tunneling is often referred to as 6to4 or 4to6 tunneling, depending on which protocol is being tunneled (IPv4 or IPv6). These are just some of the many tunneling mechanisms devised to ensure a smooth transition from IPv4 to IPv6. In fact, thanks to dual stack and tunneling features, it is very unlikely that you will see IPv4 ever completely go away in your lifetime.

Since there are so many available IPv6 addresses, network address translation (NAT) is not nearly as required in IPv6. One way it can be useful is in transition between the two versions, however. Network address translation from IPv6 to IPv4 (**NAT64**) is a technology that facilitates communication between IPv6-only clients and IPv4-only servers, bridging the gap between the two distinct IP address families. It is yet another component in the transition from the older IPv4 protocol to the newer IPv6 protocol, allowing IPv6 networks to access resources on IPv4 networks without requiring the end systems to support both protocols. NAT64 works by translating

IPv6 packets to IPv4 packets and vice versa, using a predefined prefix to generate an IPv6 address that maps to an IPv4 address.

IPv6 Address Structure

An IPv6 address has the following address format, where *X* is a hexadecimal digit in the range of 0 to F:

`XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX:XXXX`

A hexadecimal digit is 4 bits in size (4 binary bits can represent 16 values). Notice that an IPv6 address has eight fields, and each field contains four hexadecimal digits. The following formula reveals why an IPv6 address is a 128-bit address:

$4 \text{ bits per digit} \times 4 \text{ digits per field} \times 8 \text{ fields} = 128 \text{ bits in an IPv6 address}$

IPv6 addresses can be difficult to work with because of their size. Fortunately, the following rules (often collectively referred to as *shorthand notation*) exist for abbreviating these addresses:



- Leading 0s in a field can be omitted.
- Contiguous fields containing all 0s can be represented with a double colon. (Note that this can be done only once for a single IPv6 address.)

For example, consider the following IPv6 address:

`ABCD:0123:4040:0000:0000:0000:000A:000B`

Using the rules for abbreviation, the IPv6 address can be rewritten as follows:

`ABCD:123:4040::A:B`

An exciting feature of IPv6 is the Extended Unique Identifier (EUI-64) format, which permits a device to automatically populate the low-order 64 bits of an IPv6 address based on an interface's MAC address. You will read more about this capability later in this chapter.

IPv6 Address Types

The following are some of the many unique aspects of IPv6 addressing and interesting address types:

- IPv6 globally routable unicast addresses start with the first four hex characters in the range 2000 to 3999.
- An IPv6 link-local address is also used on each IPv6 interface. The link-local address begins with FE80.

- Multicast addresses begin with FF as the first two hex characters.
- IPv6 can use autoconfiguration to discover the current network and select a host ID that is unique on that network. Automatic generation of a unique host ID is made possible through a process known as *EUI-64*, which uses the 48-bit MAC address on the device to aid in the generation of the unique 64-bit host ID. Notice that the autoconfiguration capabilities described here permit you to create an IPv6 network free of DHCP-type services. The ability of IPv6 to replace the need for DHCP services like this is known as stateless address autoconfiguration (SLAAC). You will learn more about SLAAC in Chapter 16, “IPv4 and IPv6 Network Services.”
- IPv6 can also use a special version of DHCP for IPv6. Not surprisingly, this version is called *DHCPv6*.
- The protocol that is used for *network discovery*—that is, to discover the network address and learn the Layer 2 addresses of neighbors on the same network—is Neighbor Discovery Protocol (NDP).

NDP is hugely important in IPv6. It defines five ICMPv6 packet types for important jobs:

Key Topic

- **Router Solicitation:** Hosts inquire with Router Solicitation messages to locate routers on an attached link.
- **Router Advertisement:** Routers advertise their presence together with various link and Internet parameters, either periodically or in response to a Router Solicitation message.
- **Neighbor Solicitation:** Neighbor solicitation messages are used by nodes to determine the link layer address of a neighbor or to verify that a neighbor is still reachable via a cached link layer address.
- **Neighbor Advertisement:** Neighbor advertisement messages are used by nodes to respond to a Neighbor Solicitation message.
- **Redirect:** Routers may inform hosts of a better first-hop router for a destination.

IPv6 Data Flows

You might recall from our discussion of IPv4 traffic flows in Chapter 4 that there are unicast, broadcast, multicast, and anycast methods of communication possible with IP version 4. IPv6 uses just three of the four types of data flows:

**Key
Topic**

- Unicast
- Multicast
- Anycast

Just like in IPv4, IPv6 uses special address types for these data flows. The following sections summarize the characteristics of each address type.

Unicast

With unicast, a single IPv6 address is applied to a single interface, as illustrated in Figure 8-3. The communication flow can be thought of as a one-to-one communication flow.

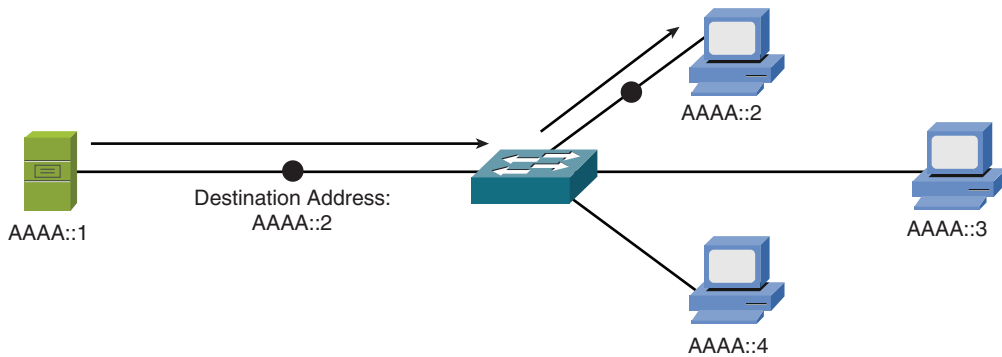


Figure 8-3 IPv6 Unicast Example

In Figure 8-3, a server (AAAA::1) is sending traffic to a single client (AAAA::2).

Multicast

With multicast, a single IPv6 address (a multicast group) can represent multiple devices on a network, as shown in Figure 8-4. The communication flow is a one-to-many communication flow.

In Figure 8-4, a server (AAAA::1) is sending traffic to a multicast group (FF00::A). Two clients (AAAA::2 and AAAA::3) have joined this group. Those clients receive the traffic from the server, and any client that did not join the group (for example, AAAA::4) does not receive the traffic.

IPv6 replaces broadcast behavior with multicast, thanks to the “all nodes” multicast group. This reserved address is FF01:0:0:0:0:0:0:1 (FF01::1). All IPv6 nodes join this group. This is a simple and efficient method for sending traffic to all nodes.

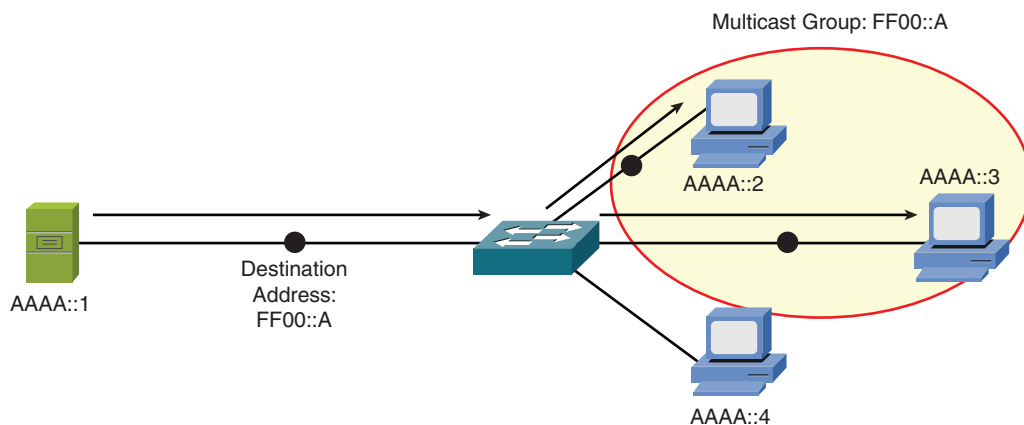


Figure 8-4 IPv6 Multicast Example

Anycast

With *anycast*, a single IPv6 address is assigned to multiple devices, as illustrated in Figure 8-5. It is a one-to-nearest (from the perspective of a router's routing table) communication flow.

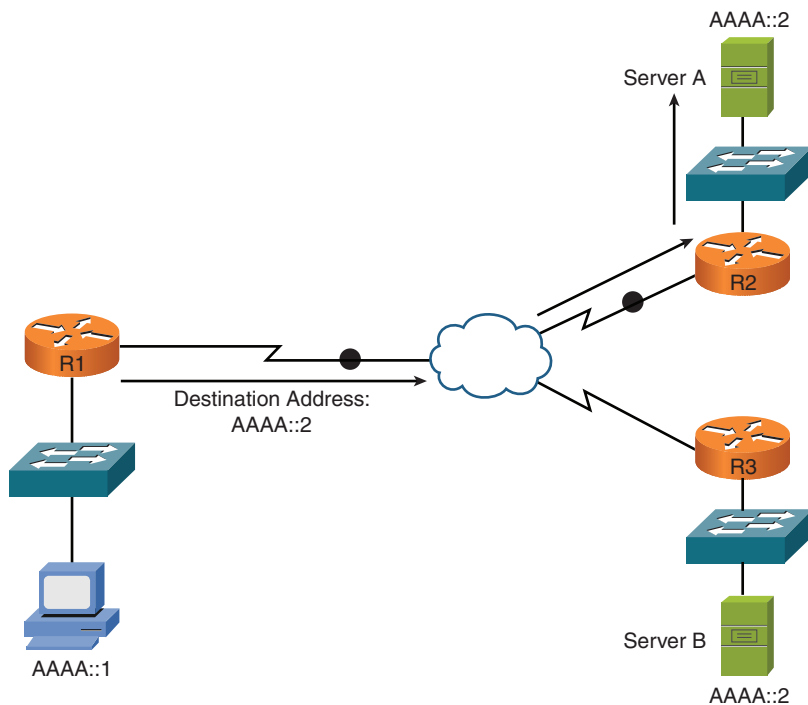


Figure 8-5 IPv6 Anycast Example

In Figure 8-5, a client with IPv6 address AAAA::1 wants to send traffic to destination IPv6 address AAAA::2. Notice that two servers (Server A and Server B) have the IPv6 address AAAA::2. In the figure, the traffic destined for AAAA::2 is sent to Server A, via router R2, because the network on which Server A resides appears to be closer than the network on which Server B resides, from the perspective of router R1's IPv6 routing table.

NOTE Remember that the dreaded broadcast frames and packets from IPv4 do not exist in an IPv6-only network. IPv6 uses only unicasts, multicasts, and anycasts, as described in this section. With IPv6, if you want to send a frame or packet to all nodes in the local network, you use the all-nodes IPv6 multicast address.

Real-World Case Study

Acme, Inc. is currently investigating the use of a software-defined wide area network (SD-WAN) to help revolutionize the legacy WAN infrastructure. Specifically, Acme is very interested in the enhanced network performance that this solution can bring. Acme would like the SD-WAN solution to dynamically route traffic over the best available path based on real-time network conditions, such as latency, jitter, and packet loss. This would ensure that critical applications, like video conferencing and cloud services, receive the bandwidth and low-latency routes they need, resulting in a better user experience.

Acme, Inc. is also actively exploring the implementation of a new zero trust architecture to enhance its cybersecurity posture and mitigate the risks associated with modern cyber threats. Unlike traditional security models that rely on a defined network perimeter, zero trust operates on the principle that no user or device, whether inside or outside the network, should be trusted by default. This approach aligns with Acme's goal of protecting sensitive data and resources in an increasingly complex and distributed IT environment. By adopting zero trust, Acme can ensure that all access requests are continuously verified and authenticated, regardless of the user's location or network. This is particularly important as Acme's workforce becomes more mobile and remote, accessing company resources from various devices and locations. Zero trust will enable Acme to enforce strict access controls and minimize the attack surface, thereby reducing the likelihood of unauthorized access and data breaches.

The move toward zero trust is also driven by Acme's desire to streamline compliance with regulatory requirements and enhance the overall resilience of its IT infrastructure. Zero trust architecture provides a comprehensive framework for implementing security policies that are consistent and enforceable across all endpoints and applications. This allows Acme to achieve greater visibility into user activities and data flows, ensuring that any suspicious behavior is promptly detected and addressed. Additionally, the granular control afforded by zero trust helps Acme to safeguard sensitive information and comply with regulations such as GDPR and HIPAA, which mandate stringent data protection measures. By integrating zero trust principles into its security strategy, Acme aims to build a robust and adaptable security model that not only protects against current threats but also evolves to address future challenges, ultimately supporting the company's growth and operational excellence.

Summary

Here are the main topics covered in this chapter:

- This chapter first provided a description of software-defined networking (SDN) and software-defined wide area networks (SD-WAN).
- This chapter next covered emerging technology of Virtual Extensible Local Area Networks (VXLAN). After describing the Layer 2 encapsulation that makes this technology function, this section described the common use case of the data center interconnect (DCI) functionality.
- This chapter then examined the emerging technology of the zero trust architecture (ZTA), including its major components of policy-based authentication, authorization, and least privilege access.
- This chapter defined the Secure Access Secure Edge (SASE) and Security Service Edge (SSE) solutions.
- This chapter covered the concept of infrastructure as code (IaC) and emphasized the features of IaC that rely on automation. This section of the chapter also discussed how source control can be critical to the IaC environment.
- The characteristics of IPv6 were highlighted, including the IPv6 address format and IPv6 data flows (unicast, multicast, and anycast).

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 8-1 lists these key topics and the page number where each is found.



Table 8-1 Key Topics for Chapter 8

Key Topic Element	Description	Page Number
Section	Software-Defined Networking (SDN)	205
Section	Virtual Extensible Local Area Network (VXLAN)	208
Section	Zero Trust Architecture (ZTA)	209
List	Key components for IaC	213
List	Steps for shorthand with IPv6 addresses	217
List	Functions of NDP	218
List	Types of IPv6 data flows	219

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

application awareness, authorization, automation, branching, central policy management, central repository, compliance, configuration drift, conflict identification, data center interconnect (DCI), dual stack, dynamic inventories, infrastructure as code (IaC), IPv6 addressing, Layer 2 encapsulation, least privilege access, NAT64, playbooks, policy-based authentication, reusable tasks, Secure Access Secure Edge (SASE), Security Service Edge (SSE), source control, software-defined networking (SDN), software-defined wide area network (SD-WAN), templates, transport agnostic, tunneling, upgrades, version control, Virtual Extensible Local Area Network (VXLAN), zero-touch provisioning (ZTP), zero trust architecture (ZTA)

Additional Resources

Software Defined Networking (SDN) Demystified: <https://www.youtube.com/watch?v=IVcUZCVvBjw>

VXLAN Simple Explanation: <https://www.youtube.com/watch?v=7Shfu9BrJP8>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. BGP is an example of a technology found in what layer/plane of operation in a software-defined network?
 - a. Management
 - b. Control
 - c. Data
 - d. Application
2. What feature of the SD-WAN allows the network to seamlessly route and manage traffic over diverse transport media without dependency on the underlying physical connections?
 - a. Central policy management
 - b. Application awareness
 - c. Zero-touch provisioning
 - d. Transport agnostic
3. What protocol serves as the transport protocol for encapsulating Layer 2 Ethernet frames with Layer 3 packets in the VXLAN solution?
 - a. UDP
 - b. TCP
 - c. ARP
 - d. FHRP

4. What is the term given to the difference in the actual state of your infrastructure compared to the state defined in an IaC implementation?
 - a. Version control
 - b. Configuration drift
 - c. Source control
 - d. Conflict identification
5. How can the following IPv6 address be condensed?
2009:0123:4040:0000:0000:000A:100B
 - a. 2009::123:404:A:100B
 - b. 2009::123:404:A:1B
 - c. 2009:123:4040::A:100B
 - d. 2009:0123:4040::0::000A:100B
6. What technology allows for the automatic assignment of the host portion of an IPv6 address?
 - a. Dual stack
 - b. EUI-64
 - c. Neighbor discovery
 - d. Anycast
7. What can IPv6 networks use to assign IP addresses?
 - a. SLAAC
 - b. CIDR
 - c. Port address translation
 - d. Classless inter-domain routing notation
8. Which of the following is a network architecture that integrates wide area networking (WAN) capabilities with comprehensive network security functions such as Secure Web Gateway (SWG), Cloud Access Security Broker (CASB), Firewall as a Service (FWaaS), and Zero Trust Network Access (ZTNA)?
 - a. SD-WAN
 - b. VXLAN
 - c. SSE
 - d. SASE

9. Which of the following involves connecting multiple data centers to provide a unified infrastructure, allowing for efficient resource sharing, workload mobility, and disaster recovery?
 - a. NAT64
 - b. SASE
 - c. DCI
 - d. SSE

10. Which of the following is a technology that facilitates communication between IPv6-only clients and IPv4-only servers, bridging the gap between the two distinct IP address families?
 - a. NAT64
 - b. Dual stack
 - c. Conflict identification
 - d. Branching

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This chapter covers the following topics related to Objective 2.1 (Explain characteristics of routing technologies) of the CompTIA Network+ N10-009 certification exam:

- Static routing
- Dynamic routing
 - Border Gateway Protocol (BGP)
 - Enhanced Interior Gateway Routing Protocol (EIGRP)
 - Open Shortest Path First (OSPF)
- Route selection
 - Administrative distance
 - Prefix length
 - Metric
- Address translation
 - NAT
 - Port address translation (PAT)
- First Hop Redundancy Protocol (FHRP)
- Virtual IP (VIP)
- Subinterfaces

Routing Technologies

In Chapter 7, “IPv4 Addressing,” you learned how Internet Protocol (IP) networks can be divided into subnets. Each subnet is its own broadcast domain, and the device that separates broadcast domains is a router (which this text considers synonymous with a multilayer switch). A multilayer switch is a network device that can perform the Layer 2 switching of frames as well as the Layer 3 routing of IP packets. Multilayer switches generally use dedicated chips to perform these functions and, as a result, may be faster than traditional routers in forwarding packets.

For traffic to flow between subnets, the traffic has to be routed; this routing is a router’s primary job. This chapter discusses how routing occurs and introduces a variety of approaches for performing routing, including dynamic routing, static routing, and default routing. The chapter also breaks down the various categories of routing protocols and provides specific examples of each.

The chapter concludes with a discussion of address translation and first hop redundancy protocols.

Foundation Topics

Routing

To understand basic routing processes, consider Figure 9-1. In this topology, PC1 needs to send traffic to Server1. Notice that these devices are on different networks. In this topology, how does a packet from the source IP address 192.168.1.2 get routed to the destination IP address 192.168.3.2?

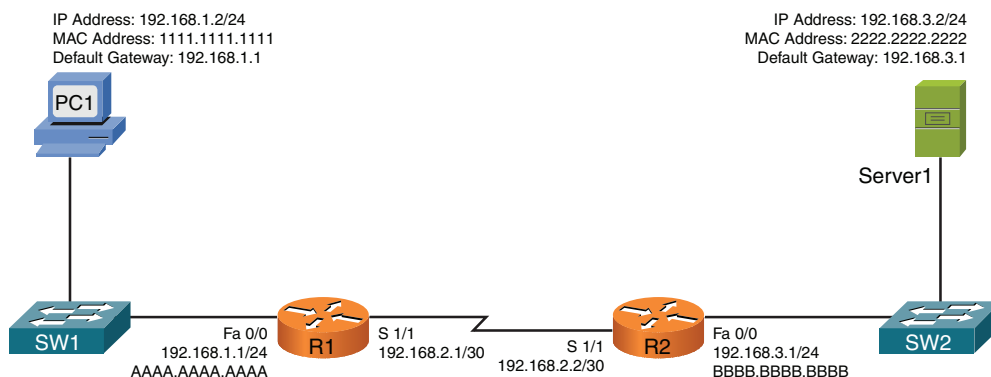


Figure 9-1 Basic Routing Topology

It might help to walk through this process systematically:

Key Topic

- Step 1.** PC1 compares its IP address and subnet mask 192.168.1.2/24 with the destination IP address and subnet mask 192.168.3.2/24. PC1 concludes that the destination IP address resides on a remote subnet. Therefore, PC1 needs to send the packet to its default gateway, which could have been manually configured on PC1 or dynamically learned via Dynamic Host Configuration Protocol (DHCP). In this example, PC1 has the default gateway 192.168.1.1 (router R1). However, to construct a Layer 2 frame, PC1 also needs the MAC address of its default gateway. PC1 sends an Address Resolution Protocol (ARP) request for router R1's MAC address. After PC1 receives an ARP reply from router R1, PC1 adds router R1's MAC address to its ARP cache. PC1 now sends its data in a frame destined for Server1, as shown in Figure 9-2.

NOTE ARP is a broadcast-based protocol and, therefore, does not travel beyond the local subnet of the sender.

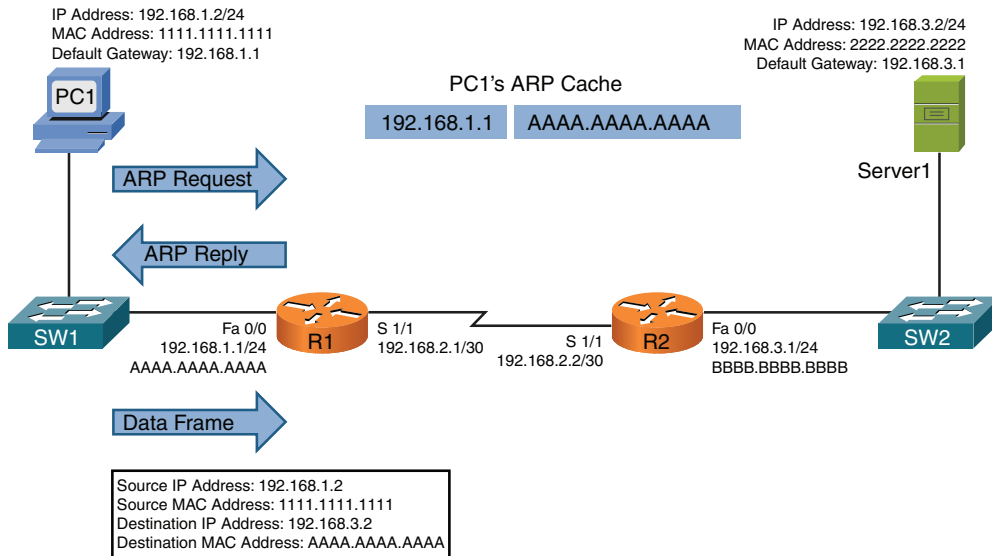


Figure 9-2 Basic Routing: Step 1

Step 2. Router R1 receives the frame sent from PC1 and interrogates the IP header. An IP header contains a Time-to-Live (TTL) field, which is decremented once for each router hop. Therefore, router R1 decrements the packet's TTL field. If the value in the TTL field is reduced to 0, the router discards the frame and sends a "time exceeded" Internet Control Message Protocol (ICMP) message back to the source. As long as the TTL has not been decremented to 0, router R1 checks its routing table to determine the best path to reach network 192.168.3.0/24. In this example, router R1's routing table has an entry stating that network 192.168.3.0/24 is accessible via interface Serial 1/1. Note that ARP is not required for serial interfaces because these interface types do not have MAC addresses. Router R1, therefore, forwards the frame out its Serial 1/1 interface, as shown in Figure 9-3.

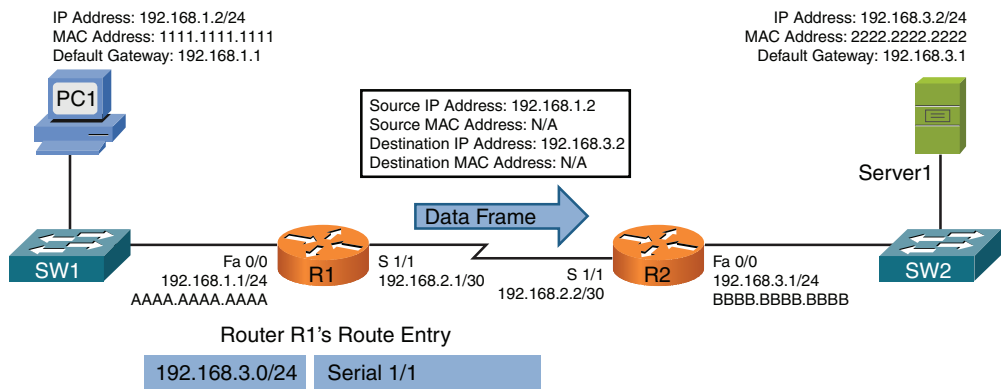


Figure 9-3 Basic Routing: Step 2

Step 3. When router R2 receives the frame, it decrements the TTL in the IP header, just as router R1 did. Again, as long as the TTL has not been decremented to 0, router R2 interrogates the IP header to determine the destination network. In this case, the destination network 192.168.3.0/24 is directly attached to router R2's Fast Ethernet 0/0 interface. Similar to the way PC1 sent out an ARP request to determine the MAC address of its default gateway, router R2 sends an ARP request to determine the MAC address of Server1. After an ARP reply is received from Server1, router R2 forwards the frame out its Fast Ethernet 0/0 interface to Server1, as illustrated in Figure 9-4.

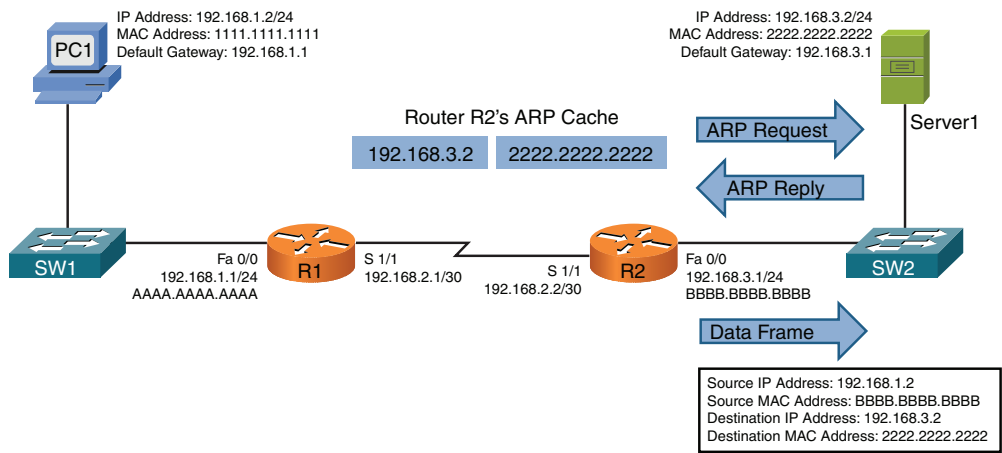


Figure 9-4 Basic Routing: Step 3

The previous steps identified two router data structures:

- **IP routing table:** When a router needed to route an IP packet, it consulted its IP routing table to find the best match. The best match is the route that has the longest prefix (referred to as *prefix length*). Specifically, a route entry with the longest prefix is the most specific network. For example, imagine that a router has an entry for network 10.0.0.0/8 and for network 10.1.1.0/24. Also, imagine that the router is seeking the best match for destination address 10.1.1.1/24. The router would select the 10.1.1.0/24 route entry as the best entry because that route entry has the longest prefix (/24 is longer than /8, which is a more specific entry).
- **Layer 3 to Layer 2 mapping:** In the previous example, router R2's ARP cache contained Layer 3 to Layer 2 mapping information. Specifically, the ARP cache had a mapping that said MAC address 2222.2222.2222 corresponded to IP address 192.168.3.2.

As shown in the preceding example, routers rely on their internal routing table to make packet-forwarding decisions. So how does a router's routing table become populated with entries? That is the focus of the next section.

Sources of Routing Information

A router's routing table can be populated from various sources. As an administrator, you could statically configure a route entry. We refer to this process as *static routing*. A route could be learned via a *dynamic routing* protocol (for example, OSPF or EIGRP), or a router could know how to get to a specific network because the router is physically attached to that network. Dynamic routing protocols are covered later in this chapter.

Directly Connected Routes

A router that has an interface directly participating in a network knows how to reach that specific destination network. For example, consider Figure 9-5.

In Figure 9-5, router R1's routing table knows how to reach the 192.168.1.0/24 and 192.168.2.0/30 networks because router R1 has an interface physically attached to each network. Similarly, router R2 has interfaces participating in the 10.1.1.0/30 and 192.168.2.0/30 networks and therefore knows how to reach those networks. The entries currently shown to be in the routing tables of routers R1 and R2 are called *directly connected routes*.

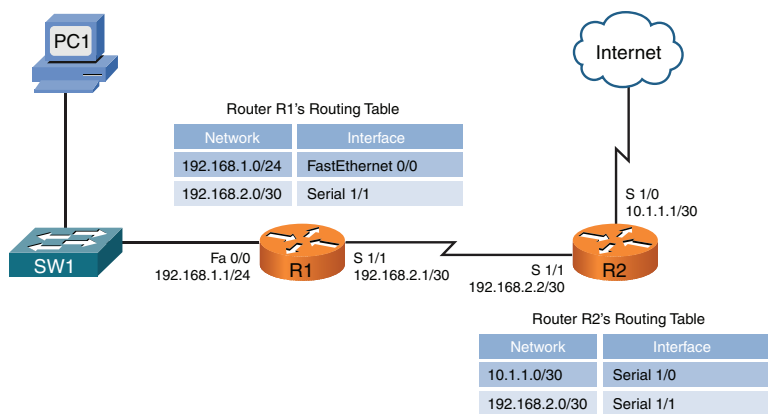


Figure 9-5 Directly Connected Routes

Static Routes

It is also possible to statically configure routes in a router's routing table. Continuing to expand on the previous example, consider router R1. As shown in Figure 9-6, router R1 does not need knowledge of each route on the Internet. Specifically, router R1 already knows how to reach devices on its locally attached networks. All router R1 really needs to know at this point is how to get out to the rest of the world. As you can see from Figure 9-6, any traffic destined for a nonlocal network (for example, any of the networks available on the public Internet) can simply be sent to router R2. Because R2 is the next router hop along the path to reach all those other networks, router R1 could be configured with a *default static route*, which says, "If traffic is destined for a network not currently in the routing table, send that traffic out interface Serial 1/1."

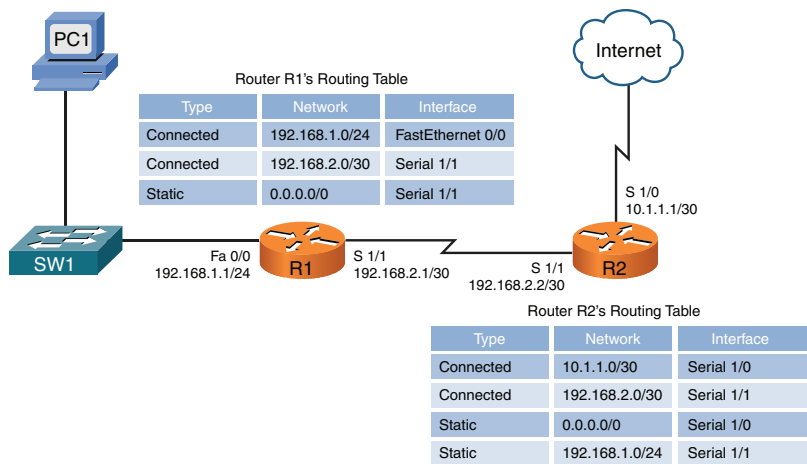


Figure 9-6 Static Routes

NOTE A static route does not always reference a local interface. Instead, a static route might point to a *next-hop IP address* (that is, an interface's IP address on the next router to which traffic should be forwarded). The network address of a default route is 0.0.0.0/0.

Similarly, router R2 can reach the Internet by sending traffic out its Serial 1/0 interface. However, router R2 does need information about how to reach the 192.168.1.0/24 network available off router R1. To educate router R2 about how this network can be reached, a network administrator can add a static route pointing to 192.168.1.0/24 to router R2's routing table.

Dynamic Routing Protocols

In complex networks, such as the topology shown in Figure 9-7, static routing does not scale well. Fortunately, a variety of dynamic routing protocols are available that allow a router's routing table to be updated as network conditions change.

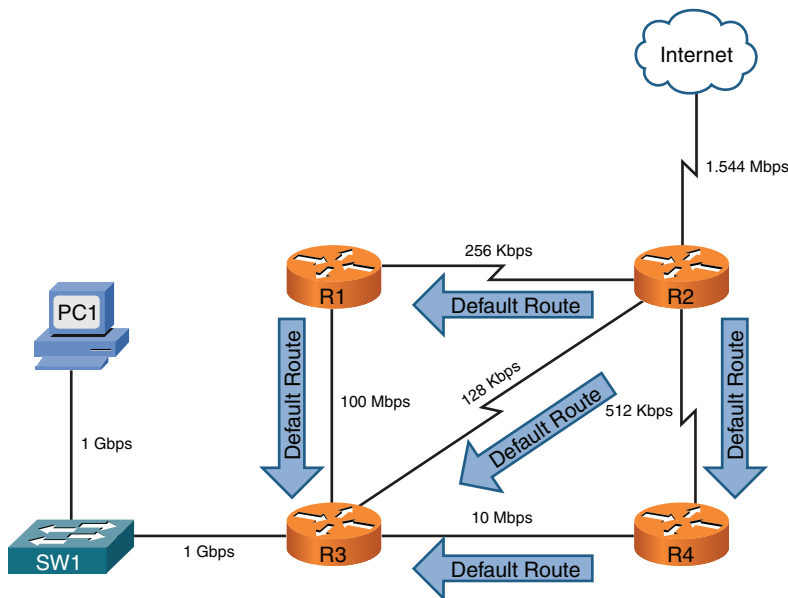


Figure 9-7 Dynamic Routes

In Figure 9-7, router R2 is advertising a default route to its neighbors (routers R1, R3, and R4). What happens if PC1 wants to send traffic to the Internet? PC1's default gateway is router R3, and router R3 has received three default routes. Which one does it use?

Router R3’s path selection depends on the dynamic routing protocol being used. As you will see later in this chapter, a routing protocol such as Routing Information Protocol (RIP) would make the path selection based on the number of routers that must be used to reach the Internet (that is, *hop count*). Based on the topology presented, router R3 would select the 128Kbps link (where Kbps stands for kilobits per second, meaning thousands of bits per second) connecting to router R2 because the Internet would be only one hop away. If router R3 instead selected a path pointing to either router R1 or R4, the Internet would be two hops away.

However, based on the link bandwidths, you can see that the path from router R3 to router R2 is suboptimal. Unfortunately, RIP does not consider available bandwidth when making its route selection. Some other protocols, such as Open Shortest Path First (OSPF), can consider available bandwidth when making their routing decisions.

Dynamic routes also allow a router to reroute around a failed link. For example, in Figure 9-8, router R3 prefers to reach the Internet via router R4. However, the link between routers R3 and R4 goes down. Thanks to a dynamic routing protocol, router R3 knows of two other paths to reach the Internet, and it selects the next-best path, which is via router R1 in this example. This process of failing over from one route to a backup route is called *convergence*.

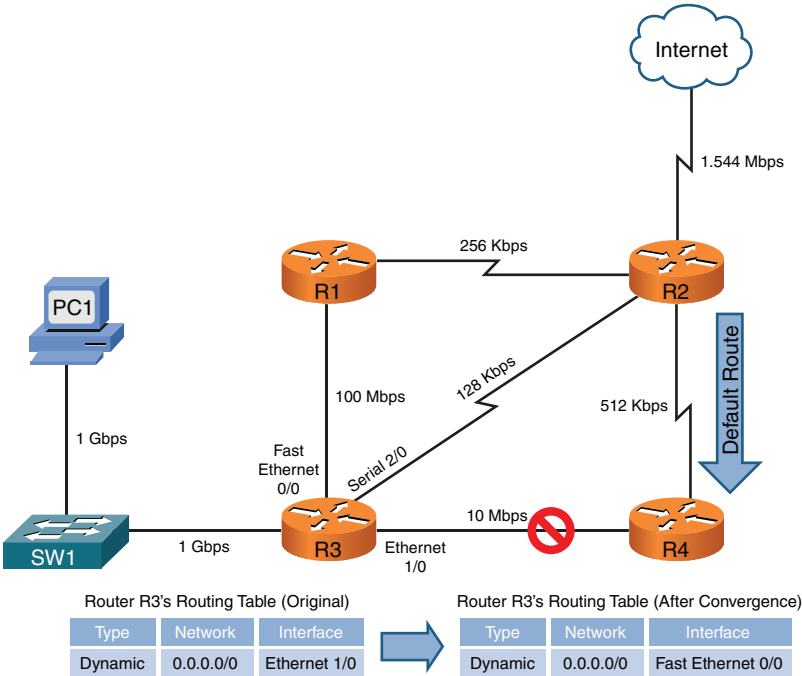


Figure 9-8 Route Redundancy

Routing Protocol Characteristics

Before examining the characteristics of routing protocols, we need to look at the important distinction between a *routing protocol* and a *routed protocol*:

- A *routing protocol* (for example, OSPF, BGP, or EIGRP) is a protocol that advertises route information between routers.
- A *routed protocol* is a protocol with an addressing scheme (for example, IP) that defines different network addresses. Traffic can then be routed between defined networks, perhaps with the assistance of a routing protocol.

This section looks at routing protocol characteristics, such as how believable a routing protocol is compared to other routing protocols. In addition, in the presence of multiple routes, different routing protocols use different *metrics* to determine the best path. A distinction is made between *interior gateway protocols (IGPs)* and *exterior gateway protocols (EGPs)*. Finally, this section discusses different approaches to making route advertisements.

Believability of a Route

If a network is running more than one routing protocol (maybe as a result of a corporate merger), and a router receives two route advertisements from different routing protocols for the same network, which route advertisement does the router believe? Interestingly, some routing protocols are considered to be more believable than others. For example, a Cisco router would consider EIGRP to be more believable than RIP.

The index of believability is called *administrative distance (AD)*. Table 9-1 shows the AD values for various sources of routing information. Note that lower AD values are more believable than higher AD values.

Key Topic

Table 9-1 Administrative Distance

Routing Information Source	AD Value
Directly connected network	0
Statically configured network	1
EIGRP	90
OSPF	110
RIP	120
External EIGRP	170
Unknown or unbelievable	255 (considered to be unreachable)

Metrics

Some networks might be reachable via more than one path. If a routing protocol knows of multiple paths to reach such a network, which route (or routes) does the routing protocol select? Actually, it varies depending on the routing protocol and what that routing protocol uses as a metric (that is, a value assigned to a route). Lower metrics are preferred over higher metrics.

Some routing protocols support load balancing across equal-cost paths; this is useful when a routing protocol knows of more than one route to reach a destination network and those routes have equal metrics. EIGRP can even be configured to do load balancing across unequal-cost paths.

Different routing protocols can use different parameters in their calculation of a metric. The specific parameters used for a variety of routing protocols are presented later in this chapter.

NOTE Remember, routers will use three main criteria when performing *route selection*. They will use the longest matching prefix length, the lowest administrative distance value, and the most preferred metric.

Interior Versus Exterior Gateway Protocols

Routing protocols can also be categorized based on the scope of their operation. Interior gateway protocols (IGPs) operate within an autonomous system, where an autonomous system is a network under a single administrative control. Conversely, exterior gateway protocols (EGPs) operate between autonomous systems.

Consider Figure 9-9. Routers R1 and R2 are in one autonomous system (AS 65002), and routers R3 and R4 are in another autonomous system (AS 65003). Within those autonomous systems, an IGP is used to exchange routing information. However, router ISP1 is a router in a separate autonomous system (AS 65001) that is run by a service provider. An EGP (typically, Border Gateway Protocol) is used to exchange routing information between the service provider's autonomous system and each of the other autonomous systems.

Route Advertisement Method

Another characteristic of a routing protocol is how it receives, advertises, and stores routing information. The two fundamental approaches are *distance vector* and *link state*.

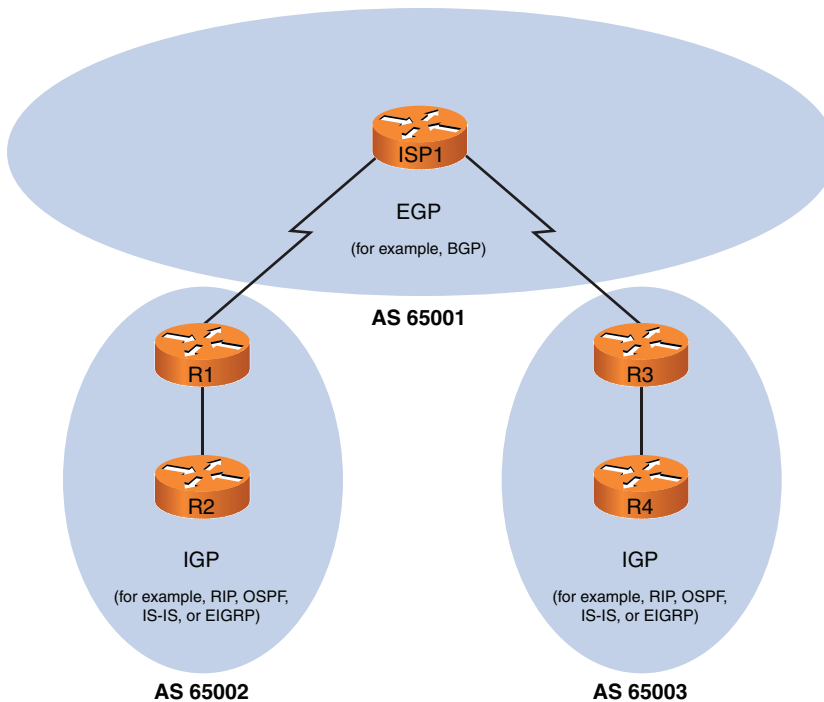


Figure 9-9 IGPs Versus EGPs

Distance Vector

A *distance-vector routing protocol* sends a full copy of its routing table to its directly attached neighbors. This is a periodic advertisement, meaning that even if there have been no topological changes, a distance-vector routing protocol will, at regular intervals, advertise again its full routing table to its neighbors.

Obviously, this periodic advertisement of redundant information is inefficient. Ideally, you want a full exchange of route information to occur only once and subsequent updates to be triggered by topological changes.

Another drawback to distance-vector routing protocols is the time they take to converge, which is the time required for all routers to update their routing tables in response to a topological change in a network. *Hold-down timers* can speed the convergence process. After a router makes a change to a route entry, a hold-down timer prevents any subsequent updates for a specified period of time. This approach helps stop flapping routes (which are routes that oscillate between being available and unavailable) from preventing convergence.

Yet another issue with distance-vector routing protocols is the potential of a routing loop. To illustrate, consider Figure 9-10. In this topology, the metric being used is *hop count*, which is the number of routers that must be crossed to reach a network. As one example, router R3’s routing table has a route entry for network 10.1.1.0/24 available off router R1. For router R3 to reach that network, two routers must be transited (routers R2 and R1). As a result, network 10.1.1.0/24 appears in router R3’s routing table with a metric (hop count) of 2.

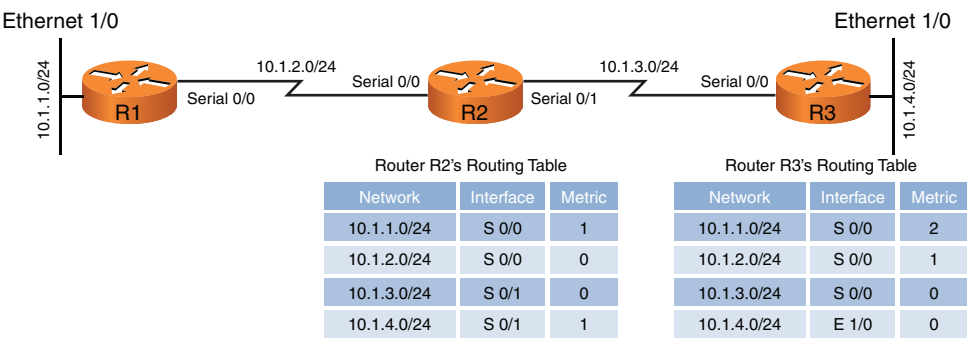


Figure 9-10 Routing Loop: Before Link Failure

Continuing with the example, imagine that interface Ethernet 1/0 on router R3 goes down. As shown in Figure 9-11, router R3 loses its directly connected route (with a metric of 0) to network 10.1.4.0/24. However, router R2 had a route to 10.1.4.0/24 in its routing table (with a metric of 1), and this route was advertised to router R3. Router R3 adds this entry for 10.1.4.0 to its routing table and increments the metric by 1.

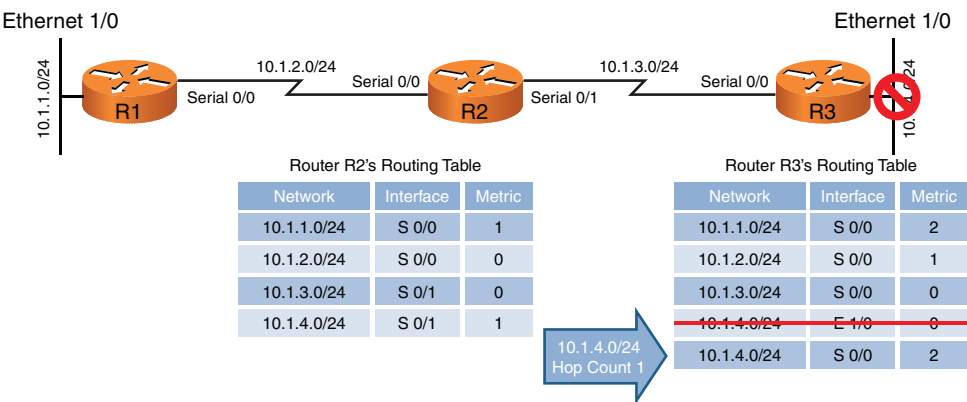


Figure 9-11 Routing Loop: After Link Failure

The problem with this scenario is that the 10.1.4.0/24 entry in router R2’s routing table was due to an advertisement router R2 received from router R3. Now, router R3 is relying on that route, which is no longer valid. The routing loop continues as router R3 advertises its newly learned route 10.1.4.0/24 with a metric of 2 to its neighbor, router R2. Because router R2 originally learned the 10.1.4.0/24 network from router R3, when it sees router R2 advertising that same route with a metric of 2, the network gets updated in router R2’s routing table to have a metric of 3, as shown in Figure 9-12.

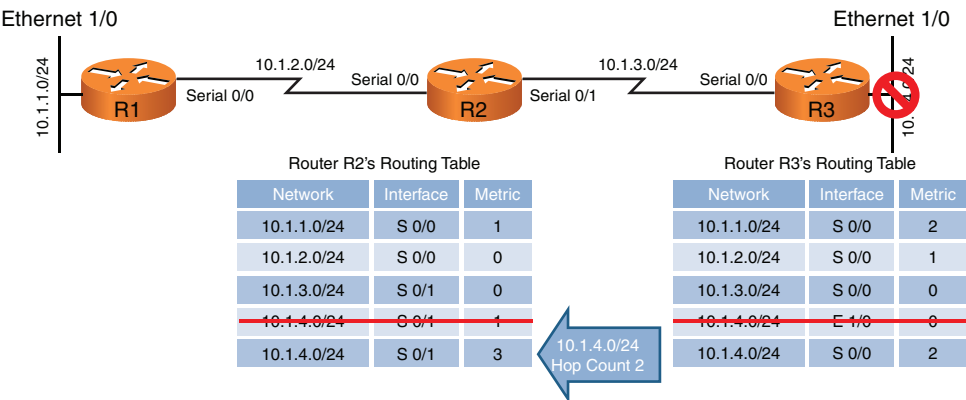


Figure 9-12 Routing Loop: Routers R2 and R3 Incrementing the Metric for 10.1.4.0/24

The metric for the 10.1.4.0/24 network continues to increment in the routing tables for both routers R2 and R3 until the metric reaches a value considered to be an unreachable value. This process is referred to as a *routing loop*.

Distance-vector routing protocols typically use one of two approaches for preventing routing loops:

Key Topic

- **Split horizon:** The split-horizon feature prevents a route learned on one interface from being advertised back out that same interface.
- **Poison reverse:** The poison-reverse feature causes a route received on one interface to be advertised back out that same interface with a metric that is considered to be infinite.

In the previous example, either approach would have prevented router R3 from adding the 10.1.4.0/24 network to its routing table based on an advertisement from router R2.

Link State

Rather than having neighboring routers exchange their full routing tables with one another, a *link-state* routing protocol allows routers to build a topological map of the network. Then, much like a Global Positioning System (GPS) device in a car, a router can execute an algorithm to calculate an optimal path (or paths) to a destination network.

Routers send *link-state advertisements (LSAs)* to advertise the networks they know how to reach. Routers then use those LSAs to construct the topological map of a network. The algorithm that runs against this topological map is *Dijkstra's shortest path first* algorithm.

Unlike distance-vector routing protocols, *link-state routing protocols* exchange full routing information only when two routers initially form their adjacency. Then routing updates are sent in response to changes in the network, as opposed to being sent periodically. Also, link-state routing protocols benefit from shorter convergence times compared to distance-vector routing protocols.

Routing Protocol Examples

Now that you understand some of the characteristics that distinguish one routing protocol from another, this section contrasts some of the most popular routing protocols used in modern networks:

Key Topic

- **Open Shortest Path First (OSPF):** OSPF is a link-state routing protocol that uses the metric *cost*, which is based on the link speed between two routers. OSPF is a popular IGP because of its scalability, fast convergence, and vendor interoperability.
- **Intermediate System-to-Intermediate System (IS-IS):** This link-state routing protocol is similar in operation to OSPF. It uses a configurable, yet dimensionless, metric associated with an interface and runs Dijkstra's shortest path first algorithm. Although IS-IS is an IGP that offers the scalability, fast convergence, and vendor-interoperability benefits of OSPF, it has not been as widely deployed as OSPF.
- **Enhanced Interior Gateway Routing Protocol (EIGRP):** EIGRP is a Cisco-proprietary protocol that is popular in Cisco-only networks but less popular in mixed-vendor environments. Like OSPF, EIGRP is an IGP that offers fast convergence and scalability. EIGRP is more challenging to classify as a distance-vector or a link-state routing protocol.

By default, EIGRP uses bandwidth and delay in its metric calculation; however, other parameters can be considered, including reliability, load, and maximum transmission unit (MTU) size. Using delay as part of the metric, EIGRP can take into consideration the latency caused by the slowest links in the path.

Some literature calls EIGRP an *advanced distance-vector* routing protocol, and some literature calls it a *hybrid routing protocol* (mixing characteristics of both distance-vector and link-state routing protocols). EIGRP uses information from its neighbors to help select an optimal route (like distance-vector routing protocols). However, EIGRP also maintains a database of topological information (like a link-state routing protocol). The algorithm EIGRP uses for its route selection is not Dijkstra's shortest path first algorithm. Instead, EIGRP uses Diffusing Update Algorithm (DUAL).

- **Border Gateway Protocol (BGP):** BGP is the only EGP in widespread use today. In fact, BGP is considered to be the routing protocol that runs the Internet, which is an interconnection of multiple autonomous systems. Although some literature classifies BGP as a distance-vector routing protocol, it can more accurately be described as a *path-vector* routing protocol, meaning that it can use as its metric the number of autonomous system hops that must be transited to reach a destination network, as opposed to a number of required router hops. BGP's path selection is not solely based on autonomous system hops, however. BGP can consider a variety of other parameters. Interestingly, none of those parameters are based on link speed. In addition, although BGP is incredibly scalable, it does not quickly converge in the event of a topological change.

NOTE When studying for the CompTIA Network+ exam, be sure to focus on OSPF, EIGRP, and BGP, as these are the dynamic routing protocols that the exam is sure to cover. Also be sure to understand the differences between static and dynamic routing.

Table 9-2 compares the key characteristics of dynamic routing protocols.

**Key
Topic**

Table 9-2 Comparing Dynamic Routing Protocols

Routing Protocol	IGP or EGP	Type	Metric
OSPF	IGP	Link state	Cost (based on bandwidth)
EIGRP	IGP	Hybrid	Composite (bandwidth and delay by default)
BGP	EGP	Path vector	Path attributes

A network can simultaneously support more than one routing protocol through the process of *route redistribution*. For example, a router could have one of its interfaces participating in an OSPF area of the network and have another interface participating in an EIGRP area of the network. This router could then take routes learned via OSPF and inject those routes into the EIGRP routing process. Similarly, EIGRP-learned routes could be redistributed into the OSPF routing process.

Address Translation

You must remember, some IP addresses are routable through the public Internet, and other IP addresses are considered private and are intended for use within an organization. **Network address translation (NAT)** allows *private IP addresses* (as defined in *RFC1918*) to be translated into Internet-routable IP addresses (that is, public IP addresses). This section examines the operation of basic NAT and a variant called **port address translation (PAT)**. Address translation can also be done for specific ports associated with an IP address. When this is done, it's often referred to as *port forwarding*.

NOTE If RFC1918 is not the most famous document in our field, it is certainly in the top five. RFC1918 was so important and had such a major impact on TCP/IP networks that we often refer to private IP addresses as simply RFC1918 addresses.

NAT

Consider Figure 9-13, which shows a basic NAT topology. Note that, even though the IP networks 172.16.1.0/24 and 192.168.1.0/24 are actually private IP networks, for this discussion, assume that they are publicly routable IP addresses. The reason for the use of these private IP addresses to represent public IP addresses is to avoid using an entity's registered IP addresses in the example.

In the topology shown in Figure 9-13, two clients with private IP addresses 10.1.1.1 and 10.1.1.2 want to communicate with a web server on the public Internet. The server's IP address is 192.168.1.1. Router R1 is configured for NAT. As an example, router R1 takes packets coming from 10.1.1.1 destined for 192.168.1.1 and changes the source IP address in the packets' headers to 172.16.1.101 (which we assume is a publicly routable IP address for the purposes of this discussion). When the server at IP address 192.168.1.1 receives traffic from the client, the server's return traffic is sent to the destination address 172.16.1.101. When router R1 receives traffic from the outside network destined for 172.16.1.101, the router translates the destination IP address to 10.1.1.1 and forwards the traffic to the inside network, where client

1 receives the traffic. Similarly, client 2's IP address 10.1.1.2 is translated into the IP address 172.16.1.102.

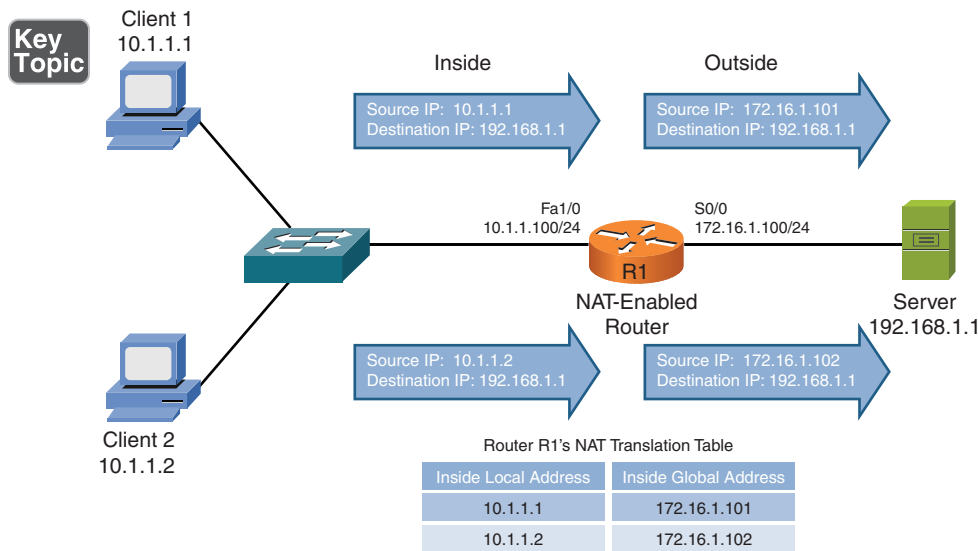


Figure 9-13 Basic NAT Topology

Table 9-3 introduces the terminology used when describing the various IP addresses involved in a translation.

Key Topic **Table 9-3** Names of NAT IP Addresses

NAT IP Address	Definition
Inside local	A private IP address referencing an inside device
Inside global	A public IP address referencing an inside device
Outside local	A private IP address referencing an outside device
Outside global	A public IP address referencing an outside device

As a memory aid, remember that *inside* always refers to an inside device (source), and *outside* always refers to an outside device (destination). Also, think of the word *local* being similar to the Spanish word *loco*, meaning crazy. That is how a local address could be thought of. It is a crazy, made-up address (a private IP address that is not routable on the Internet). Finally, let the *g* in *global* remind you of the *g* in *good*, because a global address is a good (routable on the Internet) IP address.

Based on these definitions, Table 9-4 categorizes the IP addresses shown in Figure 9-13.

Table 9-4 Classifying the NAT IP Addresses in Figure 9-13

NAT IP Address	NAT IP Address
Inside local	10.1.1.1
Inside local	10.1.1.2
Inside global	172.16.1.101
Inside global	172.16.1.102
Outside local	None
Outside global	192.168.1.1

NAT does not always have to be between private and public addresses. For example, NAT could be implemented between two private address ranges or two public address ranges.

Whether an inside local address is randomly assigned an inside global address from a pool of available addresses or is assigned an address from a static configuration determines the type of NAT in use. These two approaches to NAT are called *DNAT* and *SNAT*:

Key Topic

- **DNAT:** In the preceding example, the inside local addresses were automatically assigned an inside global address from a pool of available public addresses. This approach to NAT is referred to as Dynamic NAT (DNAT). This is often referred to as “many-to-many,” as many inside local users (a network) are mapped to a pool of inside global addresses.
- **SNAT:** Sometimes, you want to statically configure the inside global address assigned to a specific device inside your network. For example, you might have an email server inside your company and want other email servers on the Internet to send email messages to your server. Those email servers on the Internet need to point to a specific IP address, not one that was randomly picked from a pool of available IP addresses. In such a case, you can statically configure the mapping of an inside local address (the IP address of your internal email server) to an inside global address (the IP address to which email servers on the Internet will send email for your company). This approach to NAT is referred to as Static NAT (SNAT). This is often called a “one-to-one” mapping.

PAT

A challenge with basic NAT is that there is a one-to-one mapping of inside local addresses to inside global addresses, meaning that a company would need as many publicly routable IP addresses as it had internal devices needing IP addresses. This

does not scale well because, often, a service provider provides a customer with only a single IP address or a small block of IP addresses.

Fortunately, many routers support port address translation (PAT), which allows multiple inside local addresses to share a single inside global address (a single publicly routable IP address). For this reason, PAT is referred to as “many-to-one.” In Chapter 1, you learned about how IP communications rely on port numbers. As a review, when a client sends an IP packet, not only does that packet have a source and destination IP address, it has a source and destination port number. PAT leverages these port numbers to track separate communication flows.

For instance, consider Figure 9-14. Unlike in the example shown in Figure 9-13, in which each inside local address is translated to its own inside global address, the example shown in Figure 9-14 has only one inside global address. This single inside global address is shared among all the devices inside a network. The different communication flows are kept separate in router R1’s NAT translation table by considering port numbers.

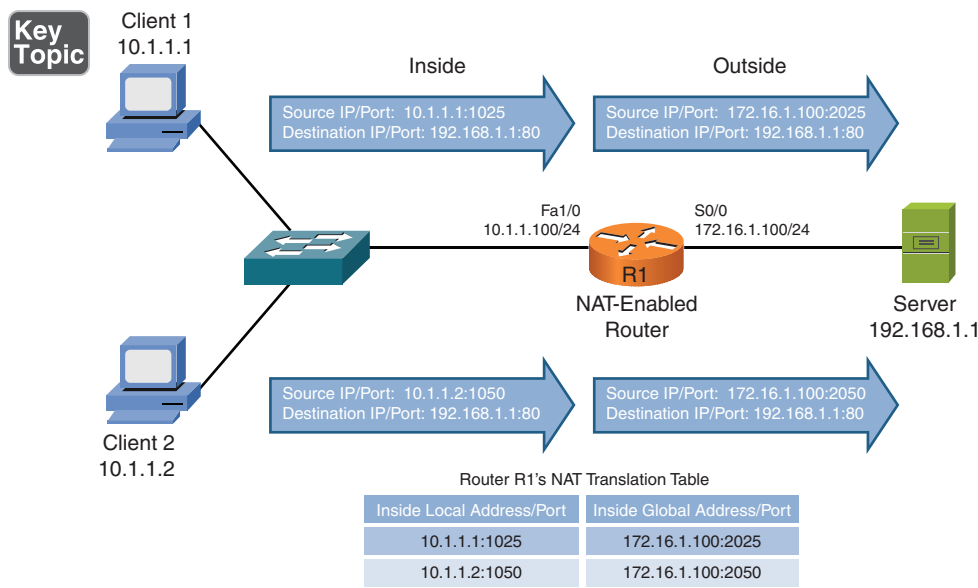


Figure 9-14 PAT Topology

When Client 1 sends a packet to the web server (with IP address 192.168.1.1), the client’s ephemeral port number (its selected source port, which is greater than 1023) is 1025. Router R1 notes that port number and translates the inside local address 10.1.1.1 with port number 1025 to the inside global address 172.16.1.100 with port

number 2025. When Client 2 sends a packet to the same web server, its inside local address 10.1.1.2 with port number 1050 is translated into the inside global address 172.16.1.100 with port number 2050.

Notice that both Client 1 and Client 2 have their inside local addresses translated into the same inside global address, 172.16.1.100. Therefore, when the web server sends packets back to Client 1 and Client 2, those packets are destined for the same IP address (172.16.1.100). However, when router R1 receives those packets, it knows to which client each packet should be forwarded based on the destination port number. For example, if a packet from the web server (192.168.1.1) arrived at router R1 with destination IP address 172.16.1.100 and destination port number 2050, router R1 would translate the destination IP address to 10.1.1.2 and port number 1050, which would be forwarded to Client 2.

Key Topic

First Hop Redundancy Protocol (FHRP)

End systems not running a routing protocol point to a default gateway. The default gateway is traditionally the IP address of a router on the local subnet. However, if the default gateway router fails, the end systems are unable to leave their subnet. There are four *First Hop Redundancy Protocol (FHRP)* technologies (which offer Layer 3 redundancy):

- **Hot Standby Router Protocol (HSRP):** HSRP is a Cisco-proprietary approach to first-hop redundancy. Figure 9-15 shows a sample HSRP topology.

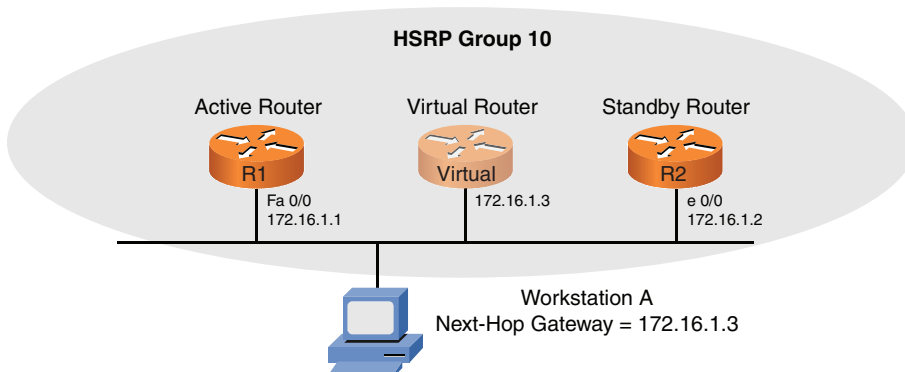


Figure 9-15 HSRP Sample Topology

In Figure 9-15, Workstation A is configured with the default gateway (that is, the next-hop gateway) 172.16.1.3. To prevent the default gateway from becoming a single point of failure, HSRP enables routers R1 and R2 to each act as the default gateway, supporting the *virtual IP (VIP)* of the HSRP group (172.16.1.3), although only one of the routers will act as the default gateway at any one time. Under normal conditions, router R1 (that is, the *active router*) forwards packets sent to virtual IP address 172.16.1.3. However, if router R1 is unavailable, router R2 (that is, the *standby router*) can take over and start forwarding traffic sent to 172.16.1.3. Notice that neither router R1 nor R2 has a physical interface with IP address 172.16.1.3. Instead, a logical router (called a *virtual router*), which is serviced by either router R1 or R2, maintains the 172.16.1.3 IP address.

- **Common Address Redundancy Protocol (CARP):** CARP is an open-standard variant of HSRP.
- **Virtual Router Redundancy Protocol (VRRP):** VRRP is an IETF open standard that operates in a similar method to Cisco's proprietary HSRP. As with HSRP, with VRRP, many routers can operate in a group and ensure that there is always a router available for the hosts to use as a default gateway. Also as with HSRP, a virtual IP address is assigned to the hosts for their default gateway setting. The routers in the VRRP group can also respond (if needed) to that virtual IP address.
- **Gateway Load Balancing Protocol (GLBP):** GLBP is another first-hop redundancy protocol that is proprietary to Cisco Systems.

With each of these technologies, the MAC address and the IP address of a default gateway can be serviced by more than one router (or multilayer switch). Therefore, if a default gateway becomes unavailable, the other router (or multilayer switch) can take over and still service the same MAC and IP addresses.

Another type of Layer 3 redundancy is achieved by having multiple links between devices and selecting a routing protocol that balances the load over the links. Link Aggregation Control Protocol (LACP) enables you to assign multiple physical links to a logical interface, which appears as a single link to a route processor. Figure 9-16 illustrates a network topology using LACP.



Figure 9-16 LACP Sample Topology

NOTE Do not confuse the LACP technology shown in Figure 9-16 with the concepts of network device *subinterfaces*. Subinterfaces are logical interfaces that can be created under the physical device interface. Notice that while this technique is a convenient way to make the interface more flexible (and service different subnets), if the physical interface fails, all subinterfaces are no longer available.

Finally, another important redundancy technique is to ensure that your default gateways can connect to multiple Internet service providers (ISPs) using diverse paths when redundant Internet connectivity is required. Note that such a solution protects against internal device failures as well as ISP failures that are beyond your control. Your network design can even incorporate periodic reachability tests to ensure that your ISPs are truly available and able to direct you to remote resources successfully.

Real-World Case Study

Acme, Inc. has decided to use a link-state routing protocol for dynamic routing between its LANs and the remote offices, which are connected over the WANs. The link-state protocol the company has chosen is OSPF. Each of the routers that has connections to the LAN and WAN subnets will learn about and advertise OSPF routes with its OSPF neighbors.

Acme selected OSPF because of its features that promote scalability. Acme plans to grow its infrastructure dramatically over the coming years and wanted a routing protocol that can easily grow with the organization. The hierarchical use of area and special area types was of particular interest to the engineers within Acme, Inc.

The branch offices will have a default route that points toward the headquarters' routers, and at the headquarters' site, they will use a default route that points toward the service provider. Acme, Inc. itself will not be using BGP, but its WAN and Internet service provider, which is interacting with other service providers, will use BGP.

The branch offices have configured their edge routers in GLBP groups to help ensure high availability connections to the HQ routers. GLBP also provides load balancing of the many connection requests from clients in the remote offices.

Summary

Here are the main topics covered in this chapter:

- This chapter discussed how routers forward traffic through a network based on source and destination IP addresses.

- This chapter also covered the sources of route information used to populate a router's routing table. These sources include directly connected routes, statically configured routes, and dynamically learned routes.
- This chapter distinguished between routed protocols (for example, IP) and routing protocols (such as OSPF or EIGRP).
- Some routing sources are more trustworthy than other routing sources, based on their administrative distances. This chapter presented the commonly used administrative distance values.
- Different routing protocols use different metrics to select the best route in the presence of multiple routes. This chapter provided many examples of dynamic routing protocol metrics.
- This chapter distinguished between IGPs (which run within an autonomous system) and EGPs (which run between autonomous systems).
- This chapter contrasted the behavior of distance-vector and link-state routing protocols and showed how split horizon and poison reverse can prevent routing loops in a distance-vector routing protocol environment.
- This chapter described today's most popular routing protocols (including OSPF, EIGRP, and BGP), along with their characteristics.
- This chapter explained NAT and PAT and the importance of these technologies for permitting communications between private and public IP addresses.
- Finally, this chapter described First Hop Redundancy Protocols and provided examples such as HSRP and VRRP.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 9-5 lists these key topics and the page number where each is found.

**Key
Topic****Table 9-5** Key Topics for Chapter 9

Key Topic Element	Description	Page Number
Step list	Basic routing process	230
Table 9-1	Administrative Distance	237
Figure 9-9	IGPs Versus EGPs	239
List	Preventing routing loops	241
List	Routing protocol examples	242
Table 9-2	Comparing Dynamic Routing Protocols	243
Figure 9-13	Basic NAT Topology	245
Table 9-3	Names of NAT IP Addresses	245
List	Types of NAT	246
Figure 9-14	PAT Topology	247
Section	First Hop Redundancy Protocol (FHRP)	248

Complete Tables and Lists from Memory

Print a copy of Appendix C, “Memory Tables,” or at least the section for this chapter and complete as many of the tables as possible from memory. Appendix D, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

administrative distance (AD), Border Gateway Protocol (BGP), dynamic routing, Enhanced Interior Gateway Routing Protocol (EIGRP), First Hop Redundancy Protocol (FHRP), metric, network address translation (NAT), Open Shortest Path First (OSPF), port address translation (PAT), prefix length, route selection, static routing, subinterfaces, virtual IP (VIP)

Additional Resources

An OSPF Review: <https://www.ajsnetworking.com/an-ospf-review/>

EIGRP's Composite Metric: <https://www.ajsnetworking.com/eigrp-metric>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. If a PC on an Ethernet network attempts to communicate with a host on a different subnet, what destination IP address and destination MAC address will be placed in the packet/frame header sent by the PC?
 - a. Destination IP: IP address of the default gateway. Destination MAC: MAC address of the default gateway.
 - b. Destination IP: IP address of the remote host. Destination MAC: MAC address of the default gateway.
 - c. Destination IP: IP address of the remote host. Destination MAC: MAC address of the remote host.
 - d. Destination IP: IP address of the remote host. Destination MAC: MAC address of the local PC.
2. What protocol is used to request a MAC address that corresponds to a known IPv4 address on the local network?
 - a. IGMP
 - b. TTL
 - c. ICMP
 - d. ARP
3. What is the network address and subnet mask of a default route?
 - a. 255.255.255.255/32
 - b. 0.0.0.0/32
 - c. 255.255.255.255/0
 - d. 0.0.0.0/0
4. What routing protocol characteristic indicates the believability of the routing protocol (compared to other routing protocols)?
 - a. Weight
 - b. Metric
 - c. Administrative distance
 - d. SPF algorithm

5. Which of the following are distance-vector routing protocol features that can prevent routing loops? (Choose two.)
 - a. Reverse path forwarding (RPF) check
 - b. Split horizon
 - c. Poison reverse
 - d. Rendezvous point
6. Which of the following is a distance-vector routing protocol with a maximum usable hop count of 15?
 - a. BGP
 - b. EIGRP
 - c. RIP
 - d. OSPF
7. Which of the following routing protocols is an EGP?
 - a. BGP
 - b. EIGRP
 - c. RIP
 - d. OSPF
8. When you are configuring PAT as part of your address translation configuration, what is the source IP address used for translation with a large potential number of inside hosts?
 - a. The IP address on the physical outside interface
 - b. The IP address on a loopback interface
 - c. The IP address on the physical interface with the lowest interface identifier
 - d. The IP address automatically assigned to the backplane on the device
9. What specifically does NAT allow to be translated into Internet-routable IP addresses?
 - a. Virtual IP addresses
 - b. RFC1918 private IP addresses
 - c. APIPA addresses
 - d. Variable-length subnet masks

10. Which of the following technologies is an example of a Cisco-proprietary FHRP?
 - a. VRRP
 - b. LACP
 - c. HSRP
 - d. CARP

11. Which of the following is a value used by routers to determine the trustworthiness or reliability of routing information received from various sources?
 - a. Virtual IP address (VIP)
 - b. Time-to-Live (TTL)
 - c. Metric
 - d. Administrative distance (AD)



This chapter covers the following topics related to Objective 2.2 (Given a scenario, configure switching technologies and features) of the CompTIA Network+ N10-009 certification exam:

- Virtual Local Area Network (VLAN)
 - VLAN database
 - Switch Virtual Interface (SVI)
- Interface configuration
 - Native VLAN
 - Voice VLAN
 - 802.1Q tagging
 - Link aggregation
 - Speed
 - Duplex
- Spanning tree
- Maximum transmission unit (MTU)
 - Jumbo frames

Ethernet Switching Technologies

Odds are, when you are working with local area networks (LANs), you are working with Ethernet as the Layer 1 technology. Back in the mid-1990s, there was tremendous competition between technologies such as Ethernet, Token Ring, and Fiber Distributed Data Interface (FDDI). Today, however, you can see that Ethernet is the clear winner of those Layer 1 wars.

Of course, over the years, Ethernet has evolved. Several Ethernet standards exist in modern LANs, with a variety of distance and speed limitations. This chapter begins by reviewing the fundamentals of Ethernet networks, including a collection of Ethernet speeds and feeds. This chapter also delves into many of the features available with some Ethernet switches.

Foundation Topics

Principles of Ethernet

Xerox Corporation developed Ethernet in 1973, with the goal of creating a technology to allow computers to connect with laser printers. A quick survey of almost any corporate network reveals that Ethernet rose well beyond its humble beginnings; today it is used to interconnect devices such as computers, printers, wireless access points, servers, switches, routers, video game systems, and more. This section discusses early Ethernet implementations and limitations as well as modern Ethernet throughput and distance specifications.

NOTE Don't worry, your CompTIA Network+ exam will not focus on the legacy equipment described in this early history. The exam focuses on current technologies covered in detail later in this chapter.

In the network industry literature, you might come upon the term *IEEE 802.3* (where IEEE refers to the Institute of Electrical and Electronics Engineers standards body). In general, you can use the term *IEEE 802.3* interchangeably with the term *Ethernet*. However, be aware that these technologies have some subtle distinctions. For example, an Ethernet frame is a fixed-length frame, whereas the 802.3 frame length can vary.

Carrier-Sense Multiple Access with Collision Detection

Ethernet was based on the philosophy that all networked devices should be eligible, at any time, to transmit on a network. In contrast, Token Ring used a deterministic media access approach. Specifically, Token Ring networks passed a token around a network in a circular fashion, from one networked device to the next. Only when a networked device was in possession of that token was it eligible to send on the network.

Figure 10-1 depicts an Ethernet network using a legacy shared bus topology.

In this topology, all devices are directly connected to the network and are free to transmit at any time, if they have reason to believe no other transmission currently exists on the wire. Ethernet permits only a single frame to be on a network segment at any one time. So, before a device in this network transmits, it listens to the wire to see if there is currently any traffic being transmitted. If no traffic is detected, the networked device transmits its data. However, what if two devices simultaneously

have data to send? If they both listen to the wire at the same time, they could simultaneously, and erroneously, conclude that it is safe to send their data. However, when both devices simultaneously send their data, a *collision* occurs (see Figure 10-2), and data corruption results.

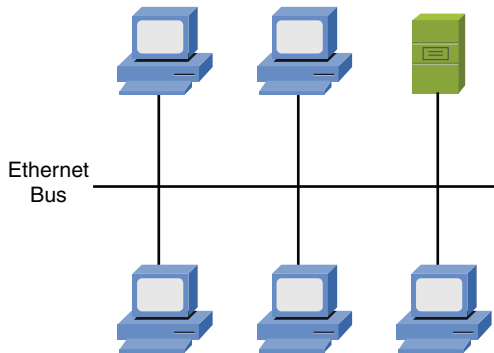


Figure 10-1 Ethernet Network Using a Shared Bus Topology

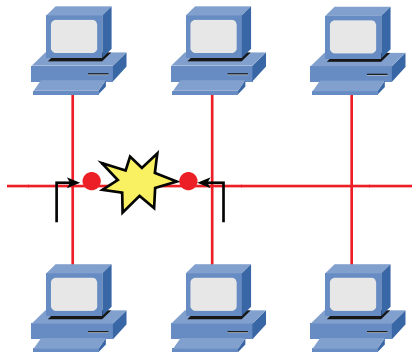


Figure 10-2 Collision on an Ethernet Segment

Fortunately, Ethernet was designed with a mechanism to detect collisions and allow the devices whose transmissions collided to retransmit their data at different times. Specifically, after the devices notice that a collision occurred, they independently set a random back-off timer. Each device waits for this random amount of time to elapse before again trying to transmit. Here is the logic: Because each device certainly picked a different amount of time to back off from transmitting, their transmissions should not collide the next time these devices transmit, as illustrated in Figure 10-3.

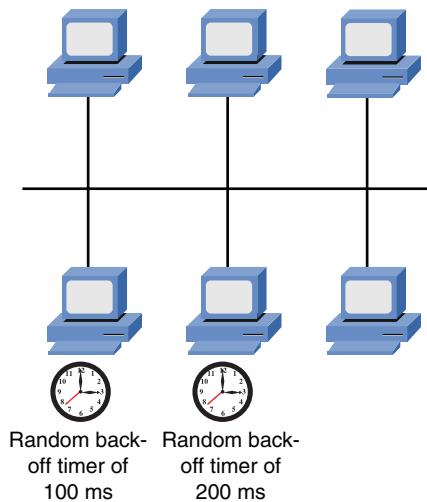


Figure 10-3 Recovering from a Collision with Random Back-off Timers

The procedure used by Ethernet to decide whether it is safe to transmit, detect collisions, and retransmit if necessary is called *carrier-sense multiple access with collision detection (CSMA/CD)*.

Let's break down CSMA/CD into its constituent components:

**Key
Topic**

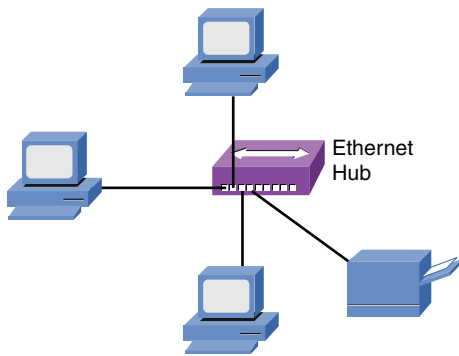
- **Carrier sense:** A device attached to an Ethernet network can listen to the wire, prior to transmitting, to make sure a frame is not being transmitted on the network segment.
- **Multiple access:** Unlike with a deterministic method of network access (for example, the method used by Token Ring), all Ethernet devices simultaneously have access to an Ethernet segment.
- **Collision detection:** If a collision occurs (perhaps because two devices were simultaneously listening to the network and simultaneously concluded that it was safe to send), Ethernet devices can detect that collision and set random back-off timers. After each device's random timer expires, the device again tries to transmit its data.

Despite Ethernet's CSMA/CD feature, Ethernet segments still suffer from scalability limitations. Specifically, the likelihood of collisions increases as the number of devices on a shared Ethernet segment increases.

An alternate approach is CSMA/CA, where CA refers to *collision avoidance*. This technology is common in wireless networks and was made famous by Token Ring in early LANs.

With wired Ethernet, devices on a shared Ethernet segment belong to the same *collision domain*. One example of a shared Ethernet segment is a 10BASE5 or 10BASE2 network with multiple devices attaching to the same cable. On that cable, only one device can send at any one time. Therefore, all devices attached to the thicknet or thinnet cable are in the same collision domain.

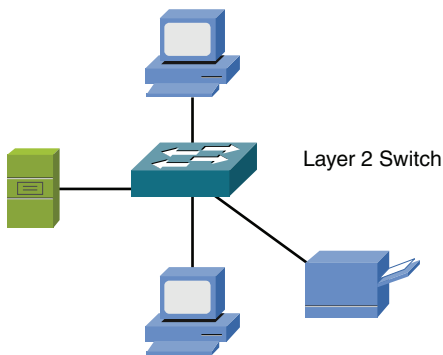
Similarly, devices connected to a legacy Ethernet hub are in the same collision domain (see Figure 10-4). A hub is a Layer 1 device and does not make forwarding decisions. Instead, a hub takes bits in on one port and sends them out all the other hub ports except the one on which the bits were received.



One Collision Domain

Figure 10-4 Shared Ethernet Hub: One Collision Domain

Ethernet switches dramatically increase the scalability of Ethernet networks by creating multiple collision domains (see Figure 10-5). In fact, every port on an Ethernet switch is in its own collision domain.



Four Collision Domains

Figure 10-5 Ethernet Switch: One Collision Domain per Port

NOTE Do switches seem familiar to you? They should. Because they are so important in today’s networks, we also covered them in Chapter 2, “Networking Appliances, Applications, and Functions.”

Ethernet switches have a less obvious but powerful benefit: Because a switch port connects to a single device, there is no chance of collision. With no chance of collision, collision detection is no longer needed, and with collision detection disabled, network devices can run in *full-duplex* mode rather than *half-duplex* mode. In full-duplex mode, a device can simultaneously send and receive at the same time.

When multiple devices are connected to the same shared Ethernet segment, such as a Layer 1 hub, CSMA/CD must be enabled. As a result, the network must work in half-duplex mode, which means that only a single networked device can transmit or receive at any one time. In half-duplex mode, a networked device cannot simultaneously send and receive, so the device makes inefficient use of the network’s bandwidth.

Another important mechanism in an Ethernet network is flow control. *Flow control* is a mechanism for temporarily stopping the transmission of data on Ethernet-based networks. The goal of this mechanism is to avoid packet loss in the presence of network congestion.

Distance and Speed Limitations

Remember, as we covered in Chapter 5, “Transmission Media and Transceivers,” the bandwidth of a network (*speed*) is measured in terms of how many bits the network can transmit during a 1-second period of time. In fact, let’s revisit the Ethernet bandwidth capabilities using Table 10-1. This table defines the common bandwidths supported on distinct types of Ethernet networks.

Key
Topic

Table 10-1 Ethernet Bandwidth Capacities

Ethernet Type	Bandwidth Capacity
Standard Ethernet	10Mbps: 10 million bits per second (that is, 10 megabits per second)
Fast Ethernet	100Mbps: 100 million bits per second (that is, 100 megabits per second)
Gigabit Ethernet	1Gbps: 1 billion bits per second (that is, 1 gigabit per second)
10-Gigabit Ethernet	10Gbps: 10 billion bits per second (that is, 10 gigabits per second)
40-Gigabit Ethernet	40 Gbps: 40 billion bits per second (that is, 40 gigabits per second)
100-Gigabit Ethernet	100Gbps: 100 billion bits per second (that is, 100 gigabits per second)

Remember, the key here is that the type of cabling used in an Ethernet network influences the bandwidth capacity and the distance limitation of the network. For example, fiber-optic cabling often has a higher bandwidth capacity and a longer distance limitation than twisted-pair cabling.

When it comes to the size of a frame the media can send, there is a *maximum transmission unit (MTU)* value the media can accommodate. You can even configure support for *jumbo frames*, which are Ethernet frames with more than 1500 bytes of payload (the limit set by the IEEE 802.3 standard).

Ethernet Switch Features

Beyond basic frame forwarding, many Layer 2 Ethernet switches offer a variety of other features to enhance such things as network performance, redundancy, security, management, flexibility, and scalability. Although the specific features offered by a switch vary, this section introduces you to some of the most common features of switches.

Virtual LANs

In a basic switch configuration, all ports on a switch belong to the same *broadcast domain*. In such an arrangement, a broadcast received on one port gets forwarded out all other ports.

Also, from a Layer 3 perspective, all devices connected in a broadcast domain have the same *network address*. Chapter 7, “IPv4 Addressing,” gets into the binary math involved in assigning IP addresses (that is, logical Layer 3 addresses) to networked devices. A portion of that address is the address of the network to which that device is attached. The remainder of that address is the address of the device itself. Devices that have the same network address belong to the same network, or *subnet*.

Say that you decide to place PCs from different departments within your company into their own subnet. One reason you might want to do this is for security purposes. For example, by having the Accounting department in a separate subnet (that is, a separate broadcast domain) from the Sales department, devices in one subnet will not see the broadcasts being sent on the other subnet.

Another reason that you might want to do this is to make the overall network segment more efficient. Remember that excessive broadcast frames can cause a network to suffer, and there are plenty of operations that rely on broadcasts to function properly. *Address Resolution Protocol (ARP)* is a great example of such an operation. ARP is a broadcast-based solution that permits a system to discover the MAC address that

coordinates to a particular IP address. IPv6 eliminates this challenge altogether with the introduction and use of *Neighbor Discovery Protocol (NDP)*. As you might guess, NDP is not broadcast based. This is fortunate, as broadcasts are not supported in IPv6.

The PCs belonging to the different departments in your company are scattered across multiple floors in a building (see Figure 10-6). The Accounting and Sales departments each have a PC on each floor of the building. Because the wiring for each floor runs back to a wiring closet on that floor, to support these two subnets using a switch's default configuration, you need to install two switches on each floor. For traffic to travel from one subnet to another subnet, that traffic has to be routed, meaning that a device such as a multilayer switch or a router forwards traffic based on a packet's destination network addresses. So, in this example, the Accounting department switches are interconnected and then connect to a router, and the Sales department switches are connected similarly.

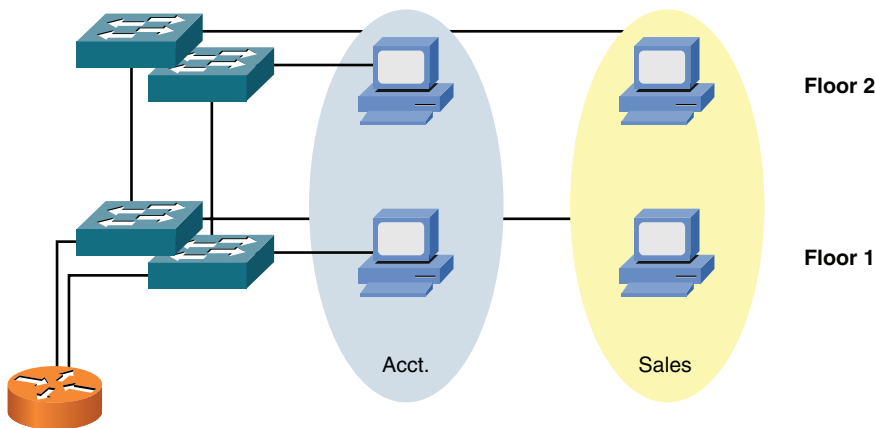


Figure 10-6 Example: All Ports on a Switch Belonging to the Same Subnet

The design described here lacks efficiency because you must install at least one switch per subnet. A more efficient design would be to logically separate a switch's ports into different broadcast domains. Then, an Accounting department PC and a Sales department PC could connect to the same switch, even though those PCs belong to different subnets. Fortunately, *virtual LANs (VLANs)* make such a design possible.

As the administrator of the network, you can easily assign VLANs to interfaces on the switch. In fact, if you need to address all the interfaces on a device that are part of a particular VLAN, you can use a *switch virtual interface (SVI)*. This interface “represents” all of the member interfaces for the specific VLAN. This special, virtual

interface has an up or down status based on whether there are physical interfaces enabled and healthy in the VLAN specified.

With VLANs, as illustrated in Figure 10-7, a switch can have its ports logically divided into more than one broadcast domain (that is, more than one subnet or VLAN). Then, devices that need to connect to those VLANs can connect to the same physical switch, but they remain logically separate from one another.

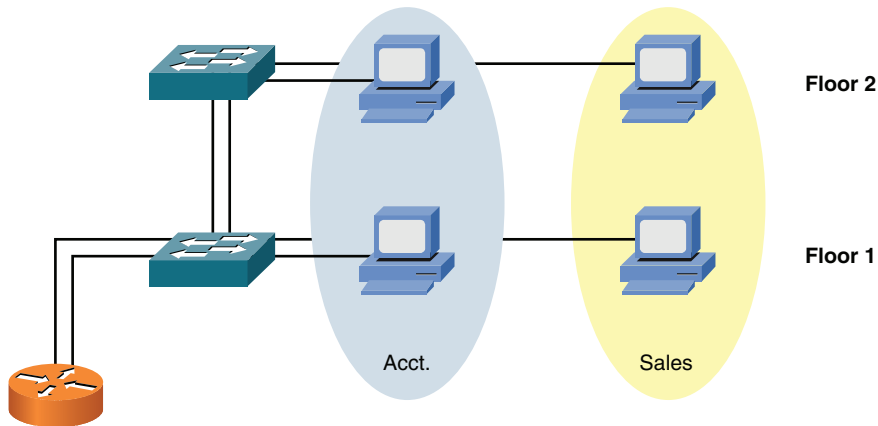


Figure 10-7 Example: Ports on a Switch Belonging to Different VLANs

NOTE You will often hear engineers speaking of the *VLAN database* on a switch as the place where the VLANs that have been configured are stored. Interestingly, some switches do use a separate storage structure to store this “database.”

VLANs are very handy with another popular configuration requirement in many modern network environments: You can use a special-purpose VLAN termed a *voice VLAN* to segment and provide network access for VoIP packets. These packets need access to the data network for transport, and they also need special priority treatment to ensure that the voice call quality always remains excellent. Using a voice VLAN is an ideal segmentation strategy for all these needs.

One challenge with VLAN configuration in large environments is the need to configure identical VLAN information on all switches. Manually performing this configuration is time-consuming and error prone. However, switches from Cisco Systems support VLAN Trunking Protocol (VTP), which allows a VLAN created on one switch to be propagated to other switches in a group of switches (that is, a VTP domain). VTP information is carried over a *trunk connection*, as discussed in the next section.

Switch Configuration for an Access Port

Configurations used on a switch port may vary, based on the manufacturer of the switch. Example 10-1 shows a sample configuration on an access port (without trunking) on a Cisco Catalyst switch. A line with a leading ! is a comment used to document the next line(s) of the configuration.

Notice in this configuration the use of port security, which is a small but useful step in securing a network. In this configuration example, the port can learn only two MAC addresses—perhaps the MAC address of a VoIP phone and the computer that connects to that phone.

Example 10-1 Switch Access Port Configuration

```
! Move into configuration mode for interface gig 0/21
SW1(config)# interface GigabitEthernet0/21

! Add a text description of what the port is used for
SW1(config-if)# description Access port in Sales VLAN 21

! Define the port as an access port, and not a trunk port
SW1(config-if)# switchport mode access

! Assign the port to VLAN 21
SW1(config-if)# switchport access vlan 21

! Enable port security
SW1(config-if)# switchport port-security

! Control the number of MAC addresses the switch may learn
! from device(s) connected to this switch port
SW1(config-if)# switchport port-security maximum 2

! Restrict any frames from MAC addresses above the 2 allowed
SW1(config-if)# switchport port-security violation restrict

! Set the speed to 1,000 Mbps (1 Gigabit per second)
SW1(config-if)# speed 1000

! Set the duplex to full
SW1(config-if)# duplex full

! Configure the port to begin forwarding without waiting the
! standard amount of time normally set by Spanning Tree Protocol
SW1(config-if)# spanning-tree portfast
```

Trunks

One challenge with carving up a switch into multiple VLANs is that several switch ports (that is, one port per VLAN) could be consumed by connecting a switch to a switch or a switch to a router. A more efficient approach is to allow traffic for multiple VLANs to travel over a single connection, as shown in Figure 10-8. This type of connection is called a *trunk*.

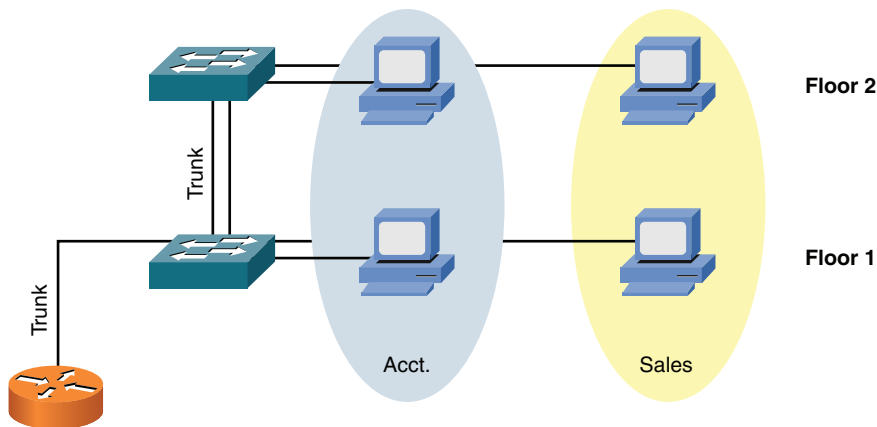


Figure 10-8 Example: Trunking Between Switches

The most popular trunking standard today is IEEE 802.1Q, which is often referred to as *dot1q*. One of the VLANs traveling over an 802.1Q trunk is called a **native VLAN**. Frames belonging to the native VLAN are sent unaltered over the trunk (untagged/no tag). However, to distinguish other VLANs from one another, the remaining VLANs are tagged.

NOTE IEEE 802.1Q is often called **802.1Q tagging** or *port tagging*.

Specifically, a nonnative VLAN has 4 tag bytes (where a *byte* is a collection of 8 bits) added to the Ethernet frame (that is, tagged frame). Figure 10-9 shows the format of an IEEE 802.1Q header with these 4 bytes.

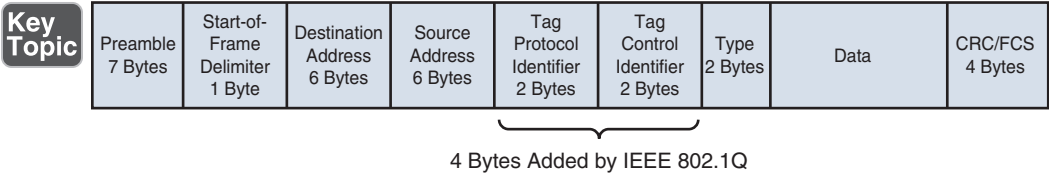


Figure 10-9 IEEE 802.1Q Header

One of these bytes contains a VLAN field, which indicates to which VLAN a frame belongs. The devices (for example, switch, multilayer switch, router) at each end of a trunk interrogate that field to determine to which VLAN an incoming frame is associated. As you can see by comparing Figures 10-6, 10-7, and 10-8, VLAN and trunking features allow switch ports to be used far more efficiently than merely relying on a default switch configuration.

NOTE What type of Ethernet media do you use for a trunk link? Well, it used to be that you needed a special Ethernet cable called a *crossover cable* for a trunk link. This is no longer the case in most networks, as switches now support *auto-medium-dependent interface crossover (MDI-X)* technology. This technology essentially permits a switch to detect the type of Ethernet cable used (straight-through versus crossover) and adjust accordingly so that communication between the two network devices (switches in this case) is successful.

Switch Configuration for a Trunk Port

Example 10-2 shows a sample configuration on a trunk port on a Cisco Catalyst switch. A line with a leading `!` is a comment used to document the next line(s) of the configuration.

Example 10-2 Sample Trunk Port Configuration

```
! Go to interface config mode for interface Gig 0/22
SW1(config)# interface GigabitEthernet0/22

! Add a text description
SW1(config-if)# description Trunk to another switch

! Specify that this is a trunk port
SW1(config-if)# switchport mode trunk

! Specify the trunking protocol to use
SW1(config-if)# switchport trunk encapsulation dot1q

! Specify the native VLAN to use for un-tagged frames
SW1(config-if)# switchport trunk native vlan 5

! Specify which VLANs are allowed to go on the trunk
SW1(config-if)# switchport trunk allowed vlan 1-50
```

Spanning Tree Protocol

Administrators of traditional corporate telephone networks often used to boast about their telephone systems—that is, private branch exchange (PBX) systems—having *five nines* availability. Five nines availability means that a system is up and functioning 99.999% of the time, which translates to only about 5 minutes of downtime per year.

Traditionally, corporate data networks struggled to compete with corporate voice networks in terms of availability. Today, however, many networks that traditionally carried only data now carry voice, video, and data. Therefore, availability is an especially important design consideration.

To improve network availability at Layer 2, many networks have redundant links between switches. However, unlike Layer 3 packets, Layer 2 frames lack a Time-to-Live (TTL) field. As a result, a Layer 2 frame can circulate endlessly through a looped Layer 2 topology. Fortunately, IEEE 802.1D ***Spanning Tree Protocol (STP)*** enables a network to physically have Layer 2 loops while strategically blocking data from flowing over one or more switch ports to prevent the looping of traffic.

In the absence of STP, if a network has parallel paths, two significant symptoms include corruption of a switch's MAC address table and broadcast storms, where frames loop over and over throughout the switched network. An enhancement to the original STP is *802.1w*, which is also called ***Rapid Spanning Tree Protocol (RSTP)*** because it does a quicker job of adjusting to network conditions, such as the addition or removal of Layer 2 links in the network. RSTP is covered in more detail later in this chapter in the section “Modern Enhancements to STP.”

Shortest Path Bridging (SPB; IEEE 802.1aq) is a protocol that is more scalable than STP in larger environments (with hundreds of switches interconnected).

Corruption of a Switch's MAC Address Table

A switch's MAC address table can dynamically learn what MAC addresses are available off its ports. However, in the case of an STP failure, a switch's MAC address table can become corrupted (see Figure 10-10).

In Figure 10-10, PC1 is transmitting traffic to PC2. When the frame sent from PC1 is transmitted on segment A, the frame is seen on the Gig 0/1 ports of switches SW1 and SW2, causing both switches to add an entry to their MAC address tables, associating MAC address AAAA.AAAA.AAAA with port Gig 0/1. Because STP is not functioning, both switches then forward the frame out on segment B. As a result, PC2 receives two copies of the frame. Also, switch SW1 sees the frame forwarded out of switch SW2's Gig 0/2 port. Because the frame has source MAC address AAAA.AAAA.AAAA, switch SW1 incorrectly updates its MAC address table,

indicating that MAC address AAAA.AAAA.AAAA resides off port Gig 0/2. Similarly, switch SW2 sees the frame forwarded on to segment B by switch SW1 on its Gig 0/2 port. Therefore, switch SW2 also incorrectly updates its MAC address table.

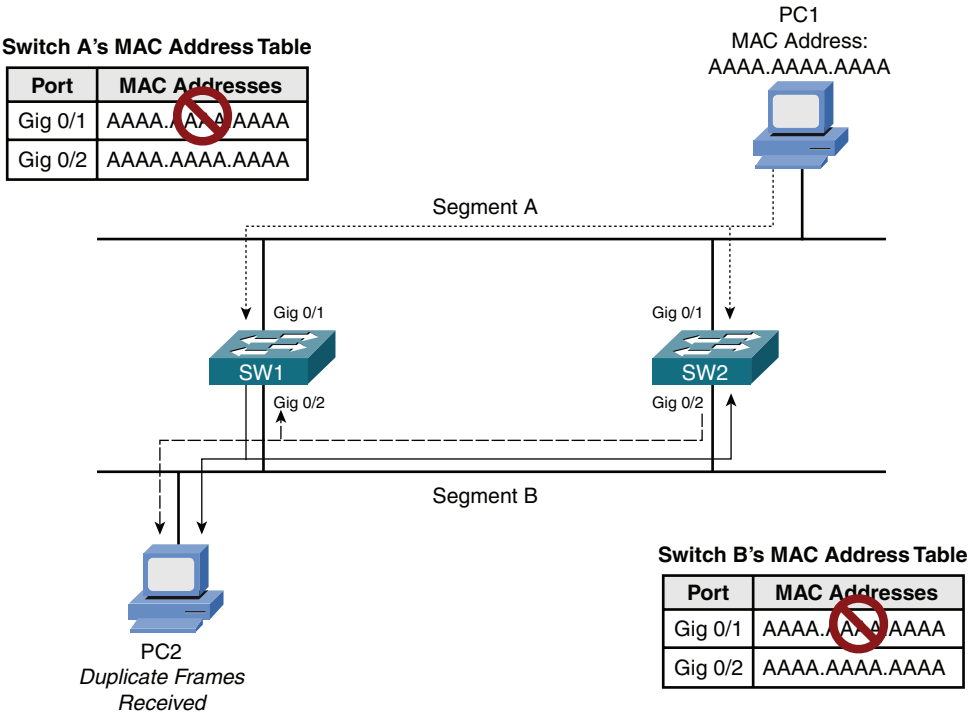


Figure 10-10 MAC Address Table Corruption

Broadcast Storms

As previously mentioned, when a switch receives a broadcast frame (that is, a frame destined for MAC address FFFF.FFFF.FFFF), the switch floods the frame out all switch ports other than the port on which the frame was received. Because a Layer 2 frame does not have a TTL field, a broadcast frame endlessly circulates through the Layer 2 topology, consuming resources on both switches and attached devices (for example, user PCs).

Figure 10-11 and the following list illustrate how a broadcast storm can form in a Layer 2 topology when STP is not functioning correctly:

- Step 1.** PC1 sends a broadcast frame on to segment A, and the frame enters each switch on port Gig 0/1.

Key Topic

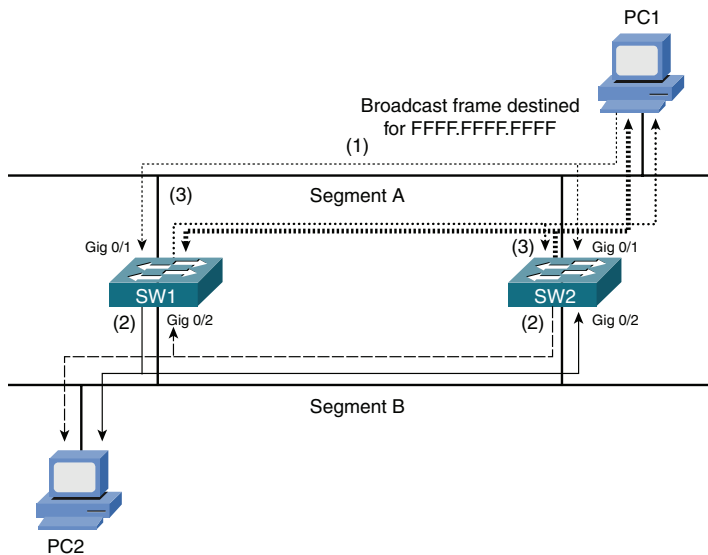


Figure 10-11 Broadcast Storm

- Step 2.** Both switches flood a copy of the broadcast frame out their Gig 0/2 ports (that is, on to segment B), causing PC2 to receive two copies of the broadcast frame.
- Step 3.** Both switches receive a copy of the broadcast frame on their Gig 0/2 ports (that is, from segment B) and flood the frame out their Gig 0/1 ports (that is, on to segment A), causing PC1 to receive two copies of the broadcast frame.

This behavior continues as the broadcast frame copies continue to loop through the network. The performance of PC1 and PC2 is affected because they also continue to receive copies of the broadcast frame.

STP Operation

STP prevents Layer 2 loops—which might result in a broadcast storm or corruption of a switch’s MAC address table—from occurring in a network. Switches in an STP topology are classified as one of the following:

Key Topic

- **Root bridge:** A root bridge is a switch elected to act as a reference point for a spanning tree. The switch with the lowest bridge ID (BID) is elected as the root bridge. The BID is made up of a priority value and a MAC address.
- **Nonroot bridge:** All other switches in the STP topology are nonroot bridges.

Figure 10-12 illustrates root bridge election in a network. Notice that in this case, the bridge priorities are both 32768; therefore, the switch with the lowest MAC address (that is, SW1) is elected as the root bridge.

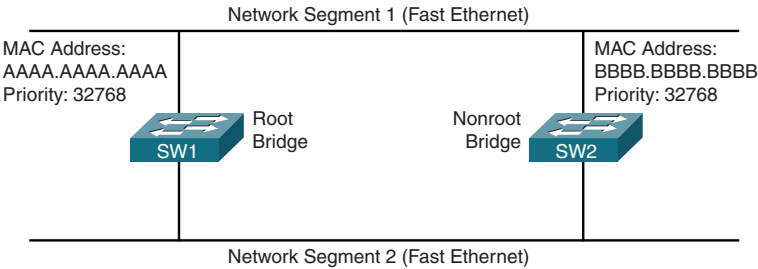


Figure 10-12 Root Bridge Election

Ports that interconnect switches in an STP topology are categorized as one of the port types described in Table 10-2.

Key Topic

Table 10-2 STP Port Types

Port Type	Description
Root port	Every nonroot bridge has a single root port, which is the port on that switch that is closest to the root bridge in terms of cost.
Designated port	Every network segment has a single designated port, which is the port on that segment that is closest to the root bridge in terms of cost. Therefore, all ports on a root bridge are designated ports.
Nondesignated port	Nondesignated ports block traffic to create a loop-free topology.

Figure 10-13 illustrates these port types. Notice that both links are equal in this case, with a cost of 19, because both links are Fast Ethernet links; therefore, the root port for switch SW2 is selected because it has the lowest port ID.

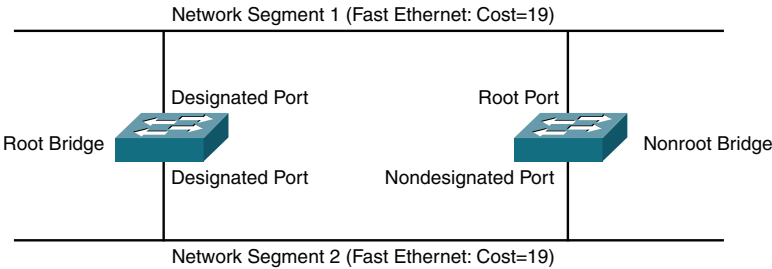


Figure 10-13 Identifying STP Port Roles

Figure 10-14 shows a similar topology to Figure 10-13. In Figure 10-14, however, the top link is running at a speed of 10Mbps, whereas the bottom link is running at a speed of 100Mbps. Because switch SW2 seeks to get back to the root bridge (that is, switch SW1) with the least cost, port Gig 0/2 on switch SW2 is selected as the root port.

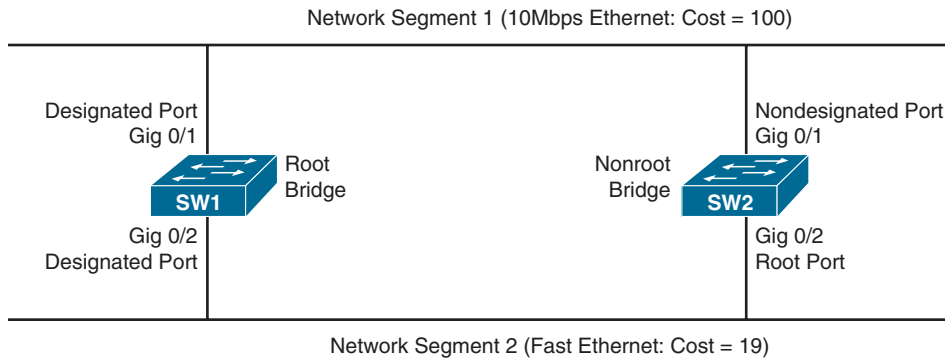


Figure 10-14 STP with Different Port Costs

Specifically, port Gig 0/1 has a cost of 100, and Gig 0/2 has a cost of 19. Table 10-3 shows the port costs for various link speeds.

**Key
Topic**

Table 10-3 STP Port Cost

Link Speed	STP Port Cost
10Mbps (Ethernet)	100
100Mbps (Fast Ethernet)	19
1Gbps (Gigabit Ethernet)	4
10Gbps (10-Gigabit Ethernet)	2
40Gbps (40-Gigabit Ethernet)	2

NOTE To make things obvious in our illustration, we used the old legacy 10Mbps Ethernet against Fast Ethernet. Remember that in many networks today, it can be hard to find connections under 1Gbps in speed.

Nondesignated ports do not forward traffic during normal operation but do receive bridge protocol data units (BPDUs). Switches exchange STP information in the form of BPDUs, which contain useful information for STP elections, path cost calculation, link suppression, and loop detection. If a link in the topology goes down,

the nondesignated port detects the link failure and determines whether it needs to transition to the forwarding state.

If a nondesignated port needs to transition to the forwarding state, it does not do so immediately. Rather, it transitions through the following states:

**Key
Topic**

- **Blocking:** The port remains in the blocking state for 20 seconds by default. During this time, the nondesignated port evaluates BPDUs in an attempt to determine its role in the spanning tree.
- **Listening:** The port moves from the blocking state to the listening state and remains in this state for 15 seconds by default. During this time, the port sources BPDUs, which inform adjacent switches of the port's intent to forward data.
- **Learning:** The port moves from the listening state to the learning state and remains in this state for 15 seconds by default. During this time, the port begins to add entries to its MAC address table.
- **Forwarding:** The port moves from the learning state to the forwarding state and begins to forward frames.

Modern Enhancements to STP

Rapid Spanning Tree Protocol (RSTP), defined in IEEE 802.1w, is an evolution of the original STP designed to provide faster convergence in our Ethernet networks. RSTP enhances the original protocol by introducing new port roles and states that allow for rapid transition to the forwarding state, thereby significantly reducing the time it takes for network topology changes to be recognized and for traffic to be rerouted. Unlike STP, which can take up to 50 seconds to respond to a topology change, RSTP can achieve convergence in a few seconds, improving the efficiency and reliability of network operations.

RSTP introduces the concept of edge ports, which are ports directly connected to end devices and can immediately transition to the forwarding state without going through the traditional listening and learning states. Additionally, RSTP defines alternative and backup ports to handle topology changes more efficiently. An alternative port provides a backup path to the root bridge in case the current path fails, while a backup port offers a redundant connection on a shared LAN segment. These enhancements enable RSTP to quickly adapt to network changes, ensuring minimal disruption to data traffic.

Multiple Spanning Tree Protocol (MSTP), specified in IEEE 802.1s, builds on RSTP to address the scalability issues associated with managing multiple VLANs in large networks. MSTP allows multiple VLANs to be mapped to a single spanning

tree instance, reducing the number of spanning tree instances that need to be maintained and managed. This approach simplifies the network design and optimizes resource utilization, as the same spanning tree can handle multiple VLANs with similar traffic patterns and requirements. By grouping VLANs into regions, MSTP enhances network efficiency and simplifies the implementation of VLAN-based network policies.

Link Aggregation

If all ports on a switch are operating at the same speed (for example, 1Gbps), the ports most likely to experience congestion are ports connecting to another switch or router. For example, imagine a wiring closet switch connected (via Fast Ethernet ports) to multiple PCs. That wiring closet switch has an uplink to the main switch for a building. Because this uplink port aggregates multiple 100Mbps connections and the uplink port is also operating at 100Mbps, it can quickly become congested if multiple PCs are transmitting traffic that needs to be sent over that uplink, as shown in Figure 10-15.

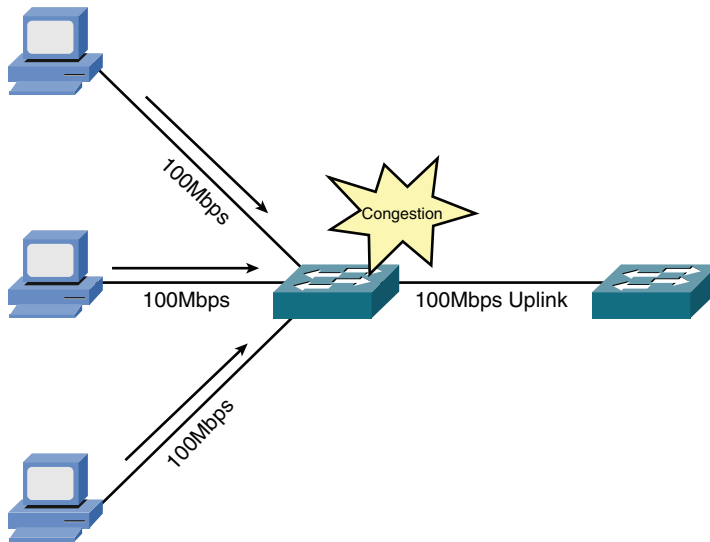


Figure 10-15 Uplink Congestion

To help alleviate congested links between switches, you can (on some) switch models logically combine multiple physical connections into a single logical connection over which traffic can be sent. This feature, which is illustrated in Figure 10-16, is called *link aggregation*.

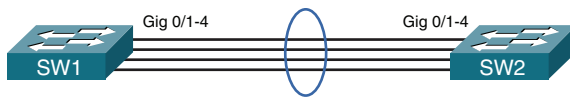


Figure 10-16 Link Aggregation

Although vendor-proprietary solutions for link aggregation have existed for some time, some solutions faced a couple of common issues:

- Each link in the logical bundle was a potential single point of failure.
- Each end of the logical bundle had to be manually configured.

In 2000, the IEEE ratified the 802.3ad standard for link aggregation. This standard supports *Link Aggregation Control Protocol (LACP)*. Unlike some of the older vendor-proprietary solutions, LACP supports automatic configuration and prevents an individual link from becoming a single point of failure. Specifically, with LACP, if a link fails, that link's traffic is forwarded over a different link. The Cisco Systems implementation of LACP is called *EtherChannel*. Groups of interfaces that make up an EtherChannel bundle are often referred to as a *link aggregation group (LAG)*. An EtherChannel group can be configured to act as a Layer 2 access port and support only a single VLAN, or it can be configured to act as a Layer 2 802.1Q trunk and support multiple VLANs of the LAG. A LAG can also be configured as a Layer 3 routed interface if the switch supports that feature. In the case of a Layer 3 EtherChannel, an IP address would be applied to the logical interface that represents the LAG. Another term related to LACP and LAGs is *port bonding*, which also refers to the same concept of grouping multiple ports and using them as a single logical interface.

LACP Configuration

Example 10-3 shows a sample configuration of LACP on a Cisco switch. A line with a leading ! is a comment used to document the next line(s) of the configuration.

Example 10-3 LACP Configuration

```
! Move to interface that will be part of the LACP group
SW1(config)# interface GigabitEthernet0/16

! Assign this interface to the LACP group 1
SW1(config-if)# channel-group 1 mode active
```

```
! Move to the other interface(s) that will be part of
! the same group
SW1(config-if)# interface GigabitEthernet0/17
SW1(config-if)# channel-group 1 mode active

! Configure the group of interfaces as a logical group
! Configuration here will also apply the individual
! interfaces that are part of the group
SW1(config-if)# interface Port-channel 1

! Apply the configuration desired for the group
! LACP groups can be access or trunk ports depending
! on how the configuration of the logical port-channel interface
! In this example the LAG will be acting as a trunk
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk encapsulation dot1q
```

Power over Ethernet

Some switches not only transmit data over a connected UTP cable but also use that cable to provide power to an attached device. For example, say that you want to mount a wireless access point (AP) on the ceiling. Although no electrical outlet is available near the AP's location, you can, as an example, run a Cat 6 UTP plenum cable above the drop ceiling and connect it to the AP. Some APs allow the switch at the other end of the UTP cable to provide power over the same wires that carry data. Examples of other devices that might benefit from receiving power from an Ethernet switch include security cameras and IP phones.

The switch feature that gives power to attached devices is called *Power over Ethernet (PoE)*, and it is defined by the IEEE 802.3af standard. As shown in Figure 10-17, the PoE feature of a switch checks for 25,000 ohms of resistance in the attached device. To check the resistance, the switch applies as much as 10V of direct current (DC) across specific pairs of wires (that is, pins 1 and 2 combine to form one side of the circuit, and pins 3 and 6 combine to form the other side of the circuit) connecting back to the attached device and checks to see how much current flows over those wires. For example, if the switch applies 10V DC across those wires and notices 0.4 mA (milliamps) of current, the switch concludes that the attached device has 25,000 ohms of resistance across those wires (based on the formula $E = IR$, where E represents voltage, I represents current, and R represents resistance). The switch could then apply power across those wires.

Key Topic

Switch applies 2.8V–10V DC to two pairs of leads to detect a 25,000-ohm resistor in the attached device



Figure 10-17 PoE

Next, the switch must determine how much power the attached device needs. The switch makes this determination by applying 15.5V–20.5V DC (making sure that the current never exceeds 100 mA) to the attached device for a brief period of time (less than one-tenth of a second). The amount of current flowing to the attached device tells the switch the *power class* of the attached device. The switch then knows how much power should be made available on the port connecting to the device requiring power, and it begins supplying an appropriate amount of voltage (in the range 44V–57V) to the attached device.

The IEEE 802.3af standard can supply a maximum of 15.4W (watts) of power. However, the later standard IEEE 802.3at offers as much as 32.4W of power, enabling PoE to support a wider range of devices, such as power-hungry IP video cameras. This newer standard for PoE is often referred to as *Power over Ethernet Plus* (PoE+).

Other Switch Features

Switch features, such as those previously described, vary widely by manufacturer, and some switches offer a variety of security features. For example, MAC filtering might be supported, which allows traffic to be permitted or denied based on a device's MAC address. Other types of traffic filtering might also be supported, based on criteria such as IP address information (for multilayer switches).

For monitoring and troubleshooting purposes, interface *diagnostics* might be accessible. This diagnostic information might include various error conditions, such as late collisions or cyclic redundancy check (CRC) errors, which might indicate a duplex mismatch.

Some switches also support *quality of service* (QoS) settings, which make it possible to forward traffic based on the traffic's priority markings. Also, some switches have the ability to perform marking and remarking of traffic priority values.

Real-World Case Study

Acme, Inc. has made some decisions regarding the setup of its LAN. For connections from the client machines to the switches in the wiring closets (the intermediate distribution frames), it will use unshielded twisted-pair Category 6a cabling with the switch ports configured as access ports and set to 1000Mbps to match the Gigabit Ethernet capabilities of the client computers that will be connecting to the switches.

Multiple VLANs will be used. The computers that are being used by the Sales department will be connected to ports on a switch that are configured as access ports for the specific VLAN for Sales. Computers used by Human Resources will connect to switch ports that are configured as access ports for the Human Resources VLAN. There will be separate IP subnetworks associated with each of the VLANs.

The fiber connections that will go vertically through the building and connect the switches in the intermediate distribution frames (IDFs) to the main distribution frame (MDF) in the basement will be running at 1Gbps each, and multiple fiber cables will be used. LACP will be used for these vertical connections to make the multiple fiber links work together as part of one logical EtherChannel interface. For the LACP connections between the IDFs and MDF to support multiple VLANs, the LAG will be configured as a trunk using 802.1Q tagging. Routing between the VLANs will be done by multilayer switches that are located near the MDF.

Spanning tree will be enabled on the switches so that in the event of parallel paths between switches, a Layer 2 loop can be prevented.

To support IP-based telephones in the offices, the switches will also provide PoE+, which can supply power to the IP phones over the Ethernet cables that run between the switch in the IDF and the IP phones.

Summary

Here are the main topics covered in this chapter:

- This chapter described the origins of Ethernet, including a discussion of Ethernet's CSMA/CD features.
- This chapter compared a variety of Ethernet standards in terms of media type, network bandwidth, and distance limitation.
- This chapter covered various features that might be available on modern Ethernet switches, including VLANs, trunking, link aggregation, PoE, and PoE+.
- This chapter also detailed Spanning Tree Protocol (STP) as it exists on modern Ethernet switches as a loop prevention mechanism.
- Finally, this chapter described the maximum transmission unit (MTU) and jumbo frames.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 10-4 lists these key topics and the page number where each is found.

Key
Topic

Table 10-4 Key Topics for Chapter 10

Key Topic Element	Description	Page Number
List	Components of CSMA/CD	260
Table 10-1	Ethernet Bandwidth Capacities	262
Figure 10-9	IEEE 802.1Q Header Tag	267
Figure 10-11	Broadcast Storm	271
List	STP operation	271
Table 10-2	STP port types	272
Table 10-3	STP port cost	273
List	STP port states	274
Figure 10-17	PoE	278

Complete Tables and Lists from Memory

Print a copy of Appendix B, “Memory Tables,” or at least the section for this chapter and complete as many of the tables as possible from memory. Appendix C, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

802.1Q tagging, full-duplex, half-duplex, jumbo frames, link aggregation, maximum transmission unit (MTU), native VLAN, Spanning Tree Protocol (STP), speed, Switch Virtual Interface (SVI), virtual LAN (VLAN), VLAN database, voice VLAN

Additional Resources

Migrating from STP to RSTP: <https://www.ajsnetworking.com/cisco-migrating-from-stp-to-rstp/>

VLAN Trunking Protocol Version 3: <https://www.ajsnetworking.com/vlan-trunking-protocol-vtp-version-3/>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. What is the distance limitation of a 1000BASE-T Ethernet network?
 - a. 100 m
 - b. 185 m
 - c. 500 m
 - d. 2 km
2. If two devices simultaneously transmit data on an Ethernet network and a collision occurs, what does each station do in an attempt to resend the data and avoid another collision?
 - a. Each device compares the other device's priority value (determined by IP address) with its own, and the device with the highest priority value transmits first.
 - b. Each device waits for a clear-to-send (CTS) signal from the switch.
 - c. Each device randomly picks a priority value, and the device with the highest value transmits first.
 - d. Each device sets a random back-off timer, and a device attempts retransmission after the timer expires.
3. Which of the following statements are true regarding VLANs? (Choose two.)
 - a. A VLAN has a single broadcast domain.
 - b. For traffic to pass between two VLANs, that traffic must be routed.
 - c. Because of a switch's MAC address table, traffic does not need to be routed to pass between two VLANs.
 - d. A VLAN has a single collision domain.

4. What name is given to a VLAN on an IEEE 802.1Q trunk whose frames are not tagged?
 - a. Native VLAN
 - b. Default VLAN
 - c. Management VLAN
 - d. VLAN 0
5. In a topology running STP, every network segment has a single _____ port, which is the port on that segment that is closest to the root bridge in terms of cost.
 - a. Root
 - b. Designated
 - c. Nondesignated
 - d. Nonroot
6. What is the IEEE standard for link aggregation?
 - a. 802.1Q
 - b. 802.3ad
 - c. 802.1d
 - d. 802.3af
7. What is the maximum amount of power a switch is allowed to provide per port, according to the IEEE 802.3af standard?
 - a. 7.7W
 - b. 15.4W
 - c. 26.4W
 - d. 32.4W
8. What is the purpose of LACP in a modern network?
 - a. To provide a backup copy of the VLAN database
 - b. To bundle interfaces together to act as a single interface
 - c. To provide additional NVRAM for modern switches
 - d. To provide additional interface buffer space

9. Which of the following represents a system in which data simultaneously transmits in two directions?
 - a. Half-duplex
 - b. MTU
 - c. Jumbo frame
 - d. Full-duplex

10. Which is a virtual interface configured on a Layer 2 switch to enable routing between VLANs?
 - a. SVI
 - b. 802.1Q
 - c. Native VLAN
 - d. VLAN tag



This chapter covers the following topics related to Objective 2.3 (Given a scenario, select and configure wireless devices and technologies) of the CompTIA Network+ N10-009 certification exam:

- Channels
 - Channel width
 - Non-overlapping channels
 - Regulatory impacts
 - 802.11h
- Frequency options
 - 2.4GHz
 - 5GHz
 - 6GHz
 - Band steering
- Service set identifier (SSID)
 - Basic service set identifier (BSSID)
 - Extended service set identifier (ESSID)
- Network types
 - Mesh networks
 - Ad hoc
 - Point to point
 - Infrastructure
- Encryption
 - Wi-Fi Protected Access 2 (WPA2)
 - WPA3
- Guest networks
 - Captive portals
- Authentication
 - Pre-shared key (PSK) vs. Enterprise
- Antennas
 - Omnidirectional vs. directional
- Autonomous vs. lightweight access point

Configure Wireless Devices and Technologies

The popularity of wireless LANs (WLANs) has exploded over the past decade, allowing users to roam within a WLAN coverage area, take their laptops and tablets with them, and maintain network connectivity as they move throughout a building or campus environment. Many other devices can also take advantage of wireless networks, such as gaming consoles, smartphones, and printers.

This chapter introduces WLAN technology, along with various wireless concepts, components, and standards. It also presents WLAN design considerations, followed by a discussion of WLAN security.

Foundation Topics

Introducing Wireless LANs

This section introduces the basic building blocks of WLANs and discusses how WLANs connect to a wired local area network (LAN). Various design options, including antenna design, frequencies, and communications *channels*, are discussed, along with a comparison of today's major wireless standards, which are all some variant of IEEE 802.11.

WLAN Concepts and Components

Wireless devices, such as laptops, tablets, and smartphones, often have built-in wireless cards that allow those devices to communicate on a WLAN. But what is the device to which they communicate? It could be another laptop or device with a wireless card. This would be an example of an *ad hoc* WLAN. However, enterprise-class WLANs, and even most WLANs in homes, are configured in such a way that a wireless client connects to some sort of wireless base station, such as a wireless access point (AP) or a wireless router. Many companies offer Wi-Fi as a service, and users in range of an AP can use the AP as a *hotspot*, indicating that Wi-Fi is available through the AP.

NOTE A much fancier term for an ad hoc network that you should be aware of is *independent basic service set (IBSS) WLAN*.

This communication might be done using a variety of antenna types, frequencies, and communication channels. The following sections consider some of these elements in more detail.

Wireless Routers

Consider the basic WLAN topology shown in Figure 11-1. Such a WLAN might be found in a residence whose Internet access is provided by a high-speed cable modem. In this topology, a wireless router and switch are shown as separate components. However, in many residential networks, a wireless router integrates switch ports and wireless routing functionality into a single device.

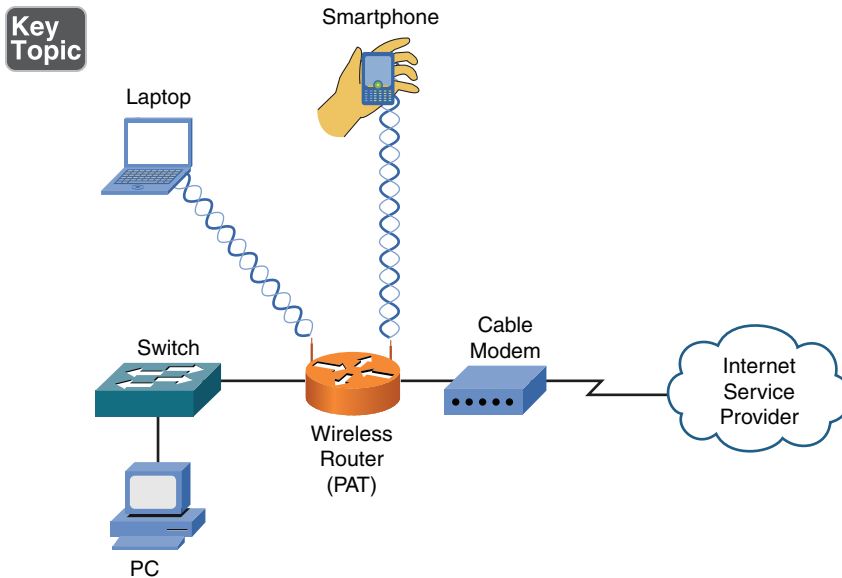


Figure 11-1 Basic WLAN Topology with a Wireless Router

In Figure 11-1, the cable modem obtains an IP address via DHCP from the Internet service provider (ISP). The wireless router behind the modem also uses DHCP to provide IP addresses to LAN devices attaching to it wirelessly or through a wired connection. The process through which a wireless client (for example, a laptop or a smartphone) attaches with a wireless router (or wireless AP) is called *association*. All wireless devices associating with a single AP share a collision domain. Therefore, for scalability and performance reasons, WLANs might include multiple APs. The router then uses port address translation to allow packets to leave the LAN and head to the Internet.

Wireless Access Point

Although a wireless access point (AP) interconnects a wired LAN with a WLAN, it does not interconnect two networks (for example, the service provider's network and an internal network). Figure 11-2 shows a typical deployment of an AP.

In Figure 11-2, the AP connects to the wired LAN, and the wireless devices that connect to the wired LAN via the AP are on the same subnet as the AP. Notice that no NAT or PAT is being performed. The access point is acting as a wireless bridge between the wireless clients connected to the AP and the wired devices connected to the switch in the same Layer 2 domain.

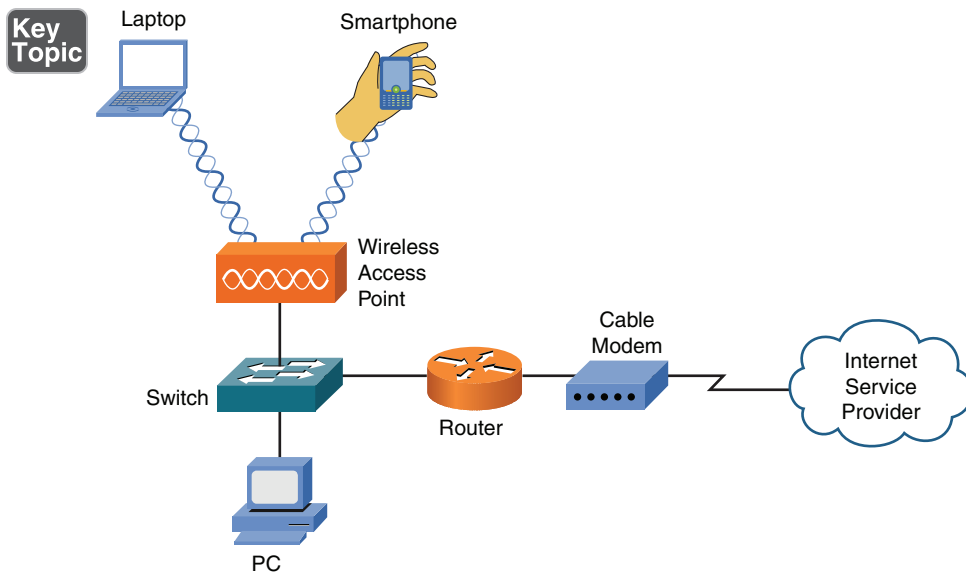


Figure 11-2 Basic WLAN Topology with a Wireless AP

NOTE NAT and PAT were covered in Chapter 9, “Routing Technologies.”

When it comes to managing access points, there are two approaches based upon the access points in use by an organization:

- **Autonomous access points:** These APs are self-contained network devices that manage all wireless network functions independently, without the need for a central controller. They are equipped with built-in capabilities to handle tasks such as service set identifier (SSID) broadcasting, client authentication, security enforcement, and traffic routing. Each autonomous access point operates as a standalone unit, making them suitable for smaller networks or environments where centralized management is not required. They offer flexibility and simplicity, as each device can be individually configured and managed, but this can lead to challenges in large-scale deployments due to the lack of centralized control and increased administrative overhead.
- **Lightweight access points (LWAPs):** These APs are designed to work in conjunction with a central wireless LAN controller (WLC). Unlike autonomous access points that operate independently, LWAPs delegate most of their functionality, such as security policies, SSID configuration, and client authentication, to the controller, simplifying network management and enhancing

scalability. They primarily handle the transmission and reception of wireless signals and rely on the controller for complex processing tasks. This architecture allows for centralized management, easier updates, and improved security across the network, making LWAPs ideal for environments with large-scale or dynamic wireless deployments.

Most wireless vendors offer both autonomous and lightweight access points. In fact, some models even allow you to switch between the modes on a single device. A Cisco model Catalyst CW9800M is an example of a wireless LAN controller for multiple APs. The protocols used to communicate between an AP and a WLC could be the older Lightweight Access Point Protocol (LWAPP) or the more current Control and Provisioning of Wireless Access Points (CAPWAP). With a WLC, VLAN pooling can be used to assign IP addresses to wireless clients from a pool of IP subnets and their associated VLANs.

Guest Networks

Today, enterprise and even home wireless networks might choose to provide *guest network* services. Guest networks provide a separate and secure Wi-Fi connection for visitors, isolating their traffic from the internal network to safeguard sensitive data and resources. These networks are typically designed to offer Internet access without compromising the security of the host network.

To manage and authenticate users, many guest networks employ *captive portals*, which are web pages that users must interact with before gaining network access. When a guest connects to the Wi-Fi, they are redirected to a captive portal, where they might need to provide credentials, accept terms of service, or enter a voucher code. This process not only enhances security by controlling and monitoring access but also allows network administrators to collect user information and enforce usage policies. Captive portals can also present opportunities for branding and communication with guests, making them a valuable tool for businesses and organizations aiming to provide a secure and user-friendly Wi-Fi experience.

Antennas

The coverage area of a WLAN is largely determined by the type of *antenna* used on a wireless AP or a wireless router. Although some lower-end, consumer-grade wireless APs have fixed antennas, higher-end, enterprise-class wireless APs often support various antenna types.

**Key
Topic**

Design goals to keep in mind when selecting an antenna include the following:

- The required distance between an AP and a wireless client
- The coverage area pattern (For example, the coverage area might radiate out in all directions, forming a spherical coverage area around an antenna, or an antenna might provide increased coverage in only one or two directions.)
- The type of environment—either indoor or outdoor
- The need to avoid interference with other APs

The strength of the electromagnetic waves being radiated from an antenna is referred to as *gain*, and it involves a measurement of both direction and efficiency of a transmission. For example, the gain measurement for a wireless AP's antenna transmitting a signal is a measurement of how efficiently the power being applied to the antenna is converted into electromagnetic waves being broadcast in a specific direction. Conversely, the gain measurement for a wireless AP's antenna receiving a signal is a measurement of how efficiently the received electromagnetic waves arriving from a specific direction are converted back into electricity leaving the antenna.

Gain is commonly measured using the dBi unit of measure, where *dB* stands for *decibels*, and *i* stands for *isotropic*. A decibel, in this context, is a ratio of radiated power to a reference value. In the case of dBi, the reference value is the signal strength (power) radiated from an isotropic antenna, which represents a theoretical antenna that radiates an equal amount of power in all directions (in a spherical pattern). An isotropic antenna is considered to have gain of 0 dBi.

Here is the most common formula used for antenna gain:

$$\text{GdBi} = 10 \times \log^{10} (G)$$

Based on this formula, an antenna with a peak power gain of 4 (*G*) would have a gain of 6.02 dBi. Antenna theory can become mathematical, heavily relying on the use of Maxwell's equations. However, generally speaking, if one antenna has 3 dB more gain than another antenna, it has approximately twice the effective power.

Antennas are classified not just by their gain but also by their coverage area. There are two broad categories of antennas, based on coverage area:

- **Omnidirectional:** An *omnidirectional antenna* (sometimes simply called an *omni*) radiates power at relatively equal power levels in all directions; it is somewhat similar to a theoretical isotropic antenna. Omnidirectional antennas, an example of which is depicted in Figure 11-3, are popular in residential WLANs and small office/home office (SOHO) locations.

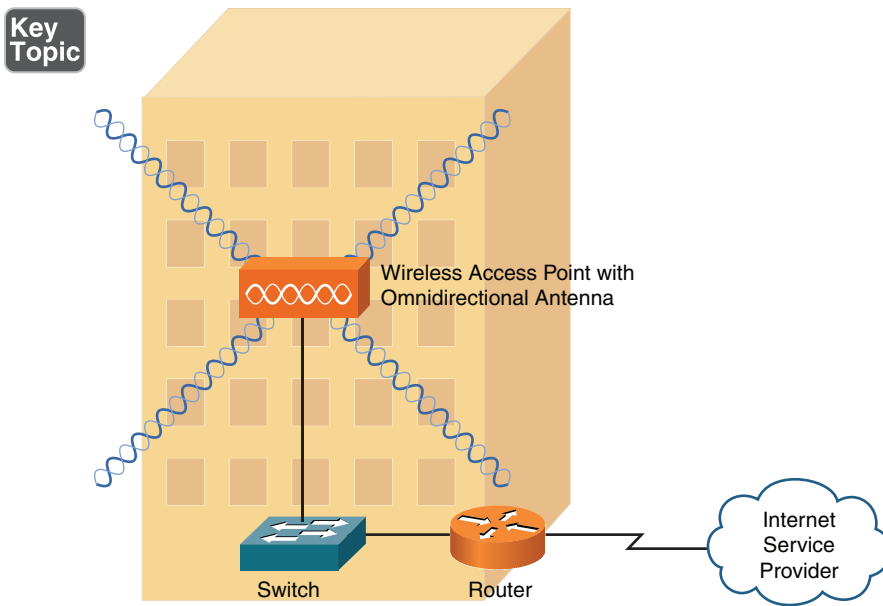


Figure 11-3 Omnidirectional Antenna Coverage

- **Unidirectional:** A *unidirectional antenna* (sometimes just termed a *directional antenna*) can focus its power in a specific direction, thus avoiding potential interference with other wireless devices and perhaps reaching greater distances than are possible with omnidirectional antennas. A Yagi antenna is a popular example of a directional antenna. One application for directional antennas is interconnecting two nearby buildings, as shown in Figure 11-4.

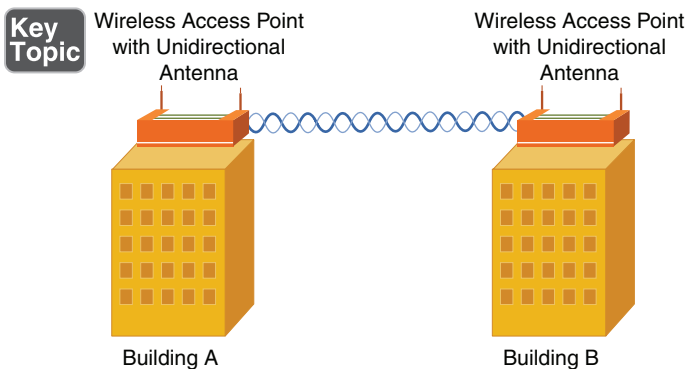


Figure 11-4 Unidirectional Antenna Coverage

Another consideration for antenna installation is the horizontal or vertical orientation of the antenna. For best performance, if two wireless APs communicate with one another, they need to have matching antenna orientations (that is, their *polarity* needs to be the same).

Channel and Frequency Options

In Chapter 5, “Transmission Media and Transceivers,” you learned the details of the channels and frequencies used in wireless networking today. In this section, we provide a targeted review of this information along with some additional information made popular in today’s enterprise networks.

Be sure to remember these key points:

- **Channel width** refers to the size of the frequency band used to transmit data between wireless devices, directly influencing the network’s data throughput and performance. Standard channel widths for Wi-Fi networks include 20MHz, 40MHz, 80MHz, and even 160MHz, with wider channels generally offering higher data rates by allowing more data to be transmitted simultaneously. However, increasing channel width can also raise the potential for interference, especially in environments with multiple Wi-Fi networks or other sources of radio frequency interference (RFI). Wider channels are more susceptible to overlap with adjacent networks, which can degrade performance. Therefore, while 80MHz and 160MHz channels are favored for their capacity to deliver faster speeds in newer Wi-Fi standards like Wi-Fi 6 (802.11ax) and Wi-Fi 6E (802.11ax operating in the 6GHz band), careful consideration must be given to network planning and environmental factors to optimize performance and minimize interference.
- As referenced in Chapter 5, **non-overlapping channels** are critical for optimizing performance and minimizing interference. In the **2.4GHz** band, which is more congested and prone to interference due to its limited spectrum, only three non-overlapping channels (typically channels 1, 6, and 11) can be used effectively without overlapping adjacent channels, thus reducing the risk of cross-channel interference. In contrast, the **5GHz** and **6GHz** bands offer a broader spectrum, allowing for more non-overlapping channels that can support wider channel widths, such as 20MHz, 40MHz, 80MHz, and 160MHz. This greater availability of non-overlapping channels facilitates higher data rates and more robust connections, as devices can operate in a cleaner spectrum with reduced likelihood of co-channel interference.
- Regulatory impacts have certainly made their mark on the field of wireless. For example, IEEE **802.11b** is an amendment to the 802.11 standard that enhances wireless networking by introducing mechanisms for dynamic

frequency selection (DFS) and transmit power control (TPC). Developed to comply with regulatory requirements, particularly in Europe, 802.11h addresses the need to coexist with radar systems and other incumbent services in the 5GHz frequency band. DFS allows wireless access points to detect radar signals and dynamically switch to a different channel if interference is detected, thereby preventing disruption to critical radar operations. TPC helps in managing the power levels of transmissions to avoid interference with other devices and to optimize energy usage.

- **Band steering** is a technique used by modern Wi-Fi access points to manage and direct client devices to the most appropriate frequency band—typically between the 2.4GHz and 5GHz bands. When a dual-band access point detects a client device that can operate on both the 2.4GHz and 5GHz bands, it can “steer” the device to the 5GHz band to alleviate congestion on the 2.4GHz band, which is more prone to interference and crowding due to its limited number of channels and higher usage by various devices. Band steering helps balance the load across both frequency bands, improving overall network performance and user experience by reducing interference and maximizing the available bandwidth.

Deploying Wireless LANs

A variety of installation options and design considerations are involved in designing and deploying WLANs. This section delves into the available options and provides you with some best-practice recommendations.

Types of WLANs

WLANs can be categorized based on their use of wireless APs. The three main categories are *independent basic service set (IBSS; aka ad hoc)*, *basic service set (BSS)*, and *extended service set (ESS)*. An IBSS WLAN operates in an ad hoc fashion, whereas BSS and ESS WLANs operate in **infrastructure mode**. The following sections describe these three types of WLANs in detail.

IBSS

As shown in Figure 11-5, a WLAN can be created without the use of an AP. Such a configuration, called an independent basic service set (IBSS), is said to work in an ad hoc fashion. An ad hoc WLAN is useful for temporary connections between wireless devices. For example, you might temporarily interconnect two laptop computers to transfer a few files.

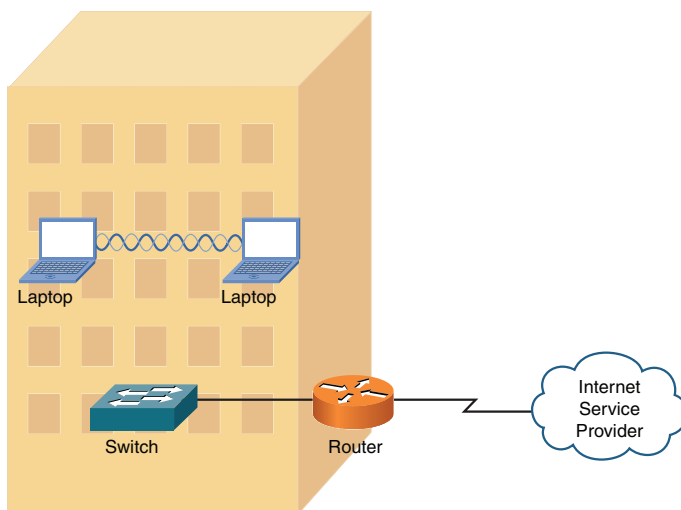
**Key
Topic**

Figure 11-5 Independent Basic Service Set (IBSS) WLAN

BSS

Figure 11-6 depicts a WLAN using a single AP. A WLAN that has just one AP is called a basic service set (BSS) WLAN. BSS WLANs are said to run in *infrastructure mode* because wireless clients connect to an AP, which is typically connected to a wired network infrastructure. BSS networks are often used in residential and SOHO locations, where the signal strength provided by a single AP is sufficient to service all the WLAN's wireless clients.

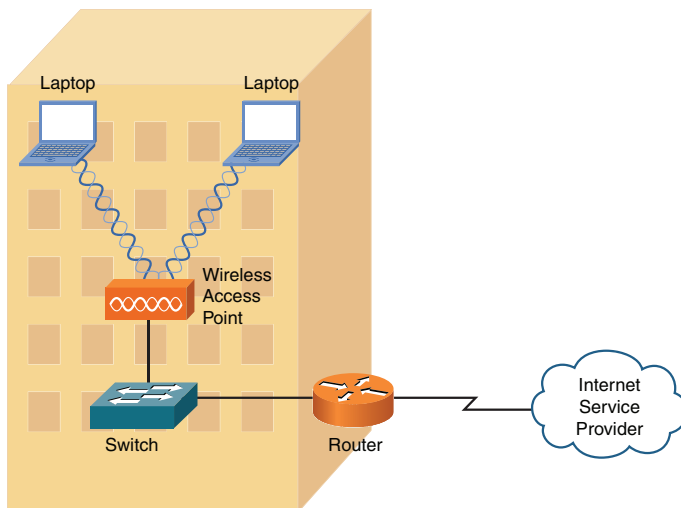
**Key
Topic**

Figure 11-6 Basic Service Set (BSS) WLAN

ESS

Figure 11-7 illustrates a WLAN using two APs. A WLAN containing more than one AP is called an extended service set (ESS) WLAN. Like BSS WLANs, ESS WLANs operate in infrastructure mode. When you have more than one AP, it is important to prevent one AP from interfering with another. Specifically, the previously discussed non-overlapping channels (channels 1, 6, and 11 for the 2.4GHz band) should be selected for adjacent wireless coverage areas.

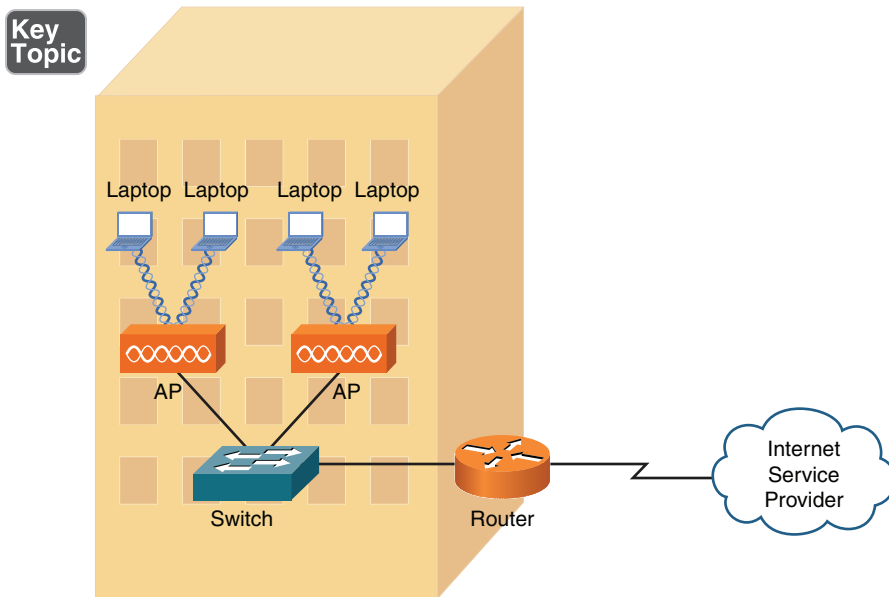


Figure 11-7 Extended Service Set (ESS) WLAN

Mesh Topology

A *mesh* wireless network is a collection of wireless devices that may not use centralized control (but instead features decentralized management). The combined wireless coverage range defines the range of the network. This could also be referred to as a *mesh cloud*. Additional wireless technologies (besides Wi-Fi) could be used to build a mesh wireless topology. This type of network could be used for hosts to communicate with other devices in the mesh, or the network could provide a gateway to the Internet or other networks.

NOTE You might also encounter an opposite type of topology to the mesh network when you are working with wireless technologies. This would be the *point-to-point* network type. A point-to-point wireless topology connects two distinct locations using a direct wireless link, effectively functioning as a wireless bridge. This setup allows for high-speed data transmission between the two points without the need for physical cabling, making it ideal for connecting buildings, campuses, or remote sites. The topology typically uses directional antennas that focus the signal in a narrow beam, ensuring a reliable and efficient connection over distances that can range from a few hundred meters to several kilometers.

Sources of Interference

A major issue for WLANs is radio frequency interference (RFI) caused by other devices using frequencies similar to those of the WLAN devices. Also, physical obstacles can impede or reflect WLAN transmissions. The following are some of the most common sources of interference:

Key Topic

- **Other WLAN devices:** Earlier in this chapter, you read about non-overlapping channels for both the 2.4GHz and 5GHz bands. If two or more WLAN devices are in close proximity and use overlapping channels, those devices could interfere with one another.
- **Cordless phones:** Several models of cordless phones operate in the 2.4GHz band and can interfere with WLAN devices. However, if you need cordless phones to coexist in an environment with WLAN devices using the 2.4GHz band, consider the use of Digital Enhanced Cordless Telecommunications (DECT) cordless phones. Although the exact frequencies used by DECT cordless phones vary based on country, DECT cordless phones do not use the 2.4GHz band. For example, in the United States, DECT cordless phones use frequencies in the range 1.92GHz–1.93GHz.
- **Microwave ovens:** Older microwave ovens, which might not have sufficient shielding, can emit relatively high-powered signals in the 2.4GHz band, resulting in significant interference with WLAN devices operating in the 2.4GHz band.
- **Wireless security system devices:** Most wireless security cameras operate in the 2.4GHz frequency range, which can cause potential issues with WLAN devices.

- **Physical obstacles:** In electromagnetic theory, radio waves cannot propagate through a perfect conductor. So, although metal filing cabinets and large appliances are not perfect conductors, they are sufficient to cause degradation of a WLAN signal. For example, a WLAN signal might hit a large air conditioning unit, causing the radio waves to be reflected and scattered in multiple directions. Not only does this limit the range of the WLAN signal, but also radio waves carrying data might travel over different paths. This *multipath issue* can cause data corruption. Concrete walls, metal studs, and even window film can reduce the quality of the wireless network signals.
- **Signal strength:** The range of a WLAN device is a function of the device's signal strength. Lower-cost consumer-grade APs do not typically allow administrative adjustment of signal strength. However, enterprise-class APs often allow signal strength to be adjusted to ensure sufficient coverage of a specific area, while avoiding interference with other APs using the same channel.

As you can see from this list, most RFI occurs in the 2.4GHz band rather than the 5GHz band. Therefore, depending on the wireless clients you need to support, you might consider using the 5GHz band, which is an option with the newer wireless standards. With the increased use of wireless, both coverage and capacity-based planning should be done to provide acceptable goodput. *Goodput* refers to the number of useful information bits that the network can deliver (not including overhead for the protocols being used). Another factor is the density (that is, the ratio of users to APs), which, if too high, could harm performance of the network. Areas expecting high density include classrooms, hotels, and hospitals. Device or bandwidth saturation could impact performance.

Wireless AP Placement

A WLAN using more than one AP (that is, an ESS WLAN) requires careful planning to prevent the APs from interfering with one another while still servicing a desired coverage area. Specifically, an overlap of coverage between APs should exist to allow uninterrupted *roaming* from one WLAN *cell* (which is the coverage area provided by an AP) to another. However, those overlapping coverage areas should not use overlapping frequencies.

Figure 11-8 shows how non-overlapping channels in the 2.4GHz band can have overlapping coverage areas to provide seamless roaming between AP coverage areas. A common WLAN design recommendation is to have 10% to 15% overlap of coverage between adjoining cells.

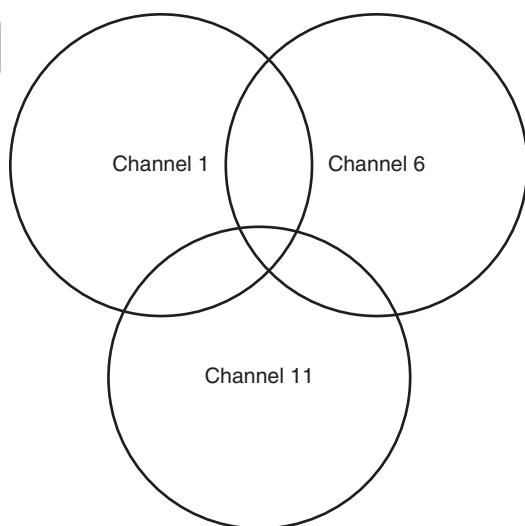
**Key
Topic**

Figure 11-8 Coverage Overlap in Coverage Areas for Non-overlapping Channels

If a WLAN has more than three APs, the APs are deployed in a honeycomb fashion to allow an overlap of AP coverage areas while avoiding an overlap of identical channels. The example in Figure 11-9 shows an approach to channel selection for adjoining cells in the 2.4GHz band. Notice that cells using the same non-overlapping channels (channels 1, 6, and 11) are separated by another cell. For example, notice that none of the cells using channel 11 overlap another cell using channel 11.

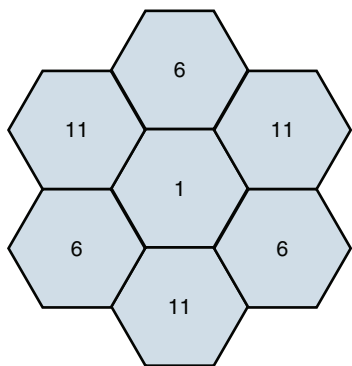
**Key
Topic**

Figure 11-9 Non-overlapping Coverage Cells for the 2.4GHz Band

NOTE Although a honeycomb channel assignment scheme can be used for the 5GHz band, identical channels should be separated by at least two cells rather than by a single cell, as shown in Figure 11-9 for the 2.4GHz band.

Securing Wireless LANs

WLANs introduce some unique concerns to a network. For example, improperly installing wireless APs is roughly equivalent to putting an Ethernet port in a building's parking lot, where someone can drive up and access your network. Fortunately, various features are available to harden the security of your WLAN, as discussed in this section.

Security Issues

In the days when dial-up modems were popular, malicious users could run a program on a computer to call all phone numbers in a certain number range. Phone numbers that answered with a modem tone became targets for later attacks. This type of reconnaissance was known as *war dialing*. A modern-day variant of war dialing is *war driving*, where potentially malicious users drive around looking for unsecured WLANs. These users might be identifying unsecured WLANs for nefarious purposes or simply looking for free Internet access. Devices such as cell phones, laptops, tablets, and gaming and media devices could act as wireless clients as well as be used in a wireless attack because they have potential Wi-Fi access to the network.

Other WLAN security threats include the following:

Key Topic

- **War chalking:** Once an open WLAN (that is, a WLAN whose SSID and authentication credentials are known) is found in a public place, a user might write a symbol on a wall (or some other nearby structure) to let others know the characteristics of the discovered network. This practice, which is a variant of the decades-old practice of hobos leaving symbols as messages to fellow hobos, is called *war chalking*. Figure 11-10 shows common war-chalking symbols.

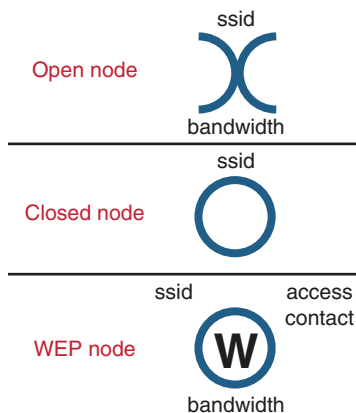


Figure 11-10 War-Chalking Symbols

NOTE Just as technologies evolve, so do computer criminals. War flying features the use of mobile devices in conjunction with drones or other aircraft to find wireless networks.

- **WEP and WPA security cracking:** As discussed later in this chapter, various security standards are available for encrypting and authenticating a WLAN client with an AP. Two of the less secure standards are Wired Equivalent Privacy (WEP) and *Wi-Fi Protected Access (WPA)*. Although WPA is considered more secure than WEP, utilities are available on the Internet for cracking each of these approaches to wireless security. By collecting enough packets transmitted by a secure AP, these cracking utilities can use mathematical algorithms to determine the *pre-shared key (PSK)* configured on a wireless AP with which an associating wireless client must also be configured. Pre-shared keys are covered in more detail in the next section.
- **Rogue access point:** A malicious user may set up an AP called a *rogue access point* to which legitimate users can connect. The malicious user might then use a packet sniffer (which displays information about unencrypted traffic, including the traffic's data and header information) to eavesdrop on communications flowing through the rogue AP. To cause unsuspecting users to connect to the rogue AP, the malicious user could configure the AP with the same **service set identifier (SSID)** used by a legitimate AP. When a rogue AP is configured with the SSID of a legitimate AP, the rogue AP is commonly referred to as an *evil twin*. Rogue access points are just one reason it is wise to change default SSIDs on wireless equipment. And remember, do not use company information (brand names, divisions, etc.) when creating the unique SSID.

NOTE An SSID is a string of characters identifying a WLAN. APs participating in the same WLAN (that is, in an ESS) can be configured with identical SSIDs. An SSID shared among multiple APs is called an **extended service set identifier (ESSID)**. Sure enough, the basic service set (BSS) wireless network uses an SSID called a **basic service set identifier (BSSID)**. While the SSID is often a “friendly name” that is relatively easy to identify, the BSSID is the MAC address of the access point.

Approaches to WLAN Security

A WLAN that does not require **authentication** or provide encryption for wireless devices (for example, a publicly available WLAN, such as the ones in many airports) is said to be using *open authentication*. To protect such a WLAN's traffic from

eavesdroppers, a variety of security standards and practices have been developed, including the following:

- **MAC address filtering:** An AP can be configured with a list of MAC addresses that are permitted to associate with the AP. If a malicious user attempts to connect via a laptop whose MAC address is not on the list of trusted MAC addresses, that user is denied access. One drawback to MAC address filtering is the administrative overhead required to keep an approved list of MAC addresses up to date. Another issue with MAC address filtering is that a knowledgeable user could falsify the MAC address of a wireless network card, making a device appear to be approved.
- **Disabling SSID broadcast:** An SSID can be broadcast by an AP to let users know the name of the WLAN. For security purposes, an AP might be configured not to broadcast its SSID. However, knowledgeable users could still determine the SSID of an AP by examining captured packets.
- ***Pre-shared key (PSK):*** To encrypt transmission between a wireless client and an AP (in addition to authenticating a wireless client with an AP), both the wireless client and the AP could be preconfigured with a matching string of characters (a PSK, as previously described). The PSK could be used as part of a mathematical algorithm to encrypt traffic, such that if an eavesdropper intercepted the encrypted traffic, they would not be able to decrypt the traffic without knowing the PSK. Although using a PSK can be effective in providing security for a small network (such as a SOHO network), it lacks scalability. For example, in a large corporate environment, the compromise of a PSK would necessitate the reconfiguration of all devices configured with that PSK. WLAN security based on PSK technology is called *Personal mode* (specifically *WPA Personal mode*).

NOTE The latest WPA security approach (WPA3) running in Personal mode replaces the PSK approach with a more secure method of authentication called Simultaneous Authentication of Equals (SAE).

- **IEEE 802.1X:** Rather than having all devices in a WLAN configured with the same PSK, a more scalable approach is to require all wireless users to authenticate using their own credentials (for example, a username and password). Allowing users to have their own credentials prevents the compromising of one password from impacting the configuration of all wireless devices. IEEE 802.1X is a technology that allows wireless clients to authenticate with an

authentication server—typically, a Remote Authentication Dial-In User Service (RADIUS) server. Figure 11-11 shows a wireless implementation of IEEE 802.1X.

NOTE WLAN security based on IEEE 802.1X and a centralized authentication server such as RADIUS is called *Enterprise mode* (or more specifically, *WPA Enterprise mode*).

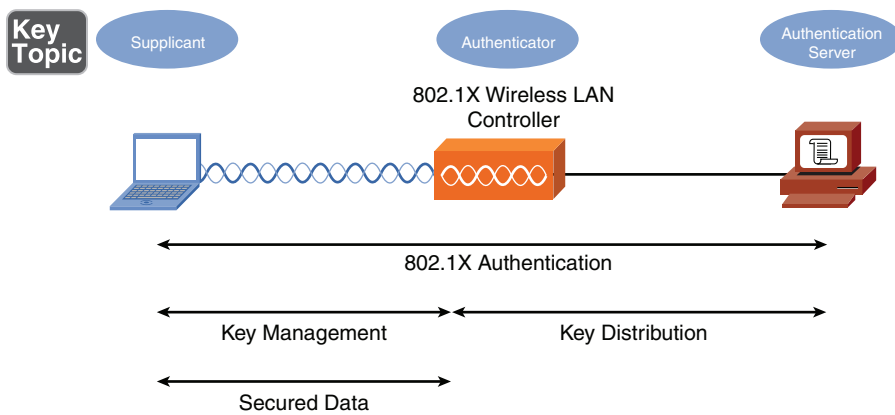


Figure 11-11 IEEE 802.1X Security for a WLAN

NOTE IEEE 802.1X works with Extensible Authentication Protocol (EAP) to perform its job of authentication. A variety of EAP types exist, including Lightweight Extensible Authentication Protocol (LEAP), EAP-Flexible Authentication via Secure Tunneling (EAP-FAST), EAP-Transport Layer Security (EAP-TLS), EAP-Tunneled Transport Layer Security (EAP-TTLS), Protected EAP–Generic Token Card (PEAP-GTC), and Protected EAP–Microsoft Challenge Handshake Authentication Protocol Version 2 (PEAP-MSCHAPv2). Although these EAP types differ in their procedures, the overriding goal for all the EAP types is to securely authenticate a supplicant and provide the supplicant and the authenticator a session key that can be used during a single session in the calculation of security algorithms (for example, encryption algorithms).

Security Standards

When you're configuring a wireless client for security, the most common security standards from which you can select are as follows:



- *Wi-Fi Protected Access 2 (WPA2)*
- *Wi-Fi Protected Access 3 (WPA3)*

The sections that follow describe these standards in detail.

WPA2

The IEEE 802.11i standard, which was approved in 2004, requires stronger algorithms for encryption and integrity checking than those used with previous WLAN security protocols, such as WEP and WPA. The requirements set forth in the IEEE 802.11i standard are implemented in the Wi-Fi Alliance's WPA Version 2 (WPA2) security standard. WPA2 uses Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) for integrity checking and *Advanced Encryption Standard (AES)* for encryption. You might find this referenced as simply CCMP-AES. Remember, as covered earlier, any version of WPA that uses a centralized server for authenticating users is referred to as WPA Enterprise mode. Any version of WPA that uses a configured password or PSK instead of a centralized server is referred to as WPA Personal mode.

WPA3

Wi-Fi Protected Access 3 (WPA3) is the latest security protocol for wireless networks, designed to enhance security and simplify connectivity compared to its predecessor, WPA2. Introduced by the Wi-Fi Alliance in 2018, WPA3 provides stronger encryption through the adoption of Simultaneous Authentication of Equals (SAE), a more secure key exchange method that mitigates the risks associated with dictionary attacks. It also offers individualized data encryption for each device on the network, ensuring that data remains protected even on open or public networks. WPA3 enhances the end-user experience with easier setup processes, especially for devices without displays, by supporting features like Wi-Fi Easy Connect. Additionally, it includes improved protections for Internet of Things (IoT) devices and ensures better overall security for modern wireless networks.

Additional Wireless Options

Other wireless technologies, such as Bluetooth, infrared (IR), and Near Field Communication (NFC), which are often integrated into smartphones, can also provide

connectivity for a personal area network (PAN) or other short-range networking applications. Many of these technologies help facilitate the IoT.

Another interesting wireless technology is *geofencing*, which often uses the Global Positioning System (GPS) or radio frequency identification (RFID) to define geographic boundaries. Geofencing allows you to define triggers so that when a device enters (or exits) the boundaries defined by the administrator, an alert is issued.

Geofence virtual barriers can be active or passive. Active geofences require an end user to opt in to location services and require a mobile app to be open. Passive geofences are always on; they rely on Wi-Fi and/or cellular data instead of GPS or RFID and can work in the background.

A practical application of geofencing might be a hospital with patient information on tablets that the hospital distributes to staff. If these tablets travel beyond the geofence, an administrative alert can trigger.

Real-World Case Study

Acme, Inc. hired an outside contractor that specializes in Wi-Fi. The consultants came in and did a needs assessment and performed a wireless site survey. Recommendations were then made about the need for 15 access points in the headquarters office spaces and 3 access points at each of the remote branch offices. Three wireless LAN controllers, one for each office, will be used to manage the respective access points. The management of the access points through the wireless LAN controllers will be done primarily through the headquarters office, using the WAN that is connecting the branch offices to the headquarters office.

Because of the high number of other Wi-Fi access points being used in the same building as the headquarters office, Acme, Inc. decided to use the 5GHz range (because of the reduced competition in that space) and to use 802.11ax.

For security, Acme, Inc. will use WPA3 in conjunction with a RADIUS server. Acme, Inc. will use Enterprise mode for authentication of each user before allowing users access on the wireless network(s). The RADIUS server is integrated with Microsoft Active Directory so that Acme, Inc. will not have to re-create every user account; the RADIUS server can check with the Active Directory server to verify user credentials and passwords.

Separate SSIDs were set up that map to the various VLANs and departments currently on the wired network. Also, a separate SSID was set up as a wireless guest network that has limited access but does provide Internet access for guest users.

Once all this was in place, a site survey was done again to verify the signal strengths and to identify any interference related to the wireless implementation. A heat map was provided to visually represent the signal strengths in the coverage areas in the respective office spaces.

Summary

Here are the main topics covered in this chapter:

- This chapter identified various components, technologies, and terms used in WLANs. This included such content as frequencies, SSIDs, network types, guest networks, and access points.
- This chapter presented WLAN design considerations, such as the selection of WLAN standards, bands, and non-overlapping channels. Potential sources of interference were also discussed.
- This chapter described some of the security risks posed by a WLAN and the technologies available for mitigating those risks.
- This chapter also presented information about the different types of antennas found in wireless networks today.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 11-1 lists these key topics and the page number where each is found.

Key Topic

Table 11-1 Key Topics for Chapter 11

Key Topic Element	Description	Page Number
Figure 11-1	Basic WLAN Topology with a Wireless Router	287
Figure 11-2	Basic WLAN Topology with a Wireless AP	288
List	Antenna selection criteria	290
Figure 11-3	Omnidirectional Antenna Coverage	291
Figure 11-4	Unidirectional Antenna Coverage	291

Key Topic Element	Description	Page Number
Figure 11-5	Independent Basic Service Set (IBSS) WLAN	294
Figure 11-6	Basic Service Set (BSS) WLAN	294
Figure 11-7	Extended Service Set (ESS) WLAN	295
List	Sources of interference	296
Figure 11-8	Coverage Overlap in Coverage Areas for Non-overlapping Channels	298
Figure 11-9	Non-overlapping Coverage Cells for the 2.4GHz Band	298
List	Wireless security threats	299
Figure 11-11	IEEE 802.1X Security for a WLAN	302
List	WLAN security standards and best practices	303

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

2.4GHz, 5GHz, 6GHz, 802.11h, ad hoc, antenna, authentication, autonomous access point, band steering, basic service set identifier (BSSID), captive portals, channel width, directional antenna, extended service set identifier (ESSID), guest network, infrastructure mode, lightweight access point (LWAP), mesh, non-overlapping channels, omnidirectional antenna, point to point, pre-shared key (PSK), service set identifier (SSID), Wi-Fi Protected Access 2 (WPA2), Wi-Fi Protected Access 3 (WPA3), WPA Enterprise mode, WPA Personal mode

Additional Resources

Wi-Fi Standards: <https://youtu.be/q64AZjPfa0Y>

Fundamentals of Wi-Fi 6: <https://www.youtube.com/watch?v=V5qLv0BtBcM>

Review Questions

The answers to these review questions are in Appendix A, “Answers to Review Questions.”

1. What type of antenna, used in wireless APs and wireless routers in SOHO locations, radiates power equally in all directions?
 - a. Unidirectional
 - b. Yagi

- c. Parabolic
 - d. Omnidirectional
2. When you're using the 2.4GHz band for multiple access points in a WLAN in the United States, which non-overlapping channels should you select? (Choose three.)
- a. 0
 - b. 1
 - c. 5
 - d. 6
 - e. 10
 - f. 11
 - g. 14
3. What is an amendment to the 802.11 standard that enhances wireless networking by introducing mechanisms for dynamic frequency selection and transmit power control?
- a. 802.11h
 - b. 802.11f
 - c. 802.11p
 - d. 802.11a
4. What type of wireless access point is used with a wireless LAN controller?
- a. Standalone access point
 - b. Rugged access point
 - c. Autonomous access point
 - d. Lightweight access point
5. What feature is often found in use with guest Wi-Fi networks in order to help enhance security?
- a. WEP
 - b. Captive portal
 - c. Band steering
 - d. 802.11h

6. A WLAN formed directly between wireless clients (without the use of a wireless AP) is referred to as what type of WLAN?
 - a. Enterprise mode
 - b. IBSS
 - c. Personal mode
 - d. BSS
7. When extending the range for a 2.4GHz WLAN, you can use non-overlapping channels for adjacent coverage cells. However, there should be some overlap in coverage between those cells (using non-overlapping channels) to prevent a connection from dropping as a user roams from one coverage cell to another. What percentage of coverage overlap is recommended for these adjacent cells?
 - a. 5% to 10%
 - b. 10% to 15%
 - c. 15% to 20%
 - d. 20% to 25%
8. If a WLAN does not need a user to provide credentials to associate with a wireless AP and access the WLAN, what type of authentication is in use?
 - a. WEP
 - b. SSID
 - c. Open
 - d. IV
9. Your access point has the ability to direct client devices into the most appropriate frequency band. What is this capability called?
 - a. Band steering
 - b. Transmit power control
 - c. Dynamic frequency selection
 - d. Simultaneous selection of equals
10. Which standard developed by the Wi-Fi Alliance implements the requirements of IEEE 802.11i?
 - a. TKIP
 - b. MIC

- c. WEP
 - d. WPA2
11. Which security technique uses wireless technologies to create an invisible boundary around some point?
- a. WPA3
 - b. LTE
 - c. War driving
 - d. Geofencing
12. Which of the following statements are true regarding the features of WPA2 and WPA3? (Select three.)
- a. The two versions of WPA2 are WPA2-Personal and WPA2-Enterprise.
 - b. WPA2 Personal uses pre-shared keys.
 - c. WPA3 prevents offline password attacks using 802.1X.
 - d. WPA3 uses 192-bit AES for the Enterprise version.
13. An administrator wants to avoid attacks such as war driving, war chalking, and war flying. What should the admin do to improve the security of the wireless network from such attacks?
- a. Change the AP SSID
 - b. Enable SSID broadcast
 - c. Blocklist all MAC addresses
 - d. Allow list all MAC addresses
14. You are studying an IEEE standard for port-based access control. This standard specifically provides for the encapsulation of EAP over the IEEE family of standards for packet-based networks and facilitates the use of various authentication methods such as RADIUS and digital certificates. What standard are you studying?
- a. 802.11
 - b. 802.1X
 - c. PSK
 - d. 802.11h



This chapter covers the following topics related to Objective 2.4 (Explain important factors of physical installations) of the CompTIA Network+ N10-009 certification exam:

- Important installation implications
 - Locations
 - Intermediate distribution frame (IDF)
 - Main distribution frame (MDF)
 - Rack size
 - Port-side exhaust/intake
 - Cabling
 - Patch panel
 - Fiber distribution panel
 - Lockable
- Power
 - Uninterruptible power supply (UPS)
 - Power distribution unit (PDU)
 - Power load
 - Voltage
- Environmental factors
 - Humidity
 - Fire suppression
 - Temperature

Physical Installations

As networking engineers, we need to give some attention to the physical characteristics of where we are locating our network equipment. In this chapter, we will take a fairly deep dive into things like the locations and cable management facilities of our networking equipment. We will also discuss the important aspects of power and environmental factors that we should consider when designing or outfitting our physical networking locations. These environmental factors include humidity, fire suppression, and temperature concerns.

Foundation Topics

Important Installation Implications

There are many physical installation aspects that you need to be concerned with. For example, you not only should carefully select the correct network media for your networking infrastructure, but also install that media as part of an organized cable distribution system. Typically, cable distribution systems are hierarchical in nature.

Consider the example profiled in Figure 12-1. In this example, cable from end-user offices runs back to common locations within the building. These locations are sometimes referred to as *wiring closets*. Cables in these locations might terminate in a **patch panel**, or *patch bay*. The patch panel might consist of some sort of cross-connect block wired into a series of ports (for example, RJ45 ports), which can be used to quickly interconnect cables coming from end-user offices with a network device, such as an Ethernet switch. A common term for cross-connect blocks is *punchdown blocks*. This term indicates how you physically connect the media—“punching” the media into the appropriate slot.

Key Topic

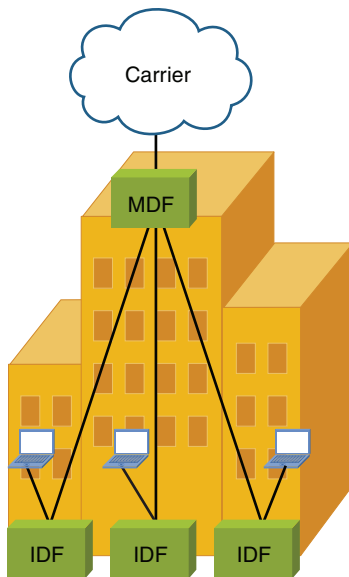


Figure 12-1 Example: Cable Distribution System

Locations

The fiber connections into a wiring closet can terminate into a ***fiber distribution panel***, also known as a ***fiber-optic patch panel***. This cable management component is mainly used for accommodating fiber cable terminations, connections, and patching. The two major categories of fiber distribution panels are wall-mount and rack-mount types.

A building might have multiple patch panels (for example, on different floors of the building). Common locations where cables from nearby offices terminate are often called ***intermediate distribution frames (IDFs)***.

The two most popular types of cross-connect blocks found in an IDF are detailed here:

- **66 block:** 66 blocks were traditionally used in corporate environments for cross-connecting phone system cabling. As 10Mbps LANs grew in popularity, in the late 1980s and early 1990s, these termination blocks were used to cross-connect Cat 3 UTP cabling. The electrical characteristics (specifically, crosstalk) of a 66 block, however, do not support higher-speed LAN technologies, such as 100Mbps or greater Ethernet networks. Figure 12-2 illustrates a 66 block.

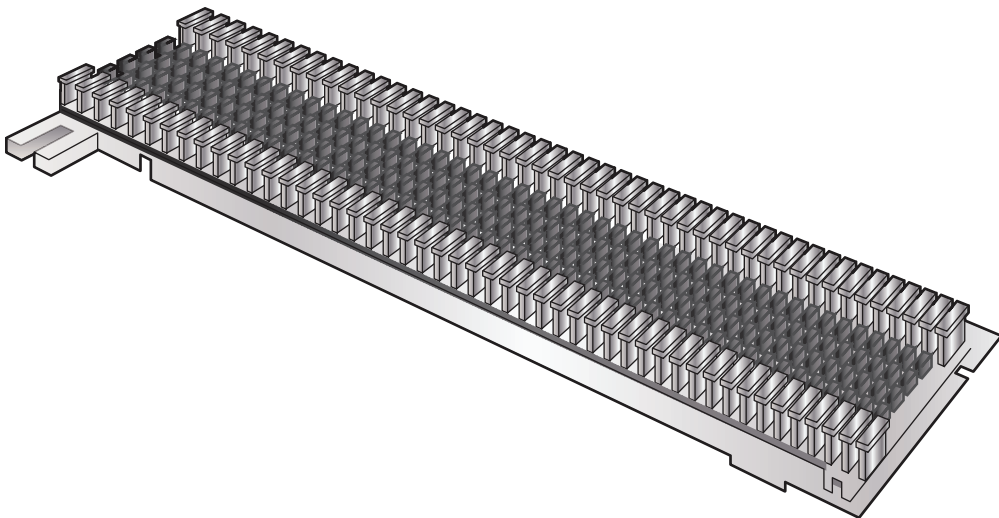


Figure 12-2 66 Block

- **110 block:** Because 66 blocks are subject to a lot of crosstalk (that is, interference between different pairs of wires) for higher-speed LAN connections, 110 blocks are often used to terminate cable (for example, a Cat 5e cable) used for higher-speed LANs. Figure 12-3 illustrates a 110 block.

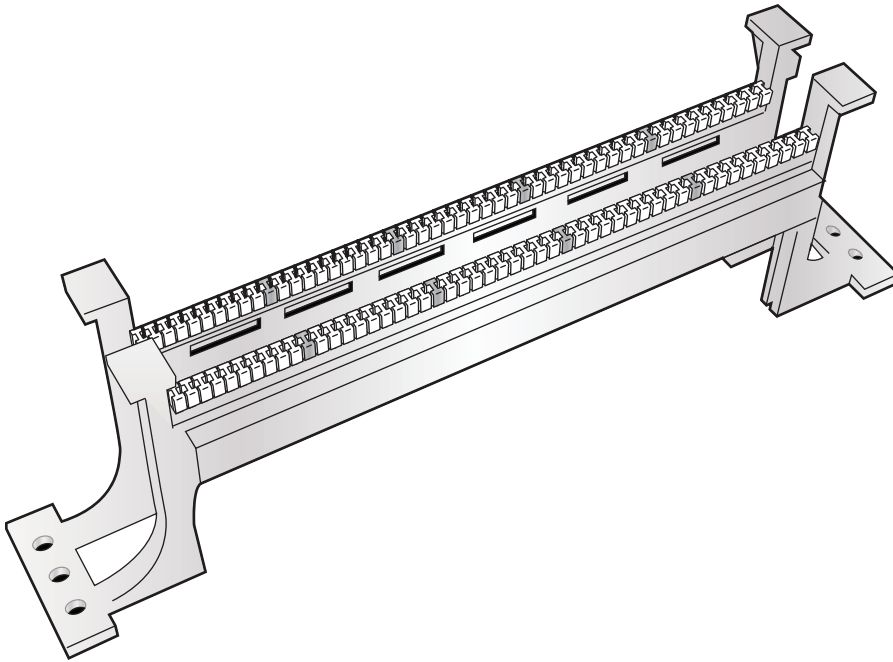


Figure 12-3 110 Block

There are two other cross-connect blocks you should be familiar with:

- **Krone (or Krone LSA-PLUS):** Krone, a proprietary European alternative to 110 block, is used not only in data environments but also in television broadcasting.
- **BIX (or Building Industry Cross-connect):** A BIX terminates 25 pairs (that is, 50 wires). The 25 pairs may be punched down to one side of a “wafer” that is then inserted into a metal frame with the punched-down side against the wall, so you only see the unused side.

The centralized distribution frame, which connects out to multiple IDFs, is called the *main distribution frame (MDF)*.

While cable management and distribution is certainly a major concern for your physical installations, another important aspect of installation is acquiring the correct equipment racks for mounting the various physical appliances that you require in your physical spaces. While it might not seem so at first glance, there are many options, with some important considerations as detailed here:

- **Rack size:** Rack sizes use a standard measure called a unit. One unit is 1.75 inches. Standard rack sizes are typically 19 inches wide and range from 42U

(units) to 48U. These standard racks are designed to house your various network devices like servers, routers, switches, and patch panels. If you find yourself short on space, you might opt for a half-height-style rack. These tend to be from 18U to 22U. You might even opt for a wall-mount rack. These are even more compact and tend to be about 12U.

- **Port-side exhausts and intakes:** Another important criterion with your equipment racks is how cooling is accommodated. Your rack design might feature a front-to-back airflow design. This is often found with your standard racks. Here, cool air is drawn in from the front and hot air is expelled from the back. Other racks might feature a side-to-side airflow. These racks accommodate devices that intake and exhaust air from the sides, requiring proper clearance and airflow management on the sides. Yet another option is a front-to-top airflow design. These more specialized racks facilitate airflow from the front intake to a top exhaust, useful in configurations where back exhausts are restricted.
- **Lockable:** To help with physical security, your network equipment racks might be lockable. Some might come with lockable front and rear doors, providing physical security to prevent unauthorized access to critical network equipment. Advanced options include electronic locks with keycard access or biometric systems, adding an extra layer of security. Some lockable racks use secure fasteners that require specific tools to open, adding another security measure for internal components.

Power

Another important physical installation factor that you should consider is power. You need to make sure you have enough of it, and you need to implement mechanisms to ensure proper power distribution and proper power redundancy.

When it comes to the distribution of power, **power distribution units (PDUs)** are devices that distribute electrical power to networking equipment within a rack or a network room. PDUs come in various configurations, including basic, metered, and intelligent PDUs, each providing different levels of power monitoring and management:

- **Basic PDUs:** Simply distribute power without any additional functionality.
- **Metered PDUs:** Allow for monitoring of power usage at the outlet level, providing insights into power consumption and enabling efficient power management.

- **Intelligent PDUs:** Go a step further by offering remote monitoring and control, allowing network administrators to manage power distribution and consumption from a central console.

Proper selection and placement of PDUs are critical for ensuring balanced power distribution and avoiding overload conditions that could lead to equipment failure.

NOTE Redundancy is often critical in networking, and it certainly is in the area of power. Therefore, redundant power sources and redundant PDUs often are used in key locations of the physical infrastructure.

When working with power in your physical installations, you will often be conscious of the **power load**, which refers to the total electrical power consumed by networking equipment in a physical installation. Understanding and managing power load is essential for ensuring that the electrical infrastructure can support the demands of the equipment without overloading circuits or causing power failures. Fortunately, there are many tools available online to assist you with determining the total power load for your networking devices.

It is important to calculate the power requirements of all devices to be installed in the network rack and ensure that the power supplies and PDUs are capable of handling the aggregate load. Overloading can lead to overheating, reduced efficiency, and increased risk of hardware failure. You should also consider future expansion and ensure that the power infrastructure has sufficient capacity to accommodate additional equipment without compromising reliability.

Voltage is another critical consideration in the physical installation of networking equipment, as it affects the performance and longevity of the devices. Networking equipment typically requires a stable voltage supply to operate correctly and to avoid damage from voltage fluctuations. Most equipment operates on standard voltages, such as 120V or 240V, depending on the region and the specific requirements of the devices. It is essential to ensure that the power supply provides the correct voltage to match the equipment's specifications.

Additionally, for Power over Ethernet (PoE) systems, specific voltage levels are required to power devices such as IP cameras and wireless access points through Ethernet cables. Consistent and proper voltage delivery ensures that networking equipment operates efficiently and reduces the risk of power-related issues.

NOTE We covered Power over Ethernet (PoE) initially in Chapter 10, “Ethernet Switching Technologies.” You will also encounter PoE again in Chapter 22, “Troubleshoot Common Cabling Problems.”

When it comes to redundancy in your power designs, the *uninterruptable power supply (UPS)* is still king. Uninterruptible power supplies are crucial for maintaining network reliability and continuity in the event of power disruptions. UPS systems provide backup power to critical networking equipment, such as servers, switches, and routers, allowing them to continue functioning during power outages. This ensures that data transmission and network services remain uninterrupted, preventing data loss and downtime. There are different types of UPS systems, including offline, line-interactive, and online double-conversion UPS, each offering varying levels of protection and power conditioning. The choice of UPS depends on the specific requirements of the network environment, including load capacity and the criticality of the equipment being protected.

Environmental Factors

When it comes to your physical computer networking facilities, it is also crucial to account for various environmental factors to ensure the reliability, efficiency, and longevity of the equipment. *Temperature* control is a primary concern, as excessive heat can significantly impact the performance and lifespan of networking devices such as servers, switches, and routers. Ideal temperature ranges for data centers typically fall between 18°C and 27°C (64°F to 80°F).

Effective cooling systems, including air conditioning and ventilation, must be in place to maintain these temperatures and prevent overheating, which can lead to system failures and downtime. Additionally, the layout of the equipment should support proper airflow, minimizing the potential for hot spots that could overheat specific components.

Humidity levels are equally important in networking facilities. Too much humidity can lead to condensation, which can cause short circuits and corrosion of electronic components. Conversely, too little humidity can result in static electricity, which poses a significant risk to sensitive equipment.

The recommended relative humidity range for data centers is typically between 40% and 60%. Monitoring and controlling humidity levels can be achieved through the use of humidifiers and dehumidifiers, ensuring that the environment remains stable and safe for the equipment.

NOTE Specific network appliances and equipment you might need to install in your physical locations will have the recommended temperatures and humidity levels specified in the literature and/or documentation that accompany the devices. Be sure to check this documentation to ensure the health and longevity of your devices. Also, to keep up with environmental standards, you can consult websites such as the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) at <https://www.ashrae.org> and Occupational Safety and Health Administration (OSHA) at <https://www.osha.gov>.

Fire suppression is another critical aspect of environmental control in networking facilities. The presence of a significant amount of electrical equipment increases the risk of fire, making it essential to have an effective fire suppression system. Traditional water-based fire sprinklers are not ideal for such environments, as water can cause extensive damage to electronic components. Instead, gas-based fire suppression systems, such as those using clean agents like FM-200 or inert gases like nitrogen and argon, are preferred. These systems are designed to suppress fires without harming the equipment or leaving residue, allowing for minimal disruption and damage in the event of a fire.

Additionally, physical security measures should be considered to protect networking facilities from environmental hazards and unauthorized access. This includes securing entry points, implementing access control systems, and monitoring for potential threats. Chapter 18, “Network Security Concepts,” will have more to say about physical security in your physical installations.

Real-World Case Study

Acme, Inc. is in the process of designing a new, state-of-the-art physical installation for its networking equipment. The company’s engineering team has undertaken a comprehensive project to design this new installation with a focus on optimal performance, reliability, and scalability.

Category 6a is the cabling media for the main Ethernet network. Cat6a cables provide superior performance with data transfer rates of up to 10Gbps and are ideal for high-bandwidth applications, ensuring that the network can handle the increasing data demands. The cabling infrastructure is being designed with a structured cabling approach, with cables neatly organized into horizontal and vertical pathways using cable trays and ladder racks. This not only facilitates easier maintenance, but also minimizes interference and signal loss. The decision to invest in high-quality,

shielded cables will help to reduce electromagnetic interference (EMI), critical for maintaining network integrity in a busy data center environment.

When it comes to server racks, Acme is opting for modular, scalable racks that can accommodate various server types and network equipment. These racks are equipped with adjustable mounting rails and cable management features, allowing for flexible configurations and efficient use of space. The engineering team is choosing racks with high airflow designs to support effective cooling and prevent overheating of the equipment. Each rack is also fitted with power distribution units (PDUs) capable of monitoring power usage in real time, providing the IT team with the necessary data to manage and optimize energy consumption. This approach not only maximizes the physical space but also enhances the manageability and scalability of the data center.

Power considerations have been crucial in Acme's planning. The company is installing an uninterruptible power supply (UPS) system to ensure that the network equipment can continue operating smoothly during power outages or fluctuations. The UPS system is complemented by a backup generator to provide additional redundancy. To further safeguard the network, Acme is implementing a dual power feed system for critical equipment, ensuring that each device has two independent power sources. This redundancy will minimize the risk of downtime and ensure continuous operation, a critical factor for maintaining the reliability and availability of Acme's services.

Acme is also paying close attention to environmental factors. To maintain optimal temperatures, the data center is to be equipped with a precision cooling system capable of maintaining temperatures between 18°C and 24°C (64°F and 75°F). This system includes hot and cold aisle containment to prevent hot air from mixing with cold air, thereby improving cooling efficiency. Humidity levels are to be controlled using advanced humidifiers and dehumidifiers to maintain a relative humidity of 45% to 55%, preventing both condensation and static electricity. For fire suppression, Acme is installing a gas-based system using FM-200, which can quickly extinguish fires without damaging electronic equipment. This comprehensive approach to environmental management will help to ensure that the data center can operate reliably and efficiently while protecting valuable equipment from environmental hazards.

Summary

Here are the main topics covered in this chapter:

- This chapter began with a look at important installation implications when it comes to physical installations of networking equipment. This included such topics as locations, equipment rack sizes, ventilation, and cabling infrastructure.
- This chapter also covered important topics regarding power implications for the physical installation. This included a discussion of uninterruptible power supplies (UPSs), power distribution units (PDUs), power load, and voltage.
- Finally, this chapter examined some of the most important environmental factors for physical installations. This included a discussion of humidity, temperature, and fire suppression techniques.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 12-1 lists these key topics and the page number where each is found.



Table 12-1 Key Topics for Chapter 12

Key Topic Element	Description	Page Number
Figure 12-1	Example: Cable Distribution System	312
List	Equipment rack considerations	314

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

fiber distribution panel, fire suppression, humidity, intake, intermediate distribution frame (IDF), lockable, main distribution frame (MDF), patch panel, port-side exhaust, power distribution unit (PDU), power load, rack size, temperature, uninterruptible power supply (UPS), voltage

Additional Resources

Real-World Network Cable Management: <https://www.youtube.com/watch?v=CV3-is8Yd8U>

Network Rack Installation – Start to Finish: <https://www.youtube.com/watch?v=QQmoZ1GrgA0>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. What is a common method of terminating cable runs in a wiring closet of an organization?
 - a. Server racks
 - b. UPS
 - c. PDU
 - d. Patch panel
2. Networking equipment racks typically use a standard measurement to help indicate the size of the rack. What is this measurement called?
 - a. Load
 - b. Unit
 - c. Area
 - d. Block
3. What is often used to disperse power to the various devices in a network equipment rack?
 - a. Punchdown block
 - b. Patch panel
 - c. PDU
 - d. UPS

4. What measurement is typically used to present the total electrical power consumed by the networking equipment in a physical installation?
 - a. Power load
 - b. Voltage
 - c. PDU
 - d. UPS

5. What networking physical installation component focuses on the redundancy of electric power?
 - a. Power load
 - b. Voltage
 - c. PDU
 - d. UPS

6. Which of the following is the primary wiring closet for a network that typically holds the majority of the network gear, including routers, switches, wiring, servers, and more?
 - a. IDF
 - b. MDF
 - c. PoE
 - d. FM-200

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This chapter covers the following topics related to Objective 3.1 (Explain the purpose of organizational processes and procedures) of the CompTIA Network+ N10-009 certification exam:

- Documentation
 - Physical vs. logical diagrams
 - Rack diagrams
 - Cable maps and diagrams
 - Network diagrams
 - Layer 1
 - Layer 2
 - Layer 3
- Asset inventory
 - Hardware
 - Software
 - Licensing
 - Warranty support
- IP address management (IPAM)
- Service-level agreement (SLA)
- Wireless survey/heat map
- Life-cycle management
 - End-of-life (EOL)
 - End-of-support (EOS)
 - Software management
 - Patches and bug fixes
 - Operating system (OS)
 - Firmware
 - Decommissioning
- Change management
 - Request process tracking/service request
- Configuration management
 - Production configuration
 - Backup configuration
 - Baseline/golden configuration

Organizational Processes and Procedures

If you are reading this text start to finish, you might have a newfound appreciation for how complex networking is today. Fortunately, there are many established processes and best practices you can use to help take much of the guesswork out of implementing and maintaining your network. This is especially beneficial in the complex and challenging area of network security.

This chapter begins by presenting many of the different forms of documentation you should consider maintaining as you design and operate modern networks. This documentation includes such items as network documentation, asset inventories, and service-level agreements (SLAs).

This chapter also discusses critical processes such as life-cycle management, change management, and configuration management. These practices are considered by many to be an absolute must when working with networks today.

Foundation Topics

Documentation

I cannot stress enough the importance of up-to-date and accurate network documentation. Documentation is useful for proper network management tasks, and it is also critical in troubleshooting, security, optimization, and other key areas of networking today. Consider these important examples of network documentation:

Key Topic

- **Physical diagrams vs. logical diagrams:** A physical diagram represents the tangible, hardware aspects of a network, such as cables, switches, routers, and their physical connections and locations within a facility. It shows how the components are physically interconnected and where they are positioned. In contrast, a logical diagram focuses on the abstract, conceptual layout of the network, illustrating the flow of data, network segmentation, IP addressing schemes, and how devices and services are logically organized and interact with each other. While a physical diagram provides a map for setting up and maintaining the actual hardware, a logical diagram is essential for understanding the network's data flow, design, and the relationships between different network entities.
- **Rack diagrams:** These days, thanks to the very small form factors of many network devices, we have more devices than ever before coexisting in a single rack in a data center. Therefore, rack diagrams are needed to assist in the management of all the various devices in the racks. Figure 13-1 shows an example of a rack diagram. This diagram was made with the free version of Lucidchart (<https://www.lucidchart.com>).
- **Cable maps and diagrams:** Another key set of documentation includes diagrams showing cabling and port locations. This documentation allows you to track cable runs from switches and map them to actual wall jacks where users connect to your network; these connections might also represent trunks to additional network devices such as wireless access points.
- **Network diagrams:** Network diagrams are visual representations that illustrate the structure, components, and connections within a network. They serve as essential tools for designing, analyzing, and managing network infrastructure. **Layer 1 diagrams**, or physical diagrams, depict the actual hardware components, such as cables, switches, and routers, highlighting how these devices are physically interconnected. **Layer 2 diagrams**, often referred to as data link layer diagrams, focus on how devices such as switches and bridges manage data frames within a local network, detailing MAC addresses and VLAN

configurations. **Layer 3 diagrams**, or network layer diagrams, emphasize the logical flow of data across networks, showcasing IP addressing, routing protocols, and the interactions between routers and other Layer 3 devices.

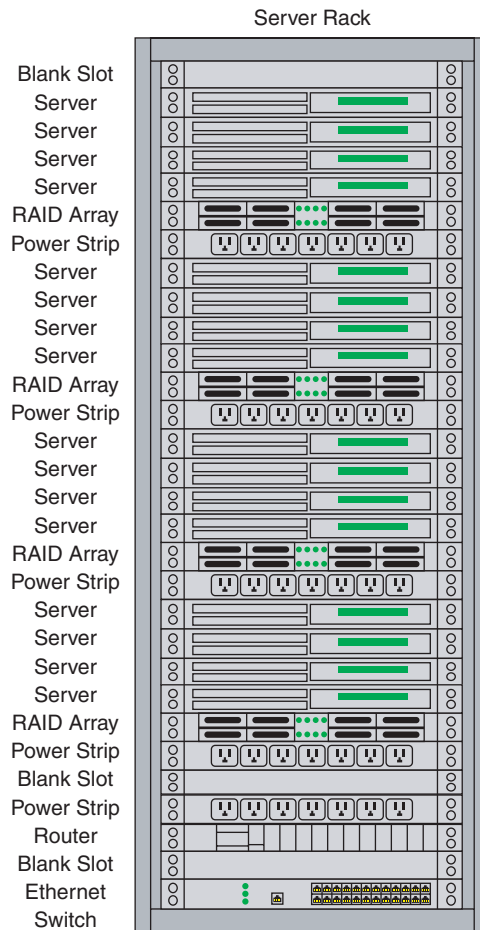


Figure 13-1 A Rack Diagram

- **Asset inventory:** An asset inventory in an IT department's documentation is a comprehensive catalog of the organization's technology assets, encompassing hardware, software, licensing, and warranty support. A **hardware asset inventory** typically lists all physical devices such as servers, desktops, laptops, network equipment, and peripherals. Such an inventory often details specifications, purchase dates, locations, and unique identifiers. A **software asset inventory** typically includes applications and operating systems, specifying versions,

installations, usage metrics, and compliance with organizational policies. The **licensing asset inventory** tracks software licenses, ensuring compliance with vendor agreements, detailing license types, quantities, expiration dates, and renewal requirements. The **warranty support asset inventory** typically records warranty coverage for hardware and software, including start and end dates, support levels, and vendor contact information, to manage maintenance and support effectively.

- **IP address management (IPAM):** Another important aspect of documentation is IP address management. IPAM is a means of planning, tracking, and managing the Internet Protocol address space used in a network. IPAM typically integrates with DNS and DHCP so that each is aware of changes in the other (for instance, DNS knowing of the IP address taken by a client via DHCP and updating itself accordingly).

NOTE Stressed out about how you are going to build your own IPAM documentation system? While you can certainly create your own using something like spreadsheet software, there are many IPAM tools available standalone and as part of broader network management systems.

- **Service-level agreements (SLAs):** Service-level agreements are often a critical component of an IT department's documentation, as they define the expected service quality and performance metrics between the IT service provider and its clients or internal stakeholders. SLAs typically outline key aspects such as service availability, response times, resolution times, and performance benchmarks, ensuring that both parties have a clear understanding of the service expectations and responsibilities. Including SLAs in IT documentation is essential for setting measurable standards for service delivery, which helps in managing and meeting client expectations, identifying areas for improvement, and maintaining accountability. SLAs also serve as a reference point for resolving disputes and managing service-related issues, ensuring that services are delivered consistently and in alignment with business goals. Furthermore, SLAs provide a framework for regular performance reviews and enable continuous improvement by highlighting areas where service levels may need to be adjusted to meet changing requirements or to enhance overall service quality.
- **Wireless survey/heat map:** Wireless surveys and heat maps are often integral components of the IT documentation in modern organizations, providing a detailed visual representation and analysis of wireless network coverage and performance across various locations. A wireless survey involves assessing the physical environment to identify optimal placement of wireless access points,

detect signal interference, and measure signal strength and quality. The resulting heat maps visually depict these findings, using color gradients to represent signal strength, coverage areas, and potential dead zones, enabling IT professionals to pinpoint areas with weak or no signal and optimize the network layout accordingly. This documentation ensures that wireless networks are robust, providing reliable connectivity and bandwidth to meet the needs of users across different locations. It also serves as a critical tool for troubleshooting, network planning, and ensuring compliance with organizational policies and standards, helping to maintain a high-quality wireless infrastructure that supports efficient and uninterrupted business operations.

Processes and Procedures

Thankfully, many important processes and procedures have become commonplace in networks today. While no one likes it when processes are overly complex due to “red tape,” well-defined and targeted initiatives can be revolutionary for organizations and their success. This section describes some of the most critical processes and procedures for your review.

Life-cycle Management

Your IT networking personnel should consider adhering to a well-planned *life-cycle management* process for the equipment and software in use. The life cycle should provide valuable guidance on best practices throughout the organization concerning the network components.

Here are some examples of phases in a system life cycle:

- Conceptual design
- Preliminary system design
- Detail design and development
- Production and construction
- Utilization and support
- Phase-out (*decommissioning*)
- Disposal

Another important aspect of life-cycle management is understanding and adhering to the support announcements of vendors regarding their systems.

These announcements will often include (at the least) the following:

**Key
Topic**

- **End-of-life (EOL):** The end-of-life status declared by a networking vendor for a product signifies that the product will no longer receive updates, support, or maintenance from the vendor. This status typically includes a timeline outlining the phases of discontinuation, such as the end of sales, end of new feature development, and the final date for support and security updates. EOL status alerts users to the necessity of planning for the transition to newer products or solutions, as continued use of EOL products may lead to increased security risks, lack of compatibility with new technologies, and potential operational disruptions. The vendor may also provide recommendations or pathways for upgrading to supported alternatives, ensuring continuity and improved performance within the network infrastructure.
- **End-of-support (EOS):** The end-of-support status assigned by a networking vendor to a product indicates that the vendor will no longer provide technical support, updates, patches, or maintenance services for that product. Once a product reaches EOS, users will not receive security patches, bug fixes, or assistance with troubleshooting issues, which may lead to vulnerabilities and operational risks over time. EOS status prompts organizations to consider upgrading to newer, supported products to maintain security, compatibility, and optimal performance.

Networking personnel often consider the life cycle of software as part of their overall **software management** processes and procedures. Considerations in this area typically include the following:

- **Patches and bug fixes:** Patches and bug fixes are often essential components of software management, ensuring that applications and systems remain secure, stable, and efficient over time. Patches typically address security vulnerabilities, performance issues, or feature updates, protecting systems from potential threats and improving functionality. Bug fixes specifically resolve defects or malfunctions in the software, preventing errors and enhancing user experience. Regularly applying patches and bug fixes helps maintain software integrity, compliance with industry standards, and compatibility with other systems.
- **Operating system (OS) updates:** Operating system updates are another critical aspect of software management, providing essential enhancements, security improvements, and new features that ensure the overall health and functionality of the IT environment. These updates address vulnerabilities that could be exploited by malicious entities, fix bugs that may affect system performance or user experience, and offer compatibility updates to support new hardware and software. Figure 13-2 shows the OS updates page of a Windows 11 system.

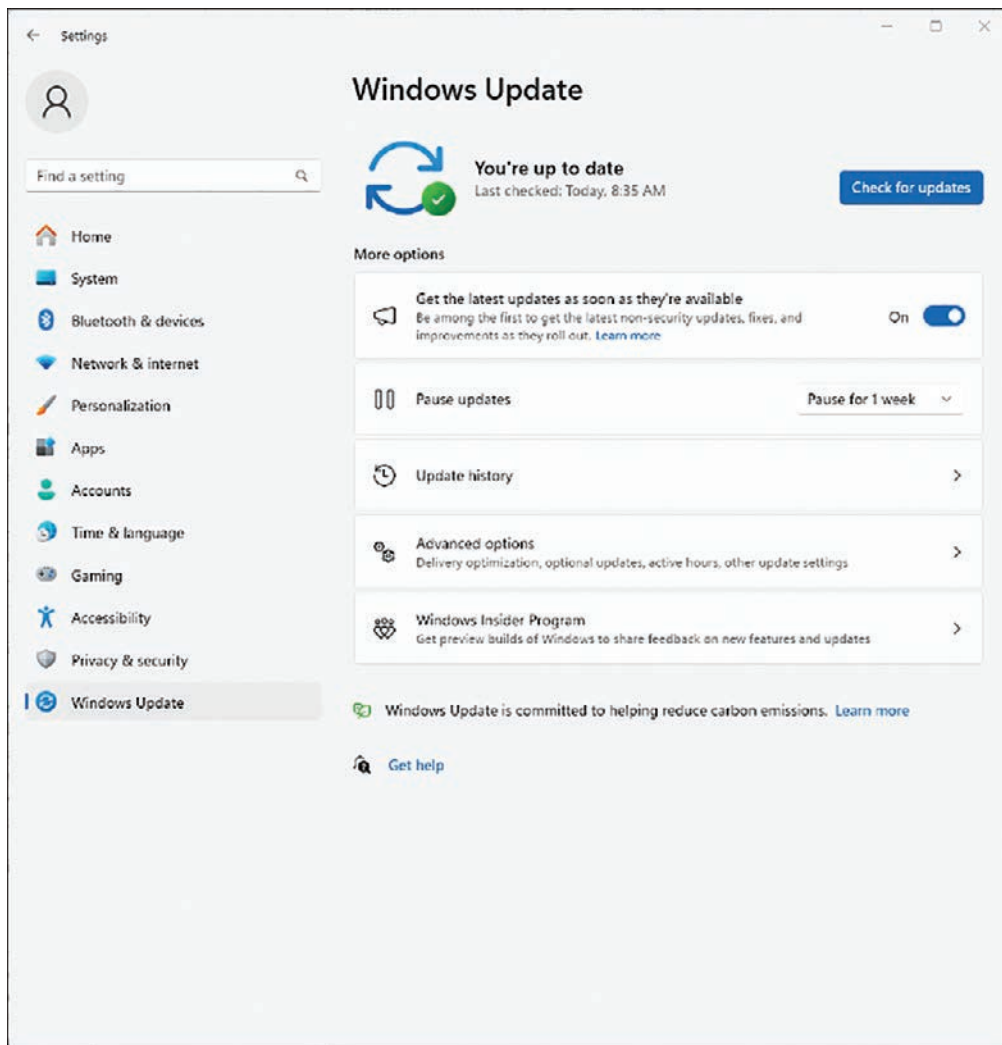


Figure 13-2 Operating System (OS) Updates

- **Firmware updates:** Firmware updates are another important element of software management, focusing on the underlying code that controls the hardware devices within a system. These updates are essential for enhancing device functionality, improving performance, and addressing security vulnerabilities that could otherwise compromise the hardware and the system as a whole. By keeping firmware up to date, organizations can ensure that their hardware operates efficiently, remains compatible with newer software, and supports the latest features and security protocols. Figure 13-3 shows an example of updating the firmware on a network access point.

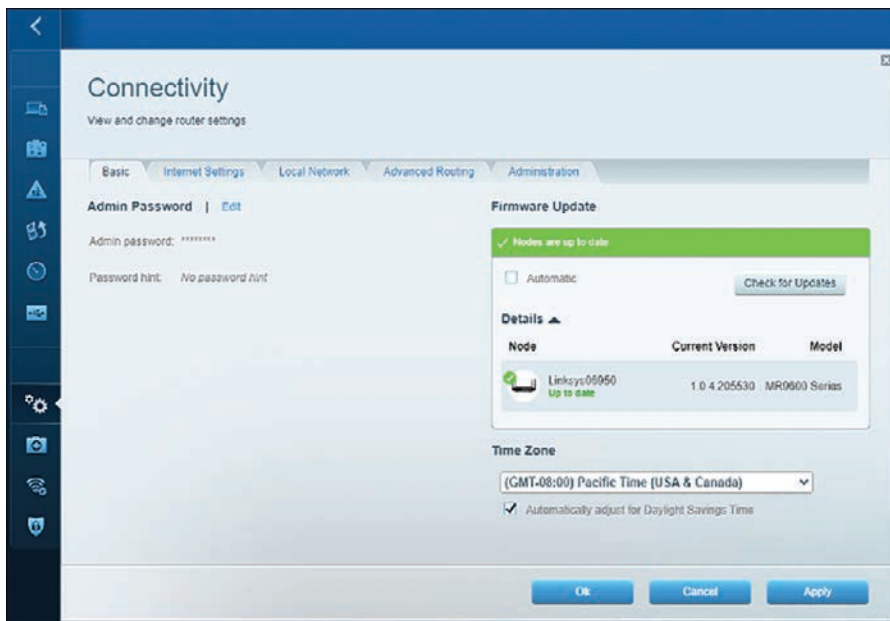


Figure 13-3 Firmware Updates

Change Management

Have you ever made a simple little change to a system and been horrified to see the entire solution self-destruct? It can happen, and because of this, you should ensure that there is a well-thought-out **change management** program in place. This change management program should include a change control policy. You should also have a plan that includes the appropriate change management documentation. Often, it is necessary to make updates to existing documentation as part of the overall change management process.

Service requests and **request process tracking** are integral to modern networking change management systems, facilitating structured and transparent management of changes to network infrastructure. Service requests, initiated by users or IT staff, encompass a range of activities, from simple troubleshooting to significant network modifications. Request process tracking ensures that each request is logged, prioritized, and systematically managed through various stages, including approval, implementation, and post-implementation review. This systematic approach helps maintain clear communication, ensures accountability, and minimizes disruptions by assessing potential impacts and risks before changes are made. By providing a comprehensive audit trail and status updates, these systems enhance efficiency, reduce

the likelihood of errors, and support the smooth and secure evolution of the network environment.

In short, change management can help ensure that your network keeps running in good health. It can also ensure that your documentation reflects the true and current state of the objects it is describing. This is very important for many aspects of your operations, including security responses and troubleshooting operations. Change management documentation often includes the following:

- A documented reason for a change
- The actual change request
- The approval process
- The required maintenance window
- The change notification process

Configuration Management

Today, *configuration management* plays a crucial role in ensuring the stability, security, and scalability of network infrastructures. It involves the systematic management of configuration changes and settings across network devices, such as routers, switches, firewalls, and servers. This process is essential for maintaining consistency, reducing errors, and facilitating efficient troubleshooting and deployment.

When working with configuration management in the modern enterprise, we are often dealing with several different types of configurations, as follows:

Key Topic

- ***Production configurations:*** Production configurations refer to the current operational settings and parameters of network devices that are actively serving traffic and fulfilling their intended roles within the network. These configurations are carefully crafted to meet specific performance, security, and operational requirements. Network administrators use configuration management tools to deploy, monitor, and update production configurations while minimizing downtime and disruptions. Automated tools often play a crucial role here, ensuring that changes are applied consistently and according to established policies.
- ***Backup configurations:*** Backup configurations are copies of production configurations stored securely to facilitate disaster recovery, rollback procedures, and historical reference. These backups are typically scheduled regularly to capture changes and updates made to production configurations over time. Having reliable backups is essential for restoring network services in case of failures, human errors, or security incidents. Configuration management

systems ensure that backups are current, complete, and easily accessible to authorized personnel.

- **Baseline/golden configurations:** A baseline or golden configuration represents standardized templates or ideal states of network device configurations that serve as benchmarks for comparison and consistency. These configurations are meticulously designed and documented to reflect best practices, security guidelines, and operational requirements. They are used as reference points for validating and aligning production configurations. When deviations occur or when updates are needed, network administrators can refer to these baseline configurations to ensure that changes are made in a controlled and compliant manner.

In practice, configuration management involves several key processes:

- **Configuration deployment:** New configurations or updates are deployed to network devices using automated tools to ensure consistency and minimize manual errors.
- **Configuration monitoring:** Continuous monitoring of configurations helps detect unauthorized changes, performance issues, or deviations from baseline configurations.
- **Change management:** Formalized processes govern how changes are proposed, reviewed, approved, and implemented to maintain stability and compliance.
- **Version control:** Keeping track of configuration versions enables rollback to previous states if changes lead to unexpected issues or failures.
- **Auditing and compliance:** Regular audits verify that configurations align with security policies, regulatory requirements, and operational standards.

Real-World Case Study

Acme, Inc. has implemented several key improvements in its IT management practices, focusing on IP address management, service-level agreements, wireless surveys, and a more robust approach to life-cycle management for the hardware and software in use at Acme.

Acme recognizes the critical importance of efficient IP address management to streamline network operations and avoid IP conflicts. By adopting IPAM tools and processes, Acme now ensures the systematic allocation, tracking, and management of IP addresses across its network infrastructure. This approach not only enhances

network reliability and scalability but also simplifies troubleshooting and enhances security by maintaining accurate IP address inventories.

Acme is now formalizing service-level agreements to define clear expectations and commitments between its IT department and internal stakeholders or external service providers. These SLAs establish benchmarks for performance, uptime, response times, and resolution targets for IT services. By aligning SLAs with business objectives, Acme aims to improve service delivery, accountability, and customer satisfaction while effectively managing expectations across the organization.

Recognizing the growing reliance on wireless connectivity, Acme has begun conducting thorough wireless surveys to optimize its Wi-Fi networks. These surveys assess signal strength, coverage areas, interference sources, and user experience metrics to ensure robust and reliable wireless connectivity throughout its facilities. By proactively addressing coverage gaps and performance issues identified through surveys, Acme enhances productivity and user satisfaction across its workforce.

Finally, Acme is adopting life-cycle management principles to systematically plan, procure, deploy, maintain, and retire IT assets and services. This approach ensures that IT resources are aligned with business needs throughout their lifespans, optimizing investments and minimizing disruptions. By proactively managing life cycles, Acme enhances cost-effectiveness, operational continuity, and technological agility in adapting to evolving business requirements.

Summary

Here are the main topics covered in this chapter:

- This chapter described some of the most important elements of documentation related to modern networking. Examples of documentation included rack diagrams, physical diagrams, logical diagrams, and much more. Not only is it important to create such documentation, but it is also critical to ensure the documentation is periodically updated.
- This chapter also discussed life-cycle management and some of the important hardware and software milestones associated with such management. These milestones included end-of-life (EOL) and end-of-support (EOS).
- Finally, this chapter examined two related topics—change management and configuration management. The chapter defined each of these related disciplines and provided example components of each.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 13-1 lists these key topics and the page number where each is found.



Table 13-1 Key Topics for Chapter 13

Key Topic Element	Description	Page Number
List	Typical types of network documentation	326
List	Vendor life-cycle announcement types	330
List	Types of configurations in configuration management	333

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

asset inventory, backup configurations, baseline/golden configurations, bug fixes, cable maps, change management, configuration management, decommissioning, end-of-life (EOL), end-of-support (EOS), firmware updates, hardware asset inventory, heat map, IP address management (IPAM), Layer 1 diagrams, Layer 2 diagrams, Layer 3 diagrams, licensing asset inventory, life-cycle management, logical diagrams, network diagrams, operating system (OS) updates, patches, physical diagrams, production configurations, rack diagrams, request process tracking, service-level agreement (SLA), service request, software asset inventory, software management, warranty support asset inventory, wireless survey

Additional Resources

Redefine Your I.T. Strategy with Proper Hardware Lifecycle Management: <https://www.youtube.com/watch?v=LhDJBtaEuR8>

What Is Configuration Management?: <https://www.redhat.com/en/topics/automation/what-is-configuration-management>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. What type of network diagram would focus on IP addresses and other network layer details?
 - a. Layer 1 network diagram
 - b. Layer 2 network diagram
 - c. Layer 3 network diagram
 - d. Layer 4 network diagram
2. What is the best definition of IPAM?
 - a. IPAM refers to the process of optimizing Internet Protocol usage through advanced monitoring.
 - b. IPAM refers to the systematic administration of IP addresses, ensuring efficient allocation, tracking, and maintenance within a network environment.
 - c. IPAM involves the physical management of network cables and their connections within a data center.
 - d. IPAM involves the automatic optimization of traffic flows based on specific quality of service (QoS) values.
3. Which of the following is the best definition of a service-level agreement?
 - a. A formal contract between a service provider and a customer that outlines the agreed-upon level of performance.
 - b. A software tool that is used to monitor network performance.
 - c. The process of negotiating software updates and maintenance.
 - d. A popular hardware specification used in network design and configuration.
4. EOS and EOL dates are often critical and are found primarily in what process discipline within IT?
 - a. Service-level agreements
 - b. Change management
 - c. Configuration management
 - d. Life-cycle management

5. What is often the last stage of a system life cycle used in a network?
 - a. Phase-out
 - b. Disposal
 - c. Support
 - d. Development
6. Which of the following is a predefined and standardized template or ideal state of configuration settings for hardware, software, or network devices?
 - a. Configuration deployment
 - b. Configuration monitoring
 - c. Version control
 - d. Baseline/golden configuration
7. Which are often integral components of the IT documentation in modern organizations, providing a detailed visual representation and analysis of wireless network coverage and performance across various locations? (Choose two.)
 - a. Wireless survey
 - b. IPAM
 - c. Layer 2 diagram
 - d. Heat map
8. Which of the following best describes a comprehensive record or database containing detailed information about all hardware, software, and resources within an organization's IT infrastructure?
 - a. Request process tracking
 - b. Asset inventory
 - c. IPAM
 - d. Service requests

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This chapter covers the following topics related to Objective 3.2 (Given a scenario, use network monitoring technologies) of the CompTIA Network+ N10-009 certification exam:

- Methods
 - SNMP
 - Traps
 - Management information base (MIB)
 - Versions
 - v2c
 - v3
 - Community strings
 - Authentication
 - Flow data
 - Packet capture
 - Baseline metrics
 - Anomaly alerting/notification
 - Log aggregation
 - Syslog collector
 - Security information and event management (SIEM)
 - Application programming interface (API) integration
 - Port mirroring
- Solutions
 - Network discovery
 - Ad hoc
 - Scheduled
 - Traffic analysis
 - Performance monitoring
 - Availability monitoring
 - Configuration monitoring

Network Monitoring

No one would argue with the statement that networks today are more critical than ever before for organizations. Fortunately, many tools and techniques are available to help us ensure that our networks are performing as we need them to and staying that way for as long as possible.

In this chapter you will learn about many metrics and statistics you can call upon to gauge the health of your network. These include simple metrics that indicate problems in the network as well as sophisticated logs that can be powerful sources of network operational information.

It is imperative to engage in careful and continuous network monitoring to create resources and reports that provide valuable information. For example, the primary network management protocol used by network management systems (NMSs) is Simple Network Management Protocol (SNMP), and this chapter discusses the various versions of SNMP. In addition, syslog servers and a variety of network monitoring reporting and logging solutions are considered.

Foundation Topics

Network Monitoring Methods

For several decades, SNMP-based software packages dominated in the enterprise monitoring space. Fortunately, SNMP evolved over several versions to include security enhancements. Even with these advancements, as you'll learn later in this chapter, there are other methods and technologies to consider when it comes to monitoring a network. This section discusses some key performance metrics and sensors that you need to monitor.

SNMP

The first Request for Comments (RFC) for *Simple Network Management Protocol (SNMP)* came out in 1988. Since then, SNMP has become the de facto standard for network management protocols. The original intent for SNMP was to manage network nodes, such as network servers, routers, and switches. SNMP Version 1 (SNMPv1) and *SNMP Version 2c (SNMPv2c)* specify three major components that makeup an SNMP solution, as detailed in Table 14-1.

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Table 14-1 Components of an SNMPv1 and SNMPv2c Network Management Solution

Component	Description
SNMP manager	An SNMP manager runs a network management application. This SNMP manager is sometimes referred to as a <i>network management system (NMS)</i> .
SNMP agent	An SNMP agent is a piece of software that runs on a managed device (for example, a server, router, or switch).
<i>Management information base (MIB)</i>	Information about a managed device's resources and activity is defined by a series of objects. The structure of these management objects is defined by a managed device's MIB. Interfaces and their details—such as errors, utilization, discards, packet drops, resets, speed and duplex, system memory, utilization of bandwidth, storage, CPU, and memory—can be monitored and reported via SNMP.

As illustrated in Figure 14-1, an SNMP manager (an NMS) can send information to, receive request information from, or receive unsolicited information from a managed device (a managed router, in this example). The managed device runs an SNMP agent and contains the MIB.

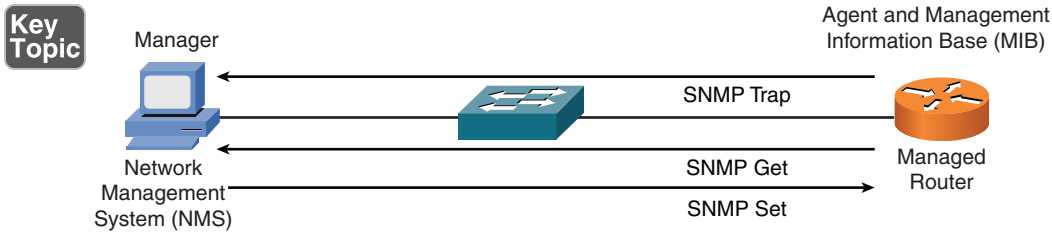


Figure 14-1 SNMPv1 and SNMPv2c Network Management Components and Messages

Clearly, the MIB is a critical component in the functionality of SNMP. It is a hierarchical collection of MIB variables that stores the data that SNMP relies on. But how are these MIB variables identified? This is done using a unique *object identifier (OID)* for each variable. The OIDs are organized in a hierarchical tree-like structure. SNMP is very efficient at scanning this tree and extracting the exact OIDs and values of these variables that are needed to monitor or even reconfigure a network device.

Multiple SNMP messages might be sent between an SNMP manager and a managed device, and there are three broad categories of SNMP message types:

- **Get:** An SNMP get message retrieves information from a managed device.
- **Set:** An SNMP set message sets a variable in a managed device or triggers an action on a managed device.
- **Trap:** An SNMP trap message is an unsolicited message sent from a managed device to an SNMP manager, which can notify the SNMP manager about a significant event that occurred on the managed device.

SNMP management software can make requests for each of the MIB objects from an SNMP agent. This can be referred to as an SNMP *walk* because the management software is logically “walking” the entire MIB (also often called the *tree*) to gather information from the agent. SNMP offers security against malicious users attempting to collect information from a managed device, change the configuration of a managed device, or intercept information being sent to an NMS. However, the security integrated with SNMPv1 and SNMPv2c is considered weak. Specifically, SNMPv1 and SNMPv2c use *community strings* to gain read-only access or read/write access to a managed device. You can think of a community string as a plaintext password. This is why there has been a big push in the networking industry as of late to move toward SNMPv3.

Also, be aware that multiple SNMP-compliant devices on the market today have the read-only community string set to *public* by default and have the read/write

community string set to *private* by default. As a result, if such devices are left at their default SNMP settings, they could be compromised.

NOTE Notice that this section refers to SNMPv2c as opposed to SNMPv2. SNMPv2 contained security enhancements as well as other performance enhancements. However, few network administrators adopted SNMPv2 because of the complexity of the newly proposed security system. Instead, Community-Based Simple Network Management Protocol (SNMPv2c) gained widespread acceptance because it included the performance enhancements of SNMPv2 without using SNMPv2's complex security solution. Instead, SNMPv2c kept the SNMPv1 concept of community strings.

Fortunately, the security weaknesses of SNMPv1 and SNMPv2c are addressed in **SNMP Version 3 (SNMPv3)**. To better understand these security enhancements, consider the concept of a security model and a security level:

- **Security model:** Defines an approach for user and group *authentication* (for example, SNMPv1, SNMPv2c, and SNMPv3).
- **Security level:** Defines the type of security algorithm performed on SNMP packets. The following are the three security levels discussed here:
 - **noAuthNoPriv:** The noAuthNoPriv (no authorization, no privacy) security level uses community strings for authorization and does not use encryption to provide privacy.
 - **authNoPriv:** The authNoPriv (authorization, no privacy) security level provides authorization using Hashed Message Authentication Code (HMAC) with Message Digest 5 (MD5) or Secure Hash Algorithm (SHA). However, no encryption is used.
 - **authPriv:** The authPriv (authorization, privacy) security level offers HMAC MD5 or SHA authentication and provides privacy through encryption. Specifically, the encryption uses the Cipher Block Chaining (CBC) Data Encryption Standard (DES) (DES-56) algorithm.

NOTE The security protocols originally featured in SNMPv3 are still considered strong enough for today's networks. Since then, additions to SNMPv3 have made it even more secure. These additions enable high-security environments to call on the latest and strongest security mechanisms for key functions, such as encryption, and feature protocols such as AES.

As summarized in Table 14-2, SNMPv3 supports all three security levels, and SNMPv1 and SNMPv2c support only the noAuthNoPriv security level.

Table 14-2 Security Models and Security Levels

Security Model	Security Level	Authentication Strategy	Encryption Type
SNMPv1	noAuthNoPriv	Community string	None
SNMPv2c	noAuthNoPriv	Community string	None
SNMPv3	noAuthNoPriv	Username	None
SNMPv3	authNoPriv	MD5 or SHA	None
SNMPv3	authPriv	MD5 or SHA	CBC-DES (DES-56)

Through the use of security algorithms, as shown in Table 14-2, SNMPv3 dramatically increases the security of network management traffic. Specifically, SNMPv3 offers three primary security enhancements over SNMPv1 and SNMPv2c:

- **Integrity:** Using hashing algorithms, SNMPv3 ensures that an SNMP message was not modified in transit.
- **Authentication:** Hashing allows SNMPv3 to validate the source of an SNMP message.
- **Encryption:** Using the CBC-DES (DES-56) encryption algorithm, SNMPv3 provides privacy for SNMP messages, making them unreadable by an attacker who might capture an SNMP packet.

In addition to its security enhancements, SNMPv3 differs architecturally from SNMPv1 and SNMPv2c. SNMPv3 defines SNMP entities, which are groupings of individual SNMP components. As shown in Figure 14-2, SNMP applications and an SNMP manager combine into an NMS SNMP entity, whereas an SNMP agent and an MIB combine into a managed node SNMP entity.

Performance Metrics/Sensors

As networks have become more and more important to the organizations that host them, metrics have increasingly become available to carefully monitor and assess the health of network components. In fact, you need to be careful not to overwhelm yourself by trying to monitor too many metrics for your devices.

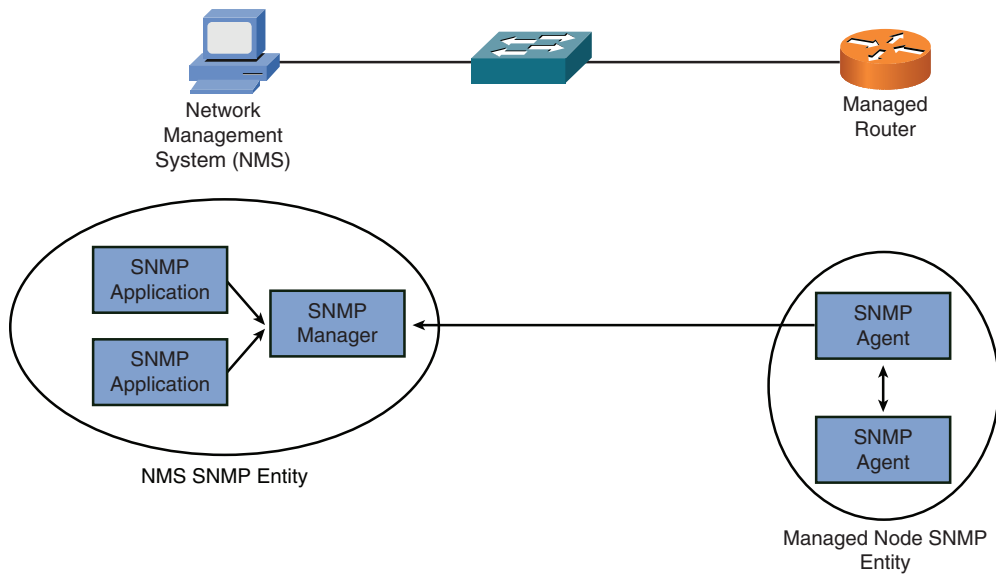


Figure 14-2 SNMPv3 Entities

The following are some key metrics that help you monitor a device and the device's chassis:

- **Temperature:** Network equipment vendors make it very clear what temperature range is supported for inside the device chassis. Of course, you will not want the temperature to exceed or go too far below this recommended range. You might run into temperature problems, for example, if your network equipment was not installed following best practices and there is not adequate distance between a device and other objects. These other objects might be blocking airflow to the network device. Maybe the warm air that is supposed to be pushed out of the chassis is getting caught and trapped inside the chassis. Sensors can provide metrics regarding the temperature inside network devices to help you pinpoint such issues. Later in this chapter, you will learn about tools called *environmental monitors* that are specialized for monitoring things like temperature, humidity, and more.
- **Central processing unit (CPU) usage:** The CPU is the “brains” of a computer, often navigating the tricky waters of the control plane processing requirements for you. It is very important to monitor CPU utilization consistently over the life of a network device. Doing so can help you not only identify network attacks but also determine the busy times for network devices during which performance is suffering. In addition, monitoring the CPU can

help you see when components are beginning to fail and causing high CPU utilization.

- **Memory:** To improve the performance of network devices, data is taken from nonvolatile storage and placed in (much faster) volatile memory. Of course, you need to make sure you have enough memory available for this at all times. Monitoring the various memory metrics can help you track down lots of current and potential performance issues.

As you can imagine, it is important to carefully monitor these metrics for a network device and its chassis. In addition, you should also carefully monitor the network itself. The following are some of the basic metrics for monitoring a network:

- **Bandwidth:** The “plumbing” of your network is the connections between the devices. Remember that these connections might be physical media that you can see, and they might also be wireless media that you cannot see. Carefully measuring the bandwidth available and bandwidth consumed by specific applications or traffic forms is imperative. Watch out for bottlenecks in the network design, which often occur when a network designer fails to consider the aggregate bandwidth required. For example, say that you have 100 10Gbps ports, and you need to plan for busy times when those 100 really fast ports all have lots of data they want to send. You might require multiple 40Gbps connections upstream to handle the aggregate bandwidth needs.
- **Latency:** Latency (that is, delay) is a fact of life. When a network device sends packets, there will be serialization delay, propagation delay, processing delay, and probably even more types of delay. You need to take baseline measurements (often termed *baseline metrics*) so you can understand the latency in your network. You should also pay attention to the requirements for certain technologies. For example, perhaps your organization wants to engage with the latest Cisco collaboration solutions, including voice and video over IP. Cisco will give you documentation on just the latency that the solutions can handle. Cisco might say something like, “For this solution to perform well, you must not exceed 120 ms of delay consistently over time.” It is important to gather data on such metrics before you even consider rolling out a solution.
- **Jitter:** Jitter is what happens when you drink too much coffee, and it also refers to large variations in latency. Jitter is especially problematic for VoIP traffic. If the latency in a network is not predictable and fairly steady, it can be very difficult for VoIP quality to remain steady and satisfactory.

NOTE Important concepts such as latency and jitter will appear again in Chapter 24, “Troubleshoot Common Performance Issues.”

For all of the major metric categories, you should ensure that you perform baselin- ing and properly store the results for future analytics. A baseline provides a look at the metrics and their values during “normal” operation. If your network tends to run at 45% capacity, then you would want to watch for this level of usage and then start capturing your metric information to create a baseline (specifically, a baseline metric). This is incredibly important. The establishment of baselines also permits for the implementation of *anomaly alerting* and *notifications*. When metric values leave baselines, you can alert administrators to examine the network for issues. This proactive approach to the network can have dramatic and positive effects.

Port Mirroring

For troubleshooting purposes, you might want to analyze packets flowing over a network. Back in the early days of networking, to capture packets (that is, store cop- ies of packets on a local hard drive) for analysis, you could attach a *network sniffer* to a hub. Because a hub sends bits received on one port out all other ports, the attached network sniffer sees all traffic entering the hub.

Several standalone network sniffers are available. However, a low-cost way to perform *packet capture* and analysis is to use software such as Wireshark (www.wireshark.org), as shown in Figure 14-3.

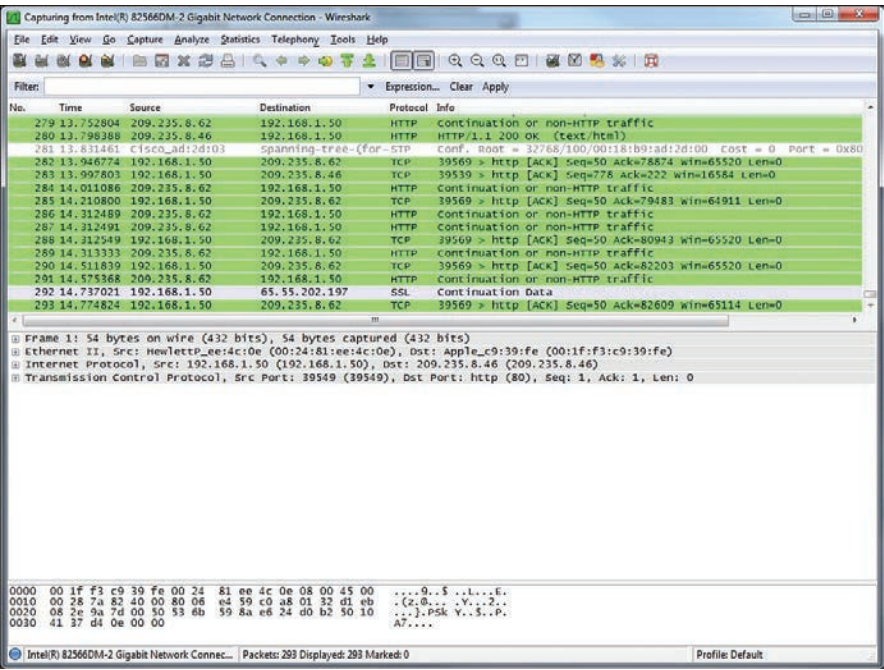


Figure 14-3 Example: Wireshark Packet-Capture Software

A challenge arises, however, if you connect a network sniffer (for example, a laptop running the Wireshark software) to a switch port rather than to a hub port. Because a switch, by design, forwards frames out ports containing the frames' destination addresses, a network sniffer attached to one port would not see traffic destined for a device connected to a different port.

For example, in Figure 14-4, traffic enters a switch on port 1 and, based on the destination MAC addresses, exits via port 2. However, a network sniffer is connected to port 3 and is unable to see (and therefore capture) the traffic flowing between ports 1 and 2.

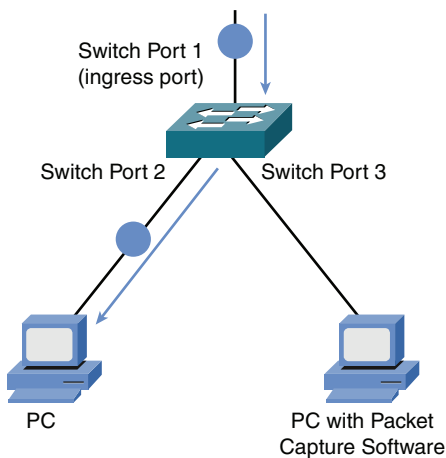


Figure 14-4 Example: Network Sniffer Unable to Capture Traffic

Fortunately, some switches support a *port mirroring* feature, which makes a copy of traffic seen on one port and sends that duplicated traffic out another port (to which a network sniffer could be attached). As shown in Figure 14-5, this switch is configured to mirror traffic on port 2 to port 3, allowing a network sniffer to capture the packets that need to be analyzed. Depending on the switch, locally captured traffic could be forwarded to a remote destination for centralized analysis of that traffic.

NOTE Does port mirroring place a higher burden on the switch that is performing it? The answer is yes! Remember, you want to engage in port mirroring only when you truly need to. This is most often when you are troubleshooting.

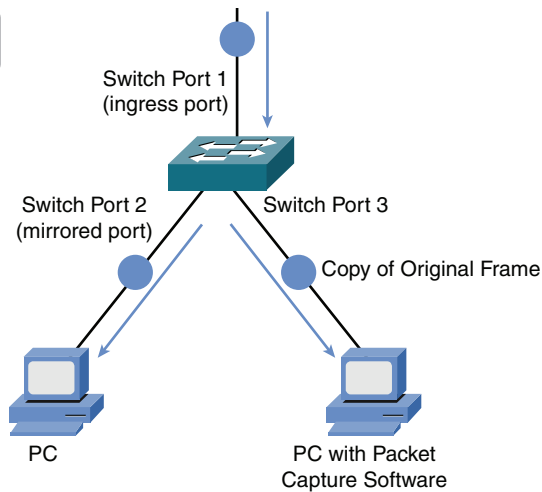
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Figure 14-5 Example: Network Sniffer with Port Mirroring Configured on the Switch

Port Mirroring Configuration

Example 14-1 shows a sample configuration from a Cisco Catalyst switch that captures all the frames coming in on port Gig 0/1 and forwards them to port Gig 0/3.

Example 14-1 Port Mirroring Configuration

```
SW1(config)# monitor session 1 source interface Gi0/1

SW1(config)# monitor session 1 destination interface Gi0/3
```

Logging

Network administrators routinely monitor network resources and review reports to be proactive in their administration. For example, a potential network issue might be averted by spotting a trend such as increasing router CPU utilization or increasing bandwidth demand on a WAN link. Monitoring resources and reports come from various sources, such as a syslog server, an SNMP server, Event Viewer logs on a Microsoft Windows server, and packet captures from a network sniffer. Remember that monitoring is also critical in the area of network security. For example, *security information and event management (SIEM)* software products and services combine security information management (SIM) and security event management (SEM). This section introduces a number of resources for monitoring network information.

An organization should consistently conduct *log reviews*. A log review should include the following, at a minimum:

- **Traffic logs:** You can use a variety of traffic logging mechanisms, many of which are built in to devices. For example, many devices today support NetFlow, which is a powerful traffic monitoring tool that can export traffic flow information to a central collector on a network. (NetFlow is covered in more detail later in this chapter.) Keep in mind that traffic logging mechanisms differ based on the network environment. For example, while you might rely on NetFlow in an on-premises network, you might be using VPC Flow Logs in your AWS infrastructure as a service environment.
- **Audit logs:** You might audit many actions in your network environment and record them in an audit log. For example, a network device might have a setting to log each time an administrator logs in to the system and uses admin powers. These logs can be very valuable in tracking and monitoring the admin activities in your network. AWS Cloud offers a tool called CloudTrail that can log every activity that takes place in your public cloud environment.



Syslog

A variety of network components (for example, routers, switches, and servers) can send their log information to a common *syslog* server. By having information for multiple devices in a common log and examining time stamps, network administrators can better correlate events occurring on one network device with events occurring on a different network device. Syslog messages and SNMP traps can be used to trigger notification messages that may be sent via email and SMS. A syslog logging solution typically consists of three primary components:

- **Syslog server:** A syslog server receives and stores log messages sent from syslog clients.
- **Syslog clients:** As shown in Figure 14-6, various types of network devices can act as syslog clients and send logging information to a syslog server.
- **Syslog collector:** Networks today will often consist of many different syslog servers for the various categories of devices. A syslog collector can gather all the various syslog messages from the servers for centralized storage and analysis.

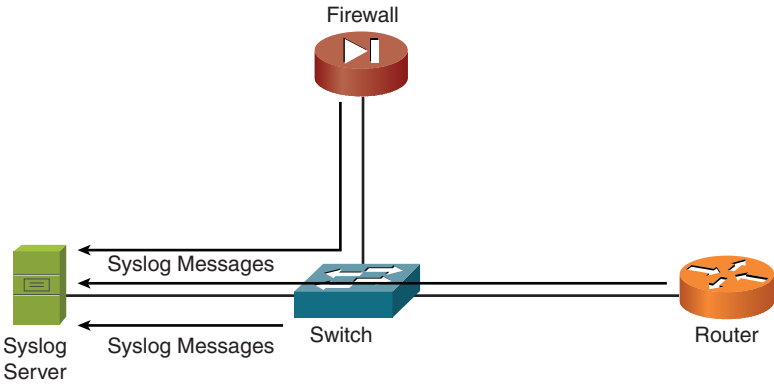


Figure 14-6 Sample Syslog Clients

Messages sent from a syslog client to a syslog server vary in their severity levels. Table 14-3 lists the eight severity levels of syslog messages. The higher the syslog level, the more detailed the logs. Keep in mind that more detailed logs require additional storage space on a syslog server.

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Table 14-3 Syslog Severity Levels

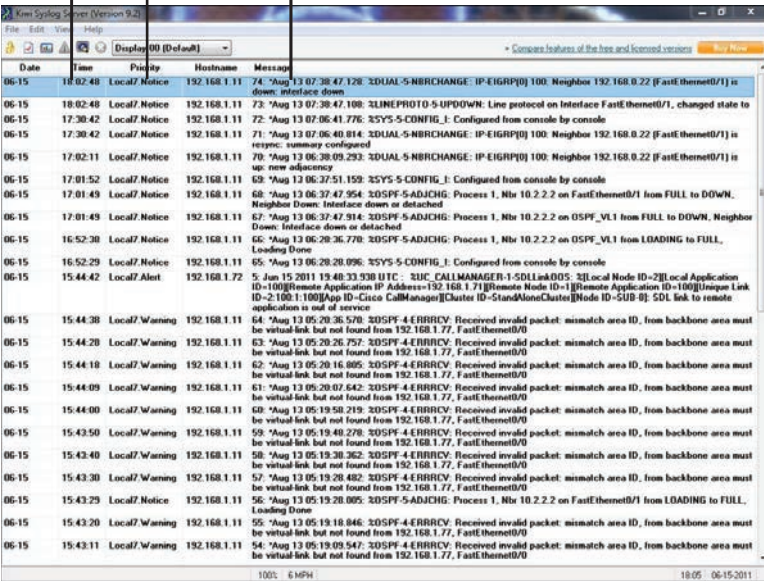
Level	Name	Description
0	Emergencies	The most severe error conditions, which render the system unusable
1	Alerts	Conditions requiring immediate attention
2	Critical	A less-severe condition than an alert that should be addressed to prevent interruption of service
3	Errors	Notifications about error conditions within the system that do not render the system unusable
4	Warnings	Notifications that specific operations failed to complete successfully
5	Notifications	Non-error notifications that alert an administrator about state changes within a system
6	Informational	Detailed information about the normal operation of a system
7	Debugging	Highly detailed information (for example, information about individual packets) that is typically used for troubleshooting purposes

Figure 14-7 shows the format of a syslog message. The syslog log entries contain time stamps, which help you understand how one log message relates to another.

The log entries also include severity level information, in addition to the text of the syslog messages.

Names of Log Message and Severity Levels

Time Stamps Text of Syslog Messages



Date	Time	Priority	Hostname	Message
06-15	18:02:48	Local7/Notice	192.168.1.11	74: *Aug 13 07:38:47.128: 3DUAL-5-NBRCHANGE: IP-ETHER[P] 100: Neighbor 192.168.0.22 [FastEthernet0/1] is down, interface down
06-15	18:02:48	Local7/Notice	192.168.1.11	73: *Aug 13 07:38:47.108: 3LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
06-15	17:30:42	Local7/Notice	192.168.1.11	72: *Aug 13 07:06:41.776: 3SYS-5-CONFIG_1: Configured from console by console
06-15	17:30:42	Local7/Notice	192.168.1.11	71: *Aug 13 07:06:40.814: 3DUAL-5-NBRCHANGE: IP-ETHER[P] 100: Neighbor 192.168.0.22 [FastEthernet0/1] is up: new adjacency
06-15	17:02:11	Local7/Notice	192.168.1.11	70: *Aug 13 06:38:09.293: 3DUAL-5-NBRCHANGE: IP-ETHER[P] 100: Neighbor 192.168.0.22 [FastEthernet0/1] is up: new adjacency
06-15	17:01:52	Local7/Notice	192.168.1.11	69: *Aug 13 06:37:51.159: 3SYS-5-CONFIG_1: Configured from console by console
06-15	17:01:49	Local7/Notice	192.168.1.11	68: *Aug 13 06:37:47.954: 3OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on FastEthernet0/1 from FULL to DOWN. Neighbor Down: Interface down or detached
06-15	17:01:49	Local7/Notice	192.168.1.11	67: *Aug 13 06:37:47.914: 3OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on OSPF_VL1 from FULL to DOWN. Neighbor Down: Interface down or detached
06-15	16:52:38	Local7/Notice	192.168.1.11	66: *Aug 13 06:28:36.778: 3OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on OSPF_VL1 from LOADING to FULL, Loading Done
06-15	16:52:29	Local7/Notice	192.168.1.11	65: *Aug 13 06:28:28.096: 3SYS-5-CONFIG_1: Configured from console by console
06-15	15:44:42	Local7/Alert	192.168.1.72	5: Jun 15 2011 19:48:33.930 UTC: 31HC CALLMANAGER-1-50Link005: 3[Local Node ID=2][Local Application ID=100][Remote Application IP Address=192.168.1.71][Remote Node ID=1][Remote Application ID=100][Unique Link ID=2:100:1:100][App ID=Cisco CallManager][Cluster ID=StandAloneCluster][Node ID=5:SUB-8] SDI link to remote application is out of service
06-15	15:44:38	Local7/Warning	192.168.1.11	64: *Aug 13 05:20:36.578: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:44:28	Local7/Warning	192.168.1.11	63: *Aug 13 05:20:26.757: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:44:18	Local7/Warning	192.168.1.11	62: *Aug 13 05:20:16.085: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:44:09	Local7/Warning	192.168.1.11	61: *Aug 13 05:20:07.642: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:44:00	Local7/Warning	192.168.1.11	60: *Aug 13 05:19:58.219: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:43:50	Local7/Warning	192.168.1.11	59: *Aug 13 05:19:48.278: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:43:40	Local7/Warning	192.168.1.11	58: *Aug 13 05:19:38.362: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:43:30	Local7/Warning	192.168.1.11	57: *Aug 13 05:19:28.482: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:43:25	Local7/Notice	192.168.1.11	56: *Aug 13 05:19:28.085: 3OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on FastEthernet0/1 from LOADING to FULL, Loading Done
06-15	15:43:20	Local7/Warning	192.168.1.11	55: *Aug 13 05:19:18.846: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0
06-15	15:43:11	Local7/Warning	192.168.1.11	54: *Aug 13 05:19:09.547: 3OSPF-4-ERRRCV: Received invalid packet: mismatch area ID, from backbone area must be virtual link but not found from 192.168.1.77, FastEthernet0/0

Figure 14-7 Structure of a Syslog Message

NOTE A variety of systems can act as syslog servers. You can download a free syslog utility from <https://solarwinds.com/downloads>.

Other Logs

In addition to logs generated by routers, switches, and other infrastructure gear, the operating systems powering network clients and servers generally have the capability to produce log output. Rather than containing general log information (that is, log information about all of a system's tracked components), Microsoft Windows incorporates the Event Viewer application, which allows you to view various log types, including application, security, and system logs. These logs can be archived for later review and can be used to spot network trends and provide data for creating baselines.

NOTE Logs are beneficial in your network management endeavors only if they are reviewed! Be sure to document standard operating procedures for periodic and careful review of the many logs incorporated into a network. Many networks today engage in **log aggregation**. This is the process of directing all or most logs to a single location or system. This permits the effective monitoring (and retention) of all log data in the network.

NetFlow

Cisco introduced *NetFlow* to the world around 1996. As discussed earlier in this chapter, NetFlow enables network administrators to collect IP network traffic statistics as traffic exits or enters interfaces. This type of network information is typically called **flow data**.

By analyzing the flow data provided by NetFlow, you can learn all kinds of valuable information about how your network is actually being used. NetFlow can also be a key tool for tracking down network bottlenecks, which tend to be somewhat inevitable. By using the data collected by NetFlow, you can often quickly determine root causes.

A typical implementation of NetFlow includes the following:

- **Flow exporter:** This component aggregates packets into flows and exports flow records toward one or more flow collectors.
- **Flow collector:** This component is responsible for reception, storage, and preprocessing of flow data received from a flow exporter.
- **Analysis application:** You use this application to analyze received flow data in various contexts. Perhaps you are most interested in performance issues during key times of the day. Or perhaps you are more concerned about security issues that NetFlow might be able to help pinpoint.

API Integration

Today, one of the most exciting areas in network monitoring is **application programming interface (API) integration**. This small revolution allows for the seamless exchange of data between different systems and network components. By leveraging APIs, network monitoring tools can access and retrieve data from a wide range of devices, including routers, switches, servers, and cloud services. This integration enables the monitoring software to collect detailed performance metrics, configuration data, and event logs, providing a comprehensive view of the network's health and performance. APIs facilitate the extraction of real-time data, which is

crucial for proactive monitoring and immediate detection of anomalies or potential issues.

One of the significant advantages of API integration in network monitoring is the ability to centralize and unify data from diverse sources into a single platform. This consolidation makes it easier to analyze the network as a whole, rather than in isolated segments. With APIs, network administrators can pull in data from various vendors and technologies, breaking down silos and ensuring that all relevant information is available in one place. This holistic view is essential for identifying trends, correlating events, and understanding the broader context of network behavior. It also simplifies the task of managing complex network environments, including those that span multiple locations or include a mix of on-premises and cloud-based resources.



Monitoring Solutions

Today, more so than ever before, network software vendors have been providing robust network monitoring suites of software that address many functions of cutting-edge management. These network monitoring solutions include the following:

- **Network discovery:** Network discovery is the process of identifying and mapping devices, services, and components within a network to understand its structure, detect new or unauthorized devices, and ensure accurate inventory management. This process can be executed in an ad hoc manner (*ad hoc network discovery*), where discovery tasks are performed as needed, typically in response to specific events or to gather immediate insights into the network's current state. Conversely, *scheduled network discovery* involves routine, automated scans that occur at regular intervals to consistently monitor changes and updates within the network infrastructure. Both approaches are critical for maintaining an up-to-date view of network assets and ensuring robust network management and security practices.
- **Traffic analysis:** Traffic analysis solutions for network monitoring are valuable tools that enable organizations to gain deep insights into their network performance, security, and efficiency. These solutions leverage various technologies and methodologies to collect, process, and analyze network traffic data in real time. By utilizing packet capture, flow analysis, and deep packet inspection, traffic analysis tools can identify and diagnose issues such as network congestion, latency, and bottlenecks. They also play a critical role in detecting security threats like intrusions, malware, and anomalous behaviors by monitoring traffic patterns and comparing them against known threat signatures and baselines. Advanced traffic analysis solutions often incorporate machine learning

and AI to enhance detection capabilities and provide predictive analytics, helping network administrators proactively manage and secure their networks.

- **Performance monitoring:** Performance monitoring solutions are designed to continuously track and evaluate the efficiency and effectiveness of various components within an IT infrastructure, including servers, applications, networks, and databases. These solutions collect and analyze data on key performance indicators (KPIs) such as response time, throughput, and resource utilization, providing real-time insights and historical trends. By identifying performance bottlenecks, anomalies, and potential issues before they impact end users, performance monitoring solutions help ensure optimal system functionality, enhance user experience, and support proactive maintenance and troubleshooting efforts. Advanced features often include automated alerts, detailed reporting, and predictive analytics to facilitate informed decision-making and maintain system reliability.
- **Availability monitoring:** Availability monitoring solutions are valuable tools that ensure IT systems, applications, and services are accessible and operational at all times. These solutions continuously check the status and uptime of various components within an infrastructure, immediately detecting and alerting administrators to any outages or disruptions. By monitoring parameters such as server responsiveness, network connectivity, and service performance, availability monitoring helps organizations maintain high levels of reliability and minimize downtime. Advanced availability monitoring tools often include features like automated incident response, root cause analysis, and detailed reporting, enabling quick resolution of issues and helping to ensure that critical systems remain consistently available to users.
- **Configuration monitoring:** In Chapter 13, “Organizational Processes and Procedures,” you learned about the importance of configuration management. It should be no surprise, therefore, that configuration monitoring is another important solution. Configuration monitoring solutions track and manage the settings and configurations of hardware, software, and network devices. These solutions continuously monitor configuration changes, ensuring compliance with predefined policies and standards while detecting unauthorized or unexpected modifications. By providing real-time visibility into the current state of configurations, they help prevent configuration drift, which can lead to security vulnerabilities, performance issues, and operational inefficiencies. Advanced configuration monitoring tools offer automated alerts, detailed audit trails, and comprehensive reporting, enabling IT teams to quickly identify and rectify configuration discrepancies, maintain system integrity, and ensure consistent and secure operations.

Real-World Case Study

Acme, Inc. has decided to improve its network and operations by enhancing the observability of its network. Acme, Inc. has decided to employ NetFlow on many of the key network nodes so that these devices can share performance data with a central repository in the network. By enabling NetFlow, Acme, Inc. can gather detailed information about traffic flows, enabling the identification of network usage patterns, potential bottlenecks, and security issues. This data-driven approach allows for more efficient network management and troubleshooting, providing a clearer understanding of how network resources are being utilized and where improvements can be made.

In addition to implementing NetFlow, Acme, Inc. is also going to start using Splunk Enterprise (a popular SIEM) to make sense of the many logs and other machine data produced in the organization. Splunk Enterprise is a powerful platform for searching, monitoring, and analyzing machine-generated data. By aggregating logs from various network devices, servers, and applications, Splunk enables the creation of comprehensive dashboards and reports. These tools will help the network operations team at Acme, Inc. to visualize network performance, detect anomalies, and gain actionable insights into the health and status of their IT infrastructure.

The combination of NetFlow and Splunk Enterprise will significantly enhance Acme, Inc.'s ability to monitor and manage its network. NetFlow provides a granular view of traffic patterns, which, when analyzed alongside log data in Splunk, can reveal deeper insights into network behavior and performance. This integration allows for more proactive network management, as potential issues can be identified and addressed before they escalate into major problems. For instance, unusual traffic patterns that might indicate a security breach can be quickly detected and mitigated, ensuring the network remains secure and reliable.

Summary

Here are the main topics covered in this chapter:

- This chapter described best practices for monitoring performance with metrics and sensors.
- This chapter discussed the operation of SNMP, providing insight into the operation of SNMPv1, SNMPv2c, and SNMPv3.
- This chapter described the operation of syslog and the syslog message severity levels.
- This chapter also discussed network monitoring solutions such as network discovery, traffic analysis, and performance monitoring.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 14-4 lists these key topics and the page number where each is found.



Table 14-4 Key Topics for Chapter 14

Key Topic Element	Description	Page Number
Table 14-1	Components of an SNMPv1 and SNMPv2c Network Management Solution	342
Figure 14-1	SNMPv1 and SNMPv2c Network Management Components and Messages	343
Figure 14-5	Example: Network Sniffer with Port Mirroring Configured on the Switch	350
Section	Syslog	351
Table 14-3	Syslog Severity Levels	352
Section	Monitoring Solutions	355

Complete Tables and Lists from Memory

Print a copy of Appendix B, “Memory Tables,” or at least the section for this chapter and complete as many of the tables as possible from memory. Appendix C, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

ad hoc network discovery, anomaly alerting, application programming interface (API) integration, authentication, availability monitoring, baseline metrics, community strings, configuration monitoring, flow data, log aggregation, management information base (MIB), network discovery, notifications, packet capture, performance monitoring, port mirroring, scheduled network discovery, security

information and event management (SIEM), Simple Network Management Protocol (SNMP), SNMP Version 2c (SNMPv2c), SNMP Version 3 (SNMPv3), syslog collector, traffic analysis, traps

Additional Resources

How Healthy Is Your Existing Network?: <https://www.ajsnetworking.com/how-healthy-is-your-existing-network>

SNMP Operation: <https://www.youtube.com/watch?v=tg47MZdtcAE>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. SNMP uses a series of objects to collect information about a managed device. What is the name of the structure, similar to a database, that contains these objects?
 - a. RIB
 - b. MIB
 - c. Syslog
 - d. Baseline
2. Which syslog level is the most severe?
 - a. Informational
 - b. Critical
 - c. Errors
 - d. Warnings
3. What are the main categories of SNMP message types? (Choose three.)
 - a. Get
 - b. Put
 - c. Set
 - d. Trap

4. As you monitor a key area of your network, you discover that the average latency spans a wide range. You are seeing some periods of 50 ms and others of 300 ms. What is the term for this type of variation in delay?
 - a. CRC
 - b. DSCP
 - c. WRED
 - d. Jitter
5. Which SNMPv3 security level is the equivalent of SNMPv2c?
 - a. noAuthNoPriv
 - b. authPriv
 - c. authNoPriv
 - d. noAuthPriv
6. What syslog level is used for informational messages?
 - a. 0
 - b. 8
 - c. 7
 - d. 6
7. Which network assurance tool would most likely feature the use of a collector?
 - a. NetFlow
 - b. SNMP
 - c. Nmap
 - d. traceroute
8. Which version of SNMP introduced significant improvements in security, offering authentication, encryption, and access control mechanisms to protect SNMP communications?
 - a. SNMPv1
 - b. SNMPv2
 - c. SNMPv2c
 - d. SNMPv3

9. Which of the following will help you better understand latency in your network?
 - a. Baseline metrics
 - b. Jitter
 - c. Bandwidth
 - d. Latency
10. What do many switches support that makes a copy of network traffic seen on one port and sends that duplicated traffic out another port (to which a network sniffer could be attached)?
 - a. Managers
 - b. Agents
 - c. Port mirroring
 - d. MIBs
11. Which of the following network monitoring solutions specifically focuses on collecting and analyzing data on key performance indicators (KPIs) such as response time, throughput, and resource utilization, providing real-time insights and historical trends?
 - a. Network discovery
 - b. Traffic analysis
 - c. Availability monitoring
 - d. Performance monitoring
12. What is the process called by which SIEM systems combine similar events into a log to reduce event volume and consolidate data so that crucial events are not missed?
 - a. Application programming interface (API) integration
 - b. Ad hoc discovery
 - c. Log aggregation
 - d. Scheduled discovery



This chapter covers the following topics related to Objective 3.3 (Explain disaster recovery (DR) concepts) of the CompTIA Network+ N10-009 certification exam:

- DR metrics
 - Recovery point objective (RPO)
 - Recovery time objective (RTO)
 - Mean time to repair (MTTR)
 - Mean time between failures (MTBF)
- DR sites
 - Cold site
 - Warm site
 - Hot site
- High-availability approaches
 - Active-active
 - Active-passive
- Testing
 - Tabletop exercises
 - Validation tests

Disaster Recovery

As we have alluded to elsewhere in this text, more and more traffic is flocking to our network infrastructures today. Networks were once relegated to the domain of data but now routinely carry voice and video traffic as well. These additional media types, as well as mission-critical data applications, need a network to be up and available for users.

It is likely that your telephone service has been unavailable much less often than your data network. Unfortunately, data networks have traditionally been less reliable than voice networks; however, today's data networks often *are* voice networks, and this convergence has contributed to the increased demand for uptime. Unified voice services such as call control and communication gateways can be integrated into one or more network devices, leveraging the bandwidth available on the LAN and WAN.

In this chapter, we take a tour of key disaster recovery techniques and concepts. We also examine high availability and tools that can help ensure high availability in networks today.

Foundation Topics

High Availability

If a network switch or router stops operating correctly (meaning that a *network fault* occurs), communication through the network could be disrupted, resulting in the network becoming unavailable to its users. Therefore, network availability, called *uptime*, is a major design consideration. This consideration might, for example, lead you to add fault-tolerant devices and fault-tolerant links between those devices. This section discusses the measurement of high availability (HA) along with a number of high-availability design considerations.

High-Availability Measurement

The availability of a network is measured by its uptime during a year. For example, if a network has *five nines* availability, it is up 99.999% of the time, which translates to a maximum of 5 minutes of downtime per year. If a network has *six nines* availability (meaning it is up 99.9999% of the time), it is down less than 30 seconds per year.

As a designer, one of your goals is to select components, topologies, and features that maximize network *availability* within certain parameters (for example, a budget). Be careful not to confuse *availability* with *reliability*. A *reliable* network, for example, does not drop many packets, whereas an *available* network is up and operational.

DR Metrics

The *mean time to repair (MTTR)* is the average time required to fix a failed component and return it to production status. This is also sometimes called mean time to recovery.

Mean time between failures (MTBF) is the average amount of time that passes between hardware component failures, excluding time spent repairing components or waiting for repairs.

Another goal in your design might be to meet the requirements set forth in a service-level agreement (SLA). An SLA is an official commitment that exists between a service provider and a client. Aspects of the IT services provided—quality, availability, specific responsibilities—are agreed upon between the service provider and the service user. Strict SLAs are becoming increasingly common in networking today as more and more cloud services find their way into the IT landscape. A cloud provider is a specialized service provider, and a cloud consumer is the client.

Two other measures that an organization should consider are the *recovery time objective (RTO)* and the *recovery point objective (RPO)*. These two measurements might sound the same, but they are very different measures:

- **RTO:** The RTO is the time in the future when you expect to restore availability after some failure (or even a disaster) has rendered your IT services unavailable. The RTO value is a very important measure of how long you expect the network to be down.
- **RPO:** As important as the RTO is the RPO, which is the point in time to which you can recover the network. For example, say that your network system failed at 4 p.m. last Monday. Your RPO might be for that very point in time. Sometimes, unfortunately, the RPO will not be the exact point of failure. This is often the case with database systems. For example, a failure might occur at 4 p.m. on a Monday, but you might only be able to recover to 11 a.m. that Monday. This gap in your data might be a result of your disaster recovery plans. Perhaps your backup design can only restore your systems to that 11 a.m. point. An organization must ensure it can accept a particular amount of potential data and service loss.

NOTE The RPO is the amount of data that will be lost or will have to be re-entered because of network downtime. The RTO is the amount of time that can pass before the disruption begins to seriously impede normal business operations.

Key Topic

Fault-Tolerant Network Design

Fault tolerance is the capability of a component, system, or network to endure a failure. It is the capability to withstand a fault (failure) without losing data. This can be accomplished through the use of Redundant Array of Independent Disks (RAID), backups, and similar technologies.

Two important concepts to know when designing a fault-tolerant network are as follows:

- **Single points of failure:** If the failure of a single network device or link (for example, a switch, router, or WAN connection) would result in a network becoming unavailable, that single device or link is a potential single point of failure. To eliminate single points of failure from your design, you might include redundant links and redundant hardware. For example, some high-end Ethernet switches support two power supplies, and if one power supply fails,

the switch continues to operate by using the backup power supply. Link redundancy, as shown in Figure 15-1, can be achieved by using more than one physical link. If a single link between a switch and a router fails, the network would not go down because of the link redundancy that is in place.

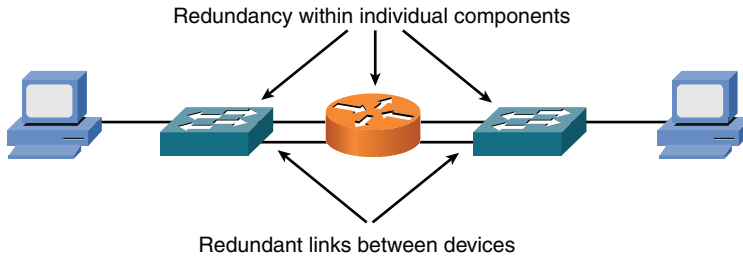


Figure 15-1 Redundant Network with Single Points of Failure

- **No single points of failure:** A network without a single point of failure contains redundant network infrastructure components (for example, switches and routers). In addition, these redundant devices are interconnected with redundant links. Although a network host could have two network interface cards (NICs), each connecting to a different switch, such a design is rarely implemented because of the increased costs. Instead, as shown in Figure 15-2, a network with no single points of failure in the backbone allows any single switch or router in the backbone to fail or any single link in the backbone to fail while maintaining end-to-end network connectivity.

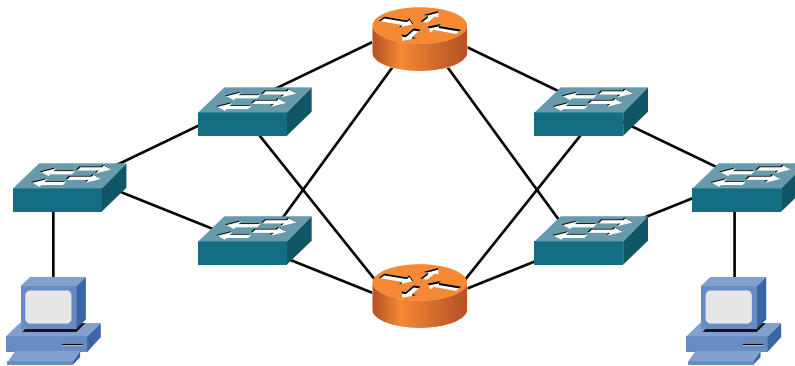


Figure 15-2 Redundant Network with No Single Point of Failure

Approaches to fault-tolerant network design can be used together to increase a network's availability even further.

NOTE Creating multiple paths that data can take in a network is often simply termed *multipathing*. Routing protocols can make multiple paths very valuable to use since they can engage in equal-cost multipathing (ECMP). In fact, Enhanced Interior Gateway Routing Protocol (EIGRP) even provides the unique capability of load sharing between paths of unequal cost.

Hardware Redundancy

Having redundant route processors in a switch or router chassis improves the reliability of the chassis. If a multilayer switch has two route processors, for example, one of the route processors could be active, and the other route processor could be standing by to take over in the event that the active processor became unavailable.

An end system can have redundant NICs. There are two modes of NIC redundancy:

- **Active-active:** Both NICs are active at the same time, and each has its own MAC address. This makes troubleshooting more complex, while giving you slightly better performance than the active-passive approach.
- **Active-passive:** Only one NIC is active at a time. This approach allows the client to appear to have a single MAC address and IP address, even in the event of a NIC failure.

NIC redundancy is most often used in strategic network hosts, rather than in end-user client computers, because of the expense and administrative overhead incurred with redundant NIC configuration.

NOTE Different vendors use different terms to refer to combining NICs for hardware redundancy. The two most commonly used terms are *network interface card (NIC) teaming* and *NIC bonding*.

Today, network engineers use the active-active and active-passive high-availability approaches with many other networking functions and components in addition to NICs. In fact, entire network sites might function in an active-active or active-passive type capacity. Generally, an active-active configuration is always preferable, but, again, the expense and administrative overhead of this configuration often make it infeasible to implement for your networking functions and components.

Another powerful method of hardware redundancy comes in the form of computer clustering. A computer cluster consists of a set of tightly connected computers that work together. In many respects, clients view them as a single system.

The servers in a cluster are usually connected to each other through fast local area networks, with each node running its own instance of an operating system. In most circumstances, all of the nodes use the same hardware and the same operating system, although in some setups, different operating systems—or even different hardware—can be used on each computer.

Computer clusters emerged as a result of the convergence of a number of computing trends, including the availability of low-cost microprocessors, high-speed networks, and software for high-performance distributed computing. Computer clusters have a wide range of applicability and deployment, ranging from small business clusters with a handful of nodes to some of the fastest supercomputers in the world.

The clustering approach has also made its way to networking devices themselves. For example, today it is not uncommon for large enterprises and data centers to feature clustered solutions for switches, routers, and firewalls.

Design Considerations for High-Availability Networks

When designing networks for high availability, it is important to consider the following questions:

Key Topic

- Where will module and chassis redundancy be used? Module redundancy provides redundancy within a chassis by allowing one module to take over in the event that a primary module fails. Chassis redundancy means having more than one chassis, thus providing a path from the source to the destination even in the event of a chassis or link failure.
- What software redundancy features are appropriate?
- What protocol characteristics affect design requirements?
- What redundancy features should be used to provide power to an infrastructure device—for example, using an *uninterruptible power supply (UPS)*, a *generator*, or dual power supplies?
- What redundancy features should be used to maintain environmental conditions (for example, dual air-conditioning units)?
- Will dual circuits be provided in the event of a loss of connection with one of the circuits?
- What backup strategy exists for infrastructure and user data? The main backup strategies are as follows:

- **Full:** A full backup is a backup of all of the data set. Although this is the safest and most comprehensive way to ensure data availability, it can be time-consuming and costly.
 - **Incremental:** An incremental backup backs up only data that has changed since the previous incremental backup. An incremental backup is incomplete for full recovery without a valid full backup and all incremental backups since the last full backup.
 - **Differential:** A differential backup is similar to an incremental backup in that it starts with a full backup, and then subsequent backups contain only data that has changed. The difference is that whereas an incremental backup only includes the data that has changed since the previous backup, a differential backup contains all of the data that has changed since the last full backup.
 - **Snapshots:** A snapshot is a read-only copy of a data set that is frozen in a point in time. This type of backup is often used with virtual machines and file system objects.
-
- What backup strategy exists for your network devices (both software images and configurations)? Are you backing up the system state information and configuration information from these devices?

High-Availability Best Practices

The following are the five best practices for designing high-availability networks:

Key Topic

- Examine technical goals.
- Identify the budget for funding high-availability features.
- Categorize business applications into profiles, each of which requires a certain level of availability.
- Establish performance standards for high-availability solutions.
- Define how to manage and measure the high-availability solution.

Although existing networks can be retrofitted to make them highly available, network designers can often reduce expenses by integrating high-availability best practices and technologies into the initial design of a network.

Content Caching

A *content engine* is a network appliance that can receive a copy of content stored elsewhere (for example, a video presentation located on a server at a corporate headquarters) and serve that content to local clients, thus reducing the bandwidth burden on an IP WAN. Figure 15-3 shows a sample topology using a content engine as a network optimization technology.

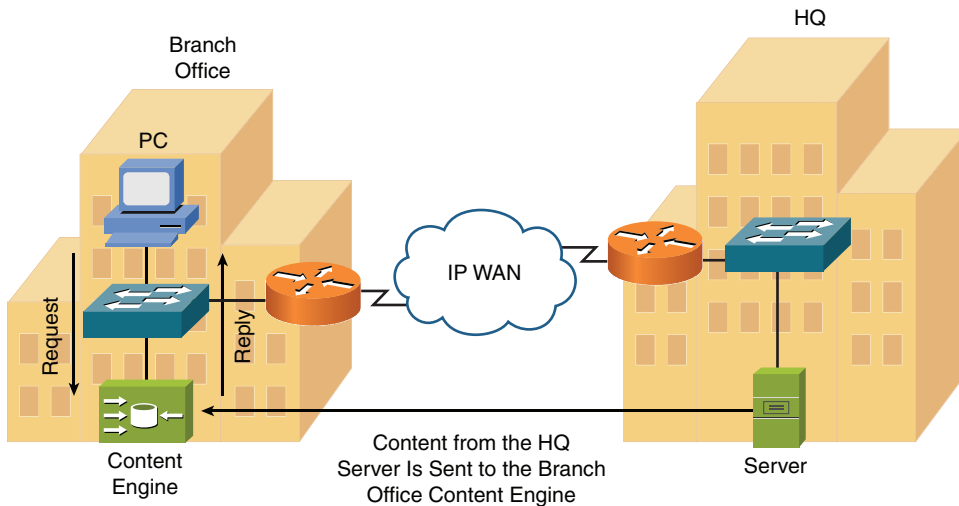


Figure 15-3 Content Engine Sample Topology

Load Balancing

Content switching allows a request coming into a server farm to be distributed across multiple servers containing identical content. This approach to *load balancing* lightens the load on individual servers in a server farm and allows servers to be taken out of the farm for maintenance without disrupting access to the server farm's data. Figure 15-4 illustrates a sample content-switching topology that performs load balancing across five servers (containing identical content) in a server farm.

Hardware Redundancy

It is possible to design site redundancy into your network infrastructure. Doing so requires redundant data and equipment located in geographically distant areas. How fast can your IT infrastructure be back up and running in the event that issues arise in your primary site? The following disaster recovery (DR) site options are possible:

- **Cold site:** A cold site is a disaster recovery location equipped with basic infrastructure but without the necessary hardware and data. Recovery is possible but is difficult and time-consuming. A cold site is the weakest of the recovery site options but also the least costly. However, keep in mind that although a cold site may be the least costly when you're planning for disaster, after a disaster occurs, equipment purchased for a cold site might be expensive or difficult to obtain.

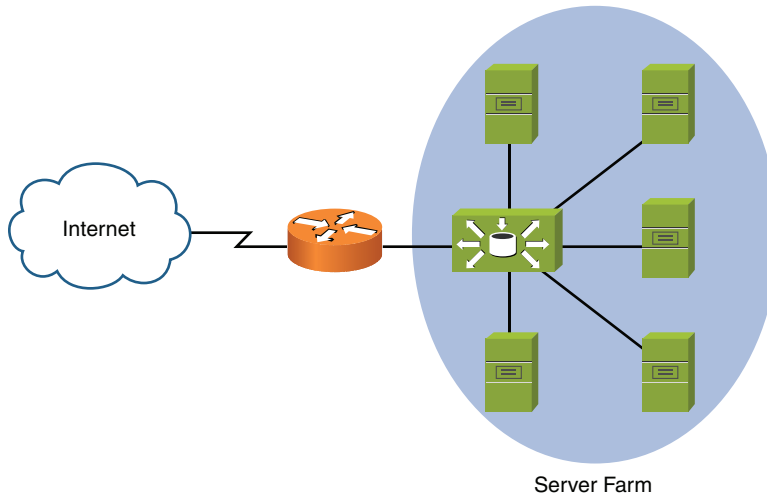


Figure 15-4 Content-Switching Sample Topology

- **Warm site:** A warm site is a backup location that is partially equipped with infrastructure and systems, allowing for a quicker recovery than a cold site but requiring some additional setup before becoming fully operational. Recovery is possible fairly quickly, but the site might not have the resources and responsiveness of the original site. A warm site is a scaled-down version of a hot site. The recovery site is often only configured with power, phone, and network jacks. It may have computers and other resources, but they are not configured and ready to go.
- **Hot site:** This type of site is most expensive, and it functions like the original site and is equipped with all necessary hardware, software, network, and Internet connectivity fully installed, configured, and ready to go. Downtime is minimal, with a service level nearly identical to that of the organization's main site.

NOTE Today, thanks to advancements in public cloud computing, *cloud sites* are also possible. You can configure cloud sites as cold, warm, or hot sites, typically with ease. This is due to the lowered costs and technological requirements made possible by cloud.

Testing

When designing high-availability and disaster recovery solutions, you should always consider a thorough *testing* phase. Common types of testing include:

- **Tabletop exercises:** Tabletop exercises are simulated, discussion-based sessions used to test and evaluate an organization's disaster recovery plans. During these exercises, key stakeholders and team members gather to walk through hypothetical disaster scenarios, such as cyberattacks, natural disasters, or system failures, and discuss their roles, responsibilities, and responses. The goal is to identify strengths and weaknesses in the DR plan, ensuring that everyone understands the procedures and can effectively communicate and collaborate during an actual disaster. These exercises provide a safe environment to practice decision-making, improve preparedness, and refine the DR strategies, ultimately enhancing the organization's resilience and ability to recover swiftly from disruptive events.
- **Validation tests:** Validation tests are crucial steps in the testing process of a new disaster recovery solution, designed to ensure that the solution meets all specified requirements and functions as intended. These tests involve systematically verifying each component of the DR plan, including data backup and restoration processes, failover procedures, and system recovery capabilities. By simulating various disaster scenarios and operational conditions, validation tests help identify any gaps, inefficiencies, or potential points of failure in the DR solution. The insights gained from these tests allow organizations to fine-tune their DR strategies, ensuring robust performance and reliability. Ultimately, validation tests provide confidence that the new DR solution can effectively protect and recover critical data and systems in the event of an actual disaster.

Real-World Case Study: Network Design

If you have read all the chapters before this one, you have learned to design an amazingly powerful network that includes redundancy and DR features. While this book does not focus on network design specifically, this case study allows you to practice with this valuable skill. Because network design is part science and part art,

multiple designs can meet the specified requirements. However, as a reference, this section presents one solution, against which you can contrast your solution.

While working through your design, consider the following:

- Meeting all requirements
- Media distance limitations
- Network device selection
- Environmental factors
- Compatibility with existing and future equipment

Case Study Scenario

The following are the design scenario and criteria for this case study:

- Company ABC leases two buildings (building A and building B) in a large office park, as shown in Figure 15-5. The office park has a conduit system that allows physical media to run between buildings. The distance (via the conduit system) between building A and building B is 1 km.
- Company ABC will use the Class B address 172.16.0.0/16 for its sites. You should subnet this classful network not only to support the two buildings (one subnet per building) but also to allow as many as five total sites in the future, as Company ABC continues to grow.
- Company ABC needs to connect to the Internet, supporting a speed of at least 30Mbps, and this connection should come into building A.
- Cost is a primary design consideration, and performance is a secondary design consideration.
- Each building contains various Wi-Fi client devices (for example, smartphones, tablets, and laptops).

Table 15-1 identifies the number of hosts contained in each building and the number of floors in each building.

Table 15-1 Case Study Information for Buildings A and B

Building	Number of Hosts	Floors (and Wireless Coverage)
A	200	Three floors, each of which can be serviced by a single wireless access point
B	100	One floor, which can be serviced by a single wireless access point

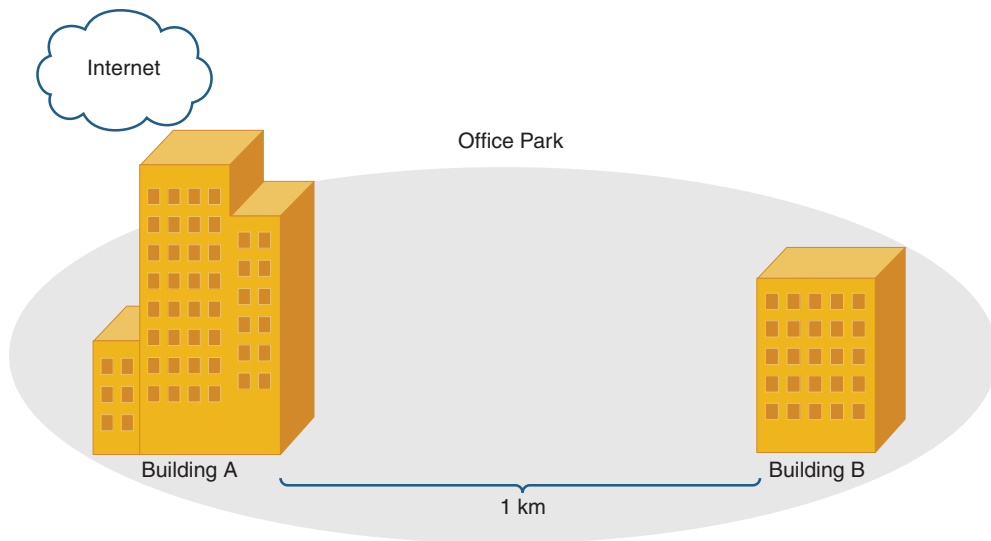


Figure 15-5 Case Study Topology

Your design should include the following information:

- Network address and subnet mask for building A
- Network address and subnet mask for building B
- Layer 1 media selection
- Layer 2 device selection
- Layer 3 device selection
- Wireless design
- Any design elements based on environmental considerations
- An explanation of where cost savings were created from performance trade-offs
- A topological diagram of the proposed design

Use multiple sheets of paper to create your network design. After your design is complete, perform a sanity check by contrasting the listed criteria against your design. Finally, while keeping in mind that multiple designs could meet the design criteria, you can review the following suggested solution. In the real world, reviewing the logic behind other designs can often give you a fresh perspective for future designs.

Suggested Solution

This suggested solution begins with IP address allocation. Then, consideration is given to the Layer 1 media, followed by Layer 2 and Layer 3 devices. Wireless design decisions are presented. Design elements based on environmental factors are discussed. The suggested solution also addresses how cost savings were achieved through performance trade-offs. Finally, a topological diagram of the suggested solution is presented.

IP Addressing

Questions you might need to consider when designing the IP addressing of a network include the following:

- How many hosts do you need to support (now and in the future)?
- How many subnets do you need to support (now and in the future)?

From the scenario, you know that each subnet must accommodate at least 200 hosts. Also, you know that you must accommodate at least 5 subnets. In this solution, the subnet mask is based on the number of required subnets. Eight subnets are supported with 3 borrowed bits, and 2 borrowed bits support only 4 subnets, based on this formula:

$$\text{Number of subnets} = 2^s$$

where s is the number of borrowed bits.

With 3 borrowed bits, you have 13 bits left for host IP addressing, which is much more than needed to accommodate 200 host IP addresses. These 3 borrowed bits yield the subnet mask 255.255.224.0. Because the third octet is the last octet to contain a binary 1 in the subnet mask, the third octet is the *interesting octet*.

The block size can be calculated by subtracting the subnet decimal value in the interesting octet from 256 (that is, $256 - 224 = 32$). Because the block size is 32 and the interesting octet is the third octet, the following subnets are created with the 255.255.224.0 (that is, /19) subnet mask:

172.16.0.0 /19
172.16.32.0 /19
172.16.64.0 /19
172.16.96.0 /19
172.16.128.0 /19

172.16.160.0 /19
172.16.192.0 /19
172.16.224.0 /19

The first two subnets are selected for the building A and building B subnets, as shown in Table 15-2.

Table 15-2 Case Study Suggested Solution: Network Addresses

Building	Subnet
A	172.16.0.0 /19
B	172.16.32.0 /19

Layer 1 Media

Questions you might need to ask when selecting the Layer 1 media types of a network include the following:

- What speeds need to be supported (now and in the future)?
- What distances between devices need to be supported (now and in the future)?

Within each building, Category 6a (Cat 6a) unshielded twisted-pair (UTP) cabling is selected to interconnect network components. The installation is based on Gigabit Ethernet. However, 10-Gigabit Ethernet devices may be installed in the future, as Cat 6a is rated for 10GBASE-T for distances as long as 100 m.

The 1 km distance between building A and building B is too great for UTP cabling. Therefore, multimode fiber (MMF) is selected. The speed of the fiber link will be 1Gbps. Table 15-3 summarizes these media selections.

Table 15-3 Case Study Suggested Solution: Layer 1 Media

Connection Type	Media Type
LAN links within buildings	Cat 6a UTP
Link between building A and building B	MMF

Layer 2 Devices

Questions you might need to consider when selecting Layer 2 devices in a network include the following:

- Where will the switches be located?

- What port densities are required on the switches (now and in the future)?
- What switch features need to be supported (for example, STP or LACP)?
- What media types are used to connect to the switches?

A collection of Ethernet switches interconnects network devices within each building. Assume that the 200 hosts in building A are distributed relatively evenly across the three floors (with each floor containing approximately 67 hosts). Therefore, each floor will have a wiring closet containing two Ethernet switches: a 48-port switch and a 24-port switch. Each switch is connected to a multilayer switch located in building A using four connections logically bundled together using LACP.

NOTE Link aggregation is also known as *port aggregation*.

Within building B, two Ethernet switches with 48 ports each and one Ethernet switch with 24 ports are installed in a wiring closet. These switches are interconnected in a stacked configuration, using four connections logically bundled together with LACP. One of the switches has an MMF port, which allows it to connect via fiber to building A's multilayer switch.

Table 15-4 summarizes the switch selections.

Table 15-4 Case Study Suggested Solution: Layer 2 Devices

Building	Quantity of 48-Port Switches	Quantity of 24-Port Switches
A	3	3
B	2	1

Layer 3 Devices

Questions you might need to consider when selecting Layer 3 devices for a network include the following:

- How many interfaces are needed (now and in the future)?
- What types of interfaces need to be supported (now and in the future)?
- What routing protocol (or protocols) needs to be supported?
- What router features (for example, HSRP or security features) need to be supported?

One Layer 3 device is used: a multilayer switch located in building A. All switches within building A home back to the multilayer switch using four LACP-bundled links. The multilayer switch is equipped with at least one MMF port, which allows a connection with one of the Ethernet switches in building B. The multilayer switch connects to a router via a 10Gbps Ethernet connection. This router contains a serial interface, which connects to the Internet via a T3 connection.

Wireless Design

Questions you might need to consider when designing the wireless portion of a network include the following:

- What wireless speeds need to be supported (now and in the future)?
- What distances need to be supported between wireless devices and wireless access points (now and in the future)?
- What IEEE wireless standards need to be supported?
- What channels should be used?
- Where should wireless access points be located?

Because the network needs to support various Wi-Fi clients, the 2.4GHz band is chosen. Within building A, a wireless access point (AP) is placed on each floor of the building. To avoid interference, the non-overlapping channels 1, 6, and 11 are chosen. The 2.4GHz band also allows compatibility with IEEE 802.11ac/ax.

Within building B, a single wireless AP accommodates Wi-Fi clients. Table 15-5 summarizes the wireless AP selection.

Table 15-5 Case Study Suggested Solution: Wireless AP Selection

AP Identifier	Building	Band	Channel
1	A (1st floor)	2.4GHz	1
2	A (2nd floor)	2.4GHz	6
3	A (3rd floor)	2.4GHz	11
4	B	2.4GHz	1

Environmental Factors

Questions you might need to consider when considering environmental factors of a network design include the following:

- What temperature or humidity controls exist in the rooms containing network equipment?
- What power redundancy systems are needed to provide power to network equipment in the event of a power outage?

Because the multilayer switch in building A could be a single point of failure for the entire network, the multilayer switch is placed in a well-ventilated room, which can help dissipate heat in the event of an air-conditioning failure. To further enhance the availability of the multilayer switch, the switch is connected to a UPS, which can help the multilayer switch continue to run for a brief time in the event of a power outage. Protection against an extended power outage could be achieved with the addition of a generator. However, no generator is included in this design for budgetary reasons.

NOTE Other common considerations in this area include the use of *power distribution units (PDUs)*, which are devices fitted with multiple outputs designed to distribute electric power. *Heating, ventilation, and air-conditioning (HVAC)* and *fire suppression systems* should also be analyzed to determine their power needs.

Cost Savings Versus Performance

When assimilating all the previously gathered design elements, you need to weigh budgetary constraints against network performance metrics. In this example, Gigabit Ethernet was chosen over 10-Gigabit Ethernet. In addition, the link between building A and building B could become a bottleneck because it runs at a speed of 1Gbps, although it transports an aggregation of multiple 1Gbps links. However, cost savings are achieved by using 1Gbps switch interfaces as opposed to 10Gbps interfaces or a bundle of multiple 1Gbps fiber links.

Topology

Figure 15-6 shows the topology of the proposed design based on design decisions described in this section.

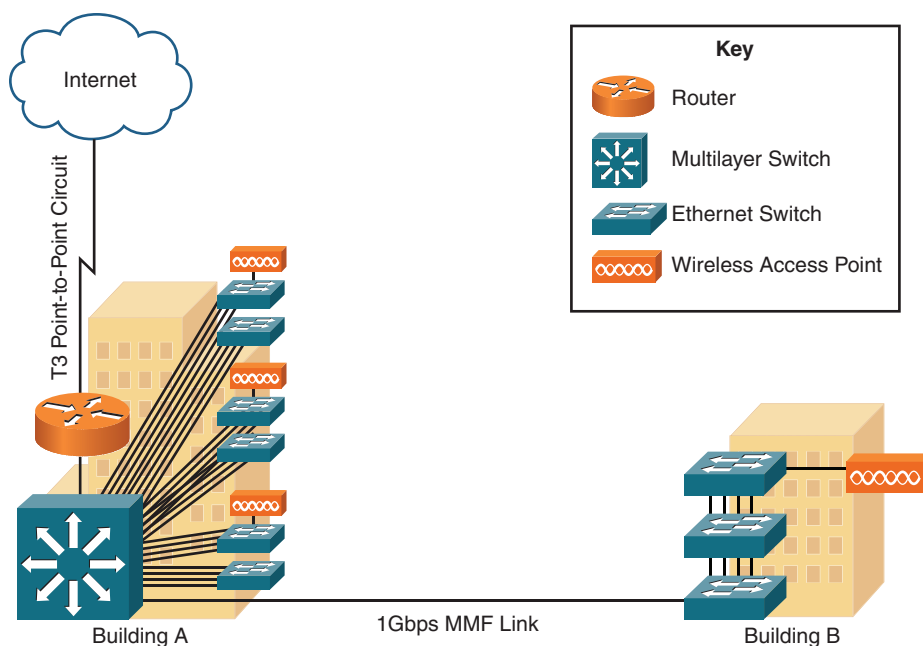


Figure 15-6 Case Study Proposed Topology

Summary

Here are the main topics covered in this chapter:

- This chapter discussed network availability, including how availability is measured and can be achieved through redundant designs.
- This chapter discussed performance optimization strategies, including the use of content caching, link aggregation, and load balancing.
- This chapter included a case study that challenged you to design a network to meet a collection of criteria.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 15-6 lists these key topics and the page number where each is found.

**Key
Topic****Table 15-6** Key Topics for Chapter 15

Key Topic Element	Description	Page Number
Section	Fault-Tolerant Network Design	365
List	Design considerations for high-availability networks	368
List	High-availability best practices	369

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

active-active, active-passive, cold site, hot site, mean time between failures (MTBF), mean time to repair (MTTR), recovery point objective (RPO), recovery time objective (RTO), tabletop exercises, validation tests, warm site

Additional Resources

High Availability/Disaster Recovery (HA/DR) Basics: <https://www.youtube.com/watch?v=cIPunZYEZHU>

What's the Difference Between RTO and RPO?: <https://www.rubrik.com/insights/rto-rpo-whats-the-difference>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. If a network has five nines availability, how much downtime does it experience per year?
 - a. 30 seconds
 - b. 5 minutes
 - c. 12 minutes
 - d. 26 minutes
2. What mode of NIC redundancy has only one NIC active at a time?
 - a. Publisher-subscriber
 - b. Client-server

- c. Active-passive
 - d. Active-subscriber
- 3. What performance optimization technology involves a network appliance that can receive a copy of content stored elsewhere (for example, a video presentation located on a server at a corporate headquarters) and serves that content to local clients, thus reducing the bandwidth burden on an IP WAN?
 - a. Content engine
 - b. Load balancer
 - c. LACP
 - d. CARP
- 4. What type of backup solution is a point-in-time, read-only copy of data?
 - a. Differential
 - b. Incremental
 - c. Snapshot
 - d. Virtual
- 5. What type of site provides a nearly identical level of service to the organization's main site, with virtually no downtime?
 - a. Warm
 - b. Cold
 - c. Hot
 - d. Remote
- 6. What capability of routing protocols allows them to help you use the bandwidth that you might have available in a multipathing design?
 - a. ECMP
 - b. Distance vector
 - c. Link state
 - d. Dual stack
- 7. In order to measure high availability, you need to know the average amount of time that passes between hardware component failures. What is this called?
 - a. MTTR
 - b. RPO

- c. RTO
 - d. MTBF
8. You are tasked with locating a backup facility for your organization to use in the event of a disaster. You have a slim budget, and the facility needs to include electricity, bathrooms, and space. Which type of recovery site suits your requirements?
- a. Hot site
 - b. Cold site
 - c. Warm site
 - d. Cloud site
9. Which of the following statements are true regarding backups? (Select three.)
- a. The difference between incremental and differential backups is that a differential backup includes all data that has changed since the last incremental backup.
 - b. A differential backup includes all data that has changed since the last full backup.
 - c. An incremental backup is incomplete for full recovery without a full backup and all incremental backups since the last full backup.
 - d. A full backup is the safest and most comprehensive way to ensure data availability, but it can be time-consuming and costly.
10. In which disaster recovery setup are multiple redundant systems simultaneously active and operational and able to serve incoming requests?
- a. Active-passive
 - b. Clustering mode
 - c. Load balancer mode
 - d. Active-active
11. Which of the following are simulated, discussion-based sessions used to test and evaluate an organization's disaster recovery plans?
- a. Validation tests
 - b. Tabletop exercises
 - c. RPO testing
 - d. RTO testing



This chapter covers the following topics related to Objective 3.4 (Given a scenario, implement IPv4 and IPv6 network services) of the CompTIA Network+ N10-009 certification exam:

- Dynamic addressing
 - DHCP
 - Reservations
 - Scope
 - Lease time
 - Options
 - Relay/IP helper
 - Exclusions
 - Stateless address autoconfiguration (SLAAC)
- Name resolution
 - DNS
 - Domain Name Security Extensions (DNSSEC)
 - DNS over HTTPS (DoH) and DNS over TLS (DoT)
 - Record types
 - Address (A)
 - Quad A (AAAA)
 - Canonical name (CNAME)
 - Mail exchange (MX)
 - Text (TXT)
 - Nameserver (NS)
 - Pointer (PTR)
 - Zone types
 - Forward
 - Reverse
 - Authoritative vs. non-authoritative
 - Primary vs. secondary
 - Recursive
 - Hosts file
- Time Protocols
 - NTP
 - Precision Time Protocol (PTP)
 - Network Time Security (NTS)

IPv4 and IPv6 Network Services

While there are many network services available today, three services are so commonplace that we encounter them at every turn in the modern network: DHCP, DNS, and NTP. In this chapter, you'll learn the important basics of these critical network components.

Foundation Topics

Dynamic Addressing

You might not be surprised to learn that *Dynamic Host Configuration Protocol (DHCP)* is the star of the show when it comes to dynamic addressing for IPv4 in our corporate networks. But what might be surprising is the fact that IPv6 has a built-in mechanism for dynamic assignments of addresses. You will learn about that in this section of the chapter as well.

DHCP

Statically assigning IP address information to individual networked devices can be time-consuming, error prone, and subject to scalability problems. Instead of using static IP address assignments, many corporate networks dynamically assign IP address parameters to their devices. This is typically referred to as *dynamic address assignment*. The alternative approach is referred to as manual assignment, or *static assignment*.

An early choice for performing automatic assignment of IP addresses was Bootstrap Protocol (BOOTP). Currently, however, the most popular approach for dynamic IP address assignment is the use of DHCP.

Engineers developed BOOTP as a method of assigning IP address, subnet mask, and default gateway information to diskless workstations. In the early days of Microsoft Windows (for example, Microsoft Windows 3.1), Microsoft Windows did not natively support TCP/IP. To include TCP/IP support, an add-on TCP/IP application (for example, Trumpet Winsock) could be run. Such an application would typically support BOOTP.

DHCP offers a more robust solution to IP address assignment than does BOOTP. DHCP does not require a statically configured database of MAC-address-to-IP-address mappings. Also, DHCP has a wide variety of options beyond basic IP address, subnet mask, and default gateway parameters. For example, a DHCP server can educate a DHCP client about the IP address of a TFTP server from which a configuration file could be downloaded.

Figure 16-1 illustrates the exchange of messages that occurs as a DHCP client obtains IP address information from a DHCP server. The list that follows describes these steps in further detail:



- Step 1.** When a DHCP client initially boots, it has no IP address, default gateway, or other such configuration information. Therefore, the way a DHCP client initially communicates is by sending a broadcast message

(that is, a DHCPDISCOVER message to the destination broadcast address 255.255.255.255) in an attempt to discover a DHCP server.

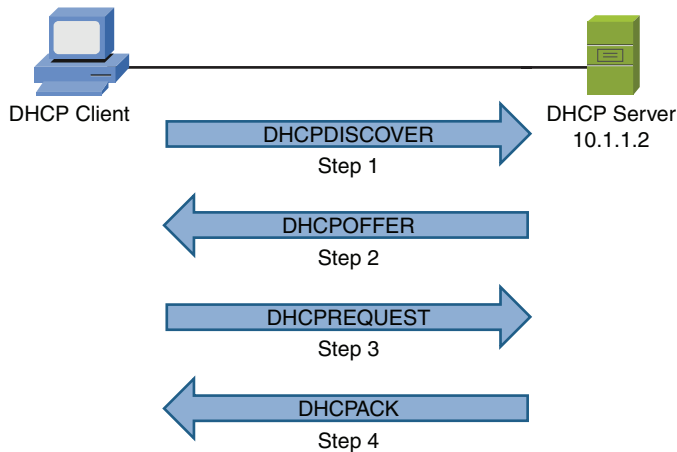


Figure 16-1 Obtaining IP Address Information from a DHCP Server

- Step 2.** When a DHCP server receives a DHCPDISCOVER message, it can respond with a unicast DHCPOFFER message. Because the DHCPDISCOVER message is sent as a broadcast, more than one DHCP server might respond to this discover request. However, the client typically selects the server that sent the first DHCPOFFER response received by the client.
- Step 3.** The DHCP client communicates with this selected server by sending a unicast DHCPREQUEST message, asking the DHCP server to provide IP configuration parameters.
- Step 4.** The DHCP server responds to the client with a unicast DHCPACK message. This DHCPACK message contains a collection of IP configuration parameters.

NOTE A method for remembering the four main steps of DHCP is to think of the acronym DORA: discover, offer, request, and acknowledge.

Notice that in step 1, the DHCPDISCOVER message is sent as a broadcast. By default, a broadcast cannot cross a router boundary. Therefore, if a client resides on a different network than the DHCP server, the client's next-hop router should be configured as a DHCP relay agent, which allows a router to relay DHCP requests

to either a unicast IP address or a directed broadcast address for a network. The DHCP relay agent is often referred to as an *IP helper*, or simply a *DHCP relay*.

Using a DHCP relay agent allows you to centrally locate a DHCP server that can simultaneously service many different TCP/IP subnets in your enterprise infrastructure. This is often a very desirable configuration because centralizing this information often makes it easier to manage. Of course, the risk in this configuration is that you must ensure that the DHCP relay agents are configured properly and that they are available; otherwise, the client broadcasts for DHCP services will not be successful.

A DHCP server can be configured to assign IP addresses to devices belonging to different subnets. Specifically, a DHCP server can determine the source subnet of a DHCP request and select an appropriate address pool from which to assign an address. One of these address pools (which typically corresponds to a single subnet) is called a *scope*.

When a network device is assigned an IP address from an appropriate DHCP scope, that assignment is not permanent. Rather, it is a temporary assignment referred to as a *lease* (the duration of your lease is the *lease time*). Although most client devices on a network work well with this dynamic addressing, some devices (for example, servers) might need to be assigned specific IP addresses. Fortunately, you can configure a *DHCP reservation*. A DHCP reservation maps a specific MAC address to a specific IP address, ensuring that a certain device in your network receives the reserved address.

Another frequent configuration you might make in a DHCP implementation is to configure an *exclusion range*. This is a portion of the address pool that you never want leased out to clients in the network. Perhaps you have numbered your servers 192.168.1.1–192.168.1.10. Because the servers are statically configured with these addresses, you exclude these addresses from the 192.168.1.0/24 pool of addresses. Your DHCP server then leases out addresses beginning at the first available in the range; in this example, that would be 192.168.1.11.

As an example of DHCP client configuration, in Microsoft Windows 11, you can select the **Obtain an IP address automatically** and **Obtain DNS server address automatically** options in the Internet Protocol Version 4 (TCP/IPv4) Properties window, as shown in Figure 16-2.

Perhaps you have noticed that DHCP dynamically assigns IP addresses and subnet masks to clients, but they need more IP addressing information than that in order to communicate properly on the network. For example, clients need DNS server addresses, they need a default gateway assigned, and they might even need the IP addresses associated with other key services, such as NTP and TFTP servers on the network. Your DHCP server can provide all this required IP addressing information.

You enable this as part of the *scope options* you can configure on the DHCP server. The scope options include the additional address information that must be distributed and can also include settings such as the lease time. Most DHCP servers permit a client to retain IP address information for 24 hours by default. If you have an environment with relatively few available addresses (and, therefore, few *available leases*), you can set the lease time to a shorter duration so that clients cannot retain the address information for excessive time periods.

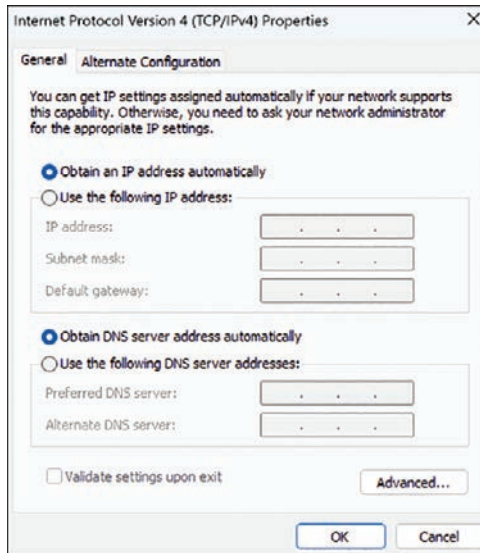


Figure 16-2 Configuring Microsoft Windows 11 to Obtain IP Address Information via DHCP

SLAAC

IPv6 *stateless address autoconfiguration (SLAAC)* is a method that allows devices on an IPv6 network to automatically configure their own IP addresses without the need for a central (DHCP) server. This makes it easier to set up and manage networks, especially as the number of connected devices continues to grow.

When a device connects to an IPv6 network, it first generates a unique link-local address, which is only valid within that local network segment. The device then uses this address to communicate with other devices on the same network and to discover routers. These routers send out Router Advertisement (RA) messages, which contain important network information, including the network prefix.

Using the information from the RA messages, the device combines the network prefix with its own unique identifier (often derived from its MAC address) to create

a global unicast address. This address can be used for communication both within the local network and with devices on other networks, including the Internet.

SLAAC simplifies network administration by reducing the need for manual IP address configuration or the deployment of a DHCP server, which is commonly used in IPv4 networks. Instead, devices can independently generate their own addresses and configure themselves, making the network more flexible and scalable.

In addition to basic address configuration, SLAAC also supports address renumbering, allowing devices to update their addresses automatically if the network prefix changes. This is particularly useful in dynamic environments where network configurations might change frequently.

Notice there are many elements to DHCP in modern networks. To help you master these components, Table 16-1 summarizes these DHCP elements for you.

Key
Topic

Table 16-1 DHCP Components

DHCP Element	Description
DHCP reservation	A configuration on a DHCP server that assigns a specific IP address to a particular device or client on a network. Reservations allow administrators to assign specific IP addresses to devices based on their MAC addresses, ensuring that a device always receives the same IP address each time it connects to the network.
DHCP scope	A range of IP addresses and configuration parameters that a DHCP server is responsible for assigning to client devices on a network.
DHCP lease time	The duration for which an IP address is allocated to a client device by a DHCP server.
DHCP relay/IP helper	A network device or service that forwards DHCP requests from clients on one subnet to a DHCP server on a different subnet.
DHCP exclusions	Specific IP addresses or ranges of IP addresses that are intentionally excluded from being dynamically assigned by a DHCP server.
SLAAC	An IPv6 feature that allows devices to automatically configure their own IP addresses without the need for a DHCP server.

Name Resolution

As humans, we love to give our network devices “friendly” names. For example, my laptop might be known as ANTMAN on the network. While this makes it obvious for me when I see that device in a list of network devices, other systems need to find this device using its IP address and MAC address. The process of resolving these friendly names to other identifiers is called *name resolution*. As you will learn in this

section, **Domain Name System (DNS)** is a critical name resolution service powering the modern Internet. This section also covers a bit of a name resolution shortcut. It is a hosts file approach on a local system.

DNS

A Domain Name System server performs the task of resolving a domain name (for example, `www.ciscopress.com`) to a corresponding IP address (for example, `159.182.74.49`). Because routers (or multilayer switches) make their forwarding decisions based on Layer 3 information (for example, IP addresses), an IP packet needs to contain IP address information, not DNS names. However, we humans recall meaningful names more readily than we recall 32-bit numbers.

As shown in Figure 16-3, an end user who wants to navigate to the `www.ciscopress.com` website enters that fully qualified domain name (FQDN) into a web browser; however, the browser cannot immediately send a packet destined for `www.ciscopress.com`. First, the end user's computer needs to resolve the FQDN `www.ciscopress.com` to a corresponding IP address, which can be inserted as the destination IP address in an IP packet. This resolution is made possible by a DNS server, which maintains a database of local FQDNs and their corresponding IP addresses, in addition to pointers to other servers that can resolve IP addresses for other domains.

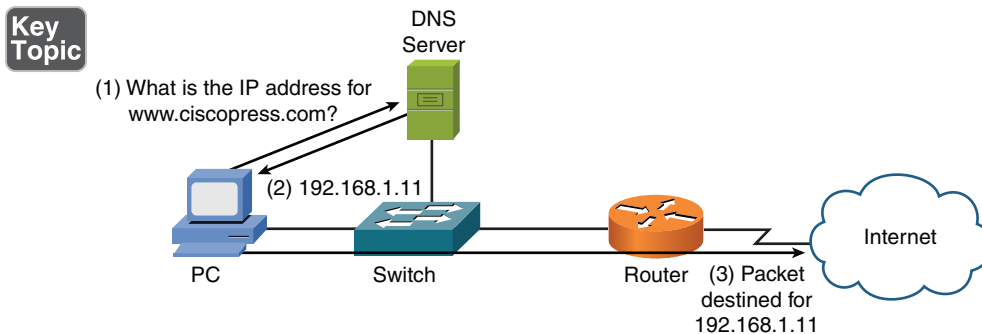


Figure 16-3 DNS Server

An FQDN is a series of strings, delimited by a period (such as `www.ciscopress.com`). The rightmost part of the FQDN is the top-level domain. Examples of top-level domains include `.com`, `.mil`, and `.edu`, as shown in Figure 16-4. Although there are many other top-level domains, these are among the most common top-level domains in the United States.

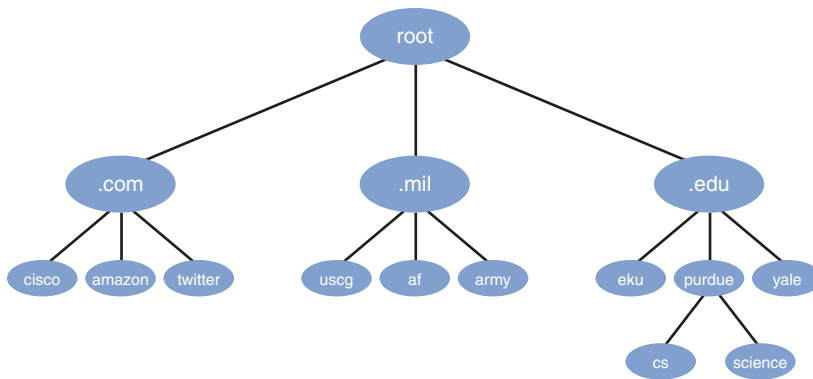
**Key
Topic**


Figure 16-4 Hierarchical Domain Name Structure

Lower-level domains can point upward to higher-level DNS servers to resolve nonlocal FQDNs, as illustrated in Figure 16-4.

DNS follows this strict hierarchy. And because public DNS is shared around the globe, it has a *global hierarchy*. In fact, the top-level domains include country references. For example, to enjoy a version of the Amazon.com domain tailored to India, you can visit www.amazon.in.

In addition to the global hierarchy of the DNS name system, the organization of DNS servers throughout the world follows a global hierarchy. The following DNS server types exist:

- **Root DNS servers:** These authoritative name servers serve the DNS root zone; they are a network of hundreds of servers in many countries around the world.
- **Internal DNS servers:** These servers exist inside an organization to resolve the names of private hosts and servers within the organization. These servers can also forward requests for outside resources to the appropriate external DNS servers when internal clients need to access external resources.
- **External DNS servers:** These servers reside outside an organization (typically on the public Internet) and can resolve names of systems that are located outside the organization.
- **Authoritative name server:** This DNS server is usually the last step in the journey for an IP address. The authoritative name server contains information specific to the domain name it serves (for example, ajsnetworking.com). The authoritative name server is able to resolve queries thanks to the records it contains.

- **Non-authoritative name server:** This DNS server provides DNS query responses based on information it has cached from previous queries rather than being the original source of the DNS records. It does not have the definitive, authoritative information but can quickly respond to requests using cached data. This type of server is typically used to reduce query times and network traffic.

A DNS server's database contains not only FQDNs and corresponding IP addresses but also **DNS record types**. For example, a mail exchange (MX) record would be the record type for an email server. As a few examples, Table 16-2 lists a collection of common DNS record types.

Key Topic

Table 16-2 Common DNS Record Types

Record Type	Description
<i>A</i>	An <i>address</i> record (that is, A record) maps a hostname to an IPv4 address.
<i>AAAA</i>	An IPv6 address record (that is, AAAA record) maps a hostname to an IPv6 address.
<i>CNAME</i>	A <i>canonical name</i> record (that is, CNAME record) is an alias of an existing record that allows multiple DNS records to map to the same IP address.
<i>MX</i>	A <i>mail exchange</i> record (that is, MX record) maps a domain name to an email (or message transfer agent) server for that domain.
<i>NS</i>	A <i>nameserver</i> record (that is, NS record) delegates a DNS zone to use the given authoritative name server.
<i>PTR</i>	A <i>pointer</i> record (that is, PTR record) points to a canonical name. A PTR record is commonly used when performing a reverse DNS lookup to determine what domain name is associated with a known IP address.
<i>SOA</i>	A <i>start of authority</i> record (that is, SOA record) provides authoritative information about a DNS zone, such as email contact information for the zone's administrator, the zone's primary name server, and various refresh timers.
<i>SRV</i>	A <i>service</i> location record (that is, SRV record) is used for newer protocols instead of creating protocol-specific records such as MX records.
<i>TXT</i>	A <i>text</i> record (that is, TXT record) originally was for arbitrary human-readable text in a DNS record. Since the early 1990s, however, this record has carried machine-readable data, such as data specified by RFC 1464, opportunistic encryption, Sender Policy Framework (SPF), or DomainKeys Identified Email (DKIM).

A potential challenge when setting up DNS records is when you want to point to the IP address of a device that might change its IP address. For example, if you have a cable modem or digital subscriber line (DSL) modem in your home, that device

might obtain its IP address from your service provider via DHCP. As a result, if you add the IP address of your cable modem or DSL modem to a DNS record (to allow users on the Internet to access one or more devices inside your network), that record could be incorrect if your device obtains a new IP address from your service provider.

To overcome this challenge, you can use dynamic DNS (DDNS). A DDNS provider supplies software you run on one of your PCs that monitors the IP address of the device referenced in the DNS record (such as your cable modem or DSL modem). If the software detects a change in the monitored IP address, that change is reported to your service provider, which is also providing DNS service.

Another option is IP address management (IPAM), which is a means of planning, tracking, and managing the Internet Protocol address space used in a network. IPAM integrates DNS and DHCP so that each is aware of changes in the other (for instance, DNS knowing of the IP address taken by a client via DHCP and updating itself accordingly).

Yet another DNS variant is Extension Mechanisms for DNS (EDNS). The original specification for DNS had size limitations that prevented the addition of certain features, such as security. EDNS supports these additional features while maintaining backward compatibility with the original DNS implementation. Rather than using new flags in the DNS header, which would negate backward compatibility, EDNS sends optional pseudo-resource records between devices supporting EDNS. These records can use 16 new DNS flags. If a legacy DNS server were to receive one of these optional records, the record would simply be ignored. Therefore, backward compatibility is maintained, while new features can be added for newer DNS servers.

When you enter a web address into your browser in the form `https://FQDN` (for example, `https://www.ajsnetworking.com`), notice that you not only indicate the FQDN of your web address but also specify that you want to access that location by using the HTTPS protocol. Such a string, which indicates both an address (for example, `www.ajsnetworking.com`) and a method for accessing that address (for example, `https://`), is called a *uniform resource locator* (URL).

Here are some other DNS concepts that you should be familiar with:

- **Zone transfers:** Because DNS servers will always need to be available (to make the Internet function), it is a common approach to use primary and secondary backup name servers. In order to keep the primary and secondary servers in sync with each other and updated with the same information, DNS zone transfers can be performed. There are multiple types of zone transfers, including full zone transfers and incremental zone transfers.

- **DNS time to live (TTL):** Do not confuse DNS TTL with the TTL that is in an IP packet header. Whereas the TTL in an IP packet header is a loop-prevention mechanism to ensure that a packet does not circulate endlessly on an IP network, the DNS TTL specifies the number of seconds a DNS server (or client) caches name resolution entries without having to perform a search of the DNS database. As you might guess, there is a lot of *DNS caching* going on in the *global DNS system* (both on servers and on clients). This is why name resolution for popular destinations like google.com appears to happen instantaneously.

NOTE The caching of DNS resolution information is another reason changes in the global DNS system can take hours, or even days, to be fully propagated. All the cached information must age out (that is, the TTL must expire), and new information must be requested.

- **Primary zone:** A primary zone is the main authoritative database for a specific domain, where all the original and authoritative DNS records for that domain are stored. It is hosted on the primary DNS server, which has read-write access to the zone file, allowing it to create, modify, and delete DNS records. Any changes to the DNS information must be made on this server, and it serves as the source of truth for the domain's DNS data. Secondary DNS servers can be set up to hold copies of the primary zone, but these are read-only and rely on the primary server for updates.
- **Secondary zone:** A secondary zone is a read-only copy of the primary zone, used to provide redundancy and load balancing for DNS queries. It is hosted on secondary DNS servers, which periodically receive updates from the primary DNS server through a process called *zone transfer*. The secondary zone contains all the same DNS records as the primary zone, ensuring that if the primary server becomes unavailable, the secondary server can still respond to DNS queries. However, since it is a read-only copy, any changes to DNS records must be made on the primary server and then propagated to the secondary zone.
- **Forward lookup zone:** You might not have realized this, but when you have a DNS server resolve a name (such as ajsnetworking.com) into an IP address, you are having the DNS server perform a *forward lookup*. A forward lookup zone in DNS is the database that maps domain names to their corresponding IP addresses. This zone is essential for directing Internet traffic to the correct servers based on domain names.

- **Reverse lookup zone:** DNS servers are also able to perform *reverse lookups*. As you might guess, this is when the client has the IP address of the resource and needs the resolution to work in reverse: the client is looking for the name that goes with the IP address. The reverse lookup zone is that database that makes these reverse lookups possible.
- **Recursive lookup:** In this type of DNS query, the client instructs the DNS server to respond with an answer (if possible), and the client does not want to be referred to another DNS server. Contrast this with the iterative (or nonrecursive) query type covered next.
- **Iterative lookup:** In this type of DNS query, the client indicates that it would like the answer from the DNS server, or it will take a referral to another DNS server that might have the answer.
- **Domain Name Security Extensions (DNSSEC):** Domain Name Security Extensions is a set of security measures added to DNS to protect it from attacks. DNSSEC works by digitally signing data to ensure that the responses to DNS queries are authentic and haven't been tampered with, helping to prevent attackers from redirecting users to malicious websites. This makes the Internet more secure by ensuring that when you type a web address into your browser, you are directed to the correct and intended website.
- **DNS over HTTPS (DoH):** DNS over HTTPS is a security protocol that encrypts DNS queries by sending them over HTTPS, the same protocol used to secure websites. This ensures that your Internet service provider and other potential eavesdroppers cannot see the websites you are trying to visit, enhancing your privacy and security online. By using DoH, your DNS requests are protected from interception and manipulation, making your Internet browsing more secure and private.
- **DNS over TLS (DoT):** DNS over TLS is a similar protocol to DoH. It differs slightly in its encryption and delivery of packets, but its job is the same: protect DNS queries as they are transmitted on the Internet.

Hosts File

A **hosts file** is a plain text file used by operating systems to map hostnames to IP addresses. This file helps your computer find and connect to other devices on a network without needing to query a DNS server.

The hosts file is usually located in a specific directory on your computer. On Windows, you can find it at C:\Windows\System32\drivers\etc\hosts, while on macOS

and Linux, it's located at `/etc/hosts`. You can open and edit this file with any text editor, like Notepad on Windows or TextEdit on macOS.

Each line in the hosts file represents a mapping between an IP address and a hostname. For example, a line might read “127.0.0.1 localhost,” which tells your computer that the hostname localhost corresponds to the IP address 127.0.0.1. You can add your own entries to this file to create custom mappings for your network.

Using the hosts file can be helpful for testing new websites or services before they go live. By adding an entry to the hosts file, you can make your computer resolve a hostname to a specific IP address, bypassing the DNS server. This allows you to see how a website will look and function without changing the actual DNS records.

Time Protocols

The correct time shared throughout network devices is an important part of networking. Why is this so important? Correct time is crucial for network devices because it ensures the proper synchronization of logs, security certificates, and scheduled tasks across the network. Accurate timekeeping is essential for troubleshooting, as logs from different devices need to be aligned to trace issues accurately. It is also vital for security, as time-sensitive protocols like Kerberos require synchronized clocks to prevent authentication failures. Furthermore, incorrect time can lead to issues with data integrity, as time stamps are used to validate the order of events, updates, and transactions. Without synchronized time, network operations can become unreliable, leading to potential vulnerabilities and operational inefficiencies. This section ensures you are knowledgeable about several different *time protocols*.

NTP

Network Time Protocol (NTP) is the final critical service we examine in this chapter. This protocol might be easy for you to overlook—or at least not think of as being very important. After all, who cares if the time is a bit off on your network devices?

For many reasons, it is actually very important that network devices possess times that are as accurate as possible. Perhaps the biggest reason is the fact that devices can then stamp Syslog and other messages with the accurate time. This can often be critical for helping you catch and prevent security attacks against your network or for helping you troubleshoot faults that are occurring as a result of failed components or configurations.

Another reason you typically need accurate time is that many services cannot properly integrate with other network services if the time is not within a certain threshold between the devices.

**Key
Topic**

The configuration of NTP involves three main components:

- **Client:** Your network device is typically the NTP client and is consuming the correct time from an NTP server system.
- **Server:** The server is the network node that provides the correct time to NTP clients. The most common servers these days are located regionally on the Internet.
- **Stratum:** The stratum is an important value in NTP operations. It is like a hop count and measures how far the client is from the accurate time source. The larger the stratum value, the more chance the time on the client could be inaccurate because it is more hops away from the accurate time source.

NTP does not require connection-oriented transport. As a result, NTP relies on UDP at the transport layer. Specifically, NTP uses UDP port 123.

PTP

Precision Time Protocol (PTP) is another protocol used to synchronize clocks across a computer network with very high accuracy. It is particularly useful in situations where precise timing is critical, such as in financial transactions, telecommunications, and industrial automation.

PTP works by exchanging timing messages between a primary clock and one or more member clocks within the network. The primary clock serves as the reference time source, and the member clocks adjust their time based on the messages received from the primary. This communication helps ensure that all devices in the network are synchronized to the same time.

The synchronization process involves several steps to account for network delays and other factors that can affect timing accuracy. PTP measures the time it takes for messages to travel between the primary and member clocks, and uses this information to correct any discrepancies. This allows PTP to achieve synchronization accuracy within nanoseconds, much higher than other protocols like NTP.

One of the key features of PTP is its ability to work in various network configurations, including Ethernet and Wi-Fi. It can also be used in both small and large networks, making it a versatile solution for different applications. Additionally, PTP supports different clock hierarchies, where multiple layers of primary and member clocks can be used to improve synchronization across complex networks.

NTS

Network Time Security (NTS) is a protocol designed to secure time synchronization over a network, specifically enhancing the security of Network Time Protocol. As you learned earlier in this section, NTP is used to synchronize the clocks of computers and devices to a precise time source, but it has vulnerabilities that can be exploited by attackers to disrupt or manipulate timekeeping.

NTS works by adding security features to NTP to protect against these threats. It uses cryptographic techniques to ensure that the time information exchanged between devices is authentic and has not been tampered with. This helps to prevent attacks such as spoofing, where an attacker could send false time information to devices, and replay attacks, where old messages are resent to disrupt the synchronization process.

When a device using NTS requests time from an NTP server, it first establishes a secure connection using Transport Layer Security (TLS), the same technology that secures websites. This secure connection is used to exchange keys and other information needed to authenticate future time requests. Once this initial setup is complete, the device and the server can communicate securely, with each time message being authenticated to ensure its integrity.

The adoption of NTS is important for critical systems that rely on precise and accurate timekeeping, such as financial systems, telecommunications, and industrial control systems. By securing the time synchronization process, NTS helps to ensure that these systems operate reliably and are protected against malicious attacks that could cause disruptions or inaccuracies.

Real-World Case Study

Acme, Inc. has been busy improving the DHCP implementation in the main end-user locations of the network. These improvements have involved increasing the availability of centralized DHCP server nodes. This increase in the availability has been made possible by the use of DHCP relay agents. These strategically placed network nodes ensure DHCP broadcasts from client systems can be routed effectively to the DHCP servers of Acme, Inc.

The DHCP enhancements also include the setting and configuration of key network parameters through the DHCP information disseminated using the scope options feature.

Acme, Inc. has also improved the dissemination of accurate time by configuring all network nodes with fallback NTP servers to use should the main, Internet-based NTP server fail.

Finally, Acme, Inc. has recently implemented Network Time Security to enhance the accuracy and reliability of its time-sensitive network operations. By deploying NTS, Acme, Inc. ensures that all its network devices, from servers to routers, are synchronized with a highly secure and accurate time source. This implementation helps protect against potential time spoofing attacks, where malicious actors could otherwise disrupt network operations by altering time synchronization.

Summary

Here are the main topics covered in this chapter:

- This chapter covered DHCP in detail, including important components of DHCP such as reservations, scopes, lease times, and many more topics.
- This chapter also covered name resolution systems. DNS is critical for a network system, especially one that is connected to the Internet and requires access to remote resources. Resolving “friendly” names against an IP address is a fundamental necessity of the modern network.
- This chapter concluded with a look at time protocols. Ensuring network nodes all agree on the correct time is another fundamental necessity in the typical network. Network Time Protocol, Precision Time Protocol, and Network Time Security were all covered.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 16-3 lists these key topics and the page number where each is found.



Table 16-3 Key Topics for Chapter 16

Key Topic Element	Description	Page Number
Step list	Four-step DHCP process	386
Figure 16-1	DHCP Components	390
Figure 16-3	DNS Server	391
Figure 16-4	Hierarchical Domain Name Structure	392
Table 16-2	Common DNS Record Types	393
List	NTP components	398

Complete Tables and Lists from Memory

Print a copy of Appendix B, “Memory Tables,” or at least the section for this chapter, and complete as many of the tables as possible from memory. Appendix C, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

AAAA, address (A), authoritative name server, canonical name (CNAME), DHCP relay, DHCP reservation, DNS over HTTPS (DoH), DNS over TLS (DoT), DNS record types, Domain Name Security Extensions (DNSSEC), Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP), exclusion range, forward lookup zone, IP helper, lease time, mail exchange (MX), name resolution, name server (NS), Network Time Protocol (NTP), Network Time Security (NTS), non-authoritative name server, pointer (PTR), Precision Time Protocol (PTP), primary zone, recursive lookup, reverse lookup zone, scope, scope options, secondary zone, stateless address autoconfiguration (SLAAC), text (TXT), time protocols

Additional Resources

What Is DHCP?: <https://www.spiceworks.com/tech/networking/articles/what-is-dhcp/>

NTP – As Easy as 1, 2, 3: <https://youtu.be/LtMWTmQqRfA>

Review Questions

The answers to these review questions are in Appendix A, “Answers to Review Questions.”

1. What is the name of the DHCP message that a client sends when it needs to obtain IP address information? This message is the first step in the four-step DHCP process.
 - a. DHCPOFFER
 - b. DHCPREQUEST
 - c. DHCPDISCOVER
 - d. DHCPACK

2. Where might you set the lease duration for a DHCP server that is servicing many clients in the network?
 - a. The zone record
 - b. Scope options
 - c. The DHCP forwarder
 - d. The relay agent
3. What type of DNS message is often used to perform reverse lookups?
 - a. PTR
 - b. SRV
 - c. TXT
 - d. MX
4. What value in DNS dictates how long a DNS server or client might cache DNS name resolution information?
 - a. Hop count
 - b. Increment duration
 - c. Backoff timer
 - d. TTL
5. What port and protocol are used by NTP? (Choose two.)
 - a. TCP
 - b. UDP
 - c. 69
 - d. 443
 - e. 123
6. Which component of NTP is a measure of how far the NTP client is from the server?
 - a. Administrative distance
 - b. Stratum
 - c. MED
 - d. Metric

7. Which is a range of IP addresses and configuration parameters that a DHCP server is responsible for assigning to client devices on a network?
 - a. Reservation
 - b. Least time
 - c. Exclusions
 - d. Scope
8. Which is a feature of IPv6 networks that allows devices to connect to the Internet without requiring any intermediate IP support from a DHCP server?
 - a. IPAM
 - b. SLAAC
 - c. Root DNS servers
 - d. Authoritative name server
9. Which DNS record maps a hostname to an IPv6 address?
 - a. A
 - b. AAAA
 - c. MX
 - d. PTR
10. Which is a protocol that encrypts DNS queries and responses using the HTTPS protocol?
 - a. DoH
 - b. DoT
 - c. DNSSEC
 - d. DNS TTL



This chapter covers the following topics related to Objective 3.5 (Compare and contrast network access and management methods) of the CompTIA Network+ N10-009 certification exam:

- Site-to-site VPN
- Client-to-site VPN
 - Clientless VPN
 - Split tunnel vs. full tunnel
- Connection methods
 - SSH
 - Graphical user interface (GUI)
 - API
 - Console
- Jump box/host
- In-band vs. out-of-band management

Network Access and Management Methods

In this chapter, we examine how many different methods there are today to access your network or networks, and also the popular methods of device management. Sure enough, you will notice that many of the methods and technologies discussed in this chapter emphasize remote access. More than ever, we want to be able to access and securely manage our networks from anywhere in the world.

This is particularly true in the case of cloud technologies. After all, one of the defining characteristics of cloud (according to the NIST) is broad network access. Fortunately, there are more options than ever before in this regard. Advancements in WAN technologies (such as SD-WAN) are making it easier than ever before to create and manage network access connections for a wide variety of purposes.

Foundation Topics

Virtual Private Networks (VPNs)

Thanks in good part to the COVID-19 virus, much of today's workforce is located outside a corporate headquarters location. But even before the pandemic, some employees worked in remote offices, and others telecommuted. Remote employees can connect to their main corporate network by using a variety of WAN technologies, such as leased lines or high-speed fiber connections. However, these WAN technologies typically cost more than widely available broadband technologies, such as cable, which might also offer faster speeds, or at least comparable speeds.

Virtual private networks (VPNs) support secure communication between sites over an untrusted network (for example, the Internet). The two primary categories of VPNs are site-to-site and client-to-site VPNs:

Key Topic

- **Site-to-site VPN:** A site-to-site VPN interconnects two sites, as an alternative to a leased line, at a reduced cost. Figure 17-1 shows an example of a site-to-site VPN.
- **Client-to-site VPN:** A client-to-site VPN (also known as a *remote access VPN*) interconnects a remote user with a site, as an alternative to dial-up or ISDN connectivity, at a reduced cost. Figure 17-2 shows an example of a client-to-site VPN.

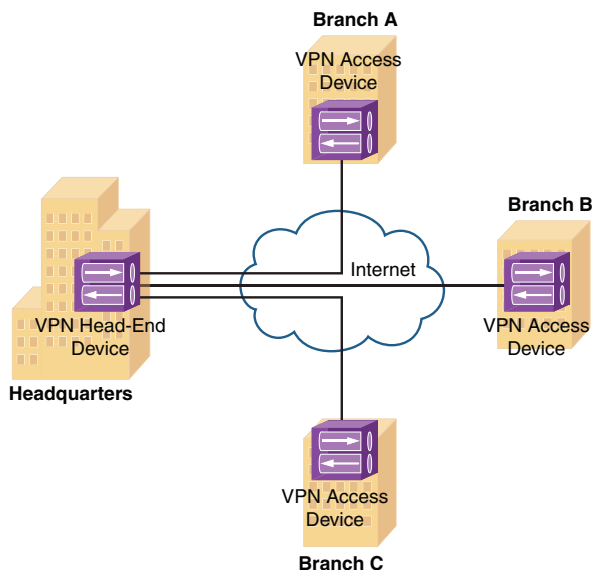


Figure 17-1 Site-to-Site VPN

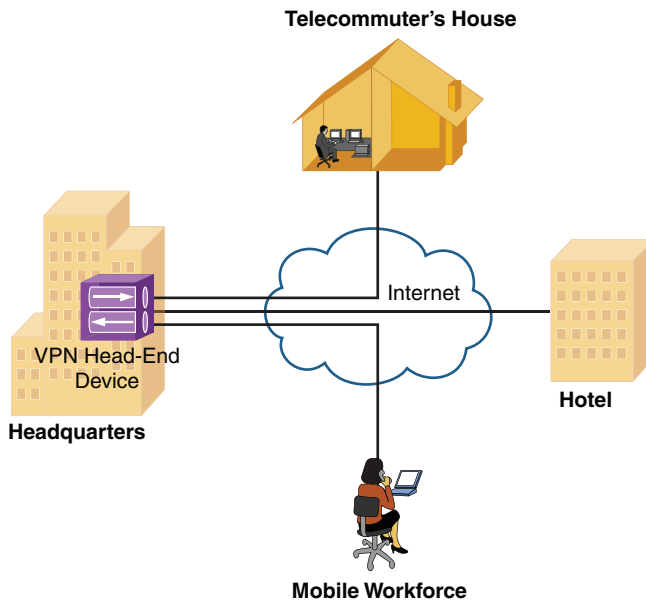


Figure 17-2 Client-to-Site VPN

Although a VPN tunnel might physically pass through multiple service provider routers, the tunnel appears to be a single router hop from the perspective of the routers at each end of the tunnel.

A client-to-site VPN allows a user with software on a client computer to connect to a centralized VPN termination device, and a site-to-site VPN interconnects two sites without requiring the computers at those sites to have any specialized VPN software installed. Client-to-site VPNs could be implemented using a VPN-compatible device, such as a router, a firewall, or a special-purpose device called a *VPN concentrator* that is custom-built for handling remote access client-to-site VPN connections. It is also possible, with the correct software, for two computers to connect to each other directly using a host-to-host IPsec VPN connection.

One very popular option in client-to-site VPNs is a *clientless VPN* (although this is a bit of a misnomer). With a clientless VPN, there is a client software piece, but it is not a separate piece of software; rather, the client's web browser acts as the VPN client software. This type of VPN connection leverages the SSL/TLS capabilities of the modern Internet and web browsers to provide a secured connection.

Another important consideration when it comes to VPN configuration is whether you will be using a split tunnel or full tunnel configuration:

- With a **full tunnel** configuration, you have all the end-user traffic go through the VPN tunnel.

- With a ***split tunnel*** configuration, you don't have all the end-user traffic go through the VPN tunnel. For example, you might choose to have just the traffic that needs to access the corporate network sent through the VPN tunnel, while allowing traffic destined for the Internet (perhaps Office 365 traffic) to bypass the tunnel. Split tunneling is often an advantageous configuration because the bandwidth required by the VPN connection (and the overhead associated with it) can be minimized.

IPsec

Broadband technologies, such as cable, in addition to other VPN transport mechanisms, often traverse an untrusted network, such as the Internet. Therefore, a primary concern with using a broadband technology as a VPN transport is security.

VPN technologies such as IP Security (IPsec), Generic Routing Encapsulation (GRE), Layer 2 Tunneling Protocol (L2TP), and Layer 2 Forwarding (L2F) offer a variety of features, but IPsec VPNs offer strong security features. Specifically, IPsec offers CIA protection for traffic. CIA is considered the foundation for a secure network and consists of the following:

- **Confidentiality:** Data confidentiality is provided by encrypting data. A third party who intercepts the encrypted data will not be able to interpret it.
- **Integrity:** Data integrity ensures that data is not modified in transit. For example, routers at each end of a tunnel can calculate a checksum value or a hash value for the data, and if both routers calculate the same value, the data has most likely not been modified in transit.
- **Authentication:** Data authentication allows parties involved in a conversation to verify that the other party is the party they claim to be.

IPsec also scales to a wide range of networks. IPsec operates at Layer 3 of the OSI model (the network layer). As a result, IPsec is transparent to applications, which means that applications do not require any sort of integrated IPsec support.

IKE

IPsec uses a collection of protocols to provide its features. One of the primary protocols that IPsec uses is Internet Key Exchange (IKE). Specifically, IPsec can provide encryption between authenticated peers using encryption keys that are periodically changed. IKE, however, allows an administrator to manually configure keys.

As outlined in Table 17-1, IKE can use three modes of operation to set up a secure communication path between IPsec peers.

Table 17-1 IKE Modes

Mode	Description
Main mode	Main mode involves three exchanges of information between the IPsec peers: Exchange 1: The responder selects a proposal it received from the initiator. Exchange 2: Diffie-Hellman (DH) is used to securely establish a shared secret key over the unsecured medium. Exchange 3: An Internet Security Association and Key Management Protocol (ISAKMP) session is established. This secure session is then used to negotiate an IPsec session. One peer, called the <i>initiator</i>, sends one or more proposals to the other peer, called the <i>responder</i>. The proposals include supported encryption and authentication protocols and key lifetimes. In addition, the proposals indicate whether perfect forward secrecy (PFS) should be used. PFS ensures that a session key remains secure, even if one of the private keys used to derive the session key becomes compromised.
Aggressive mode	Aggressive mode more quickly achieves the same results as main mode, using only three packets. The initiator sends the first packet, which contains all the information necessary to establish a security association (SA)—that is, an agreement between the two IPsec peers about the cryptographic parameters to be used in the ISAKMP session. The responder sends the second packet, which contains the security parameters selected by the responder (the proposal, the keying material, and its ID). The responder uses this second packet to authenticate the session. The third and final packet, which is sent by the initiator, finalizes the authentication of the ISAKMP session.
Quick mode	Quick mode negotiates the parameters (the SA) for the IPsec session. This negotiation occurs within the protection of an ISAKMP session.

The IKE modes reflect the two primary phases of establishing an IPsec tunnel. For example, during IKE Phase 1, a secure ISAKMP session is established, using either main mode or aggressive mode. During IKE Phase 1, the IPsec endpoints establish transform sets (which are collections of encryption and authentication protocols), hash methods, and other parameters needed to establish a secure ISAKMP session (sometimes called an ISAKMP tunnel or an IKE Phase 1 tunnel). This collection of parameters is called a *security association* (SA). With IKE Phase 1, the SA is bidirectional, which means that the same key exchange is used for data flowing across the tunnel in either direction.

IKE Phase 2 occurs within the protection of an IKE Phase 1 tunnel, using the previously described *quick mode* of parameter negotiation. A session formed during IKE

Phase 2 is sometimes called an *IKE Phase 2 tunnel* or simply an *IPsec tunnel*. However, unlike IKE Phase 1, IKE Phase 2 performs unidirectional SA negotiations, which means that each data flow uses a separate key exchange.

Although an IPsec tunnel can be established using just IKE Phase 1 and IKE Phase 2, an optional IKE Phase 1.5 can be used. IKE Phase 1.5 uses the Extended Authentication (XAuth) protocol to perform user authentication of IPsec tunnels. Like IKE Phase 2, IKE Phase 1.5 is performed within the protection of an IKE Phase 1 tunnel. The user authentication provided by this phase adds a layer of authentication for VPN clients. Also, parameters such as IP, WINS, and DNS server information can be provided to a VPN client during this optional phase.

A newer version of IKE called IKEv2 (discussed more a bit later in this chapter) combines many of the same functions of IKEv1 and uses an initial IKEv2 tunnel (instead of IKEv1 Phase 1) and child security associations (SAs/tunnels) for the IPsec tunnels instead of calling them IKE Phase 2 tunnels.

Authentication Header and Encapsulating Security Payload

In addition to IKE, which establishes the IPsec tunnel, IPsec relies on either the Authentication Header (AH) protocol (IP protocol number 51) or the Encapsulating Security Payload (ESP) protocol (IP protocol number 50). Both AH and ESP offer origin authentication and integrity services, which ensure that IPsec peers are who they claim to be and that the data was not modified in transit.

However, the main distinction between AH and ESP is encryption support. ESP encrypts the original packet, whereas AH does not offer encryption. As a result, ESP is much more popular on today's networks.

Both AH and ESP can operate in one of two modes: transport mode or tunnel mode. Figure 17-3 illustrates the structure of an ESP transport mode packet versus an ESP tunnel mode packet.

NOTE You might be concerned that transport mode allows the IP address of the IPsec peers to remain visible during transit because the original packet's IP header is used to route a packet. However, IPsec is often used in conjunction with the *Generic Routing Encapsulation (GRE)* tunneling protocol. With transport mode, the original IP packet is encapsulated inside a GRE tunnel packet, which adds a new GRE tunnel header. The GRE packet is then sent over an IPsec tunnel. Even if the IPsec tunnel were running in transport mode, the original packet's IP header would still not be visible. Instead, the GRE packet's header would be visible.

**Key
Topic****Transport Mode**

ESP Auth	ESP Trailer	Payload	ESP Header	Original IP Header
-------------	----------------	---------	---------------	-----------------------

Tunnel Mode

ESP Auth	ESP Trailer	Payload	Original IP Header	ESP Header	New IP Header
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Figure 17-3 Transport Mode Versus Tunnel Mode

Following is a detailed description of these two modes:

- **Transport mode:** Uses a packet's original IP header, as opposed to adding an additional tunnel header. This approach works well in networks where increasing a packet's size might cause an issue. Also, transport mode is often used for client-to-site VPNs, where a PC running VPN client software connects back to a VPN termination device at a headquarters location.
- **Tunnel mode:** Unlike transport mode, tunnel mode encapsulates an entire packet. As a result, the encapsulated packet has a new header (an IPsec header). This new header has source and destination IP address information that reflects the two VPN termination devices at different sites. Therefore, tunnel mode is often used in an IPsec site-to-site VPN.

One reason a GRE tunnel might be used with an IPsec tunnel is as a limitation on the part of IPsec. Specifically, an IPsec tunnel can only transmit unicast IP packets. The challenge is that large enterprise networks might have a significant amount of broadcast or multicast traffic (for example, routing protocol traffic). GRE can take any traffic type and encapsulate the traffic in a GRE tunnel packet, which is a unicast IP packet that can then be sent over an IPsec tunnel. Take, for example, a multicast packet used by a routing protocol. Although IPsec cannot directly transport the multicast packet, if the packet is first encapsulated by GRE, the GRE packet can then be sent over an IPsec tunnel, thereby securing the transmission of the multicast packet.

The Five Steps in Setting Up and Tearing Down an IPsec Site-to-Site VPN Using IKEv1

The process of establishing, maintaining, and tearing down an IPsec site-to-site VPN consists of five primary steps, which are illustrated in Figure 17-4 and described in detail in the list that follows:

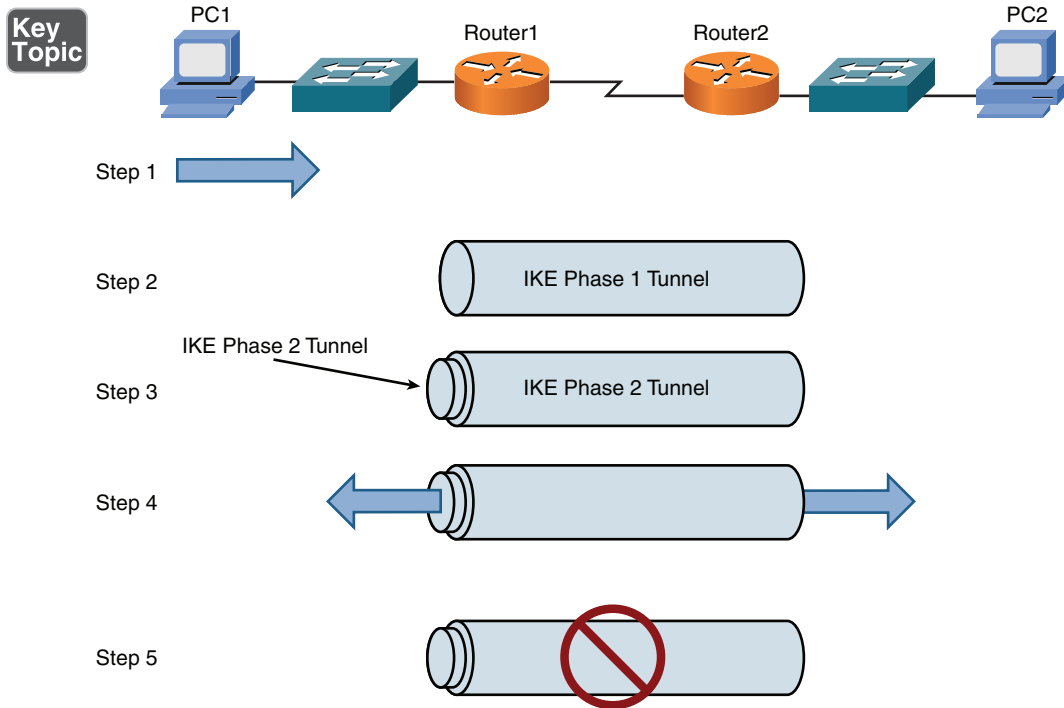


Figure 17-4 IPsec VPN Steps

- Step 1.** PC1 sends traffic destined for PC2. Router1 classifies the traffic as “interesting” traffic, and this classification initiates the creation of an IPsec tunnel.
- Step 2.** Router1 and Router2 negotiate an SA used to form an IKE Phase 1 tunnel, which is also known as an ISAKMP tunnel.
- Step 3.** Within the protection of the IKE Phase 1 tunnel, an IKE Phase 2 tunnel is negotiated and set up. An IKE Phase 2 tunnel is also known as an IPsec tunnel.
- Step 4.** After the IPsec tunnel is established, interesting traffic (for example, traffic classified by an ACL) flows through the protected IPsec tunnel. Note

that traffic not deemed interesting can still be sent between PC1 and PC2. However, the noninteresting traffic is transmitted outside the protection of the IPsec tunnel.

Step 5. After no interesting traffic is seen for a specified amount of time, the IPsec tunnel is torn down and the IPsec SA is deleted.

This example describes an IPsec site-to-site VPN, but the procedure is similar for a client-to-site VPN. IPsec is typically deployed using IKEv1, with its two phases (Phase 1 and Phase 2).

IKEv2

As you might guess, IKE Version 2 (IKEv2) offers many great improvements over IKEv1. IKEv2 uses fewer packets in setting up the SAs between VPN peers and does not use the terms *Phase 1* and *Phase 2*. Instead, the initial tunnel is called the *IKEv2 SA*, and the IPsec SA is referred to as a *child tunnel* (instead of being called an IKE Phase 2 tunnel).

Additional features that are integrated into IKEv2 include the following:

- Extensible Authentication Protocol (EAP)
- NAT traversal (that is, the ability to detect NAT in the path between the peers)
- The ability to validate the tunnel

IKEv1 requires additional configuration and vendor add-ons to implement similar types of features.

NOTE Is there an IKE version 3? There sure is. IKEv3 simplifies the negotiation of security associations and improves security with stronger cryptographic algorithms. It introduces features like built-in NAT traversal and support for multiple authentication methods. IKEv3 also streamlines the protocol, reducing the number of message exchanges required to establish secure connections.

Other VPN Technologies

Although IPsec VPNs are popular for securely interconnecting sites or connecting a remote client to a site, you need to be aware of other VPN protocols, examples of which are provided in Table 17-2.

**Key
Topic**

Table 17-2 Examples of VPN Protocols

Protocol	Description
SSL	Secure Socket Layer (SSL) provides cryptography and reliability for the upper layers (Layers 5–7) of the OSI model. SSL, which was introduced in 1995, has largely been replaced by Transport Layer Security (TLS). However, recent versions of SSL (for example, SSL 3.3) have been enhanced to be more comparable with TLS. Both SSL and TLS provide secure web browsing via Hypertext Transfer Protocol Secure (HTTPS).
L2TP	Layer 2 Tunneling Protocol (L2TP) is a VPN protocol that lacks security features, such as encryption. However, L2TP can still be used for a secure VPN connection if it is combined with another protocol that does provide encryption.
L2F	Layer 2 Forwarding (L2F) is a VPN protocol designed (by Cisco Systems) with the goal of providing a tunneling protocol for PPP. Like L2TP, L2F lacks native security features.
PPTP	Point-to-Point Tunneling Protocol (PPTP) is an older VPN protocol (which supported the dial-up networking feature in older versions of Microsoft Windows). Like L2TP and L2F, PPTP lacks native security features. However, Microsoft’s versions of PPTP bundled with various versions of Microsoft Windows were enhanced to offer security features.
TLS	Transport Layer Security (TLS) has largely replaced SSL as the VPN protocol of choice for providing cryptography and reliability to upper layers of the OSI model. For example, when you securely connect to a website using HTTPS, you are probably using TLS.
SSTP	Secure Socket Tunneling Protocol (SSTP) is a VPN tunnel that transports PPP traffic through an SSL/TLS channel. SSL/TLS provides transport-level security. SSTP’s use of SSL/TLS over TCP port 443 allows SSTP to pass through virtually all firewalls and proxy servers.
OpenVPN	OpenVPN is a software VPN system that creates secure point-to-point or site-to-site connections in routed or bridged configurations and remote access facilities. It implements both client and server applications.

Other Network Access Technologies

When it comes to network access solutions, it is not surprising that varying amounts of security are possible today and can be implemented in a wide variety of ways. Your organization might require the strongest possible protections over one network access connection, while another might require the mildest possible security treatment. Fortunately, remote network access technologies abound today, and these varying degrees of security can be accommodated with no problem.

Although ACLs can be used to permit or deny specific connections flowing *through* a router (or switch), you also need to control connections *to* network devices (for

example, routers, switches, or servers). Table 17-3 summarizes many of these network access technologies.

**Key
Topic**

Table 17-3 Network Access Technologies

Method	Description
Virtual desktop	Virtual desktop technology is growing at a fast pace, thanks in large part to the adoption of cloud technologies and increased bandwidth between data centers and end users. Using this approach, a user accesses a desktop that is housed on a server in the data center. The user might not even realize that the operating system is not hosted on the local PC or laptop. Note that the virtual desktop concept discussed here is different from other types of virtual desktops in IT. For example, in both Windows and macOS systems, you can create virtual desktops. These are just running “spaces” on the local system that have different windows and programs running within them, which makes it easier to multitask. Notice how very different this technology is from the virtual desktops referenced here.
SSH	Secure Shell (SSH) is a protocol used to securely connect to a remote host (typically via a terminal emulator). Secure Shell has enjoyed such widespread success that it is even used as the basis for several other security protocols and techniques. For example, SCP calls on SSH for some of its functionality. SSH uses port 22. As mentioned earlier, Linux administrators often prefer to use the command line for all operation and work in the operating system, and SSH is a perfect secure remote access tool for such users. Administrators can use the CLI to connect securely with SSH, and then they can operate in a BASH CLI environment to manage or work within the system.
Graphical user interface (GUI)	Many systems still provide a graphical user interface (GUI) for management access, no matter if this is local or remote access. Always be sure that appropriate bandwidth exists for a seamless experience with these GUIs.
API	Application programming interfaces (APIs) are increasingly being used to access and manage networks by providing a standardized way for software applications to communicate with network devices and services. Through APIs, developers can programmatically configure network settings, monitor network performance, and automate network management tasks. This allows for greater flexibility, scalability, and efficiency in network operations, as tasks that once required manual intervention can now be automated.
Console	Most network devices permit local access through what is termed a console port. Notice this console port should be protected with the appropriate logical and physical security controls.

Method	Description
AAA	Authentication, authorization, and accounting (AAA, pronounced “triple A”) allows a network to have a single repository of user credentials. A network administrator can, for example, supply the same credentials to log in to various network devices (for example, routers and switches). RADIUS and TACACS+ are protocols commonly used to communicate with a AAA server.
RADIUS	Remote Authentication Dial-In User Service (RADIUS) is a UDP-based protocol used to communicate with a AAA server. Unlike TACACS+, RADIUS does not encrypt an entire authentication packet; rather, it encrypts only the password. However, RADIUS does offer more robust accounting features than TACACS+. Also, RADIUS is a standards-based protocol, whereas TACACS+ is a Cisco-proprietary protocol.
TACACS+	Terminal Access Controller Access-Control System Plus (TACACS+) is a Cisco-proprietary TCP-based AAA protocol. TACACS+ has three separate and distinct sessions or functions for authentication, authorization, and accounting.
NAC	Network access control (NAC) can be used to permit or deny access to a network based on characteristics of the device seeking admission rather than just checking user credentials. For example, a client’s operating system and version of antivirus software could be checked against a set of requirements before allowing the client to access a network. This process of checking a client’s characteristics is called <i>posture assessment</i> .
IEEE 802.1X	IEEE 802.1X is a type of NAC that can permit or deny a wireless or wired LAN client access to a network. If IEEE 802.1X is used to permit access to a LAN via a switch port, then IEEE 802.1X is being used for port security. While covered elsewhere in this text, to summarize, in 802.1X, the device seeking admission to the network is called the <i>supplicant</i>. The device to which the supplicant connects (either wirelessly or through a wired connection) is called the <i>authenticator</i>. The device that checks the supplicant’s credentials and permits or denies the supplicant to access the network is called an <i>authentication server</i>. Usually, an authentication server is a RADIUS server.
EAP	Extensible Authentication Protocol (EAP) specifies how authentication is performed by IEEE 802.1X. A variety of EAP types exist (see Chapter 11), such as Extensible Authentication Protocol-Flexible Authentication via Secure Tunneling (EAP-FAST), Extensible Authentication Protocol-Message Digest 5 (EAP-MD5), and Extensible Authentication Protocol-Transport Layer Security (EAP-TLS).

Method	Description
Two-factor authentication	Two-factor authentication (TFA) requires two types of authentication from a user seeking admission to a network. For example, a user might have to <i>know</i> something (for example, a password) and <i>have</i> something (such as a specific fingerprint, which can be checked with a biometric authentication device).
Multifactor authentication	Multifactor authentication requires two or more types of successful authentication factors before granting access to a network.
Single sign-on	Single sign-on (SSO) allows a user to authenticate only once to gain access to multiple systems rather than needing to independently authenticate with each system.
Local authentication	Local authentication refers to a network device authenticating the user with a database of user account information stored on the device itself; this is often an important fallback method of authentication used when another external method fails.
Captive portal	A captive portal is a web page that appears before a user is able to access the network resource; this web page accepts the credentials of the user for authentication and presents them to the authentication server.
Jump box/host	A jump box , or jump host , is a secure intermediary server used to access and manage devices within a protected network. It acts as a gateway, allowing administrators to connect to the internal network from an external or less secure network. By requiring all remote connections to go through the jump box, organizations can enforce strict access controls, monitor activity, and reduce the attack surface. This added layer of security helps protect sensitive network resources and ensures that only authorized users can perform administrative tasks within the network, enhancing overall network security.

Authentication and Authorization Considerations

There is a strong emphasis in IT on *authentication* and *authorization* for remote access as this area continues to explode in popularity and usage. For example, multifactor authentication (MFA) is currently surging in popularity due to its ability to help secure networks and their data. You will learn more about MFA in Chapter 18.

Technology has also progressed in the specific area of authorization. New technologies such as data loss prevention (DLP) enable mechanisms for the protection of data throughout its lifetime and use in all locations related to the enterprise. This type of technology is especially useful in heavily mobile environments. In fact, DLP functions are often a key aspect of mobile device management (MDM) solutions.

DLP systems are designed to detect and prevent unauthorized use and transmission of confidential information based on one of the three states of data:

- In use
- In motion/transit
- At rest

A well-designed DLP strategy allows control over sensitive data, reduces the cost of data breaches, and achieves greater insight into organizational data use.

In-Band vs. Out-of-Band Management

One reason for the explosion in remote access technologies and their adoption is the growing number of solutions emphasizing out-of-band management. ***Out-of-band management*** means that the network management traffic never mingles with the user data traffic. Your network device can have a separate network interface for this purpose. This separate network interface connects to a separate WAN connection, and this separate connection is used only for the network management traffic. Consider the advantages of out-of-band versus ***in-band network management***. With in-band management, security issues are possible because the network management traffic shares the same path. There is also a contention for bandwidth in this scenario.

While out-of-band management may sound like the de facto answer and solution you should implement, keep in mind that not all designs are optimal for all different topologies, technologies, and solutions. There are plenty of in-band management solutions that run securely and optimally. In fact, well-designed network management solutions tend to involve low bandwidth consumption as part of their excellent design.

Real-World Case Study

Acme, Inc. has implemented a new site-to-site VPN between the main HQ location and a large branch office that has just been brought online. Acme, Inc. decided on a site-to-site VPN because of the high volume of sensitive corporate data that needs to be sent frequently between these two locations. Having the VPN always available for these transfers is a major requirement.

Acme, Inc. has also decided to implement out-of-band network management traffic whenever possible. This includes the conversion of several types of management traffic from the current use of in-band management traffic.

An important exception to this new rule has been established. The IT Support Team will still be permitted to use SSH to several key systems using the main data network. Note that this is only permitted due to the strong encryption that is used with SSH.

Summary

Here are the main topics covered in this chapter:

- This chapter discussed VPN technologies in general.
- This chapter discussed client-based and site-to-site based VPNs. Technologies covered included clientless VPNs, as well as split tunnel and full tunnel configurations.
- This chapter covered a wide variety of network access technologies and approaches, including SSH, GUIs, APIs, console ports, and more.
- Finally, this chapter covered both in-band and out-of-band network management.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 17-4 lists these key topics and the page number where each is found.



Table 17-4 Key Topics for Chapter 17

Key Topic Element	Description	Page Number
List	Two primary categories of VPNs	406
Figure 17-3	Transport Mode Versus Tunnel Mode	411
Figure 17-4	IPsec VPN Steps	412
Table 17-2	Examples of VPN Protocols	414
Table 17-3	Network Access Technologies	415

Complete Tables and Lists from Memory

Print a copy of Appendix C, “Memory Tables,” or at least the section for this chapter, and complete as many of the tables as possible from memory. Appendix D, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

application programming interface (API), client-to-site VPN, clientless VPN, console, full tunnel, graphical user interface (GUI), in-band management, jump box, jump host, out-of-band management, Secure Shell (SSH), site-to-site VPN, split tunnel

Additional Resources

Remote Desktop Protocol (RDP) Using an SSL VPN: <https://www.youtube.com/watch?v=NOytvWA0ZQwIPsec>

Virtual Private Network (VPN) – Deep Dive: <https://www.youtube.com/watch?v=7WhEk7Ga-Bw>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. What type of VPN might feature the use of a clientless VPN solution?
 - a. Site-to-site
 - b. Client-to-site
 - c. Client-to-client
 - d. Server-to-server
2. What VPN method permits only certain traffic to flow over the secure VPN connection while other traffic flows directly over the Internet?
 - a. Split tunnel
 - b. Full tunnel
 - c. In-band
 - d. Out-of-band

3. What remote access technology is considered a secure alternative to Telnet for making a secure connection to a remote network device and operating at the CLI?
 - a. SCP
 - b. SFTP
 - c. SSH
 - d. SSL
4. What is an out-of-band physical port that an administrator can use to configure a network device?
 - a. Loopback
 - b. API
 - c. Console
 - d. Monitor
5. A user clicks Accept, views an advertisement, provides an email address, or performs some other required action, and the network grants access to the user. Which of the following is being described here?
 - a. Virtual network computing (VNC)
 - b. Remote desktop connection (RDC)
 - c. SSH
 - d. Captive portal
6. What data state describes data stored on a hard drive or tape drive?
 - a. Data in use
 - b. Data at rest
 - c. Data in an inconsistent state
 - d. Data in transit
7. Which protocol has largely replaced SSL as the VPN protocol of choice for providing cryptography and reliability to upper layers of the OSI model?
 - a. PPTP
 - b. L2TP
 - c. L2F
 - d. TLS

8. Which is a method of managing and monitoring network devices using the same communication path or network infrastructure that carries user data traffic?
 - a. Jump box/host
 - b. In-band management
 - c. Out-of-band management
 - d. SSO

9. What is a secure intermediary server used to access and manage devices within a protected network?
 - a. Jump box
 - b. VPC
 - c. SSH
 - d. API

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This chapter covers the following topics related to Objective 4.1 (Explain the importance of basic network security concepts) of the CompTIA Network+ N10-009 certification exam:

- Logical security
 - Encryption
 - Data in transit
 - Data at rest
 - Certificates
 - Public key infrastructure (PKI)
 - Self-signed
 - Identity and access management (IAM)
 - Authentication
 - Multifactor authentication (MFA)
 - Single sign-on (SSO)
 - Remote Authentication Dial-in User Service (RADIUS)
 - LDAP
 - Security Assertion Markup Language (SAML)
 - Terminal Access Controller Access Control System Plus (TACACS+)
 - Time-based authentication
 - Authorization
 - Least privilege
 - Role-based access control
 - Geofencing
- Physical security
 - Camera
 - Locks
- Deception technologies
 - Honeypot
 - Honeynet
- Common security terminology
 - Risk
 - Vulnerability
 - Exploit
 - Threat
 - Confidentiality, Integrity, and Availability (CIA) triad
- Audits and regulatory compliance
 - Data locality
 - Payment Card Industry Data Security Standards (PCI DSS)
 - General Data Protection Regulation (GDPR)
- Network segmentation enforcement
 - Internet of Things (IoT) and Industrial Internet of Things (IIoT)
 - Supervisory control and data acquisition (SCADA), industrial control system (ICS), operational technology (OT)
 - Guest
 - Bring your own device (BYOD)

Network Security Concepts

Today's networks are increasingly dependent on connectivity with other networks. However, connecting an organization's trusted network to untrusted networks, such as the Internet, introduces security risks. Security risks exist even internally within an organization.

To protect your organization's data from malicious users, you need to understand the types of threats against which you might have to defend. Then you need to know your options for defending the network. A key security concept to understand is that you need multiple layers of security (*defense in depth*) for your network, not just a single solution, such as a firewall. You might, for example, combine user training, security policies, remote access security protocols, firewalls, VPNs, and intrusion prevention systems to provide overlapping layers of network protection.

This chapter begins by introducing the fundamentals of security, including a discussion of both logical and physical security mechanisms. This chapter also presents the most common authentication methods used in networks today, network segmentation methods, and regulatory compliance.

Foundation Topics

Core Security Concepts

This section examines several very important core security concepts that are critical for you to understand in the current cybersecurity landscape—starting with *confidentiality, integrity, and availability (CIA)*. This is often termed the *CIA Triad*.

Key Topic

Confidentiality, Integrity, and Availability (CIA)

For most of today's corporate networks, the demands of e-commerce and customer contact require connectivity between internal corporate networks and the outside world. Today's corporate networks tend to be large and interconnected with other networks, and they typically run both standards-based and proprietary protocols. Of course, another massive trend has been cloud technologies, where once again, networks connecting to other networks is critical for proper functionality of systems.

In addition, the devices and applications connecting to and using corporate networks are continually increasing in complexity. It would be unusual for a corporate network not to need network security.

CIA refers to the three primary goals of network security:

- Confidentiality
- Integrity
- Availability

The following sections explain these goals in more detail.

Confidentiality

Data confidentiality involves keeping data private by physically or logically restricting access to sensitive data or encrypting traffic traversing a network. While much of this chapter deals with *logical security* mechanisms, we will also be examining *physical security* as well.

A network that provides confidentiality would, for example, do the following:

- Use network security mechanisms, such as firewalls and access control lists (ACLs), to prevent unauthorized access to network resources.
- Require appropriate credentials (such as usernames and passwords) to access specific network resources.

- Encrypt traffic such that any traffic captured off the network by an attacker could not be deciphered by the attacker.

Confidentiality can be provided by using *encryption*. Encryption allows a packet to be encoded in such a way that it can be decoded by an intended party. However, a malicious user who intercepted an encrypted packet in transit would not be able to decrypt the packet. The way most modern encryption algorithms prevent decryption by a third party is through the use of a *key*. Because the encryption or decryption algorithm uses a key in its mathematical calculation, a third party who does not possess the key cannot interpret intercepted data that is encrypted.

When you are planning for encryption in your network, it is important to remember the three states of data introduced in Chapter 17, “Network Access and Management Methods.” The states of data are *data in transit*, *data at rest*, and *data in use*. An example of data in transit is when you are downloading information from the cloud. Data at rest is when you are storing the data, and data in use is when you are actually working with the data. While encryption is not typically used with data in use, we include this state of data here for completion’s sake and so we do not forget to consider it when we are planning our network security.

Encryption has two basic forms: *symmetric encryption* and *asymmetric encryption*. As you will learn next, symmetric key algorithms are faster and are often used for bulk encryption of data. Asymmetric key algorithms are slower and are often used for initial security transactions such as authentication.



Symmetric Encryption

Symmetric encryption is faster than asymmetric encryption. The word symmetric in symmetric encryption implies that the same key is used by both the sender and the receiver to encrypt or decrypt a packet. Examples of symmetric encryption algorithms include the following:

- **DES:** Data Encryption Standard (DES) is an older encryption algorithm (developed in the mid-1970s) that uses a 56-bit key. It is considered weak by today’s standards and has been deprecated.
- **3DES:** Triple DES (3DES), developed in the late 1990s, uses three 56-bit DES keys (for a total of 168 bits) and was originally considered a strong encryption algorithm. However, the security of 3DES varies based on the way it is implemented. Specifically, 3DES has three keying options: All three keys may be different (keying option 1), two of the three keys may be the same (keying option 2), or all three keys may be the same (keying option 3) to maintain backward compatibility with DES. Because of the high computational

overhead of 3DES, many companies have moved directly to the much faster AES instead of using 3DES.

- **AES:** Advanced Encryption Standard (AES), released in 2001, is the preferred symmetric encryption algorithm. AES, which is a variant of the Rijndael family of symmetric ciphers, is available in 128-bit key, 192-bit key, and 256-bit key versions. This technology is being used more and more for many different applications, including wireless, mobile, VPNs, and web security implementations. It is popular because of its overall strength and its ability to provide various levels of security based on key length. For lower-security environments, you can use 128-bit key AES; a 256-bit key implementation is considered almost unbreakable.

Figure 18-1 provides an example of symmetric encryption, where both parties have a shared key to be used during a session (called a *session key*).

NOTE Another widely deployed encryption algorithm is Pretty Good Privacy (PGP), which is often used to encrypt email traffic. PGP uses both symmetric and asymmetric algorithms. A free variant of PGP is GNU Privacy Guard (GPG).

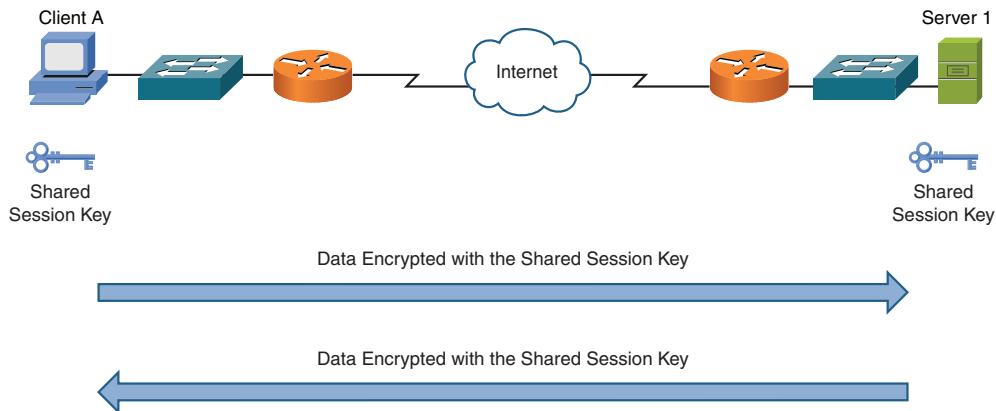


Figure 18-1 Symmetric Encryption Example

Key Topic

Asymmetric Encryption

Asymmetric encryption is slow compared to symmetric encryption but provides better security. As its name suggests, asymmetric encryption uses asymmetric (different) keys for the sender and the receiver of a packet. Because of its speed, asymmetric encryption is not typically used to encrypt large quantities of real-time data. Rather,

asymmetric encryption might be used to encrypt a small chunk of data used, for example, to authenticate the other party in a conversation or to exchange a shared key to be used during a session (after which the parties in the conversation could start using symmetric encryption). One of the most popular asymmetric encryption algorithms in use today is RSA; its name comes from the last initials of its inventors: Ronald L. Rivest, Adi Shamir, and Leonard M. Adleman.

RSA is commonly used as part of a **public key infrastructure (PKI)** system. Specifically, PKI uses digital **certificates** and a certificate authority (CA) for authentication and encryption services.

For example, in the example shown in Figure 18-2, when client A wants to communicate securely with server 1, the following steps occur:

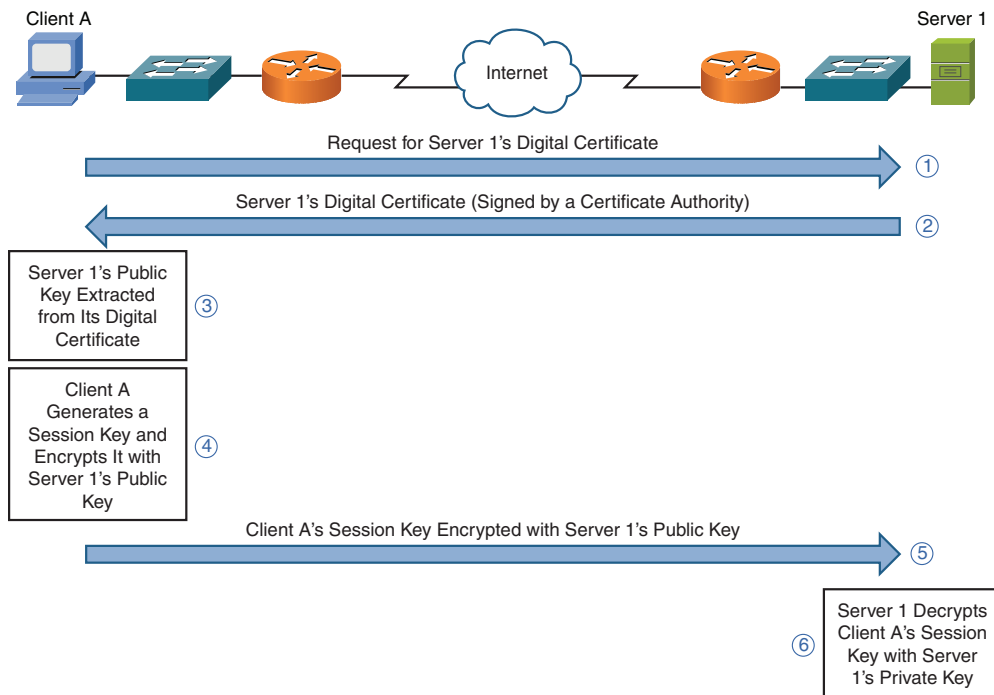


Figure 18-2 Asymmetric Encryption Example

- Step 1.** Client A requests server 1's digital certificate.
- Step 2.** Server 1 sends its digital certificate, and client A knows the received certificate is really from server 1 because the certificate has been authenticated (*signed*) by a trusted third party, called a *certificate authority*.

- Step 3.** Client A extracts server 1's public key from server 1's digital certificate. Data encrypted using server 1's public key can only be decrypted with server 1's private key, which only server 1 has.
- Step 4.** Client A generates a random string of data called a *session key*.
- Step 5.** The session key is encrypted using server 1's public key and sent to server 1.
- Step 6.** Server 1 decrypts the session key by using its private key.

At this point, both client A and server 1 know the session key, which can be used to symmetrically encrypt traffic during the session.

NOTE While PKI is amazing and relied upon for almost all Internet security today, there is also the option to use digital certificates that are *self-signed*. This permits the technology to protect internal corporate communications without involving the trusted third party.

Key Topic

Integrity

Ensuring data *integrity* involves making sure data has not been modified in transit. Also, a data integrity solution might perform origin authentication to verify that traffic is originating from the source that should send the traffic.

Examples of integrity violations include the following:

- Modifying the appearance of a corporate website
- Intercepting and altering an e-commerce transaction
- Modifying financial records that are stored electronically

Hashing is one approach to providing integrity to data transmissions crossing a network. Specifically, hashing takes a string of data (such as a password) and runs it through an algorithm. The result of the algorithm is called a *hash* or a *hash digest*. If the sender of that data runs a hashing algorithm on the data and sends the hash digest along with the data, when the recipient receives the data, she can also run the data through the same hashing algorithm. If the recipient calculates the same hash digest, she might conclude that the data has not been modified in transit (that is, she has confirmed the integrity of the data). Note that a hashing algorithm produces hash digests of the same length, regardless of the size of the data being hashed.

Two hashing algorithms are commonly used:

- **Message Digest 5 (MD5):** Creates 128-bit hash digests

- **Secure Hash Algorithm 1 (SHA-1):** Creates 160-bit hash digests
- **Secure Hash Algorithm 2 (SHA-2):** Creates 224-, 256-, 384-, and 512-bit hash digests

Hashing by itself, however, does not guarantee data integrity because an attacker could intercept a string of data, manipulate it, and recalculate the hash value based on the manipulated data. The victim would then determine that the hash is valid based on the data. To overcome this limitation of pure hashing, Hashed Message Authentication Code (HMAC) uses an additional secret key in the calculation of a hash value. Therefore, an attacker would not be able to create a valid hash value because he would not know the secret key. Other variants of hashing algorithms, such as SHA-256, involve longer digests. In cryptography, a bigger digest implies better security.

NOTE Challenge-Response Authentication Mechanism Message Digest 5 (CRAM-MD5) was a common variant of HMAC frequently used in email systems. Today, even this enhancement has been deprecated in favor of more secure technologies such as SSL and TLS.

Availability

The availability of data is a measure of the data's accessibility. For example, as we covered back in Chapter 15, "Disaster Recovery," if a server were down only 5 minutes per year, the server would have an availability of 99.999% (that is, *five nines availability*).

Here are a couple of examples of how an attacker could attempt to compromise the availability of a network:

- Send improperly formatted data to a networked device, resulting in an unhandled exception error.
- Flood a network system with an excessive amount of traffic or requests, which would consume a system's processing resources and prevent the system from responding to many legitimate requests. This type of attack is commonly referred to as a denial-of-service (DoS) attack.

Threats, Vulnerabilities, Risks, and Exploits

To effectively guard a network against attacks, you should be able to identify a number of security concepts within your own environment, as described in this section.

Threats

You should work with your team to carefully identify the realistic security *threats* that your enterprise could face. This might mean dealing with hypotheticals, but it is nonetheless a very important exercise. While the focus is often on external threats to an organization, you cannot forget that internal threats exist. In fact, a network often faces more internal threats than external ones.

Vulnerabilities

One of the reasons there are so many potential threats against organizations today is the fact that *vulnerabilities* are constantly discovered in networks and network components. New code installed on a network device (such as a router) might cause a vulnerability. For example, this code could introduce a new feature that creates a backdoor into the device for unauthorized persons.

Because vulnerabilities are a major issue for IT staff, a number of tools have been developed to deal with them. The *Common Vulnerabilities and Exposures (CVE)* system is free to use and can be incredibly helpful. This online resource provides excellent search tools and can leverage a large database of publicly known information security vulnerabilities and exposures. CVE, which was officially launched in September 1999, is sponsored by US-CERT within the U.S. Department of Homeland Security (DHS). The MITRE Corporation maintains a CVE dictionary at <https://cve.mitre.org>.

NOTE The two catalogs of known vulnerabilities you should be familiar with are CVE and CVSS. CVE is a list of publicly known vulnerabilities containing an ID number, description, and reference for each vulnerability. The Common Vulnerability Scoring System (CVSS) provides a score from 0 to 10 that indicates the severity of a vulnerability, with a score of 10 being the most severe. The CVSS is maintained by the Forum of Incident Response and Security Teams (FIRST), at <https://www.first.org/cvss/>. Also of great value is the Open Web Application Security Project (OWASP), which is a foundation that works to improve software security. Each year, OWASP publishes the OWASP Top 10 for web application security exploits. For more information, visit <https://owasp.org>.

NOTE *Zero-day attacks* are cybersecurity attacks that use techniques or vulnerabilities that have not already been discovered by cybersecurity specialists or anti-malware companies. While it might sound hopeless, remember that today's sophisticated security appliances can typically prevent or mitigate damage from zero-day attacks by using machine learning (ML) to stop activities in a network that are outside the normal baseline activities.

Risks

Risks refer to the potential threats and vulnerabilities that could compromise the confidentiality, integrity, and availability of your network resources, data, and services. These risks can arise from various sources, including cyberattacks like malware, hacking, and phishing, as well as insider threats, physical breaches, and system malfunctions. Understanding and managing these risks is crucial for safeguarding sensitive information, maintaining operational continuity, and protecting an organization from financial loss, legal repercussions, and reputational damage.

Exploits

As mentioned a bit earlier, when we discuss threats, we are typically talking about hypotheticals—events that might possibly occur in a network. When we discuss **exploits**, however, we are talking about facts. An exploit consists of a (hopefully detailed) description of what exactly occurred with a security breach. An exploit often describes the most likely attacker, the technology used, and any vulnerabilities or misconfigurations that made the attack possible.



Least Privilege

An important security best practice that you should implement everywhere possible in your network is **least privilege**. You and every other person who interacts with the network should always be using a user account that has the least number of privileges required to do a job. It is important to carefully audit the required permissions to complete tasks and then create accounts (or groups) that have just those permissions. Then, if a computer criminal manages to begin operating on the network using your account, the criminal will only be able to accomplish minimal tasks.

NOTE What privileges you can carry out on the network using your user account (and often group membership) are referred to as your **authorization**.

Implementing least privilege might sound easy, but human nature often opposes this commonsense measure. For example, an administrator might prefer to use an administrator or root account for all administrative tasks because switching between accounts can be tedious.

It can also be very easy to violate this best practice without even knowing it. This often happens in public cloud environments. An administrator may start a free-tier Amazon Web Services (AWS) account using his email address. He may have not studied much about AWS and might not realize that there is a service called AWS Identity and Access Management (IAM) designed for the creation of accounts and

permissions in AWS. The administrator may continue to use that administrative account email address for day-to-day operations in AWS, even though AWS makes it clear that using such a high-level account is not safe. The only time you should use that high-level admin account is when you are performing some specific change that requires it, such as changing the credit card on the account or other billing configurations.

Role-Based Access Control

Most network device designers strive to implement *role-based access control (RBAC)* for administrators who must manage devices. In addition, more and more network access models for users are emphasizing RBAC for network access.

The idea with RBAC is to create permissions around roles that you can then assign to users. For example, you might use RBAC for guest access; to do so, you would create a guest role and assign that role to visitors to your organization when you want to give them limited privileges. Perhaps you just want such visitors to be able to access your public Internet connection for limited Internet access. You might also use RBAC to create network administrator accounts for users who need to have the ability to access the network and also reconfigure it.

NOTE A scalable method for assigning roles (permissions) to user accounts is to include those user accounts in roles (also called groups) that have the appropriate permissions assigned. The users inherit the permissions thanks to their group membership. On some network devices, RBAC is more limited, and you can assign permissions only to the user accounts themselves.

Defense in Depth

Cisco Systems is one of the networking giants that helped to popularize the great idea of *defense in depth* for network security. Defense in depth involves striving to secure each layer of a solution. For example, you might use strong security on containers and virtual machines that provide software as a service (SaaS) for your organization. You can then use strong security controls for the host systems that provide these virtual servers. You can also have strong security controls in place on the network switch that connects to these hosts. This process continues, building layer after layer of security into the network and the systems that connect to it.

As with the zero trust security approach introduced in Chapter 8, “Evolving Use Cases,” there is no single technique that makes defense in depth what it is. This section describes just some of the security approaches you might employ when you are using a defense-in-depth approach.

Screened Subnet

A *screened subnet*—previously called a *demilitarized zone (DMZ)*—often contains servers that should be accessible from the public Internet. With a screened subnet, for example, you could allow users on the Internet to initiate an email or web session coming into your organization's email or web server; however, you could block other protocols for those users.

Separation of Duties

A popular security best practice is to implement a *separation of duties* policy in the network. This policy ensures that user accounts in the network do not have too much power. You can also ensure that your IT staff are mapped to these various duties and even rotated through them. This helps to ensure that internal attacks against your network do not occur. It is important to include separation of duties when planning for security policy compliance. Without this separation, all areas of control and compliance could end up in the hands of a single individual.

Network Access Control

Network access control (NAC) involves using a set of protocols to ensure that users are identified immediately when they try to access the network. A NAC system might integrate automatic remediation processes to fix the issues with host systems so that they are brought in compliance with corporate security policies.

A common approach to building a NAC system is to incorporate 802.1X in the network. 802.1X is covered later in this chapter.

Honeypot

A *honeypot* acts as a deception technology. Specifically, a system designated as a honeypot appears to be an attractive attack target. One school of thought on the use of honeypots is to place one or more honeypot systems in a network to entice attackers into thinking a system is real. The attackers then use their resources to attack the honeypot, and in doing so, they leave the real servers alone.

A honeypot can also be used to see what attackers are attempting to do on the system. A honeypot could, for example, be a UNIX- or Linux-based system configured with a weak password. After an attacker logs in, surveillance software could log what the attacker does on the system. This knowledge could then be used to protect real servers in the network.

NOTE For larger networks, a network administrator might deploy multiple honeypots, forming a *honeynet*.

Network Segmentation Enforcement

Logical segmentation of networks helps to enforce strong security designs. For example, VLANs can be used to constrain broadcasts and permit the definition of VLAN access control lists (ACLs) and router ACLs for controlling communication between VLANs.

Network segmentation enforcement can also help to minimize the effects of security attacks by limiting them to the local subnet. At Layer 3, you can segment by using virtual routing and forwarding (VRF) tables for network devices. These tables, which are like virtual routers running within the main router, can be separate and distinct from the main routing table. The main routing table is termed the *global routing table*.

Of course, network segmentation enforcement can also be based around function. For example, a guest network should most certainly be segmented from non-guest network functions and resources. What follows are some important examples of segmentation based on function.

IoT and IIoT

The *Internet of Things (IoT)* refers to the network of everyday objects that are connected to the Internet, allowing them to send and receive data. These objects can include anything from household appliances like refrigerators and thermostats to wearable devices like fitness trackers and smartwatches. By connecting to the Internet, these devices can be controlled remotely, share information with each other, and provide useful data to users, making daily tasks more convenient and efficient. For example, a smart thermostat can learn your temperature preferences and adjust itself automatically, saving energy and improving comfort. Robust security mechanisms should be in place for the IoT infrastructure and it should be properly segmented.

The *Industrial Internet of Things (IIoT)* is a network of industrial devices connected to the Internet, allowing them to collect, share, and analyze data. These devices can include machinery, sensors, and equipment used in manufacturing, energy, and other industrial sectors. By being connected, these devices can improve efficiency, reduce downtime, and enhance safety. For example, sensors on a factory machine can detect potential issues before they cause a breakdown, allowing for timely maintenance and avoiding costly interruptions. Overall, IIoT helps industries run more smoothly and effectively by using real-time data and smart technologies.

Like any other network segment, IIoT should be secured as best as possible, and should be properly segmented as much as possible.

SCADA, ICS, and OT

Supervisory control and data acquisition (SCADA) is a system used to monitor and control industrial processes. It involves computers and software that gather real-time data from equipment like sensors and machines in various industries, such as manufacturing, energy, and water treatment. Operators use SCADA systems to see what's happening in their industrial processes, make adjustments, and ensure everything runs smoothly and safely. For instance, in a water treatment plant, SCADA can monitor water levels and chemical usage, allowing operators to maintain the right balance and ensure clean water.

An **industrial control system (ICS)** is a group of technologies used to control and automate industrial processes. These systems include devices like sensors, controllers, and software that work together to manage operations in industries such as manufacturing, power plants, and chemical processing. ICS helps ensure that machinery and processes run smoothly, efficiently, and safely. For example, in a power plant, ICS can regulate the flow of electricity, monitor equipment performance, and alert operators to any issues. By automating many tasks and providing real-time data, ICS improves productivity and reduces the risk of errors and accidents.

Operational technology (OT) systems refer to the hardware and software used to detect or cause changes through direct monitoring and control of physical devices, processes, and events within an organization. These systems are critical in industries like manufacturing, energy, and transportation, where they help manage machinery, infrastructure, and facility operations. OT systems include things like programmable logic controllers (PLCs), distributed control systems (DCS), and human-machine interfaces (HMIs).

Guest Networks

Some networks offer **guest networks** for, you guessed it, guests. Some of these guest networks even feature open authentication methods that make access simple for users. Properly segmenting guest networks from other, more critical segments is important.

BYOD

Another optional aspect to the modern network is the concept that some organizations permit **bring your own device (BYOD)**. BYOD is a policy that allows

employees to use their personal devices, such as smartphones, tablets, and laptops, for work purposes. Instead of using company-provided equipment, employees can access company networks, applications, and data on their own devices. This approach can increase convenience and flexibility for employees, as they can work from anywhere and use devices they are already comfortable with. However, it also requires a company to implement strong security measures to protect sensitive information and ensure that personal devices do not pose a risk to the organization's network.

NOTE Some organizations opt for choose your own device (CYOD) instead of BYOD. In the CYOD policy, the employees needing mobile access to corporate resources get to choose their mobile device from a list of approved (and often preconfigured) mobile devices. This gives the organization more control and typically stronger security over the mobile devices. Often, organizations use mobile device management (MDM) software to ensure strict policies governing and managing the approved devices.

Authentication Methods

Authentication is the process of verifying the identity of a user, system, or entity to ensure that it is who or what it claims to be. Because authentication is such an important part of network security, it is little surprise that, today, there are many different approaches you can use. This section describes several authentication methods that all might be found as part of an organization's *identity and access management (IAM)* strategy.

Multifactor

Authentication that requires multiple factors is called *multifactor authentication (MFA)*. The following are some options for components of multifactor authentication:

- Something you know (such as a passcode or pin)
- Something you have (such as a smart card or badge)
- Something you are (such as a fingerprint or retinal scan)
- Something you do (such as a swipe pattern or puzzle completion)
- Somewhere you are (such as a geolocation or IP address)

To ensure that you understand the simplicity and beauty of multifactor authentication, consider this simple example: John goes to the store and puts his bank card into an ATM and enters his PIN. What examples of multifactor authentication has he exhibited?

This automated teller machine (ATM) provides a common example of a multifactor authentication system, requiring both of the following:

- A “something you have” physical key (your ATM card)
- A “something you know” personal identification number (PIN)

Today, smartphones are often used in multifactor authentication. A smartphone is something you possess, and a network can take advantage of this by sending a one-time password (OTP) to your smartphone to make sure you are who you claim to be.

TACACS+

Terminal Access Controller Access Control System Plus (TACACS+) is a Cisco-proprietary TCP-based protocol. TACACS+ has three separate and distinct sessions or functions for authentication, authorization, and accounting (AAA). It is similar to Remote Authentication Dial-In User Service (RADIUS) but uses TCP instead of UDP as a transport method; it uses port 49 as the default port. TACACS+ takes a client/server model approach.

Single Sign-On

Single sign-on (SSO) allows a user to authenticate only once to gain access to multiple systems or applications, without requiring the user to independently authenticate with each system or application.

RADIUS

Remote Authentication Dial-In User Service (RADIUS) is a UDP-based protocol used to communicate with a AAA server. Unlike TACACS+, RADIUS does not encrypt an entire authentication packet; it encrypts only the password. However, RADIUS does offer more robust accounting features than TACACS+. Also, RADIUS is a standards-based protocol, whereas TACACS+ is a Cisco-proprietary protocol. RADIUS uses UDP port 1812 for authentication and authorization and UDP port 1813 for accounting.

LDAP

Lightweight Directory Access Protocol (LDAP) permits a set of standards for the storage and access of user account information. Many proprietary user stores support LDAP for ease of access, including Microsoft's Active Directory. By default, LDAP traffic is unsecured (over port 389). LDAP over TLS/SSL (LDAPS) is a secure form of LDAP with secured communications using port 636.

Kerberos

Kerberos is a client/server authentication protocol that supports mutual authentication between a client and a server. With Kerberos, a trusted third party (a key distribution center [KDC]) hands out tickets that are used instead of username and password combinations.

Here is a look at this process:

- Step 1.** The client contacts a CA.
- Step 2.** The CA creates a time-stamped session key with a limited duration (by default, eight hours) by using the client's key and a randomly generated key that includes the identification of the target service.
- Step 3.** The CA sends this information back to the client in the form of a ticket-granting ticket (TGT).
- Step 4.** The client submits the TGT to a ticket-granting server (TGS).
- Step 5.** The TGS generates a time-stamped key encrypted with the service's key and returns both keys to the client.
- Step 6.** The client uses its key to decrypt its ticket, contacts the server, and offers the encrypted ticket to the TGS.
- Step 7.** The TGS uses its key to decrypt the ticket and verify that the time stamps match and the ticket remains valid.
- Step 8.** The service contacts the KDC and receives a time-stamped session-keyed ticket that it returns to the client.
- Step 9.** The client decrypts the keyed ticket by using its key. When the client and the server agree that they are the proper accounts and that the keys are within their valid lifetimes, communication is initiated.

SAML

Security Assertion Markup Language (SAML) is a standard for securely sharing information about user identities and authentication between different systems. It allows a user to log in once and access multiple, separate applications without needing to log in again for each one. SAML works by passing information (called assertions) about the user from an identity provider (like a company's login system) to a service provider (like a web application). This makes it easier and more secure for users to access different services with a single set of login credentials.

Time-based Authentication

Time-based authentication is a security method that uses a temporary, time-sensitive code for verifying a user's identity. This code is typically generated by an app or sent via SMS and changes every 30 to 60 seconds. When a user logs in, they must enter this code along with their regular password. Because the code is constantly changing, it adds an extra layer of security, making it much harder for unauthorized users to gain access even if they know the password. This method is commonly used in two-factor authentication (2FA) to enhance online account security.

Local Authentication

Local authentication refers to a network device authenticating the user with a database of user account information stored on the device itself. This is often an important fallback authentication method used when another external method fails.

Risk Management and SIEM

It is time to tackle a new subject, risk management, and revisit one that we first saw in Chapter 14, "Network Monitoring"—security information and event management (SIEM). Every organization should consider risk management, and many would argue that every organization should also consider SIEM. Both of these areas can help an organization save money, save time, and, ultimately, be more secure.

Risk Management

It is important for an organization to take a formal approach to *risk management* in order to secure the organization and also help ensure that the organization is spending its time and resources on the best possible security solutions for the situation.

Risk management involves carefully identifying, evaluating, and prioritizing risks. It typically also involves the coordinated application of resources to minimize, monitor, and control the probability or impact of cybersecurity-related events.

Security Risk Assessments

Cybersecurity concerns continue to increase in number and importance. Fortunately, there are many assessments you can conduct to help improve the overall security of your network infrastructure. This section details some of the ones you should be familiar with.

Threat Assessment

Often, as part of risk management, you need to engage in *threat assessment*. This process involves evaluating the credibility and seriousness of potential threats and analyzing the probability that particular threats will become a reality.

Vulnerability Assessment

Another common part of risk management is *vulnerability assessment*. When you perform a vulnerability assessment, you identify, quantify, and prioritize the vulnerabilities in a system.

NOTE Vulnerability assessments can be performed for a wide variety of systems within an IT department and in a network as a whole.

Penetration Testing

Penetration testing can be an important part of the risk management processes for an organization. Penetration testing involves trying to find vulnerabilities and problems with the security settings of a network or network component. Penetration testing can help you find major problems with systems and mitigate the risk of cyberattack.

Posture Assessment

Another common part of risk management is *posture assessment*, which involves examining the state of a network and the network nodes from a security perspective. You can use the results of posture assessment to implement new security rules through a AAA system. With posture assessment, you can design requirements related to patch levels, software updates, or things like firewall rules. If an incoming system does not meet those requirements, you can deny it access to the network.

Business Risk Assessment

Thorough risk management processes might include business risk assessment. With this type of assessment, the focus may be on the business processes and vendors the

organization has partnered with. Process and vendor assessment are just two of the many types of business risk assessment an organization might perform.

Process Assessment

A thorough *process assessment* can help determine the security-consciousness of the various business processes that an organization engages in. Process assessments often concentrate on many other areas beyond security. For example, an organization may find many inefficiencies in the business processes in use daily.

One of the many reasons that business process assessments are necessary is that companies can get complacent and fail to update their process documentation even though many small changes have been implemented over time.

Vendor Assessment

A *vendor assessment* is a type of risk assessment in which a team analyzes the many vendors an organization uses in its business processes. This type of assessment considers how much organizational data vendors have access to, and the security practices of the vendors. There are a number of issues you should consider in assessing the vendors your business relies on.



Security Information and Event Management (SIEM)

Monitoring is a critical part of network security. For example, *security information and event management (SIEM)* software products and services combine security information management (SIM) and security event management (SEM).

SIEM systems provides real-time analysis of security alerts generated by applications and network hardware. These systems can log security data and generate reports for compliance purposes.

A SIEM system can take many forms; it can be implemented as software, as appliances, or as managed services.

Here are just some of the focuses of various SIEM products:

- **Log management:** This aspect of the SIEM helps administrators back up and archive the many log files in the network, along with many other management tasks appropriate for this data.
- **Security information management (SIM):** SIM focuses on long-term storage as well as analysis and reporting of log data.
- **Security event management (SEM):** SEM involves real-time monitoring, correlation of events, notifications, and console views.

- **Managed security service (MSS) or managed security service provider (MSSP):** The most common managed services are related to connectivity and bandwidth, network monitoring, security, virtualization, and disaster recovery.
- **Security as a service (SECaaS):** These security services often include authentication, antivirus, anti-malware/spyware, intrusion detection, penetration testing, and security event management, among others.

Physical Security

Have you ever had (the rather depressing) thought that all the hard work you have done in the operating system of a network device to secure it is worthless if an attacker gains physical access to your device? Someone who has physical access to a device might press a reset button in order to reset the device to factory defaults (allowing them to log in and reconfigure the device) or they might just take a less sophisticated approach and smash the device using a sledgehammer! If an attacker has access to your network equipment, you are in trouble. This section discusses physical security within the framework of two main principles, detection and prevention.

Detection Methods

When you think about detecting an attack against your network, you likely envision monitoring packets traveling into and out of your network devices. When you think about physical security, you can think in similar terms. You will certainly need to monitor the flow of humans into and out of the area where the network devices exist.

Today, there are many affordable options in the physical security detection realm of network security. Here are just some of the ones you should be aware of:

Key Topic

- **Motion detection:** Using either mechanical or electronic methods, you can install motion detection devices that alert you when objects change position in a physical network location.
- **Cameras:** Cameras enable you to view key network areas by using video surveillance. Video footage is often recorded and stored locally or in the cloud. Modern camera systems have been combined with more advanced intelligence to detect weapons and other threats.
- **Asset tracking tags:** You can use various wireless technologies to track the physical locations of network objects and personnel by using tag technologies attached to the entities. For example, radio frequency identification (RFID) uses electromagnetic fields and allows one-way communication of information from a chip to an RFID reader.

- **Tamper detection:** You can take steps to ensure that network equipment is not tampered with. For example, you might place key network cable runs in clearly viewable conduits and make appliances readily viewable through glass enclosures. You could also use IP camera systems to send alerts when certain areas containing hardware are infiltrated and tamper panels to detect when the cover of an all-in-one system or computer case is opened or tampered with.

Prevention Methods

There are many options you can use to assist in the area of physical attack prevention.

Many security professionals today will beg you to accept the fact that the most important prevention method you can take is complete and thorough *employee training*. One of the reasons for this common opinion is that many attacks cannot be carried out without user intervention. For example, social engineering requires a user to give sensitive information (such as username and password credentials) to an attacker in order for the attacker to access the user's account. As a result, a number of potential cybersecurity-related attacks can be thwarted through effective user training. For instance, users could be trained on using policies such as the following:

- Never give out your password to anyone, even if that person claims to be from IT.
- Do not open email attachments from unknown sources.
- Select strong passwords, consisting of at least 8 to 12 characters and containing a mixture of alphabetic (upper- and lowercase), numeric, and special characters.
- Do not visit unauthorized websites.
- Report suspicious activity.
- Do not run or install any applications not provided directly by the company.
- Change your passwords monthly.

This list is only an example, and you should develop a collection of best practices for your users based on your network's specific circumstances. Users should also know whom to contact in the event of a suspected data breach or compromise of the computers and systems the users are responsible for. Users should also know never to run penetration testing tools or other network discovery tools that may unintentionally lead to denial of service (DoS) or other harm to the network and its devices. Technical controls such as web/content filtering, port filtering, IP filtering, and access control lists (ACLs) that deny specific traffic can be used to assist in the enforcement of the policies agreed to by the users.

As part of user training and for the safety of human life, emergency procedures should also be communicated and verified with each user, including the

- Building layout
- Fire-escape plan
- Safety and emergency exits
- Doors that automatically fail closed or fail open based on their purpose to contain or allow access
- Emergency alert systems
- Fire suppression systems
- Heating, ventilation, and air conditioning (HVAC) operations
- Emergency shutoff procedures

In a data center, procedures for safety related to electrostatic discharge (ESD), grounding, rack installation, lifting, tool safety, and the correct placement of devices should also be planned, communicated, and verified. If there are dangerous substances in or near the work area, a material safety data sheet (MSDS) should be created to identify procedures for handling and working with those substances.

While employee training is undeniably important, keep in mind that it transcends just physical security concerns. Here are just some of the additional prevention methods you can engage in for physical security:

Key Topic

- **Badge readers:** Identification badges assist with physical security regarding employees. Badge readers are specialized hardware devices that read identification badges, and the data they collect is valuable for physical security. Whereas identification might be limited to name and photo, more sophisticated approaches can include electronic swipes and asset tags.
- **Biometrics:** Although at one time biometrics were reserved for the most sophisticated of environments, now using a thumbprint reader, facial recognition, or a retina scan is a reality with such common devices as cell phones.

NOTE It is common to refer to badge readers and biometrics as belonging to a category of devices called *access control hardware*.

- **Smart cards:** A smart card looks like a credit card but possesses circuitry that allows it to authenticate a user against network systems; this aids security and constitutes multifactor authentication (MFA) because the user must possess something (the smart card) and know something (the password). This MFA approach has worked well for ATMs in the banking system for decades.

- **Key fobs:** A key fob is an electronic attachment to a key ring that might provide access to a network system or might simply lock or unlock a secure door.
- **Locks:** Sometimes the simplest of security mechanisms are the most effective; this is often the case with locks on network areas and equipment. Your enterprise might feature racks for network equipment that have locking cages over them. These are often simply referred to as *locking racks*. You can also use *locking cabinets* for physical security.
- **Smart lockers:** It is always wonderful when one technology can assist another. This is certainly the case with smart lockers. These lockers might look like typical lockers from the outside, but they include network connectivity intelligence and might even be configured for Internet access. Smart lockers can accurately log user access and provide a level of security never before seen with basic lockers.
- **Access control vestibule:** There are at least two forms of social engineering attacks—tailgating and piggybacking—that require an attacker to follow an authorized user through a secured entrance or exit. An access control vestibule (formerly known as a mantrap), which is a small entry area with two interlocking doors, provides assurance that these forms of social engineering cannot take place.
- **Geofencing:** Geofencing is a technology that creates a virtual boundary around a real-world geographic area. When a person or device enters or exits this defined area, an action is triggered, such as sending a notification or starting an app. Businesses use geofencing for various purposes (not just security), like sending promotional messages to customers near their stores or tracking the location of employees in a specific work zone. It relies on GPS, Wi-Fi, or cellular data to determine the location and ensure the actions are taken based on the geographic boundaries.

Audits and Regulatory Compliance

Another important aspect of modern network security is conducting audits and ensuring compliance with regulations applicable to your organization. Here are just several important considerations in this area:

- **Data locality:** Where your data is located on the globe is really important when it comes to regulatory compliance. This is because different countries and regions have varying laws about where and how data can be stored and processed to protect privacy and security.

- ***Payment Card Industry Data Security Standards (PCI DSS)***: PCI DSS is a set of security standards designed to ensure that all companies that accept, process, store, or transmit credit card information maintain a secure environment. It was created to protect cardholder data from theft and fraud. The standards include requirements for managing security, policies, procedures, network architecture, software design, and other critical protective measures. Compliance with PCI DSS helps businesses safeguard sensitive payment information and build customer trust.
- ***General Data Protection Regulation (GDPR)***: GDPR is a set of rules created to protect the privacy and personal data of individuals in the European Union. It gives people more control over their personal information and requires organizations to be transparent about how they collect, store, and use this data. GDPR applies to any business, regardless of location in the world, that processes the data of EU citizens. It includes rights for individuals such as the right to access their data, the right to have it deleted, and the right to be informed about data breaches. The regulation aims to enhance privacy and trust in the digital age.

Real-World Case Study

Acme, Inc. is currently enhancing the security posture of the organization to safeguard against the evolving landscape of cyber threats. As part of this effort, Acme, Inc. is providing comprehensive training to the IT team, focusing on the most important network security concepts. The concepts include the most important principles of logical security and some of the most important physical security concepts.

In addition to bolstering the team's knowledge, the company is also deploying a well-regarded security information and event management (SIEM) system. This SIEM solution will play a crucial role in the day-to-day monitoring, detection, and management of potential security exploits. By centralizing and analyzing logs from various sources, the SIEM will enable the IT team to identify suspicious activities more swiftly and respond to threats in real time, thereby reducing the window of opportunity for malicious actors.

Furthermore, Acme, Inc. is making significant changes to its authentication methods to strengthen access controls across the enterprise network. The company is planning to implement multifactor authentication (MFA) for all users of its enterprise software. This move is intended to add an extra layer of security by requiring users to provide two or more verification factors to gain access, making it significantly harder for unauthorized individuals to breach the system.

Recognizing the importance of audits and regulatory compliance, Acme, Inc. is also providing recurring training on some of the most critical regulations that must be followed. This part of the training is also considering industry best practices in addition to those local and federal regulations that must be followed.

Ultimately, these efforts are part of a broader strategy by Acme, Inc. to create a robust and adaptive cybersecurity framework. By investing in both technology and training, the company is taking a holistic approach to security, ensuring that it is not only well prepared to handle current threats but also agile enough to respond to new challenges as they arise.

Summary

Here are the main topics covered in this chapter:

- This chapter reviewed key security concepts that you should be aware of in networking today. These include important concepts like CIA, symmetric and asymmetric encryption, and threats, vulnerabilities, and exploits.
- This chapter examined the various authentication methods that are popular today. These include technologies like MFA (multifactor authentication).
- This chapter also examined risk management and SIEMs. These can be important aspects of the security operations for the organization.
- Finally, this chapter also discussed physical security techniques as well as audits and regulatory compliance.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 18-1 lists these key topics and the page number where each is found.



Table 18-1 Key Topics for Chapter 18

Key Topic Element	Description	Page Number
Section	Confidentiality, Integrity, and Availability (CIA)	426
Section	Symmetric Encryption	427
Section	Asymmetric Encryption	428

Key Topic Element	Description	Page Number
Section	Integrity	430
Section	Least Privilege	433
Section	Security Information and Event Management (SIEM)	443
List	Detection methods	444
List	Prevention methods	446

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

authentication; authorization; bring your own device (BYOD); camera; certificates; confidentiality, integrity, and availability (CIA); data at rest; data in transit; data in use; data locality; encryption; exploit; General Data Protection Regulation (GDPR); geofencing; guest network; honeynet; honeypot; identity and access management (IAM); industrial control system (ICS); Industrial Internet of Things (IIoT); Internet of Things (IoT); least privilege; Lightweight Directory Access Protocol (LDAP); locks; logical security; multifactor authentication (MFA); network segmentation enforcement; operational technology (OT); Payment Card Industry Data Security Standards (PCI DSS); physical security; public key infrastructure (PKI); Remote Authentication Dial-In User Service (RADIUS); risk; role-based access control (RBAC); Security Assertion Markup Language (SAML); self-signed; single sign-on (SSO); supervisory control and data acquisition (SCADA); Terminal Access Controller Access Control System Plus (TACACS+); threat; time-based authentication; vulnerability

Additional Resources

8 Most Common Cybersecurity Threats: <https://www.youtube.com/watch?v=Dk-ZqQ-bfy4>

10 Hacking Tactics You Should Know Of: <https://www.youtube.com/watch?v=kOGSPHI-Ok4>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. Which of the following is a symmetric encryption algorithm available in 128-bit, 192-bit, and 256-bit key versions?
 - a. RSA
 - b. 3DES
 - c. AES
 - d. TKIP
2. What aspect of modern cybersecurity focuses on ensuring that data has not been manipulated in transit?
 - a. Integrity
 - b. Confidentiality
 - c. Authentication
 - d. Availability
3. What security approach involves creating multiple accounts for your own access to the network and to its devices?
 - a. 802.1X
 - b. Least privilege
 - c. Network access control
 - d. SIEM
4. Which of the following is a device that is meant to attract security attacks?
 - a. SIEM
 - b. Geofencing
 - c. IPS
 - d. Honeypot
5. What network policy might permit company employees to use their own phones and tablets in the corporate data setting?
 - a. Honeynet
 - b. BYOD

- c. IIoT
 - d. OT
6. Which of the following provides excellent search tools to leverage a large database of publicly known information security vulnerabilities and exposures?
 - a. ACL
 - b. AWS
 - c. SHA-256
 - d. CVE
 7. Which of the following provides real-time analysis of security alerts generated by applications and network hardware and can log security data and generate reports for compliance purposes?
 - a. SIEM system
 - b. Screened subnet
 - c. VRF instance
 - d. Defense in depth
 8. Which of the following is a client/server authentication protocol that supports mutual authentication between a client and a server and hands out tickets that are used instead of a username and password combination?
 - a. TACACS+
 - b. RADIUS
 - c. Kerberos
 - d. LDAP
 9. You go to the store and put your bank card into an ATM and enter your PIN. Which of the following factors of multifactor authentication have you exhibited? (Choose two.)
 - a. Something you are
 - b. Something you have
 - c. Something you know
 - d. Somewhere you are

10. Which of the following would best help you ensure that all areas of control and compliance don't end up in the hands of a single individual?
 - a. Role-based access control
 - b. Zero trust
 - c. Posture assessment
 - d. Separation of duties
11. Which of the following is not a common physical security prevention method?
 - a. Motion detection
 - b. Employee training
 - c. Locking racks
 - d. Access control vestibule
12. Which of the following is not a common physical security detection method or tool?
 - a. Camera
 - b. Asset tag
 - c. Tamper detection
 - d. Biometrics
13. What does a public key infrastructure (PKI) system use for authentication and encryption services? (Choose two.)
 - a. Digital certificates
 - b. Certificate authority
 - c. SHA-1
 - d. SHA-2
14. Which is a computer threat that tries to exploit computer application vulnerabilities that are unknown to others—even the software developer?
 - a. CVSS
 - b. NAC
 - c. Zero-day
 - d. Least privilege

15. Which is a small entry area with two interlocking doors that provides assurance that tailgating and piggybacking forms of social engineering cannot take place?
 - a. Geofencing
 - b. Access control vestibule
 - c. ESD
 - d. MSDS

16. Which is a set of rules created to protect the privacy and personal data of individuals in the European Union (EU)?
 - a. PCI DSS
 - b. ICS
 - c. OT
 - d. GDPR

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This chapter covers the following topics related to Objective 4.2 (Summarize various types of attacks and their impact to the network) of the CompTIA Network+ N10-009 certification exam:

- Denial-of-service (DoS)/distributed denial-of-service (DDoS)
- VLAN hopping
- Media Access Control (MAC) flooding
- Address Resolution Protocol (ARP) poisoning
- ARP spoofing
- DNS poisoning
- DNS spoofing
- Rogue devices and services
 - DHCP
 - AP
- Evil twin
- On-path attack
- Social engineering
 - Phishing
 - Dumpster diving
 - Shoulder surfing
 - Tailgating
- Malware

Types of Network Attacks

Have you ever heard the expression “Know your enemy”? It is a very fitting expression for this chapter, which focuses on the attacks that are very common in today’s networked environment. And while you might think that all network attacks focus deeply on technology, remember that many attacks focus on the human element. As you will learn later in this chapter, we call attacks leveraging the human social engineering attacks.

It is important to note that many of the successful security attacks carried out against a wide variety of organizations are successful because they involve using multiple attack types in strategic ways. Very successful cybercriminals realize that they can use specific attacks to achieve specific goals as part of an overall strategy.

Foundation Topics

Technology-Based Attacks

When you think about cybersecurity attacks, you are likely to think about attacks that make use of technology. After all, you are in the field of technology. But these are not the only attacks that affect networks. Social engineering attacks are also a major problem today and are often the first step in more sophisticated attacks that take place. Before we launch into a look at these social engineering attacks, let's take a detailed look at a variety of common technology-based attacks.

Key Topic

Denial-of-Service (DoS)

An attacker can launch a *denial-of-service (DoS)* attack on a system by sending the target system a flood of data or requests that consumes the target system's resources. In addition, some operating systems and applications crash when they receive specific strings of improperly formatted data, and the attacker can leverage such OS/application vulnerabilities to render a system or an application inoperable. The attacker often uses IP spoofing to conceal their identity when launching a DoS attack, as illustrated in Figure 19-1.

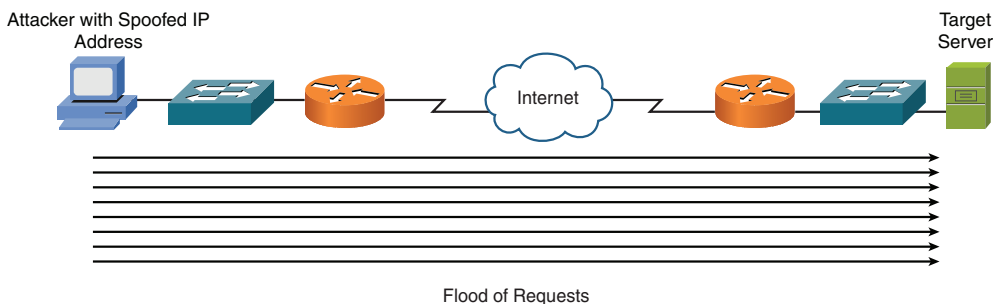


Figure 19-1 DoS Attack

DoS attacks are often categorized as follows:

- **Reflective:** A third-party system is used to help carry out this type of attack. Often this third party is not compromised, which makes a reflective attack very difficult to track down.
- **Amplified:** A DNS server is often used in an amplification attack, but other services could be used in such an exploit as well. With these attacks, legitimate servers are tricked into flooding responses at a target system; the forged

request tends to be small but results in large responses hitting the target. It can be difficult to mitigate amplified attacks because the server involved (called the *reflector server*) is a legitimate device.

**Key
Topic****Distributed Denial-of-Service (DDoS)**

A *distributed denial-of-service (DDoS)* attack builds upon the DoS attack and makes it one of the most feared of all the attacks that exist. These attacks can dramatically increase the amount of traffic flooded to a target system. A DDoS attack is a coordinated attack. Specifically, an attacker compromises multiple systems and then instructs those compromised systems, called *zombies or bots*, to simultaneously launch a DDoS attack against a target system. The attacker uses *command and control* software to instruct the zombies to do their job. These zombies are even scarier than the ones in *The Walking Dead* because they are remotely controlled.

NOTE A *botnet* is an entire network segment that is filled with bots.

A significant traffic spike (as compared to the baseline) could provide an early indication that a DDoS attack is taking place. An intrusion prevention system (IPS) is designed to recognize and alert when attacks or malicious traffic is present on a network.

NOTE Remember that a DoS attack originates from a single system, whereas a DDoS attack originates from multiple systems simultaneously and the attacker distributes zombie software or infects multiple (even into the thousands) hosts, providing the attacker partial or full control of the infected computer system through one or more command and control servers. Finally, the army of bots or compromised machines attacks the victim by overwhelming it, making it slow to respond or unable to respond to legitimate requests.

On-Path Attack

An attacker who can get in the direct path between a client and a server can then eavesdrop on the conversation between the client and the server; this type of attack is called an *on-path attack*. This attack used to go by the name of a man-in-the-middle attack.

If cryptography is being used and the attacker fools the client and server both into building VPNs to the attacker instead of to each other, the attacker can see all the data in plaintext. On a local Ethernet network, methods such as Address Resolution

Protocol (ARP) spoofing, ARP cache poisoning, Dynamic Host Configuration Protocol (DHCP) spoofing, and Domain Name System (DNS) spoofing and poisoning may be used to redirect a client's traffic through the attacker instead of directly to the server.

NOTE The newer terminology *on-path attack* is much more accurate than the older term *man-in-the-middle attack*, which does not properly convey that the attacker can carry out an attack from any point on the path (not just the middle).

DNS Poisoning

Another potential for attacks against cybersecurity involves the DNS service and systems, and many advancements have therefore been made in DNS security mechanisms. One simple approach to DNS attacks is the injection of false entries into the DNS system, known as **DNS poisoning**. This might be done in order to deny service to systems or to redirect them to websites that might be designed to disseminate malware or collect credentials. DNS poisoning is also known as **DNS spoofing**, hijacking, or redirection.

VLAN Hopping

802.1Q trunk links support a feature called Q-in-Q tunneling. This approach, which involves encapsulating 802.1Q traffic inside other 802.1Q traffic, can be very beneficial for service providers that might take trunk traffic from customers and need to tunnel that traffic through their own service provider trunks.

Unfortunately, the use of a native VLAN can be exploited with this Q-in-Q tunneling technology. Specifically, an attacker can use the combination of Q-in-Q tunneling and the native VLAN feature to send traffic into a VLAN that the attacker would not normally be able to send traffic into. This type of attack is called **VLAN hopping**.

Common security changes to prevent VLAN hopping attacks include ensuring that the native VLAN is set to a completely unused VLAN or, for switches that support it, tagging the native VLAN. Both measures eliminate VLAN hopping attacks.

ARP Poisoning

In an **Address Resolution Protocol (ARP) poisoning** attack, the attacker changes the MAC address table to contain incorrect entries. Oftentimes, this is done to redirect traffic to the attacker, allowing them to eavesdrop on communications, capture

sensitive data like login credentials, inject malicious data, or even disrupt network services.

ARP Spoofing

As you might expect, *ARP spoofing* involves falsifying identity. With ARP spoofing, the attacker misrepresents the Layer 2 MAC address. An attacker might use MAC spoofing to bypass simple MAC-based security mechanisms, such as port security.

Rogue DHCP

Rogue DHCP servers might be accidental or malicious in a network. When they are malicious, the attacker is seeking to disrupt (or redirect) end-user traffic by providing false DHCP leased IP address information.

Today, DHCP snooping provides excellent protection against such attacks. This technology seeks to ensure that DHCP server traffic is legitimate in the network at all times.

Rogue Access Point

Rogue access points are unauthorized access points (APs) that prevent legitimate network access through intentional misconfiguration. This approach is very similar to the rogue DHCP server approach.

Evil Twin

As you learned in Chapter 11, “Configure Wireless Devices and Technologies,” an *evil twin* is a special type of rogue access point attack. An evil twin attack is designed to capture authentication information from an unsuspecting network user. This attack is so named because a rogue AP is configured in this scenario to appear just like the legitimate AP that the client should be connecting with.

Ransomware

Ransomware has been an extremely popular type of cybersecurity attack for the past several years. With this style of exploit, the attacker locks access to a system or files (often using encryption) or at least pretends to have locked access. The attacker demands the computer user to pay a ransom to get the system or files unlocked. Attackers commonly require the ransom to be paid in bitcoin or some other cryptocurrency to make the attack less traceable.

Password Attacks

For as long as computer systems have existed, *password attacks* have also existed. With this type of attack, the attacker seeks to gain access to systems or files by using the actual password required for that access. There are many different forms of password attacks, including the following:

- **Brute-force password attack:** In this type of attack, the attacker tries all possible password combinations until a match is made. For example, a brute-force attack might start with the letter a and go through the letter z, and then the attacker might attempt the letters aa through zz, continuing to try combinations until the password is determined. Using complicated passwords—with a mixture of upper- and lowercase letters as well as special characters and numbers—can help prevent brute-force attacks.
- **Dictionary password attack:** In this type of attack, the attacker tries multiple password guesses. However, a dictionary attack is based on a dictionary of commonly used words rather than trying all possible combinations, as in a brute-force attack. Picking a password that is not a common word helps thwart dictionary attacks.

MAC Spoofing

MAC spoofing is a simple attack in which a cybercriminal pretends to possess a different MAC address than is actually on the criminal's system. MAC spoofing is often the first step in a much larger and more sophisticated attack. Because MAC spoofing is relatively easy to accomplish, port security is not often viewed as a sophisticated and comprehensive defense in a network. More sophisticated security solutions, such as dynamic ARP inspection (DAI), are more robust and more effective mechanisms to mitigate issues like MAC spoofing in the network.

MAC Flooding

A *Media Access Control (MAC) flooding* attack is a network-based exploit in which an attacker overwhelms a network switch by sending many packets with spoofed MAC addresses. The attacker does this to cause the switch's MAC table to exceed its capacity. Once the table is full, the switch is forced to direct traffic out all ports. This results in severe network performance degradation and creates a significant security vulnerability, as sensitive data intended for specific devices is broadcast across the entire network. This makes it easier for the attacker to intercept and capture sensitive information, facilitating further attacks such as eavesdropping, data theft, or on-path attacks.

IP Spoofing

It did not take long in the history of TCP/IP and the Internet for attackers to realize that they could fool many IP security mechanisms by simply pretending to possess a different IP address. Think about how many security mechanisms (such as access control lists [ACLs]) focus on the IP address that a source of traffic is using. Faking this address is often all that is required to breach an IP address-based security solution.

It is relatively easy to modify a source IP address, but there are many mechanisms you can use in modern networks to guard against *IP address spoofing*. One Cisco solution is IP Source Guard, which is a tool that relies on DHCP snooping and coordinates DHCP-assigned IP addresses to systems in a network based on the MAC addresses of those systems.

Deauthentication

In a *deauthentication* attack, the attacker sends a deauthentication frame to the victim to disconnect that client from the wireless LAN. While continuously disconnecting a client from Wi-Fi would do a great job frustrating the end user (and the local help desk), it would not in and of itself cause much damage. Typically, the attacker carries out further attacks against the client. For example, an attacker who has installed an evil twin on the network might first carry out a deauthentication attack and then, when the client is reconnecting to the network, attempt to ensure that the end user connects to the evil twin.

Malware

Malware has become a catchall term for the various types of viruses, worms, spyware, Trojans, and other problematic and malicious software or code on a network. Malware can vary from a mere inconvenience to a catastrophic type of security incident. For example, after a single machine in an organization is compromised and is running malicious software, the attacker might use that single computer to proceed further into the internal network and use the compromised host as a pivot point.

Malware in your organization may have been implemented by an outside attacker or by an inside disgruntled employee. Antivirus and anti-malware software should be run on all systems, and users should be given very limited rights related to installation of any software on the computers they use.



Social Engineering Attacks

Attacks that leverage human nature and human behavior are called *social engineering* attacks. These types of attacks are very common and are often the first step or steps in much larger attacks. The following are some specific types of social engineering attacks:

- **Phishing:** This variation of a social engineering attack involves sending an email to a user that appears to be legitimate in an attempt to have that user input authentication information that is then captured. For example, the email may ask the reader to click a website link to claim a package from FedEx. The attacker constructs a website at the false address that looks just like the actual FedEx website.
- **Tailgating:** With tailgating, the attacker physically follows a valid employee through a secured area of the organization. For example, the attacker might notice that a security badge swipe opens a door and that the door, once opened, does not lock again for 30 seconds. The attacker will wait for a valid entry and then sneak into the area after that entry and before 30 seconds have elapsed. An access control vestibule, formerly known as a mantrap, is a physical access control system consisting of a small area and two interlocking doors used to prevent tailgating.
- **Piggybacking:** Piggybacking is similar to tailgating and, in fact, the two terms are often used interchangeably. However, piggybacking is often distinguished from tailgating in that the valid employee knows and is cooperating with the cybercriminal. In addition, the term *piggybacking* is used to refer to a purely electronic transaction. In this form of piggybacking, data flows through some key point in the network, and while that channel is open, the criminal sneaks their data through. As previously noted, access control vestibules are implemented to stop piggybacking/tailgating.
- **Shoulder surfing:** In this classic form of social engineering attack, the cybercriminal simply watches over the shoulder of an authorized employee to learn passwords and other inputs to gain access at a later time. In addition, a shoulder surfer might look for sensitive information such as employees' Social Security numbers on the screen of an HR employee. Special screen filters are available for computer displays and ATMs to prevent someone from shoulder surfing or seeing the screen at an angle.
- **Dumpster diving:** Dumpster diving is an attack where individuals scavenge through trash or discarded materials to extract sensitive information that can be used for malicious purposes, such as identity theft or corporate espionage. Attackers target dumpsters or trash bins near homes, businesses, and public

places to retrieve documents, old computers, or storage devices that may contain confidential data like financial records, passwords, personal details, or proprietary business information. This method exploits the tendency of individuals and organizations to improperly dispose of sensitive items.

Other Miscellaneous Attacks

Table 19-1 describes other miscellaneous types of attacks that you need to be familiar with in today's networked environment.

Table 19-1 Miscellaneous Attacks

Attack	Description
Packet capture	An attacker can use a packet capture (or <i>packet sniffing</i>) utility such as Wireshark (https://wireshark.org) to capture packets after placing a PC's network interface card (NIC) in promiscuous mode. Some protocols, such as Telnet and HTTP, are sent in plaintext, which means packets sent with these protocols can be read by an attacker, perhaps allowing the attacker to see confidential information.
Confidentiality attacks (ping sweep and port scan)	A confidentiality attack might begin with a scan of network resources to identify attack targets on a network. A ping sweep could be used to ping a series of IP addresses. Ping replies might indicate to an attacker that network resources were reachable at those IP addresses. After a collection of IP addresses is identified, the attacker might scan a range of UDP or TCP ports to see what services are available on the hosts at the specified IP addresses. Also, port scans often help attackers identify the operating system running on a target system. These attacks are also commonly referred to as <i>reconnaissance attacks</i> .
Electromagnetic interference (EMI) interception	Data is often transmitted over wires (for example, unshielded twisted pair), and attackers can sometimes copy information traveling over a wire by intercepting the EMI <i>emanations</i> being emitted by the transmission medium. A government project called Tempest studied the ability to understand the data traveling through a network by listening to emanations. A Tempest room is designed to keep emanations contained within that room to increase the security of data communications happening there.
Wiretapping	An attacker who gains physical access to a wiring closet might physically tap into telephone cabling to eavesdrop on telephone conversations or might insert a shared media hub inline with a network cable to connect to the hub and receive copies of packets flowing through the network cable.

Attack	Description
Information sent over overt channels	An attacker might send or receive confidential information over a network by using an <i>overt channel</i> . An example of using an overt channel is tunneling one protocol inside another (for example, sending instant-messaging traffic via HTTPS). Steganography is another example of sending information over an overt channel. For example, an attacker might send a digital image made up of millions of pixels with secret information encoded in specific pixels, where only the sender and the receiver know which pixels represent the encoded information.
Information sent over covert channels	An attacker might send or receive confidential information over a network by using a covert channel, which can communicate information as a series of codes/events. For example, an attacker could represent binary data by sending a series of pings to a destination. A single ping within a certain period of time could represent a binary 0, and two pings within that same time period could represent a binary 1.
FTP bounce	FTP supports a variety of commands for setting up a session and managing file transfers. An attacker may use one of these commands, the port command, to access a system that would otherwise deny the attacker. Specifically, an attacker connects to an FTP server by using the standard port 21. However, FTP uses a secondary connection to send data. The client issues a port command to specify the destination port and destination IP address for the data transmission. Normally, the client would send its own IP address and an ephemeral port number. The FTP server would then use the source port 20 and the destination port specified by the client when sending data to the client. However, an attacker might issue a port command specifying the IP address of a device to access, along with an open port number on that device. As a result, the targeted device might allow an incoming connection from the FTP server's IP address but reject a connection coming in from the attacker's IP address. Fortunately, most modern FTP servers do not accept the port command coming from a device that specifies a different IP address than the client's IP address.
Session hijacking	An attacker could hijack a TCP session, for example, by completing the third step in the three-way TCP handshake process between an authorized client and a protected server. An attacker who successfully hijacks the session of an authorized device might be able to maliciously manipulate data on the protected server.
Salami attack	A salami attack is a collection of small attacks that result in a larger attack when combined. For example, an attacker who has a collection of stolen credit card numbers could withdraw small amounts of money using each credit card, and the combination of the multiple small withdrawals would add up to a significant sum for the attacker.

Attack	Description
Data diddling	Data diddling involves changing data before it is stored in a computing system. Malicious code in an input application or a virus could perform data diddling. For example, a virus, Trojan horse, or worm could be written to intercept keyboard input, and while the appropriate characters are displayed onscreen (so that the user does not see an issue), manipulated characters could be entered into a database application or sent over a network.
Trust relationship exploitation	Different devices in a network might share a trust relationship. For example, a certain host might be trusted to communicate through a firewall using specific ports, while other hosts are denied passage through the firewall using those same ports. If an attacker were able to compromise the host that had a trust relationship with the firewall, the attacker could use the compromised host to pass normally denied data through a firewall.
<i>Logic bomb</i>	In this type of attack, malicious code is hidden in a system and can be triggered by the author or by another attacker. For example, a programmer might hide malicious code that starts deleting files if the programmer's employment is terminated.
TCP SYN flood	In this variant of a DoS attack, an attacker initiates multiple TCP sessions by sending SYN segments but then never completing the three-way TCP handshake. The attacker can send multiple SYN segments to a target system with false source IP addresses in the header of the SYN segments. Because many servers limit the number of TCP sessions they can have open simultaneously, a SYN flood can render a target system incapable of opening a TCP session with a legitimate user.
Buffer overflow	A computer program may be given a <i>buffer</i> , which is a dedicated area of memory to which it can write. If the program attempts to write more information than the buffer can accommodate, a buffer overflow may occur. If permitted to do so, the program can fill up its buffer and then have its output spill over into the memory area being used for a different program. This can potentially cause the other program to crash. Some programs are known to have this vulnerability (that is, be able to overrun their memory buffers), which can be exploited by attackers.
ICMP attack	Many networks permit the use of Internet Control Message Protocol (ICMP) traffic (for example, ping traffic) because pings can be useful for network troubleshooting. However, attackers can use ICMP for DoS attacks. One ICMP DoS attack variant, called the <i>ping of death</i> , uses ICMP packets that are too big. Another variant sends ICMP traffic as a series of fragments in an attempt to overflow the fragment reassembly buffers on the target device. Also, a <i>Smurf attack</i> can use ICMP traffic directed to a subnet to flood a target system with ping replies.

Attack	Description
Electrical disturbance	At a physical level, an attacker may launch an availability attack by interrupting or interfering with the electrical service available to a system. For example, an attacker who gains physical access to a data center's electrical system might be able to cause a variety of electrical disturbances, such as power spikes, electrical surges, blackouts, and brownouts.
Supply chain attack	This type of attack takes advantage of the weakest cybersecurity link and often begins with an advanced persistent threat (APT) during the manufacturing process of an electronic or digital product in order to ultimately cause harm to a target customer or company.
Cryptojacking	In this attack, criminals use malware to deliver malicious cryptomining software in order to use distributed resources of others (often as part of a botnet) to mine for cryptocurrency.
Keylogger trojans	Keylogger trojans monitor and send keystrokes typed from an infected machine.
Advanced persistent threat (APT)	APTs are a "low and slow" style of attack executed to infiltrate a network and remain inside while going undetected.
Fileless malware	This attack works similarly to a memory-resident virus but is more insidious. While the latter still requires some components of the virus to be written to disk, a fileless virus does not.
Birthday attack	A birthday attack is a cryptographic method of attack against a secure hash. It is based on what is known as the birthday paradox.
Downgrade attack	The downgrade attack is often a result of security configurations not being updated. Often this stems from the desire to maintain backward compatibility.
Password spraying	Password spraying is an attack that attempts to access a large number of user accounts with a very few number of commonly used passwords.
Skimming	Skimming essentially copies data from the card (ATM or other) using a specialized terminal. As a result, the card can subsequently be cloned by taking a blank card and encoding the stolen data.
USB drop attack	A USB drop attack occurs when an attacker drops USB flash drives loaded with malware in a public place in the hopes the target will pick it up and, out of curiosity, plug it into a system to see what's on it. Once plugged in, the malware can automatically run and infect the system. This can be mitigated by enabling Turn Off AutoPlay in the Windows Local Group Policy Editor AutoPlay Policies settings.
SQL injection attack	A SQL injection attack is malicious code that is inserted into strings that are later passed to a database server. The SQL server then parses and executes this code.

Attack	Description
LDAP injection attack	The LDAP injection attack is similar to the SQL injection attack, but instead malicious input is applied to a directory server, which may result in unauthorized queries, granting of permissions, and even password changes.
Privilege escalation	This attack occurs when an attacker is able to gain elevated access to areas that otherwise should be restricted.
Bluejacking	This attack occurs when Bluetooth-enabled smart devices receive photos, messages, or other file broadcasts sent from a nearby Bluetooth-enabled transmitting device.
Bluesnarfing	A bluesnarfing attack is a type of unauthorized access where an attacker exploits a Bluetooth-enabled device to steal sensitive information, such as contacts, messages, and other data, without the device owner's knowledge. Notice that this is a more aggressive attack than bluejacking.

NOTE The attack types listed in this chapter are just some of the many attack types being used today. New attacks and even new categories of attacks are being created all the time. Also, keep in mind that many attacks are actually perpetrated by employees of an organization. They are called *insider attacks*.

Real-World Case Study

Acme, Inc. has recognized the growing threat landscape and decided to enhance its security training for both IT staff members and all end users across the organization. This comprehensive approach acknowledges that cybersecurity is a shared responsibility and aims to equip everyone with the necessary skills to protect the organization against emerging threats. By focusing on tailored training sessions for different groups, Acme, Inc. ensures that both technical and non-technical employees are well prepared to respond to various security challenges.

The decision to prioritize social engineering in the training for end users highlights the company's awareness of the increasing sophistication of such attacks. Social engineering, particularly phishing, remains one of the most common and effective tactics used by cybercriminals to breach organizational defenses. As the organization has experienced a rise in successful phishing attempts, the training will specifically address these threats, teaching end users how to recognize and respond to phishing attempts, thereby reducing the likelihood of successful attacks.

For the IT staff, Acme, Inc. is implementing ongoing training that not only covers social engineering but also dives deep into the latest advancements in cybersecurity threats. The focus on continuous education ensures that the IT team stays ahead of the curve, with particular attention to detecting and mitigating distributed denial-of-service (DDoS) attacks. Given the potential damage that DDoS attacks can cause, the training will emphasize rapid detection at the network perimeter, enabling the IT staff to respond swiftly and minimize disruption to the organization's operations.

By dedicating resources to both end-user and IT staff training, Acme, Inc. is taking a proactive stance against cyber threats. The company understands that while technical defenses are essential, human error remains a significant vulnerability. Therefore, empowering employees with knowledge and awareness is a critical component of the organization's overall cybersecurity strategy.

This dual-focused training approach not only strengthens the immediate defenses against specific threats like phishing and DDoS attacks but also fosters a culture of security awareness throughout the organization. As a result, Acme, Inc. is better positioned to handle current and future challenges, ensuring the security and resilience of its operations.

Summary

Here are the main topics covered in this chapter:

- This chapter described some of the most common technology-based security attacks.
- This chapter described some of the most common human-based attacks.
- This chapter described some of the common miscellaneous attacks that may be carried out against a network.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 19-2 lists these key topics and the page number where each is found.

**Table 19-2** Key Topics for Chapter 19

Key Topic Element	Description	Page Number
Section	Denial-of-Service (DoS)	458
Section	Distributed Denial-of-Service (DDoS)	459
Section	Social Engineering Attacks	464

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

Address Resolution Protocol (ARP) poisoning, ARP spoofing, denial-of-service (DoS), distributed denial-of-service (DDoS), DNS poisoning, DNS spoofing, dumpster diving, evil twin, logic bomb, malware, Media Access Control (MAC) flooding, on-path attack, phishing, rogue access point, rogue DHCP server, shoulder surfing, social engineering, tailgating, VLAN hopping

Additional Resources

The Cyber Kill Chain for Aspiring Ethical Hackers: <https://www.youtube.com/watch?v=HVwI1KbrhOU>

What Is a Cyberattack?: <https://www.cisco.com/c/en/us/products/security/common-cyberattacks.html>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. Which type of attack exploits the native VLAN of 802.1Q?
 - a. Evil twin
 - b. Tailgating
 - c. Deauthentication
 - d. VLAN hopping

2. In what type of attack does the attacker compromise multiple systems and then instruct those compromised systems, called *zombies*, to simultaneously flood a target system with traffic?
 - a. DoS attack
 - b. TCP SYN flood attack
 - c. Buffer overflow
 - d. DDoS attack
3. Which of the following is an example of a social engineering attack?
 - a. DDoS attack
 - b. DoS attack
 - c. Tailgating
 - d. On-path attack
4. In what type of attack does the attacker try all possible password combinations until a match is made?
 - a. Dictionary attack
 - b. MAC spoofing
 - c. IP spoofing
 - d. Brute-force attack
5. What type of attack often seeks payment in bitcoin or other cryptocurrency?
 - a. Malware
 - b. DDoS
 - c. Ransomware
 - d. DNS poisoning
6. What type of attack originates from multiple systems simultaneously?
 - a. DoS
 - b. DDoS
 - c. VLAN hopping
 - d. Media Access Control (MAC) flooding

7. Which of the following is a type of rogue access point attack designed to capture authentication information from an unsuspecting network user?
 - a. Address Resolution Protocol (ARP) poisoning
 - b. Brute-force
 - c. Dictionary
 - d. Evil twin

8. Which of the following is not considered a type of social engineering attack?
 - a. Shoulder surfing
 - b. Dumpster diving
 - c. Phishing
 - d. Password spraying



This chapter covers the following topics related to Objective 4.3 (Given a scenario, apply network security features, defense techniques, and solutions) of the CompTIA Network+ N10-009 certification exam:

- Device hardening
 - Disable unused ports and services
 - Change default passwords
- Network access control (NAC)
 - Port security
 - 802.1X
 - MAC filtering
- Key management
- Security rules
 - Access control list (ACL)
 - Uniform Resource Locator (URL) filtering
 - Content filtering
- Zones
 - Trusted vs. untrusted
 - Screened subnet

Network Security Features

While there are numerous security tools available to us today, this chapter examines some of the most popular. We will discuss the importance and techniques involved in device hardening. We then examine network access control (NAC), key management, security rules, and zones. These topics help you build a strong foundation for further network security studies. Note that some of these topics have been touched on in previous chapters, but in this chapter, they receive the security-centric focus they deserve.

Foundation Topics

Device Hardening

For a long time, many network vendors emphasized using a defense-in-depth strategy. And while they would still stand behind this approach, today plenty of other security initiatives are promoted, such as zero trust. If you think about the defense-in-depth concept for a moment, you'll realize the importance of network *device hardening* in this strategy. After all, it's wise to tightly secure the network devices themselves in an attempt to build a very secure network.

In addition, network (device) hardening is an excellent idea because many devices ship from the manufacturer with default configurations that can be quite dangerous. For example, you might receive a network router that has a web server process enabled inside it. This web server software might permit HTTP (not HTTPS) connections, and there may be no access control list or firewall of any kind protecting this web server. Perhaps this security vulnerability is meant to permit web-based management of the device. Your security policy probably does not permit such a vulnerability. Hardening this device by disabling or even eliminating that web server is an excellent idea. In fact, because you should always try to adhere to your security policy, in this scenario, the web server must be uninstalled or at least disabled.

Best Practices

While the hardening of a network through the hardening of its network devices can be a daunting task, the great news is that there are many standard best practices in this regard. Just remember that whenever you see a list like this, you should be prepared to encounter plenty of best practices that will not be applicable for your particular network or your particular network device.

The following is an impressive list of network hardening techniques you should at least consider:

Key Topic

- **Change default passwords:** At a very minimum, you should always do this step! The first thing an attacker is likely to do as an initial step in a larger-scale attack is try using default username and password credentials to access network devices. Plenty of websites provide lists of default username and password credentials for common systems from popular vendors. If you take none of the other steps described in this chapter, at least take this one as it is crucial.

- **Avoid using common passwords and usernames on devices:** Avoid using usernames like **admin** and passwords like **password123**. You also need to follow your own enforced password complexity and length policy. Today, it is recommended that passwords be at least 8 to 12 characters and include a certain combination of special characters, numbers, and upper- and lowercase letters.

NOTE Always be sure to keep up on the latest industry recommendations. For example, NIST’s “Digital Identity Guidelines” includes the latest password recommendations. This document is available at <https://pages.nist.gov/800-63-3/sp800-63b.html>.

- **Upgrade firmware:** By installing updated firmware, you gain new functionality on a device, and oftentimes the device gets much-needed security patches and enhancements. Figure 20-1 shows the firmware upgrade process for a small office/home office (SOHO) wireless router.

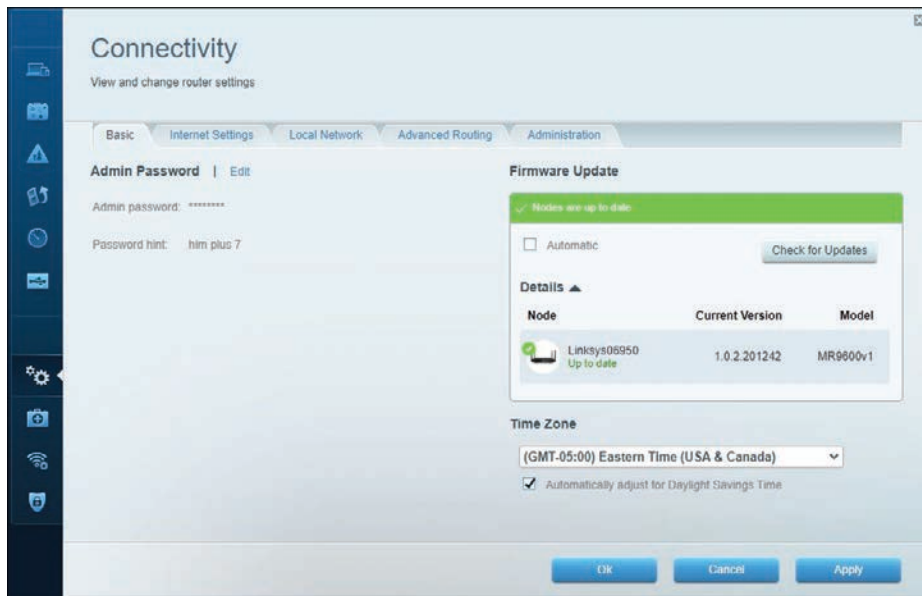


Figure 20-1 Upgrading the Firmware on a SOHO Wireless Router

- **Patch and update:** Just as your firmware needs consistent attention to patches and updates, so do your operating system and applications. Many network devices from popular vendors today are so sophisticated with their OS that they can run plug-in types of applications or even full-blown apps. Patches and updates might bring advanced features to a device.

- **Disable unneeded network services:** Modern network devices tend to have many network services enabled by default. To reduce the attack surface, you should trim off any unnecessary services by disabling them. Reducing the attack surface by not running services means that there is less that the attacker can target. Attackers commonly target Bluetooth and remote desktop services as methods of carrying out access for their subsequent attacks.
- **Disable unneeded switch ports:** Ports on switches that are not connected to anything need to be shut down. One primary reason to disable unneeded switch ports is that if someone gains physical access to a network switch and connects to a switch port, you want to ensure that the person cannot access the network through that port. You can use port blocking as well as physical security practices to prevent such access.
- **Use secure protocols:** Use secure protocols over unsecure protocols whenever possible. For example, use Secure Shell (SSH) rather than the insecure Telnet for remote management using terminals. Other unsecure protocols to remember include HTTP, SLIP, FTP, Trivial FTP (TFTP), and Simple Network Management Protocol Version 1/2 (SNMPv1/v2).
- **Secure SNMP:** Be sure to use the security features of the relatively new SNMP Version 3. “SNMP means Security Is Not My Problem” does not apply to Version 3 of SNMP, which provides encryption.
- **Generate new keys/credentials:** Rotate, rotate, and then rotate again those important credentials that are used to guard your corporate systems and data. Fortunately, today there are many tools that enterprises can turn to for assistance in this area. IT departments can call upon cloud-based solutions for key management, and corporate employees can use password management systems to help ensure strong and rotated network passwords.
- **Ensure spanning tree protections:** If you are stuck with Spanning Tree Protocol (STP) in your network, you need to be sure to use any protections offered by a network device. These protections might include Root Guard, Bridge Protocol Data Unit (BPDU) Guard, and Flood Guard.
- **Enable DHCP snooping:** This involves preventing rogue DHCP servers and DHCP pool exhaustion attacks by restricting ports from accepting certain DHCP messages.
- **Use VLAN segmentation:** Doing so inherently protects systems from accidental or malicious attacks from systems in other broadcast domains. VLAN segmentation also forces inter-VLAN communications to pass through a router, allowing you to easily enforce security policy and security checks on the traffic.

- **Use Router Advertisement (RA) Guard:** This helps protect against attacks in IPv6 environments, where routers can advertise the network prefix information to end systems. These end systems can configure their own unique host IDs. Therefore, you can address end systems without requiring DHCP implementation. While these types of features are excellent, you need security controls such as RA Guard to make sure these features are not exploited.
- **Use DAI:** *Dynamic ARP inspection (DAI)* is a security mechanism that guards against MAC address spoofing. It can pair nicely with port security.
- **Use CoPP:** *Control plane policing (CoPP)* is an excellent security feature that can control the rate of packets to and from the control plane of a network device. Think about life without CoPP: Attackers could easily cause denial of service (DoS) on a network device by simply flooding fake Border Gateway Protocol (BGP) updates to the device (or updates from another control plane protocol).
- **Use private VLANs:** You will often implement networks for special types of deployments. For example, say that you are creating a network to be shared by a row of stores in a shopping mall. Each store needs to be segmented from every other store. *Private VLANs* are excellent for such a design. Each store can have its own private VLAN. All of these private VLANs can reach the Internet, but they cannot enter the other private VLANs. All of the VLANs in this case can be in the same IP subnet. This type of topology would confuse an attacker who is unaware of the private VLAN feature. The attacker would see that the systems are clearly in the same IP subnet, but they would be unable to communicate with each other. This would be very strange in a VLAN topology.
- **Change the default VLAN:** Most network devices that offer switch ports place all of these ports in a default VLAN. For example, Cisco Systems switches have all ports in VLAN 1 by default. Attackers know this and seek out ports in the default configuration to try to exploit them.
- **Use access control lists (ACLs):** ACLs are a set of rules used to control network traffic and restrict access to network resources by specifying which users or system processes can access certain resources and the operations they are allowed to perform. Use ACLs wherever possible to help strengthen security. For example, a network device might actually be using the built-in web server for network management. If this device offers an ACL feature to protect the internal web server from unauthorized access, you should take advantage of this feature.
- **Use role-based access:** Consider using *role-based access control (RBAC)* by creating groups or roles on network devices. You can map these groups to

different permission levels. This would allow you to permit junior engineers read-only access to devices to help you with monitoring. At the same time, you can have groups with much higher levels of privileges to help with configuration and optimization.

- **Use firewall rules:** Many network devices have full-blown firewall services built in. It is important to consider how a firewall operates with traffic that does not match any permit statement in the firewall configuration. Does the device have an *implicit deny* rule for all of this traffic, or do you need to configure an *explicit deny* for this traffic? As traffic passes through the firewall rules, it is permitted or denied based on those rules. When it reaches the end of the explicitly stated rules, is it dropped due to an implicit deny? Or do you need to configure an explicit deny rule at the end? The most common configuration is to have an implicit deny in place for all the unmatched traffic in the firewall rules.

NOTE For a firewall that consists of all DENY entries, you might need to add an explicit *permit all* entry at the end of the rules to override an implicit *deny all*.

Network Access Control (NAC)

There are a number of security best practices to consider for our wired networks, our wireless networks, and the ever-growing *Internet of Things (IoT)*. Notice that most of these are in the category of **network access control (NAC)**. It is the job of NAC to manage and control the access of devices to a network, ensuring that only authorized and compliant devices can connect. Here are some popular techniques:

Key Topic

- **Use port security:** Port security is a network security feature that helps control access to specific ports on a switch based on the MAC address of the device requesting access. It prevents unauthorized devices from connecting to the network by allowing only approved devices to use those ports. Locking down switch ports to MAC addresses is a good first step in hardening a network. Doing so guards against MAC flood attacks nicely, but note that the network is still vulnerable to MAC spoofing attacks.
- **Use MAC filtering:** MAC filtering, which is similar to port security, is important in a wireless environment. With MAC filtering, you lock down access to the Wi-Fi network by allowing only specific MAC addresses.
- **Use wireless client isolation:** Wireless client isolation is a security feature that prevents wireless clients from communicating with one another. This feature is useful for guest and BYOD wireless networks.

- **Use guest network isolation:** If you do not care if your guest Wi-Fi nodes can reach each other, you might consider implementing isolation for the entire guest Wi-Fi network. This is a very common practice. The guest Wi-Fi permits the nodes inside it to access the public Internet, but it does not permit any of the network devices in the enterprise hosting the guest Wi-Fi.
- **Rotate and secure the use of strong pre-shared keys (PSKs):** If you are relying on PSKs in your Wi-Fi security environment, be sure to rotate and protect these credentials just as you do with the more traditional passwords for user accounts in your enterprise. See Chapter 11, “Configure Wireless Devices and Technologies,” for more details about PSKs.
- **Use Extensible Authentication Protocol (EAP):** EAP has become the de facto standard for carrying security credentials in a Wi-Fi network. In the case of the popular *802.1X*, the EAP client (called the *supplicant*) sends the security credentials to the WLC wrapped in an EAP packet. The WLC wraps this packet in a RADIUS packet and sends the information to an authentication server on the network. This system checks the security credentials and then instructs the WLC whether it should permit the device or user on the network.
- **Use geofencing where needed:** This technology often uses the Global Positioning System (GPS) or radio frequency identification (RFID) to define geographical boundaries. *Geofencing* allows you to define triggers so that when a device enters (or exits) the boundaries defined by the administrator, an alert is issued. Geofence virtual barriers can be active or passive. An active geofence requires an end user to opt in to location services and requires a mobile app to be open. Passive geofences are always on; they rely on Wi-Fi and/or cellular data instead of GPS or RFID and can work in the background. For example, say that a hospital keeps patient information on tablets that the hospital distributes to staff. If these tablets travel beyond the geofence, an administrative alert can trigger.
- **Use captive portals:** As introduced in Chapter 11, *captive portals* can obtain information about guests that are attempting to access guest networks or other network resources. With this approach, you send the user to a web page to enter credential information and/or accept agreements on usage. Most public networks, including Wi-Fi hotspots, use a captive portal, which requires users to agree to some condition before they use the network or Internet.
- **Control IoT access:** You need to be concerned about breaches of security in your IoT infrastructures, which often rely on your Wi-Fi (or at least portions of it). Therefore, many of the other best practices listed in this section are appropriate for IoT infrastructures. It is a good idea to consistently update

your IoT devices for the new security enhancements that inevitably come. You should also segment the IoT traffic as much as possible in the enterprise.

Other Network Security Features

Among the numerous other network security features, here are just some of the more popular features in use today. Notice that many of these are key topics for the CompTIA Network+ exam:

- **Key management:** Key management involves the processes and systems used to create, distribute, store, and manage cryptographic keys. These keys are essential for encrypting and decrypting data, ensuring that only authorized parties can access sensitive information. Effective key management ensures that keys are kept secure, regularly updated, and properly destroyed when no longer needed. This helps prevent unauthorized access and protects data from being compromised, playing a crucial role in maintaining the overall security of a network. Public cloud providers are now offering key management as one of the many managed services that are available. This is a clearcut attempt to lure to the cloud those IT departments buckling under the network security workloads often required today.
- **Security rules:** In addition to access control lists (described earlier in this chapter), there are numerous other security rules that might assist the modern network engineer. For example, an organization might engage in ***Uniform Resource Locator (URL) filtering***. URL filtering is a network security measure that restricts access to certain websites based on their URLs, allowing organizations to block or allow specific web content to protect users from harmful sites and enforce Internet usage policies. Another example of a security rule approach is ***content filtering***. Content filtering is a technique used to monitor and control the types of content that can be accessed or transmitted over a network. It works by examining data such as emails, web pages, and files to block or allow specific content based on predefined criteria, helping to protect users from inappropriate, harmful, or malicious material and ensuring compliance with organizational policies.
- **Zones:** Today, it is quite common for network security engineers to divide the network into zones to benefit the overall security posture. For example, a network might be divided into ***trusted zones*** and ***untrusted zones***. Trusted zones are network segments that contain resources and systems deemed secure and reliable, typically with strict access controls and security measures in place. Untrusted zones, on the other hand, are network segments that include external or less secure networks, such as the Internet or guest networks, where the risk of threats is higher. By separating these zones, organizations can

implement different security policies and controls, ensuring that sensitive data and critical systems in the trusted zones are protected from potential threats originating from the untrusted zones. Yet another type of zone that might be identified is a **screened subnet**. A screened subnet, also known as a demilitarized zone (DMZ), is a network segment that sits between an internal trusted network and an external untrusted network, providing an additional layer of security by isolating and protecting internal systems from external threats while allowing controlled access to public-facing services. Figure 20-2 shows an example of a screened subnet.



Figure 20-2 An Example of a Screened Subnet

Real-World Case Study

Acme, Inc. has decided to perform a security audit of the network. As part of this audit, Acme, Inc. is finding many improvements that can be made very easily. Most of these center on network and device hardening.

Acme, Inc. has identified many services running on network nodes that can safely be disabled. An audit of network access also revealed many unused network ports that can be disabled. Also, Acme identified an area of the organization where the password policy in effect was not appropriately following best practices.

Finally, Acme, Inc. is changing a portion of the network design where in-band network management traffic was discovered. This traffic, if captured, would have revealed several parameters regarding the network configuration that are considered private. This management traffic is now flowing through an encrypted tunnel so that the chance of an on-path attack is now greatly reduced.

Summary

Here are the main topics covered in this chapter:

- This chapter covered best practices for hardening a network and network devices.

- This chapter discussed many security features related to network access controls in networks today.
- This chapter also presented various other security measures commonly found in secure networking, including key management, the use of security rules, and the use of security-based zones in the network.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 20-1 lists these key topics and the page number where each is found.



Table 20-1 Key Topics for Chapter 20

Key Topic Element	Description	Page Number
List	Network hardening best practices	476
List	Network access control techniques	480
Figure 20-2	An Example of a Screened Subnet	483

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

802.1X, access control list (ACL), content filtering, device hardening, key management, MAC filtering, network access control (NAC), port security, screened subnet, trusted zone, Uniform Resource Locator (URL) filtering, untrusted zone

Additional Resources

Hardening Network Devices: https://media.defense.gov/2020/Aug/18/2002479461/-1/-1/0/HARDENING_NETWORK_DEVICES.PDF

What Is RBAC?: <https://digitalguardian.com/blog/what-role-based-access-control-rbac-examples-benefits-and-more>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. Your primary concern when hardening your network is the fact that you are vulnerable to several DoS attacks that involve your IGP and EGP protocols. What hardening technique addresses this challenge most directly?
 - a. Control plane policing
 - b. Geofencing
 - c. SNMP
 - d. Dynamic ARP inspection
2. What protocol makes 802.1X possible?
 - a. SSH
 - b. Telnet
 - c. SNMPv3
 - d. EAP
3. What device hardening technique might be found in a row of stores in a shopping mall to ensure that the different stores are segmented from each other?
 - a. Default VLAN
 - b. DHCP snooping
 - c. Private VLAN
 - d. Dynamic ARP inspection
4. Which of the following is not a network hardening best practice?
 - a. Use SNMPv3
 - b. Disable unneeded services
 - c. Implement role-based access
 - d. Change to default passwords
5. Which of the following means that if you have not been explicitly granted access, then access is denied?
 - a. Implicit deny
 - b. Explicit deny

- c. Allow
 - d. BPDU
6. What do most public networks, including Wi-Fi hotspots, use to require users to agree to some condition before they use the network or Internet?
 - a. PSKs
 - b. Proper antenna placement
 - c. Appropriate signal power levels
 - d. Captive portal
 7. Which of the following is a commonly used network zone indicating the highest level of risk where multiple threats are most likely?
 - a. Trusted
 - b. Untrusted
 - c. Screened subnet
 - d. Backbone
 8. Your supervisor is interested in full automation for the rotation of passwords securing encrypted files at rest belonging to your organization. What security category does this most closely align with?
 - a. Content filtering
 - b. Device hardening
 - c. Network access control
 - d. Key management
 9. Which of the following are considered secure protocols? (Choose two.)
 - a. SSH
 - b. FTP
 - c. SNMPv1
 - d. SNMPv3

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This chapter covers the following topics related to Objective 5.1 (Explain the troubleshooting methodology) of the CompTIA Network+ N10-009 certification exam:

- Identify the problem
 - Gather information
 - Question users
 - Identify symptoms
 - Determine if anything has changed
 - Duplicate the problem, if possible
 - Approach multiple problems individually
- Establish a theory of probable cause
 - Question the obvious
 - Consider multiple approaches
 - Top-to-bottom/bottom-to-top OSI model
 - Divide and conquer
- Test the theory to determine the cause
 - If theory is confirmed, determine next steps to resolve problem
 - If theory is not confirmed, establish a new theory or escalate
- Establish a plan of action to resolve the problem and identify potential effects
- Implement the solution or escalate as necessary
- Verify full system functionality and implement preventive measures if applicable
- Document findings, actions, outcomes, and lessons learned throughout the process

A Network Troubleshooting Methodology

As you perform your day-to-day tasks of administering a network, a significant percentage of your time will be dedicated to resolving network issues. Whether the issues you are troubleshooting were reported by an end user or were issues you discovered, you need an effective plan to respond to them. Specifically, you need a systematic approach to clearly articulate an issue, gather information about the issue, hypothesize the underlying cause of the issue, validate your hypothesis, create an action plan, implement that action plan, observe results, and document your resolution. Without a plan, your efforts might be inefficient, as you try one thing after another, possibly causing other issues in the process.

Although your troubleshooting efforts can most definitely benefit from a structured approach, realize that troubleshooting is part art and part science. Specifically, your intuition and instincts play a huge role in isolating an issue. Of course, those skills are developed over time and come with experience and exposure to more and more scenarios.

To help you start developing or continue honing your troubleshooting skills, this chapter presents a troubleshooting methodology that can act as a guide for addressing almost any network issue. This chapter also presents a collection of common network issues to consider in your real-world troubleshooting efforts (and issues to consider as you prepare for the CompTIA Network+ exam).

Foundation Topics

Troubleshooting Basics

Troubleshooting network issues is implicit in the responsibilities of a network administrator. Such issues could arise as a result of human error (for example, a misconfiguration), equipment failure, software bugs, or traffic patterns (for example, high utilization or a network being under attack by malicious traffic).

Many network issues can be successfully resolved using a variety of approaches. This section begins by introducing you to troubleshooting fundamentals and then presents a structured troubleshooting methodology you should know for the Network+ exam.

Troubleshooting Fundamentals

The process of troubleshooting, at its essence, is the process of responding to a problem report (sometimes in the form of a *trouble ticket*), diagnosing the underlying cause of the problem, and resolving the problem. Although you normally think of the troubleshooting process beginning when a user reports an issue, through effective network monitoring, you might detect a situation that could become a troubleshooting issue and resolve that situation before it impacts users.

After an issue is reported, the first step toward resolution is clearly defining the issue. After you have clearly defined the troubleshooting target, you can begin gathering information related to that issue. Based on the information collected, you might be able to better define the issue. Then you can hypothesize the likely causes of the issue. Evaluation of these likely causes leads to the identification of the suspected underlying root cause of an issue.

After a suspected underlying cause is identified, you define approaches to resolve an issue and select what you consider to be the best approach. Sometimes the best approach to resolving an issue cannot be implemented immediately. For example, a piece of equipment might need to be replaced. However, implementing such an approach during working hours might disrupt a business's workflow. In such situations, a troubleshooter might use a temporary fix until a permanent fix can be put in place.

As a personal example, when helping troubleshoot a connectivity issue for a resort hotel at a major theme park, my coworkers and I discovered that a modular Ethernet switch had an issue causing Spanning Tree Protocol (STP) to fail, resulting in a Layer 2 loop. This loop flooded the network with traffic, preventing the hotel from issuing keycards for guest rooms. The underlying cause was clear: The Ethernet switch had a bad module. However, the issue was pinpointed around 4 p.m., a peak

time for guest registration. So, instead of immediately replacing the faulty module, we disconnected one of the redundant links, thus breaking the Layer 2 loop. The logic was that it was better to have the network function at this time without STP than for the network to experience an even longer outage while the bad module was replaced. Late that night, someone came back to the switch and swapped out the module, resolving the underlying cause while minimizing user impact.

Figure 21-1 provides a simplified model of the troubleshooting steps just described, which consists of three steps:

- Step 1.** Problem report
- Step 2.** Problem diagnosis
- Step 3.** Problem resolution

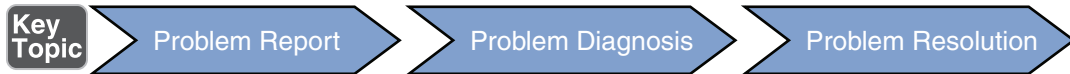


Figure 21-1 Simplified Troubleshooting Flow

Of these three steps, the majority of a troubleshooter's efforts are spent in the *problem diagnosis* step. Table 21-1 describes key components of this diagnosis step.

Key Topic **Table 21-1** Steps to Diagnose a Problem

Step	Description
Gather information.	Because a typical problem report lacks sufficient information to give a troubleshooter insight into a problem's underlying cause, the troubleshooter should collect additional information, perhaps using network maintenance tools or interviewing impacted users.
Duplicate the problem, if possible.	Testing to see if you can duplicate the problem is often a key step in problem diagnosis.
Question users.	Although it can be difficult to gather information from your end users, doing so is often critical in correctly pinpointing the exact problem. Oftentimes, finding out user actions prior to the problem is critical.
Identify symptoms.	What symptoms has the problem has created?
Determine if anything has changed.	Perhaps your end users will provide valuable clues if they accurately indicate what changes they might have made to systems.
Approach multiple problems individually.	Unfortunately, you might discover that there are multiple issues. Be sure to approach each one individually.

Structured Troubleshooting Methodology

Troubleshooting skills vary from administrator to administrator. Therefore, although most troubleshooting approaches include the collection and analysis of information, elimination of potential causes, hypothesis of likely causes, and testing of the suspected cause, different troubleshooters might spend different amounts of time performing these tasks.

If a troubleshooter does not follow a structured approach, the temptation is to move between the previously listed troubleshooting tasks in a fairly random way, often based on instinct. Although such an approach might well lead to problem resolution, it can become confusing to remember what you have tried and what you have not tried. Also, if another administrator comes to assist you, communicating to that other administrator the steps you have already gone through could be challenging. Therefore, following a structured troubleshooting approach not only helps prevent you from trying the same thing more than once and inadvertently skipping a task but also aids in communicating to someone else the possibilities you have already eliminated.

You might encounter a variety of structured troubleshooting methodologies in networking literature. However, for the Network+ exam, the methodology shown in Figure 21-2 is the one you should memorize.

Key Topic

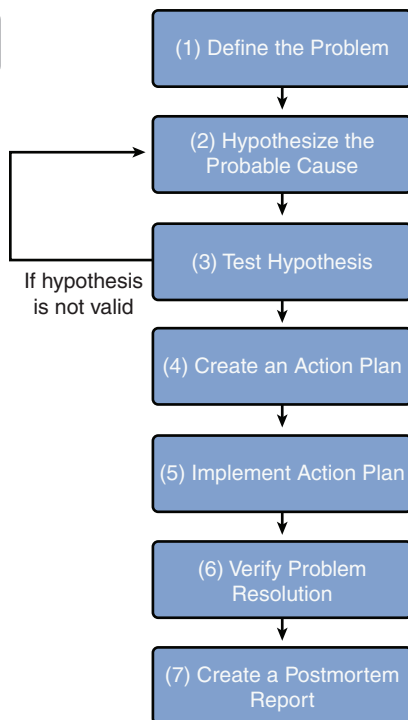


Figure 21-2 Structured Troubleshooting Approach

The following is an elaboration on this seven-step methodology:



- Step 1. Identify the problem.** Effective troubleshooting begins with a clear problem definition. This definition might include specific symptoms. Here's an example: "User A's computer is unable to communicate with server 1 (as verified by a ping test). However, user A can communicate with all other servers. Also, no other user seems to have an issue connecting to server 1." This problem definition might come from questioning the impacted user(s) and doing your own testing (for example, seeing if you can ping from user A's computer to server 1). If possible, determine whether anything has changed in the network (or in the computer) configuration. Also, find out whether this is a new installation that has failed to work in the past.
- Step 2. Establish a theory of probable cause.** This is the point in the troubleshooting process where your experience and intuition can be extremely helpful because it is when you brainstorm a list of possible causes. As you brainstorm, be sure to question the obvious. Also, think in terms of *top-to-bottom troubleshooting* (moving from top to bottom within the OSI model) or *bottom-to-top troubleshooting*. Alternatively, you could use a *divide-and-conquer troubleshooting* approach. When examining your collected data (for example, output from the **ipconfig /all** command), question everything. For example, you might think that the issue described in step 1 could result from causes such as an ACL blocking traffic to or from the PC, a connectivity issue with the PC or server, or an incorrect IP address configuration on the PC. From your list of possible causes, select the one you consider the most likely. From the previous list, you might believe that an incorrect IP address configuration on the PC is the most likely cause of the problem. Specifically, you might conclude that the issue is not related to connectivity because other PCs can get to the server, and user A's PC can get to other servers. Also, you might conclude that it is more likely that user A's PC has a bad IP address configuration than that an ACL has been administratively added to the router to block traffic only between user A's PC and server 1.
- Step 3. Test the theory to determine the cause.** Before taking action on what you consider to be the most likely cause of a problem, do a *sanity check* on your theory. Would your hypothesized cause lead to the observed symptoms? In the example presented in the preceding steps, you might examine the subnet mask assigned to user A's computer and determine that it is incorrect. Specifically, the subnet mask makes user A's computer think that server 1 is on the same subnet as user A's computer. As a result, user A's computer does not forward traffic to its default gateway when

attempting to reach server 1. If your hypothesis is technically sound, you can proceed to step 4. However, if you notice a flaw in your logic, you need to formulate an alternate hypothesis. The formation of an alternate hypothesis might involve escalating the problem to someone more familiar with the device(s) in question.

Step 4. Establish a plan of action to resolve the problem and identify potential effects. Once you have confirmed that your theory makes sense technically, you need to develop an action plan. If time permits, you should document your action plan. The documentation of your action plan can be used as a *back-out plan* if your hypothesis is incorrect. In the example we have been building on throughout these steps, an action plan might be to change the subnet mask on user A's computer from 255.255.0.0 to 255.255.255.0.

Step 5. Implement the solution or escalate as necessary. Based on your documented plan of action, you should schedule an appropriate time to implement the action plan. The selection of an appropriate time is a balance between the severity of a problem and the impact your action plan will have on other users. Sometimes, when attempting to implement an action plan, you realize that you do not have sufficient administrative privileges to perform a task in your action plan. In such cases, you should escalate the issue to someone who has appropriate administrative rights. In this example, changing the subnet mask on one computer should not impact any other devices. So, you might immediately make the configuration change on user A's computer.

Step 6. Verify full system functionality and implement preventive measures if applicable. After implementing an action plan, you need to verify that the symptoms listed in your original problem definition are gone. You also need to attempt to determine whether your action plan has caused any other issues on the network. A mistake many troubleshooters make at this point is believing that the issue has been resolved because the specific symptom (or symptoms) they were looking for is gone. However, the user who originally reported the issue might still be having a problem. Therefore, troubleshooters should live by the mantra "A problem isn't fixed until the user believes it's fixed." You should get confirmation from the person reporting an issue that, from her perspective, the reported issue has indeed been resolved. In this example, you could attempt to ping server 1 from user A. If the ping is successful, you can check with user A to see whether she agrees that the problem is resolved.

Step 7. Document findings, actions, outcomes, and lessons learned throughout the process. Notice how this step really emphasizes documentation. That's right—you should be documenting throughout the entire process. This often includes the creation of a postmortem report. A *postmortem* report is a document that describes the reported issue, its underlying causes, and what was done to resolve the issue. This report might be useful when troubleshooting similar issues in the future.

Keep in mind when working your way through the previous steps that you might encounter an issue that you do not have sufficient information to solve. When that happens, you might need to further research the issue yourself. However, if time is of the essence, you might need to immediately escalate the issue to someone else within your organization, to an equipment vendor, or to an outside consultant.

Real-World Case Study

The Acme, Inc. networking team and some key IT stakeholders are analyzing the efficiency of the support desk. During this analysis, it was discovered that the previous attempts at an enforcement of a troubleshooting methodology were not being followed. The methodology also had several weaknesses to it that did not warrant following to begin with. Finally, it was discovered that the documentation of the desired troubleshooting methodology was sorely lacking.

The networking team is now implementing training on a new and improved troubleshooting methodology and is introducing several new software tools that will make it easier to follow and more effective overall. This new troubleshooting approach begins with a careful identification of the problem and establishment of a theory of probable cause. The process then involves the following steps: testing the theory to determine the cause; establishing a plan of action to resolve the problem; implementing the solution; verifying full system functionality; and finally, ensuring documentation has been created or updated throughout the process.

Summary

Here are the main topics covered in this chapter:

- This chapter covered the basics of troubleshooting and looked at the fundamentals of this vast subject.
- This chapter described a structured troubleshooting methodology.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 21-2 lists these key topics and the page number where each is found.

Key
Topic

Table 21-2 Key Topics for Chapter 21

Key Topic Element	Description	Page Number
Figure 21-1	Simplified Troubleshooting Flow	491
Table 21-1	Steps to Diagnose a Problem	491
Figure 21-2	Structured Troubleshooting Approach	492
Step list	Steps in the CompTIA Network+ structured troubleshooting methodology	493

Complete Tables and Lists from Memory

Print a copy of Appendix B, “Memory Tables,” or at least the section for this chapter, and complete as many of the tables as possible from memory. Appendix C, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

bottom-to-top troubleshooting, divide-and-conquer troubleshooting, top-to-bottom troubleshooting

Additional Resource

Use a Troubleshooting Methodology for More Efficient IT Support:
<https://www.comptia.org/blog/troubleshooting-methodology>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. Which of the following is often the first step in a structured troubleshooting methodology?
 - a. Test possible causes.
 - b. Create an action plan.
 - c. Establish a theory of probable cause.
 - d. Identify the problem.
2. Which of the following comprise a simplified troubleshooting flow? (Choose three.)
 - a. Problem resolution
 - b. Problem monitoring
 - c. Problem diagnosis
 - d. Problem report
3. What step is most likely to follow the “establish a plan of action” step in a structured troubleshooting methodology?
 - a. Implement the solution.
 - b. Test the hypothesis.
 - c. Define the problem.
 - d. Hypothesize the possible cause.
4. Which networking troubleshooting methodology step would include establishing a new theory or escalating if the theory is not confirmed?
 - a. Step 1: Identify the problem.
 - b. Step 4: Establish a plan of action to resolve the problem and identify potential effects.
 - c. Step 2: Establish a theory of probable cause.
 - d. Step 3: Test the theory to determine the cause.

5. You have verified full system functionality and implemented preventive measures. What should you do next?
 - a. Question the obvious and duplicate the problem.
 - b. Create a differential backup plan.
 - c. Document findings, actions, outcomes, and lessons learned.
 - d. Gather information and create a baseline.
6. In the networking troubleshooting methodology, which of the following are included in Step 2, “Establish a theory of probable cause”? (Choose two.)
 - a. Top-to-bottom/bottom-to-top OSI model
 - b. Divide and conquer
 - c. If theory is confirmed, determine next steps to resolve problem
 - d. If theory is not confirmed, establish a new theory or escalate

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This chapter covers the following topics related to Objective 5.2 (Given a scenario, troubleshoot common cabling and physical interface issues) of the CompTIA Network+ N10-009 certification exam:

- Cable issues
 - Incorrect cable
 - Single mode vs. multimode
 - Category 5/6/7/8
 - Shielded twisted pair (STP) vs. unshielded twisted pair (UTP)
 - Signal degradation
 - Crosstalk
 - Interference
 - Attenuation
 - Improper termination
 - Transmitter (TX)/Receiver (RX) transposed
- Interface issues
 - Increasing interface counters
 - Cyclic redundancy check (CRC)
 - Runt
 - Giant
 - Drop
 - Port status
 - Error disabled
 - Administratively down
 - Suspended
- Hardware issues
 - Power over Ethernet (PoE)
 - Power budget exceeded
 - Incorrect standard
 - Transceivers
 - Mismatch
 - Signal strength

Troubleshoot Common Cabling Problems

A lot can go wrong when it comes to the physical media (cables) that make up a network. But the great news is that there are more tools and techniques than ever before to proactively head off physical layer issues.

This chapter ensures that you understand the common problems to watch out for, as well as the tools that the best in the industry commonly use to solve these many potential issues.

Foundation Topics

Specifications and Limitations

There are some very basic requirements when it comes to thinking about the physical layer media of your infrastructure. You need to focus on at least these three aspects of the specifications and limitations for the media:

- **Throughput:** This is a measure of the total amount of data that media can send over a given time period.
- **Speed:** This is a measure of the data send rate; for example, Cat 6 media supports a maximum speed of 10Gbps.
- **Distance:** For both wired and wireless media, as you get farther away from the sender over the media, you start to suffer the effects of attenuation (that is, a weakening of the signal over distance). You need to pay attention to the maximum distances supported by various physical layer media options. For example, Cat 6 has become the minimum standard for new network cable installations. And while the speed of Cat 6 (10Gbps) is exciting, Cat 6 supports distances only up to 55 meters.

Common Cable Issues

The following are some considerations you should keep in mind when considering physical cable:

Key Topic

- **Single mode versus multimode:** Remember to select the appropriate type of fiber cable, with your main two options being single mode and multimode. *Single mode* fiber features a smaller core diameter supporting a single light mode. *Multimode* fiber, with its larger core diameter, supports multiple light modes and is ideal for shorter distances and bandwidth-hungry applications. Note that these types of applications are often found in the LAN portion of enterprise networks today.
- **Shielded versus unshielded:** Physical cable often needs to be twisted or shielded (or both) in order not to suffer from electromagnetic interference (EMI). For example, in *shielded twisted-pair (STP)* cables, a metallic shielding covers the twisted pair inside the cable. This shielding helps protect against EMI, but it does so at a cost: These types of cables can be much more expensive than unshielded cables due to the shielding.

With *unshielded twisted-pair (UTP)* cables, the twists in the cable are tighter to reduce EMI. This eliminates the need for the expensive shielding.

- **Category 5/6/7/8:** Choosing the correct category of UTP cable is crucial for optimizing network performance, ensuring signal integrity, minimizing interference, and future-proofing the infrastructure. Different UTP categories (5/6/7/8) offer varying levels of bandwidth, data transmission speeds, and resistance to crosstalk and EMI.
- **Plenum and riser rated:** If a twisted-pair cable is to be installed under raised flooring or in an open-air return, fire codes must be considered. For example, imagine that a fire breaks out in a building. If the outer insulation of a twisted-pair cable catches fire or starts to melt, it could release toxic fumes. If those toxic fumes were released in a location such as an open-air return, those fumes could be spread throughout a building, posing a huge health risk.

To mitigate the concern of pumping poisonous gas throughout a building's heating, ventilation, and air-conditioning (HVAC) system, plenum-rated cabling can be used. The outer insulation of a plenum twisted-pair cable is fire retardant; in addition, some plenum cabling uses a fluorinated ethylene polymer (FEP) or a low-smoke polyvinyl chloride (PVC) to minimize dangerous fumes.

NOTE Check your local fire codes before installing network cabling.

A *riser* is a vertical area that passes from one floor to another floor inside a building. Riser-rated cables are cables that run through risers; they should be fire-proof to prevent flames from traveling up the cable. However, the fire rating requirements for riser areas are less strict than the requirements for plenum areas.

- **Attenuation:** As mentioned earlier in this chapter, distance is one of the enemies of cabling. *Attenuation* is the loss of signal experienced as data transmits over distance and across the network medium. The longer the signal travels through the media, the more it weakens (known as *signal degradation*). It is very important to know the maximum distances for the various media in your enterprise network. Attenuation is measured in decibels (dB), which is covered shortly.
- **Interference:** While they are not as sensitive to *interference* as wireless media, cables can suffer from EMI and other types of interference that cause signal degradation. Just as with wireless networks, you need to do proper testing for interference issues before deploying physical media. In addition, to reduce interference, you should follow the recommendations of the cable manufacturer.

- **Crosstalk:** *Crosstalk* refers to the unwanted transfer of signals between communication channels, often causing interference and degradation of the signal quality. This phenomenon is prevalent in various communication systems, including telephony, networking, and audio equipment. Crosstalk occurs when a signal transmitted on one circuit or channel creates an undesired effect on another circuit or channel. It can be caused by physical proximity of the wires, EMI, or poor shielding and insulation. Effective crosstalk management is essential to ensure clear and reliable communication, particularly in environments with high data transmission rates and dense wiring.
- **Decibel (dB) loss:** If physical cable has lots of different splits or splices in it, there might be a great deal of dB loss. This loss is also more prone to happen as the signal needs to go farther and farther in the media.
- **Bad cables, incorrect pinouts, or bent pins:** Faulty cables (with electrical characteristics preventing successful transmission) or faulty connectors (which do not properly make connections) can prevent successful data transmission at Layer 1. A bad cable could simply be an incorrect category of cable being used for a specific purpose. For example, using a Cat 5 cable (instead of a Cat 6 or higher cable) to connect two 1000BASE-TX devices would result in data corruption. Bent pins in a connector or incorrect pinouts could also cause data to become corrupted.
- **Bad ports:** A physical port on a network device might be bad. You can often easily confirm whether a port is bad by looking at the network connection LED and status indicators.
- **Open/short:** An *open* is a broken strand of copper that prevents current from flowing through a circuit. A *short* occurs when two copper connectors touch each other, resulting in current flowing through that short rather than through the attached electrical circuit because the short has lower resistance.
- **Light-emitting diode (LED) status indicators:** Many network devices are very specialized and require special types of indicators. Oftentimes, indicators are implemented in the form of special LED status indicators. Thanks to such indicators, with a single glance, a network administrator can ascertain whether there is a problem with the network device. For example, a solid green LED for a network card normally indicates that the card is connected or receiving a signal. If there are no lights present or if the lights are orange or red, the network may not be connected properly or may be receiving a signal from the network.
- **Speed and duplex issues:** Speed and duplex mismatches can be tricky to troubleshoot in a network, especially considering that connectivity is often maintained (although at unacceptable levels). Auto-negotiation of speed and

duplex between network devices should be tested and verified before production deployment.

- **Transmitter (TX) and receiver (RX) transposed:** Some Ethernet switches support *media-dependent interface crossover (MDIX)*, which allows a switch port to properly configure its leads as transmit (TX) or receive (RX) leads. You can interconnect such switches by using a straight-through cable (as opposed to a crossover cable). However, if a network device does not support MDIX, it needs a crossover cable in order for its TX leads to connect to the RX leads on a connected device and vice versa. Therefore, you must take care when selecting cable types for interconnecting network components.
- **Dirty optical cables:** An often-overlooked aspect of fiber-optic media care and maintenance is keeping the fiber-optic connector end faces clean. A dirty fiber connection can either slow down or completely inhibit network traffic. One method to clean optical cables is to use a combination of wet and dry cleaning approaches. For example, you might use a small amount of solvent on a wiping material and immediately dry the surface.
- **Improper termination:** Improper termination occurs when the ends of a cable are not correctly connected to the connectors or devices, leading to poor signal transmission and potential data loss. Symptoms of improper termination include intermittent connectivity, slow network performance, and frequent data errors. This problem can result from incorrect wiring, loose connections, or the use of inappropriate connectors. Ensuring proper termination involves following standardized wiring schemes, using the right tools, and double-checking connections to maintain the integrity and performance of the cabling system.

Common Interface Issues

Although many things can go wrong with your cables and interfaces, fortunately, network operating systems will typically help you by displaying *interface counters* to signal problems. Here are some of the more common interface metrics that you should be aware of:

Key Topic

- **Cyclic redundancy check:** *Cyclic redundancy check (CRC)* errors on a network device indicate that data packets have been corrupted during transmission. These errors occur when the receiving device's calculated CRC value does not match the value sent with the data packet, signaling that the data has been altered or damaged. What can cause these errors? Typically, CRC errors are the result of faulty cables, EMI, or hardware problems.

- **Runts:** Is your network interface reporting *runts* on the network? Runts refer to data packets that are smaller than the minimum allowed size. For Ethernet networks, this is typically less than 64 bytes. What could cause such packet sizes? Typically, the issue is with collisions, faulty hardware, or improper network configurations.
- **Giants:** As you might guess, *giant* errors are essentially the opposite of runt errors. Giants occur when data packets exceed the maximum allowed size. This is typically 1518 bytes for Ethernet networks. These oversized packets can result from faulty hardware, misconfigurations, or software failures.
- **Drops:** Interface *drop* errors occur when a network interface discards incoming or outgoing packets due to congestion, buffer overflows, or other issues. These drops most often result from excessive network traffic, insufficient bandwidth, or hardware limitations.

Another important characteristic of your interfaces that you should monitor is the *port status*. The following are three important port statuses that you should monitor:

- **Error disabled:** A port might enter the *error disabled* status when some violation has occurred on the network. For example, a switch port might enter the error disabled state when it is configured for BPDU Guard and a BPDU arrives on the interface. This is desired behavior, since the introduction of a rogue switch could have disastrous impacts on the Layer 2 infrastructure.
- **Administratively down:** Network interfaces will often default to this status, requiring network administrators to manually enable those interfaces they plan to use. On a Cisco router, for example, you would use the **no shutdown** command to enable an interface and force it to leave the *administratively down* status.
- **Suspended:** While an error disabled state is common when there has been a violation based on a network engineer's configuration of some protection mechanism, a *suspended* port is often the result of an error in configuration or a hardware problem. Like error disabled ports, a suspended port will not send or receive traffic.

Common Hardware Issues

At this point in the chapter, you might be thinking, "When is this list of potential problems going to end?" Well, let's round out the list of potential issues with a couple of items that are more hardware-centric:

**Key
Topic**

- **Power over Ethernet (PoE):** Power over Ethernet (PoE) is a technology that makes it simple to connect devices such as cameras and access points to the network and have these devices get power from the single network connection. This relieves you from having to find an outlet for plugging in a number of smart devices. For PoE to work, you must use Cat 5e or better media. You also must ensure that you are providing enough total power for the device to operate properly. A common error you might see should you violate this principle is the *power budget exceeded* error. Finally, there are several PoE standards in use in networking today. Be sure you are following the correct standard in your hardware and software configurations.
- **Incorrect transceivers:** A *transceiver* is a device that is able to both send and receive analog or digital signals. You must be sure to select transceivers carefully. The transceiver you select for a network device must match the cable type used and the wavelengths (*signal strength*) that are in use on the network. A *transceiver mismatch* is often best avoided through careful documentation review.

Common Tools

As we have seen in this chapter, there is a lot that can go wrong in your modern network. We have examined cabling issues, interface issues, and hardware issues. Fortunately, this chapter does contain some great news. There are many tools you can use to attack these problems and even prevent them from occurring. In this chapter, we present the tools that you can use to help identify or solve problems related to your interfaces, cables, and hardware. In Chapter 25, “Network Troubleshooting Tools,” you will learn about many more tools at your disposal for more general network troubleshooting. Some of our cable-centric tools include

- **Cable crimper:** A crimper, as pictured in Figure 22-1, can be used to attach a connector (for example, an RJ45 connector) to the end of a UTP cable. To accompany a crimper, you might want to purchase a spool of cable (for example, Category 6 or higher UTP cable) and a box of RJ45 connectors. You will then be equipped to make your own Ethernet patch cables, which might be less expensive than buying pre-terminated UTP cables. Making your own is also convenient when you need a patch cable of a nonstandard length or when you need a nonstandard pinout on the RJ45 connectors (for example, when you need a T1 crossover cable). Many crimpers have a built-in wire stripper and wire snip function as well.



Figure 22-1 Crimper

- **Punchdown tool:** When terminating wires on a punchdown block (for example, a 110 block), you insert an insulated wire between two contact blades. These blades cut through the insulation and make electrical contact with the inner wire. As a result, you do not have to strip off the insulation. However, if you attempt to insert the wire between the two contact blades using a screwdriver, for example, the blades might be damaged to the point where they will not make a good connection. Therefore, you should use a punchdown tool, which is designed to properly insert an insulated wire between the two contact blades without damaging the blades.
- **Loopback plug:** When troubleshooting a network device, you might want to confirm that a network interface is functional (for example, ensure that it can transmit and receive traffic). One way to perform such a test is to attach a loopback plug to a network interface and run diagnostic software designed to use the loopback plug. A loopback plug takes the transmit pins on an Ethernet connector and connects them to the receive pins, such that everything that is transmitted is received back on the interface. Similarly, a fiber-optic loopback plug, as shown in Figure 22-2, interconnects a fiber connector's transmit fiber with a connector's receive fiber. The diagnostic software can then transmit traffic out of a network interface and confirm its successful reception on that same interface.
- **Optical time-domain reflectometer (OTDR) and time-domain reflectometer (TDR):** Suppose that you have been troubleshooting a network cable (either copper or fiber optic), and you determine that there is a break in or physical damage to the cable. Identifying exactly where a break or damage exists in a long length of cable can be problematic. Fortunately, you can use a time-domain reflectometer (TDR) for copper cabling or an optical time-domain reflectometer (OTDR) for fiber-optic cabling to locate a cable fault. Both light and electricity travel at speeds approaching 3×10^8 meters

per second (approximately 186,000 miles per second), although the speeds are a bit slower and vary depending on the media. A TDR can send an electrical signal down a copper cable or an OTDR can send light down a fiber-optic cable, and when the electrical signal or light encounters a cable fault, a portion of the electric signal or light reflects back to the source. Based on the speed of electricity or light in the medium and the amount of time required for the reflected electric signal or light to be returned to the source, a TDR or an OTDR can mathematically determine where the cable fault lies. Figure 22-3 shows an example of an OTDR.



Figure 22-2 Fiber-Optic Loopback Plug (Photo Courtesy of DigiKey Corporation, <https://www.digikey.com>)



Figure 22-3 Optical Time-Domain Reflectometer (Photo Courtesy of Coral-i Solutions, <https://www.coral-i.co.za>)

- **Multimeter:** When working with copper cabling (as opposed to fiber-optic cabling), you can use a multimeter to check a variety of electrical characteristics in a cable, including resistance (in ohms), current (in amps), and voltage (in volts). Figure 22-4 shows an example of a multimeter.



Figure 22-4 Multimeter

For example, you could use the ohmmeter function of a multimeter (the resistance feature) to check the continuity of an Ethernet cable. If you connect the two leads of a multimeter to two pins of a cable, the resulting resistance is approximately 0 ohms if those two pins are connected, and the resulting resistance approaches an infinite number of ohms if the pins do not connect with one another.

Another common use of a multimeter is for measuring voltage. For example, you could check the leads of an Ethernet cable to see whether DC voltage is being applied to a device requiring PoE.

- **Wire map tester:** Wire map testing enables you to determine whether a wire has continuity and whether each conductor of a four-pair cable is correctly connected to the corresponding pin at the far end. Wire map testing tests for opens, shorts, reversed pairs, crossed pairs, and split pairs.

- **Fusion splicer:** Fusion splicing is a way to join two fibers by using heat. Prepared fiber ends are placed in the splicer and automatically aligned and then fused together. Fusion splicing ensures greater reliability with less light being scattered or reflected back by the splice. If you follow the instructions for a fusion splicer, you should end up with an optical cable that is as strong as the original cable.
- **Spectrum analyzer:** A spectrum analyzer measures the magnitude of an input signal versus frequency within the full frequency range of the instrument. The primary use is to measure the power of the spectrum of known and unknown signals.
- **Snips/cutters:** As a network administrator, you often need to precisely cut or snip cables and cabling accessories. It is important that you have a pair of well-maintained snips or cutters in your supply bag at all times.
- **Cable stripper:** Cable strippers, which allow you to remove the plastic shielding from cables, are often built in to cable crimpers.
- **Fiber light meter:** A fiber light meter is a device used to measure the power in an optical signal. It is a device for testing average power in fiber-optic systems.

Real-World Case Study

At Acme, Inc., the IT department has recently embarked on a comprehensive training program focused on mastering the most common cable, interface, and hardware issues that the company encounters. Recognizing that even small, seemingly insignificant problems can lead to significant downtime and disruption, the leadership team decided that it was essential to equip the IT staff with the skills to quickly diagnose and resolve these issues. The training covers a wide range of topics, from identifying faulty cables and connectors to troubleshooting interface problems that can occur between different types of hardware.

The program includes hands-on workshops where IT staff can practice identifying and resolving these issues in a controlled environment. For instance, technicians learn to differentiate between various types of cables, such as Ethernet and fiber optics, and understand how to spot signs of wear and tear that could lead to connectivity problems. Interface training covers the intricacies of network interface cards (NICs), USB ports, and other connection points, ensuring that IT personnel are adept at recognizing and resolving compatibility and performance issues.

Finally, the workshops facilitate training with the latest in network troubleshooting tools. IT staff members get to practice what they have learned with hands-on tool usage in simulated failure scenarios.

Summary

Here are the main topics covered in this chapter:

- This chapter covered cabling considerations and common issues.
- This chapter examined potential issues you should be aware of when it comes to network device interfaces, and the network device hardware itself.
- Finally, this chapter also covered common tools used to support the network media.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 22-1 lists these key topics and the page number where each is found.



Table 22-1 Key Topics for Chapter 22

Key Topic Element	Description	Page Number
List	Common cable issues	502
List	Common interface issues	505
List	Common hardware issues	507

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

administratively down, attenuation, crosstalk, cyclic redundancy check (CRC), drops, error disabled, giants, interference, multimode, Power over Ethernet (PoE), runts, shielded twisted pair (STP), single mode, suspended, unshielded twisted pair (UTP)

Additional Resources

Network Cable Testers: <https://www.flukenetworks.com/expertise/learn-about/cable-testing>

How to Use a Multimeter: <https://www.youtube.com/watch?v=ts0EVc9vXcs>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. What type of cable must be used for PoE with network devices?
 - a. Cat 5e or greater
 - b. Cat 3 or greater
 - c. Cat 5 or greater
 - d. Cat 4 or greater
2. Which of the following can be a major issue for fiber-optic media?
 - a. Incorrect pinouts
 - b. Dirty optical media
 - c. EMI
 - d. dB loss
3. What type of cable testing device permits a network device to communicate with itself in order to test its ability to send and receive traffic?
 - a. Cable tester
 - b. Loopback plug
 - c. Cable crimper
 - d. Fusion splicer
 - e. Tone generator
4. Which types of cables would you install to meet most building codes today and prevent fires and poisonous gases from spreading? (Choose two.)
 - a. UTP rated
 - b. Rollover rated
 - c. Plenum rated
 - d. Riser rated

5. Which of the following terms are directly related to the weakening or loss of a signal as it travels farther down the network media? (Choose two.)
 - a. Attenuation
 - b. Throughput
 - c. OTDR
 - d. Interference
 - e. Decibel (dB) loss
6. You have configured your switch with a Spanning Tree Protocol protection feature. There has been a violation of this feature due to a new switch that has been introduced to your topology. What is the most likely status of the switch port where the violation occurred?
 - a. Administratively down
 - b. Error disabled
 - c. Suspended
 - d. Normal
7. You have a link between two network interfaces (devices) that is operating inefficiently. What would most likely be the cause of this problem?
 - a. Incorrect pinout
 - b. Bad ports
 - c. Duplex mismatch
 - d. Failed NIC
8. Which statements are true regarding fiber-optic cabling? (Choose two.)
 - a. It is immune to EMI.
 - b. Troubleshooting equipment is less expensive than UTP.
 - c. Multimode fiber is used for longer distances.
 - d. It is more difficult to install than UTP.
9. What occurs when a network interface discards incoming or outgoing packets due to congestion, buffer overflows, or other issues?
 - a. Giants
 - b. Runts
 - c. Drops
 - d. Improper termination

10. What occurs when a signal transmitted on one circuit or channel creates an undesired effect on another circuit or channel?
- a. Attenuation
 - b. Crosstalk
 - c. dB loss
 - d. Suspended port



This chapter covers the following topics related to Objective 5.3 (Given a scenario, troubleshoot common issues with network services) of the CompTIA Network+ N10-009 certification exam:

- Switching issues
 - STP
 - Network loops
 - Root bridge selection
 - Port roles
 - Port states
 - Incorrect VLAN assignment
 - ACLs
- Route selection
 - Routing tables
 - Default routes
- Address pool exhaustion
- Incorrect default gateway
- Incorrect IP address
 - Duplicate IP address
- Incorrect subnet mask

Troubleshoot Common Issues with Network Services

In the previous chapter, our troubleshooting focus was entrenched in the physical layer. We examined just some of the many things that can go wrong with things like cables, interfaces, and even the wireless radio frequency signals traveling through the air.

This chapter details many of the network service issues you might encounter today, why they occur, and what you can often do about them. It is worth noting that sometimes you need to troubleshoot and verify that things are working properly at the lower layers (like the physical layer) before you begin looking at the potential for network service problems.

Foundation Topics

Considerations for General Network Troubleshooting

How much can go wrong in a network today? *A lot!* In fact, the list of potential issues grows all the time as new services and data types are provided by networks. Fortunately, there are some common considerations that can help you greatly as you approach this vast and challenging topic. Here are some common considerations and best practices that you should keep in mind:

Key Topic

- **Device configuration:** You should consistently review the configurations of network devices to ensure that your network is not suffering from *configuration drift* (that is, configurations gradually changing from their original documented defaults for a network). Many software packages and tools, such as Ansible, can assist you in making sure all your network devices remain in a constant, stable, and known configuration state.
- **Routing tables:** It is important to carefully monitor and analyze the routing tables of Layer 3 devices. Forwarding decisions (*route selection*) are critical to the operation of a network, and you want to ensure that packets are flowing exactly as you want and expect them to. Remember, these routing tables will also typically include **default routes** that are of critical importance. The default route is used when no specific route exists for a network destination. Always remember to check routing tables to make certain that the routes within are the most cost effective in terms of hops and routes taken.
- **Interface status:** One embarrassing network issue that you do not want to experience is the administratively down interface scenario. In this scenario, you try desperately to reach a key network device, only to discover that you failed to enable the interface that you are trying to contact. Remember that there are simple commands you can use, and even GUI-based web pages that can give you status information on your critical network interfaces. Knowing which interfaces are healthy and which are not can be critical in network troubleshooting.
- **Incorrect network layer settings:** Often the status of an interface in a network is contingent on the network layer settings. The physical and data link layer settings might be fine (for example, a Cisco device displays the status as UP/UP), but if the IP settings are incorrect, the physical and data link layer health of the interface is not of great benefit to you. The following settings should be perfect: default gateway IP address, subnet mask, IP address, and DNS address or addresses. Think about the symptoms you might see for each of these potential misconfigurations. For example, if you have an *incorrect*

default gateway IP address, you might be able to communicate with your local subnet peers just fine, but you will not be able to successfully reach any remote destinations—which would not be good! What if you have an *incorrect subnet mask* assignment? Now the system will have issues communicating with its local subnet peers, and possibly the remote systems as well. Notice this is basically the same scenario if your IP address assignment is incorrect. If the address is far off from what it should be, you will not be able to communicate with local or remote hosts. You must ensure proper IP addresses, subnet masks, default gateways, and DNS assignments. There is simply no question about this.

- **VLAN assignment:** Remember that a virtual local area network (VLAN) allows you to create groups of users and systems and segment them on the network. This segmentation lets you hide segments of the network from other segments and thereby control access. As a result, *incorrect VLAN assignments* can cause communication issues within the network. In fact, these types of misconfigurations can be more difficult to pinpoint because the VLAN assignment is “behind the scenes” in the network. These issues also present no obvious signs of trouble outside of communication difficulties.
- **Network performance baselines:** As you have probably noticed throughout this book, it is critical that you know the details about how your network performs under “normal” conditions and what its speeds and throughput really are. (How can you know that you are having a problem if you do not even know what your network should be performing like when there are no problems?) Fortunately, as this book has discussed, there are many tools and techniques for capturing accurate and detailed network performance baselines and using the data you collect to keep tabs on the network. Remember that creating baselines is only a part of the equation; you also have to analyze them. Unfortunately, too many network administrators collect information that they never do anything with. Be sure to examine the baseline data on a regular basis and track current conditions to see what needs to be improved upon.

Common Network Service Issues

The following list of common general network issues may be a bit intimidating, and it is not even an exhaustive list. As a network administrator and as a candidate for the CompTIA Network+ certification, you need to understand the following common issues:



- **Duplicate IP address:** Hosts on a subnet should have unique IP addresses. If two hosts are configured with the same IP address, unpredictable traffic patterns can occur for those hosts. One reason (of many) this might happen is

because a DHCP server is inadvertently leasing out an address that has been statically assigned on the network. These statically assigned addresses should be excluded from the DHCP scope.

- **Duplicate MAC address:** Although it's rarer than a duplicate IP address on a network, a MAC address could be duplicated. A MAC address on a network is typically taken from the address that is "burned in" to the hardware by the device vendor. To ensure uniqueness between vendors, each network device vendor is assigned a MAC prefix, and the vendor can control the remaining bits. Therefore, when a MAC address is duplicated on a network, it is most often a human error that occurs when MAC addresses are software defined to override the hardware addresses that are burned in.
- **Expired IP address:** Perhaps a leased DHCP address has expired for a workstation. This might be the case, for example, if the lease duration has been set to a small value, perhaps because of an IP address shortage in a particular area of the network.
- **DHCP scope exhaustion:** If your end-user network device population has grown, you might have issues with *address pool exhaustion* that are innocent in nature. DHCP scope exhaustion refers to the scope being completely out of IP addresses that it can lease to clients. If your population is outgrowing your design, a redesign is expected, and you should not be surprised by this. Sadly, issues are not always innocent in nature. Cybercriminals can launch attacks that intentionally exhaust the address space in the DHCP pool. DHCP scope exhaustion attacks can be very disruptive because you will not know it is coming. Suddenly, clients in your network cannot communicate properly because they have lost their IP address configuration.
- **Rogue DHCP server:** This security concern could certainly cause troubleshooting headaches on a network due to the incorrect IP configurations that could result. When rogue DHCP servers appear on a network, it is often because users have brought in their own devices, either maliciously or accidentally. Fortunately, as you learned in our chapters regarding network security, there are many features to help guard against this problem, including DHCP snooping, port security, and even rogue AP detection mechanisms in the latest wireless LAN technologies.
- **Untrusted SSL certificate:** If you are using certificates for authentication, you might have an issue with an SSL certificate preventing such authentication. Perhaps a certificate is invalid due to time, or perhaps it has been manually revoked for some reason.
- **Incorrect time:** Because many network configurations rely on the correct time, you might have problems in a network due to a workstation, server, or

network appliance having the wrong time set. Remember that you should strongly consider using Network Time Protocol (NTP) or a competing technology to help ensure that your network devices are all set to the correct time.

- **Blocked TCP/UDP ports, services, or addresses:** A trouble ticket may be about blocked TCP or UDP ports. This type of blockage is often the result of misconfigured security devices in the network path. There might be other security mechanisms that are blocking specific IP addresses or services. These are all worthy of inspection when you are experiencing unexpected connectivity issues in a network.
- **Incorrect firewall settings:** Host-based firewall settings and network-based firewall settings may cause problems. Firewalls are great, but you need to make sure they are not blocking traffic that you need to pass through the network.
- **Incorrect access control list (ACL) settings:** ACLs can have effects on network traffic, and if their settings are incorrect, you could end up with trouble tickets. Imagine an ACL setting that accidentally blocks TCP port 443. This would end up stopping almost all Internet traffic as a result of the HTTPS protocol being blocked.
- **Unresponsive service:** Services that have failed in a network are common sources of problems.
- **Collisions:** You should not see any collisions taking place if you have built a network using modern techniques and standards. Remember that microsegmentation of a network means keeping single devices in collision domains so that you do not experience collisions on the network. Often, if you are seeing collisions on a network, it is an indicator that you have a misconfiguration or a hardware device that is malfunctioning.
- **Broadcast storms:** It might sound crazy, but it is possible to go to a Layer 2 switch and turn off *Spanning Tree Protocol (STP)*. This is *never* recommended. Even in cases where you don't need STP because you have used technologies like port channels, you need to make sure STP is still enabled and running in the background. If STP isn't running, a broadcast storm may occur as a result of a Layer 2 loop (often just termed *network loops*). You might experience broadcast storms for other reasons as well, such as a NIC malfunctioning and flooding the network with erroneous broadcast frames.
- **STP misconfigurations:** While there are many potential Spanning Tree Protocol tuning configurations you can make, one of the most critical configurations is the *root bridge* selection. This selection influences all of the *port roles* and *port states* of the spanning tree topology. If this root bridge is incorrect or suboptimal, it can have devastating effects on the network. Imagine how slow a

network might perform if the root bridge ends up being a very underpowered, access layer, basic Layer 2 switch. Most administrators will be sure to configure the best possible root bridge and then place STP enhancements like Root Guard and BPDU Guard to ensure this device remains the root bridge.

- **Multicast flooding:** You need to be concerned about broadcast, unknown unicast, and multicast (BUM) traffic in your network. This is the traffic that Layer 2 switches must flood. You want to minimize the amount of traffic flooding your network must perform, and you therefore often need to closely analyze the amount of BUM traffic flooding. In addition, you can use protection mechanisms such as network storm control to guard against excessive amounts of BUM traffic hampering network operations.
- **Asymmetrical routing:** Sometimes packets take very different paths back to you than they took toward the destination. This situation, called asymmetrical routing, is not always a bad thing, and sometimes you even engineer it to happen for some specific reason. Often, however, asymmetrical routing is not wanted and can cause issues.
- **Switching loops:** If STP is not running correctly—or not running at all—it will not be long before you have broadcast storms and switching loops. These are the enemies that STP is engineered to protect you from.
- **Routing loops:** The dynamic routing protocols in use today do an excellent job of protecting themselves against loops. In fact, a loop could never get terribly bad because of the Time-to-Live (TTL) mechanism at Layer 3, which prevents a packet from circulating on the network endlessly in a loop. Layer 3 routing loops can still happen, though, especially when redistribution between routing protocols is configured incorrectly, and they are still awful. Be very careful when performing redistribution.
- **Missing routes:** When a user indicates that a remote destination must be down, be sure to run through your methodical troubleshooting steps; in many cases you may discover that the issue is simply a missing route in the local routing table. Remember that networks change, and you need to ensure that your dynamic routing protocols (or even static routes) are being updated properly. When you are troubleshooting, don't forget the excellent tools at your disposal, such as **ping**, **tracert**, and the other useful utilities that are covered in Chapter 25, "Network Troubleshooting Tools."
- **Hardware failure:** Network devices or parts sometimes fail, and components can burn out. Such failures are rare, but you need to consider them in your troubleshooting flow.

- **Low optical link budget:** A fiber-optic link budget, also known as a *loss budget*, indicates the total acceptable amount of optical power loss (expressed in decibels) that a fiber-optic link can withstand. Such power loss results from cables, connectors, splices, and couplers in the installed system.
- **BYOD challenges:** Perhaps the investigation of a trouble ticket reveals that an end user is trying to interact with your network by using a tablet the user brought in from home. Whether or not your organization's bring your own device (BYOD) policy allows this type of use and how you control this potential nightmare—such as through a mobile device management (MDM) policy—is an area of networking that deserves your attention.
- **Licensed feature issues:** Sometimes a network does not function properly because the organization lacks the appropriate licensing. Licensing of network devices can get quite complicated. You might need a license for a network device itself, and then you might also need additional licenses for certain features or levels of scalability.
- **Network performance issues:** Sometimes you know exactly why a network is not performing well. For example, you might have to fail over to another service provider and a very slow backup WAN link when the main service provider is down for maintenance. It is always an excellent idea to inform end users about these kinds of issues up front to prevent the disruption that would be caused by a flood of trouble tickets.

Real-World Case Study

To be better prepared for network issues, Acme, Inc. has introduced a training program for its support team that focuses on common issues that have been occurring in the network. This training focuses on the following areas:

- Spanning Tree Protocol (STP) issues
- Layer 2 configurations such as VLAN assignments
- Route selection issues
- IP addressing issues

IT personnel are receiving training on the most common symptoms and causes for each of these areas. The team is also receiving training on the most effective tools and techniques that can be used for each area.

The IT team is also receiving recurring training on the overall troubleshooting methodology, with a focus on documentation maintenance and improvements. Since initiating this training program, Acme, Inc. has reported far fewer hours committed to troubleshooting, and much more up-to-date and useful documentation.

Summary

Here are the main topics covered in this chapter:

- This chapter focused on the considerations you frequently need to make while engaged in general network troubleshooting.
- This chapter examined many common problematic issues in networks today.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 23-1 lists these key topics and the page number where each is found.



Table 23-1 Key Topics for Chapter 23

Key Topic Element	Description	Page Number
List	Common considerations for general network troubleshooting	518
List	Common network service issues	519

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

default route, network loops, root bridge, routing table, Spanning Tree Protocol (STP)

Additional Resources

How to Troubleshoot Issues in Computer Networks: <https://www.youtube.com/watch?v=kthHizueMiY>

What Is STP (Spanning Tree Protocol)?: https://www.youtube.com/watch?v=i_q-kIgz9Wk

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. What might you want to investigate if you can reach a web server by using its IP address but not its name?
 - a. NTP
 - b. DHCP
 - c. DNS
 - d. ARP
2. What device would most likely be unreachable if your default gateway IP address is misconfigured on the client?
 - a. A remote web server
 - b. A local printer
 - c. Your neighbor’s laptop
 - d. Your loopback
3. In the absence of STP, what issues might result from a Layer 2 loop in a network? (Choose two.)
 - a. A router interface’s MTU decrementing
 - b. MAC address table corruption
 - c. Broadcast storms
 - d. Packet fragmentation
4. If you successfully ping from host A to host B, what can you conclude about host A?
 - a. OSI Layers 1–4 are functional.
 - b. OSI Layers 1–3 are functional.

- c. OSI Layers 1–7 are functional.
 - d. You must have a fully functional default gateway.
- 5. What configuration on a Layer 2 switch is used to define broadcast domains where they would not exist naturally on their own?
 - a. ARP
 - b. STP
 - c. VTP
 - d. VLAN
- 6. To what subnet does a host with the IP address 172.16.155.10/18 belong?
 - a. 172.16.0.0 /18
 - b. 172.16.96.0 /18
 - c. 172.16.128.0 /18
 - d. 172.16.154.0 /18
- 7. What will likely occur if STP is not running correctly—or not running at all? (Choose two.)
 - a. Broadcast storms
 - b. Microsegmentation
 - c. Low optical link budget
 - d. Switching loops
- 8. Which of the following refers to a lack of IP addresses that can be leased to clients?
 - a. Incorrect time
 - b. Address pool exhaustion
 - c. Untrusted SSL certificates
 - d. Incorrect ACL settings
- 9. An administrator excludes statically assigned addresses from a DHCP scope. What is the administrator trying to avoid?
 - a. Duplicate IP addresses
 - b. Duplicate MAC addresses
 - c. Expired IP address
 - d. Rogue DHCP server

10. In STP, what is the central reference point of the network topology, selected based on the lowest bridge ID, to which all other network switches determine their best path to ensure a loop-free network?
- a. Core switch
 - b. Root port
 - c. Root bridge
 - d. Designated port



This chapter covers the following topics related to Objective 5.4 (Given a scenario, troubleshoot common performance issues) of the CompTIA Network+ N10-009 certification exam:

- Congestion/contention
- Bottlenecking
- Bandwidth
 - Throughput capacity
- Latency
- Packet loss
- Jitter
- Wireless
 - Interference
 - Channel overlap
 - Signal degradation or loss
 - Insufficient wireless coverage
 - Client disassociation issues
 - Roaming misconfiguration

Troubleshoot Common Performance Issues

Our networks today are so important to the organization that poor performance is considered almost as big an issue as a lack of network functionality. In this chapter, we take a close look at some of the major factors in this important networking aspect. You will quickly notice that this chapter has a definite bias toward wireless networking performance. Troubleshooting wireless might be a close second to supporting local network printers—although typically it is not as labor intensive. Wireless has become an expected part of the enterprise network that is a fairly consistent source of troubleshooting for you and your department.

Foundation Topics

Network Performance Considerations

What are the enemies of a highly performant network? There are many. Let's begin this chapter by tackling some of the most important issues when it comes to your network's performance. By the way, you might recognize some of these issues as they have been mentioned in previous chapters.

Key Topic

- **Bandwidth:** Bandwidth refers to the maximum amount of data that can be transmitted over the network in a given amount of time, usually measured in bits per second (bps). Think of it as the width of a highway: the wider the highway, the more cars can travel on it at once. Similarly, higher bandwidth allows more data to flow through the network quickly and efficiently. If a network has low bandwidth, it can lead to slow Internet speeds and long loading times for websites and applications, affecting overall network performance.

NOTE Later in this chapter, we discuss network throughput capacity. This is often confused with the concept of bandwidth. Remember, bandwidth refers to the maximum amount of data that can be transmitted over a network in a given amount of time, whereas throughput is the actual amount of data successfully transmitted and received over the network during a specific time period, often affected by various factors such as network congestion, latency, and packet loss.

- **Congestion/contention:** Congestion and contention are important factors that affect how well a network performs. Congestion happens when there is too much traffic on the network, causing delays and data loss because the network can't handle all the data at once. Contention occurs when multiple devices or applications try to use the same network resources at the same time, leading to competition for bandwidth and slower performance. Both congestion and contention can make a network less efficient and slower, which is why managing them is crucial for maintaining a fast and reliable network.
- **Bottlenecks:** A network bottleneck is a point in the network where the flow of data is slowed down or restricted, much like a narrow section of a road that causes traffic to back up. This restriction can occur because of limited bandwidth, outdated hardware, or excessive network traffic. When a bottleneck happens, it can cause delays, slow down data transfer, and reduce the overall performance of the network.

- **Latency:** Latency is the time it takes for a data packet to travel from its source to its destination across a network and is one of the components that leads to network delays that frustrate users. Wireless networks may experience more delay than their wired counterparts. One reason for the increased delay is the use of carrier-sense multiple access with collision avoidance (CSMA/CA) in WLANs; in an attempt to avoid collisions, CSMA/CA introduces a random delay before data is transmitted. Another reason for the increased delay is the fact that all wireless devices associated with a single wireless AP are in the same collision domain, introducing the possibility of collisions (retransmissions), which can increase delay.
- **Packet loss:** Packet loss occurs when data packets traveling across a network fail to reach their destination, causing parts of the transmitted information to be missing. This can happen due to network congestion, faulty hardware, or poor signal quality. When packet loss occurs, it can lead to slow Internet speeds, disrupted voice or video calls, and incomplete data transfers.
- **Jitter:** Jitter refers to the variation in the time it takes for data packets to travel across a network. Instead of arriving at a consistent pace, packets can arrive irregularly, causing delays and disruptions. This can be especially problematic for activities that require a steady stream of data, like video calls or online gaming, where jitter can result in poor video quality, lag, and choppy audio. Notice that jitter can be far worse than simple latency. If there is a large latency in the network that is predictable, various mechanisms can be used to deal with this. On the other hand, when there is high jitter, it can be very problematic.

Wireless Performance Considerations

Chapter 5, “Transmission Media and Transceivers,” examines wireless standards. As you should recall from that chapter, there are big differences in the 802.11 standards when it comes to the specifications and limitations of each standard.

This section reviews just some of the key values and concepts that you need to be concerned about when troubleshooting WLAN deployments:

Key Topic

- **Speed:** Typically, when the speed of the wireless technology is described, it is actually the theoretical maximum speed that is actually being described. For example, the speed of 802.11n is typically said to be 300Mbps. But is this true? In short, the answer is no. A more accurate way to describe how a WLAN is actually performing is to look at throughput, described next.
- **Throughput:** Throughput (or *throughput capacity*) is typically used to indicate the amount of data you are able to send on the network media. In this case,

of course, the media is the radio frequencies that power wireless. Although the stated speed of 802.11n is 300Mbps, when you actually test the throughput, you might discover that the network is averaging 225Mbps. This actual throughput measurement provides a much more accurate picture of the capabilities of the network segment.

- **Distance:** Attenuation is a fact of life with network media. As the signal gets farther and farther away from the source, the signal weakens, and speed (and throughput) suffer. We often call this *signal degradation* or *loss*. Notice that in a wired infrastructure, the effect of distance can be much easier to predict. When you are troubleshooting wireless environments, it's important to keep in mind that all distances are not created equal. For example, there might be only half the distance between your access point (AP) and your client as between your AP and another client in your network. If there is a large source of WLAN interference in the shorter path, this might be the much more problematic client system.

NOTE There are actually many places in the wireless infrastructure where attenuation can be an issue. A careful analysis of the Wi-Fi design might find attenuation/*signal loss* resulting from the antenna cable in use. This might add to the existing and known problems of RF attenuation and signal loss.

- **Received signal strength indication (RSSI) signal strength:** When you are performing site surveys or just performing typical wireless troubleshooting, it is important to use metric information that you can trust and interpret accurately. The RSSI might be presented in different formats, depending on the software you are working with, but comparing your internal testing results can make this measurement extremely valuable. No matter how exactly you see the value, the RSSI is a measure of the power level being received by the local client you are testing with. In most software packages, the greater the RSSI value, the stronger the signal, and the better the chances for higher throughput.
- **Effective isotropic radiated power (EIRP)/power settings:** Another very valuable metric used with wireless today, EIRP, measures the maximum amount of power that could be radiated from an antenna. Of course, this value must incorporate the antenna gain and the transmitter power of the RF system. You most often see EIRP presented as decibels over isotropic, or dBi.

Other Wireless Considerations

Troubleshooting wireless networks might seem overwhelming compared with troubleshooting wired infrastructures. As described in the following sections, there are certainly many unique considerations when troubleshooting Wi-Fi networks. However, through study and practice, you will soon be as efficient at troubleshooting Wi-Fi as you are at troubleshooting a pesky network printer.

Antennas

As discussed in Chapter 11, antenna choices are often critical for the proper functioning of a wireless LAN. These choices will often directly impact the challenge of *insufficient wireless coverage*. You need to ensure that the most critical design goals were met:

- Double-check distances from the clients and APs.
- Test the coverage area of the antenna type and form factor selected.
- Test both the indoor and outdoor coverages.
- Test for interference with other APs.

Recall from Chapter 11 that there are multiple types of wireless antennas, and they fall into two broad categories, based on coverage area:

- **Omnidirectional:** The signal is spread in nearly a 360-degree radius around the antenna.
- **Unidirectional:** The signal is projected in a single, relatively precise direction.

NOTE The direction in which an antenna emits signal is termed *polarization*.

Frequencies and Channels

Keep in mind these key facts regarding frequencies and channels when troubleshooting wireless networks:



- WLANs use frequencies in the 2.4GHz–2.5GHz range or in the 5.725GHz–5.875GHz range.
- Within each band are specific frequencies (or *channels*) at which wireless devices operate.

- In the 2.4GHz band, channel frequencies are separated by 5MHz (with the exception of channel 14, which has 12MHz of separation from channel 13).
- A single channel's transmission can spread over a frequency range of 22MHz. As a result, five consecutive channels tend to overlap one another. We refer to this challenge as **channel overlap**. Remember, in the United States, the non-overlapping channels are 1, 6, and 11. To avoid **interference**, nearby wireless APs should use frequencies that do not overlap with one another.
- More and more 802.11ax (Wi-Fi 6) devices are currently available. This technology extends Wi-Fi into the 6GHz band.

More Considerations

An exhaustive list of WLAN considerations is beyond the scope of this book, but there are a few more you might encounter in a network and on the Network+ exam:

- **AP association time:** In larger enterprise environments, wireless LAN controllers (WLCs) often provide the control plane intelligence that APs lack. Key metrics to consider in this environment are how long it takes for a typical client to properly associate with an AP and how long it takes an AP to associate with a WLC. These are both points of latency as well as potential failures.
- **Site survey:** There is no substitute for a thorough, well-planned, well-documented site survey when it comes to planning and deploying wireless successfully. When implementing a production WLAN, planning is a much better option than simply trying and failing. A proper site survey can set you up for great success. It will help you determine where to place antennas and what types of antennas you need, and it will help you learn about potential sources of interference and other enemies of the WLAN. Fortunately, there are many options to choose from when it comes to site survey blueprints and software packages.

In a site survey, you use Wi-Fi and other wireless analyzers to understand and map out the wireless infrastructure. One output is a wireless heat map, which provides a visual method for understanding coverage and signal strength.

Common Wireless Issues

Network engineers are consistently thinking about what can go wrong. Fortunately, this type of doomsday thinking can be very valuable in networking. The following are some of the common issues you need to be aware of when it comes to wireless networking:

**Key
Topic**

- **Interference:** Wireless communication can be interrupted due to radio frequency interference (RFI). Common RFI sources that impact wireless networks include 2.4GHz cordless phones, microwave ovens, baby monitors, and gaming consoles.
- **Signal strength:** The RSSI value measures the power of a wireless signal. RSSI values vary based on distance from a wireless antenna and physical objects interfering with line-of-sight communication with a wireless antenna (for example, drywall, metal filing cabinets, elevator shafts). Some wireless networks automatically drop their wireless transmission rate when an RSSI value drops below a certain value.
- **Misconfiguration of wireless parameters:** For communication to occur, a variety of wireless parameters must match between a wireless client and a wireless AP. For example, the client needs to be using a wireless standard supported by the wireless AP (for example, IEEE 802.11g/n/ac/ax). Wireless channels must also match. However, a wireless client usually automatically sets its channel based on the wireless AP's channel. Encryption standards must match. In addition, the service set identifier (SSID) of a wireless AP must be selected by the wireless client. In many cases, a wireless AP broadcasts its SSID, and a wireless client can select that SSID from a list of visible SSIDs. In other cases, a wireless AP does not broadcast its SSID, and a wireless client must have a matching SSID manually configured.
- **Multiple paths of propagation:** An electromagnetic waveform cannot pass through a perfect conductor. Admittedly, perfect conductors do not exist in most office environments. However, very good conductors, such as metal filing cabinets, are commonplace in offices. As a result, if the waveform of a wireless transmission encounters one of these conductive objects, most of the signal bounces off the object, creating multiple paths (modes) of propagation. These multiple modes of propagation can cause data (specifically, bits) to arrive at uneven intervals, possibly corrupting data. This problem is similar to multimode delay distortion, which occurs in multimode fiber-optic cabling.
- **Incorrect AP placement:** Wireless APs should be strategically located in a building to provide sufficient coverage to all desired coverage areas. However, the coverage areas of wireless APs using overlapping channels should not overlap. To maintain coverage between coverage areas, you should have overlapping coverage areas among wireless APs using non-overlapping channels (for example, channels 1, 6, and 11 for wireless networks using the 2.4GHz band of frequencies). A common design recommendation is that overlapping coverage areas (using non-overlapping channels) should have an overlap of approximately 10% to 15%. Keep in mind that the 5GHz band and the 6 GHz band of Wi-Fi 6e offer many more non-overlapping channels.

- **Captive portal issues:** Many wireless network environments make use of a captive portal. A captive portal is a web page that often presents an acceptable use policy (AUP) that the user must accept in order to access the Internet via a wireless connection. The web page also typically collects profile information from end users. It might even market products to wireless users. The captive portal is a key element of the wireless association process, and so it must perform as expected. Most wireless networks fail closed, which means if the captive portal or any other step of the process is not functioning, the wireless network remains inaccessible to new clients.
- **Client disassociation issues:** Clients appreciate a wireless network much more if they are able to remain connected. While you might be able to immediately rule out many categories of issues that could cause disassociation, there are plenty that might require additional inspection and analysis. Remember that there is an entire category of attacks that target client disassociation, and these attacks are often initial steps in much larger attacks; an attacker may disassociate a client and then associate it to a rogue device.
- **Roaming misconfiguration:** Roaming misconfigurations happen when devices like smartphones or laptops have trouble smoothly switching from one Wi-Fi access point to another as they move around. Instead of seamlessly staying connected, these devices might drop the connection or experience interruptions. This can lead to slow Internet speeds, dropped calls, and delays in data transfer. Ensuring proper configuration of roaming settings is crucial for maintaining consistent and reliable network performance, especially in environments like offices or campuses where users frequently move between different network areas.

Wireless Network Troubleshooting

As a practice troubleshooting scenario for wireless networks, consider Figure 24-1. Based on this topology, can you spot a design issue with the wireless network?

Wireless Network Troubleshooting Solution

The wireless network presented in Figure 24-1 includes two wireless APs. Although these wireless APs have a matching wireless standard, encryption type, and SSID, the channels being used (channels 1 and 5) interfere with one another. Recall from Chapter 5 that channels in the 2.4GHz band need at least five channels of separation (for overlapping coverage areas), but the channels used in this example have only four channels of separation. A fix for this *channel overlap* issue is to assign AP-2 to channel 6, thus providing five channels of separation between AP-1 and AP-2.

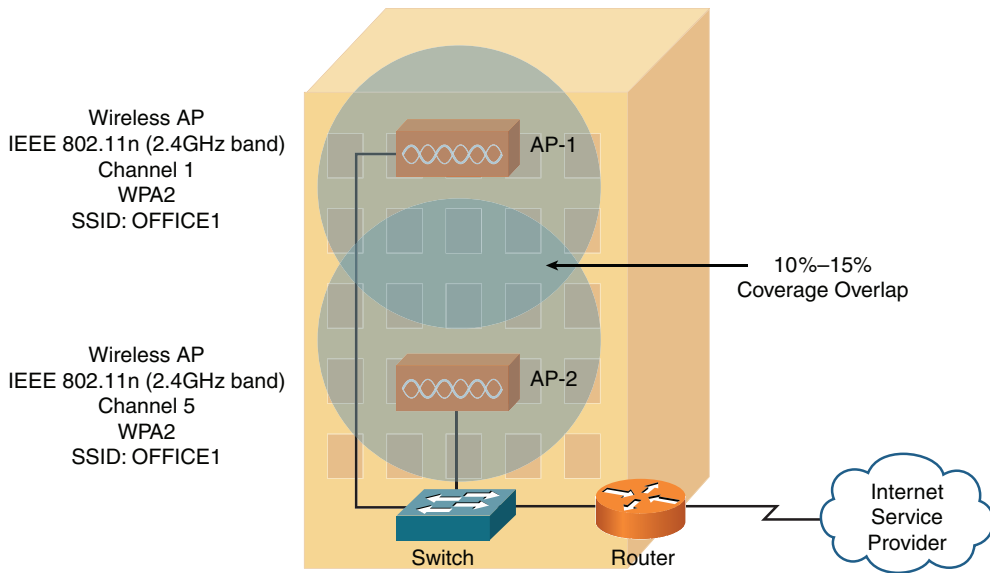
**Key
Topic**

Figure 24-1 Wireless Network Troubleshooting: Sample Topology

A wireless analyzer may be needed to identify problems such as signal loss, overlapping or mismatched channels, unacceptable signal-to-noise ratios, rogue APs, and power levels. Breaking down a problem into smaller pieces allows you to identify the fault domain or the area that is causing the problem. For example, if a user cannot access the wireless network, the pieces involved may be the user connecting to an incorrect SSID or problems with the AP, the switch, the WLC, the RADIUS server, the Active Directory (AD) server, or the user account and password. By testing the individual components, where possible, you can isolate the problem and then correct it.

Real-World Case Study

Acme, Inc. has been troubleshooting poor performance issues that have been reported throughout the infrastructure of the main campus HQ. The networking team has been hard at work improving the wireless performance in the main Enterprise campus. As part of these improvements, the IT team is performing an analysis of the existing Wi-Fi signals. A big metric in this study is the RSSI and EIRP values throughout the location. The team has improved its heat map creation and has been studying the data collected through the use of a Wi-Fi analyzer.

Another big aspect of the improvements is the replacement of several access points and the introduction of a new and improved wireless LAN controller (WLC). Networking is also ensuring its new IT team members are properly trained in

supporting and understanding the Wi-Fi infrastructure. This includes training on channel overlap, signal degradation, and interference issues.

Summary

Here are the main topics covered in this chapter:

- This chapter described some of the important metrics associated with specifications and limitations in wired and wireless networking, including such topics as congestion, contention, and bottlenecks.
- This chapter examined several important considerations that are critical in wireless troubleshooting.
- This chapter concluded by examining some of the most common issues in wireless networks today, including such topics as channel overlap, signal degradation, and roaming misconfigurations.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 24-1 lists these key topics and the page number where each is found.



Table 24-1 Key Topics for Chapter 24

Key Topic Element	Description	Page Number
List	Network performance considerations	530
List	Wireless specifications and limitations	531
List	Key facts for frequencies and channels	533
List	Common issues for wireless networking	535
Figure 24-1	Wireless Network Troubleshooting: Sample Topology	537

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

bandwidth, bottleneck, channel overlap, client disassociation, congestion, contention, interference, jitter, latency, packet loss, roaming, signal degradation, throughput

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. Which of the following is a value measuring the power of a wireless signal?
 - a. RSSI
 - b. SSID
 - c. RFI
 - d. CSMA/CA
2. Which of the following are the most common sources of radio frequency interference (RFI) in a wireless network? (Choose three.)
 - a. Gaming consoles
 - b. Fax machines
 - c. Microwave ovens
 - d. Baby monitors
 - e. Bluetooth
3. Which metric provides the best idea of how a wireless network is actually performing?
 - a. Throughput
 - b. Speed
 - c. Relative jitter
 - d. Availability
4. A satellite dish antenna is an example of what type of wireless antenna?
 - a. Omnidirectional
 - b. Integrated
 - c. Beam-formed
 - d. Unidirectional

5. What is the name for a web page that is presented to wireless clients seeking access to a network?
 - a. Authentication server
 - b. Captive portal
 - c. Self-registration gateway
 - d. Supplicant client
6. Which of the following should you consider conducting before designing and deploying a high-performance WLAN infrastructure?
 - a. Site survey
 - b. Reconnaissance
 - c. Nmapping
 - d. Port scan
7. A target wireless client has been disconnected from the company AP and associated with a rogue device. What has most likely occurred?
 - a. Disassociation
 - b. Propagation
 - c. Incorrect AP placement
 - d. Misconfigured parameter
8. To maintain coverage between areas, you need to have overlapped coverage areas among wireless APs using non-overlapping channels. In the United States, what channels in the 2.4GHz band of frequencies should you use?
 - a. CSMA/CA channels
 - b. 1, 6, and 11
 - c. EIRP channels
 - d. 2, 7, and 12
9. Which of the following refers to the maximum amount of data that can be transmitted over the network in a given amount of time?
 - a. Latency
 - b. Jitter
 - c. Throughput
 - d. Bandwidth

10. What term best describes multiple devices or applications simultaneously competing for the same network resources, leading to potential delays and reduced performance?
- a. Congestion
 - b. Bottleneck
 - c. Contention
 - d. Packet loss



This chapter covers the following topics related to Objective 5.5 (Given a scenario, use the appropriate tool or protocol to solve networking issues) of the CompTIA Network+ N10-009 certification exam:

- Software tools
 - Protocol analyzer
 - Command line
 - ping
 - traceroute/tracert
 - nslookup
 - tcpdump
 - dig
 - netstat
 - ip/ifconfig/ipconfig
 - arp
 - Nmap
 - Link Layer Discovery Protocol (LLDP)/Cisco Discovery Protocol (CDP)
 - Speed tester
- Hardware tools
 - Toner
 - Cable tester
 - Taps
 - Wi-Fi analyzer
 - Visual fault locator
- Basic networking device commands
 - show mac-address-table
 - show route
 - show interface
 - show config
 - show arp
 - show vlan
 - show power

Network Troubleshooting Tools

Troubleshooting your network and proactively stopping issues before they become trouble tickets are made much easier for you by a wide variety of software, hardware, and command-line tools that you can use with ease.

In this chapter, you will learn about a number of these tools that can make maintenance and troubleshooting of your network much easier. This chapter also reviews several commands that are very important on common network devices.

Foundation Topics

Software Tools

An array of software tools are designed to help with managing networks. This section provides an excellent starting point, introducing you to some of the commonly used tools, many of which are freely available.

Protocol Analyzer/Packet Capture

If you understand the characteristics of the protocols running on your network (such as the fields in a protocol's header), a **protocol analyzer** (also known as a *network sniffer* or simply *packet capture* tool) can be a tremendous troubleshooting asset. A protocol analyzer can be a standalone device or software running on a laptop or other mobile computer. You can use a protocol analyzer to capture traffic flowing through a network switch, using the port mirroring feature of the switch. By examining the captured packets, you can discern the details of communication flows (sessions) as they are being set up, maintained, and torn down. The examination of these captured packets, referred to as *traffic analysis*, provides an administrator with valuable insights about the nature of traffic flowing through a network.

Protocol analyzers come with a wide range of features and at a variety of price points. Wireshark is a free software program that can make your laptop act like a protocol analyzer. Protocol analyzers can assist in identifying details such as top talkers, top destinations, top protocols in use, and quantity of traffic on the network. You can download a free copy of Wireshark from <https://www.wireshark.org>. Figure 25-1 shows the Wireshark protocol analyzer application.

Bandwidth Speed Tester

Many bandwidth **speed testers** are available to assist you in verifying throughput from a local computer to an Internet site. One example is <https://www.speedtest.net>. Using sites such as this can assist you in determining whether an overall connection to the Internet is slow or whether just a specific site or server is slow to respond.

Port Scanner

A **port scanner** is a useful software tool that probes a network device to determine which ports are open on the device. A port scanner can send port scans to a single device, or it can perform a port sweep, where it checks for open ports on a number of network devices at one time.

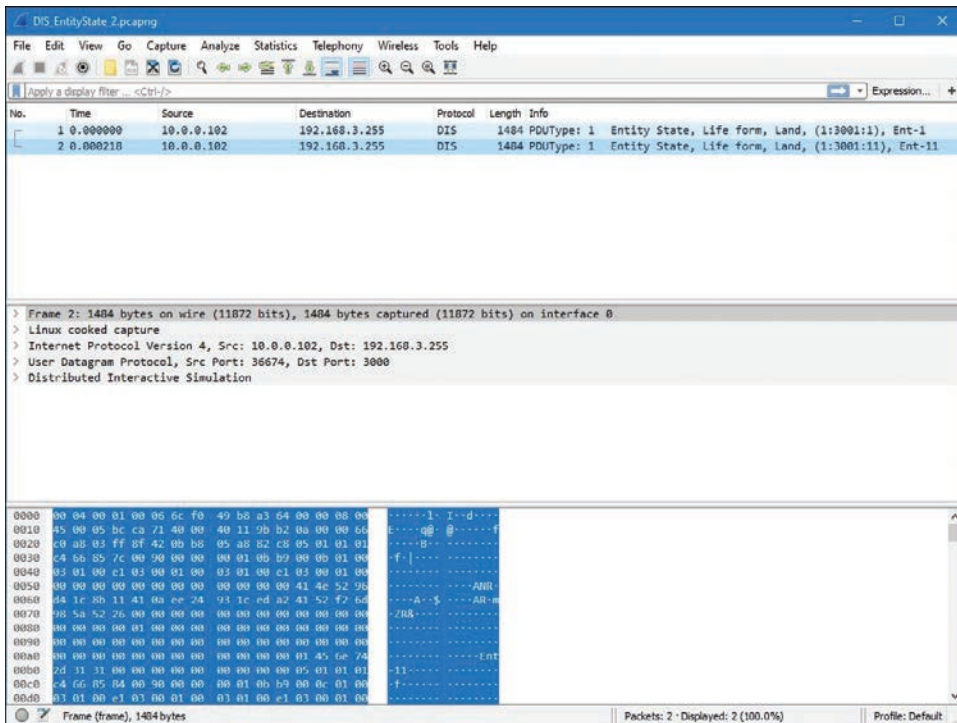
Key
Topic

Figure 25-1 Wireshark Protocol Analyzer Software

Many port scanners today are considered IP scanners. IP scanners are superb at running port sweeps against entire subnets of IP addressed network hosts or systems.

In addition to being helpful to network administrators, port scanners can be used to carry out harm. Computer criminals might use port scanners in launching attacks on open ports.

A very famous port scanner tool is a simple command-line tool called netstat. This tool is covered in detail later in this chapter.

iperf

iperf is a software tool for networking that obtains active measurements of the maximum bandwidth available on IP networks. This software tool supports tuning of various parameters related to timing, buffers, and protocols. The protocols you can tune include Transmission Control Protocol (TCP), User Datagram Protocol (UDP), and Stream Control Transmission Protocol (SCTP) with Internet Protocol Version 4 (IPv4) and Version 6 (IPv6).

iperf reports on bandwidth, packet loss, and other important network parameters. A nice feature of iperf is that it can be run in Linux and Windows environments.

NetFlow Analyzers

NetFlow, which is a software component included in many network devices today, collects statistics about the traffic flows through a device. However, the data that NetFlow collects is not useful unless you can analyze it and see it in a way that makes sense. You can use *NetFlow analyzers* (often also called *collectors*) to examine NetFlow collections and get valuable and insightful representations of the data.

TFTP Server

For network maintenance, it is helpful to have a *Trivial File Transfer Protocol (TFTP) server* or two on your network. These low-overhead-producing servers allow you to store backups of software images that power network devices, as well as backup and baseline configurations. Many network devices today can install their software or configurations from a TFTP server directly over the network. Be sure to remember security when working with these servers. This is critical because the TFTP server itself does not provide security mechanisms like secure authentication or encryption.

Terminal Emulator

A software tool you are likely to find yourself using all the time is a *terminal emulator*. This program allows you to access a command-line interface (CLI) to monitor and configure devices, which is often a very efficient way to carry out these networking tasks.

IP Scanner

Many tools can provide IP address information for systems in a network, but *IP scanners* can do this precise function better than any alternative. IP scanners can acquire information more accurately and faster than the alternatives. Many IP scanners can also provide additional information from nodes, such as operating system, software application, and physical resource usage information.

LLDP/CDP

Link Layer Discovery Protocol (LLDP) is a network protocol used by devices to identify and share information about themselves and their neighbors within a local area network. Think of it as a way for network devices, like switches and routers, to introduce themselves and exchange details, such as their names, capabilities, and

connections. This helps network administrators map out the network topology, making it easier to manage and troubleshoot connections between devices.

LLDP operates at the data link layer (Layer 2) of the OSI model. It works directly with the hardware that connects devices on a network. It sends out small packets of information, called LLDP Data Units (LLDPDU)s, at regular intervals. These packets are received by neighboring devices, which store the information in a special database. By using LLDP, administrators can quickly gather crucial information about the network, such as which devices are connected, how they are connected, and their specific capabilities, all without having to physically inspect the network hardware.

While LLDP is an open standard defined by the Institute of Electrical and Electronics Engineers (IEEE), *Cisco Discovery Protocol (CDP)* is an invention by Cisco Systems. It essentially strives to do what LLDP does. CDP helps Cisco devices, like switches and routers, automatically discover and share information with each other within a local network. CDP allows Cisco devices to exchange details such as device identity, capabilities, and the status of interfaces. This helps network administrators easily map out the network, troubleshoot issues, and optimize performance. By regularly sending out small packets of information, CDP ensures that devices can keep each other updated about their presence and capabilities, simplifying network management in environments using Cisco equipment.

NOTE Cisco is such a giant in the networking industry that you might encounter non-Cisco devices that also speak CDP. A great example is non-Cisco IP phones that speak CDP so they can seamlessly integrate with Cisco collaboration solutions.



Command-Line Tools

You will often find yourself firing open a terminal emulator to access a command-line interface when you are maintaining or troubleshooting a network. This section ensures that you are very familiar with many of the command-line tools in use today.

ping

The *ping* command is one of the most commonly used command-line commands. You can use it to check IP connectivity between two network devices. Multiple platforms (for example, routers, switches, and hosts) support the *ping* command. We will focus on the Windows version of this command, but keep in mind there are versions for just about every operating system you can think of.

The **ping** command uses Internet Control Message Protocol (ICMP), which is a Layer 4 protocol. If you issue a **ping** command from your PC, your PC sends an ICMP echo message to the specified destination host. If the destination host is reachable, the host responds with an ICMP echo reply message. Other ICMP messages can be returned to your PC, from your PC’s default gateway, to indicate that a destination host is unreachable, that an ICMP echo timed out, or that a Time-to-Live (TTL) value (which is decremented by 1 at each router hop) has expired (that is, has been decremented to a value of 0).

The syntax of the **ping** command, along with some of its commonly used options, is as follows:

```
ping [-t] [-n count] [-l size] [-f] [-i TTL] [-S srcaddr] target_name
```

Table 25-1 explains the **ping** command options.

Table 25-1 Parameters for the Windows **ping** Command

Parameter	Purpose
-t	This option repeatedly sends pings (ICMP echo messages) until you stop it by pressing Ctrl-C.
-n <i>count</i>	This option specifies the number of pings to send.
-l <i>size</i>	This option enables you to define the size of the ICMP echo requests in bytes; the default value is 32.
-f	This option sets the “don’t fragment” bit in a packet’s header. If the packet tries to cross a router that attempts to fragment the packet, the packet is dropped, and an ICMP error message is returned.
-i <i>TTL</i>	This option sets the TTL value in a packet’s header. The TTL value is decremented for each router hop. A packet is discarded when its TTL value reaches 0.
-S <i>srcaddr</i>	If the PC from which you are issuing the ping command has more than one IP address, this option allows you to specify the source IP address from which the ICMP echo messages should be sent.
<i>target_name</i>	This option specifies the name or the IP address of the device to which you are sending ICMP echo messages.

A Windows **ping** command specifying only the *target_name* parameter sends four ICMP echo messages to the specified target, as shown in Example 25-1. In the output, notice that none of the packets were dropped.

**Key
Topic****Example 25-1** Sample Output from the Windows **ping** Command

```
C:\> ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=2ms TTL=64
Reply from 192.168.1.2: bytes=32 time=1ms TTL=64
Reply from 192.168.1.2: bytes=32 time=1ms TTL=64
Reply from 192.168.1.2: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

If the specified target address is unreachable, output from the **ping** command indicates that the target cannot be reached, as shown in Example 25-2.

Example 25-2 Windows **ping** Command Indicating an Unreachable Destination

```
C:\> ping 192.168.1.200

Pinging 192.168.1.200 with 32 bytes of data:
Reply from 192.168.1.50: Destination host unreachable.
Reply from 192.168.1.50: Destination host unreachable.
Reply from 192.168.1.50: Destination host unreachable.
Reply from 192.168.1.50: Destination host unreachable.

Ping statistics for 192.168.1.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

ping with IPv6

Depending on the operating system, **ping** can natively work to test connectivity using IPv6 when an IPv6 destination address is part of the **ping** command. On some systems, the command **ping -6 IPv6-destination-address**, **ping6 IPv6-destination-address**, or some variant specific to that operating system may be available for testing IPv6 connectivity.

ipconfig

You can use the **ipconfig** command to display IP address configuration parameters on a Windows PC. In addition, if the PC uses Dynamic Host Configuration Protocol (DHCP), you can use the **ipconfig** command to release and renew a DHCP lease, which is often useful when troubleshooting.

The syntax of the **ipconfig** command, along with some of its most commonly used parameters, is as follows:

```
ipconfig [ /all | /renew | /release | /renew6 | /release6 ]
```

Table 25-2 describes the parameters for the **ipconfig** command.

Table 25-2 Parameters for the Windows **ipconfig** Command

Parameter	Purpose
/all	The ipconfig command entered by itself displays summary information about a PC's IP address configuration. This parameter gives more verbose information, including DNS, MAC address, and IPv6 address information.
/release and /release6	These options release a DHCP lease for an IPv4 and IPv6 address, respectively.
/renew and /renew6	These options renew a DHCP lease for an IPv4 and IPv6 address, respectively.

Example 25-3 shows the **ipconfig** command, without any parameters, being issued on a PC. The PC contains an Ethernet network interface card (NIC) and a wireless NIC. From the output, you can conclude that one of the NICs has the IP address 172.16.202.129, and the other NIC has the IP address 172.16.202.128. Also, you can see that these two NICs share the common default gateway 172.16.202.2.

**Key
Topic**

Example 25-3 Sample Output from the Windows **ipconfig** Command

```
C:\> ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 3:

    Connection-specific DNS Suffix . : localdomain
    Link-local IPv6 Address . . . . . : fe80::5101:b420:4354:d496%20
    IPv4 Address. . . . . : 172.16.202.129
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.202.2

Ethernet adapter Local Area Connection :

    Connection-specific DNS Suffix . : localdomain
    Link-local IPv6 Address . . . . . : fe80::a10f:cff4:15e4:aa6%11
    IPv4 Address. . . . . : 172.16.202.128
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.202.2

OUTPUT OMITTED...
```

Example 25-4 shows the **ipconfig /all** command being issued on a PC. Notice the additional output from this command beyond what is shown for the **ipconfig** command in Example 25-3. For example, in Example 25-4 you can see the MAC address (labeled as the *physical address*) for each NIC and the DNS server's IP address, 172.16.202.2.

Example 25-4 Sample Output from the Windows **ipconfig /all** Command

```
C:\> ipconfig /all

Windows IP Configuration

    Host Name . . . . . : WIN-OD1IG7JF47P
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : localdomain

Ethernet adapter Local Area Connection 3:

    Connection-specific DNS Suffix . . . : localdomain
    Description . . . . . : Intel(R) PRO/1000 MT Network
                             Connection #2
    Physical Address. . . . . : 00-0C-29-3A-21-67
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::5101:b420:4354:d496%20
                                         (Preferred)
    IPv4 Address. . . . . : 172.16.202.129 (Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Saturday, May 29, 2026 6:28:08
                                         PM
    Lease Expires . . . . . : Saturday, May 29, 2026 9:28:08
                                         PM
    Default Gateway . . . . . : 172.16.202.2
    DHCP Server . . . . . : 172.16.202.254
    DHCPv6 IAID . . . . . : 419433513
    DHCPv6 Client DUID. . . . . : 00-01-00-01-14-A6-11-77-00-0C-
                                         29-3A-21-5D
    DNS Servers . . . . . : 172.16.202.2
    Primary WINS Server . . . . . : 172.16.202.2
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . . . : localdomain
```

```
Description . . . . . : Intel(R) PRO/1000 MT Network
                        Connection
Physical Address. . . . . : 00-0C-29-3A-21-5D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a10f:cff4:15e4:aa6%11
                                      (Preferred)
IPv4 Address. . . . . : 172.16.202.128(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, May 29, 2026
                        6:27:56 PM
Lease Expires . . . . . : Saturday, May 29, 2026 9:28:08
                        PM
Default Gateway . . . . . : 172.16.202.2
DHCP Server . . . . . : 172.16.202.254
DHCPv6 IAID . . . . . : 234884137
DHCPv6 Client DUID. . . . . : 00-01-00-01-14-A6-11-77-00-0C-
                        29-3A-21-5D
DNS Servers . . . . . : 172.16.202.2
Primary WINS Server . . . . . : 172.16.202.2
NetBIOS over Tcpi. . . . . : Enabled
OUTPUT OMITTED...
```

If you are troubleshooting a PC and suspect that IP addressing might be an issue, you can release the PC's current DHCP lease by issuing the **ipconfig /release** command, as shown in Example 25-5. Then you can renew the DHCP lease with the **ipconfig /renew** command, as shown in Example 25-6.

Example 25-5 Sample Output from the Windows **ipconfig /release** Command

```
C:\> ipconfig /release
Windows IP Configuration
Ethernet adapter Local Area Connection 3:
    Connection-specific DNS Suffix . . . :
    Link-local IPv6 Address . . . . . : fe80::5101:b420:4354:d496%20
    Default Gateway . . . . . :
Ethernet adapter Local Area Connection :
    Connection-specific DNS Suffix . . . :
    Link-local IPv6 Address . . . . . : fe80::a10f:cff4:15e4:aa6%11
    Default Gateway . . . . . :
OUTPUT OMITTED...
```

Example 25-6 Sample Output from the Windows **ipconfig /renew** Command

```

C:\> ipconfig /renew
Windows IP Configuration

Ethernet adapter Local Area Connection 3:

    Connection-specific DNS Suffix . . . : localdomain
    Link-local IPv6 Address . . . . . : fe80::5101:b420:4354:d496%20
    IPv4 Address. . . . . : 172.16.202.129
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.202.2

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . . . : localdomain
    Link-local IPv6 Address . . . . . : fe80::a10f:cff4:15e4:aa6%11
    IPv4 Address. . . . . : 172.16.202.128
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.202.2
OUTPUT OMITTED...

```

ifconfig

The rough equivalent in the UNIX/Linux world for the **ipconfig** utility is **ifconfig**. Example 25-7 demonstrates the use of this command to learn IP address information configured on an interface.

Example 25-7 Sample Output from the Linux **ifconfig** Command

```

anthony@DESKTOP-91165JO:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.86.29 netmask 255.255.255.0 broadcast
192.168.86.255
    inet6 fe80::6044:94be:583f:4f82 prefixlen 64 scopeid
0xofd<compat,link,site,host>
    ether 84:8f:69:f5:5f:3d (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 1500
    inet 127.0.0.1 netmask 255.0.0.0

```

```

    inet6 ::1 prefixlen 128 scopeid 0xfe<compat,link,site,host>
    loop (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
anthony@DESKTOP-91165JO:~$

```

NOTE This might sound crazy, but I captured Example 25-7 on my Windows 11 desktop PC. It is simple now to run Linux inside Windows.

ip

Many Linux systems are moving to the **ip** command instead of even preloading the **ifconfig** command. The powerful **ip** command functions with many additional keywords, including the **address** keyword, which displays the same information as **ifconfig**.

nslookup

Although the **nslookup** command offers various command options, this section focuses on the most common use for the command: resolving a fully qualified domain name (FQDN) to an IP address. This can, for example, help you determine whether a DNS record is correct and verify that your DNS server is operating.

The **nslookup** command can be issued along with an FQDN, or it can be used in an interactive mode, where you are prompted to enter command parameters. The syntax of this command can be summarized as follows:

```
nslookup [fqdn]
```

In noninteractive mode, you issue the **nslookup** command followed by an FQDN to display the IP address corresponding to the FQDN. To illustrate, consider Example 25-8, where the **nslookup** command is issued to resolve the IP address of the website `ajsnetworking.com`, which appears to be 172.31.194.74.

NOTE A private IP address is used here for illustrative purposes. In a real-world example, a public IP address would be displayed.

**Key
Topic****Example 25-8** Sample Output from the Windows **nslookup** Noninteractive Command

```
C:\> nslookup ajsnetworking.com
Server: UnKnown
Address: 192.168.1.1

Non-authoritative answer:
Name: ajsnetworking.com
Address: 172.31.194.74
```

In interactive mode, you enter the **nslookup** command and then, from the **>** prompt, enter command parameters. Example 25-9 shows `ajsnetworking.com` entered at the prompt to see the IP address corresponding to that FQDN. Also, notice that entering a question mark (?) displays a help screen that shows command options. By entering **quit**, you exit interactive mode.

Example 25-9 Sample Output from the Windows **nslookup** Interactive Command

```
C:\> nslookup
Default Server: UnKnown
Address: 192.168.1.1

> ajsnetworking.com
Server: UnKnown
Address: 192.168.1.1

Non-authoritative answer:
Name: ajsnetworking.com
Address: 172.31.194.74

> ?
Commands:      (identifiers are shown in uppercase, [] means optional)
NAME           - print info about the host/domain NAME using default
server
NAME1 NAME2    - as above, but use NAME2 as server
help or ?      - print info on common commands
set OPTION     - set an option
    all                - print options, current server
and host
    [no]debug         - print debugging information
```



```

[no]d2                                - print exhaustive debugging
information
[no]defname                            - append domain name to each query
[no]recurse                           - ask for recursive answer to query
[no]search                             - use domain search list
[no]vc                                - always use a virtual circuit
domain=NAME                           - set default domain name to NAME
srchlist=N1[/N2/.../N6]               - set domain to N1 and search list to
N1,N2, etc.
root=NAME                             - set root server to NAME
OUTPUT OMITTED...
> quit
C:\>

```

dig

As you have seen here, the Windows **nslookup** command is used to resolve a given FQDN to its IP address. UNIX has a similar **nslookup** command, which you can also use for FQDN to IP address resolution.

You can also use the **dig** command to resolve FQDNs to IP addresses. Unlike the **nslookup** command, however, the **dig** command is entirely a command-line command. (That is, **dig** lacks the interactive mode of the **nslookup** command.)

Example 25-10 compares the output of the **nslookup** and **dig** commands. Notice that the **dig** command offers more information than the **nslookup** command. For example, the *A* in the QUESTION SECTION part of the output of the **dig** command identifies the DNS record type (an A record, which is an alias record). If you peruse the output, you can find a few other pieces of information present in the **dig** command output that are not available in the **nslookup** command output; however, the **dig** command is rarely used to glean these more subtle pieces of information. Rather, many UNIX administrators use the **dig** command simply as an alternative way of resolving FQDNs to IP addresses. Notice in Example 25-10 that both the **nslookup** and **dig** commands indicate that the IP address corresponding to the FQDN `www.pearsonitcertification.com` is `159.182.74.51`.

Key Topic

Example 25-10 Comparing Output from the Windows **nslookup** and UNIX **dig** Commands

```

C:\> nslookup www.pearsonitcertification.com
Server: 192.168.1.1
Address: 192.168.1.1#53

```

Non-authoritative answer:

Name: www.pearsonitcertification.com

Address: 159.182.74.51

HOST# **dig www.pearsonitcertification.com**

```
; <<>> DiG 9.6.0-APPLE-P2 <<>> www.pearsonitcertification.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 10821
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;www.pearsonitcertification.com. IN A

;; ANSWER SECTION:
www.pearsonitcertification.com. 10791 IN A 159.182.74.51

;; Query time: 5 msec
;; SERVER: 192.168.1.1#53(192.168.1.1)
;; WHEN: Mon May 30 13:36:11 2026
;; MSG SIZE rcvd: 64
```

traceroute

You can use the UNIX command *traceroute* to determine which router hop along the path from a source device to a destination device is having issues. Also, based on the round-trip response time information reported for each hop, you can better determine which network segment might be causing excessive delay due to congestion. Example 25-11 provides sample output from the **traceroute** command, which is identifying the 13 router hops a UNIX host must transit to reach pearsonitcertification.com.

Key Topic

Example 25-11 Sample Output from the UNIX **traceroute** Command

```
HOST# traceroute www.pearsonitcertification.com
traceroute to pearsonitcertification.com (64.28.85.25), 64 hops max,
52 byte packets
1  192.168.1.1 (192.168.1.1) 3.480 ms 2.548 ms 2.404 ms
```

```

 2  cpe-76-177-16-1.natcky.res.rr.com (76.177.16.1) 22.150 ms
11.300 ms 9.719 ms
 3  gig2-0-0.rcmdky-mx41.natcky.rr.com (65.28.199.205) 9.242 ms
19.940 ms 11.735 ms
 4  tge0-2-0.chcgileq-rtr1.kc.rr.com (65.28.199.97) 38.459 ms
38.821 ms 36.157 ms
 5  ae-4-0-cr0.chi10.tbone.rr.com (66.109.6.100) 41.903 ms 37.388 ms
31.966 ms
 6  ae-0-0.pr0.chi10.tbone.rr.com (66.109.6.153) 75.757 ms 46.287 ms
35.031 ms
 7  if-4-0-0.core1.ct8-chicago.as6453.net (66.110.14.21) 48.020 ms
37.248 ms 45.446 ms
 8  if-1-0-0-1878.core2.ct8-chicago.as6453.net (66.110.27.78)
108.466 ms 55.465 ms 87.590 ms
 9  63.243.186.25 (63.243.186.25) 64.045 ms 63.582 ms 69.200 ms
10  cr2-pos-0-8-0-3.nyr.savvis.net (208.173.129.29) 64.933 ms
65.113 ms 61.759 ms
11  hrl-tengig-13-0-0.waltham2bo2.savvis.net (204.70.198.182)
71.964 ms 65.430 ms 74.397 ms
12  das3-v3038.bo2.savvis.net (209.202.187.182) 65.777 ms 64.483 ms
82.383 ms
13  blhosting.bridgelinesw.com (64.14.81.46) 63.448 ms !X * 68.879 ms
!X

```

The Windows **tracert** command can be used for the same purpose as the UNIX **traceroute** command.

traceroute for IPv6

You can verify the IPv6 path through a network by using **traceroute** for IPv6. Depending on the vendor and platform, this may be done by using **traceroute destination-IPv6-address**, **traceroute6 destination-IPv6-address**, **traceroute -6 destination-IPv6-address**, or some variant specific to the vendor and product being used.

arp

You can use the **arp** command to view the MAC address to IP address name resolution that has succeeded and been entered in the ARP cache. In addition, you can use the **arp** command to statically add a MAC address to IP address mapping to a PC's Address Resolution Protocol (ARP) MAC address lookup table.

The syntax of the **arp** command is as follows:

```

arp -s inet_addr eth_addr [if_addr]
arp -d inet_addr [if_addr]
arp -a [inet_addr] [-N if_addr] [-v]

```

Table 25-3 describes the **arp** command’s *switches* (for example, **-s**, **-d**, and **-a**) and *arguments* (for example, *inet_addr* and *if_addr*).

Table 25-3 Parameters for the Windows **arp** Command

Parameter	Purpose
-a or -g	These options display current entries in a PC’s ARP table.
-v	This option, which stands for <i>verbose</i> , includes any invalid and loopback interface entries in an ARP table.
<i>inet_addr</i>	This option is a specific IP address.
-N <i>if_addr</i>	This option shows ARP entries learned for a specified network.
-d	This option, in combination with the <i>inet_addr</i> parameter, can delete an ARP entry for a host. With the wildcard character *, it can delete all host entries.
-s	This option, used in conjunction with the <i>inet_addr</i> and <i>eth_addr</i> parameters, statically adds a host entry to the ARP table.
<i>eth_addr</i>	This parameter is a 48-bit MAC address.
<i>if_addr</i>	If a host has multiple interfaces, an ARP entry might be associated with a specific interface. This option can be used to statically add or delete an ARP entry in a specified interface.

Example 25-12 shows the **arp -a** command being issued on a PC. The output shows what MAC addresses have been learned for the listed IP addresses. The dynamically learned addresses have *dynamic* listed in the *Type* column, and statically configured addresses (which are addresses configured by a user or the OS) are listed with *static* in the *Type* column. From the output, you can determine, for example, that the network device with the IP address 172.16.202.1 has the MAC address 00-50-56-c0-00-08, which could alternatively be written as 0050.56c0.0008. Also, you can determine from the output that this information was dynamically learned, as opposed to being statically configured.

Key
Topic

Example 25-12 Sample Output from the Windows **arp -a** Command

```
C:\> arp -a
Interface: 172.16.202.128 --- 0xb
    Internet Address      Physical Address      Type
    172.16.202.1          00-50-56-c0-00-08    dynamic
    172.16.202.2          00-50-56-fd-65-2c    dynamic
```

172.16.202.254	00-50-56-e8-84-fc	dynamic	
172.16.202.255	ff-ff-ff-ff-ff-ff	static	
224.0.0.22	01-00-5e-00-00-16	static	static
224.0.0.252	01-00-5e-00-00-fc	static	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static	

Interface: 172.16.202.129 --- 0x14

Internet Address	Physical Address	Type	
172.16.202.1	00-50-56-c0-00-08	dynamic	dynamic
172.16.202.2	00-50-56-fd-65-2c	dynamic	dynamic
172.16.202.254	00-50-56-e8-84-fc	dynamic	
172.16.202.255	ff-ff-ff-ff-ff-ff	static	
224.0.0.22	01-00-5e-00-00-16	static	static
224.0.0.252	01-00-5e-00-00-fc	static	static
224.0.1.60	01-00-5e-00-01-3c	static	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static	

From a troubleshooting perspective, keep in mind that static ARP entries tend to be more problematic than dynamic entries. For example, a static entry might be added to a laptop computer, and the computer might later connect to a different network. If a PC then attempts to reach the IP address specified in the static ARP entry, the Layer 2 frame would have the incorrect destination MAC address (which should then be the MAC address of the PC’s default gateway) in its header.

netstat

You can use the *netstat* command to display various information about IP-based connections on a PC. For example, you can view information about current sessions, including source and destination IP addresses and port numbers. You can also display protocol statistics, which might be useful for troubleshooting purposes. For example, you might issue the *netstat* command and see that your PC has sessions open to an unknown host on the Internet. These sessions might prompt further investigation to determine why the sessions are open and if they might be resulting in performance issues on the PC or possibly posing a security risk.

The following is the syntax for the *netstat* command and some of its commonly used options:

```
netstat [-a] [-b] [-e] [-f] [-p proto] [-r] [-s]
```

Table 25-4 explains these command options.

Table 25-4 Parameters for the Windows **netstat** Command

Parameter	Purpose
-a	This option displays all of a PC's active IP-based sessions, along with the TCP and UDP ports of each session.
-b	This option shows the name of the program that opened a session.
-e	This option shows statistical information for an interface's IP-based traffic, such as the number of bytes sent and received.
-f	This option displays FQDNs of destination addresses appearing in a listing of active sessions.
-p <i>proto</i>	This option displays connections for a specific protocol, which might be icmp , icmpv6 , ip , ipv6 , tcp , tcpv6 , udp , or udpv6 .
-r	This option displays a PC's IP routing table. (Note that netstat with this option generates the same output as the route print command.)
-s	This option displays statistical information for the following protocols: ICMPv4, ICMPv6, IPv4, IPv6, TCPv4, TCPv6, UDPv4, and UDPv6.

The **netstat** command issued without any options lists source and destination IP addresses and port numbers for all IP-based sessions. Example 25-13 shows sample output from this command.

**Key
Topic**
Example 25-13 Sample Output from the Windows **netstat** Command

```

C:\> netstat
OUTPUT OMITTED...
  TCP  127.0.0.1:27015      LIVE-DELIVERY:1309      ESTABLISHED
  TCP  192.168.1.50:1045     172.16.224.200:https     CLOSE_WAIT
  TCP  192.168.1.50:1058     THE-WALLACES-TI:microsoft-ds ESTABLISHED
  TCP  192.168.1.50:1079     tcpep:https             ESTABLISHED
  TCP  192.168.1.50:1081     174:http                ESTABLISHED
  TCP  192.168.1.50:1089     by2msg4020609:msnp      ESTABLISHED
  TCP  192.168.1.50:1111     HPB81308:netbios-ssn    ESTABLISHED
  TCP  192.168.1.50:1115     10.65.228.81:https      ESTABLISHED
  TCP  192.168.1.50:1116     10.65.228.81:https      ESTABLISHED
  TCP  192.168.1.50:1117     10.65.228.81:https      ESTABLISHED
  TCP  192.168.1.50:1118     10.65.228.81:https      ESTABLISHED
  TCP  192.168.1.50:1126     10.65.228.81:https      ESTABLISHED
  TCP  192.168.1.50:1417     vip1:http               CLOSE_WAIT
  TCP  192.168.1.50:1508     208:https               CLOSE_WAIT
  TCP  192.168.1.50:1510     208:https               CLOSE_WAIT
  TCP  [::1]:2869           LIVE-DELIVERY:1514      TIME_WAIT
  TCP  [::1]:286            LIVE-DELIVERY:1515      ESTABLISHED
OUTPUT OMITTED...

```

You might notice an open connection using a specific port and be unsure what application opened that connection. As you can see in Example 25-14, the **netstat -b** command shows which application opened a specific connection. In this example, Dropbox.exe, iTunes.exe, firefox.exe, and OUTLOOK.exe are applications that have currently open connections.

**Key
Topic**

Example 25-14 Sample Output from the Windows **netstat -b** Command

```
C:\> netstat -b
Active Connections
OUTPUT OMITTED...
Proto          Local Address      Foreign Address     State
TCP            127.0.0.1:1068      LIVE-DELIVERY:19872 ESTABLISHED
[Dropbox.exe]
TCP            127.0.0.1:1309      LIVE-DELIVERY:27015 ESTABLISHED
[iTunes.exe]
TCP            127.0.0.1:1960      LIVE-DELIVERY:1961  ESTABLISHED
[firefox.exe]
TCP            192.168.1.50:1115    10.1.228.81:https    ESTABLISHED
[OUTLOOK.EXE]
TCP            192.168.1.50:1116    10.1.228.81:https    ESTABLISHED
[OUTLOOK.EXE]
OUTPUT OMITTED...
```

hostname

The **hostname** command is available on both UNIX/Linux and Windows systems. On a Linux system, you use this command to change the computer name or the domain name of the system. When you run the command on a Windows system, the command simply displays the computer name.

route

The **route** command can display a PC's current IP routing table. In addition, you can use the **route** command to add or delete entries in the routing table. The syntax of the **route** command, with a collection of commonly used options, is as follows:

```
C:\> route [-f] [-p] command [destination] [mask netmask] [gateway]
[metric metric] [if interface]
```

Table 25-5 explains these command options.

Table 25-5 Parameters for the Windows **route** Command

Parameter	Purpose
-f	This option clears gateway entries from the routing table. If this option is used with another option, the clearing of gateways from the routing table occurs before any other specified action.
-p	This option can be used with the add command to make a statically configured route persistent, which means the route will remain in a PC's routing table even after a reboot.
<i>command</i>	Supported commands include print , add , delete , and change . The print command lists entries in a PC's routing table. The add command adds a route entry. The delete command removes a route from the routing table, and the change command can modify an existing route.
<i>destination</i>	This option specifies the destination host or subnet to add to a PC's routing table.
mask <i>netmask</i>	This option, used in conjunction with the <i>destination</i> option, specifies the subnet mask of the destination. If the destination is the IP address of a host, the <i>netmask</i> parameter is 255.255.255.255.
<i>gateway</i>	This option specifies the IP address of the next-hop router used to reach the specified destination.
metric <i>metric</i>	This option specifies the cost to reach a specified destination. If a routing table contains more than one route to reach the destination, the route with the lowest cost is selected.
if <i>interface</i>	This option forwards traffic to a specified destination out a specific interface.

Example 25-15 illustrates the use of the **route print** command, which displays the contents of a PC's routing table. Notice that the output identifies a listing of the PC's interfaces, along with IPv4 routes and IPv6 routes. From the output, you can see that the 10.0.0.0 255.0.0.0 network is reachable via two gateways (192.168.1.77 and 192.168.1.11). Also, notice that there is a persistent route (a route entry that survives a reboot) to act as a default gateway for the PC, which is 192.168.1.1.

Example 25-15 Sample Output from the Windows **route print** Command

```
C:\> route print
=====
Interface List
  11...00 24 81 ee 4c 0e .....Intel(R) 82566DM-2 Gigabit Network
Connection
  1.....Software Loopback Interface 1
```



```
12...00 00 00 00 00 00 00 e0 Microsoft ISATAP Adapter
13...00 00 00 00 00 00 00 e0 Teredo Tunneling Pseudo-Interface
=====
```

IPv4 Route Table

=====+=====

Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.50	276
10.0.0.0	255.0.0.0	255.0.0.0	192.168.1.77	192.168.1.50	21
10.0.0.0	255.0.0.0	255.0.0.0	192.168.1.11	192.168.1.50	21
127.0.0.0	255.0.0.0	255.0.0.0	On-link	127.0.0.1	306
127.0.0.1	255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
127.255.255.255	255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
172.16.0.0	255.255.0.0	255.255.0.0	192.168.1.1	192.168.1.50	21
192.168.0.0	255.255.255.0	255.255.255.0	192.168.1.11	192.168.1.50	21
192.168.1.0	255.255.255.0	255.255.255.0	On-link	192.168.1.50	276
192.168.1.50	255.255.255.255	255.255.255.255	On-link	192.168.1.50	276
192.168.1.255	255.255.255.255	255.255.255.255	On-link	192.168.1.50	276
224.0.0.0	240.0.0.0	240.0.0.0	On-link	127.0.0.1	306
224.0.0.0	240.0.0.0	240.0.0.0	On-link	192.168.1.50	276
255.255.255.255	255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
255.255.255.255	255.255.255.255	255.255.255.255	On-link	192.168.1.50	276

=====

Persistent Routes:

Network	Address	Netmask	Gateway Address	Metric
0.0.0.0	0.0.0.0	0.0.0.0	192.168.1.1	Default

=====

IPv6 Route Table

=====

Active Routes:

If	Metric	Network	Destination	Gateway
13	58 :	:	0	On-link
1	306 :	:	1/128	On-link
13	58	2001::	/32 On-link	
13	306	2001:0:4137:9e76:10e2:614f:b34e:ea84	/128	On-link

```
11    276    fe80::/64                On-link
13    306    fe80::/64                On-link
13    306    fe80::10e2:614f:b34e:ea84/128
                                On-link

11    276    fe80::f46d:4a34:a9c4:51a0/128
                                On-link

1     306    ff00::/8                On-link
13    306    ff00::/8                On-link
11    276    ff00::/8                On-link

=====
Persistent Routes:
None
```

Say that you want to remove one of the route entries for the 10.0.0.0 255.0.0.0 network. Example 25-16 shows how one of the two entries (specifically, the entry pointing to 192.168.1.11) can be removed from the routing table. Notice from the output that after the **route delete 10.0.0.0 mask 255.0.0.0 192.168.1.11** command is issued, the route no longer appears in the routing table.

Example 25-16 Sample Output from the Windows **route delete** Command

```
C:\> route delete 10.0.0.0 mask 255.0.0.0 192.168.1.11
OK!

C:\> route print
OUTPUT OMITTED...
IPv4 Route Table

=====++++=====

Active Routes:
Network Destination  Netmask          Gateway          Interface        Metric
-----
0.0.0.0              0.0.0.0          192.168.1.1     192.168.1.50     276
10.0.0.0             255.0.0.0        192.168.1.77    192.168.1.50     21
127.0.0.0            255.0.0.0        On-link         127.0.0.1        306
127.0.0.1            255.255.255.255  On-link         127.0.0.1        306
127.255.255.255      255.255.255.255  On-link         127.0.0.1        306
172.16.0.0           255.255.0.0      192.168.1.11    192.168.1.50     21
192.168.0.0          255.255.255.0    192.168.1.11    192.168.1.50     21
192.168.1.0          255.255.255.0    On-link         192.168.1.50     276
```

192.168.1.50	255.255.255.255	On-link	192.168.1.50	276
192.168.1.255	255.255.255.255	On-link	192.168.1.50	276
224.0.0.0	240.0.0.0	On-link	127.0.0.1	306
224.0.0.0	240.0.0.0	On-link	192.168.1.50	276
255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
255.255.255.255	255.255.255.255	On-link	192.168.1.50	276
=====				
OUTPUT OMITTED...				

You can add a route by using the **route add** command. Example 25-17 shows and confirms the addition of a route pointing to the 10.2.1.0 255.255.255.0 network, with the next-hop route (gateway) 192.168.1.1.

Example 25-17 Sample Output from the Windows **route add** Command

C:\> route add 10.2.1.0 mask 255.255.255.0 192.168.1.1					
OK!					
C:\> route print					
OUTPUT OMITTED...					
IPv4 Route Table					
=====					
Active Routes:					
Network	Netmask	Gateway	Interface	Metric	Destination
0.0.0.0	0.0.0.0	192.168.1.1	192.168.1.50	276	
10.0.0.0	255.0.0.0	192.168.1.77	192.168.1.50	21	
10.2.1.0	255.255.255.0	192.168.1.1	192.168.1.50	21	
127.0.0.0	255.0.0.0	On-link	127.0.0.1	306	
127.0.0.1	255.255.255.255	On-link	127.0.0.1	306	
127.255.255.255	255.255.255.255	On-link	127.0.0.1	306	
172.16.0.0	255.255.0.0	192.168.1.11	192.168.1.50	21	
192.168.0.0	255.255.255.0	192.168.1.11	192.168.1.50	21	
192.168.1.0	255.255.255.0	On-link	192.168.1.50	276	
192.168.1.50	255.255.255.255	On-link	192.168.1.50	276	
192.168.1.255	255.255.255.255	On-link	192.168.1.50	276	
224.0.0.0	240.0.0.0	On-link	127.0.0.1	306	
224.0.0.0	240.0.0.0	On-link	192.168.1.50	276	
255.255.255.255	255.255.255.255	On-link	127.0.0.1	306	
255.255.255.255	255.255.255.255	On-link	192.168.1.50	276	
=====					
OUTPUT OMITTED...					

telnet

For many years, **telnet** was the command-line tool of choice for making remote access connections to systems for management. Today, this command is frowned upon because the Telnet protocol offers no security for the information transmitted. Telnet is now found only in lab and practice environments. Secure Shell (SSH) is the technology of choice today for remote access to systems because it is secure.

tcpdump

You can use the **tcpdump** command to print out the headers of packets on a network interface that match a Boolean expression. You can also run the command with the **-w** flag to save the packet data to a file for later analysis and/or with the **-r** flag to read from a saved packet file rather than to read packets from a network interface. The syntax and options for the **tcpdump** command are as follows:

```
tcpdump [ -adeflnNOpqRStuvxX ] [ -c count ] [ -C file_size ] [ -F
file ]
[ -i interface ] [ -m module ] [ -r file ]
[ -s snaplen ] [ -T type ] [ -U user ] [ -w file ]
[ -E algo:secret ] [ expression ]
```

nmap

The **nmap** command (short for Network Mapper) is an open source and very versatile tool for network administrators. While **nmap** was first introduced with Linux systems, this command is now available on Linux, macOS, and Windows systems. You use **nmap** to explore networks, perform security scans, create network audits, and find open ports on remote machines. The tool can scan for live hosts, operating systems, packet filters, and open ports. Here is an example of the syntax for this command:

```
nmap [Scan Type(s)] [Options] {target specification}
```



Basic Networking Device Commands

Although the CompTIA Network+ exam is vendor neutral, you need to be aware of several Cisco network device commands that are incredibly popular and used on equipment from a number of vendors:

- **show interface:** The **show interface** command enables you to examine the statistics and the status of the interfaces on a network system.

- **show config:** The *show config* command (or some variation of it) is used to examine the configuration of a network device. For example, on a Cisco router, the **show running-configuration** command permits you to see the current configuration of the device, which is stored in the RAM of the device. To view the saved configuration that is loaded when the system is rebooted, you can use the **show startup-configuration** command.
- **show route:** The *show route* command (or some variation of it) is used to view the routing table configuration of the network device. On a Cisco router, you can use **show ip route** to view the IPv4 routing table.
- **show mac-address-table:** The *show mac-address-table* command allows you to see the MAC address table that currently exists on a Layer 2 or multilayer switch. The MAC address table maintains the mappings of MAC addresses to interfaces on the device. This is the critical “brains” of the switch functionality, allowing it to effectively direct traffic to interfaces using the table mappings.
- **show arp:** Earlier in this chapter, you learned about the **arp** command. On Cisco, and many other networking devices, the *show arp* command is the rough equivalent of the **arp** command. The **show arp** command allows you to view the current ARP resolution protocol table of MAC to IP address mappings.
- **show vlan:** On switches, the *show vlan* command enables you to view the current VLAN database.
- **show power:** The *show power* command on a network device displays information about the device’s power status and usage. It helps administrators check how much power the device is consuming and whether there are any power issues or shortages. This is especially important on network devices that are providing Power over Ethernet (PoE) via the device interfaces.

Hardware Tools

This chapter would not be complete without a discussion of some of the more popular hardware tools that can assist when troubleshooting the network.

Wi-Fi Analyzer

A *Wi-Fi analyzer* is software that runs on a general-purpose computer or on a specialized device that can perform wireless analysis of Wi-Fi signals. This type of tool would be used as part of a wireless site survey after Wi-Fi has been implemented to

create a heat map of the wireless airspace. Such analysis can be critical to ensure network administrators have a thorough knowledge of the wireless infrastructure. Such tools can directly assist with the proper placement of wireless access points and the type and strength of antennas that are needed.

Tone Generator

If you are working on a punchdown block and attempting to identify which pair of wires connect back to an end user's location (for example, someone's office), you can use a *toner* probe. A toner probe allows you to place a tone generator at one end of a connection (for example, someone's office) and use a probe on a punchdown block to audibly detect to which pair of wires the tone generator is connected. A toner probe, therefore, comes in two pieces: the tone generator and the probe. Another common name for a toner probe is a fox and hound, where the tone generator is the fox, and the probe (which searches for the tone) is the hound. Some network devices have built-in troubleshooting tools; for example, a voice-enabled Cisco router can produce test tones.

Cable Tester

A *cable tester* can test the conductors in an Ethernet cable. By connecting the two parts of a cable tester to the ends of a cable under test, you can check the wires in the cable for continuity (that is, check to make sure there are no opens, or breaks, in a conductor). In addition, you can verify an RJ45 connector's pinouts (that is, ensure that wires are connected to the appropriate pins on an RJ45 connector).

Tap

A traffic access point (also called a test access point or *tap*) is a hardware device inserted at a specific point in a network where data can be accessed for testing or troubleshooting purposes. Network taps are mainly used to monitor the network traffic between two points in a network infrastructure.

Visual Fault Locator

A *visual fault locator* is a small, handheld device used to find breaks or problems in fiber-optic cables. It works by sending a bright red laser light through the cable. If there is a break or a fault, the light will escape and become visible at the damaged spot. This makes it easy to see where the problem is so it can be fixed quickly.

Real-World Case Study

Acme, Inc. has invested in several new software packages to help support the network. These include enterprise-level versions of protocol analyzers and port scanners.

Acme, Inc. is also providing training to the junior technicians that includes command-line tool training. This training includes common network device command-line work, including the monitoring commands used in the wide array of Cisco and Juniper network devices in use.

Acme is also updating and expanding on the number of physical troubleshooting kits for the network engineers onsite. These kits include cable testers and Wi-Fi analyzers.

Summary

The main topics covered in this chapter are the following:

- This chapter covered many different software tools that can assist greatly in the support of the network. This coverage included tools such as Wi-Fi analyzers and port scanners.
- This chapter described command-line tools that you can use to help troubleshoot and maintain modern networks. These included the commonly used tools like **ipconfig** and **tracert**.
- This chapter covered some basic commands found on some network devices.
- Finally, this chapter covered some of the commonly used hardware tools used to troubleshoot networks.

Exam Preparation Tasks

Review All the Key Topics

Review the most important topics from this chapter, noted with the Key Topic icon in the outer margin of the page. Table 25-6 lists these key topics and the page number where each is found.

Key
Topic**Table 25-6** Key Topics for Chapter 25

Key Topic Element	Description	Page Number
Figure 25-1	Wireshark Protocol Analyzer Software	545
Section	Command-Line Tools	547
Example 25-1	Sample Output from the Windows ping Command	549
Example 25-3	Sample Output from the Windows ipconfig Command	550
Example 25-8	Sample Output from the Windows nslookup Noninteractive Command	555
Example 25-10	Comparing Output from the Windows nslookup and UNIX dig Commands	556
Example 25-11	Sample Output from the UNIX traceroute Command	557
Example 25-12	Sample Output from the Windows arp -a Command	559
Example 25-13	Sample Output from the Windows netstat Command	561
Example 25-14	Sample Output from the Windows netstat -b Command	562
Section	Basic Network Platform Commands	567

Complete Tables and Lists from Memory

Print a copy of Appendix B, “Memory Tables,” or at least the section for this chapter, and complete as many of the tables as possible from memory. Appendix C, “Memory Tables Answer Key,” includes the completed tables and lists so you can check your work.

Define Key Terms

Define the following key terms from this chapter and check your answers in the Glossary:

arp, cable tester, Cisco Discovery Protocol (CDP), **dig**, **ifconfig**, **ip**, **ipconfig**, Link Layer Discovery Protocol (LLDP), **netstat**, **nmap**, **nslookup**, **ping**, protocol analyzer, **show arp**, **show config**, **show interface**, **show mac-address-table**, **show power**, **show route**, **show vlan**, speed tester, tap (test access point), **tcpdump**, toner, **traceroute**, **tracert**, visual fault locator, Wi-Fi analyzer

Additional Resources

Networking Tools – Hardware: <https://www.youtube.com/watch?v=HAceoquch1c>

Troubleshooting from the CLI: <https://www.youtube.com/watch?v=EYAnaNnWqBg>

Review Questions

The answers to these review questions appear in Appendix A, “Answers to Review Questions.”

1. Consider the following output:

```
C:\> arp -a
Interface: 172.16.202.128 --- 0xb
Internet Address Physical Address Type
172.16.202.2 00-50-56-fd-65-2c dynamic
172.16.202.255 ff-ff-ff-ff-ff-ff static
224.0.0.22 01-00-5e-00-00-16 static
224.0.0.252 01-00-5e-00-00-fc static
255.255.255.255 ff-ff-ff-ff-ff-ff static
```

In this example, what is the MAC address corresponding to the IP address 172.16.202.2?

- a. ff-ff-ff-ff-ff-ff
 - b. 00-50-56-fd-65-2c
 - c. 01-00-5e-00-00-16
 - d. 01-00-5e-00-00-fc
2. What option would you specify after the **ipconfig** command to display the IP address of a Windows PC's DNS server?
 - a. No option is needed because **ipconfig** displays DNS server information by default.
 - b. /full
 - c. /fqdn
 - d. /all

3. What protocol is used by the **ping** command?
 - a. IGMP
 - b. PIM
 - c. ICMP
 - d. RTP
4. Which of the following commands is used on a Linux host to generate information about each router hop along the path from a source to a destination?
 - a. **ping -t**
 - b. **tracert**
 - c. **ping -r**
 - d. **tracertoute**
5. Which of the following Linux commands can be used to check FQDN to IP address resolution? (Choose three.)
 - a. **nslookup**
 - b. **netstat**
 - c. **dig**
 - d. **host**
6. What command produced the following snippet of output?

```
OUTPUT OMITTED...  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 62169  
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0,  
ADDITIONAL: 0  
;; QUESTION SECTION:  
;pearsonitcertification.com. IN A  
;; ANSWER SECTION:  
  pearsonitcertification.com. 10800 IN A 64.28.85.25  
;; Query time: 202 msec  
;; SERVER: 192.168.1.1#53(192.168.1.1)  
;; WHEN: Wed Jun 1 20:41:57 2011  
;; MSG SIZE rcvd: 60  
OUTPUT OMITTED...
```

- a. **tracert -d pearsonitcertification.com**
 - b. **dig pearsonitcertification.com**
 - c. **netstat -a pearsonitcertification.com**
 - d. **nbtstat pearsonitcertification.com**

- 7. Which tool would be used as part of a wireless site survey and produces a heat map?
 - a. Bandwidth tester
 - b. Wi-Fi analyzer
 - c. port scanner
 - d. iperf

- 8. What command is used to view the routing table configuration of a network device?
 - a. **show config**
 - b. **show interface**
 - c. **show route**
 - d. **show tcpdump**

- 9. Which of the following are used by devices to identify and share information about themselves within a local area network? (Choose two.)
 - a. STP
 - b. IDS
 - c. LLDP
 - d. CDP

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Final Preparation

The first 25 chapters of this book cover the technologies, protocols, design concepts, and considerations you need to understand for the CompTIA Network+ N10-009 exam. This chapter details a set of tools and a study plan to help you complete your preparation for the CompTIA Network+ N10-009 exam.

This brief chapter has three main sections. The first section lists exam preparation tools. The second section provides a suggested study plan you can use now that you have completed all the earlier chapters in this book. The third section describes some strategies that can help you to take and pass the exam.

Tools for Final Preparation

The digital resources associated with the book are available on the book's companion website. To access the companion website, follow these steps:

- Step 1.** Go to <http://www.pearsonitcertification.com/register>.
- Step 2.** Either log in to your account (if you have an existing account) or create a new account.
- Step 3.** Enter the ISBN of this book (9780135367889) and click **Submit**.
- Step 4.** Answer the challenge questions to validate your purchase.
- Step 5.** On your account page, click the **Registered Products** tab and then click the **Access Bonus Content** link.

NOTE You must access the companion website and register the book ISBN by December 31, 2027.

Resources available on the companion website include the following:

- Interactive glossary flash card application
- The Pearson Test Prep practice test software
- Video training on key exam topics
- Memory table review exercises and answer keys

- A study planner tool
- Your CompTIA Network+ certification exam voucher, which provides a 10% discount on the exam

Video Training

The companion website for this book makes available ten training videos that address a couple of the most misunderstood concepts in the CompTIA Network+ curriculum—specifically the OSI model and IP addressing:

- **Video 1:** Introduction to the OSI Model
- **Video 2:** Layer 1—The Physical Layer
- **Video 3:** Layer 2—The Data Link Layer
- **Video 4:** Layer 3—The Network Layer
- **Video 5:** Layer 4—The Transport Layer
- **Video 6:** Layers 5–7—The Upper Layers
- **Video 7:** IP Addressing—Part 1 (Binary Numbering Review)
- **Video 8:** IP Addressing—Part 2 (Basic Subnetting)
- **Video 9:** IP Addressing—Part 3 (Advanced Subnetting)
- **Video 10:** IP Addressing—Part 4 (Introduction to IPv6)

These training videos are presented by Kevin Wallace, who is a dear friend and industry great. You can find more of his great work at his training company website, <https://www.kwtrain.com>.

Memory Tables

Like most other Cert Guides from Pearson IT Certification, this book purposefully organizes information into tables and lists for easier study and review. Rereading these tables and lists can be very useful before the exam. However, it is easy to skim over tables without paying attention to every detail, especially when you remember having seen the table's contents when reading the chapter.

To help you more effectively learn the information in the tables throughout this book, the companion website provides two appendixes:

- Appendix B, “Memory Tables,” lists partially completed versions of many of the tables in this book. You can open Appendix B and print the appendix. For review, you can attempt to complete the tables.

- Appendix C, “Memory Tables Answer Key,” which is also a PDF, lists the completed tables so that you can check your answers. You can also just refer to the tables printed in the book.

End-of-Chapter Review Tools

Chapters 1–25 all have several features in the “Exam Preparation Tasks” and “Review Questions” sections at the end of the chapter. You might have already worked through these tools in each chapter. However, it can help to use them again as you make your final preparations for the exam.

Suggested Plan for Final Review and Study

This section lists a suggested study plan from the point at which you finish reading Chapter 25 until you take the CompTIA Network+ exam. You can ignore this plan, use it as is, or modify it to better meet your needs.

The plan involves these steps:

- Step 1. Review key topics.** You can use the table at the end of each chapter that lists the key topics of the chapter, or you can just flip through the pages looking for key topics icons.
- Step 2. Complete memory tables.** Open Appendix B on the companion website, print the entire appendix, and then complete the tables.
- Step 3. Study the “Review Questions” sections.** Go through the “Review Questions” section at the end of each chapter to identify areas in which you need more study. All of these questions are available to you in the Pearson Test Prep software. Simply select the **Book Questions** exam bank to run a practice exam using just these questions from the book.
- Step 4. Use the Pearson Test Prep (PTP) app to practice.** You can use the PTP app to study using a bank of unique, exam-realistic questions available only with this book. Run through the questions the first time in study mode and assess your areas of weakness. Then work through the questions again in practice exam mode to assess your exam readiness. Go through the questions a final time in flash card mode to really challenge yourself to answer the questions without being able to see the multiple-choice answers.

The database of questions used by the Pearson Test Prep app was created specifically for this book. You can use the Pearson Test Prep app either in study mode, practice exam mode, or flash card mode, as follows:

- **Study mode:** Study mode is most useful when you want to use the questions for learning and practicing. In study mode, you can select options such as randomizing the order of the questions and answers, automatically viewing answers to the questions as you go, and testing on specific topics.
- **Practice exam mode:** Practice exam mode presents questions in a timed environment, providing you with an exam-realistic experience. It also restricts your ability to see your score as you progress through the exam and view answers to questions as you are taking the exam. These timed exams not only allow you to study for the actual CompTIA Network+ exam but also help you simulate the time pressure that can occur during an exam.
- **Flash card mode:** Flash card mode simply removes the multiple-choice answers, challenging you to answer each question in your head, without the benefit of seeing the possible answers. Using this mode is a great way to drill on topics in which you are particularly weak. Note that in this mode, you will not be able to record questions as right or wrong, and a grade report is not available; you will have to assess yourself on your own when you view the correct answers.

When doing your final preparation, you can use study mode, practice exam mode, and flash card mode. However, after you have seen each question a few times, you are likely to remember the questions, and the usefulness of the exam database might decrease. Therefore, consider the following options when using the exam engine:

- Use the exam bank 1 question database for review. Use study mode to study the questions by chapter, just as with the other final review steps listed in this chapter. When you are ready to do final review, use the other exam bank that you have not seen before.
- Save the question database by not using it during your review of each part of this book so that you will not have seen the questions before. Then, use practice exam mode to simulate the exam.

Picking the correct mode from the exam engine's user interface is straightforward. The following steps show you how to move to the screen where you can select the mode:

- Step 1.** Click the **My Products** tab if you are not already in that screen.
- Step 2.** From the list of available products, select the product you want to use.
- Step 3.** Click the **Exams** button.

You can then choose study mode, practice exam mode, or flash card mode. When in study mode or flash card mode, you can further choose the book chapters, thus limiting the questions to those explained in the specified chapters of this book.

The PTP app also allows you to mark questions, thus creating a custom database of questions for topics on which you might need extra review. It also allows you to enter notes on questions and review those notes.

Strategies for Taking the Exam

Are you ready for some strategy tips for your Network+ exam attempt? This section provides several of them. Of course, you can apply many of these to other certification exams as well:

- You can use one of two overall strategies for scheduling your exam. One approach is to schedule for an exam slot when you are fully ready; this is my favorite approach. You might have to wait as much as a week if all of the exam slots at your desired time are full, but typically if you are near a large city, you can find a slot for the very next day. The other approach is to schedule for several months out and then study, study, study to be ready for the exam slot you booked. Some people need to use this strategy to avoid procrastinating. If you adopt this strategy, be sure to check into the cancellation or rescheduling policy just in case your life gets in the way of your studies!
- Schedule the exam for a time that works great for you. If you are not a morning person, do not schedule the exam for early morning! Instead, schedule the exam for a time slot that works for you, even if it means waiting an extra day or two.
- Plan for and get a good night's rest before the exam. Also be sure to eat a good meal (breakfast or lunch) before the exam. Drink plenty of water to stay hydrated. Your mind will perform better when your body has what it needs. With that said, be sure not to experiment the day of the exam. I have talked to students who decided to suddenly try consuming a bunch of caffeine before the exam. They had a terrible exam experience because they were doing something their bodies were not used to.
- Map out where the testing center is and ensure that you know how to get there and how long the trip takes. Of course, if you are testing at home, you do not have this to worry about. Instead, be sure to follow the test at home instructions provided to you. For more information about testing at home, visit <https://www.pearsonvue.com> and click the **Online Testing** tab.
- Have the proper identification required for the exam.

- Be ready at least 30 minutes before your exam time. Take a few moments to relax and take several healthy breaths to increase your ability to do well on the exam. You might want to use the restroom before taking the exam so that your focus can be in the correct place during the exam.
- Manage your time in the exam. It is hard to answer a question correctly if you have run out of time and never see that question! If you have questions that you are unsure about, or if there is a simulation or lengthy question that may take several minutes, consider marking that question for review and come back to it after you have had an opportunity to answer the other questions. Often, other questions will refresh your memory and provide insight into questions that you have previously marked for review.
- Get ready for the performance-based questions to be grouped together and to be near the front of the exam. CompTIA has stuck to this formula pretty consistently. Doing this gives the students a little better chance at proper time management because these performance-based questions tend to take a bit more time than the multiple-choice questions.
- Smile at questions that are puzzling to you. The exam may give you an opportunity to learn a couple new things, and smiling when you see a tricky question can assist you in staying positive while you continue to answer the other questions that you know. It is likely that you will want to mark any puzzling questions for review and come back to them.
- Use your scratch paper in the exam (or the digital notepad) to note the topics of questions that you are forced to take a best guess at. If you don't pass the exam, memorize this list of topics. These are areas you should spend extra time on in your preparation for your second (and hopefully final!) attempt. Focusing on these areas, repeat the preparation steps outlined in this chapter.
- Have fun with this process. If you are enjoying the journey, everything regarding this certification process will go much more smoothly.

Summary

The tools and suggestions listed in this chapter are designed with one goal in mind: to help you develop the skills required to pass the CompTIA Network+ exam. This book has been developed from the beginning to not just present you with a collection of facts but also to help you learn how to apply those facts. Regardless of your experience level before reading this book, it is my hope that the broad range of preparation tools and the structure of the book will help you pass the exam with ease. I wish you success with your exam and hope that our paths will cross again as you continue to grow in your networking career and certification endeavors.

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CompTIA Network+ N10-009 Cert Guide Exam Updates

Over time, reader feedback allows Pearson to gauge which topics give our readers the most problems when taking the exams. To assist readers with those topics, the authors create new materials clarifying and expanding on those troublesome exam topics. As mentioned in the Introduction, the additional content about the exam is contained in a PDF on this book's companion website, at <https://www.pearsonITcertification.com/title/9780135367889>.

This chapter is intended to provide you with updated information if CompTIA makes minor modifications to the exam upon which this book is based. When CompTIA releases an entirely new exam, the changes are usually too extensive to provide in a simple update chapter. In those cases, you might need to consult the new edition of the book for the updated content. This chapter attempts to fill the void that occurs with any print book. In particular, this chapter does the following:

- Mentions technical items that might not have been mentioned elsewhere in the book
- Covers new topics if CompTIA adds new content to the exam over time
- Provides a way to get up-to-the-minute current information about content for the exam

Always Get the Latest at the Book's Product Page

You are reading the version of this chapter that was available when your book was printed. However, given that the main purpose of this chapter is to be a living, changing document, it is important that you look for the latest version online at the book's companion website. To do so, follow these steps:

- Step 1.** Browse to www.pearsonITcertification.com/title/9780135367889.
- Step 2.** Click the **Updates** tab.
- Step 3.** If there is a new Chapter 27 document on the page, download the latest Chapter 27 document.

NOTE The downloaded document has a version number. Comparing the version of the print Chapter 27 (Version 1.0) with the latest online version of this chapter, you should do the following:

- **Same version:** Ignore the PDF that you downloaded from the companion website.
- **Website has a later version:** Ignore this Chapter 27 in your book and read only the latest version that you downloaded from the companion website.

Technical Content

The current Version 1.0 of this chapter does not contain additional technical coverage.

Answers to Review Questions

Chapter 1

1. **d.** The data link layer of the OSI model is the only layer of the famous model that is typically divided into sublayers (the MAC and LLC sublayers).
2. **b.** Baseband technology uses the entire medium to transmit. In contrast, broadband technology can divide the medium into different channels. A great example of broadband is the use of the coaxial cable you might have in your home, which carries cable television signals as well as high-speed Internet.
3. **c.** The transport layer offers TCP and UDP. With TCP, a connection-oriented protocol, windowing can be used to dictate how much data is sent at one time.
4. **a.** An IP address is a typical ingredient at the network layer (Layer 3) of the OSI model. Routers use these addresses to route traffic through an internetwork.
5. **c.** User Datagram Protocol (UDP) sacrifices reliability for speed and efficiency in the transport of data.
6. **b.** The maximum transmission unit (MTU) is the network interface setting that defines the largest packet that may be sent onto the network.
7. **b.** The well-known port numbers are all below 1024. An example would be HTTP (WWW) at port 80.
8. **b.** HTTPS uses TCP port 443 in its operation. Contrast this to HTTP (that offers no security) and uses port 80.
9. **c.** The Time-to-Live (TTL) value of the IP packet is decremented for each router hop. This is a loop prevention mechanism.
10. **d.** The connection-oriented TCP uses several techniques to ensure reliability in the communications. Windowing is one of those.

- 11. **b.** As data moves down through the OSI model layers (on a host), it is encapsulated with a header added to the beginning and a trailer to the end. When the data arrives at the receiving host, it moves up the model and is decapsulated in that the header and trailer are stripped off as it moves up through the OSI layers.
- 12. **d.** The data link layer is concerned with packaging data into frames and ensuring that frames do not exceed the maximum transmission unit (MTU) of the physical media.

Chapter 2

- 1. **c.** Routers operate at the network layer (Layer 3) of the OSI model.
- 2. **b.** The best answer here is switch because, unlike a multilayer switch, this device must use MAC addressing for frame forwarding.
- 3. **d.** Routers make their forwarding decisions using IP addresses at Layer 3 of the OSI model. Specifically, the destination IP address is the key piece of information used by default.
- 4. **a.** When a switch receives a frame destined for an unknown MAC address, the switch floods the frame out of all the ports on the device, except for the port that received the frame.
- 5. **d.** A router creates a broadcast domain on each of the router (Layer 3) ports.
- 6. **d.** A switch creates a collision domain on each port by default.
- 7. **d.** An access point allows wireless clients to connect to the wireless (and potentially) wired network.
- 8. **d.** An intrusion prevention system (IPS) typically uses signatures to identify common network attacks.
- 9. **c.** A proxy server caches content and improves performance. It can also include permit and deny lists for Internet locations.
- 10. **b.** A stateful firewall has the ability to examine outbound connections and dynamically permit the appropriate inbound responses.
- 11. **b.** An intrusion prevention system (IPS) is a network device that can continually scan the network looking for attack activity. An IPS can stop potential threats. Note that an IDS can identify potential attacks, but it responds by notifying administrators of the attack activity.
- 12. **a.** Network-attached storage (NAS) is a specialized type of storage device that is designed to provide centralized storage and file sharing capabilities to multiple users and devices within a network. Unlike traditional storage

solutions that are directly attached to individual computers, NAS devices connect to a local area network or wide area network, allowing them to be accessed by any device connected to the network.

13. **b.** Quality of service (QoS) includes strategies used to manage and increase the flow of network traffic. QoS features enable administrators to predict bandwidth use, monitor that use, and control it to ensure that bandwidth is available to the applications that need it.

Chapter 3

1. **c.** A network security group (or simply security group) seeks to control traffic into and out of virtual machines in the cloud. These security groups are assigned to virtual network interfaces of virtual machines and function as simple, yet powerful, firewalls.
2. **a.** A public cloud provider services many clients from a single large infrastructure.
3. **c.** Platform as a service (PaaS) is targeted at developers. This cloud service seeks to provide everything the developers need to test and deploy applications.
4. **b.** Elasticity refers to the ability to scale resources based on demand. For example, virtual machines might be dynamically added to a pool of servers or removed, based on demand at different times.
5. **b, c.** Companies that are concerned about data security as they send and receive data from the cloud often use a VPN or a direct private connection to provide strong security for the transfer of this data.
6. **b.** Multitenancy refers to physical servers in the public cloud infrastructure hosting workloads for multiple customers.
7. **a.** The AWS Virtual Private Cloud (VPC) lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment, including the selection of your IP address range, creation of subnets, and configuration of route tables and network gateways.
8. **d.** With software as a service (SaaS), the details of the servers are hidden from the customer, and the customer's experience is similar to the experience of using a web-based application. Examples of SaaS include Gmail and Microsoft 365.

Chapter 4

1. **b.** The Secure Shell (SSH) protocol allows you to make secure remote connections to network systems. This protocol is specialized for terminal connections. For graphical user interface connections, you can use technologies such as Remote Desktop Protocol (RDP) and VNC Viewer.
2. **d.** Dynamic Host Configuration Protocol (DHCP) is used to dynamically assign IP address information to network systems (typically end-user devices).
3. **d.** Domain Name System (DNS) is a global hierarchy system that resolves domain names to IP addresses.
4. **a, c.** Hypertext Transfer Protocol Secure (HTTPS) uses TCP port 443 in its operation.
5. **b, d.** Syslog produces machine data that you can use to monitor and understand the state of services on a device. Syslog runs on UDP port 514.
6. **c.** Internet Control Message Protocol (ICMP) is used by many troubleshooting and monitoring tools. ping and traceroute are two such ICMP-based utilities.
7. **c.** The port number typically used for SSH (Secure Shell) is 22.
8. **c.** The Internet Key Exchange (IKE) protocol is used to negotiate cryptographic keys and security parameters for establishing secure IPsec connections, ensuring that both parties agree on the encryption and authentication methods before data transmission.
9. **c.** In IPv4 networking, broadcast traffic is used to send data to all devices within a local network segment, using the special destination address 255.255.255.255. This address directs the packet to all devices on the network, ensuring that every node receives the information. However, in IPv6, broadcast traffic is no longer supported due to the inefficiency and security concerns associated with broadcasting.
10. **d.** Remote Desktop Protocol (RDP) is a Microsoft protocol that allows a user to view and control the desktop of a remote computer. RDP uses TCP port 3389.
11. **b.** Multicast is a mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group.

Chapter 5

1. **b, c.** Cat 5e and Cat 6 are often used for Fast Ethernet networks.
2. **b.** Plenum cable is a special type of cable for use in the plenum spaces.

3. **b.** The RJ45 connector uses eight pins and is the most common connector used in Ethernet networks.
4. **b, d.** The two major categories of fiber-optic media are single-mode and multimode.
5. **b.** Fast Ethernet operates at 100Mbps.
6. **d.** Fibre Channel is a network technology often found in storage area networks (SANs). Its focus is fast and reliable transfers of data in the SAN.
7. **c.** Cat 5e is an enhanced version of Cat 5. Cat 5e was the first standard for 1Gbps Ethernet.
8. **c.** The 802.11ax standard has also been named Wi-Fi 6 to help typical consumers better identify the latest Wi-Fi technologies. 802.11ac is also known as Wi-Fi 5.
9. **b, d.** To define industry-standard pinouts and color coding for twisted-pair cabling, the TIA/EIA-568 standard was developed. The first iteration of the TIA/EIA-568 standard has come to be known as the TIA/EIA-568A standard, and since then, an updated standard called TIA/EIA-568B has been released. TIA/EIA-568B is the more commonly used standard in the United States today.
10. **a.** 1000BASE-LX uses single-mode fiber (SMF), has a bandwidth capacity of 1Gbps, and has a distance limitation of 5 km.
11. **d.** Nonoverlapping channels are a set of wireless frequency channels that do not interfere with each other. For example, in the 2.4GHz band channels 1, 6, and 11 are commonly considered nonoverlapping channels for Wi-Fi networks.
12. **b.** MU-MIMO is a set of advanced MIMO technologies included with IEEE 802.11ac and 802.11ax that dramatically enhances wireless throughput.
13. **a.** Single-mode fiber (SMF) is a type of fiber that uses a single direct beam of light, thus allowing for greater distances and increased transfer speeds.
14. **d.** Cat 8 is capable of 40Gbps speeds. It supports distances of only 30 to 36 m, depending on the patch cables used. These short distances and very high speeds are ideal for connections in a data center between high-speed multilayer switches.

Chapter 6

1. **d.** A local area network (LAN) most often consists of Ethernet connections as defined in IEEE 802.3.
2. **c.** Actual traffic flows dictate the logical topology that is in use within a network.
3. **d.** The core layer is most concerned with raw speed.
4. **b.** Each leaf device connects to each and every spine device to provide a full mesh of all the leaves for every spine.
5. **d.** A metropolitan area network is superb for connecting various buildings in a citywide area. Metro Ethernet is one example of connectivity technology for this design.
6. **c.** A personal area network features fewer nodes than are typically found in a LAN. This design also features a small geographic area, such as a car or a single-room office.
7. **c.** The star topology is common today in switched LANs. However, it does feature quite a bit of cabling, as the switches require media to each and every node (wired) that needs to communicate. There are also typically trunk connections between the switches for an extended star topology, and this means even more cabling.
8. **a.** The full-mesh topology might be complex and expensive to implement, but it does provide excellent levels of redundancy. In the event that one or more links fail, it is often possible to reroute traffic around the failures.
9. **b.** The formula for the number of connections required for the full-mesh topology is $n(n - 1) / 2$, where n is the number of nodes. So, in this case, $5(5 - 1) / 2$ is 10.
10. **a, c.** The hub-and-spoke topology is actually a form of a partial-mesh topology. This topology has the advantage of being less complex, less costly, and, overall, more scalable than a full-mesh design.
11. **c.** A peer-to-peer network features clients sharing information directly with each other.
12. **b.** Peer-to-peer networks are often celebrated for their simplicity and low cost. Unfortunately, these networks are not scalable and can actually cause administrative nightmares when scaling is required.
13. **d.** The storage area network (SAN) assists when large amounts of data must be stored in the network and transferred efficiently from various nodes to other nodes.

14. **a.** East–west traffic flows occur primarily between servers or services deployed within the same data center or cloud environment. North–south traffic flows involve traffic moving between clients or end users and servers or services hosted in the data center or cloud.
15. **b, c, d.** The three-tier hierarchical model is built with three layers: the access layer (which is the layer closest to the end users), the distribution layer, and the core layer, which is responsible for moving massive amounts of data in an enterprise network—often between a data center (or even data centers) and the main employee locations. There is no intermediate layer in the three-tier hierarchical model.

Chapter 7

1. **b.** The binary representation of 117 is 01110101 (based on $117 = 64 + 32 + 16 + 4 + 1$).
2. **d.** 10110100 is the equivalent of 180 because $128 + 32 + 16 + 4 = 180$.
3. **a.** Because the decimal value 10 appears in the first octet, this is a Class A address.
4. **a.** The classful subnet mask for a Class B IPv4 address features 16 bits and is represented as 255.255.0.0 in dotted-decimal notation.
5. **d.** The Dynamic Host Configuration Protocol (DHCP) is a dynamic method of IP address assignment.
6. **a.** There are 5 bits left here for host address assignment, and 2 raised to the fifth power is 32. You then subtract 2 for the network ID and broadcast address. That is, $2^5 - 2 = 30$.
7. **c.** /28 is the prefix notation for 255.255.255.240. Notice that the mask consists of 8 bits + 8 bits + 8 bits + 4 bits.
8. **c.** Using a /27 mask gives you 3 bits for subnetting. This permits the creation of eight subnets and satisfies your requirement of seven subnets and gives you the most possible host addresses per subnet.
9. **a.** The host address and subnet mask combination of 172.16.18.5/18 references the 172.16.0.0/18 subnet. The usable host IP range is 172.16.0.1–172.16.63.254.
10. **c.** Classless inter-domain routing (CIDR) shortens a classful subnet mask by removing 1s from the mask.
11. **a.** The 127.x.x.x network range is reserved for the loopback function. It is not one of the recognized private address ranges. The private address ranges as defined in RFC1918 are 10.x.x.x, 172.16.x.x to 172.31.x.x, and 192.168.x.x.

Chapter 8

1. **b.** Border Gateway Protocol (BGP) is an exterior gateway protocol used to share prefixes across the Internet. It can also be used to route internally for very large organizations, and it is an example of a control plane technology in the SDN design.
2. **d.** Transport agnostic refers to the capability of the network to seamlessly route and manage traffic over diverse transport media such as MPLS, broadband, LTE, and satellite, without dependency on the underlying physical connections.
3. **a.** User Datagram Protocol (UDP) plays a crucial role in Virtual Extensible LAN (VXLAN) environments by serving as the transport protocol for encapsulating Layer 2 Ethernet frames within Layer 3 IP packets, enabling the creation of scalable and flexible virtual networks over existing IP infrastructure.
4. **b.** Configuration drift occurs when the actual state of the infrastructure deviates from the desired state defined in the IaC. This can happen due to manual changes, system updates, or environmental factors. Automation in IaC helps mitigate configuration drift by regularly applying the desired state to the infrastructure. Tools like Terraform and Ansible can perform periodic checks and reapply configurations to ensure consistency.
5. **c.** Remember, you can drop the leading zeros in any field, and one time within an address, you can use the shorthand symbol :: to represent continuous sections of zeros.
6. **b.** EUI-64 in IPv6 permits the automatic generation of host portions of addresses.
7. **a.** IPv6 networks use stateless address autoconfiguration (SLAAC) to assign IP addresses. With SLAAC, a device sends the router a request for the network prefix, and then the device uses the prefix along with its own MAC address to create an IP address.
8. **d.** Secure Access Secure Edge (SASE) is a network architecture that integrates wide area networking (WAN) capabilities with comprehensive network security functions such as Secure Web Gateway (SWG), Cloud Access Security Broker (CASB), Firewall as a Service (FWaaS), and Zero Trust Network Access (ZTNA) into a unified, cloud-native service model to securely connect users, systems, and endpoints to applications and services.
9. **c.** One of the primary applications of VXLAN is in data center interconnect (DCI). DCI involves connecting multiple data centers to provide a unified infrastructure, allowing for efficient resource sharing, workload mobility, and

disaster recovery. VXLAN is particularly suited for DCI because it enables the extension of Layer 2 networks over Layer 3 distances, thus facilitating the seamless migration of VMs and applications between data centers.

10. a. NAT64 (Network Address Translation from IPv6 to IPv4) is a technology that facilitates communication between IPv6-only clients and IPv4-only servers, bridging the gap between the two distinct IP address families. It is yet another component in the transition from the older IPv4 protocol to the newer IPv6 protocol, allowing IPv6 networks to access resources on IPv4 networks without requiring the end systems to support both protocols.

Chapter 9

1. b. Note that the destination IP address does not change in the packet. However, as the router sends the frame out the exit interface heading toward the destination, it will update the destination MAC address to the next hop/device.
2. d. Address Resolution Protocol (ARP) is one of the many name resolution protocols we deal with. In this case, the names are Layer 2 and Layer 3 addresses.
3. d. The default route is 0.0.0.0/0.
4. c. The better the administrative distance, the more believable the protocol. Lower is better for the AD score.
5. b, c. Split horizon and poison reverse are critical loop prevention techniques for early routing protocols.
6. c. Routing Information Protocol (RIP) cannot be used in large networks due to a maximum hop count of 15.
7. a. Border Gateway Protocol (BGP) is the exterior gateway protocol (EGP) that makes the Internet a reality.
8. a. The IP address assigned to the physical outside interface is the IP address that is often the basis for the port address translation (PAT) configuration. Many different internal IP addresses (inside) can be translated into this single address.
9. b. Network address translation (NAT) allows private IP addresses (as defined in RFC1918) to be translated into Internet-routable IP addresses (public IP addresses).
10. c. The Hot Standby Router Protocol (HSRP) is a Cisco-proprietary approach to first hop redundancy. This technology permits the identification of an active

and standby router or routers that all respond to a virtual IP address on the network used as the default gateway.

11. **d.** Administrative distance (AD) is a value used by routers to determine the trustworthiness or reliability of routing information received from various sources. It is a numerical value assigned to routing protocols or static routes, where a lower administrative distance indicates higher preference or priority.

Chapter 10

1. **a.** A 1000BASE-T Ethernet network has a distance limitation of 100 m.
2. **d.** A random back-off timer is used in a CSMA/CD network when a collision is detected.
3. **a, b.** Virtual LANs (VLANs) are also IP subnets. A router must route between these subnets. The VLAN is also a broadcast domain.
4. **a.** The 802.1Q native VLAN is the only VLAN in a Layer 2 domain that is not tagged.
5. **b.** The designated port is forwarding, and there must be one on every network segment. Remember that all root ports are designated by default.
6. **b.** 802.3ad is the open-standard version of link aggregation.
7. **b.** The IEEE 802.3af standard specifies 15.4W as the maximum amount of power a switch is allowed to provide per port.
8. **b.** The Link Aggregation and Control Protocol (LACP) permits administrators to bundle interfaces on the switch together, adding to the redundant capabilities of modern switches.
9. **d.** Full-duplex is a system in which data simultaneously transmits in two directions. In half-duplex, data is transmitted in both directions but not simultaneously.
10. **a.** A Switch Virtual Interface (SVI), also known as a VLAN interface, is a virtual interface configured on a Layer 2 switch to enable routing between VLANs.

Chapter 11

1. **d.** The key to this question is the fact that power is being radiated in all directions relatively equally.
2. **b, d, f.** The non-overlapping channels in the United States are 1, 6, and 11.

3. **a.** The 802.11h amendment enhances wireless networking by introducing mechanisms for dynamic frequency selection (DFS) and transmit power control (TPC).
4. **d.** Lightweight access points are used in conjunction with wireless LAN controllers by offloading their management functions to the WLCs, allowing the APs to handle only the transmission and reception of wireless signals, while the WLCs centrally manage tasks such as security policies, configuration, and network management, leading to simplified deployment and maintenance of large-scale wireless networks.
5. **b.** A captive portal on a guest Wi-Fi network is a web page that users are automatically redirected to upon connecting, where they must authenticate or accept terms of service before gaining access to the Internet.
6. **b.** An independent basic service set (IBSS) is formed directly between wireless clients.
7. **b.** The recommended amount of overlap is 10% to 15%.
8. **c.** Open authentication permits the use of the wireless LAN with no credentials. This is useful in a public, free Wi-Fi hotspot or in a guest area in an enterprise.
9. **a.** Band steering is a feature in Wi-Fi networks that directs dual-band-capable devices to connect to the less congested 5GHz band instead of the more crowded 2.4GHz band, enhancing overall network performance and reducing interference.
10. **d.** The IEEE 802.11i requirements are found in WPA2.
11. **d.** Geofencing permits the creation of a boundary for administrative alerts and actions.
12. **a, b, d.** Two versions of WPA2 exist: WPA2 Personal and WPA2 Enterprise. WPA2 Personal protects unauthorized network access via a password. WPA2 Enterprise verifies network users through a server. WPA2 Personal uses pre-shared keys. WPA3 also includes both a Personal and Enterprise version. WPA3 maintains equivalent cryptographic strength through the required use of 192-bit AES for the Enterprise version, and optional 192-bit AES for the Personal version.
13. **a.** To improve the security of your network, change the service set identifiers (SSIDs) on your access points (APs). Using the default SSID poses a security risk even if the AP is not broadcasting it. When changing default SSIDs, do not change the SSID to reflect your company's main names, divisions,

products, or address. This simply makes you an easy target for attacks such as war driving, war chalking, and war flying.

14. **b.** 802.1X is an Institute of Electrical and Electronics Engineers (IEEE) standard for port-based access control and provides a method for authenticating a device to another system via an authentication server. 802.1X specifically provides for the encapsulation of Extensible Authentication Protocol (EAP) over the IEEE family of standards for packet-based networks. The standard essentially facilitates the use of various authentication methods such as RADIUS and digital certificates.

Chapter 12

1. **d.** A patch panel is a hardware device that provides a central point for organizing and connecting incoming and outgoing network cables, allowing for easy management and reconfiguration of network connections.
2. **b.** Equipment rack sizes are typically expressed in units, where one unit is 1.75 inches.
3. **c.** When it comes to the distribution of power, power distribution units (PDUs) are devices that distribute electrical power to networking equipment within a rack or a network room.
4. **a.** Power load refers to the total electrical power consumed by networking equipment in a physical installation.
5. **d.** Uninterruptible power supplies are crucial for maintaining network reliability and continuity in the event of power disruptions. UPS systems provide backup power to critical networking equipment, such as servers, switches, and routers, allowing them to continue functioning during power outages.
6. **b.** The main distribution frame (MDF) is the primary wiring closet for a network that typically holds the majority of the network gear, including routers, switches, wiring, servers, and more. This is also typically the wiring closet where outside lines run into the network.

Chapter 13

1. **c.** The Layer 3 network diagram focuses on details associated with the network layer of the OSI model. This layer would include such details as IP addresses and IP connectivity.
2. **b.** IP address management (IPAM) refers to the systematic administration of IP addresses, ensuring efficient allocation, tracking, and maintenance within a network environment.

3. **a.** A service-level agreement (SLA) is a formal contract between a service provider and a customer that outlines the agreed-upon level of service, including performance metrics and responsibilities.
4. **d.** EOL (end-of-life) and EOS (end-of-support) dates are often critical components in life-cycle management, as they provide clear timelines for when products or systems will no longer receive updates, security patches, or vendor support, prompting organizations to plan for replacements or upgrades to maintain operational efficiency and security compliance.
5. **b.** Disposal is often the last stage of a system life cycle, where outdated or decommissioned hardware and software are securely and responsibly discarded or recycled to ensure data protection and environmental compliance.
6. **d.** A baseline/golden configuration is a predefined and standardized template or ideal state of configuration settings for hardware, software, or network devices, used as a reference to ensure consistency, security, and compliance across an organization's IT environment.
7. **a, d.** Wireless surveys and heat maps are often integral components of the IT documentation in modern organizations, providing a detailed visual representation and analysis of wireless network coverage and performance across various locations. A wireless survey involves assessing the physical environment to identify optimal placement of wireless access points, detect signal interference, and measure signal strength and quality. The resulting heat maps visually depict these findings, using color gradients to represent signal strength, coverage areas, and potential dead zones, enabling IT professionals to pinpoint areas with weak or no signal and optimize the network layout accordingly.
8. **b.** An asset inventory in an IT department's documentation is a comprehensive catalog of the organization's technology assets, encompassing hardware, software, licensing, and warranty support.

Chapter 14

1. **b.** The management information base (MIB) is a database filled with variables that represent the metrics of the managed network devices.
2. **b.** The Critical syslog level is Level 2, the most severe level.
3. **a, c, d.** The put method is used with HTTP. SNMP uses get, set, and trap messages—among others.
4. **d.** Jitter, which is variation in delay, can be a large problem for VoIP networks.
5. **a.** The noAuthNoPriv security level of SNMPv3 is like the community string approach of SNMPv2c.

- 6. **d.** Syslog level 6 is for informational messages. Note that the lowest level, level 7, is used for debugging messages.
- 7. **a.** NetFlow often uses a flow collector, which can be a single-purpose device on a network for storing, backing up, and making logs available to the authorized staff.
- 8. **d.** SNMPv3 introduced significant improvements in security, offering authentication, encryption, and access control mechanisms to protect SNMP communications. SNMPv3 is the latest and most secure version of SNMP and allows for the use of strong cryptographic algorithms, such as HMAC-SHA, HMAC-MD5, and AES, to ensure data integrity and confidentiality.
- 9. **a.** When a network device sends packets, there will be serialization delay, propagation delay, processing delay, and probably even more types of delay. You need to take baseline measurements so you can understand the latency in your network.
- 10. **c.** Some switches support a port mirroring feature, which makes a copy of traffic seen on one port and sends that duplicated traffic out another port (to which a network sniffer could be attached). This allows a network sniffer to capture the packets that need to be analyzed.
- 11. **d.** Performance monitoring solutions are designed to continuously track and evaluate the efficiency and effectiveness of various components within an IT infrastructure, including servers, applications, networks, and databases. These solutions collect and analyze data on key performance indicators (KPIs) such as response time, throughput, and resource utilization, providing real-time insights and historical trends.
- 12. **c.** Log aggregation is the process by which security information and event management (SIEM) systems combine similar events to reduce event volume. SIEMs can aggregate data into log format from many network sources and consolidate the data so that crucial events are not missed.

Chapter 15

- 1. **b.** Five nines availability means 99.999% uptime and translates into about 5 minutes of downtime per year.
- 2. **c.** The active-passive design uses one NIC that is actually active and able to pass traffic. The other NIC is in a passive (often called standby) state.
- 3. **a.** A content engine permits the caching of key data to which clients need low-latency access.
- 4. **c.** A snapshot is typically a complete point-in-time, read-only copy of data.

5. **c.** A hot site is often the ultimate goal of a disaster recovery design. A hot site would be able to provide full-service levels with virtually no downtime after a disaster occurs.
6. **a.** Equal-cost multipathing (ECMP) enables routing protocols to distribute traffic among different available paths in the network. This is very advantageous, especially compared to classic Layer 2 designs that feature STP, which blocks redundant paths in order to prevent loops.
7. **d.** The mean time between failures (MTBF) is the average amount of time that passes between hardware component failures, excluding time spent repairing components or waiting for repairs.
8. **b.** A cold site is the weakest of the recovery sites but also the least costly. However, although a cold site may be the least costly when you're planning for disaster, after a disaster occurs, equipment purchased for a cold site might be expensive or difficult to obtain.
9. **b, c, d.** The difference between incremental and differential backups is that differential backups include all data that has changed since the full backup. Therefore, answer a is the only false statement.
10. **d.** In an active-active, high-availability disaster recovery setup, multiple redundant systems are simultaneously active and operational, serving incoming requests or processing tasks. In this configuration, each system is capable of independently handling the workload, providing load balancing and fault tolerance.
11. **b.** Tabletop exercises are simulated, discussion-based sessions used to test and evaluate an organization's disaster recovery plans. During these exercises, key stakeholders and team members gather to walk through hypothetical disaster scenarios, such as cyberattacks, natural disasters, or system failures, and discuss their roles, responsibilities, and responses.

Chapter 16

1. **c.** The DHCP DISCOVER message is the first step in the four-step DHCP process (aka DORA). A client broadcasts these messages on the local subnet in an attempt to find a DHCP server.
2. **b.** You often use the scope options in DHCP to set things like lease duration, as well as other pieces of IP configuration information that the client might need.
3. **a.** A pointer record in DNS points to a canonical name. This record type is typically used to perform reverse lookups when required.

- 4. **d.** The DNS TTL (time to live) dictates how long devices can cache the results of name resolutions.
- 5. **b, e.** NTP relies on UDP port 123 in its operation.
- 6. **b.** The stratum is a value used in NTP to provide an indicator of how many hops a client is from the NTP server.
- 7. **d.** A DHCP scope is a range of IP addresses and configuration parameters that a DHCP server is responsible for assigning to client devices on a network.
- 8. **b.** Stateless address autoconfiguration (SLAAC) is a method that allows devices to automatically configure their own IPv6 addresses and network settings without the need for a central DHCP server.
- 9. **b.** AAAA is a DNS record that maps a hostname to an IPv6 address.
- 10. **a.** DNS over HTTPS (DoH) is a protocol that encrypts Domain Name System (DNS) queries and responses using the HTTPS protocol, providing privacy and security enhancements for DNS resolution and helping to prevent DNS-related attacks, such as DNS spoofing or DNS hijacking.

Chapter 17

- 1. **b.** The client-to-site VPN type often features the use of a client system and its web browser and SSL/TLS to make a VPN connection. This is often considered “clientless” because the web browser is built in to the operating system, and there is no need for any separate client software installation.
- 2. **a.** Split tunneling is a VPN configuration that enables a user to access a public network (like the Internet) and a local private network (such as a corporate network) simultaneously, directing some traffic through the VPN and other traffic directly to the Internet.
- 3. **c.** The Secure Shell (SSH) protocol is the secure replacement technology for Telnet. Unlike Telnet, SSH offers strong security mechanisms that are in wide use today.
- 4. **c.** A console port is a physical interface on a network device, such as a router or switch, used for direct local management and configuration via a terminal or computer, typically connected using a console cable.
- 5. **d.** Captive portals are common in public places such as airports and coffee shops. The user simply clicks Accept, views an advertisement, provides an email address, or performs some other required action, and the network then grants access to the user and no longer holds the user captive to that portal.

- 6. **b.** Data at rest is data that is stored or retained in a persistent storage medium, such as a hard disk drive (HDD), solid-state drive (SSD), tape drive, or optical disc, and is not actively being accessed or transmitted.
- 7. **d.** Transport Layer Security (TLS) has largely replaced SSL as the VPN protocol of choice for providing cryptography and reliability to upper layers of the OSI model. For example, when you securely connect to a website using HTTPS, you are probably using TLS.
- 8. **b.** In-band management is a method of managing and monitoring network devices, such as routers, switches, firewalls, and servers, using the same communication path or network infrastructure that carries user data traffic.
- 9. **a.** A jump box, also known as a jump host or bastion host, is a secure, isolated server used to provide controlled access to devices or networks in a different security zone.

Chapter 18

- 1. **c.** Advanced Encryption Standard (AES) permits the configuration of various strength levels, including 128-, 192-, and 256-bit key versions.
- 2. **a.** Integrity involves ensuring that data has not been manipulated in transit.
- 3. **b.** The principle of least privilege involves giving a user account the fewest possible permissions required to do a job.
- 4. **d.** A honeypot is a network device that tries to attract security attacks, allowing a network administrator to analyze the attackers and their strategies.
- 5. **b.** A BYOD policy permits employees to bring their own devices like smartphones and tablets in order to make them more productive.
- 6. **d.** The Common Vulnerabilities and Exposures (CVE) system is a free online resource that provides excellent search tools to leverage a large database of publicly known information security vulnerabilities and exposures.
- 7. **a.** A security information and event management (SIEM) system provides real-time analysis of security alerts generated by applications and network hardware. SIEM systems can log security data and generate reports for compliance purposes.
- 8. **c.** Kerberos is a client/server authentication protocol that supports mutual authentication between a client and a server. With Kerberos, a trusted third party (a key distribution center) hands out tickets that are used instead of username and password combinations.
- 9. **b, c.** An automated teller machine (ATM) provides a common example of a multifactor authentication system. It requires both a “something you

have” physical key (your ATM card) and a “something you know” personal identification number (PIN).

10. **d.** It is important to include separation of duties when planning for security policy compliance. Without this separation, all areas of control and compliance could end up in the hands of a single individual.
11. **a.** Motion detection can certainly help with physical security, but it is a technique used in detection, not prevention.
12. **d.** Biometrics is an excellent form of a physical security control; however, this control is considered a prevention method and not a detection method.
13. **a, b.** PKI uses digital certificates and a certificate authority (CA) for authentication and encryption services.
14. **c.** A zero-day attack or threat is a computer threat that tries to exploit computer application vulnerabilities that are unknown to others—even the software developer. Those weaknesses are also called zero-day vulnerabilities. Zero-day exploits involve using software to exploit security holes to carry out attacks; attackers carry out these exploits or share information about them before the developer of the target software knows about the vulnerability.
15. **b.** An access control vestibule (formerly known as a mantrap) is a small entry area with two interlocking doors that provides assurance that tailgating and piggybacking forms of social engineering cannot take place.
16. **d.** The General Data Protection Regulation (GDPR) is a set of rules created to protect the privacy and personal data of individuals in the European Union. It gives people more control over their personal information and requires organizations to be transparent about how they collect, store, and use this data. GDPR applies to any business, regardless of location in the world, that processes the data of EU citizens.

Chapter 19

1. **d.** A VLAN hopping attack leverages two technologies in its operation: Q-in-Q tunneling and the native VLAN feature.
2. **d.** A distributed denial-of-service (DDoS) attack involves leveraging many systems to compromise the availability of a system.
3. **c.** Tailgating is a social engineering attack in which an authorized user permits an unauthorized user to access an area or systems. The unauthorized user may follow closely behind the authorized user through some type of security checkpoint.

4. **d.** In a brute-force password attack, the attacker tries all possible password combinations until a match is made. For example, a brute-force attack might start with the letter a and go through the letter z, and then the attacker might attempt the letters aa through zz, continuing to try combinations until the password is determined. Using complicated passwords—with a mixture of upper- and lowercase letters as well as special characters and numbers—can help prevent brute-force attacks.
5. **c.** Ransomware is an attack that locks access to a system or files (often using encryption) and demands payment of a ransom (often in cryptocurrency) for access to files or systems to be restored.
6. **b.** A DoS attack originates from a single system, whereas a DDoS attack originates from multiple systems simultaneously and the attacker distributes zombie software or infects multiple (even into the thousands) hosts, providing the attacker partial or full control of the infected computer system through one or more command and control servers.
7. **d.** An evil twin is a special type of rogue access point attack. This attack is designed to capture authentication information from an unsuspecting network user. This attack is so named because a rogue AP is configured in this scenario to appear just like the legitimate AP that the client should be connecting with.
8. **d.** Password spraying is a type of password attack that attempts to access a large number of user accounts with a very few number of commonly used passwords.

Chapter 20

1. **a.** Control plane policing (CoPP) can help in this situation. Because your IGP and EGP routing protocols are part of the control plane, you can use CoPP to watch the amount of traffic that is permitted to your CPU. In this way, you can prevent many different types of DoS attacks that target the control plane.
2. **d.** Extensible Authentication Protocol (EAP) is a flexible solution that is used in many network environments to support a wide variety of authentication and authorization scenarios. EAP is the featured technology of 802.1X.
3. **c.** Private VLANs add segmentation capabilities beyond what is typical for VLAN communication. You can create segmentation within an IP subnet by using this technology.
4. **d.** Network hardening best practices include using SNMPv3 instead of earlier versions, disabling unneeded ports and unneeded services, and changing default passwords to something other than the known default passwords.

5. **a.** An implicit deny clause (in a firewall rule) means that if the proviso in question has not been explicitly granted, then access is denied.
6. **d.** A public network or a Wi-Fi hotspot may use a captive portal, which requires users to agree to some condition before they can use the network or Internet.
7. **b.** An untrusted zone is a network segment with higher risk, such as external networks, where security threats are more likely, requiring stringent controls to protect internal systems.
8. **d.** Key management involves the processes and systems for generating, distributing, rotating, storing, and handling cryptographic keys to ensure secure encryption and decryption of data.
9. **a, d.** Use secure protocols over unsecure protocols whenever possible. For example, use Secure Shell (SSH) rather than the insecure Telnet for remote management using terminals, and use SNMPv3 over previous versions if possible.

Chapter 21

1. **d.** Identifying or defining the problem is often the first step in a troubleshooting methodology.
2. **a, c, d.** Identifying, fixing, and reporting the problem are often the three steps in a simplified troubleshooting flow.
3. **a.** One of the final steps in a structured troubleshooting methodology would be to implement the solution. If you and your team are unable to implement a solution, you should escalate the problem.
4. **d.** Step 3 is “Test the theory to determine the cause.” This step includes two substeps: If the theory is confirmed, determine the next steps to resolve the problem. If the theory is not confirmed, establish a new theory or escalate. If you answered this question correctly, you have a solid grasp of the CompTIA network troubleshooting methodology!
5. **c.** After you have verified full system functionality and implemented preventive measures (Step 6), you should complete the final step (Step 7) by documenting findings, actions, outcomes, and lessons learned.
6. **a, b.** Substeps include top-to-bottom/bottom-to-top OSI model and divide and conquer.

Chapter 22

1. **a.** To use Power over Ethernet (PoE), your physical media must be Cat 5e or higher.

2. **b.** You must ensure that you do not allow dirt to enter fiber-optic cable or cable connectors.
3. **b.** Using a loopback plug is a simple method of testing a network device's capability to send and receive traffic.
4. **c, d.** To mitigate the concern of pumping poisonous gas throughout a building's heating, ventilation, and air-conditioning (HVAC) system, you can use plenum-rated cabling. The outer insulation of a plenum twisted-pair cable is fire retardant; in addition, some plenum cabling uses a fluorinated ethylene polymer (FEP) or a low-smoke polyvinyl chloride (PVC) to minimize dangerous fumes. Cables that run through risers should also be fireproof to prevent flames from traveling up the cable.
5. **a, e.** Attenuation is the weakening of a signal with greater distance between two devices on network media. Attenuation is measured in decibels (dB). dB loss is also more prone to happen as the signal needs to go farther and farther in the media.
6. **b.** An error disabled state is often the result of a violation of some protection mechanism on the network device. For example, the BPDU Guard feature of Spanning Tree Protocol would produce an error disabled state if there were to be a violation of this protection mechanism.
7. **c.** Duplex mismatches can cause links between devices to run inefficiently. A common scenario is one device operating in half-duplex while the other operates in full-duplex. You can check this setting in Windows 11 for a NIC, for example. Open Device Manager and right-click the suspect network adapter. Select Properties and then Speed & Duplex and look at the Value setting.
8. **a, d.** Fiber-optic cabling is immune to electromagnetic interference (EMI) and radio frequency interference (RFI) and it is more difficult to install and troubleshoot than unshielded twisted pair (UTP).
9. **c.** Interface drop errors occur when a network interface discards incoming or outgoing packets due to congestion, buffer overflows, or other issues. These drops most often result from excessive network traffic, insufficient bandwidth, or hardware limitations.
10. **b.** Crosstalk occurs when a signal transmitted on one circuit or channel creates an undesired effect on another circuit or channel. It can be caused by physical proximity of the wires, electromagnetic interference, or poor shielding and insulation.

Chapter 23

1. **c.** This problem sounds like it could be related to DNS-based name resolution.
2. **a.** A misconfigured default gateway address can lead to issues with reaching remote destinations.
3. **b, c.** Spanning Tree Protocol (STP) protects against three main issues: MAC address table corruption, broadcast storms, and loops at Layer 2.
4. **b.** Being able to successfully ping another host is a simple method of confirming that Layers 1 through 3 of the OSI model are working.
5. **d.** The configuration of virtual LANs (VLANs) permits the creation of additional broadcast domains.
6. **c.** This IP address and subnet mask combination means that this host lives on the 172.16.128.0 subnet.
7. **a, d.** If STP is not running correctly—or not running at all—it will not be long before you have broadcast storms and switching loops. These are the enemies that STP is engineered to protect you from.
8. **b.** DHCP scope exhaustion refers to the scope being completely out of IP addresses that can be leased to clients.
9. **a.** Failure to exclude statically assigned IP addresses in the network can lead to duplicate IP address issues. When systems have duplicate IP addresses, their ability to communicate successfully on the network can be compromised.
10. **c.** The root bridge in Spanning Tree Protocol (STP) is the central reference point of the network topology, selected based on the lowest bridge ID, to which all other network switches determine their best path to ensure a loop-free network.

Chapter 24

1. **a.** Received signal strength indication (RSSI) is an excellent metric for measuring the strength of the received wireless signal.
2. **a, c, d.** Fax machines and Bluetooth are the technologies in this list that are least likely to cause interference with wireless networks. Gaming consoles, microwaves, and baby monitors are the most common sources of RFI in wireless networks.
3. **a.** Throughput is a measure of the actual data that can be sent through the network. Speed is the theoretical maximum data rate for wireless.

4. **d.** A unidirectional antenna directs the bulk of its signal in a specific direction. In contrast, an omnidirectional antenna sends the signal in a 360-degree coverage area around the antenna.
5. **b.** A captive portal is a web page that asks clients seeking access to agree to organizational policies and typically captures information from clients.
6. **a.** A site survey can be a critical aid in implementing wireless network infrastructure. During a site survey, key issues can be tested for, such as excessive interference in certain areas.
7. **a.** Disassociation is often an initial step in a much larger attack. Once the client is disassociated, it can be associated to a rogue device.
8. **b.** The coverage areas of wireless APs using overlapping channels should not overlap. To maintain coverage between coverage areas, you should have overlapping coverage areas among wireless APs using non-overlapping channels (for example, channels 1, 6, and 11 for wireless networks using the 2.4GHz band of frequencies).
9. **d.** Bandwidth refers to the maximum amount of data that can be transmitted over the network in a given amount of time, usually measured in bits per second (bps).
10. **c.** Contention occurs when multiple devices or applications simultaneously compete for the same network resources, leading to potential delays and reduced performance.

Chapter 25

1. **b.** The **arp** command permits you to see the IP address to MAC address mappings. You can read them from left to right.
2. **d.** You can use the **/all** switch to learn many additional details about the IP configuration, including DNS details.
3. **c.** The **ping** command uses Internet Control Message Protocol (ICMP) in its operation. If you issue a **ping** command from your PC, your PC sends an ICMP echo message to the specified destination host. If the destination host is reachable, the host responds with an ICMP echo reply message.
4. **d.** You use **traceroute** in Linux to follow the hop-by-hop path that a packet takes.
5. **a, c, d.** You can use the Linux commands **nslookup**, **dig**, and **host** to verify DNS operations.
6. **b.** This output is an example of output from the **dig** command.

7. **b.** A Wi-Fi analyzer is a tool that would be used as part of a wireless site survey after Wi-Fi has been implemented to create a heat map of the wireless airspace.
8. **c.** The **show route** command is used to view the routing table configuration of a network device.
9. **c, d.** Link Layer Discovery Protocol (LLDP) and Cisco Discovery Protocol (CDP) are protocols used by devices to identify and share information about themselves within a local area network. While LLDP is an open standard defined by the IEEE, CDP is an invention by Cisco Systems.

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Memory Tables

Use this appendix to test your memory on what you’ve learned in the chapters. Fill in the missing text in the tables, and check your answers in Appendix C, “Memory Tables Answer Key.”

Chapter 2

Table 2-1 Networking Device Summary

Device	Description	Key Points
Router		
Switch		
Multilayer switch		
Firewall		
IDS/IPS		
Load balancer		
Proxy		
Network-attached storage (NAS)		

Device	Description	Key Points
Storage area network (SAN)		
Access point (AP)		
Controller		

Table 2-2 Three Categories of Quality Issues

Issue	Description
Delay	
Jitter	
Drops	

Chapter 4

Table 4-1 Port Assignments for Commonly Used Protocols

Protocol	Description	Port Assignment
DHCP (Dynamic Host Configuration Protocol)		
DNS (Domain Name System)		
FTP (File Transfer Protocol)		

Protocol	Description	Port Assignment
HTTP (Hypertext Transfer Protocol)		
HTTPS (Hypertext Transfer Protocol Secure)		
LDAP (Lightweight Directory Access Protocol)		
LDAPS (Lightweight Directory Access Protocol over SSL)		
NTP (Network Time Protocol)		
RDP (Remote Desktop Protocol)		
SFTP (Secure File Transfer Protocol)		
SIP (Session Initiation Protocol)		
SMB (Server Message Block)		
SMTP (Simple Mail Transfer Protocol)		
SMTPS (Simple Mail Transfer Protocol Secure)		
SNMP (Simple Network Management Protocol)		
SQL (Structured Query Language) Server		

Protocol	Description	Port Assignment
SSH (Secure Shell)		
Syslog		
Telnet		
TFTP (Trivial File Transfer Protocol)		

Table 4-2 Internet Protocol (IP) Types Summary

Protocol	Complete Protocol Name	Description
		A complex suite of protocols that is used to create secured connections between network systems.
		A tunneling protocol used to encapsulate a wide variety of network layer protocols inside virtual point-to-point links or point-to-multipoint links over an IP network.
		A network layer protocol, documented in RFC 791, that offers a connectionless internetwork service. IP provides features for addressing, packet fragmentation and reassembly, type-of-service specification, and security.
		A connection-oriented protocol that offers flow control, sequencing, and retransmission of dropped packets.
		A connectionless alternative to TCP used for applications that do not require the functions offered by TCP.
		Sends error messages and operational information indicating issues with packet delivery in IP networks.
		Provides for encryption in the IPsec suite of protocols.
		Provides for authentication in the IPsec suite of protocols.
		Negotiates cryptographic keys and security parameters for IPsec.

Table 4-3 Traffic Types Summary

Traffic type	Description
	Traffic is one-to-one communication between a single sender and a single receiver, and this is the most common type of traffic in IP networks. It is used for point-to-point communication between devices, such as client-server interactions and most Internet browsing.
	A mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group.
	_____ traffic is one-to-nearest communication from a sender to the nearest of multiple receivers. Multiple devices share the same anycast IP address, but packets are routed to the nearest (or best) destination based on routing metrics such as shortest path or lowest latency.
	A _____ is at the opposite end of the spectrum from a unicast. _____ traffic is one-to-all communication from a single sender to all devices within a specific network segment.

Chapter 5

Table 5-3 Characteristics of 802.11 Standards

Standard	Band	Maximum Bandwidth	Transmission Method	Maximum Range
802.11			DSSS or FHSS	
802.11a			OFDM	
802.11b			DSSS	
802.11g			OFDM or DSSS	
802.11n			OFDM	
802.11ac			OFDM	
802.11ax			OFDMA	

Table 5-4 Ethernet Bandwidth Capacities

Ethernet Type	Bandwidth Capacity
	10Mbps: 10 million bits per second (that is, 10 megabits per second)
	100Mbps: 100 million bits per second (that is, 100 megabits per second)
	1Gbps: 1 billion bits per second (that is, 1 gigabit per second)
	10Gbps: 10 billion bits per second (that is, 10 gigabits per second)
	40Gbps: 40 billion bits per second (that is, 40 gigabits per second)
	100Gbps: 100 billion bits per second (that is, 100 gigabits per second)

Table 5-5 Types of Ethernet

Ethernet Standard	Media Type	Bandwidth Capacity	Distance Limitation
	Coax (thicknet)	10Mbps	500 m
	Coax (thinnet)	10Mbps	185 m
	Cat 3 (or higher) UTP	10Mbps	100 m
	Cat 5 (or higher) UTP	100Mbps	100 m
	MMF	100Mbps	2 km
	MMF	100Mbps	300 m
	Cat 5e (or higher) UTP	1Gbps	100 m
	Cat 6 (or higher) UTP	1Gbps	100 m
	MMF	1Gbps	550 m
	SMF	1Gbps	5 km
	SMF	1Gbps	10 km
	SMF	1Gbps	70 km
	MMF	10Gbps	26–400 m
	SMF	10Gbps	10–25 km
	SMF	10Gbps	40 km
	MMF	10Gbps	300 m

Ethernet Standard	Media Type	Bandwidth Capacity	Distance Limitation
	SMF	10Gbps	10 km
	SMF	10Gbps	40 km
	Cat 6a (or higher)	10Gbps	100 m
	Cat 8	40Gbps	30 m
	MMF	100Gbps	125 m
	SMF	100Gbps	10 km
	SMF	100Gbps	40 km

Chapter 6

Table 6-1 Characteristics, Benefits, and Drawbacks of Client/Server Networks

Characteristics	Benefits	Drawbacks
Client devices (for example, PCs) share a common set of resources (for example, file or print resources) located on one or more dedicated servers.		Because multiple clients might rely on a single server for their resources, the single server can become a single point of failure in the network.
Resource sharing is made possible via dedicated server hardware and network operating systems.		Client/server networks can cost more than peer-to-peer networks. For example, client/server networks might require the purchase of dedicated server hardware and a network OS with an appropriate number of licenses.

Table 6-2 Characteristics, Benefits, and Drawbacks of a Peer-to-Peer Network

Characteristics	Benefits	Drawbacks
Client devices (for example, PCs) share their resources (for example, file and printer resources) with other client devices.		Scalability is limited because of the increased administration burden of managing multiple clients.

Characteristics	Benefits	Drawbacks
Resource sharing is made available through the clients' operating systems.		Performance might not be as strong as in a client/server network because the devices providing network resources might be performing other tasks not related to resource sharing (for example, word processing).

Table 6-3 Characteristics, Benefits, and Drawbacks of a Star Topology

Characteristics	Benefits	Drawbacks
Devices have independent connections to a central device (for example, a hub or a switch).		More cable is required for a star topology because each device requires its own cable to connect back to the central device.
Star topologies are commonly used with Ethernet technologies (described in Chapter 5, "Transmission Media and Transceivers").		Installation can take longer for a star topology because more cable runs must be installed.

Table 6-4 Characteristics, Benefits, and Drawbacks of a Hub-and-Spoke WAN Topology

Characteristics	Benefits	Drawbacks
Each remote site (that is, a spoke) connects to a main site (that is, the hub) via a WAN link.		Suboptimal routes must be used between remote sites because all intersite communication must travel via the main site.
Communication between two remote sites travels through the hub site.		Because all remote sites converge on the main site, this hub site is potentially a single point of failure.
—		Because each remote site is reachable by only a single WAN link, the hub-and-spoke topology lacks redundancy.

Table 6-5 Characteristics, Benefits, and Drawbacks of a Full-Mesh WAN Topology

Characteristics	Benefits	Drawbacks
Every site has a direct WAN connection to every other site.		A full-mesh network can be difficult and expensive to scale because the addition of one new site requires a new WAN link between the new site and every other existing site.
The number of required WAN connections can be calculated with the formula $w = n \times (n - 1) / 2$, where w = the number of WAN links and n = the number of sites. For example, a network with 10 sites would require 45 WAN connections to form a fully meshed network: $45 = 10 \times (10 - 1) / 2$.		—
—		—

Table 6-6 Characteristics, Benefits, and Drawbacks of a Partial-Mesh Topology

Characteristics	Benefits	Drawbacks
Selected sites (that is, sites with frequent intersite communication) are interconnected via direct links, whereas sites that have less-frequent communication can communicate via another site.		A partial-mesh topology is less fault tolerant than a full-mesh topology.
A partial-mesh topology uses fewer links than a full-mesh topology and more links than a hub-and-spoke topology for interconnecting the same number of sites.		A partial-mesh topology is more expensive than a hub-and-spoke topology.

Chapter 7

Table 7-19 IP Address Classes

Address Class	Value in First Octet	Classful Mask (Dotted Decimal)	Classful Mask (Prefix Notation)
Class A			/8
Class B			/16
Class C			/24
Class D		—	—
Class E		—	—

Table 7-20 Private IP Networks

Address Class	Address Range	Default Subnet Mask
Class A		255.0.0.0
Class B		255.255.0.0
Class B		255.255.0.0
Class C		255.255.255.0

Table 7-22 Dotted-Decimal and Prefix-Notation Representations for IPv4 Subnets

Dotted-Decimal Notation	Prefix Notation
255.0.0.0	
255.128.0.0	
255.192.0.0	
255.224.0.0	
255.240.0.0	
255.248.0.0	
255.252.0.0	
255.254.0.0	
255.255.0.0	
255.255.128.0	
255.255.192.0	
255.255.224.0	
255.255.240.0	
255.255.248.0	

Dotted-Decimal Notation	Prefix Notation
255.255.252.0	
255.255.254.0	
255.255.255.0	
255.255.255.128	
255.255.255.192	
255.255.255.224	
255.255.255.240	
255.255.255.248	
255.255.255.252	

Table 7-23 Subnet Octet Values

Subnet Octet Value	Number of Contiguous Left-Justified 1s
0	
128	
192	
224	
240	
248	
252	
254	
255	

Table 7-24 Number of Subnets Created by a Specified Number of Borrowed Bits

Borrowed Bits	Number of Subnets Created (2^s , Where s Is the Number of Borrowed Bits)
0	
1	
2	
3	
4	
5	
6	
7	

Borrowed Bits	Number of Subnets Created (2^s, Where s Is the Number of Borrowed Bits)
8	
9	
10	
11	
12	

Table 7-25 Number of Supported Hosts, Given a Specified Number of Host Bits

Host Bits	Number of Supported Hosts ($2^h - 2$, Where h Is the Number of Host Bits)
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	

Table 7-26 Usable IP Address Ranges for the 192.168.10.0/27 Subnets

Subnet Address	Directed Broadcast Address	Usable IP Addresses
192.168.10.0		
192.168.10.32		
192.168.10.64		
192.168.10.96		
192.168.10.128		
192.168.10.160		
192.168.10.192		
192.168.10.224		

Table 7-27 Usable IP Address Ranges for the 192.168.0.0/26 Subnets

Subnet Address	Directed Broadcast Address	Usable IP Addresses
192.168.0.0		
192.168.0.64		
192.168.0.128		
192.168.0.192		

Chapter 9

Table 9-1 Administrative Distance

Routing Information Source	AD Value
Directly connected network	
Statically configured network	
EIGRP	
OSPF	
RIP	
External EIGRP	
Unknown or unbelievable	

Table 9-2 Comparing Dynamic Routing Protocols

Routing Protocol	IGP or EGP	Type	Metric
OSPF			
EIGRP			
BGP			

Table 9-3 Names of NAT IP Addresses

NAT IP Address	Definition
	A private IP address referencing an inside device
	A public IP address referencing an inside device
	A private IP address referencing an outside device
	A public IP address referencing an outside device

Chapter 10

Table 10-1 Ethernet Bandwidth Capacities

Ethernet Type	Bandwidth Capacity
Standard Ethernet	
Fast Ethernet	
Gigabit Ethernet	
10-Gigabit Ethernet	
40-Gigabit Ethernet	
100-Gigabit Ethernet	

Table 10-2 STP Port Types

Port Type	Description
Root port	
Designated port	
Nondesignated port	

Table 10-3 STP Port Cost

Link Speed	STP Port Cost
10Mbps (Ethernet)	
100Mbps (Fast Ethernet)	
1Gbps (Gigabit Ethernet)	
10Gbps (10-Gigabit Ethernet)	
40Gbps (40-Gigabit Ethernet)	

Chapter 14

Table 14-1 Components of an SNMPv1 and SNMPv2c Network Management Solution

Component	Description
SNMP manager	
SNMP agent	
Management information base (MIB)	

Table 14-2 Security Models and Security Levels

Security Model	Security Level	Authentication Strategy	Encryption Type
SNMPv1		Community string	None
SNMPv2c		Community string	None
SNMPv3		Username	None
SNMPv3		MD5 or SHA	None
SNMPv3		MD5 or SHA	CBC-DES (DES-56)

Table 14-3 Syslog Severity Levels

Level	Name	Description
0		The most severe error conditions, which render the system unusable
1		Conditions requiring immediate attention
2		A less-severe condition than an alert that should be addressed to prevent interruption of service
3		Notifications about error conditions within the system that do not render the system unusable
4		Notifications that specific operations failed to complete successfully
5		Non-error notifications that alert an administrator about state changes within a system

Level	Name	Description
6		Detailed information about the normal operation of a system
7		Highly detailed information (for example, information about individual packets) that is typically used for troubleshooting purposes

Chapter 16

Table 16-1 DHCP Components

DHCP Element	Description
	A configuration on a DHCP server that assigns a specific IP address to a particular device or client on a network. Reservations allow administrators to assign specific IP addresses to devices based on their MAC addresses, ensuring that a device always receives the same IP address each time it connects to the network.
	A range of IP addresses and configuration parameters that a DHCP server is responsible for assigning to client devices on a network.
	The duration for which an IP address is allocated to a client device by a DHCP server.
	A network device or service that forwards DHCP requests from clients on one subnet to a DHCP server on a different subnet.
	Specific IP addresses or ranges of IP addresses that are intentionally excluded from being dynamically assigned by a DHCP server.
	An IPv6 feature that allows devices to automatically configure their own IP addresses without the need for a DHCP server.

Table 16-2 Common DNS Record Types

Record Type	Description
A	
AAAA	
CNAME	
MX	
NS	

Record Type	Description
PTR	
SOA	
SRV	
TXT	

Chapter 17

Table 17-1 IKE Modes

Mode	Description
Main mode	

Mode	Description
Aggressive mode	
Quick mode	

Table 17-2 Examples of VPN Protocols

Protocol	Description
SSL	
L2TP	
L2F	
PPTP	
TLS	
SSTP	
OpenVPN	

Table 17-3 Network Access Technologies

Method	Description
Virtual desktop	
SSH	
Graphical user interface (GUI)	
API	
Console	
AAA	

Method	Description
RADIUS	
TACACS+	
NAC	
IEEE 802.1X	
EAP	
Two-factor authentication	
Multifactor authentication	

Method	Description
Single sign-on	
Local authentication	
Captive portal	
Jump box/host	

Chapter 21

Table 21-1 Steps to Diagnose a Problem

Step	Description
	Because a typical problem report lacks sufficient information to give a troubleshooter insight into a problem’s underlying cause, the troubleshooter should collect additional information, perhaps using network maintenance tools or interviewing impacted users.
	Testing to see if you can duplicate the problem is often a key step in problem diagnosis.
	Although it can be difficult to gather information from your end users, doing so is often critical in correctly pinpointing the exact problem. Oftentimes, finding out user actions prior to the problem is critical.
	What symptoms has the problem created?
	Perhaps your end users will provide valuable clues if they accurately indicate what changes they might have made to systems.
	Unfortunately, you might discover that there are multiple issues. Be sure to approach each one individually.

Chapter 25

Table 25-1 Parameters for the Windows **ping** Command

Parameter	Purpose
-t	
-n <i>count</i>	
-l <i>size</i>	
-f	
-i <i>TTL</i>	
-S <i>srcaddr</i>	
<i>target_name</i>	

Table 25-2 Parameters for the Windows **ipconfig** Command

Parameter	Purpose
/all	
/release and /release6	
/renew and /renew6	

Table 25-3 Parameters for the Windows **arp** Command

Parameter	Purpose
-a or -g	
-v	
<i>inet_addr</i>	

Parameter	Purpose
-N <i>if_addr</i>	
-d	
-s	
<i>eth_addr</i>	
<i>if_addr</i>	

Table 25-4 Parameters for the Windows **netstat** Command

Parameter	Purpose
-a	
-b	
-e	
-f	
-p <i>proto</i>	
-r	
-s	

Table 25-5 Parameters for the Windows **route** Command

Parameter	Purpose
-f	
-p	

Parameter	Purpose
<i>command</i>	
<i>destination</i>	
mask <i>netmask</i>	
<i>gateway</i>	
metric <i>metric</i>	
if <i>interface</i>	

Memory Tables

Answer Key

Chapter 2

Table 2-1 Networking Device Summary

Device	Description	Key Points
Router	Connects networks	A router uses the software-configured network address to make forwarding decisions.
Switch	Connects devices on a twisted-pair network	A switch forwards data to its destination by using the MAC address embedded in each packet. It forwards data only to nodes that need to receive it.
Multilayer switch	Functions as a switch or router	Operates on Layers 2 and 3 of the OSI model as a switch and can perform router functionality.
Firewall	Provides controlled data access between networks	Firewalls can be hardware or software based. They are an essential part of a network's security strategy.
IDS/IPS	Detects and prevents intrusions	Monitors the network and attempts to detect/prevent intrusion attempts.
Load balancer	Distributes network load	Load balancing increases redundancy and performance by distributing the load to multiple servers.
Proxy	Sends resource requests on behalf of network clients	Can add to the security and bandwidth efficiency of the network.
Network-attached storage (NAS)	Centralized storage device that connects to a network	Provides scalability and flexibility for the large amounts of storage today.

Device	Description	Key Points
Storage area network (SAN)	A high-speed network that connects storage devices to servers	Allows multiple servers to easily and efficiently access large amounts of storage
Access point (AP)	Enables wireless devices to connect to wireless and wired networks	Should provide the latest in security standards to help reduce security breaches
Controller	Used with branch/remote office deployments for wireless authentication	When an AP boots, it authenticates with a controller before it can start working as an AP

Table 2-2 Three Categories of Quality Issues

Issue	Description
Delay	Delay is the time required for a packet to travel from source to destination. You might have witnessed delay on the evening news when the news anchor is talking via satellite to a foreign news correspondent. Because of the satellite delay, the conversation begins to feel unnatural.
Jitter	Jitter is the uneven arrival of packets. For example, imagine a VoIP conversation where packet 1 arrives at a destination router. Then, 20 ms later, packet 2 arrives. After another 70 ms, packet 3 arrives, and then packet 4 arrives 20 ms behind packet 3. This variation in arrival times (that is, <i>variable delay</i>) is not due to dropped packets, but the jitter might be interpreted by the listener as dropped packets.
Drops	Packet drops occur when a link is congested and a router's interface queue overflows. Some types of traffic, such as UDP traffic carrying voice packets, are not retransmitted if packets are dropped.

Chapter 4

Table 4-1 Port Assignments for Commonly Used Protocols

Protocol	Description	Port Assignment
DHCP (Dynamic Host Configuration Protocol)	Dynamically assigns IP address information (for example, IP address, subnet mask, DNS server IP address, and default gateway IP address) to a network device	UDP 67, 68
DNS (Domain Name System)	Resolves domain names to corresponding IP addresses	TCP/UDP 53
FTP (File Transfer Protocol)	An insecure protocol used to transfer files with a remote host (typically requires authentication of user credentials)	TCP 20, 21

Protocol	Description	Port Assignment
HTTP (Hypertext Transfer Protocol)	An insecure protocol used to retrieve content from a web server	TCP 80
HTTPS (Hypertext Transfer Protocol Secure)	Used to securely retrieve content from a web server	TCP 443
LDAP (Lightweight Directory Access Protocol)	Provides directory services (for example, a user directory that includes username, password, email, and phone number information) to network clients	TCP 389
LDAPS (Lightweight Directory Access Protocol over SSL)	The secure version of LDAP	TCP 636
NTP (Network Time Protocol)	Used by a network device to synchronize its clock with a time server (NTP server)	UDP 123
RDP (Remote Desktop Protocol)	A Microsoft protocol that allows a user to view and control the desktop of a remote computer	TCP/UDP 3389
SFTP (Secure File Transfer Protocol)	Provides FTP file transfer service over an SSH connection	TCP 22
SIP (Session Initiation Protocol)	Used to create and end sessions for one or more media connections, including Voice over IP (VoIP) calls	TCP/UDP 5060, 5061
SMB (Server Message Block)	Used to share files, printers, and other network resources	TCP 445
SMTP (Simple Mail Transfer Protocol)	Used for sending email throughout the network	TCP 25
SMTPS (Simple Mail Transfer Protocol Secure)	Allows for secure communication between email clients and mail servers, ensuring that email messages are encrypted during transmission	TCP 587
SNMP (Simple Network Management Protocol)	Used to monitor and manage network devices; SNMPv3 is the latest and most secure version of SNMP and allows for the use of strong cryptographic algorithms, such as HMAC-SHA, HMAC-MD5, and AES, to ensure data integrity and confidentiality	UDP 161, 162
SQL (Structured Query Language) Server	A server that specializes in SQL and is used to run powerful queries against data that is stored in databases	TCP/UDP 1433

Protocol	Description	Port Assignment
SSH (Secure Shell)	Used to securely connect to a remote host (typically via a terminal emulator)	TCP 22
Syslog	The standard used by network devices (and other computer systems) to report on status information and events	UDP 514
Telnet	An insecure protocol used to connect to a remote host (typically via a terminal emulator)	TCP 23
TFTP (Trivial File Transfer Protocol)	Used to transfer files with a remote host (does not require authentication of user credentials)	UDP 69

Table 4-2 Internet Protocol (IP) Types Summary

Protocol	Complete Protocol Name	Description
IPsec	Internet Protocol Security	A complex suite of protocols that is used to create secured connections between network systems.
GRE	Generic Routing Encapsulation	A tunneling protocol used to encapsulate a wide variety of network layer protocols inside virtual point-to-point links or point-to-multipoint links over an IP network.
IP	Internet Protocol	A network layer protocol, documented in RFC 791, that offers a connectionless internetwork service. IP provides features for addressing, packet fragmentation and reassembly, type-of-service specification, and security.
TCP	Transmission Control Protocol	A connection-oriented protocol that offers flow control, sequencing, and retransmission of dropped packets.
UDP	User Datagram Protocol	A connectionless alternative to TCP used for applications that do not require the functions offered by TCP.
ICMP	Internet Control Message Protocol	Sends error messages and operational information indicating issues with packet delivery in IP networks.
ESP	Encapsulating Security Payload	Provides for encryption in the IPsec suite of protocols.
AH	Authentication Header	Provides for authentication in the IPsec suite of protocols.
IKE	Internet Key Exchange	Negotiates cryptographic keys and security parameters for IPsec.

Table 4-3 Traffic Types Summary

Traffic type	Description
Unicast	Traffic is one-to-one communication between a single sender and a single receiver, and this is the most common type of traffic in IP networks. It is used for point-to-point communication between devices, such as client-server interactions and most Internet browsing.
Multicast	A mechanism by which groups of network devices can send and receive data between the members of the group at one time (one-to-many), instead of separately sending messages to each device in the group.
Anycast	Anycast traffic is one-to-nearest communication from a sender to the nearest of multiple receivers. Multiple devices share the same anycast IP address, but packets are routed to the nearest (or best) destination based on routing metrics such as shortest path or lowest latency.
Broadcast	A broadcast is at the opposite end of the spectrum from a unicast. Broadcast traffic is one-to-all communication from a single sender to all devices within a specific network segment.

Chapter 5

Table 5-3 Characteristics of 802.11 Standards

Standard	Band	Maximum Bandwidth	Transmission Method	Maximum Range
802.11	2.4GHz	1Mbps or 2Mbps	DSSS or FHSS	20 m indoors/100 m outdoors
802.11a	5GHz	54Mbps	OFDM	35 m indoors/120 m outdoors
802.11b	2.4GHz	11Mbps	DSSS	32 m indoors/140 m outdoors
802.11g	2.4GHz	54Mbps	OFDM or DSSS	32 m indoors/140 m outdoors
802.11n	2.4GHz or 5GHz (or both)	> 300Mbps (with channel bonding)	OFDM	70 m indoors/250 m outdoors
802.11ac	5GHz	> 3Gbps (with MU-MIMO and several antennas)	OFDM	70 m indoors/250 m outdoors
802.11ax	2.4GHz, 5GHz, 6GHz	9.6Gbps	OFDMA	70 m indoors/250 m outdoors

Table 5-4 Ethernet Bandwidth Capacities

Ethernet Type	Bandwidth Capacity
Standard Ethernet	10Mbps: 10 million bits per second (that is, 10 megabits per second)
Fast Ethernet	100Mbps: 100 million bits per second (that is, 100 megabits per second)
Gigabit Ethernet	1Gbps: 1 billion bits per second (that is, 1 gigabit per second)
10-Gigabit Ethernet	10Gbps: 10 billion bits per second (that is, 10 gigabits per second)
40-Gigabit Ethernet	40Gbps: 40 billion bits per second (that is, 40 gigabits per second)
100-Gigabit Ethernet	100Gbps: 100 billion bits per second (that is, 100 gigabits per second)

Table 5-5 Types of Ethernet

Ethernet Standard	Media Type	Bandwidth Capacity	Distance Limitation
10BASE5	Coax (thicknet)	10Mbps	500 m
10BASE2	Coax (thinnet)	10Mbps	185 m
10BASE-T	Cat 3 (or higher) UTP	10Mbps	100 m
100BASE-TX	Cat 5 (or higher) UTP	100Mbps	100 m
100BASE-FX	MMF	100Mbps	2 km
100BASE-SX	MMF	100Mbps	300 m
1000BASE-T	Cat 5e (or higher) UTP	1Gbps	100 m
1000BASE-TX	Cat 6 (or higher) UTP	1Gbps	100 m
1000BASE-SX	MMF	1Gbps	550 m
1000BASE-LX	SMF	1Gbps	5 km
1000BASE-LH	SMF	1Gbps	10 km
1000BASE-ZX	SMF	1Gbps	70 km
10GBASE-SR	MMF	10Gbps	26–400 m
10GBASE-LR	SMF	10Gbps	10–25 km
10GBASE-ER	SMF	10Gbps	40 km
10GBASE-SW	MMF	10Gbps	300 m

Ethernet Standard	Media Type	Bandwidth Capacity	Distance Limitation
10GBASE-LW	SMF	10Gbps	10 km
10GBASE-EW	SMF	10Gbps	40 km
10GBASE-T	Cat 6a (or higher)	10Gbps	100 m
40GBASE-T	Cat 8	40Gbps	30 m
100GBASE-SR10	MMF	100Gbps	125 m
100GBASE-LR4	SMF	100Gbps	10 km
100GBASE-ER4	SMF	100Gbps	40 km

Chapter 6

Table 6-1 Characteristics, Benefits, and Drawbacks of Client/Server Networks

Characteristics	Benefits	Drawbacks
Client devices (for example, PCs) share a common set of resources (for example, file or print resources) located on one or more dedicated servers.	Client/server networks can easily scale, which might require the purchase of additional client licenses.	Because multiple clients might rely on a single server for their resources, the single server can become a single point of failure in the network.
Resource sharing is made possible via dedicated server hardware and network operating systems.	Administration is simplified because parameters such as file-sharing permissions and other security settings can be administered on a server as opposed to on multiple clients.	Client/server networks can cost more than peer-to-peer networks. For example, client/server networks might require the purchase of dedicated server hardware and a network OS with an appropriate number of licenses.

Table 6-2 Characteristics, Benefits, and Drawbacks of a Peer-to-Peer Network

Characteristics	Benefits	Drawbacks
Client devices (for example, PCs) share their resources (for example, file and printer resources) with other client devices.	Peer-to-peer networks can be installed easily because resource sharing is made possible by the clients' operating systems, and knowledge of advanced networking operating systems is not required.	Scalability is limited because of the increased administration burden of managing multiple clients.

Characteristics	Benefits	Drawbacks
Resource sharing is made available through the clients' operating systems.	Peer-to-peer networks usually cost less than client/server networks because there is no requirement for dedicated server resources or advanced NOS software.	Performance might not be as strong as in a client/server network because the devices providing network resources might be performing other tasks not related to resource sharing (for example, word processing).

Table 6-3 Characteristics, Benefits, and Drawbacks of a Star Topology

Characteristics	Benefits	Drawbacks
Devices have independent connections to a central device (for example, a hub or a switch).	A cable break impacts only the device connected via the broken cable and not the entire topology.	More cable is required for a star topology because each device requires its own cable to connect back to the central device.
Star topologies are commonly used with Ethernet technologies (described in Chapter 5, "Transmission Media and Transceivers").	Troubleshooting is relatively simple because a central device in the star topology acts as the aggregation point for all the connected devices.	Installation can take longer for a star topology because more cable runs must be installed.

Table 6-4 Characteristics, Benefits, and Drawbacks of a Hub-and-Spoke WAN Topology

Characteristics	Benefits	Drawbacks
Each remote site (that is, a spoke) connects to a main site (that is, the hub) via a WAN link.	Costs are reduced (as compared to a full-mesh or partial-mesh topology) because a minimal number of links is used.	Suboptimal routes must be used between remote sites because all intersite communication must travel via the main site.
Communication between two remote sites travels through the hub site.	Adding one or more additional sites is easy (compared to a full-mesh or partial-mesh topology) because only one link needs to be added per site.	Because all remote sites converge on the main site, this hub site is potentially a single point of failure.
—	—	Because each remote site is reachable by only a single WAN link, the hub-and-spoke topology lacks redundancy.

Table 6-5 Characteristics, Benefits, and Drawbacks of a Full-Mesh WAN Topology

Characteristics	Benefits	Drawbacks
Every site has a direct WAN connection to every other site.	An optimal route exists between any two sites.	A full-mesh network can be difficult and expensive to scale because the addition of one new site requires a new WAN link between the new site and every other existing site.
The number of required WAN connections can be calculated with the formula $w = n \times (n - 1) / 2$, where w = the number of WAN links and n = the number of sites. For example, a network with 10 sites would require 45 WAN connections to form a fully meshed network: $45 = 10 \times (10 - 1) / 2$.	A full-mesh network is fault tolerant because one or more links can be lost, and reachability between all sites might still be maintained.	—
—	Troubleshooting a full-mesh network is relatively easy because each link is independent of the other links.	—

Table 6-6 Characteristics, Benefits, and Drawbacks of a Partial-Mesh Topology

Characteristics	Benefits	Drawbacks
Selected sites (that is, sites with frequent intersite communication) are interconnected via direct links, whereas sites that have less-frequent communication can communicate via another site.	A partial-mesh topology provides optimal routes between selected sites with higher intersite traffic volumes while avoiding the expense of interconnecting every site to every other site.	A partial-mesh topology is less fault tolerant than a full-mesh topology.
A partial-mesh topology uses fewer links than a full-mesh topology and more links than a hub-and-spoke topology for interconnecting the same number of sites.	A partial-mesh topology is more redundant than a hub-and-spoke topology.	A partial-mesh topology is more expensive than a hub-and-spoke topology.

Chapter 7

Table 7-19 IP Address Classes

Address Class	Value in First Octet	Classful Mask (Dotted Decimal)	Classful Mask (Prefix Notation)
Class A	1–126	255.0.0.0	/8
Class B	128–191	255.255.0.0	/16
Class C	192–223	255.255.255.0	/24
Class D	224–239	—	—
Class E	240–255	—	—

Table 7-20 Private IP Networks

Address Class	Address Range	Default Subnet Mask
Class A	10.0.0.0–10.255.255.255	255.0.0.0
Class B	172.16.0.0–172.31.255.255	255.255.0.0
Class B	169.254.0.0–169.254.255.255	255.255.0.0
Class C	192.168.0.0–192.168.255.255	255.255.255.0

Table 7-22 Dotted-Decimal and Prefix-Notation Representations for IPv4 Subnets

Dotted-Decimal Notation	Prefix Notation
255.0.0.0	/8 (classful subnet mask for Class A networks)
255.128.0.0	/9
255.192.0.0	/10
255.224.0.0	/11
255.240.0.0	/12
255.248.0.0	/13
255.252.0.0	/14
255.254.0.0	/15
255.255.0.0	/16 (classful subnet mask for Class B networks)
255.255.128.0	/17
255.255.192.0	/18
255.255.224.0	/19
255.255.240.0	/20
255.255.248.0	/21

Dotted-Decimal Notation	Prefix Notation
255.255.252.0	/22
255.255.254.0	/23
255.255.255.0	/24 (classful subnet mask for Class C networks)
255.255.255.128	/25
255.255.255.192	/26
255.255.255.224	/27
255.255.255.240	/28
255.255.255.248	/29
255.255.255.252	/30

Table 7-23 Subnet Octet Values

Subnet Octet Value	Number of Contiguous Left-Justified 1s
0	0
128	1
192	2
224	3
240	4
248	5
252	6
254	7
255	8

Table 7-24 Number of Subnets Created by a Specified Number of Borrowed Bits

Borrowed Bits	Number of Subnets Created (2^s , Where s Is the Number of Borrowed Bits)
0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128

Borrowed Bits	Number of Subnets Created (2^s, Where s Is the Number of Borrowed Bits)
8	256
9	512
10	1024
11	2048
12	4096

Table 7-25 Number of Supported Hosts, Given a Specified Number of Host Bits

Host Bits	Number of Supported Hosts ($2^h - 2$, Where h Is the Number of Host Bits)
2	2
3	6
4	14
5	30
6	62
7	126
8	254
9	510
10	1022
11	2046
12	4094

Table 7-26 Usable IP Address Ranges for the 192.168.10.0/27 Subnets

Subnet Address	Directed Broadcast Address	Usable IP Addresses
192.168.10.0	192.168.10.31	192.168.10.1–192.168.10.30
192.168.10.32	192.168.10.63	192.168.10.33–192.168.10.62
192.168.10.64	192.168.10.95	192.168.10.65–192.168.10.94
192.168.10.96	192.168.10.127	192.168.10.97–192.168.10.126
192.168.10.128	192.168.10.159	192.168.10.129–192.168.10.158
192.168.10.160	192.168.10.191	192.168.10.161–192.168.10.190
192.168.10.192	192.168.10.223	192.168.10.193–192.168.10.222
192.168.10.224	192.168.10.255	192.168.10.225–192.168.10.254

Table 7-27 Usable IP Address Ranges for the 192.168.0.0/26 Subnets

Subnet Address	Directed Broadcast Address	Usable IP Addresses
192.168.0.0	192.168.0.63	192.168.0.1–192.168.0.62
192.168.0.64	192.168.0.127	192.168.0.65–192.168.0.126
192.168.0.128	192.168.0.191	192.168.0.129–192.168.0.190
192.168.0.192	192.168.0.255	192.168.0.193–192.168.0.254

Chapter 9

Table 9-1 Administrative Distance

Routing Information Source	AD Value
Directly connected network	0
Statically configured network	1
EIGRP	90
OSPF	110
RIP	120
External EIGRP	170
Unknown or unbelievable	255 (considered to be unreachable)

Table 9-2 Comparing Dynamic Routing Protocols

Routing Protocol	IGP or EGP	Type	Metric
OSPF	IGP	Link state	Cost (based on bandwidth)
EIGRP	IGP	Hybrid	Composite (bandwidth and delay by default)
BGP	EGP	Path vector	Path attributes

Table 9-3 Names of NAT IP Addresses

NAT IP Address	Definition
Inside local	A private IP address referencing an inside device
Inside global	A public IP address referencing an inside device
Outside local	A private IP address referencing an outside device
Outside global	A public IP address referencing an outside device

Chapter 10

Table 10-1 Ethernet Bandwidth Capacities

Ethernet Type	Bandwidth Capacity
Standard Ethernet	10Mbps: 10 million bits per second (that is, 10 megabits per second)
Fast Ethernet	100Mbps: 100 million bits per second (that is, 100 megabits per second)
Gigabit Ethernet	1Gbps: 1 billion bits per second (that is, 1 gigabit per second)
10-Gigabit Ethernet	10Gbps: 10 billion bits per second (that is, 10 gigabits per second)
40-Gigabit Ethernet	40 Gbps: 40 billion bits per second (that is, 40 gigabits per second)
100-Gigabit Ethernet	100Gbps: 100 billion bits per second (that is, 100 gigabits per second)

Table 10-2 STP Port Types

Port Type	Description
Root port	Every nonroot bridge has a single root port, which is the port on that switch that is closest to the root bridge in terms of cost.
Designated port	Every network segment has a single designated port, which is the port on that segment that is closest to the root bridge in terms of cost. Therefore, all ports on a root bridge are designated ports.
Nondesignated port	Nondesignated ports block traffic to create a loop-free topology.

Table 10-3 STP Port Cost

Link Speed	STP Port Cost
10Mbps (Ethernet)	100
100Mbps (Fast Ethernet)	19
1Gbps (Gigabit Ethernet)	4
10Gbps (10-Gigabit Ethernet)	2
40Gbps (40-Gigabit Ethernet)	2

Chapter 14

Table 14-1 Components of an SNMPv1 and SNMPv2c Network Management Solution

Component	Description
SNMP manager	An SNMP manager runs a network management application. This SNMP manager is sometimes referred to as a <i>network management system (NMS)</i> .
SNMP agent	An SNMP agent is a piece of software that runs on a managed device (for example, a server, router, or switch).
Management information base (MIB)	Information about a managed device's resources and activity is defined by a series of objects. The structure of these management objects is defined by a managed device's MIB. Interfaces and their details—such as errors, utilization, discards, packet drops, resets, speed and duplex, system memory, utilization of bandwidth, storage, CPU, and memory—can be monitored and reported via SNMP.

Table 14-2 Security Models and Security Levels

Security Model	Security Level	Authentication Strategy	Encryption Type
SNMPv1	noAuthNoPriv	Community string	None
SNMPv2c	noAuthNoPriv	Community string	None
SNMPv3	noAuthNoPriv	Username	None
SNMPv3	authNoPriv	MD5 or SHA	None
SNMPv3	authPriv	MD5 or SHA	CBC-DES (DES-56)

Table 14-3 Syslog Severity Levels

Level	Name	Description
0	Emergencies	The most severe error conditions, which render the system unusable
1	Alerts	Conditions requiring immediate attention
2	Critical	A less-severe condition than an alert that should be addressed to prevent interruption of service
3	Errors	Notifications about error conditions within the system that do not render the system unusable
4	Warnings	Notifications that specific operations failed to complete successfully
5	Notifications	Non-error notifications that alert an administrator about state changes within a system

Level	Name	Description
6	Informational	Detailed information about the normal operation of a system
7	Debugging	Highly detailed information (for example, information about individual packets) that is typically used for troubleshooting purposes

Chapter 16

Table 16-1 DHCP Components

DHCP Element	Description
DHCP reservation	A configuration on a DHCP server that assigns a specific IP address to a particular device or client on a network. Reservations allow administrators to assign specific IP addresses to devices based on their MAC addresses, ensuring that a device always receives the same IP address each time it connects to the network.
DHCP scope	A range of IP addresses and configuration parameters that a DHCP server is responsible for assigning to client devices on a network.
DHCP lease time	The duration for which an IP address is allocated to a client device by a DHCP server.
DHCP relay/IP helper	A network device or service that forwards DHCP requests from clients on one subnet to a DHCP server on a different subnet.
DHCP exclusions	Specific IP addresses or ranges of IP addresses that are intentionally excluded from being dynamically assigned by a DHCP server.
SLAAC	An IPv6 feature that allows devices to automatically configure their own IP addresses without the need for a DHCP server.

Table 16-2 Common DNS Record Types

Record Type	Description
A	An address record (that is, A record) maps a hostname to an IPv4 address.
AAAA	An IPv6 address record (that is, AAAA record) maps a hostname to an IPv6 address.
CNAME	A canonical name record (that is, CNAME record) is an alias of an existing record that allows multiple DNS records to map to the same IP address.
MX	A mail exchange record (that is, MX record) maps a domain name to an email (or message transfer agent) server for that domain.
NS	A nameserver record (that is, NS record) delegates a DNS zone to use the given authoritative name server.

Record Type	Description
PTR	A pointer record (that is, PTR record) points to a canonical name. A PTR record is commonly used when performing a reverse DNS lookup to determine what domain name is associated with a known IP address.
SOA	A <i>start of authority</i> record (that is, SOA record) provides authoritative information about a DNS zone, such as email contact information for the zone's administrator, the zone's primary name server, and various refresh timers.
SRV	A <i>service</i> location record (that is, SRV record) is used for newer protocols instead of creating protocol-specific records such as MX records.
TXT	A text record (that is, TXT record) originally was for arbitrary human-readable text in a DNS record. Since the early 1990s, however, this record has carried machine-readable data, such as data specified by RFC 1464, opportunistic encryption, Sender Policy Framework (SPF), or DomainKeys Identified Email (DKIM).

Chapter 17

Table 17-1 IKE Modes

Mode	Description
Main mode	Main mode involves three exchanges of information between the IPsec peers: Exchange 1: The responder selects a proposal it received from the initiator. Exchange 2: Diffie-Hellman (DH) is used to securely establish a shared secret key over the unsecured medium. Exchange 3: An Internet Security Association and Key Management Protocol (ISAKMP) session is established. This secure session is then used to negotiate an IPsec session. One peer, called the <i>initiator</i>, sends one or more proposals to the other peer, called the <i>responder</i>. The proposals include supported encryption and authentication protocols and key lifetimes. In addition, the proposals indicate whether perfect forward secrecy (PFS) should be used. PFS ensures that a session key remains secure, even if one of the private keys used to derive the session key becomes compromised.

Mode	Description
Aggressive mode	Aggressive mode more quickly achieves the same results as main mode, using only three packets. The initiator sends the first packet, which contains all the information necessary to establish a security association (SA)—that is, an agreement between the two IPsec peers about the cryptographic parameters to be used in the ISAKMP session. The responder sends the second packet, which contains the security parameters selected by the responder (the proposal, the keying material, and its ID). The responder uses this second packet to authenticate the session. The third and final packet, which is sent by the initiator, finalizes the authentication of the ISAKMP session.
Quick mode	Quick mode negotiates the parameters (the SA) for the IPsec session. This negotiation occurs within the protection of an ISAKMP session.

Table 17-2 Examples of VPN Protocols

Protocol	Description
SSL	Secure Socket Layer (SSL) provides cryptography and reliability for the upper layers (Layers 5–7) of the OSI model. SSL, which was introduced in 1995, has largely been replaced by Transport Layer Security (TLS). However, recent versions of SSL (for example, SSL 3.3) have been enhanced to be more comparable with TLS. Both SSL and TLS provide secure web browsing via Hypertext Transfer Protocol Secure (HTTPS).
L2TP	Layer 2 Tunneling Protocol (L2TP) is a VPN protocol that lacks security features, such as encryption. However, L2TP can still be used for a secure VPN connection if it is combined with another protocol that does provide encryption.
L2F	Layer 2 Forwarding (L2F) is a VPN protocol designed (by Cisco Systems) with the goal of providing a tunneling protocol for PPP. Like L2TP, L2F lacks native security features.
PPTP	Point-to-Point Tunneling Protocol (PPTP) is an older VPN protocol (which supported the dial-up networking feature in older versions of Microsoft Windows). Like L2TP and L2F, PPTP lacks native security features. However, Microsoft's versions of PPTP bundled with various versions of Microsoft Windows were enhanced to offer security features.
TLS	Transport Layer Security (TLS) has largely replaced SSL as the VPN protocol of choice for providing cryptography and reliability to upper layers of the OSI model. For example, when you securely connect to a website using HTTPS, you are probably using TLS.
SSTP	Secure Socket Tunneling Protocol (SSTP) is a VPN tunnel that transports PPP traffic through an SSL/TLS channel. SSL/TLS provides transport-level security. SSTP's use of SSL/TLS over TCP port 443 allows SSTP to pass through virtually all firewalls and proxy servers.
OpenVPN	OpenVPN is a software VPN system that creates secure point-to-point or site-to-site connections in routed or bridged configurations and remote access facilities. It implements both client and server applications.

Table 17-3 Network Access Technologies

Method	Description
Virtual desktop	Virtual desktop technology is growing at a fast pace, thanks in large part to the adoption of cloud technologies and increased bandwidth between data centers and end users. Using this approach, a user accesses a desktop that is housed on a server in the data center. The user might not even realize that the operating system is not hosted on the local PC or laptop. Note that the virtual desktop concept discussed here is different from other types of virtual desktops in IT. For example, in both Windows and macOS systems, you can create virtual desktops. These are just running “spaces” on the local system that have different windows and programs running within them, which makes it easier to multitask. Notice how very different this technology is from the virtual desktops referenced here.
SSH	<i>Secure Shell (SSH)</i> is a protocol used to securely connect to a remote host (typically via a terminal emulator). Secure Shell has enjoyed such widespread success that it is even used as the basis for several other security protocols and techniques. For example, SCP calls on SSH for some of its functionality. SSH uses port 22. As mentioned earlier, Linux administrators often prefer to use the command line for all operation and work in the operating system, and SSH is a perfect secure remote access tool for such users. Administrators can use the CLI to connect securely with SSH, and then they can operate in a BASH CLI environment to manage or work within the system.
Graphical user interface (GUI)	Many systems still provide a <i>graphical user interface (GUI)</i> for management access, no matter if this is local or remote access. Always be sure that appropriate bandwidth exists for a seamless experience with these GUIs.
API	<i>Application programming interfaces (APIs)</i> are increasingly being used to access and manage networks by providing a standardized way for software applications to communicate with network devices and services. Through APIs, developers can programmatically configure network settings, monitor network performance, and automate network management tasks. This allows for greater flexibility, scalability, and efficiency in network operations, as tasks that once required manual intervention can now be automated.
Console	Most network devices permit local access through what is termed a console port. Notice this console port should be protected with the appropriate logical and physical security controls.
AAA	Authentication, authorization, and accounting (AAA, pronounced “triple A”) allows a network to have a single repository of user credentials. A network administrator can, for example, supply the same credentials to log in to various network devices (for example, routers and switches). RADIUS and TACACS+ are protocols commonly used to communicate with a AAA server.

Method	Description
RADIUS	Remote Authentication Dial-In User Service (RADIUS) is a UDP-based protocol used to communicate with a AAA server. Unlike TACACS+, RADIUS does not encrypt an entire authentication packet; rather, it encrypts only the password. However, RADIUS does offer more robust accounting features than TACACS+. Also, RADIUS is a standards-based protocol, whereas TACACS+ is a Cisco-proprietary protocol.
TACACS+	Terminal Access Controller Access-Control System Plus (TACACS+) is a Cisco-proprietary TCP-based AAA protocol. TACACS+ has three separate and distinct sessions or functions for authentication, authorization, and accounting.
NAC	Network access control (NAC) can be used to permit or deny access to a network based on characteristics of the device seeking admission rather than just checking user credentials. For example, a client's operating system and version of antivirus software could be checked against a set of requirements before allowing the client to access a network. This process of checking a client's characteristics is called <i>posture assessment</i> .
IEEE 802.1X	IEEE 802.1X is a type of NAC that can permit or deny a wireless or wired LAN client access to a network. If IEEE 802.1X is used to permit access to a LAN via a switch port, then IEEE 802.1X is being used for port security. While covered elsewhere in this text, to summarize, in 802.1X, the device seeking admission to the network is called the <i>supplicant</i>. The device to which the supplicant connects (either wirelessly or through a wired connection) is called the <i>authenticator</i>. The device that checks the supplicant's credentials and permits or denies the supplicant to access the network is called an <i>authentication server</i>. Usually, an authentication server is a RADIUS server.
EAP	Extensible Authentication Protocol (EAP) specifies how authentication is performed by IEEE 802.1X. A variety of EAP types exist (see Chapter 11), such as Extensible Authentication Protocol-Flexible Authentication via Secure Tunneling (EAP-FAST), Extensible Authentication Protocol-Message Digest 5 (EAP-MD5), and Extensible Authentication Protocol-Transport Layer Security (EAP-TLS).
Two-factor authentication	Two-factor authentication (TFA) requires two types of authentication from a user seeking admission to a network. For example, a user might have to <i>know</i> something (for example, a password) and <i>have</i> something (such as a specific fingerprint, which can be checked with a biometric authentication device).
Multifactor authentication	Multifactor authentication requires two or more types of successful authentication factors before granting access to a network.

Method	Description
Single sign-on	Single sign-on (SSO) allows a user to authenticate only once to gain access to multiple systems rather than needing to independently authenticate with each system.
Local authentication	Local authentication refers to a network device authenticating the user with a database of user account information stored on the device itself; this is often an important fallback method of authentication used when another external method fails.
Captive portal	A captive portal is a web page that appears before a user is able to access the network resource; this web page accepts the credentials of the user for authentication and presents them to the authentication server.
Jump box/host	A <i>jump box</i> , or <i>jump host</i> , is a secure intermediary server used to access and manage devices within a protected network. It acts as a gateway, allowing administrators to connect to the internal network from an external or less secure network. By requiring all remote connections to go through the jump box, organizations can enforce strict access controls, monitor activity, and reduce the attack surface. This added layer of security helps protect sensitive network resources and ensures that only authorized users can perform administrative tasks within the network, enhancing overall network security.

Chapter 21

Table 21-1 Steps to Diagnose a Problem

Step	Description
Gather information.	Because a typical problem report lacks sufficient information to give a troubleshooter insight into a problem's underlying cause, the troubleshooter should collect additional information, perhaps using network maintenance tools or interviewing impacted users.
Duplicate the problem, if possible.	Testing to see if you can duplicate the problem is often a key step in problem diagnosis.
Question users.	Although it can be difficult to gather information from your end users, doing so is often critical in correctly pinpointing the exact problem. Oftentimes, finding out user actions prior to the problem is critical.
Identify symptoms.	What symptoms has the problem created?
Determine if anything has changed.	Perhaps your end users will provide valuable clues if they accurately indicate what changes they might have made to systems.
Approach multiple problems individually.	Unfortunately, you might discover that there are multiple issues. Be sure to approach each one individually.

Chapter 25

Table 25-1 Parameters for the Windows **ping** Command

Parameter	Purpose
-t	This option repeatedly sends pings (ICMP echo messages) until you stop it by pressing Ctrl-C.
-n count	This option specifies the number of pings to send.
-l size	This option enables you to define the size of the ICMP echo requests in bytes; the default value is 32.
-f	This option sets the “don’t fragment” bit in a packet’s header. If the packet tries to cross a router that attempts to fragment the packet, the packet is dropped, and an ICMP error message is returned.
-i TTL	This option sets the TTL value in a packet’s header. The TTL value is decremented for each router hop. A packet is discarded when its TTL value reaches 0.
-S srcaddr	If the PC from which you are issuing the ping command has more than one IP address, this option allows you to specify the source IP address from which the ICMP echo messages should be sent.
<i>target_name</i>	This option specifies the name or the IP address of the device to which you are sending ICMP echo messages.

Table 25-2 Parameters for the Windows **ipconfig** Command

Parameter	Purpose
/all	The ipconfig command entered by itself displays summary information about a PC’s IP address configuration. This parameter gives more verbose information, including DNS, MAC address, and IPv6 address information.
/release and /release6	These options release a DHCP lease for an IPv4 and IPv6 address, respectively.
/renew and /renew6	These options renew a DHCP lease for an IPv4 and IPv6 address, respectively.

Table 25-3 Parameters for the Windows **arp** Command

Parameter	Purpose
-a or -g	These options display current entries in a PC’s ARP table.
-v	This option, which stands for <i>verbose</i> , includes any invalid and loopback interface entries in an ARP table.
<i>inet_addr</i>	This option is a specific IP address.

Parameter	Purpose
-N <i>if_addr</i>	This option shows ARP entries learned for a specified network.
-d	This option, in combination with the <i>inet_addr</i> parameter, can delete an ARP entry for a host. With the wildcard character *, it can delete all host entries.
-s	This option, used in conjunction with the <i>inet_addr</i> and <i>eth_addr</i> parameters, statically adds a host entry to the ARP table.
<i>eth_addr</i>	This parameter is a 48-bit MAC address.
<i>if_addr</i>	If a host has multiple interfaces, an ARP entry might be associated with a specific interface. This option can be used to statically add or delete an ARP entry in a specified interface.

Table 25-4 Parameters for the Windows **netstat** Command

Parameter	Purpose
-a	This option displays all of a PC's active IP-based sessions, along with the TCP and UDP ports of each session.
-b	This option shows the name of the program that opened a session.
-e	This option shows statistical information for an interface's IP-based traffic, such as the number of bytes sent and received.
-f	This option displays FQDNs of destination addresses appearing in a listing of active sessions.
-p <i>proto</i>	This option displays connections for a specific protocol, which might be icmp , icmpv6 , ip , ipv6 , tcp , tcpv6 , udp , or udpv6 .
-r	This option displays a PC's IP routing table. (Note that netstat with this option generates the same output as the route print command.)
-s	This option displays statistical information for the following protocols: ICMPv4, ICMPv6, IPv4, IPv6, TCPv4, TCPv6, UDPv4, and UDPv6.

Table 25-5 Parameters for the Windows **route** Command

Parameter	Purpose
-f	This option clears gateway entries from the routing table. If this option is used with another option, the clearing of gateways from the routing table occurs before any other specified action.
-p	This option can be used with the add command to make a statically configured route persistent, which means the route will remain in a PC's routing table even after a reboot.

Parameter	Purpose
<i>command</i>	Supported commands include print , add , delete , and change . The print command lists entries in a PC's routing table. The add command adds a route entry. The delete command removes a route from the routing table, and the change command can modify an existing route.
<i>destination</i>	This option specifies the destination host or subnet to add to a PC's routing table.
mask <i>netmask</i>	This option, used in conjunction with the <i>destination</i> option, specifies the subnet mask of the destination. If the destination is the IP address of a host, the <i>netmask</i> parameter is 255.255.255.255.
<i>gateway</i>	This option specifies the IP address of the next-hop router used to reach the specified destination.
metric <i>metric</i>	This option specifies the cost to reach a specified destination. If a routing table contains more than one route to reach the destination, the route with the lowest cost is selected.
if <i>interface</i>	This option forwards traffic to a specified destination out a specific interface.

Study Planner

Appendix D

Practice Test	Reading	Task
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Element	Task	Goal Date	First Date Completed	Second Date Completed (Optional)	Notes
Introduction	Read Introduction				
1. The OSI Model and Encapsulation	Read Foundation Topics				
1. The OSI Model and Encapsulation	Review Key Topics				
1. The OSI Model and Encapsulation	Define Key Terms				
1. The OSI Model and Encapsulation	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 1 in practice test software				
2. Networking Appliances, Applications, and Functions	Read Foundation Topics				
2. Networking Appliances, Applications, and Functions	Review Key Topics				
2. Networking Appliances, Applications, and Functions	Complete Memory Tables				
2. Networking Appliances, Applications, and Functions	Define Key Terms				
2. Networking Appliances, Applications, and Functions	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 2 in practice test software				

3. Cloud Concepts	Read Foundation Topics				
3. Cloud Concepts	Review Key Topics				
3. Cloud Concepts	Define Key Terms				
3. Cloud Concepts	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 3 in practice test software				
4. Networking Ports, Protocols, Services, and Traffic Types	Read Foundation Topics				
4. Networking Ports, Protocols, Services, and Traffic Types	Review Key Topics				
4. Networking Ports, Protocols, Services, and Traffic Types	Complete Memory Tables				
4. Networking Ports, Protocols, Services, and Traffic Types	Define Key Terms				
4. Networking Ports, Protocols, Services, and Traffic Types	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 4 in practice test software				
5. Transmission Media and Transceivers	Read Foundation Topics				
5. Transmission Media and Transceivers	Review Key Topics				
5. Transmission Media and Transceivers	Complete Memory Tables				
5. Transmission Media and Transceivers	Define Key Terms				
5. Transmission Media and Transceivers	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 5 in practice test software				

6. Network Topologies, Architectures, and Types	Read Foundation Topics				
6. Network Topologies, Architectures, and Types	Review Key Topics				
6. Network Topologies, Architectures, and Types	Complete Memory Tables				
6. Network Topologies, Architectures, and Types	Define Key Terms				
6. Network Topologies, Architectures, and Types	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 6 in practice test software				
7. IPv4 Addressing	Read Foundation Topics				
7. IPv4 Addressing	Review Key Topics				
7. IPv4 Addressing	Complete Memory Tables				
7. IPv4 Addressing	Define Key Terms				
7. IPv4 Addressing	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 7 in practice test software				
8. Evolving Use Cases	Read Foundation Topics				
8. Evolving Use Cases	Review Key Topics				
8. Evolving Use Cases	Define Key Terms				
8. Evolving Use Cases	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 8 in practice test software				
9. Routing Technologies	Read Foundation Topics				
9. Routing Technologies	Review Key Topics				

9. Routing Technologies	Complete Memory Tables				
9. Routing Technologies	Define Key Terms				
9. Routing Technologies	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 9 in practice test software				
10. Ethernet Switching Technologies	Read Foundation Topics				
10. Ethernet Switching Technologies	Review Key Topics				
10. Ethernet Switching Technologies	Complete Memory Tables				
10. Ethernet Switching Technologies	Define Key Terms				
10. Ethernet Switching Technologies	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 10 in practice test software				
11. Configure Wireless Devices and Technologies	Read Foundation Topics				
11. Configure Wireless Devices and Technologies	Review Key Topics				
11. Configure Wireless Devices and Technologies	Define Key Terms				
11. Configure Wireless Devices and Technologies	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 11 in practice test software				
12. Physical Installations	Read Foundation Topics				
12. Physical Installations	Review Key Topics				
12. Physical Installations	Define Key Terms				
12. Physical Installations	Complete Review Questions section				

Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 12 in practice test software				
13. Organizational Processes and Procedures	Read Foundation Topics				
13. Organizational Processes and Procedures	Review Key Topics				
13. Organizational Processes and Procedures	Define Key Terms				
13. Organizational Processes and Procedures	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 13 in practice test software				
14. Network Monitoring	Read Foundation Topics				
14. Network Monitoring	Review Key Topics				
14. Network Monitoring	Complete Memory Tables				
14. Network Monitoring	Define Key Terms				
14. Network Monitoring	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 14 in practice test software				
15. Disaster Recovery	Read Foundation Topics				
15. Disaster Recovery	Review Key Topics				
15. Disaster Recovery	Define Key Terms				
15. Disaster Recovery	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 15 in practice test software				

16. IPv4 and IPv6 Network Services	Read Foundation Topics				
16. IPv4 and IPv6 Network Services	Review Key Topics				
16. IPv4 and IPv6 Network Services	Complete Memory Tables				
16. IPv4 and IPv6 Network Services	Define Key Terms				
16. IPv4 and IPv6 Network Services	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 16 in practice test software				
17. Network Access and Management Methods	Read Foundation Topics				
17. Network Access and Management Methods	Review Key Topics				
17. Network Access and Management Methods	Complete Memory Tables				
17. Network Access and Management Methods	Define Key Terms				
17. Network Access and Management Methods	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 17 in practice test software				
18. Network Security Concepts	Read Foundation Topics				
18. Network Security Concepts	Review Key Topics				
18. Network Security Concepts	Define Key Terms				
18. Network Security Concepts	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 18 in practice test software				
19. Types of Network Attacks	Read Foundation Topics				
19. Types of Network Attacks	Review Key Topics				
19. Types of Network Attacks	Define Key Terms				

19. Types of Network Attacks	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 19 in practice test software				
20. Network Security Features	Read Foundation Topics				
20. Network Security Features	Review Key Topics				
20. Network Security Features	Define Key Terms				
20. Network Security Features	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 20 in practice test software				
21. A Network Troubleshooting Methodology	Read Foundation Topics				
21. A Network Troubleshooting Methodology	Review Key Topics				
21. A Network Troubleshooting Methodology	Complete Memory Tables				
21. A Network Troubleshooting Methodology	Define Key Terms				
21. A Network Troubleshooting Methodology	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 21 in practice test software				
22. Troubleshoot Common Cabling Problems	Read Foundation Topics				
22. Troubleshoot Common Cabling Problems	Review Key Topics				
22. Troubleshoot Common Cabling Problems	Define Key Terms				
22. Troubleshoot Common Cabling Problems	Complete Review Questions section				

Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 22 in practice test software				
23. Troubleshoot Common Issues with Network Services	Read Foundation Topics				
23. Troubleshoot Common Issues with Network Services	Review Key Topics				
23. Troubleshoot Common Issues with Network Services	Define Key Terms				
23. Troubleshoot Common Issues with Network Services	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 23 in practice test software				
24. Troubleshoot Common Performance Issues	Read Foundation Topics				
24. Troubleshoot Common Performance Issues	Review Key Topics				
24. Troubleshoot Common Performance Issues	Define Key Terms				
24. Troubleshoot Common Performance Issues	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 24 in practice test software				
25. Network Troubleshooting Tools	Read Foundation Topics				
25. Network Troubleshooting Tools	Review Key Topics				
25. Network Troubleshooting Tools	Complete Memory Tables				
25. Network Troubleshooting Tools	Define Key Terms				
25. Network Troubleshooting Tools	Complete Review Questions section				
Practice Test	Take practice test in study mode using Exam Bank 1 questions for Chapter 25 in practice test software				
26. Final Preparation					

26. Final Preparation	Take practice test in study mode for all book questions in practice test software				
26. Final Preparation	Review all Key Topics in all chapters				
26. Final Preparation	Take practice test in practice exam mode using Exam Bank #1 questions for all chapters				
26. Final Preparation	Take practice test in practice exam mode using Exam Bank #2 questions for all chapters				

Glossary of Key Terms

Numerics

2.4GHz The radio frequency band used by Wi-Fi networks for wireless communication, known for its longer range and better penetration through obstacles but susceptible to interference from other devices operating in the same frequency.

3G The third generation of wireless mobile telecommunications technology.

4G The fourth generation of wireless mobile technology, which was designed as a complete replacement for earlier 3G technology. The first 4G networks were widely deployed starting in 2009.

5G The fifth generation of wireless mobile technology, which seeks to eventually replace 4G technology. This technology enhances performance once again and began to be deployed in 2019.

5GHz The radio frequency band used for wireless networking that offers higher data rates and less interference compared to the 2.4GHz band, due to its broader spectrum and more non-overlapping channels.

6GHz The frequency range from 5.925GHz to 7.125GHz, offering additional spectrum for Wi-Fi networks to reduce congestion and improve performance.

10BASE-T An Ethernet standard that transmits at 10Mbps over twisted wire pairs (telephone wire).

10GBASE-LR A 10-Gigabit Ethernet standard specified to transmit data over long distance through single-mode fiber. (The letters “LR” stand for “long range.”)

10GBASE-SR A multimode fiber-optic medium. (The letters “SR” stand for “short range.”)

10GBASE-T A type of Ethernet signaling providing speeds over twisted-pair cabling that go beyond 1Gbps for distances up to 100 meters.

40GBASE-T 40-Gigabit Ethernet over copper twisted-pair cabling.

66 block A block traditionally used in corporate environments for cross-connecting phone system cabling. As 10Mbps LANs started to grow in popularity in the late 1980s and early 1990s, these termination blocks were used to cross-connect Category 3 UTP cabling. The electrical characteristics (specifically, crosstalk) of a 66 block, however, do not support higher-speed LAN technologies, such as 100Mbps Ethernet networks.

100BASE-FX A version of Fast Ethernet over optical fiber.

100BASE-SX A version of Fast Ethernet over optical fiber standardized in TIA/EIA-785-1-2002. It is a lower-cost, shorter-distance alternative to 100BASE-FX.

100BASE-TX The predominant form of Fast Ethernet, which runs over two wire pairs inside a Cat 5 or later cable. Each network segment can have a maximum cabling distance of 100 meters.

110 block A block that can be used to terminate a cable (such as a Cat 5 cable) used for high-speed LANs.

802.11a A standard, ratified in 1999, that supported speeds as high as 54Mbps. Other supported data rates (which could be used if conditions were not suitable for the 54Mbps rate) included 6Mbps, 9Mbps, 12Mbps, 18Mbps, 24Mbps, 36Mbps, and 48Mbps. The 802.11a standard used the 5GHz band and the OFDM transmission method.

802.11ac An IEEE wireless networking standard that operates in the 5GHz range and provides increased throughput compared to previous Wi-Fi IEEE standards.

802.11ax A modern high-speed WLAN technology that uses the 6GHz spectrum in addition to the 2.4GHz and 5GHz bands. Also known as Wi-Fi 6.

802.11b A standard, ratified in 1999, that supported speeds as high as 11Mbps. However, 5.5Mbps was another supported data rate. The 802.11b standard used the 2.4GHz band and the DSSS transmission method.

802.11g A standard, ratified in 2003, that supported speeds as high as 54Mbps. As with 802.11a, 802.11g also supported data rates of 6Mbps, 9Mbps, 12Mbps, 18Mbps, 24Mbps, 36Mbps, and 48Mbps. However, like 802.11b, 802.11g operated in the 2.4GHz band, which allowed it to offer backward compatibility to 802.11b devices. 802.11g used either the OFDM or DSSS transmission method.

802.11h 802.11h is an amendment to the IEEE 802.11 standard that provides enhancements for spectrum and transmit power management in the 5GHz band, primarily to ensure compliance with European regulatory requirements.

802.11n A standard, ratified in 2009, that supports a variety of speeds, depending on the implementation. Although the speed of an 802.11n network could approach 300Mbps (through the use of channel bonding), many 802.11n devices on the

market have speed ratings in the 130Mbps–150Mbps range. An 802.11n WLAN can operate in the 2.4GHz band, the 5GHz band, or both simultaneously. 802.11n uses the OFDM transmission method.

802.11 standards A set of protocols developed by the IEEE for implementing wireless local area network (WLAN) communication in various frequency bands, including specifications for speed, range, and encryption.

802.1Q tagging A networking standard that adds a 4-byte tag to Ethernet frames to identify and manage traffic within multiple VLANs on a single physical network.

802.1X A type of network access control (NAC) that can permit or deny a wireless or wired LAN client access to a network. If IEEE 802.1X is used to permit access to a LAN via a switch port, then IEEE 802.1X is being used for port security.

802.3 standards A collection of protocols developed by the IEEE that defines the physical and data link layers for wired Ethernet networks, including specifications for media access control (MAC), transmission speeds, and network topologies.

1000BASE-LX An optical-fiber Gigabit Ethernet standard specified in IEEE 802.3 to work over a distance of up to 5 km over 10 micrometer single-mode fiber.

1000BASE-SX An optical-fiber Gigabit Ethernet standard for operation over multimode fiber up to a distance of 220 to 550 meters, depending on the type of multimode fiber.

1000BASE-T A standard for Gigabit Ethernet over copper wiring designed to work over a maximum length of 100 meters.

A

AAAA A DNS record that maps a domain name to its corresponding IPv6 address, allowing devices to locate and connect to the domain over an IPv6 network.

acceptable use policy (AUP) A policy that identifies what users of a network are and are not allowed to do on that network. For example, retrieving sports scores during working hours via an organization's Internet connection might be deemed inappropriate by an AUP.

access control list (ACL) Rules typically applied to router interfaces to specify permitted and denied traffic.

access control vestibule A small entry area with two interlocking doors that prevents tailgating/piggybacking. Formerly called a mantrap.

access layer A layer in the three-tier hierarchical network design that is concerned with permitting end users fast and secure access to the network.

access point (AP) In wireless networks, a device that permits wireless clients to access the network. Access points tend to fall into two categories: lightweight and autonomous. A lightweight AP cannot perform control plane functions and requires a wireless LAN controller (WLC) for the control plane. An autonomous AP does not require a WLC.

active-active A configuration where two or more systems or network devices are simultaneously online and sharing the load, providing redundancy and improving performance by ensuring that all components are actively participating in processing tasks.

active-passive A configuration where one system or network device is actively handling tasks while a secondary system remains on standby, ready to take over if the active system fails, ensuring redundancy and high availability.

ad hoc A peer-to-peer network configuration where devices communicate directly with each other without relying on a central access point.

ad hoc network discovery The process of performing network discovery when initiated by a network administrator based on some network event of perceived need.

address (A) A DNS record that maps a hostname to an IPv4 address.

Address Resolution Protocol (ARP) poisoning A network attack where an attacker sends falsified ARP messages to associate their MAC address with the IP address of a legitimate device, enabling them to intercept, modify, or disrupt network traffic.

Address Resolution Protocol (ARP) request A broadcast asking for the MAC address corresponding to a known IP address. An ARP reply contains the requested MAC address.

administrative distance (AD) A routing protocol's index of believability. Routing protocols with a smaller AD are considered more believable than routing protocols with a higher AD.

administratively down The state of a network interface or device that has been manually disabled or turned off by an administrator, typically through software configuration, rather than due to a technical fault or failure.

Advanced Encryption Standard (AES) A symmetric block cipher implemented in hardware and software that protects sensitive or confidential data through encryption. AES offers various levels of strength.

angled physical contact (APC) A type of fiber connector that minimizes back reflection due to a 5° to 15° angle polish applied to end faces.

anomaly alerting A process in network management and cybersecurity where automated systems detect and flag unusual patterns or behaviors that deviate from normal operations, indicating potential issues or threats.

antenna A device that transmits or receives electromagnetic waves for communication purposes.

anycast A communication that has a one-to-nearest (from the perspective of a router's routing table) flow.

AP association time The amount of time it takes to associate a client with an AP. This can also be a measure of the delay when the AP is associating to a wireless LAN controller.

application awareness The capability of a network to identify, prioritize, and optimize traffic for specific applications, enhancing performance and efficiency based on application-specific requirements. Application awareness is a significant feature of software-defined networking (SD-WAN).

application layer In software-defined networking (SD-WAN), the logical layer that presents the applications to the network manager. These applications allow simple control over the environment, permitting the engineer to configure and monitor the network as needed.

application layer (OSI model) Layer 7 of the OSI model. This top layer of the model provides applications access to network services.

application layer (TCP/IP stack) The topmost layer of the TCP/IP model, which provides protocols and services for end-user applications to communicate over the network, facilitating data exchange directly with software applications.

application programming interface (API) A set of rules and protocols that allows different software applications to communicate and interact with each other.

application programming interface (API) integration The process of connecting different software systems and devices using APIs to enable automated, seamless communication and control across the network infrastructure.

arp A command that can be used in either a Microsoft Windows or UNIX environment to see which Layer 2 MAC addresses correspond to Layer 3 IP addresses.

ARP spoofing An attack that involves falsifying a Layer 2 MAC address in order to bypass simple MAC-based security mechanisms, such as port security.

asset inventory A comprehensive record or database containing detailed information about all hardware, software, and resources within an organization's IT infrastructure.

asset tag A tag applied to a physical network asset to permit the monitoring of the location of this network asset.

asymmetrical routing A type of routing in which an exit path does not match an entry path of traffic. This is sometimes very undesirable and can cause issues for applications.

attenuation The weakening of a signal as a result of distance between two network devices.

audit and assessment report A report that helps identify and address critical issues that might be occurring in the network infrastructure. This report often identifies performance and security issues that need to be addressed to help ensure that the network continues to function as desired.

audit logs Log files that contain records of actions that were performed in the network. For example, audit logs could contain records of the user accounts that successfully (or unsuccessfully) tried to access a key network resource.

auditing (AAAA) A DNS IPv6 address record that maps a hostname to an IPv6 address.

authentication The process of verifying the identity of a user, system, or entity to ensure that it is who or what it claims to be.

Authentication Header (AH) An IPsec protocol that provides authentication and integrity services but that does not provide encryption services.

authentication server In a network using 802.1X user authentication, a server (typically, a RADIUS server) that checks a supplicant's credentials. If the credentials are acceptable, the authentication server notifies the authenticator that the supplicant is allowed to communicate on the network. The authentication server also gives the authenticator a key that can be used to securely transmit data during the authenticator's session with the supplicant.

authenticator In a network using 802.1X user authentication, a device that forwards a supplicant's authentication request on to an authentication server. After the authentication server authenticates the supplicant, the authenticator receives a key that is used to communicate securely during a session with the supplicant.

authoritative name server A DNS server that holds and provides the definitive, original records for a specific domain, responding to queries with accurate and authoritative information.

authorization In a network, a definition of what users are permitted to do. Note that authorization often follows an authentication phase.

Automatic Private IP Addressing (APIPA) A feature that allows a networked device to self-assign an IP address from the 169.254.0.0/16 network. Note that this address is usable only on the device's local subnet (meaning that the IP address is not routable).

automation The automated completion of a task or tasks.

autonomous access point A standalone Wi-Fi device that independently handles all wireless networking functions, including management, configuration, and data traffic, without relying on a central controller.

availability monitoring The process of continuously checking the uptime and accessibility of systems, applications, or services to ensure they are operational and reachable for users.

available leases The number of remaining IP addresses available to be leased to clients in a DHCP scope.

B

backbone Another name for the core layer of the classic three-tier network design. This is also the name given to area 0 in Open Shortest Path First (OSPF).

backup configuration A duplicate or copy of the production configuration that is stored securely, serving as a contingency measure to restore operational settings in case of failures, data loss, or necessary rollbacks within an IT infrastructure.

badge reader A device used to read ID badges worn by employees of an organization to assist with physical security.

band steering A technique used in Wi-Fi networks to automatically guide dual-band client devices to connect to the less congested and higher performance 5GHz band instead of the 2.4GHz band.

bandwidth The measure of network throughput possible on network media or a network path.

bandwidth speed tester *See* speed tester.

baseline A collection of data portraying the characteristics of a network under normal operating conditions. Data collected while troubleshooting can be contrasted against baseline data.

baseline configuration documentation Documentation that makes it possible to return network equipment to the exact configurations that existed at a specific point in time. Various stages of the network device life cycles can be captured in baselines.

baseline/golden configuration A predefined and standardized template or ideal state of configuration settings for hardware, software, or network devices, used as a reference to ensure consistency, security, and compliance across an organization's IT environment.

baseline metrics Initial measurements used to establish a reference point for comparing future performance or progress.

basic service set identifier (BSSID) A unique 48-bit identifier, usually the MAC address of the access point's radio, that distinguishes each wireless access point in a Wi-Fi network.

basic service set (BSS) WLAN A wireless LAN that has just one access point (AP). BSS WLANs are said to run in infrastructure mode because wireless clients connect to an AP, which is typically connected to a wired network infrastructure. BSS networks are often used in residential and SOHO locations, where the signal strength provided by a single AP is sufficient to service all of the WLAN's wireless clients.

Bayonet Neill–Concelman (BNC) A type of coaxial connector commonly used for radio frequency signals, characterized by its quick connect and disconnect bayonet mount mechanism.

bidirectional wavelength-division multiplexing (WDM) The transmission of optical channels on a fiber propagating simultaneously in both directions. This is accomplished by use of either a WDM technique on a single fiber or unidirectional transmission over two fibers.

biometrics The use of unique characteristics of the human body to provide access credentials and security. For example, a thumbprint can be used to access a mobile device.

block size The number of IP addresses in a subnet, including the subnet's address and the subnet's directed broadcast address.

Bootstrap Protocol A legacy broadcast-based protocol used by networked devices to obtain IP address information.

Border Gateway Protocol (BGP) A path-vector routing protocol that can use as its metric the number of autonomous system hops that must be transited to reach a destination network, as opposed to the number of required router hops. BGP is the only exterior gateway protocol (EGP) in widespread use today, and it is considered to be the routing protocol that runs the Internet.

borrowed bits Bits added to a classful subnet mask.

botnet/command and control A collection of compromised systems used in a distributed denial-of-service (DDoS) attack. The attacker uses command and control software to control the compromised systems.

bottleneck A point in the network where the data flow is significantly restricted, causing a slowdown in overall network performance.

bottom-to-top troubleshooting A troubleshooting methodology in which the engineer starts at the physical layer of the OSI model and works up the stack, trying to determine the problem.

branching The practice of creating separate lines of development within a project, allowing parallel work on different features or fixes without affecting the main codebase.

bridge The predecessor technology to switches. Bridges could connect different types of networks. While this was important so that the two networks could communicate, it also ended up creating additional broadcast domains.

bring your own device (BYOD) A policy that specifies the types of permitted devices, security, data, and applications as well as offloading procedures when an employee leaves a company.

broadcast Traffic that travels from a single source to all destinations on a network (that is, a *broadcast domain*).

broadcast storm A problem that results when excessive broadcast packets flood a network area. It might be caused by an issue with Spanning Tree Protocol (STP), which is designed to protect against broadcast storms due to redundancy in the design.

brute-force A form of password attack in which the attacker tries all possible password combinations until a match is made.

bug fixes Specific changes made to software to correct errors and improve stability and performance.

Building Industry Cross-connect (Bix) A commonly used networking panel that terminates 25 pairs (50 wires) that are punched down to one side of a wafer. The wafer is then inserted into the metal frame with the punched-down side against the wall, so only the unused side is visible.

bus topology A topology that typically uses a cable running through the area requiring connectivity; devices to be networked can tap into that cable.

business continuity plan (BCP) A comprehensive policy that includes a disaster recovery plan. It may be broken down into supporting plans and policies that encompass topics such as business resumption plans and incident management.

C

cable Broadband Internet connection provided by a cable TV company. This infrastructure might include either coaxial or fiber-optic cabling—or both.

cable crimper A tool used to attach a connector (for example, an RJ45 connector) to the end of an unshielded twisted-pair (UTP) cable.

cable map A visual or diagrammatic representation detailing the physical connections and layouts of cables within a network infrastructure.

cable modem A device that attaches to the same coaxial cable (typically in a residence) that provides television programming. A cable modem can use predetermined frequency ranges to transmit and receive data over that coaxial cable.

cable speeds The data transfer rates achievable over coaxial cable networks, commonly used for Internet access, which can vary based on the technology and network infrastructure, with typical speeds ranging from several Mbps to Gbps.

cable stripper A tool used to remove the plastic shielding from cables; this feature is often built in to cable crimpers.

cable tester A tool that can check the conductors in an Ethernet cable. It contains two parts. By connecting these parts of the cable tester to each end of a cable under test, it is possible to check the wires in the cable for continuity (that is, check to make sure that there are no opens, or breaks, in a conductor). In addition, it is possible to verify an RJ45 connector's pinouts (which are wires connected to the appropriate pins on an RJ45 connector).

camera A device that permits the viewing of key network areas using video surveillance. Video footage is often recorded and stored locally or in the cloud. Modern camera systems have been combined with more advanced intelligence and even have the ability to detect weapons and other threats.

campus area network (CAN) An interconnection of networks located in nearby buildings (for example, buildings on a college campus).

canonical name (CNAME) A DNS record that aliases an existing record, thus allowing multiple DNS records to map to the same IP address.

captive portal Typically, a web page designed to collect the username and password of a user trying to gain access to a network or an application. Such a portal might require the user to agree to an acceptable use policy (AUP).

carrier-sense multiple access with collision detection (CSMA/CD) A media access control method used on an Ethernet network to help prevent collisions from occurring and to recover if collisions do occur. CSMA/CD is only needed on half-duplex connections.

Cat 5 Category 5 twisted-pair cable. Supports 10BASE-T, 100BASE-TX (Fast Ethernet), 1000BASE-T (Gigabit Ethernet), 2.5GBASE-T, and 5GBASE-T.

Cat 5e Category 5e twisted-pair cable. Supports Gigabit Ethernet (10/100/1000Mbps).

Cat 6 Category 6 twisted-pair cable. Supports 10G Ethernet (10/100/1000/10000Mbps) and reduces crosstalk for more reliable connections at Gigabit speeds.

Cat 6a Category 6a twisted-pair cable. Support frequencies up to 500MHz and speeds up to 10Gbps.

Cat 7 Category 7 twisted-pair cable used in high-speed Ethernet-based computer networks of 1Gbps or higher.

Cat 8 Category 8 twisted-pair cable. Differs greatly from the previous cables in that it supports a frequency of up to 2GHz (2000MHz) and is limited to a 30-meter two-connector channel. Supports speeds of 25Gbps or even 40Gbps.

cellular A wireless communication technology that uses a network of distributed cell sites, or base stations, to provide mobile connectivity and enable devices to communicate over long distances through radio waves.

central policy management A feature of the SD-WAN architecture that provides a unified platform for defining and enforcing network policies across all connected devices and locations. *See also* software-defined wide area network (SD-WAN).

central repository In a source control system, acts as the single source of truth for all infrastructure code. This repository stores the master copies of the code and provides a central location where all team members can access, contribute to, and collaborate on the infrastructure codebase.

certificates Digital documents that use cryptographic techniques to validate the identity of entities and secure communications over networks by binding a public key to the identity of an individual, organization, or device.

change management In networking, careful planning and policy creation to accommodate thoughtful and warranted changes to the network infrastructure and configurations.

channel bonding A process in which two wireless bands are logically bonded together, forming a band with twice the bandwidth of an individual band. Some literature calls channel bonding 40MHz mode, which refers to the bonding of two adjacent 20MHz bands into a 40MHz band.

channel overlap A wireless issue in the 2.4GHz band that causes interference. In order to avoid this issue, adjacent WLAN segments should be placed on

non-overlapping channels—that is, channels 1, 6, and 11 in the 2.4GHz band. Notice channel overlap is not an issue in the 5GHz band.

channel width In Wi-Fi networks, the bandwidth or frequency range used by the wireless signal, which affects data transmission speed and the potential for interference.

Cisco Discovery Protocol (CDP) A Cisco-proprietary protocol that enables network devices to discover information about directly connected Cisco equipment. *See also* Link Layer Discovery Protocol (LLDP).

Class A Used for very large networks; they have a default subnet mask of 255.0.0.0, with the first octet representing the network portion.

Class B Used for medium-sized networks; they have a default subnet mask of 255.255.0.0, with the first two octets representing the network portion.

Class C Used for small networks; they have a default subnet mask of 255.255.255.0, with the first three octets representing the network portion.

Class D Reserved for multicast groups, enabling the delivery of data to multiple destinations simultaneously. They do not have a subnet mask.

Class E Reserved for experimental purposes and are not used in general network operations.

classful mask The default subnet mask applied to Class A, B, and C IPv4 networks. Specifically, Class A networks have the classful mask 255.0.0.0, Class B networks have the classful mask 255.255.0.0, and Class C networks have the classful mask 255.255.255.0.

classless interdomain routing (CIDR) A process that involves shortening a classful subnet mask by removing right-justified 1s from a classful mask. As a result, CIDR allows contiguous classful networks to be aggregated. This process is sometimes called *route aggregation*.

client With NTP, a network device that consumes the correct time from an NTP server system.

client disassociation The process of intentionally or unintentionally disconnecting a connected device, or client, from a network or access point.

client/server network A network in which a dedicated server (for example, a file server or a print server) provides shared access to a resource (for example, files or a printer). Clients (for example, PCs) on the network with appropriate privilege levels can gain access to those shared resources.

client-to-site VPN Also known as a remote-access VPN, a virtual private network that interconnects a remote user with a site. The remote user is typically using a

broadband Internet connection and software running in the OS that configures and maintains the VPN connection.

clientless VPN A virtual private network that has a client software piece that is not a separate piece of software. In this arrangement, the client's web browser acts as the VPN client software. This type of VPN connection leverages the SSL/TLS capabilities of the modern Internet and web browsers in order to provide a secured connection.

cloud gateway A network service or device that acts as an intermediary, facilitating secure data transfer between an organization's on-premises infrastructure and cloud services.

cloud site A recovery site in the cloud that can be configured for cold, warm, or hot behavior, typically with ease. Cloud sites offer lower costs and relaxed technological requirements compared to physical recovery sites.

coaxial/RG6 A network cable that is composed of two conductors. One of the conductors is an inner insulated conductor. This inner conductor is surrounded by another conductor. This second conductor is sometimes made of a metallic foil or woven wire. RG6 is used to distribute HDTV signals in a home or office. RG6QS uses quad shielding for better signal quality over long runs. Also known as coax.

code-division multiple access (CDMA) A technology that allows several transmitters to send information simultaneously over a single communication channel so that several users can share a band of frequencies. CDMA is used as the access method in many mobile phone standards.

cold site A barebones redundant site for a network that requires time and effort to bring online.

collapsed core design A network architecture where the core and distribution layers are combined into a single layer to simplify the network, reduce costs, and enhance performance in smaller or less complex environments.

collision A problem that occurs when two devices on an Ethernet network simultaneously transmit frames. Because an Ethernet segment cannot handle more than one frame at a time, both frames in a collision become corrupted.

Common Vulnerabilities and Exposures (CVE) A free online resource that provides excellent search tools to leverage a large database of publicly known information security vulnerabilities and exposures.

community cloud Cloud services used by individuals, companies, or entities with similar interests. The community cloud is typically a partnership between many different cloud vendors or organizations.

community string A cleartext string that serves as a password in SNMP and determines the type of access (read-only or read-write) permitted for network monitoring and management.

compliance Ensures that infrastructure configurations adhere to predefined standards and policies, maintaining consistency and security across systems.

confidentiality, integrity, and availability (CIA) A security principle that helps in securing a network. It entail using confidentiality (often using encryption), integrity (often using hashing algorithms), and availability with a wide variety of security mechanisms.

configuration drift Occurs when the actual state of the infrastructure deviates from the desired state defined in the IaC. This drift can happen due to manual changes, system updates, or environmental factors. *See also* infrastructure as code (IaC).

configuration management The systematic process of maintaining and controlling the settings, specifications, and configurations of hardware, software, and network devices to ensure consistency, security, and optimal performance throughout their life cycle.

configuration monitoring The continuous oversight and tracking of system and network settings to ensure compliance, detect unauthorized changes, and maintain optimal performance and security.

conflict identification The process of detecting changes made to the same lines of code by different contributors, requiring manual resolution to ensure consistency and correctness. Conflict identification is an essential feature of source control systems, especially in collaborative IaC environments, where multiple team members may work on the same code simultaneously. *See also* infrastructure as code (IaC).

congestion A state where network traffic exceeds the available bandwidth, leading to delays, packet loss, and reduced quality of service.

connection-oriented Refers to transport protocols that provide reliable transport, in that if a segment is dropped, the sender can detect that drop and retransmit the dropped segment. Specifically, a receiver acknowledges segments that it receives. Based on those acknowledgments, a sender can determine which segments were successfully received.

connectionless Refers to transport protocols that are sent unreliably. There is no session created in which flow control and error checking types of behaviors take place. UDP is the connectionless transport protocol of the transport layer of the OSI model.

console A physical interface on a network device that allows direct access for configuration and management, typically using a serial connection.

console cable A cable used to make a console connection to a network device such as a router.

content delivery network (CDN) A geographically distributed network of servers designed to deliver web content and digital media to users more efficiently by reducing latency and offloading traffic from the origin server.

content filtering Monitors and controls data transmitted over a network, blocking or allowing content based on set criteria to protect users and enforce policies.

contention Occurs when multiple devices or applications simultaneously compete for the same network resources, leading to potential delays and reduced performance.

control layer In software-defined networking, the layer of abstraction that defines the control plane operations, which might include a centralized controller and both southbound and northbound APIs. *See also* software-defined networking (SDN).

control plane policing (CoPP) An implementation of policing for the control plane of a network device that protects the device from many different potential DoS attacks targeting control plane protocols such as OSPF and BGP.

core layer A layer in the three-layer network design that is concerned with speed. This layer can be responsible for moving massive amounts of data in an enterprise network.

course wavelength-division multiplexing (CWDM) A technology that allows up to 18 channels to be connected over a dark fiber pair.

crossover cable A cable that can be used to directly connect two devices—such as two computer systems—or as a means to expand networks that use devices such as hubs or switches. A traditional crossover cable is a UTP cable in which the wires are crossed for the purposes of placing the transmit line of one device on the receive line of the other.

crosstalk The unwanted transfer of signals between communication channels, causing interference and signal degradation.

current state modulation A way to electrically or optically represent binary digits by using a binary 1 for the presence of voltage on a copper cable or the presence of light on a fiber-optic cable and using a binary 0 for the absence of light or voltage.

cutter A tool used to cut cables and other cabling accessories.

cyclic redundancy check (CRC) A mathematical algorithm that is executed on a data string by both the sender and the receiver of the data string. If the calculated CRC values match, the receiver can conclude that the data string was not corrupted during transmission.

D

data as a service (DaaS) A cloud-based service that uses software tools for working with data. Of course, all of these software tools are cloud-based.

data at rest Refers to inactive data that is stored physically on any digital medium, such as hard drives, databases, or backups and is not actively being used, processed, or transmitted.

data center interconnect (DCI) A feature of VXLAN that enables the extension of Layer 2 networks across geographically dispersed data centers, facilitating seamless connectivity and workload mobility. *See also* Virtual Extensible Local Area Network (VXLAN).

data in transit Digital information actively moving between locations, such as across the Internet or within private networks, often requiring encryption to protect it from interception or unauthorized access.

data in use Data actively being processed, accessed, or updated by applications, systems, or users within a computing environment.

data link layer Layer 2 of the OSI model, which is concerned with packaging data into frames and transmitting those frames on a network, performing error detection/correction, uniquely identifying network devices with an address, and handling flow control.

data locality The practice of storing data within specific geographic boundaries, which is crucial for regulatory compliance.

data loss prevention (DLP) The practice of detecting and preventing data breaches, exfiltration, or unwanted destruction of sensitive data in an organization.

deauthentication A type of attack in which the attacker sends a deauthentication frame to the victim in order to disconnect the user from the wireless LAN. Such an attack is usually followed up with another form of attack, such as an evil twin.

decibel (dB) loss A loss of signal power. If a transmission's dB loss is too great, the transmission cannot be properly interpreted by the intended recipient.

decommissioning The systematic process of safely retiring and dismantling a system, asset, or facility from active service, ensuring proper disposal or repurposing and mitigation of any associated risks.

default gateway The IP address of a router (or multilayer switch) to which a network device sends traffic destined for a subnet other than the device's local subnet.

default route A network route used by a router to forward packets when no specific route to the destination is found in the routing table.

default static route An administratively configured entry in a router's routing table that specifies where traffic for all unknown networks should be sent.

default VLAN The VLAN assigned to interfaces on a device when no explicit VLAN assignment has been made.

defense in depth A technique that involves striving to secure each layer of a solution.

demarcation point (demarc) The point in a telephone network where the maintenance responsibility passes from a telephone company to a subscriber (unless the subscriber purchased an inside wiring plan). This demarc is typically a box mounted to the outside of a customer's building (for example, a residence).

denial-of-service (DoS) A type of attack that floods a system with an excessive amount of traffic or requests, consuming the system's processing resources and preventing the system from responding to many legitimate requests.

dense wavelength-division multiplexing (DWDM) An optical multiplexing technology used to increase bandwidth over existing fiber networks. DWDM works by combining and transmitting multiple signals simultaneously at different wavelengths on the same fiber.

designated port In an STP topology, a single port on a network segment that is closest to the root bridge, in terms of cost. All ports on a root bridge are designated ports. *See also* Spanning Tree Protocol (STP).

desktop as a service (DaaS) A cloud computing offering that delivers virtual desktop environments to users over the Internet, allowing access to applications and data from any device, anywhere.

device hardening The process of securing a device by reducing its attack surface through the implementation of security measures such as disabling unnecessary services, applying patches, and configuring security settings.

DHCP relay A device (often a router or a server) that can convert broadcast DHCP DISCOVER messages to unicast packets so that they can be forwarded to a DHCP server on another TCP/IP subnet. Normally these broadcast messages would be dropped by a router on a segment.

DHCP reservation A configuration that assigns a specific IP address to a particular device based on its MAC address, ensuring that the device always receives the same IP address from the DHCP server.

DHCP scope exhaustion A situation that occurs when there are no available IP addresses to lease out to clients.

DHCP snooping The process of securing a network against a rogue DHCP server attack or other types of DHCP security attacks.

DHCPv6 The IPv6 version of Dynamic Host Configuration Protocol (DHCP).

dictionary attack An attack, similar to a brute-force password attack, that uses commonly used words (rather than every combination) to determine a password.

dig A command that can resolve a fully qualified domain name (FQDN) to an IP address on UNIX hosts.

digital subscriber line (DSL) A group of technologies that provides high-speed data transmission over existing telephone wiring. DSL has several variants that differ in terms of data rates and distance limitations.

direct attach copper (DAC) cable A high-speed, cost-effective cable used to connect network devices in data centers and enterprise networks directly, typically within short distances up to 7 meters. It combines transceivers and copper cabling in a single fixed assembly, eliminating the need for separate optical transceivers and fiber-optic cables.

Direct Connect A dedicated network service that establishes private, high-speed, and reliable connections between an organization's on-premises data center or network and the AWS cloud. Direct Connect enables secure and consistent access to AWS resources with reduced latency and increased data transfer efficiency.

directional antenna A type of antenna that focuses its signal in a specific direction, providing increased signal strength and range in that direction while reducing interference from other directions.

disaster recovery plan (DRP) A documented process to execute an organization's disaster recovery processes. A DRP is designed to help recover and protect IT infrastructure in the event of a disaster.

distance In reference to wired and wireless media, how far a signal can travel over media before it weakens and suffers from the effects of attenuation.

distance-vector A type of routing protocol that determines the best route for data packets based on distance.

distributed denial-of-service (DDoS) An attack that can increase the amount of traffic flooded to a target system. Specifically, an attacker compromises multiple systems, and those compromised systems, called zombies, can be instructed by the attacker to simultaneously launch a DDoS attack against a target system.

distribution layer A layer in the three-tier hierarchical network design that is concerned with routing traffic to remote destinations.

divide-and-conquer troubleshooting A troubleshooting methodology in which an engineer starts at a precise layer of the OSI model based on information collected. The engineer might move up or down from that layer after discovering more information.

DMZ *See* screened subnet.

DNS caching A process that keeps the name-to-IP address mapping in memory for subsequent usage. A DNS TTL ensures that entries time out after some time period, which helps ensure that cached information has not gone stale. *See also* DNS Time-to-Live (TTL).

DNS over HTTPS (DoH) A protocol that encrypts DNS queries by sending them over HTTPS, enhancing privacy and security by preventing eavesdroppers from seeing the websites you are trying to visit.

DNS over TLS (DoT) A protocol that encrypts DNS queries by sending them over Transport Layer Security (TLS), enhancing privacy and security by preventing eavesdroppers from viewing or tampering with your DNS requests.

DNS poisoning The introduction of false data into the DNS system.

DNS record types Specific kinds of data entries in the DNS system that define various aspects of a domain, such as its IP address (A record), mail servers (MX record), or alias names (CNAME record).

DNS server A server that resolves a domain name (for example, `www.ciscopress.com`) to a corresponding IP address (for example, `10.1.2.3`).

DNS spoofing A cyberattack where an attacker alters DNS records to redirect users to fraudulent websites, often to steal sensitive information or distribute malware.

DNS Time-to-Live (TTL) The amount of time DNS clients or servers are permitted to cache resolution entries for quick recall when needed on the network. The shorter the DNS TTL, the more current the DNS information that is available, but this might come at the cost of more “cache misses,” where the information cannot be quickly recalled from system memory.

Domain Name Security Extensions (DNSSEC) A security protocol that adds digital signatures to DNS data to ensure its authenticity and integrity, protecting against attacks like spoofing and redirecting users to malicious sites.

Domain Name System (DNS) A system that resolves a domain name (for example, `www.ciscopress.com`) to a corresponding IP address (for example, `159.182.74.49`).

dotted-decimal notation A method of writing an IPv4 address or subnet mask, where groups of 8 bits (called octets) are separated by periods.

drops Errors that occur when packets are discarded by a network interface due to congestion, buffer overflow, or other issues, leading to packet loss and reduced performance.

DSL modem A local device that makes data transfers possible in DSL WAN environments. DSL is a relatively high-speed WAN (Internet) option that operates over the plain old telephone service (POTS).

dual stack The ability of a network interface to run multiple protocols, such as IPv4 and IPv6.

dumpster diving A social engineering tactic where individuals sift through discarded materials to obtain sensitive information for malicious use, such as identity theft or corporate espionage.

Dynamic ARP Inspection (DAI) A feature that ensures that MAC addresses coordinate to the Layer 3 addresses as they should.

dynamic assignment The automatic assignment of IP address information in a network. In a TCP/IPv4 environment, this is most often accomplished with DHCP. *See also* static assignment.

Dynamic Host Configuration Protocol (DHCP) A protocol that dynamically assigns IP address information (for example, IP address, subnet mask, DNS server's IP address, and default gateway's IP address) to network devices.

dynamic inventories A feature of IaC automation that allows the infrastructure to be dynamically discovered and managed based on current configurations and states. *See also* infrastructure as code (IaC).

dynamic routing The process of distributing network prefix information through a network so that traffic can be forwarded to network destinations. The opposite approach to dynamic routing is static routing, where an administrator trains a network device how to reach network destinations.

E

east-west traffic flows The horizontal flow of traffic between devices in a data center.

effective isotropic radiated power (EIRP)/power settings A value that seeks to measure the maximum amount of power that could be radiated from an antenna. This value must incorporate the antenna gain and the transmitter power of the RF system.

elasticity The ability of a cloud solution to grow and shrink as needed.

employee training A proactive way to ensure that employees are aware of company policies and educated on cybersecurity threats such as social engineering.

Encapsulating Security Payload (ESP) An IPsec protocol that provides authentication, integrity, and encryption services.

encryption The process of converting data into a coded format to prevent unauthorized access, ensuring that only those with the correct decryption key can read the information.

end-of-life (EOL) The point at which a product, service, or system is no longer supported, updated, or sold by the manufacturer, often leading to its phase-out or replacement.

end-of-support (EOS) The point at which a vendor ceases to provide maintenance, updates, or technical support for a product or service, making it more vulnerable to security risks and operational issues.

enhanced form-factor pluggable (SFP+) A compact, hot-swappable transceiver designed to support 100/1000Mbps Ethernet, 10Gbps Ethernet, Fibre Channel, and SONET, among other communication standards.

Enhanced Interior Gateway Routing Protocol (EIGRP) A Cisco-proprietary protocol that is popular in Cisco-only networks but less popular in mixed-vendor networks. Like OSPF, EIGRP is an interior gateway protocol (IGP) with very fast convergence and high scalability. EIGRP is considered to be an advanced distance-vector or a hybrid routing protocol.

enhanced quad small form-factor pluggable (QSFP+) A compact, hot-swappable transceiver that supports Ethernet, Fibre Channel, InfiniBand, and SONET/SDH standards with different data rate options.

error disabled A state in which a network device automatically disables a port or interface due to the detection of a critical error or security violation, preventing further network issues or unauthorized access.

Ethernet A Layer 1 technology developed by Xerox that encompasses a variety of standards that specify various media types, speeds, and distance limitations.

EUI-64 A method used to automatically configure IPv6 host addresses.

evil twin A device that is postured to appear like a legitimate access point on a network to carry out a wireless attack.

exclusion range A range of IP addresses that are not assigned to clients of a network. It is often necessary to configure these ranges to avoid inadvertently leasing out IP addresses that are already statically assigned to servers or key routers in an organization.

exploit A detailed description of what exactly occurred with a security breach. An exploit often describes the most likely attacker, the technology used, and any vulnerabilities or misconfigurations that made it possible.

extended service set identifier (ESSID) A unique identifier assigned to a Wi-Fi network to distinguish it from other networks and is used by devices to connect to the desired wireless network within a larger network infrastructure.

extended service set (ESS) WLAN A wireless LAN containing more than one access point (AP). Like basic service set (BSS) WLANs, ESS WLANs operate in infrastructure mode. With more than one AP, it is important to prevent one AP from interfering with another. Specifically, non-overlapping channels (that is, channels 1, 6, and 11 for the 2.4GHz band) should be selected for adjacent wireless coverage areas.

Extensible Authentication Protocol (EAP) A security protocol for networking that expands the authentication methods used by the early Point-to-Point Protocol (PPP). EAP is used on encrypted networks to provide a secure way to send identifying information to provide network authentication.

exterior gateway protocol (EGP) A routing protocol that operates between autonomous systems, which are networks under different administrative control. Border Gateway Protocol (BGP) is the only EGP in widespread use today.

external DNS server A DNS server that is external to an organization (most often located on the Internet). An organization might not control (at all) the external DNS servers.

F

F-type connector A connector used for cable, satellite, and fixed wireless Internet and TV service. F-connectors can be crimped or attached via compression to coaxial cable.

factory reset The process of restoring a network device to factory settings to ensure that it contains no data or settings that are specific to the organization.

fiber distribution panel A networking panel used for accommodating fiber cable terminations, connections, and patching.

fiber light meter A device used to measure the power in an optical signal. The term usually refers to a device for testing average power in fiber-optic systems.

Fibre Channel (FC) A high-speed network technology primarily used for storage area networks (SANs) to facilitate fast, reliable data transfer between computer data storage devices and servers. It supports data rates up to 128Gbps and is known for its low latency and robust performance in enterprise environments.

File Transfer Protocol (FTP) A standard network protocol used to transfer files between a client and a server over a TCP/IP network. Notice there is no focus on security with the FTP protocol.

fire suppression The systems and methods used to extinguish or prevent the spread of fires, typically involving the use of water, chemical agents, or inert gases.

firewall Primarily a network security appliance that can protect a trusted network (for example, a corporate LAN) from an untrusted network (for example, the Internet) by allowing the trusted network to send traffic into the untrusted network and receive the return traffic from the untrusted network while blocking traffic for sessions that were initiated on the untrusted network.

firmware updates Enhancements or corrections applied to the software embedded within hardware devices to improve functionality and security.

First Hop Redundancy Protocol (FHRP) A protocol that permits redundancy for the Layer 3 gateway function. There are several protocols to choose from in this category.

floor plan A drawing to scale, typically showing a view from above, of the relationships between rooms, spaces, traffic patterns, and other physical features at one level of a structure.

flow data A collection of details about conversations happening across your network; for example, systems that are communicating and how much data they're exchanging.

forward lookup A standard DNS query in which the client has the name of the network resource and is requesting the IP address associated with this name.

forward lookup zone A database that maps domain names to their corresponding IP addresses.

full-duplex A connection that allows a device to simultaneously transmit and receive data.

full-mesh topology A network topology that directly connects every site to every other site.

full tunnel A type of VPN design in which all traffic is sent down the VPN connection. *See also* split tunnel.

fusion splicer A tool used to join two fibers together by using heat.

G

General Data Protection Regulation (GDPR) A comprehensive data protection law in the European Union that sets guidelines for the collection, processing, and storage of personal data to ensure privacy and security.

generator A device that converts motive power (mechanical energy) into electrical power for use in an external circuit. Sources of mechanical energy include steam turbines, gas turbines, water turbines, internal combustion engines, wind turbines, and even hand cranks.

Generic Routing Encapsulation (GRE) A tunneling protocol used to encapsulate a wide variety of network layer protocols inside virtual point-to-point links or point-to-multipoint links over an IP network.

geofencing A virtual perimeter of a geographic area. For example, a wireless geofence boundary around a data center could send an alarm if equipment leaves the perimeter.

giants Data packets that exceed the maximum allowable size for the network media. This is typically 1518 bytes for Ethernet networks. Giants can result from faulty hardware, misconfigurations, or software failures.

global DNS hierarchy In the DNS database system, the strict hierarchy in domain naming or in servers that make up the DNS system.

Global System for Mobile Communications (GSM) A standard developed by the European Telecommunications Standards Institute (ETSI) to describe the protocols for second-generation digital cellular networks used by mobile devices such as tablets.

graphical user interface (GUI) A user interface that allows interaction with software through graphical elements like buttons, icons, and menus, rather than text-based commands.

guest network A separate, isolated Wi-Fi network created within an existing network to provide Internet access to visitors without granting them access to the main network's resources and devices.

H

half-duplex A connection that allows a device to either receive or transmit data at any one time. However, a half-duplex device cannot simultaneously transmit and receive.

hardware asset inventory A comprehensive list of all physical hardware devices within a network, including servers, switches, routers, and end-user devices.

heat map A visual representation that uses color gradients to illustrate wireless data intensity or distribution across a geographic area or space, often used to depict metrics like network signal strength or user activity.

heating, ventilation, and air conditioning (HVAC) Technology used for indoor and vehicular environmental control of temperature and air quality.

hold-down timer A timer that can speed the convergence process of a routing protocol. After a router makes a change to a route entry, the hold-down timer prevents subsequent updates for a specified period of time. This approach can help stop flapping routes (which are routes that oscillate between being available and unavailable) from preventing convergence.

honeynet A network of decoy servers and systems designed to attract, detect, and analyze cyber attackers by mimicking real network environments, helping to study attack methods and gather intelligence.

honeypot A system that appears to be an attractive attack target to entice attackers into thinking the system is real. The attackers then use their resources to attack the honeypot and leave the real servers alone.

hostname On a Windows system, the command that simply displays the computer name. On a Linux system, a command used to change the computer name or the domain name of a system.

hosts file A plain text file used by operating systems to map hostnames to IP addresses, enabling local name resolution without needing a DNS server.

hot site A redundant data center location that is ready to replace a failed data center with little to no time or effort.

hub An older technology used to interconnect network components, such as clients and servers. Hubs vary in the number of available ports. A hub does not

perform inspection of the traffic it passes. Rather, a hub simply receives traffic in a port and repeats that traffic out all of its other ports.

hub-and-spoke topology A network topology used to interconnect multiple sites (for example, multiple corporate locations) via WAN links, with a WAN link from each remote site (a spoke site) to the main site (the hub site).

humidity The concentration of water vapor present in the air; network devices specify optimum humidity values to ensure they are not damaged.

HVAC sensor A sensor that facilitates connectivity of an HVAC system to a network for maintenance and configuration purposes.

hybrid A network structure that combines two or more different types of network topologies, such as star, ring, or bus, to take advantage of their respective strengths and minimize weaknesses.

hybrid cloud An environment that features private cloud services, some of which interact with public clouds.

hybrid routing protocol A routing protocol that features aspects of different routing protocol techniques. For example, EIGRP has aspects that are taken from both the distance-vector and link-state routing protocol approaches.

Hypertext Transfer Protocol (HTTP) An insecure protocol for retrieving HTML (and other files) over an IP network. HTTP is the basis for the Internet as we know it today.

Hypertext Transfer Protocol Secure (HTTPS) A protocol for securing web traffic over the Internet by using Secure Socket Layer (SSL) and Transport Layer Security (TLS) technology.

hypervisor Software that makes the virtualization of underlying hardware possible. Hypervisors are divided into Type 1 and Type 2 variants. Type 1 hypervisors install directly on “bare metal,” and Type 2 hypervisors install within an OS.

I

identity and access management (IAM) A framework of policies, processes, and technologies that ensures the right individuals have appropriate access to technology resources, protecting and managing user identities and privileges within an organization.

ifconfig A command used to display IP address configuration parameters on a UNIX/Linux system.

IMAP over SSL The secure version of IMAP, which uses Secure Socket Layer (SSL) for security, much as HTTPS uses SSL/TLS. *See also* Internet Message Access Protocol (IMAP).

in-band management Management traffic that is sent on the same paths and equipment as data traffic.

incident response policies Policies designed to deal with any encountered security issues, typically consisting of the following phases: prepare, identify, contain, eradicate, recover, and review.

incorrect transceiver A transceiver selected for a network device that does not match the cable type used and the wavelengths in use in a network.

independent basic service set (IBSS) ad hoc A WLAN that is created without the use of an AP. Such a configuration, called an IBSS, is said to work in an ad hoc fashion. An ad hoc WLAN is useful for temporary connections between wireless devices. For example, it might be necessary to temporarily interconnect two laptop computers to transfer a few files.

industrial control system (ICS) An integrated set of hardware and software used to monitor, control, and automate industrial processes such as manufacturing, power generation, and water treatment.

Industrial Internet of Things (IIoT) The use of interconnected sensors, devices, and machinery within industrial settings to collect, exchange, and analyze data, improving operational efficiency, productivity, and decision-making.

infrastructure A network configuration where wireless devices communicate with each other through an access point, which serves as a central hub for data transmission and network management.

infrastructure as a service (IaaS) A cloud service model in which the company rents virtualized servers (which are hosted by a service provider) and runs specific applications on those servers.

infrastructure as code (IaC) A cloud capability that makes it possible to easily automate, and even orchestrate, common networking tasks that used to take weeks or months to carry out on physical servers.

infrastructure layer In software-defined networking, the layer where the networking devices are defined. *See also* software-defined networking (SDN).

intake A vent or opening through which cool air is drawn into the device to dissipate heat generated by its components, ensuring proper thermal management and preventing overheating.

interface status An indication of the operational and administrative characteristics of an interface. For example, on a multilayer switch, part of the interface status can be whether the port will function as a router port (Layer 3) or a switch port (Layer 2).

interference In networking, the disruptive modification of a signal as it travels along a communication channel between source and receiver. The term is often used to refer to the addition of unwanted signals during a useful and expected signal. A common example is radar impacting a Wi-Fi LAN that receives the unwanted radar signals.

interior gateway protocol (IGP) A routing protocol that operates within an autonomous system, which is a network under a single administrative control. OSPF and EIGRP are popular examples of IGPs.

intermediate distribution frame (IDF) A network distribution point where cables from end-user devices connect to the main distribution frame, facilitating the organization and management of network connections within a building or campus.

internal DNS server A DNS server that is part of an organization's network and equipment.

Internet Control Message Protocol (ICMP) A transport layer protocol used by utilities such as **ping** and **traceroute**.

Internet gateway A horizontally scalable and highly available virtual gateway service that facilitates secure bidirectional communication between resources within an AWS virtual private cloud and the Internet, allowing for seamless connectivity and data transfer while enforcing customizable access control policies.

Internet Key Exchange (IKE) A protocol used to set up a secure, authenticated communication channel between two devices by negotiating, creating, and managing encryption keys for IPsec VPNs.

Internet layer The layer of the TCP/IP stack that maps to Layer 3 (the network layer) of the OSI model. Although multiple routed protocols (for example, IPv4 and IPv6) may reside at the OSI model's network layer, the Internet layer of the TCP/IP stack focuses on IP as the protocol to be routed through a network.

Internet Message Access Protocol (IMAP) A protocol that retrieves email from an email server. It is only one option of many for the retrieval of email from such servers. IMAP uses TCP port 143 in its operation.

Internet of Things (IoT) A network of physical objects (things) that possess embedded sensors, software, and other technologies for the purpose of exchanging data with other devices and systems. Often, the connectivity is with devices over the Internet. These devices are often called smart devices.

Internet Protocol (IP) A set of rules governing the format of data sent over the Internet or other networks, enabling devices to communicate with each other by routing packets of information to their destination addresses.

Internet Protocol Security (IPsec) A complex suite of protocols that is used to create secured connections between network systems.

intrusion detection system (IDS) device A system that seeks to alert administrators regarding a cybersecurity attack but that typically does not prevent the attack through its own actions.

intrusion prevention system (IPS) A system that seeks to recognize attack traffic and prevent that traffic from entering a network or devices.

ip A command used to assign an address to a network interface or configure network interface parameters in Linux systems. It replaces the **ifconfig** command in modern Linux versions.

IP address management (IPAM) The systematic administration of IP addresses, ensuring efficient allocation, tracking, and maintenance within a network environment.

IP address spoofing Falsifying the IP address information in a packet.

IP header Information at the beginning of an Internet Protocol (IP) packet that enables proper addressing and routing.

IP helper A DHCP relay agent, which can forward DHCP DISCOVER messages as unicast packets to the DHCP server housed on a remote subnet. Also known as UDP forwarder.

IP scanner A tool that is designed to efficiently gather IP address information for the nodes in a network.

ipconfig A Microsoft Windows command that can be used to display IP address configuration parameters on a PC. In addition, if the PC uses DHCP, the **ipconfig** command can be used to release and renew a DHCP lease, which can be useful during troubleshooting.

iperf A tool for network performance measurement and tuning. It is a cross-platform tool that can produce standardized performance measurements for any network.

IPv4 address classes Predefined ranges of IP addresses (Classes A, B, C, D, and E) designed to allocate addresses based on network size, with Class A for large networks, Class B for medium-sized networks, Class C for small networks, Class D for multicast, and Class E reserved for experimental use.

IPv6 addressing An updated version of IPv4 that uses 128-bit addresses to provide a vastly larger pool of unique IP addresses compared to IPv4, enabling more efficient routing and addressing in modern networks.

iterative lookup A DNS lookup in which the client indicates it will accept the answer, or a referral to another DNS server that might have the answer.

J-K-L

jitter The uneven arrival of packets.

jumbo frames Ethernet frames with a payload size significantly larger than the standard 1500 bytes, typically up to 9000 bytes, used to improve network efficiency by reducing overhead.

jump box A secure, intermediary server that administrators use to access and manage devices on a network, often isolated in a different security zone.

jump host A different term for a jump box in a network topology.

Kerberos A client/server authentication protocol that supports mutual authentication between a client and a server. Kerberos uses the concept of a trusted third party (a key distribution center) that hands out tickets to be used instead of a username and password combination.

key management The processes and systems for generating, distributing, storing, and handling cryptographic keys to ensure secure encryption and decryption of data.

Krone A proprietary European alternative to 110 block. Krone is used in data environments and also television broadcasting.

latency A measure of the delay in a network.

Layer 1 diagrams Diagrams that represent the physical layout of a network, including cables, hardware devices, and their physical connections.

Layer 2 diagrams Diagrams that illustrate data link layer information, detailing network switches, MAC addresses, and VLAN configurations.

Layer 2 encapsulation A feature of VXLAN that encapsulates a Layer 2 Ethernet frame from a virtual machine (VM) or host into a Layer 3 UDP packet. *See also* Virtual Extensible Local Area Network (VXLAN).

Layer 2 Ethernet switch A transparent switch that is not capable of routing. This switch tends to be lower cost and is designed for access layers in Ethernet networks.

Layer 3-capable switch *See* multilayer switch.

Layer 3 diagrams Diagrams that show the network layer components, including routers, IP addresses, subnets, and routing protocols.

lease time The amount of time that a DHCP client gets to maintain its DHCP-learned information before it must surrender or renew this information.

leased line Typically a point-to-point connection interconnecting two sites. All the bandwidth on that dedicated leased line is available to those sites. Often referred to as a dedicated leased line.

least privilege access An important security best practice that dictates that every user who interacts with the network should always be using a user account that has the least privileges required to their job.

LED status indicators Indicators on a networking device that signal whether operations are functioning as expected.

licensing asset inventory An inventory that tracks the procurement, deployment, and usage of software licenses to ensure compliance with vendor agreements.

life-cycle management The process of overseeing the planning, acquisition, deployment, maintenance, and disposal of assets or systems to ensure they meet organizational needs and optimize their value and performance throughout their life spans.

lightweight access point A Wi-Fi access point that relies on a central controller for configuration and management, simplifying deployment and network maintenance.

Lightweight Directory Access Protocol (LDAP) An open standard for storing directory information for the network, such as usernames and passwords and other user and computer parameters.

Lightweight Directory Access Protocol over SSL (LDAPS) A secure version of LDAP in which Secure Socket Layer (SSL) provides protection.

link aggregation As defined by the IEEE 802.3ad standard, a process that allows multiple physical connections to be logically bundled into a single logical connection.

Link Layer Discovery Protocol (LLDP) A vendor-neutral protocol used by network devices to advertise their identity and capabilities to neighbors. *See also* Cisco Discovery Protocol (CDP).

link-local A nonroutable IP address that is usable only on a local subnet.

link-state A category of routing protocol that maintains a topology of a network and uses an algorithm to determine the shortest path to a destination network.

link-state advertisement (LSA) A message sent by a router in a network to advertise the networks the router knows how to reach. Routers use LSAs to construct a topological map of a network. The algorithm run against this topological map is Dijkstra's shortest path first algorithm.

load balancer A networking device that distributes traffic from clients (or other sources) to multiple nodes to increase performance and availability.

load balancing The process of distributing client requests among different network resources that can provide the same service or data.

local area network (LAN) A network that interconnects network components within a local region (for example, within a building).

local authentication The process in which a network device authenticates a user with a database of user account information stored on the device itself; this is often an important fallback method of authentication used when another external method fails.

local connector (LC) A terminating device used with fiber-optic cabling.

localhost Refers to the default hostname that a computer uses to identify itself, typically corresponding to the IP address 127.0.0.1 on a client system; this address is commonly used to test network services locally.

locking cabinets/racks Cabinets or racks for network equipment that have locking cages over them to assist with physical security.

locks Mechanical or electronic devices used to restrict access to a building, room, or container, ensuring that only authorized individuals with the proper key, code, or credential can gain entry.

log aggregation The process of collecting and centralizing log data from various sources into a single location for easier monitoring, analysis, and troubleshooting.

logic bomb An attacker's malicious code that resides in a software system and will be triggered when certain conditions are met.

logical diagram A diagram that focuses on how information flows through a network infrastructure. Such diagrams tend to omit details used in physical network diagrams because the concern is with the overall information flows through many underlying network devices.

logical security The protection of digital systems and data through software-based measures, such as authentication, encryption, and access controls, to prevent unauthorized access or breaches.

logical topology A topology of the network that is based on the flow of traffic.

Long-Term Evolution (LTE) A standard for wireless broadband communication for mobile devices based on the GSM/EDGE and UMTS/HSPA technologies. It increases capacity and speed by using a different radio interface together with core network improvements.

loopback A logical (virtual) interface on a network device that is often used for testing purposes.

loopback plug A device that connects the transmit pins on an Ethernet connector to the receive pins, such that everything that is transmitted is received back on the interface.

M

MAC filtering A security method in which the MAC address assigned to each network card is used to determine access to the network. It is often used as an additional security measure in Wi-Fi environments.

MAC spoofing An attack in which the cybercriminal pretends to possess a different MAC address from what they actually have on their system. This is often the first step in a much larger and more sophisticated attack.

mail exchange (MX) A DNS record that maps a domain name to an email (or message transfer agent) server for that domain.

main distribution frame (MDF) A central point in a computer network where incoming lines from external networks connect to internal network wiring, serving as the primary hub for distributing and managing network connections.

malware A catchall term for the various types of viruses, worms, spyware, Trojans, and other malicious software or code on a network.

management information base (MIB) A database of variables that exists on an SNMP-managed device.

management layer In software-defined networking, the layer used to describe the various network management protocols that might be in use to manage the various other layers of the solution. *See also* software-defined networking (SDN).

maximum transmission unit (MTU) The largest size of a data packet that can be transmitted over a network medium without requiring fragmentation. It is typically measured in bytes and can help efficient data transmission by preventing packets from being too large for the network to handle.

mean time between failures (MTBF) The predicted elapsed time between inherent failures in a mechanical system.

mean time to repair (MTTR) The predicted time it takes to repair a network or network object.

mechanical transfer registered jack (MTRJ) A type of fiber-optic connector that features a small, rectangular design with two optical fibers and a latching mechanism, commonly used in high-density network environments.

Media Access Control (MAC) flooding A network-based exploit in which an attacker overwhelms a network switch by sending many packets with spoofed MAC addresses.

media converter A device that connects different types of network devices together, such as a fiber-optic device that needs to communicate with a twisted-pair device in a network segment or an adapter used to facilitate connectivity in an environment with a wide variety of copper and fiber cabling used by different network devices.

memorandum of understanding (MOU) An agreement between two parties that outlines the roles and responsibilities that each will play.

mesh A network topology where multiple wireless nodes work together to provide seamless, extended coverage and robust connectivity by dynamically routing data through the most efficient path.

metric A value assigned to a route. Lower metrics are preferred over higher metrics.

metro-optical The optical-based WAN portion of a metropolitan area network. This is often implemented with SONET.

metropolitan area network (MAN) A network that interconnects locations scattered throughout a metropolitan area.

motion detection A physical security approach that involves using sensors to detect motion in a secured area.

multicast A one-to-many communication flow.

multicast flooding The process of sending a multicast packet to all ports of a VLAN.

multifactor authentication (MFA) A type of authentication that requires two or more types of successful authentication before access to a network is granted.

multi-fiber push on (MPO) connector A type of fiber-optic connector that can simultaneously connect multiple optical fibers, typically used for high-density network applications. It facilitates quick and efficient connections in data centers and telecommunication networks.

multilayer switch A switch that can make traffic forwarding decisions based on Layer 3 information. These powerful devices are often modular, so it is possible to equip them with the “perfect” number of interfaces for Layer 2 and Layer 3 functionality.

multimeter A tool used to check the electrical characteristics of cabling, including resistance (in ohms), current (in amps), and voltage (in volts).

multimode fiber (MMF) A type of fiber-optic cabling that has a core with a diameter large enough to permit the injection of light at multiple angles. The different paths (that is, modes) that light travels can lead to multimode delay distortion, which causes bits to be received out of order because the pulses of light representing the bits traveled different paths (and therefore different distances).

multipathing The process of creating multiple paths that data can take in a network. Routing protocols can make these multiple paths very valuable to use since they can engage in what is called equal-cost multipathing (ECMP).

multi-user MIMO (MU-MIMO) A technology for wireless communication that relies on multiple users or terminals, each radioing over one or more antennas to communicate with one another.

multiple-input multiple-output (MIMO) A technology that uses multiple antennas for transmission and reception. These antennas do not interfere with one another, thanks to the use of spatial multiplexing, which encodes data based on the antenna from which the data will be transmitted. Both reliability and throughput can be increased with MIMO’s simultaneous use of multiple antennas.

Multipoint Generic Routing Encapsulation (mGRE) A version of the GRE encapsulation protocol that permits a local device to make multiple GRE connections to various remote devices. This technology is a key ingredient in the Cisco Dynamic Multipoint VPN (DMVPN) solution.

Multiprotocol Label Switching (MPLS) A WAN technology popular among service providers that performs label switching to forward traffic within an MPLS cloud by inserting a 32-bit header (which contains a 20-bit label) between a frame’s Layer 2 and Layer 3 headers and making forwarding decisions based on the label within an MPLS header.

multitenancy A situation in which physical servers in the public cloud infrastructure are hosting workloads for many different customers.

MySQL An open source relational database management system (RDBMS) based on a client/server model.

N

name resolution The process of translating human-readable domain names into their corresponding IP addresses so that networked devices can communicate with each other.

name server (NS) A DNS record that delegates a DNS zone to use the given authoritative name servers.

NAT64 A technology that facilitates communication between IPv6-only clients and IPv4-only servers, helping to bridge the gap between the two distinct IP address families.

native VLAN The default VLAN on a trunk port that carries untagged traffic between switches in an 802.1Q-tagged network.

neighbor discovery A process whereby network elements can discover each other on the network. In the case of EIGRP, hello packets are used, for example.

NetFlow A software system that seeks to collect key network traffic information (statistics) and, most commonly, send these statistics to a central network collector.

NetFlow analyzer Software that can consume NetFlow data from remote systems and provide analysis of the information.

netstat A command that can display a variety of information about IP-based connections on a Windows or UNIX host.

network access control (NAC) A process that uses a set of protocols to ensure that users are identified immediately when they try to access the network. A NAC system might integrate automatic remediation processes that can fix issues with host systems so that they are brought into compliance with the corporate security policies.

network address translation (NAT) A technology that allows private IP addresses (as defined in RFC 1918) to be translated into Internet-routable IP addresses (that is, public IP addresses).

network address translation (NAT) gateway A managed service that enables private network instances within a virtual private cloud to access the Internet while providing secure outbound traffic routing and NAT functionality, ensuring seamless connectivity without exposing internal IP addresses.

network-attached storage (NAS) A centralized storage device that connects to a network, enabling multiple users and devices to access and share files.

network diagram A graphical representation that illustrates the structure, components, connections, and relationships within a computer network.

network discovery The automated process of identifying and mapping devices, services, and connections within a network to provide a comprehensive overview of the network's topology and assets.

network functions virtualization (NFV) A process for virtualizing aspects of a network such as security, storage, compute, and monitoring services.

network interface card (NIC) teaming A technology that permits multiple network interface cards to function as a single interface to the network. NIC teaming provides excellent redundancy.

network interface layer The layer of the TCP/IP stack (also known as the network access layer) that encompasses the technologies addressed by Layers 1 and 2 (that is, the physical and data link layers) of the OSI model.

network layer Layer 3 of the OSI model, which is primarily concerned with forwarding data based on logical addresses.

network loops Occur when multiple paths exist between two network devices, causing data to circulate indefinitely and potentially leading to network congestion or failure.

network security group An access control list that protects virtual machine resources in the cloud. This list specifies the type and source of traffic that is permitted to flow into and out of the cloud-based virtual machine.

network security list An access control list that protects entire subnets in a virtual private cloud. These lists seek to control traffic flows into and out of and between subnets in the VPC.

network segmentation enforcement The practice of dividing a network into smaller, isolated segments to limit access and control traffic flow, thereby enhancing security and reducing the risk of unauthorized access or the spread of threats.

Network Time Protocol (NTP) A protocol that strives to ensure that network devices (including client systems) have the correct time and date. NTP uses port 123.

Network Time Protocol (NTP) server With Network Time Protocol (NTP), a device that provides the correct time to NTP clients. The most common servers these days are located regionally on the Internet.

Network Time Security (NTS) A security protocol designed to protect Network Time Protocol (NTP) communications from various types of attacks, ensuring the integrity and authenticity of time synchronization.

next-hop IP address An IP address on the next router to which traffic should be forwarded.

nmap A management tool that permits the scanning of a network for hosts and services.

non-authoritative name server Provides DNS responses based on cached information rather than being the original source of the DNS records.

nondesignated port In an STP topology, a port that blocks traffic to create a loop-free topology. *See also* Spanning Tree Protocol (STP).

nondisclosure agreement (NDA) A legal contract used to create a confidential relationship between two parties to protect any type of confidential and proprietary information or trade secret.

non-overlapping channels Frequency bands in Wi-Fi networks that do not interfere with each other, allowing multiple devices to communicate simultaneously without cross-channel interference. Non-overlapping channels in the 2.4GHz band are channels 1, 6, and 11. Channel overlap is not an issue in the 5GHz band.

north-south traffic flows The flow of user traffic in a data center from the top layer of the hierarchy to the bottom layer and vice versa.

notifications Automated alerts or messages generated by a system to inform users or administrators about specific events, changes, or conditions that require attention.

nslookup A command that can resolve a fully qualified domain name (FQDN) to an IP address on Microsoft Windows and UNIX hosts.

O

object identifier (OID) A unique identifier for an SNMP management information base (MIB) variable. OIDs are organized in a tree-like hierarchy within a database.

octet A grouping of 8 bits. An IPv4 address consists of four octets (that is, a total of 32 bits).

omnidirectional antenna An antenna that radiates power at relatively equal power levels in all directions (somewhat similar to the theoretical isotropic antenna). Omnidirectional antennas are popular in residential WLANs and SOHO locations.

onboarding/offboarding procedures Procedures for the hiring and termination of employees that relate to IT and the network.

on-path attack An attack in which an attacker attempts to get in the direct path between a client and a server in order to eavesdrop. If cryptography is being used and the attacker fools the client and server both into building VPNs to the attacker instead of to each other, the attacker can see all the data in plaintext. Formerly known as man-in-the-middle attack.

open A broken strand of copper that prevents current from flowing through a circuit.

Open Shortest Path First (OSPF) A link-state routing protocol that uses the metric cost, which is based on the link speed between two routers. OSPF is a popular interior gateway protocol (IGP) because of its scalability, fast convergence, and vendor interoperability.

Open Systems Interconnection (OSI) reference model A seven-layer model that categorizes various network technologies. Commonly referred to as the OSI model or the OSI stack.

operating system (OS) updates Periodic releases from the operating system vendor that include security patches, new features, and improvements to system performance.

operational technology (OT) The hardware and software used to detect or cause changes through direct monitoring and control of physical devices, processes, and events in industrial settings.

optical time-domain reflectometer (OTDR) A device that detects the location of a fault in a fiber cable by sending light down the fiber-optic cable and measuring the time required for the light to bounce back from the cable fault. The OTDR can then mathematically calculate the location of the fault.

orchestration The scheduling and monitoring of many different automations.

out-of-band management Management traffic that is sent outside the networks used for data traffic. It tends to be more secure than in-band management.

P

packet capture A software tool (or sometimes hardware tool) used to intercept, record, and analyze network traffic by capturing data packets traveling across a network.

packet loss The failure of data packets to reach their intended destination across a network, resulting in incomplete or missing information.

partial-mesh topology A hybrid of a hub-and-spoke topology and a full-mesh topology that can be designed to provide an optimal route between selected sites while avoiding the expense of interconnecting every site to every other site.

password attack An attack that aims to compromise a password in order to gain access to systems and files.

password policy A policy that stipulates the required complexity (length, mix of uppercase/lowercase letters, symbols, duration before expiration, and so on) of passwords in order to ensure security.

patch panel A box designed as a junction point for twisted-pair (TP) cable and fiber-optic cable used in networks. Also known as a patch bay.

patches Updates to software that fix security vulnerabilities, fix bugs, or add minor enhancements to improve functionality.

payload A simple and generic method of describing data itself, which is separate and distinct from any of the other information required for proper transmission.

Payment Card Industry Data Security Standards (PCI DSS) A set of security standards designed to ensure that all companies that accept, process, store, or transmit credit card information maintain a secure environment.

peer-to-peer network A network that allows interconnected devices (for example, PCs) to share their resources (such as files or printers) with one another.

penetration testing A type of testing that tries to find vulnerabilities and problems with the security settings of a network or network component. Penetration tests can help find major problems with systems and mitigate those risks before they are the subject of cyberattack.

performance monitoring The continuous tracking and analysis of system metrics to ensure optimal functioning and identify potential issues in hardware, software, or network operations.

personal area network (PAN) A network with a smaller scale than a LAN. A connection between a PC and a digital camera via a USB cable is an example of a PAN.

phishing An attack that uses email or other messages to attempt to capture authentication information (or other information) from an end user.

physical access control device A device used to enforce the security of the physical resources that make up a network and infrastructure.

physical diagram A diagram of specific parts of a network, which typically includes documentation of physical device names, locations, and ports.

physical layer Layer 1 of the OSI model, which is concerned with the transmission of bits on a network.

physical security The protection of personnel, hardware, software, networks, and data from physical actions and events that could cause damage or unauthorized access.

physical topology A network topology (typically a diagram) based on how the components are physically interconnected.

piggybacking Following an authorized employee through a security checkpoint. It is similar to tailgating except that the two individuals moving through the checkpoint are cooperating with each other.

ping One of the most commonly used command-line commands, which can check IP connectivity between two network devices. Multiple platforms (for example, routers, switches, and hosts) support the **ping** command.

platform as a service (PaaS) A cloud service model that can provide a development platform for companies that are developing applications and want to focus on creating the software without having to worry about the servers and infrastructure being used for that development.

playbooks A fundamental tool in IaC automation that provide a structured way to define a series of tasks that automate the provisioning, configuration, and management of infrastructure. *See also* infrastructure as code (IaC).

plenum Fire-retardant cable that minimizes toxic fumes if it catches on fire. Local fire codes often require plenum cabling in raised flooring or in open-air return ducts.

point to point A wireless communication link established directly between two devices, typically for the purpose of connecting two locations over a distance without the need for intermediate devices.

point-to-point network topology A direct connection between two network nodes, allowing data to travel exclusively between these two points.

pointer (PTR) A DNS record that points to a canonical name. A PTR record is commonly used when performing a reverse DNS lookup, which is a process used to determine what domain name is associated with a known IP address.

poison reverse A feature of a distance-vector routing protocol that causes a route received on one interface to be advertised back out that same interface with a metric that is considered to be infinite.

polarization The direction in which an antenna radiates wavelengths. This direction can be vertical, horizontal, or circular.

policy-based authentication A feature of the zero trust architecture (ZTA) that ensures all access requests are dynamically authenticated using predefined security policies that consider a variety of factors.

POP3 over SSL A secure version of Post Office Protocol Version 3 (POP3) that uses Secure Socket Layer (SSL) for security.

port address translation (PAT) A variant of NAT in which multiple inside local IP addresses share a single inside global IP address. PAT can distinguish between different flows based on port numbers.

port mirroring A network feature that duplicates the traffic from one or more network ports to a designated monitoring port for analysis or troubleshooting purposes.

port scanner Software that seeks to discover port states (that is, open or closed) on a network device. In this case, ports can be TCP or UDP ports.

port security 802.1X protection of a port or a Layer 2 security mechanism that can lock down a port to learning specific MAC addresses.

port-side exhaust Network equipment design in which the cooling system directs hot air out through the front (port side) of the device, typically used in data centers to manage airflow and temperature more efficiently.

Post Office Protocol Version 3 (POP3) A protocol used to retrieve email from an email server.

posture assessment The process of examining the state of a network and network nodes from a security perspective.

power distribution unit (PDU) A device fitted with multiple outputs designed to distribute electric power.

power load The total amount of electrical power consumed by all the devices and equipment connected to a power source or system at a given time.

Power over Ethernet (PoE) A network feature defined by IEEE 802.3af and 802.3at that allows an Ethernet switch to provide power to an attached device (for example, a wireless access point, security camera, or IP phone) by applying power to the same wires in a UTP cable that are used to transmit and receive data.

Power over Ethernet Plus (PoE+) A network feature defined by IEEE 802.3at that offers as much as 32.4W of power, enabling PoE to support a wider range of devices than PoE.

Precision Time Protocol (PTP) A protocol used to synchronize clocks across a computer network with high accuracy, often achieving synchronization within nanoseconds.

prefix length A critical criteria for a router making a best path decision; the router will select the route that has the longest matching prefix length.

prefix notation A method of indicating how many bits are in a subnet mask. For example, /24 is prefix notation for a 24-bit subnet mask. Prefix notation is also known as slash notation.

presentation layer Layer 6 of the OSI model, which is responsible for the formatting of data being exchanged and securing the data with encryption.

pre-shared keys (PSKs) A method of performing authentication in a network environment in which each device has a complex string configured and stored locally that must match the string on the remote device.

primary zone The authoritative database for a domain, where all original and writable DNS records are stored and managed.

private cloud Cloud services that include systems that only have interactions and communications with other devices inside that same private cloud or system.

private IP addresses A range of Class A, B, and C addresses that are designed for private use. Although these networks are routable within an organization, service providers do not route these private networks over the public Internet.

private VLAN A sub-VLAN within a main VLAN in a LAN, which can have special rules regarding access. For example, an isolated sub-VLAN cannot communicate with any other private VLAN members. *See also* virtual LAN (VLAN).

process assessment An assessment that looks for flaws in common business processes for an organization. Process assessments often concentrate on many other areas beyond security. For example, an organization may find many inefficiencies in the business processes in use daily.

production configuration The specific set of settings, parameters, and configurations that are actively deployed and operational within a live environment, serving the primary functions and services of an organization's IT infrastructure.

protocol analyzer A software (or perhaps hardware) tool that captures packets and analyzes protocol information from that captured data.

protocol data unit (PDU) Data at different layers of the OSI model. Specifically, a Layer 4 PDU is a segment, a Layer 3 PDU is a packet, a Layer 2 PDU is a frame, and a Layer 1 PDU is a bit.

proxy server A server that intercepts requests being sent from a client and forwards those requests to their intended destination. The proxy server then sends any return traffic to the client that initiated the session. This provides address hiding for the client. Also, some proxy servers conserve WAN bandwidth by offering a content-caching function. In addition, some proxy servers offer URL filtering to, for example, block users from accessing social networking sites during working hours.

public cloud Cloud services that interact with devices on public networks such as the Internet and potentially other public clouds.

public key infrastructure (PKI) A framework of policies, technologies, and services that enables secure communication and authentication over a network by managing digital certificates and public-private key pairs.

punchdown block A mechanism used to cross-connect sets of wires through a metal peg system in a local area network (LAN).

punchdown tool A tool that is designed to properly insert an insulated wire between two contact blades in a punchdown block without damaging the blades.

Q–R

quad small form-factor pluggable (QSFP) A compact, hot-pluggable transceiver also used for data communications applications.

quality of service (QoS) An approach to ensuring that network traffic receives the correct treatment by network devices to foster high performance.

rack diagram A two-dimensional elevation drawing showing the organization of specific equipment on a rack. It is often drawn to scale and typically shows the layout of components from both the front and the back of the rack.

rack size The standard dimensions, typically measured in units (U), of equipment racks used to mount servers and other hardware in data centers and IT environments.

ransomware An exploit in which the attacker locks out access to a system or to files (often using encryption) or at least pretends to have locked out access. The attacker then demands the computer user to pay a ransom to unlock the system or the files.

Rapid Spanning Tree Protocol (RSTP) An enhancement to the original STP standard that improves convergence time and adds other operational improvements. It is defined in IEEE 802.1W.

received signal strength indication (RSSI) signal strength A measure of the power level being received by a local client.

recovery point objective (RPO) The point in time to which an organization can recover a network.

recovery time objective (RTO) The time in the future when an organization expects to restore availability after some failure (or even a disaster) has rendered IT services unavailable.

recursive lookup A DNS lookup in which the client instructs the DNS server that it must provide an answer, and if it cannot, it should not respond with a referral to another DNS server.

registered jack (RJ) A connector found in most networks that is either RJ45 (for Ethernet) or RJ11 (for telephony).

remote access policy A policy that typically applies to remote connections to an organization, including reading or sending email and viewing intranet web resources. Such a policy also tends to cover every remote access option, including dial-up, VPN, and web portal access.

Remote Authentication Dial-In User Service (RADIUS) A UDP-based protocol used to communicate with an AAA server. Unlike TACACS+, RADIUS does not encrypt an entire authentication packet; it encrypts only the password. However, RADIUS offers more robust accounting features than TACACS+. Also, RADIUS is a standards-based protocol, whereas TACACS+ is a Cisco-proprietary protocol.

remote desktop connection A connection to a remote system using RDP (or a similar remote desktop technology).

remote desktop gateway A service available in Windows server systems. It is a very clever use of RDP in which the server becomes a gateway to permit access to the server for management purposes for administrators who need to connect.

Remote Desktop Protocol (RDP) A protocol that permits remote access to an OS desktop remotely from another device on the network. For example, this is the default solution for accessing a Windows system hosted in the cloud via Amazon Web Services (AWS).

repeater *See* hub.

request process tracking The systematic monitoring and documentation of change requests from submission through approval and implementation, ensuring compliance with procedures and providing transparency throughout the change life cycle.

reusable tasks A key aspect of IaC automation that promotes efficiency and maintainability. *See also* infrastructure as code (IaC).

reverse lookup A DNS query in which the client has the IP address of the resource and is trying to resolve to the name that the IP address is associated with.

reverse lookup zone A DNS zone that maps IP addresses back to their corresponding domain names, allowing for the reverse translation from IP address to name.

RF attenuation The weakening of radio frequencies over distance.

RFC1918 A document that defines the range of private IP addresses.

ring topology A network topology in which traffic flows in a circular fashion around a closed network loop (that is, a ring). Typically, a ring topology sends data,

in a single direction, to each connected device in turn, until the intended destination receives the data.

riser-rated Cable that is designed to pass from one floor to another floor inside a building.

risk The potential for loss, damage, or harm resulting from vulnerabilities that can be exploited by threats, impacting assets, operations, or individuals.

risk management A process that involves carefully engaging in the identification, evaluation, and prioritization of risks. It is typically followed by a coordinated application of resources to minimize, monitor, and control the probability or impact of cybersecurity-related events.

RJ11 A telephone interface that uses a cable of twisted wire pairs and a modular jack with two, four, or six contacts. RJ11 is the common connector for plugging a telephone into a wall and a handset into a telephone.

RJ45 The most common Ethernet cable connector, which connects network interface cards on PCs to network switches and SOHO routers.

roaming A feature in modern WLANs that permits users to move throughout an organization without losing Wi-Fi access as they transition to different Wi-Fi cells and equipment.

rogue access point An access point that is not permitted on a network.

rogue DHCP An attack in which the attacker seeks to disrupt (or redirect) end-user traffic by providing false DHCP-leased IP address information.

rogue DHCP server A server that is not authorized to exist on a network.

role-based access control (RBAC) An approach that involves creating permissions around roles that are then assigned to users. An excellent example of the use of RBAC is with guest access.

rollover cable A flattened cable that comes with a network device from the factory. Also called a console cable.

root bridge In an STP topology, the central reference point of the network topology, selected based on the lowest bridge ID, to which all other network switches determine their best path to ensure a loop-free network. *See also* Spanning Tree Protocol (STP).

root DNS server An authoritative DNS server for a root domain in the DNS system. Root DNS servers are located all over the public Internet to ensure proper resolution of the root domains.

root port In an STP topology, the port on a switch that is closest to the root bridge, in terms of cost. Every nonroot bridge has a single root port. *See also* Spanning Tree Protocol (STP).

route A command that can add, modify, or delete routes in the IP routing tables of Microsoft Windows and UNIX hosts. In addition, the **route** command can be used to view the IP routing tables of Microsoft Windows hosts.

route redistribution A process that allows routes learned by one routing protocol to be injected into the routing process of another routing protocol.

route selection The process of determining the optimal path for data packets to travel from a source to a destination within a network.

routed protocol A protocol with an addressing scheme (for example, IP) that defines different network addresses.

router A Layer 3 device that makes forwarding decisions based on logical network addresses.

router advertisement A message that a router sends to advertise its presence together with various link and Internet parameters, either periodically or in response to a router solicitation message.

Router Advertisement (RA) Guard A feature that can filter IPv6 router advertisement messages that the protocol uses to simplify IPv6 address assignments such as the default gateway address.

Routing Information Protocol (RIP) A distance-vector routing protocol that uses hop count as the metric. The maximum number of hops between two routers in an RIP-based network is 15. Therefore, a hop count of 16 is considered to be infinite. RIP is considered to be an interior gateway protocol (IGP).

routing loop A problem that occurs when routers send packets to each other endlessly. The Time-to-Live (TTL) in a Layer 3 packet is one guard against this behavior.

routing protocol A protocol (such as RIP, OSPF, or EIGRP) that advertises route information between routers to describe how to reach specified destination networks.

routing table A table that stores the routing entries that dictate how traffic is forwarded through a network.

runts Ethernet frames that are smaller than the IEEE 802.3's minimum length of 64 octets.

S

sanitize a device To purge a device of all existing data before disposing of the device. This might involve passing powerful magnets over the surface of the storage to ensure that the data can never be recovered.

satellite A device that provides WAN access to sites where terrestrial WAN solutions are unavailable. Satellite WAN connections can suffer from long round-trip delay (which can be unacceptable for latency-sensitive applications) and are susceptible to poor weather conditions.

scalability Refers to a solution's ability to grow with need or demand.

scheduled network discovery The process of performing network discovery based on a predefined schedule.

scope An address pool assigned by a DHCP server.

scope options In DHCP, various additional pieces of information or settings that can be applied to clients of that scope.

screened subnet A subnet that often contains servers that should be accessible from the public Internet. This approach allows users on the Internet to initiate an email or web session coming into an organization's email or web server, but other protocols are blocked. Previously known as DMZ.

secondary zone A read-only copy of a primary DNS zone, used to provide redundancy and load balancing by replicating DNS records from the primary server.

Secure Access Secure Edge (SASE) A network architecture that integrates wide-area networking (WAN) capabilities with comprehensive network security functions such as Secure Web Gateway (SWG), Cloud Access Security Broker (CASB), Firewall as a Service (FWaaS), and Zero Trust Network Access (ZTNA) into a unified, cloud-native service model to securely connect users, systems, and endpoints to applications and services.

Secure File Transfer Protocol (SFTP) A protocol that provides FTP file transfer service over a Secure Shell (SSH) connection.

Secure Shell (SSH) A cryptographic network protocol for operating network services securely over an unsecured network. SSH is used for remote login to computer systems.

Secure Sockets Layer (SSL) A cryptographic protocol designed to provide secure communication over a computer network by encrypting data transmitted between a web server and a client.

Security Assertion Markup Language (SAML) A protocol that enables secure, single sign-on (SSO) by allowing identity providers to pass authorization credentials to service providers.

security information and event management (SIEM) Software products and services that combine security information management (SIM) and security event management (SEM). SIEM systems provide real-time analysis of security alerts generated by applications and network hardware. SIEM systems can log security data and generate reports for compliance purposes.

security policy A continually changing document that dictates a set of guidelines for network use. These guidelines complement organizational objectives by specifying rules for how a network is used.

Security Service Edge (SSE) A cloud-based security framework that delivers essential security services like Secure Web Gateway (SWG), Cloud Access Security Broker (CASB), and Zero Trust Network Access (ZTNA) to protect data and applications, ensuring secure access from any location.

self-signed A digital certificate that is signed by the entity creating it rather than by a trusted certificate authority, often used for internal testing or encryption purposes without third-party validation.

separation of duties policy A network policy that ensures that user accounts do not have too much power. It may also ensure that IT staff are mapped to these various duties and even rotated through them to help prevent internal attacks against the network.

Server Message Block (SMB) A protocol used by Microsoft clients to share files, printers, and other network resources.

service (SRV) A generalized service location DNS record that is used for newer protocols instead of creating protocol-specific records such as MX.

service-level agreement (SLA) A formal contract between a service provider and a customer that outlines the expected level of service, performance standards, and responsibilities, including metrics for measuring service quality and penalties for non-compliance.

service request A formal request for modifications or enhancements to IT services or systems, initiated to address user needs, improve functionality, or resolve issues within an organization.

service set identifier (SSID) A string of characters that identifies a WLAN. APs participating in the same WLAN can be configured with identical SSIDs. An SSID shared among multiple APs is called an extended service set identifier (ESSID).

Session Initiation Protocol (SIP) A Voice over IP (VoIP) signaling protocol that is used to set up, maintain, and tear down VoIP phone calls.

session layer Layer 5 of the OSI model, which is responsible for setting up, maintaining, and tearing down sessions.

shielded twisted pair (STP) Cabling that is shielded to prevent the wires in the cable from acting as an antenna, which might receive or transmit electromagnetic interference (EMI). STP cable might have a metallic shielding, similar to the braided wire that acts as an outer conductor in a coaxial cable.

short A problem that occurs when two copper connectors touch each other, resulting in current flowing through the short rather than through the attached electrical circuit because the short has lower resistance.

shoulder surfing A social engineering attack in which a cybercriminal watches over the shoulder of an authorized employee to learn information that can be used at a later time.

show arp A command that shows the ARP table, which maps IP addresses to their corresponding MAC addresses.

show config A command that is used to examine the configuration of a network device.

show interface A command that permits the examination of the statistics and the status of the interfaces on a network system.

show mac-address-table A command that displays the MAC address table, listing all MAC addresses and their associated interfaces on a switch.

show power A command that displays information about the power consumption and status of power supplies in network devices.

show route A command that is used to view the routing table configuration of a network device.

show vlan A command that provides information about VLANs configured on a switch, including VLAN IDs and associated ports.

signal degradation The loss of quality or strength of a signal as it travels through a medium or over distance.

signal loss A condition that occurs when a wireless device is suffering from interference or other issues, in which the RF signals are not strong enough to allow association or data transfers.

Simple Mail Transfer Protocol (SMTP) A protocol used for sending email throughout a network.

Simple Mail Transfer Protocol Secure (SMTPS) A method of securing Simple Mail Transfer Protocol (SMTP) traffic through the use of Transport Layer Security (TLS).

Simple Network Management Protocol (SNMP) A protocol used to monitor and manage network devices, such as routers, switches, and servers.

single-mode fiber (SMF) Cabling that has a core with a diameter large enough to permit only a single path for light pulses (that is, only one mode of propagation). By having a single path for light to travel, SMF eliminates the concern of multimode delay distortion.

single sign-on (SSO) A process that allows a user to authenticate once to gain access to multiple systems without requiring the user to independently authenticate with each system.

site survey An analysis of an area where wireless is to be deployed. A site survey can identify major issues supporting wireless, such as interference from certain objects, devices, and/or technologies.

site survey report A report that describes the Wi-Fi capabilities and potential risk points in a given installation. Site survey reports are critical during wireless site surveys.

site-to-site VPN A virtual private network (VPN) that connects to another organization site (for example, a headquarters location connecting to a remote branch).

slash notation See prefix notation.

small form-factor pluggable (SFP) A compact, hot-pluggable network interface module used for both telecommunication and data communications applications.

smart locker A locker that accurately logs user access to provide increased security.

smartjack A network interface device that adds circuitry for features such as converting between framing formats on a digital circuit (for example, a T1 circuit), supporting remote diagnostics, and regenerating a digital signal.

snips A tool used to cut cables and other cabling accessories.

SNMP Version 2 (SNMPv2c) A network management protocol used for monitoring and managing network devices, featuring improvements over Simple Network Management Protocol Version 1 (SNMPv1), such as enhanced performance and security, but it still relies on community strings for authentication.

SNMP Version 3 (SNMPv3) A version of Simple Network Management Protocol that provides enhanced security features, including authentication, encryption, and access control, to ensure secure and efficient management of network devices.

social engineering A technique (which often leverages people's desire to be helpful) that attackers use to obtain confidential information. For example, an attacker might pose as a member of an IT department and ask a company employee for their login credentials in order for the IT staff to test the connection.

software as a service (SaaS) A cloud service model in which the details of the servers are hidden from the customer, and the customer's experience is similar to using a web-based application.

software asset inventory A detailed record of all software applications and licenses installed and used within an organization.

software-defined networking (SDN) An approach to computer networking that allows network administrators to programmatically initialize, control, change, and manage network behavior dynamically via open interfaces and provide abstraction of lower-level functionality.

software-defined wide area network (SD-WAN) An overlay solution for a traditional WAN that provides more flexibility, visibility, and manageability to the enterprise WAN.

software management The process of overseeing the acquisition, deployment, maintenance, and life cycle of software applications to ensure they are effectively utilized, updated, and aligned with organizational requirements and standards.

source control A system that manages changes to code, documents, or other data, enabling version tracking, collaboration, and rollback to previous states.

Spanning Tree Protocol (STP) Defined by the IEEE 802.1D standard, a protocol that allows a network to have redundant Layer 2 connections while logically preventing loops, which could lead to symptoms such as broadcast storms and MAC address table corruption.

spectrum analyzer A tool used to measure the magnitude of an input signal versus frequency within the full frequency range of the instrument.

speed A measure of the data send rate; for example, Cat 6 media supports a maximum speed of 10Gbps.

speed tester Software or hardware that is designed to test the true speed of a link, which will vary from the advertised theoretical maximum bandwidth of the technology.

spine and leaf A newer network topology that consists of just two layers: a spine layer and a leaf layer.

split horizon A feature of a distance-vector routing protocol that prevents a route learned on one interface from being advertised back out that same interface.

split tunnel A type of VPN design that has some traffic (only the required traffic) sent to another enterprise location and the remainder of traffic sent out an Internet connection (or other destination). *See also* full tunnel.

SQLnet Oracle's networking software that allows remote data access between programs and the Oracle database or among multiple Oracle databases.

standard operating procedures (SOP) A set of step-by-step instructions compiled by an organization to help workers carry out routine operations.

star topology A network topology that has a central point (for example, a switch) from which all attached devices radiate.

start of authority (SOA) A DNS record that provides authoritative information about a DNS zone, such as email contact information for the zone's administrator, the zone's primary name server, and various refresh timers.

state transition modulation An approach of representing binary digits in which the transition between a voltage level (for example, going from a state of no voltage to a state of voltage, or vice versa, on a copper cable) or the transition of having light or no light on a fiber-optic cable is represented as a binary 1; having no transition in a voltage level or light level from one time period to the next is represented as a binary 0.

stateless address autoconfiguration (SLAAC) A method that allows devices to automatically configure their own IPv6 addresses and network settings without the need for a central server.

static assignment Manual assignment of IP address information for servers or hosts. *See also* dynamic assignment.

static routing A method of manually configuring a network route by specifying fixed paths for data packets to travel within a network.

storage area network (SAN) A specialized network for the storage of data.

straight tip (ST) connector A connector used with fiber-optic cables that utilizes a bayonet-style plug and socket.

stratum An important value in Network Time Protocol (NTP) operations that is like a hop count and measures how far the client is from the accurate time source. The larger the stratum value, the more chance that time on the client is inaccurate because it is more hops away from the accurate time source.

Structured Query Language (SQL) A language used to run powerful queries against the data stored in local or remote databases.

Structured Query Language (SQL) Server A relational database management system designed to store, retrieve, and manage data for various applications using Structured Query Language (SQL).

subinterfaces Logical interfaces that can be created under a physical device interface.

subnet mask A 32-bit value (in IPv4) that indicates what portion of the IP address is the network ID versus what portion is the host ID.

subnetting The process of dividing a larger IP network into smaller, more manageable subnetworks (subnets) to optimize routing and improve network performance and security.

subscriber connector (SC) A fiber-optic connector with a push-pull latching mechanism that provides quick insertion and removal while ensuring a positive connection.

supervisory control and data acquisition (SCADA) A system used for monitoring and controlling industrial processes, infrastructure, and facilities by collecting real-time data from sensors and devices, enabling remote management and automation.

supplicant In a network using 802.1X user authentication, the device that wants to gain access to a network.

suspended A state in which a network port has been temporarily disabled or blocked, preventing any data traffic from passing through until it is reactivated or reset.

switch A network device that learns which devices reside off which ports, interrogates traffic to see where it's destined, and forwards the traffic out the appropriate port and not out all the other ports.

Switch Virtual Interface (SVI) A logical interface on a network switch that allows for the routing and management of traffic between VLANs.

switching loop A problem that occurs when switches forward frames endlessly back and forth. Spanning Tree Protocol (STP) was designed to permit switches to eliminate switching loops caused by redundancy.

Syslog A logging solution that consists of two primary components: syslog servers (which receive and store log messages sent from syslog clients) and syslog clients (which can be a variety of network devices that send logging information to a syslog server).

syslog collector A centralized server or application that gathers, stores, and manages log messages generated by network devices, servers, and applications for monitoring, analysis, and troubleshooting purposes.

system life cycle A model that provides valuable guidance on best practices throughout an organization concerning the network components from design through disposal.

T

tabletop exercises Simulated, discussion-based sessions in which team members review and practice their roles and responses to hypothetical disaster scenarios to test and improve an organization's disaster recovery plan.

tailgating A social engineering attack in which the attacker follows a valid employee through a secured area of an organization.

tamper detection The use of a device to sense when an active attempt to compromise the device integrity or the data associated with the device is in progress; the detection of the threat may enable the device to initiate appropriate defensive actions.

tap A hardware device that provides a way to monitor and capture data traffic on a network by copying packets passing through a specific point, allowing for analysis without interrupting the flow of traffic.

TCP flag A field in TCP headers used to indicate a particular connection state or provide additional information. It is often used for troubleshooting purposes or to control how a particular connection is handled.

TCP header A data packet header that contains ten mandatory fields totaling 20 bytes (or octets). The header holds information about the connection and the data currently being sent.

TCP/IP stack A four-layer model (as opposed to the seven-layer OSI model) that targets the suite of TCP/IP protocols. Also known as the DoD model because it was developed by the U.S. Department of Defense.

tcpdump A common packet analyzer that is run at the command line. It allows the user to display TCP/IP and other packets being transmitted or received over a network to which the computer is attached.

Telnet A method of remote access for network devices that does not provide any security mechanisms.

temperature A measure of the average kinetic energy of the particles in a substance, indicating how hot or cold the substance is; network devices specify optimum temperature values to ensure they are not damaged.

templates Used in IaC to define infrastructure resources in a reusable and consistent manner. Templates can include variables, allowing for parameterization and flexibility in resource configurations. *See also* infrastructure as code (IaC).

Temporal Key Integrity Protocol (TKIP) A security protocol used in the IEEE 802.11 wireless networking standard that was designed to make improvements over security flaws in WEP. TKIP is no longer considered secure and was deprecated in the 2012 revision of the 802.11 standard.

Terminal Access Controller Access Control System Plus (TACACS+) A TCP-based protocol used to communicate with an authentication, authorization, and accounting (AAA) server. Unlike RADIUS, TACACS+ encrypts an entire authentication packet rather than just the password. TACACS+ is a Cisco-proprietary protocol.

terminal emulator A computer program that emulates a video terminal within some other display architecture.

text (TXT) A DNS record that was originally meant for arbitrary human-readable text. Since the early 1990s, however, this record has carried machine-readable data.

threat A realistic attack that might possibly occur against a network. Threats often target network vulnerabilities.

threat assessment The process of evaluating the credibility and seriousness of potential threats. A threat assessment involves analyzing the probability that a threat will become a reality.

three-tier hierarchical model A classic network design that consists of a core layer, a distribution layer, and an access layer.

throughput A measure of the successful data send rate between systems.

TIA/EIA-568A A wiring standard that uses the following wires from pin 1 to 8: green stripe, green, orange stripe, blue, blue stripe, orange, brown stripe, brown.

TIA/EIA-568B A wiring standard that uses the following wires from 1 to 8: orange stripe, orange, green stripe, blue, blue stripe, green, brown stripe, brown.

time-based authentication A security method that uses temporary, time-sensitive codes to verify a user's identity.

time-division multiplexing (TDM) A method of transmitting different communication sessions (for example, different telephone conversations in a telephony

network) on the same physical medium by allowing sessions to take turns. For a brief period of time, defined as a time slot, data from the first session is sent, followed by data from the second session. This continues until every session has had a turn, and the process repeats.

time protocols Standardized methods used by network devices to synchronize their clocks accurately across a network, ensuring consistent timing for operations, logging, and security functions.

Time-to-Live (TTL) A field in an IP header that is decremented once for each router hop. If the value in a TTL field is reduced to 0, a router discards the frame and sends a time exceeded Internet Control Message Protocol (ICMP) message back to the source.

tone generator A tool that is used in conjunction with a toner probe on a punch-down block to audibly detect to which pair of wires the tone generator is connected.

toner A tool used in network cabling to trace and identify individual wires or cables by sending a tone signal through the cable, which is then detected by a hand-held probe; this tool is sometimes called a toner probe.

top-of-rack switching A form of data center switching in which the switching occurs at the top of each rack of servers.

top-to-bottom troubleshooting A troubleshooting methodology in which the engineer starts at the application layer of the OSI model and works down, trying to determine the problem.

traceroute/tracert A UNIX (**traceroute**) or Windows (**tracert**) command that displays every router hop along the path from a source host to a destination host on an IP network. Information about the router hop can include the IP address of the router hop and the round-trip delay of that router hop.

traffic access point (TAP) A hardware device inserted at a specific point in a network where data can be accessed for testing or troubleshooting purposes. Also called a test access point.

traffic analysis The process of examining data packets and network flows to monitor, diagnose, and optimize network performance and security.

traffic log A log file in a network that details traffic statistics for key areas of the infrastructure.

traffic policing/shaping Tools that make it possible to limit available bandwidth.

transceiver A device that makes it possible to uplink one Ethernet switch to another if different connectors (for example, MMF, SMF, or UTP) for different installations are required.

Transmission Control Protocol (TCP) A connection-oriented transport protocol that provides reliable transport, in that if a segment is dropped, the sender can detect that drop and retransmit that dropped segment. Specifically, a receiver acknowledges segments that it receives. Based on those acknowledgments, a sender can determine which segments were successfully received.

transport agnostic A feature of the SD-WAN that means it can leverage any available connectivity option, such as broadband, MPLS, LTE, or even satellite links.

transport layer (OSI model) Layer 4 of the OSI model, which acts as a dividing line between the upper layers and the lower layers. Specifically, messages are taken from the upper layers (Layers 5–7) and encapsulated into segments for transmission to the lower layers (Layers 1–3). Similarly, data streams coming from lower layers are decapsulated and sent to Layer 5 (the session layer) or some other upper layer, depending on the protocol.

transport layer (TCP/IP stack) Similar to the transport layer of the OSI model, the transport layer of the TCP/IP stack is responsible for end-to-end communication between applications, providing services like error recovery, flow control, and data segmentation. It uses protocols like TCP (connection-oriented, reliable) and UDP (connectionless, lightweight). This layer ensures proper data delivery to the appropriate application processes on the destination device.

Transport Layer Security (TLS) A security protocol designed to ensure privacy between communicating client/server applications.

traps An unsolicited SNMP message sent from a managed device to an SNMP manager that notifies the SNMP manager about a significant event that occurred on the managed device.

Trivial File Transfer Protocol (TFTP) A file transfer protocol that does not have the security or error checking of FTP. TFTP uses UDP as a transport protocol and therefore is connectionless.

trunk (1) In the context of an Ethernet network, a single physical or logical connection that simultaneously carries traffic for multiple VLANs. (2) In the context of telephony, an interconnection between telephone switches.

trusted zone A network segment deemed secure, with strict access controls to protect sensitive resources and data.

tunneling The process of transmitting traffic with additional encapsulation.

twinaxial A type of cable similar to coaxial cable but with two inner conductors instead of one.

twisted-pair Today's most popular media type, in which individually insulated copper strands are intertwined into a twisted-pair cable. Two categories of twisted-pair cable are shielded twisted pair (STP) and unshielded twisted pair (UTP).

U

UDP header In a UDP datagram, a header that contains four fields (source port, destination port, length, and checksum) totaling 8 bytes.

ultra-physical contact (UPC) A style of fiber-optic connector with back reflection around -55 dB.

unicast A one-to-one communication flow.

unidirectional antenna A style of Wi-Fi antenna that directs the signal in a specific direction. *See also* omnidirectional antenna.

Uniform Resource Locator (URL) filtering Restricts access to specific websites by allowing or blocking URLs based on predefined security policies.

uninterruptible power supply (UPS) An appliance that provides power to networking equipment in the event of a power outage.

unshielded twisted pair (UTP) Cabling that blocks electromagnetic interference (EMI) from the copper strands making up a twisted-pair cable by having more tightly twisted strands (that is, more twists per centimeter). Because these strands are wrapped around each other, the wires insulate each other from EMI.

untrusted zone A network segment with higher risk, such as external networks, where security threats are more likely, requiring stringent controls to protect internal systems.

upgrades The process of updating infrastructure components, such as software versions, configurations, or hardware specifications, to newer versions or improved states, while maintaining the infrastructure's code-based management and deployment.

URL filtering *See* Uniform Resource Locator (URL) filtering.

User Datagram Protocol (UDP) A connectionless transport protocol that provides unreliable transport, in that if a segment is dropped, the sender is unaware of the drop, and no retransmission occurs.

V

validation tests Systematic evaluations conducted to ensure a new disaster recovery solution meets specified requirements and functions correctly under simulated disaster conditions.

Variable-Length Subnet Masking (VLSM) The process of assigning various subnetwork IDs in a network to issue the appropriate number of IP addresses.

vendor assessment An assessment in which a team analyzes the many vendors the organization might use in their business processes. It considers how much data of the organization the vendors have access to and what security practices the vendors follow.

version control A feature of IaC that enables teams to track changes to infrastructure code over time, maintaining a history of modifications, additions, and deletions. *See also* infrastructure as code (IaC).

virtual desktop A solution that allows a user to store data in a centralized data center rather than on the hard drive of the local computer. With appropriate authentication credentials, the user can access their data from various remote devices (for example, their smartphone or another computer).

Virtual Extensible Local Area Network (VXLAN) A network virtualization technology that extends Layer 2 networks over a Layer 3 infrastructure by encapsulating Ethernet frames within UDP packets, enabling scalable and flexible network segmentation.

virtual IP (VIP) In load-balancing environments, an address that is used to provide a single IP address for accessing a shared resource.

virtual LAN (VLAN) A single broadcast domain, representing a single subnet. Typically, a group of ports on a switch is assigned to a single VLAN. For traffic to travel between two VLANs, that traffic needs to be routed.

virtual network computing (VNC) An alternative to RDP that allows the access of a desktop from another system on the network. *See also* Remote Desktop Protocol (RDP).

virtual network interface card (vNIC) A virtualized network interface that permits virtual machines to connect to virtual and physical networks beyond the local host.

virtual private cloud (VPC) A virtual network environment within a cloud computing platform, such as AWS, that enables users to provision and manage a logically isolated section of the cloud where they can deploy and run their resources with control over IP addressing, routing, and security configurations.

virtual private network (VPN) A set of security protocols that, when implemented by two devices on either side of an insecure network such as the Internet, can allow the devices to send data securely. VPNs provide privacy, device authentication, anti-replay services, and data integrity services.

Virtual Router Redundancy Protocol (VRRP) A protocol that increases availability by providing automatic assignment of available IP routers to hosts.

visual fault locator A handheld device that emits a visible red laser to detect breaks and faults in fiber-optic cables.

VLAN database A centralized repository on a network device that stores the configuration details and information about all VLANs on a network.

VLAN hopping An attack in which the attacker uses Q-in-Q tunneling in conjunction with the native VLAN feature to send packets into a VLAN from an unauthorized VLAN.

voice gateway Often considered the most critical component in a VoIP network, a gateway that connects the VoIP of an enterprise to the global public telephone network.

Voice over IP (VoIP) Convergence of voice (telephone) traffic and the data networks of today. VoIP introduces new and exciting features for communication that were not possible under the legacy voice networks.

voice VLAN A dedicated VLAN specifically configured to prioritize and manage voice traffic separately from data traffic on a network to ensure better quality of service for VoIP communications.

voltage The electric potential difference applied to network cables and devices to ensure proper signal transmission and power delivery. Voltage is crucial for maintaining signal integrity and the functioning of Power over Ethernet (PoE) systems, where specific voltage levels are required to power network devices such as IP phones, IP cameras, and wireless access points through Ethernet cables.

VPN headend A network device typically found in the headquarters of an enterprise network that terminates VPN connections from remote clients and/or branch offices.

vSwitch A virtual switch that performs Layer 2 functions (for example, VLAN separation and filtering) between various server instances running on a single physical server.

vulnerability A weakness in a component of software that might be exploited by an attacker. New code might be installed on a network device (a router, for example), and this code could cause a vulnerability. For example, the code could introduce a new feature that permits a backdoor into the device for unauthorized persons.

vulnerability assessment An assessment that seeks to identify, quantify, and prioritize the vulnerabilities in a system. Vulnerability assessments can be performed for a wide variety of systems within the IT department and the network as a whole.

W

warm site A redundant site that can be brought online with minimal time and effort.

warranty support asset inventory An inventory that monitors the warranty status and expiration dates of hardware assets to manage replacements and repairs.

wide area network (WAN) A network that interconnects geographically separated network components.

Wi-Fi analyzer Software that runs on a general-purpose computer or on a specialized device that can perform wireless analysis of Wi-Fi signals. This type of tool can be used as part of a wireless site survey after Wi-Fi has been implemented to create a heat map of the wireless airspace.

Wi-Fi Protected Access (WPA) A security standard developed by Wi-Fi Alliance (a nonprofit organization formed to certify interoperability of wireless devices) to address the weaknesses of Wired Equivalent Privacy (WEP). This new security standard was called Wi-Fi Protected Access (WPA) Version 1.

Wi-Fi Protected Access 2 (WPA2) A security protocol for wireless networks that uses Advanced Encryption Standard (AES) to provide strong encryption and improved security over its predecessor, WPA.

Wi-Fi Protected Access 3 (WPA3) A security protocol for Wi-Fi networks that provides enhanced encryption and authentication mechanisms to improve protection against unauthorized access and cyber threats. Note this version provides many improvements to its predecessor, WPA2.

wipe configuration The standard procedure used to wipe the configuration from a networking device when disposal is needed. In some cases, this can be done remotely for a device that is lost or stolen.

wire map testing A type of testing used with cables to find opens, shorts, reversed pairs, crossed pairs, and split pairs.

wireless bands The specific frequencies that are capable of carrying modern Wi-Fi. IEEE 802.11 uses various frequency bands, including the 2.4GHz, 5GHz, 6GHz, and 60GHz bands.

wireless channel An indication of how the Wi-Fi frequency bands are divided up in order to carry the Wi-Fi signal. Countries apply their own regulations to the available channels to define such factors as maximum power levels within these frequency ranges.

wireless LAN controller (WLC) A specialized network device that permits the central control and management of large numbers of lightweight access points.

wireless local area network (WLAN) A wireless network that interconnects network components that are geographically separated.

wireless survey A detailed assessment of a physical area to measure and analyze the coverage, performance, and signal strength of wireless networks to optimize connectivity and identify potential issues.

wiring diagram A key piece of documentation that details wiring and port locations. This documentation makes it possible to track cable runs from switches and map them to the wall jacks where users connect to the network; these connections might also represent trunks to additional network devices such as wireless access points.

WPA Enterprise mode A security mode of WPA2 designed for corporate environments, using 802.1X authentication with a RADIUS server to provide robust security and individual user credentials.

WPA Personal mode Also known as WPA-PSK (Pre-Shared Key), a security protocol for Wi-Fi networks that uses a shared password for authentication without requiring an enterprise authentication server.

WPA3 A security protocol for wireless networks that enhances encryption, authentication, and privacy compared to its predecessors, WPA2 and WPA, to protect Wi-Fi communications against more advanced cyber threats.

X-Z

Zeroconf A technology that performs three basic functions: assigning link-local IP addresses, resolving computer names to IP addresses, and locating network services.

zero-day Cybersecurity attacks that take advantage of techniques or vulnerabilities that have not already been discovered by cybersecurity specialists or anti-malware companies.

zero-touch provisioning (ZTP) A significant feature of SD-WAN, simplifying the deployment and management of network devices. With ZTP, network administrators can configure and deploy new branch devices without manual intervention. *See also* software-defined wide area network (SD-WAN).

zero trust architecture (ZTA) A security model centered on the principle that no entity, whether inside or outside the network, should be trusted by default. Instead, verification is required from everyone trying to access resources within the network, ensuring robust and granular security.

zone transfer A process by which different DNS servers can sync their databases. Zone transfers are often performed when a primary DNS server updates one or more secondary (backup) DNS servers.

zones In networking, zones are defined segments within a network that separate and isolate different types of traffic or devices, enhancing security by controlling access between areas with varying levels of trust.

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